Mechanical Engineering Series

Ronald N. Miles

Physical Approach to Engineering Acoustics

Second Edition



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The Matlab programs will be available on the https://www.springer.com/in/book/9783030226756

ISSN 0941-5122 ISSN 2192-063X (electronic)
Mechanical Engineering Series
ISBN 978-3-031-33008-7 ISBN 978-3-031-33009-4 (eBook)
https://doi.org/10.1007/978-3-031-33009-4

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With love to Carol, Amy, and Rebecca, by far the best people I know.

Preface

Acoustics is a subject that intersects with nearly every academic discipline. This results from the fact that hearing is a primary means of human communication, and hence, is of central importance. No attempt is made in this book to cover or even mention the broad range of topics that overlap with acoustics. Instead, our attention is limited to some of the topics that touch the field of mechanical engineering. Significant attention is paid to the analysis and design of electro-acoustic devices such as loudspeakers (or ear buds) and microphones. These devices play a major role in billions of electronic products since sound provides an increasingly important interface between humans and technology.

The material included here was developed for two graduate courses, ME522 Acoustics, and ME622 Advanced Acoustics, offered in the Department of Mechanical Engineering at the State University of New York at Binghamton (also known as Binghamton University). These courses introduce graduate students (and advanced undergraduates) to some of the tools needed in engineering acoustics. Students in engineering, mathematics, or physics who have had a course on partial differential equations should be well-prepared for the material. The approach taken in these courses follows from a physical (or physics-based) view of the governing principles of acoustics. This differs from a common, popular view of acoustical systems using a representation based on equivalent electrical circuits.

The equivalent circuit approach can be extremely valuable for analyzing the complexities of coupled acoustical systems and networks. It proved particularly helpful to the earliest acoustics researchers during the first half of the twentieth century, most likely because the field was dominated by an interest in electro-acoustic systems. Many of the pioneers during this period were electrical engineers so the idea of equivalent electrical circuits was a very natural way to model these complicated systems. While arguments over the advantages or disadvantages of a particular approach will be left for others, the approach taken here is to rely on a Newtonian, or physics-based approach to model acoustical systems rather than construct analogous equivalent circuits.

The chapters are organized as follows: Chapter 1 introduces what I believe are essential definitions and principles that any acoustical engineer should know in order

to properly calculate and interpret acoustical data. Because acoustical signals tend to be complicated, much of the first chapter deals with spectral analysis. The analysis of time domain signals is something that is often not emphasized in mechanical engineering curricula (unlike in electrical engineering), so it is hoped that some of the material here can help fill the gap. Chapter 2 introduces the basic equations of acoustical wave propagation in one dimensional systems. Chapters 3 and 4 use the analysis of one dimensional waves introduced in Chap. 2 to examine two important applications in engineering acoustics, sound transmission through walls and the design of mufflers. Chapter 5 introduces methods for analyzing sound propagation in three dimensions, with emphasis on understanding how sound is radiated by vibrating surfaces. This chapter includes an analysis of the sound radiated by loudspeakers, which can also be applied to analyze and design miniature sound sources such as ear buds.

Chapter 6 deals with computational methods for predicting sound fields. The approach here is based on a boundary element method which follows from the use of Green's functions to solve the acoustic wave equation. Chapter 7 covers the solution of the wave equation for bounded domains through expansions in eigenfunctions which is shown to lead to essentially the same result as obtained in Chap. 6 for a particular rectangular enclosure example. In many engineering applications, particularly in noise control, one does not have the luxury of performing a detailed analysis of a sound field using advanced computational methods which leads one to rely on the approximate, geometrical acoustic methods introduced in Chap. 8. These methods are extremely helpful in providing insight into how to reduce the unwanted noise in a space.

Chapter 9 presents a modification to the basic differential equations to account for viscous effects, which can be important for devices such as miniature microphones. When sound interacts with very small structures in air, viscosity can have a marked influence and greatly complicates the analysis and design. Chapter 10 deals with the sensing of sound and provides a detailed description of the working principles in microphones. Nearly all current microphones are designed to detect acoustic pressure through the use of a thin, flexible diaphragm or membrane. A package, or enclosure is normally included in the design to ensure that sound primarily drives only one side of the diaphragm. The enclosure has a marked influence on the performance of the microphone. The vast majority of current microphones (particularly miniature microphones) employ capacitive sensing and must be designed with the electronic interface in mind. The electronic transduction of the motion of the microphone diaphragm is treated in Chap. 11. Because capacitive transduction plays a central role in nearly all current miniature microphones, a detailed review of the estimation of capacitance is presented in Chap. 12. Nearly all current capacitive sensors may be designed using a very simple algebraic formula for the capacitance of parallel plates. It is likely that the existence of this simple formula has caused designers to gravitate toward configurations that can be analyzed by this simple formula. As a result, a host of other possible capacitive sensing configurations have been ignored. Extending one's view to unconventional electrode configurations may enable higher performance designs that avoid the common drawbacks of parallel plate sensors.

Finally, Chap. 13 introduces methods of creating models of acoustical devices based on measured data. This is an extension of well-known least squares curve fitting which is applied to all sorts of measured data. The identification of parameters in a model by processing measured results is a vexing problem since countless models can be created from the same set of data. An empirical model can, however, prove quite useful in helping the designer understand the underlying principles in a given device and to provide a convenient means of calculating performance.

The material in this book can be divided into introductory and more advanced courses in acoustics. A beginning course which includes material that I believe is essential for practicing acoustical engineers would cover most of Chaps. 1–5, 8 and 10. Chapters 6, 7, 9, 11–13 could be covered in a second course.

I would like to express my appreciation to some of those who I've learned from: Michael Carroll, who taught my first courses in acoustics and served as my mentor and advisor while I was an undergraduate at Berkeley and later as a Research Engineer after graduate school. His joyful pursuit of truth was infectious and has had a deep enduring influence on me. Leo Butzel who patiently introduced me to random vibrations and orthogonal functions while I was on the Noise Staff at The Boeing Company. Per Reinhall, my graduate advisor at the University of Washington, who taught me linear and nonlinear vibrations and continues as my friend and mentor. I owe a great deal to Prof. Ronald Hoy at Cornell University, who has served as my mentor for three decades. He has taught me a great deal about acoustics, hearing, and research. Finally, I'd like to thank my colleagues and my students at Binghamton University who have helped make the department a great place to learn.

Binghamton, NY, USA Ronald N. Miles

Contents

1 Analysis of Acoustic Signals 1
- 1.1 Some Basics—Measures of Sound 1
- 1.2 One-Third Octave Band Levels 3
- 1.3 A-Weighted Sound Levels 3
- 1.4 Narrowband Signal Analysis 5
- 1.5 Power Spectral Density by the Finite Fourier Transform 7
- 1.6 Spectral Analysis of Measured Time Series 9
- 1.7 Sound Level Calculations from Narrowband Power Spectra 13
- 1.8 The "Slow" Fourier Transform: Least Squares Extraction of a Harmonic Signal 15
- 1.9 Frequency Response of Linear Systems with Random Input 18
- 1.10 Input–Output Relationships for a Linear System 21
- 1.11 Spectral Approach to Evaluating the Convolution Integral 23
- 1.12 Example: FFT for Response of a Spring/Mass/Damper 26
- 1.13 Example: Numerical Differentiation and Integration Using the FFT 26
- 1.14 FFT and iFFT to Estimate the Fourier Transform for Complex Functions 29
- 1.15 Problems 31

2 One Dimensional Sound Fields 35
- 2.1 Newton's Second Law 35
- 2.2 Conservation of Mass 37
- 2.3 Equation of State 38
- 2.4 Solutions for Some Simple Fields 38
- 2.5 Sound Intensity and the Sound Absorption Coefficient 41
- 2.6 d'Alembert's Solution 44
- 2.7 Sound in an Infinite Tube 47
- 2.8 Problems 49

3 Sound Transmission Loss 53
3.1 The Mass Law 53
3.2 Random Incidence Sound Transmission Loss 56
3.3 Transmission Loss Calculations Using Transfer Matrices 62
3.4 Transfer Matrix for a Solid Element 65
3.5 Transfer Matrix for an Air Gap 66
3.6 Air Gap with Non-normal Incident Sound 68
3.7 Double Walls 70
3.7.1 Simple Air Gap 73
3.7.2 Low Frequencies 74
3.7.3 Slightly Higher Frequencies—Double Wall Resonance 74
3.7.4 Very High Frequencies 75
3.8 Analysis of Double Walls with Mechanical Coupling 76
3.9 Problems 80
References for Chapter 3 82

4 Analysis of Mufflers and Ducts 83
4.1 The Junction of Two Pipes 83
4.2 The Expansion Muffler 84
4.3 The Helmholtz Resonator 86
4.4 Side Branches 88
4.5 General Side Branch 90
4.6 Simple Pipe Side Branch 90
4.7 Sound in Tubes of Varying Cross-Sectional Area 92
4.7.1 Conical Horn 94
4.7.2 Assemble Conical Elements to Model a Tube with Arbitrary Cross-Sectional Area 95
4.7.3 Solving for the Pressure and Velocity Within the Domain 96
4.7.4 The Exponential Horn 98
4.7.5 Numerical Example for a Non-uniform Duct 100
4.8 Problems 101
Reference for Chapter 4 102

5 Sound Radiation in Three Dimensions 103
5.1 Wave Equation in Three Dimensions 103
5.2 Sound Intensity of Spherical Waves 106
5.3 Sound Radiation by a Simple Source 107
5.4 Sound Field Inside a Pulsating Sphere 108
5.5 Simple Model of Sound Radiation from a Loudspeaker 109
5.6 The Image Source 114
5.6.1 Periodic Frequency Response Due to Reflections 118
5.7 The Acoustic Dipole 119
5.8 The Line Source 121

5.9 Surface Source . . . 123
5.10 Sound Radiation from a Piston in a Baffle . . . 124
5.11 Single Piston . . . 124
5.12 Piston Vibration . . . 127
5.13 Multiple Piston Radiators . . . 129
5.14 Analysis and Design of Loudspeakers in Vented Boxes . . . 131
5.14.1 Loudspeaker Diaphragm in a Closed Box . . . 132
5.14.2 Diaphragm Force Due to Back Volume of Air . . . 132
5.14.3 Effect of the Air in the Vent . . . 134
5.14.4 Response Due to Harmonic Sound Fields . . . 135
5.14.5 Sound Radiation from Loudspeakers in Vented Boxes . . . 136
5.15 Problems . . . 138

6 Computer-Aided Acoustics . . . 141
6.1 The Green's Function . . . 142
6.2 Green's Function for Infinite Half-Space Bounded by an Infinite Rigid Plane . . . 144
6.3 Integral Equation for the General Sound Radiation Problem . . . 147
6.4 Numerical Solution . . . 149
6.5 Effect of Boundary Impedance . . . 151
6.6 Sound Field Due to a Point Source in an Enclosure . . . 152
6.7 Calculation of the Pressure at an Arbitrary Point in Space . . . 154
6.8 Approximate Evaluation of Integrals . . . 156
6.8.1 Elements that Are Sufficiently Separated: Estimation Using Midpoints . . . 157
6.8.2 Estimation of B_{ll} . . . 159
6.8.3 Estimation of C_{ll} . . . 160
6.8.4 Computer-Aided Design Model to Specify Geometry . . . 161
6.8.5 Evaluation of Singular Integrals for C_{ll} . . . 162
6.9 Numerical Results . . . 163
6.9.1 Sound Inside a Rectangular Tube . . . 163
6.9.2 Sound Inside a Radially Oscillating Sphere . . . 166
6.9.3 Sound Radiation from a Radially Oscillating Cylinder . . . 167
6.9.4 Sound Radiation from a Radially Oscillating Cylinder . . . 172
6.10 Problems . . . 173
References for Chapter 6 . . . 173

7 Modal Solutions for the Sound in Enclosures 175
7.1 Natural Frequencies and Eigenfunctions for the Rectangular Enclosure 176
7.2 Solution for the Pressure Field 180
7.3 Numerical Results 182

8 Geometrical Room Acoustics 185
8.1 Relation Between Energy Density and Intensity 186
8.2 Effect of Sound Absorption Coefficient on Steady-State Sound Fields 188
8.3 Effect of Sound Absorption Coefficient on Decaying Sound Fields 190
8.4 Problems 191

9 Effects of Viscosity 193
9.1 Basic Equations for Sound in a Viscous Fluid 193
9.2 Viscous Flow in One Dimension 197
9.3 Squeeze Film Damping in a Compressible Fluid in Two Dimensions 199
9.3.1 Limiting Cases at Low and High Frequencies 203
9.3.2 Numerical Results 205
9.4 Viscous Force on an Oscillating Cylinder 206
9.5 Viscous Acoustic Excitation of a Thin Beam Having Circular Cross Section 210
9.6 Solutions for the Acoustic Response 213
9.6.1 Response of an Infinitely Long Uniform Fiber 213
9.6.2 Response of a Fiber of Finite Length 214
9.7 Pressure and Flow Around a Periodic Array of Thin Beams Including Viscosity and Compressibility 217
9.7.1 Fourier Transform Solution for the Velocities and Pressure 219
9.7.2 Numerical Evaluation 222
9.7.3 Numerical Example 225
9.8 Sound Pressure and Viscous Flow Around a Periodic Array of Thin Beams 226
9.8.1 Numerical Results 230
9.9 Problems 234
References for Chapter 9 234

10 Acoustic Sensing 237
10.1 Mechanical Sensitivity of a Microphone Diaphragm 237
10.2 Diaphragm with No Vent 237
10.3 Diaphragm Force Due to Back Volume of Air 238
10.4 Effect of the Air in the Vent 240
10.5 Response Due to Harmonic Sound Fields 242
10.6 The Microphone Package as a Helmholtz Resonator 243

10.7 An Important Note on the Ideal Diaphragm Properties 244
10.8 Damping Due to the Vent 245
10.9 Acoustic Radiation Loading on the Diaphragm 246
10.10 Numerical Examples 246
10.11 Analysis of Thermal-Mechanical Noise in a Pressure-Sensing Microphone 249
10.12 Limiting Case of Thermal-Mechanical Noise in an Ideal Pressure-Sensing Microphone 252
10.13 Thermal Response of a Velocity Driven Spring/Mass 254
10.13.1 Application to a Hair for Sensing Sound 257
10.13.2 Thermal Noise Response Increases with the Number of Degrees of Freedom 258
10.14 Viscous Flow-Induced Response to Sound and Signal to Noise Ratio, SNR 261
10.15 Effect of Sound Angle of Incidence on a Pressure Microphone 262
10.16 Acceleration Sensitivity 265
10.17 Basics of Directional Microphones 266
10.17.1 First-Order Directional Microphones 267
10.17.2 Second-Order Directional Microphones 269
10.18 The Ribbon Microphone 270
10.19 Effect of Gain Errors on Directivity Index 272
10.20 Empirical Microphone Compensation for Desired Directivity 276
10.21 Noise in Directional Microphones 279
10.22 Directivity Index by Integration in the Wave Vector Domain 282
10.23 Acoustic Particle Velocity Estimation Using a Pressure Differential Microphone 285
10.23.1 Measurement of Acoustic Particle Velocity 285
10.23.2 Pressure Gradient and Velocity Microphones 286
10.23.3 Measurement Error 287
10.24 Two-Microphone Intensity Measurement 288
10.25 Problems 291
References for Chapter 10 292

11 Electronic Transduction for Acoustic Sensors 293
11.1 Electrical Sensitivity of a Microphone 293
11.1.1 Microphone Sensitivity Using a Constant Charge, High Impedance Voltage Amplifier 294
11.1.2 Effect of Buffer Amplifier Input Impedance—Parasitic Capacitance 295
11.1.3 Microphone Sensitivity Using a Constant Voltage Bias 298

11.2 Effect of the Electronic Circuit on the Diaphragm Mechanical Response 299
11.2.1 Electric Force Due to Constant Charge 300
11.2.2 Electric Force Due to Constant Voltage 302
11.3 Optimum Capacitance for Sensitivity and Stability 304
11.4 Sound Input-Referred Noise 306
11.5 Effects of Electronic Noise and Stray Capacitance 307
11.6 Noise Analysis of a Charge Amplifier 308
11.7 Microphone Equivalent Input Noise Including Electronics 310
11.8 Electrodynamic Microphones 312

12 Estimation of Capacitance 315
12.1 Coulomb's Law and the Electric Potential 315
12.2 Integral Equation Relating Charge and Potential 318
12.3 Numerical Evaluation for Three-Dimensional Domains 320
12.4 Potential Energy for a Distribution of Charge Densities 321
12.5 Equation of Motion for a System Having a Single Degree of Freedom 322
12.6 Example of a Softening Electrode: Parallel Plate 325
12.7 Example of a Stiffening Electrode: Electrostatic Pendulum 328
12.8 Example of an Adjustable Electrode: Electrostatic Pendulum with Reduced Stiffness at Equilibrium 331
12.9 Example of an Adjustable Electrode: A Bistable Electrostatic Pendulum 333
12.10 Electronic Sensitivity 333
12.11 Compliant Repulsive Actuator 336
References for Chapter 12 337

13 Parameter Identification of Acoustic Systems 339
13.1 Least Squares Model of a Complex Transfer Function with Real Unknowns 341
13.2 Accounting for an Unknown Time Delay in H 346
13.3 Examples 346
13.4 Problems 352
References for Chapter 13 353

Appendix A: The Use of Complex Notation 355

Appendix B: Introduction to Probability and Random Processes 359

Appendix C: The Mean Square Response of a Spring/Mass/Damper 375

Appendix D: Analysis of Circuit Noise 381
Appendix E: Some Useful Formulas 387
Index 389

Nomenclature

SPL	Sound pressure level in decibels (dB)
P_{ref}	Reference pressure for SPL (pascals)
SPL_{OA}	Overall sound pressure level (dB)
SPL_A	A-weighted sound pressure level (dB)
$S_{pp}(\omega)$	Power spectral density of sound pressure (pascal^2/(radian/second))
ω	Frequency in radians/second
f	Frequency in cycles/second (Hertz)
$R_{pp}(\tau)$	Autocorrelation function of pressure (pascal^2)
τ	Separation time in the autocorrelation function (seconds)
E[·]	Expected value
$< \cdot >$	Expected value
FFT	Fast Fourier transform
$G_{pp}(f)$	Power spectral density of sound pressure (pascal^2/(cycles/second))
DC	'Direct Current', i.e. a constant, or zero frequency component in a signal
\	The back-slash operator in Matlab used to solve an overdetermined system of equations
$h(\tau)$	The impulse response of a linear system
$H(\omega)$	The frequency response of a linear system
$\Re[\cdot]$	The real part of a complex quantity
$i\sqrt{-1}$	Imaginary number
$iFFT$	The inverse fast Fourier transform
$P(x,t)$	The sound pressure as a function of location, x (meters), and time t (seconds), (pascals)
∂	Partial derivative
ρ	Density (kg/m^3)
ρ_a	The acoustic fluctuating density (kg/m^3)
ρ_0	The nominal static air density (kg/m^3)
P_0	Nominal, static air pressure (pascals)

c	Speed of propagation of a sound wave (meters/second)
p_1	Amplitude of a wave traveling in the positive x direction (pascals)
p_2	Amplitude of a wave traveling in the negative x direction (pascals)
I	Acoustic intensity (pascal meters/second)
Z	Acoustic impedance (pascal seconds/meter)
α	Sound absorption coefficient (dimensionless)
r	Sound reflection coefficient (dimensionless)
D	Bending stiffness (flexural rigidity) of a wall (pascal meters)
W	Deflection of a wall (meters)
k_x, k_y, and k_z	x, y, and z components of the acoustic wave vector (1/meter)
k	Acoustic wave number (1/meter)
ρ_w	Density of wall material (kg/m^3)
E	Young's modulus of elasticity (pascals)
τ	Sound transmission coefficient (dimensionless)
h	Thickness of a wall (meters)
TL	Sound transmission loss (decibels)
∇	Gradient differential operator (1/meters)
*	Complex conjugate
D_0	Constant to indicate the strength of a simple point sound source (pascal meters)
ln Base e	(natural) logarithm
log_{10}	Base 10 logarithm
Q	Sound source strength, i.e. volume rate of fluid displacement meter^3/second
BL	Product of magnetic flux density (teslas) and coil length (meters)
i_c	Current in a coil in an electrodynamic transducer (amperes)
Hz	Cycles per second (Hertz)
$\mathcal{E}$	Sound energy density (joule/meter^3)
Π	Acoustic power (watts)
μ	Dynamic viscosity (pascal seconds)
ν	Kinematic viscosity (meter^2/second)
S_d	Diaphragm area (meter^2)
K_d	Effective spring constant of air in an enclosed volume (newton/meter)
S_v	Area of the vent in an acoustic transducer (meter^2)
m_v	Mass of the air in the vent in an acoustic transducer (kg)
c_v	Equivalent viscous damping coefficient of the air in the vent in an acoustic transducer (newton second/meter)
ω_{cut}	Low cut-off frequency of an acoustic transducer (radians/second)
K_B	Boltzmann constant (joule/kelvin)
ζ	Damping ratio (ratio of damping constant to critical damping constant) in a one degree of freedom system (dimensionless)
v_o	Microphone output voltage (volts)
S_m	Microphone mechanical sensitivity (meter/pascal)
S_e	Microphone electrical sensitivity (volts/meter)

S_t	Microphone total sensitivity (volts/pascal)
C_p	Microphone parasitic capacitance (farads)
V_b	Microphone bias voltage (volts)
ϵ	Electrical permittivity of free space (farad/m)

Chapter 1
Analysis of Acoustic Signals



In the following, we will give a very brief introduction to methods used to analyze acoustical signals. This topic on its own constitutes a huge field and one can devote a lifetime to mastering it. Much current research in acoustics involves the creation of algorithms for compressing digitized acoustical signals to facilitate their transmission and storage. A large community of researchers devote their efforts to the analysis of speech signals to enable more efficient and effective means of human–machine and human–human communication.

The focus of the following sections is to write down the bare essentials of acoustic signal analysis. Much of the material consists of topics that every acoustical engineer must know.

1.1 Some Basics—Measures of Sound

Sound levels are typically expressed in decibels. This is a logarithmic measure that makes it convenient to quantify the incredibly wide range of sound levels that our ears can detect. The sound pressure level (SPL) is defined by

$$SPL = 10 \log_{10} \left(\frac{< p^2(t) >}{P_{ref}^2} \right), \tag{1.1.1}$$

where $< p^2(t) >$ is the mean of the square of the fluctuating sound pressure and $P_{ref} = 20 \times 10^{-6}$ Pa is the standard reference pressure. The sound pressure level is dimensionless but it is normally stated with the units of decibels (dB). The mean of

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_1.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_1

the square of the fluctuating pressure (or mean square pressure) may be determined from

$$< p^2(t) >= \lim_{T\to\infty} \frac{1}{T} \int_0^T p^2(t)dt. \tag{1.1.2}$$

It is helpful to remember that if the mean square pressure is equal to 1 Pa, the corresponding sound pressure level is

$$SPL = 10 \log_{10} \left(\frac{1}{(20 \times 10^{-6})^2} \right) \approx 94 \,\text{dB}. \tag{1.1.3}$$

Humans with good hearing can experience (and detect) sound levels over a range of zero dB up to about 130 dB. This corresponds to mean square pressures that cover a range from 20×10^{-6} to about 63 Pa.

In nearly all acoustical environments, sound sources tend to be independent of each other so that their outputs can be considered to be uncorrelated signals. Suppose we have two uncorrelated sound pressures $p_1(t)$ and $p_2(t)$ and we'd like to know the sound pressure level when both are present. Since the total pressure will be $p(t) = p_1(t) + p_2(t)$, applying Eq. (1.1.2) gives

$$\begin{aligned}
&< p^2(t) >= \lim_{T\to\infty} \frac{1}{T} \int_0^T p^2(t)dt \\
&= \lim_{T\to\infty} \frac{1}{T} \int_0^T (p_1(t) + p_2(t))^2 dt \\
&= \lim_{T\to\infty} \frac{1}{T} \int_0^T p_1(t)^2 dt + \lim_{T\to\infty} \frac{1}{T} \int_0^T p_2(t)^2 dt \\
&+ 2 \lim_{T\to\infty} \frac{1}{T} \int_0^T p_1(t) p_2(t) dt \\
&= < p_1^2(t) > + < p_2^2(t) > +2 < p_1(t) p_2(t) > .
\end{aligned} \tag{1.1.4}$$

The last term in Eq. (1.1.4), $2 < p_1(t) p_2(t) >$, is twice the cross-correlation between $p_1(t)$ and $p_2(t)$. Since these signals are typically uncorrelated, this term will equal zero. The sound pressure level of the combined signals is thus obtained by first finding the mean square of each signal, and applying Eq. (1.1.1),

$$SPL_{1+2} = 10 \log_{10} \left(\frac{< p_1^2(t) > + < p_2^2(t) >}{P_{ref}^2} \right). \tag{1.1.5}$$

If the sound pressure levels SPL_1 and SPL_2 that correspond to $p_1(t)$ and $p_2(t)$ are known, then Eq. (1.1.1) can be manipulated to determine the corresponding mean squares. This result can then be combined with Eq. (1.1.5) to calculate the sound pressure level when both signals are present,

$$SPL_{1+2} = 10\log_{10}(10^{SPL_1/10} + 10^{SPL_2/10}). \tag{1.1.6}$$

Note that the reference pressure, P_{ref}, cancels out in this expression. One can, of course, extend this approach to any number of uncorrelated sound pressures.

1.2 One-Third Octave Band Levels

Because acoustic pressures are normally rather complicated signals, it is common to quantify them in the frequency domain. The sound pressure levels discussed above are usually referred to as "overall" levels, which represent the sum of the contributions from all frequencies. It is common to represent sound levels in either octave or one-third octave frequency bands. An octave band spans a frequency range that varies from some frequency to twice that frequency. The one-third octave frequency bands contain three frequency bands for each octave. The center of each frequency band is determined approximately from $10^{n/10}$ where n is the band number. The thirtieth ($n = 30$) third octave frequency band is thus centered at 1000 Hz. Table 1.1 gives approximate values for the third octave band center frequencies and their upper and lower frequency ranges.

Since the signals at different frequencies will surely be uncorrelated, knowing m octave band or one-third octave band sound pressure levels SPL_n for $n = 1, \ldots, m$ for a given field, one can calculate the overall sound pressure level, SPL_{OA}, by an extension of Eq. (1.1.6),

$$SPL_{OA} = 10\log_{10}\left(\sum_{n=1}^{m} 10^{SPL_n/10}\right). \tag{1.2.1}$$

1.3 A-Weighted Sound Levels

While the overall sound pressure level provides a convenient single number to represent the amplitude of a given sound, it often doesn't adequately characterize sound fields. This is because different frequencies create significantly different levels of annoyance along with hearing damage. To better represent the perception and hearing damage due to a given noise environment, it can be helpful to use a frequency-weighted measure of the sound pressure level rather than the overall sound pressure level (OASPL). The most important (and commonly used) frequency-weighted measure of sound is the dBA scale. To apply this weighting, one effectively passes the fluctuating pressure $p(t)$ through a frequency-dependent filter before calculating sound pressure level. The A-weighting filter is defined by the standard, ANSI S1.42-2001, Design Response of Weighting Networks for Acoustical Measurements.

Table 1.1 Approximate one-third octave band center and cutoff frequencies with dBA weightings

Band	Lower freq	Center freq	Upper freq	dBA
12	14	16	18	−56.4
13	18	20	22	−50.4
14	22	25	28	−44.8
15	28	31.5	35	−39.5
16	36	40	45	−34.5
17	45	50	56	−30.3
18	56	63	71	−26.2
19	71	80	90	−22.4
20	89	100	112	−19.1
21	111	125	140	−16.2
22	143	160	180	−13.2
23	178	200	224	−10.8
24	223	250	281	−8.67
25	281	315	354	−6.64
26	356	400	449	−4.77
27	445	500	561	−3.25
28	561	630	707	−1.91
29	713	800	898	−0.795
30	891	1000	1122	0
31	1114	1250	1403	0.576
32	1425	1600	1796	0.993
33	1782	2000	2245	1.2
34	2227	2500	2806	1.27
35	2806	3150	3536	1.2
36	3564	4000	4490	0.963
37	4454	5000	5612	0.554
38	5613	6300	7072	−0.116
39	7127	8000	8980	−1.15
40	8909	10000	11225	−2.49
41	11136	12500	14031	−4.25
42	14254	16000	17959	−6.71
43	17960	20000	22628	−9.45

The A-weighting filter shape as a function of frequency, f in Hertz is defined to be

$$H(f) = \frac{\sqrt{3.5041384 \times 10^{16}}(if)^4}{\alpha(f)}, \tag{1.3.1}$$

where,

$$\alpha(f) = (if + 20.598997)^2(if + 107.65265) \times (if + 737.86223)(if + 12194.217)^2, \tag{1.3.2}$$

and $i = \sqrt{-1}$. The filter shape may be plotted versus frequency on a decibel scale by plotting

$$dBAWeight(f) = 10\log_{10}(H(f)H^*(f)), \tag{1.3.3}$$

where * denotes the complex conjugate.

Knowing m one-third octave band sound pressure levels, SPL_i for $i = 1, \ldots, m$ as in Eq. (1.2.1), the A-weighted sound pressure level may be calculated from

$$SPL_{dBA} = 10\log_{10}\left(\sum_{n=1}^{m} 10^{(SPL_n + dBAWeight(f_n))/10}\right), \tag{1.3.4}$$

where $dBAWeight(f_n)$ is the dBA weighting that corresponds to the center frequency f_n. Approximate values for these weighting factors are given in Table 1.1.

1.4 Narrowband Signal Analysis

While overall, A-weighted and octave band levels provide excellent measures of loudness in many situations, it is also often necessary to resort to frequency domain analysis in narrow frequency bands. This is often provided in the form of a power spectral density with units of pascals^2/Hz or pascals^2/(rad/s). These data are normally provided for frequency bands that have the same width Δf over the entire frequency range rather than the logarithmic spacing used in octave bands.

The power spectral density is an extremely important statistic for the characterization of complex, random signals as are often encountered in acoustics. The power spectral density, $S_{pp}(\omega)$, is defined to be the Fourier transform of the autocorrelation function for a weakly stationary ergodic random process, $R_{pp}(\tau) = E[p(t)p(t+\tau)]$, where E[] denotes the expected value. Weak stationarity means that certain first-order statistics of the process are independent of time, t which is why the argument of R_{pp} depends only on τ.

It is important to note that the autocorrelation function is an even function of τ. This is a result of the fact that it is independent of a shift in the time origin. Let $t_0 = t + \tau$ so that the autocorrelation function becomes

$$R_{pp}(\tau) = E[p(t)p(t+\tau)] = E[p(t_0 - \tau)p(t_0)] = R_{pp}(-\tau). \tag{1.4.1}$$

The Fourier transform of this gives the power spectral density,

$$S_{pp}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{pp}(\tau)e^{-i\omega\tau}d\tau. \tag{1.4.2}$$

Although the Fourier transform of a real function often results in a complex quantity, the power spectral density is always entirely real. This is a result of the fact that the autocorrelation function is an even function of τ. To see this, take the complex conjugate of the power spectral density,

$$S^*_{pp}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{pp}(\tau)e^{i\omega\tau}d\tau. \tag{1.4.3}$$

Let $\tau' = -\tau$,

$$S^*_{pp}(\omega) = \frac{-1}{2\pi}\int_{\infty}^{-\infty} R_{pp}(-\tau')e^{-i\omega\tau'}d\tau' = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{pp}(\tau')e^{-i\omega\tau'}d\tau' = S_{pp}(\omega), \tag{1.4.4}$$

where we have used the fact that $R_{pp}(\tau') = R_{pp}(-\tau')$. Equation (1.4.4) also shows that the power spectral density is an even function of ω, $S_{pp}(\omega) = S_{pp}(-\omega)$. This fact leads us to refer to $S_{pp}(\omega)$ as the double-sided power spectral density.

Since the power spectral density is obtained via the Fourier transform of the autocorrelation function, these two quantities are also related through the inverse Fourier transform,

$$R_{pp}(\tau) = \int_{-\infty}^{\infty} S_{pp}(\omega)e^{i\omega\tau}d\omega. \tag{1.4.5}$$

Equations (1.4.2) and (1.4.5) thus form a Fourier transform pair.

An extremely important feature of the power spectral density is that the integral over all frequencies is equal to the mean square response. To see this, set $\tau = 0$ in Eqs. (1.4.1) and (1.4.5),

$$R_{pp}(0) = E[p^2(t)] = \int_{-\infty}^{\infty} S_{pp}(\omega)d\omega. \tag{1.4.6}$$

The integral of the power spectral density between two frequencies, ω_1 and ω_2 will then give the contribution to the mean square associated with that frequency range.

While the double-sided power spectral density $S_{pp}(\omega)$ follows from its definition in terms of the transform of the autocorrelation function, engineers generally prefer the single-sided density $G_{pp}(f)$, which is expressed in units of pascals^2/Hz. This is defined so that, again,

$$< p^2(t) >= \int_0^\infty G_{pp}(f)df, \tag{1.4.7}$$

where the integration is now taken over only positive frequencies.

If one has a pressure power spectral density, $G_{pp}(f)$, the corresponding dBA weighted sound pressure level is calculated from

$$SPL_A = 10\log_{10}\left(\int_0^\infty \frac{G_{pp}(f)}{P_{ref}^2} H(f)H^*(f)df\right), \tag{1.4.8}$$

where $P_{ref}^2 = 20 \times 10^{-6}$ Pa is the mean square reference pressure.

In some cases, the pressure power spectral density is independent of frequency so that it can be taken outside the integral,

$$\begin{aligned} SPL_A &= 10\log_{10}\left(G_{pp}\int_0^\infty \frac{1}{P_{ref}^2} H(f)H^*(f)df\right) \\ &\approx 10\log_{10}(G_{pp}) + 135.2423, \end{aligned} \tag{1.4.9}$$

where the integration has been carried out numerically.

1.5 Power Spectral Density by the Finite Fourier Transform

Thus far, we have computed the power spectral density by first determining the autocorrelation function and then using the Fourier transform. In the analysis of measured data, it is far more efficient to employ the Fast Fourier Transform (FFT) to compute spectra. To do this we need to first compute the power spectrum in terms of a finite Fourier transform.

Given a stationary random process, $x(t)$, we will show that the power spectral density, $S_{xx}(\omega)$ may be computed by

$$S_{xx}(\omega) = \lim_{T\to\infty} \frac{\pi}{T} \mathrm{E}[|X(\omega, T)|^2], \tag{1.5.1}$$

where the finite Fourier transform of $x(t)$ is,

$$X(\omega, T) = \frac{1}{2\pi}\int_{-T}^{T} x(t)e^{-i\omega t}dt.. \tag{1.5.2}$$

Multiplying Eq. (1.5.2) by its complex conjugate and taking the expected value gives

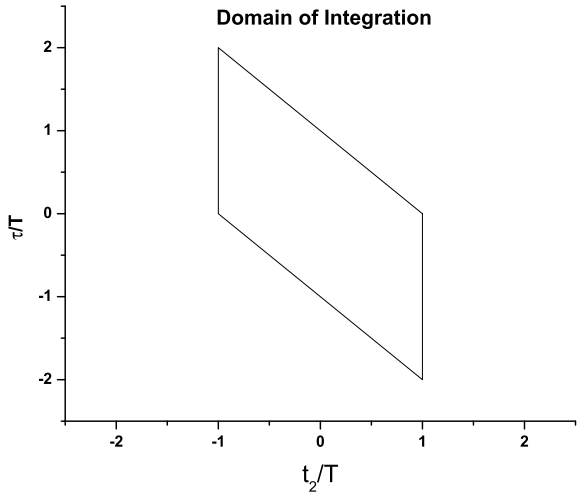


Fig. 1.1 Domain of integration for computing the Fourier transform of the autocorrelation function in terms of the Finite Fourier transform

$$\mathrm{E}[|X(\omega,T)|^2] = \frac{1}{(2\pi)^2}\mathrm{E}\left[\int_{-T}^{T}\int_{-T}^{T} x(t_1)x(t_2)e^{-i\omega(t_1-t_2)}dt_1dt_2\right]. \tag{1.5.3}$$

Bringing the expected value operation inside the integral gives

$$\mathrm{E}[|X(\omega,T)|^2] = \frac{1}{(2\pi)^2}\int_{-T}^{T}\int_{-T}^{T} R_{xx}(t_1-t_2)e^{-i\omega(t_1-t_2)}dt_1dt_2. \tag{1.5.4}$$

To carry out the double integration in Eq. (1.5.4) make a change of variables from t_1 to τ, where $\tau = t_1 - t_2$. Eliminating t_1 in Eq. (1.5.4) gives

$$\mathrm{E}[|X(\omega,T)|^2] = \frac{1}{(2\pi)^2}\int_{-T}^{T}\int_{-T-t_2}^{T-t_2} R_{xx}(\tau)e^{-i\omega(\tau)}d\tau dt_2. \tag{1.5.5}$$

The integration range is now over the domain shown in Fig. 1.1.

It is convenient to first integrate over t_2 and then over τ. When changing the order of the integration we must integrate over separate triangles for positive and negative values of τ. For the upper triangle we will integrate over τ for $0 \le \tau \le 2T$. The range for t_2 for this triangle will be from $-T$ to the line described by $\tau = T - t_2$. For the lower triangle, τ will be integrated from $-2T$ to 0, or, $-2T \le \tau \le 0$, and the range for t_2 will be from the line described by $\tau = -T - t_2$ to T. The integral in Eq. (1.5.5) may then be written as

$$\begin{aligned}\mathrm{E}[|X(\omega,T)|^2] = \frac{1}{(2\pi)^2}\Bigg(&\int_{0}^{2T}\int_{-T}^{T-\tau} R_{xx}(\tau)e^{-i\omega(\tau)}dt_2d\tau \\ &+ \int_{-2T}^{0}\int_{-T-\tau}^{T} R_{xx}(\tau)e^{-i\omega(\tau)}dt_2d\tau\Bigg).\end{aligned} \tag{1.5.6}$$

Since the integrands in Eq. (1.5.6) do not depend on t_2, the integrals over t_2 are easily evaluated which gives

$$\mathrm{E}[|X(\omega,T)|^2] = \frac{1}{(2\pi)^2}\left(\int_0^{2T} R_{xx}(\tau)e^{-i\omega(\tau)}(2T-\tau)d\tau + \int_{-2T}^{0} R_{xx}(\tau)e^{-i\omega(\tau)}(2T+\tau)d\tau\right). \tag{1.5.7}$$

Combining the integrals in Eq. (1.5.7) gives

$$\mathrm{E}[|X(\omega,T)|^2] = \frac{1}{(2\pi)^2}\int_{-2T}^{2T} R_{xx}(\tau)e^{-i\omega(\tau)}(2T-|\tau|)d\tau. \tag{1.5.8}$$

To examine the validity of Eq. (1.5.1) we must multiply by $\frac{\pi}{T}$ and take the limit,

$$\begin{aligned}
&\lim_{T\to\infty}\frac{\pi}{T}\mathrm{E}[|X(\omega,T)|^2] = \\
&\lim_{T\to\infty}\frac{1}{4\pi T}\int_{-2T}^{2T} R_{xx}(\tau)e^{-i\omega(\tau)}(2T-|\tau|)d\tau \\
&= \lim_{T\to\infty}\left(\frac{1}{2\pi}\int_{-2T}^{2T} R_{xx}(\tau)e^{-i\omega(\tau)}d\tau\right. \\
&\left.-\frac{1}{4\pi}\int_{-2T}^{2T} R_{xx}(\tau)e^{-i\omega(\tau)}\frac{|\tau|}{T}\right)d\tau.
\end{aligned} \tag{1.5.9}$$

It can be shown that

$$\lim_{T\to\infty}\frac{1}{4\pi}\int_{-2T}^{2T} R_{xx}(\tau)e^{-i\omega(\tau)}\frac{|\tau|}{T}d\tau = 0. \tag{1.5.10}$$

Taking the limit in Eq. (1.5.9) then produces the Fourier transform of the autocorrelation function which gives the power spectrum as in Eq. (1.5.1).

1.6 Spectral Analysis of Measured Time Series

In the following, we will use our ability to compute the power spectral density from the finite Fourier transform shown in the previous section to show how to approximate the power spectral density of measured signals by using the Fast Fourier Transform (FFT). This provides an extremely important tool for the analysis of time domain data. The FFT algorithm is the basis of virtually all methods of examining the frequency content of signals.

Suppose a measured time series, $x(t)$ is sampled at discrete values of time, t, then let

$$x_n = x(t_n) = x(n\Delta t), \quad n = 0, 1, \ldots, N-1, \tag{1.6.1}$$

where the time duration between the equally spaced sample times is Δt. To satisfy the Nyquist sampling criterion, the time series, $x(t)$ must be sampled at a high enough rate to obtain two samples during one period of the highest frequency component in the signal. If ω_{max} is the highest frequency component in the signal, the Nyquist criterion is

$$\Delta t \leq \frac{2\pi}{\omega_{max}} \frac{1}{2}. \tag{1.6.2}$$

If a total of N data points are obtained, then the time duration of the time series is $T = N\Delta t$.

In the previous section we have shown that the power spectral density may be computed from

$$S_{xx}(\omega) = \lim_{\mathcal{T}\to\infty} \frac{\pi}{\mathcal{T}} \mathrm{E}[|X(\omega, \mathcal{T})|^2], \tag{1.6.3}$$

where $X(\omega, \mathcal{T})$ is the finite Fourier transform of $x(t)$,

$$X(\omega, \mathcal{T}) = \frac{1}{2\pi} \int_{-\mathcal{T}}^{\mathcal{T}} x(t) e^{-i\omega t} dt. \tag{1.6.4}$$

The finite Fourier transform in Eq. (1.6.4) is expressed as an integral over both negative and positive times, t. In the analysis of measured data it is convenient to consider the beginning of the time record as being at $t = 0$ rather than at $t = -\mathcal{T}$. Since our random process is assumed to be stationary, the statistics are independent of a shift in the time origin. Equation (1.6.4) may then be rewritten as

$$X(\omega, \mathcal{T}) = \frac{1}{2\pi} \int_{0}^{2\mathcal{T}} x(t) e^{-i\omega t} dt. \tag{1.6.5}$$

In the notation of Eq. (1.6.5), the time duration of the integration is $2\mathcal{T}$. We have assumed above that data has been obtained for $0 \leq t \leq T$. Let $T = 2\mathcal{T}$, or $\mathcal{T} = T/2$ which allows us to write Eq. (1.6.3) as

$$S_{xx}(\omega) = \lim_{T/2\to\infty} \frac{\pi}{T/2} \mathrm{E}[|X(\omega, T/2)|^2], \tag{1.6.6}$$

and the Fourier transform in Eq. (1.6.5) may be written as

$$X(\omega, T/2) = \frac{1}{2\pi} \int_{0}^{T} x(t) e^{-i\omega t} dt. \tag{1.6.7}$$

If $x(t)$ is known only at discrete times as in Eq. (1.6.1), then Eq. (1.6.7) must be approximated by a finite summation,

$$X(\omega, T/2) \approx \frac{1}{2\pi} \sum_{n=0}^{N-1} x_n e^{-i\omega t_n} \Delta t. \tag{1.6.8}$$

To employ the FFT algorithm to evaluate Eq. (1.6.8), suppose we wish to evaluate Eq. (1.6.8) only at discrete values of ω, $\omega_j = j\Delta\omega$, $j = 0, 1, \ldots, N-1$, where N is, as above, the total number of time points in the record x_n. $\Delta\omega$ will be the lowest frequency component in the signal in radians/second,

$$\Delta\omega = \frac{2\pi}{T}. \tag{1.6.9}$$

After substituting ω_j for ω, using Eq. (1.6.9), and using the fact that $\Delta t = T/N$, Eq. (1.6.8) becomes

$$\begin{aligned} X(\omega_j, T/2) &= \frac{1}{2\pi} \sum_{n=0}^{N-1} x_n e^{-i\omega_j t_n} \Delta t \\ &= \frac{1}{2\pi} \sum_{n=0}^{N-1} x_n e^{-ij\Delta\omega n\Delta t} \Delta t \\ &= \frac{1}{2\pi} \sum_{n=0}^{N-1} x_n e^{-ijn2\pi/N} \Delta t. \end{aligned} \tag{1.6.10}$$

Equation (1.6.10) may be evaluated using a Fast Fourier Transform (FFT) algorithm by noting that given a discrete sequence, a_n, the FFT provides an efficient means of computing A_k, where

$$A_k = FFT(a_n) = \sum_{n=0}^{N-1} a_n e^{-i2\pi kn/N}, \tag{1.6.11}$$

for $k = 0, 1, 2, \ldots, N-1$. The FFT algorithm requires that N be a power of 2. By comparing Eqs. (1.6.10) and (1.6.11) we get

$$X(\omega_j, T/2) = \frac{\Delta t}{2\pi} FFT(x_n), \tag{1.6.12}$$

where $FFT(x_n)$ is equivalent to

$$\begin{aligned} FFT(x_n) &= \mathcal{X}_j \\ &= \sum_{l=0}^{N-1} x_l e^{-i2\pi jl/N}, \end{aligned} \tag{1.6.13}$$

for $j = 0, 1, 2, \ldots, N-1$. The FFT of the sequence x_n is a sequence $\mathcal{X}_j$ for $j = 0, 1, 2, \ldots, N-1$.

Now that we have an efficient method of evaluating the finite Fourier transform in Eq. (1.6.7), we need to substitute our result into Eq. (1.6.6) to compute the power spectral density. To do this we need to compute the expected value in Eq. (1.6.6). This may be accomplished by evaluating a FFT for each of a set of time records and averaging the squared magnitude of the transforms. Suppose we store a sequence of time records, x_{nl}, where n denotes the time increment ($t_n = n\Delta t$), and l denotes the number of the record. Equation (1.6.12) can be used to compute $\mathcal{X}_j$ for each l, which we will denote by $\mathcal{X}_{jl}$,

$$\begin{aligned} FFT(x_n, l) &= \mathcal{X}_{jl} \\ &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi jn/N}, \end{aligned} \tag{1.6.14}$$

for $j = 0, 1, 2, \ldots, N-1$. Equation (1.6.12) now takes the form

$$X(\omega_j, T/2, l) = \frac{\Delta t}{2\pi} FFT(x_n, l) = \frac{\Delta t}{2\pi} \mathcal{X}_{jl}. \tag{1.6.15}$$

By using Eq. (1.6.15), the expected value in Eq. (1.6.6) may be approximated by the average over the records,

$$\mathrm{E}[|X(\omega_j, T/2)|^2] \approx \sum_{l=1}^{M} \frac{1}{M} \left(\frac{\Delta t}{2\pi}\right)^2 |\mathcal{X}_{jl}|^2, \tag{1.6.16}$$

where M is the total number of records. Substituting Eq. (1.6.16) into Eq. (1.6.6) gives

$$\begin{aligned} S_{xx}(\omega_j) &= \lim_{T/2 \to \infty} \frac{\pi}{T/2} \mathrm{E}[|X(\omega_j, T/2)|^2], \\ &\approx \frac{\pi}{T/2} \sum_{l=1}^{M} \frac{1}{M} \left(\frac{\Delta t}{2\pi}\right)^2 |\mathcal{X}_{jl}|^2, \end{aligned} \tag{1.6.17}$$

where in the second of Eq. (1.6.17), T, denotes the duration of each time record which remains finite. Because $\Delta t = T/N$, Eq. (1.6.17) may be simplified as

$$S_{xx}(\omega_j) \approx \sum_{l=1}^{M} \frac{1}{M} \frac{T}{2\pi N^2} |\mathcal{X}_{jl}|^2. \tag{1.6.18}$$

It is important to note that Eq. (1.6.18) can be evaluated for any j. It turns out, however, that not all of these values are unique. Plotting the result of Eq. (1.6.18) versus

ω_j shows that the estimated spectrum is symmetric about $\omega_{N/2}$. Unique results are obtained only for $j = 0, 1, \ldots, N/2$. To see this let's evaluate the sum in Eq. (1.6.14) for $j = N/2 - J$ and $j = N/2 + J$. Setting $j = N/2 - J$ in Eq. (1.6.14) gives

$$\begin{aligned}\mathcal{X}_{(N/2-J)l} &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi(N/2-J)n/N} \\ &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi(-J)n/N} e^{-i2\pi(N/2)n/N} \\ &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi(-J)n/N} e^{-i\pi n}. \end{aligned} \tag{1.6.19}$$

Setting $j = N/2 + J$ in Eq. (1.6.14) gives

$$\begin{aligned}\mathcal{X}_{(N/2+J)l} &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi(N/2+J)n/N} \\ &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi(+J)n/N} e^{-i2\pi(N/2)n/N} \\ &= \sum_{n=0}^{N-1} x_{nl} e^{-i2\pi(+J)n/N} e^{-i\pi n}. \end{aligned} \tag{1.6.20}$$

Since $e^{-i\pi n} = (-1)^n$, Eqs. (1.6.19) and (1.6.20) are complex conjugates of each other. When the magnitudes of these are used in Eq. (1.6.18) we will get

$$S_{xx}(\omega_{N/2-J}) = S_{xx}(\omega_{N/2+J}), \tag{1.6.21}$$

so the spectrum is symmetric about $j = N/2$. The highest value of j for which Eq. (1.6.18) may be evaluated is then $j = N/2$ which corresponds to a frequency of $\omega_{N/2} = \pi/\Delta t$, or, half the sampling frequency.

1.7 Sound Level Calculations from Narrowband Power Spectra

In the following, a procedure is outlined to compute one-third octave band levels from the double-sided power spectral density of a sound pressure, $S_{pp}(\omega)$. Knowing the one-third octave levels one can then estimate the equivalent dBA level etc. The one-third octave band level at the frequency f_3 of a weakly stationary random signal, $p(t)$, is given by

$$SPL_3(f_3) = 10\log_{10}\left(\frac{< p^2 >_{f_3}}{p_{ref}^2}\right). \tag{1.7.1}$$

where $p_{ref} = 20\,\mu\text{Pa}$ is the reference pressure and $< p^2 >_{f_3}$ is the mean square pressure in the frequency range of the one-third octave band, i.e., for frequencies f that satisfies

$$\frac{f_3}{\sqrt{2^{1/3}}} \le f \le f_3\sqrt{2^{1/3}}. \tag{1.7.2}$$

To determine $< p^2 >_{f_3}$ it is helpful to first convert the double-sided power spectral density, $S_{pp}(\omega)$, to a single-sided power spectral density, $G_{pp}(f)$, with units of pascals^2/Hz. Given the double-sided power spectral density of a random process, $S_{pp}(\omega)$ and the single-sided spectral density $G_{pp}(f)$, it is useful to recall that the total mean square response, $E[p^2]$ is obtained by integrating the spectral density over all frequencies,

$$\begin{aligned} E[p^2] &= \int_0^\infty G_{pp}(f)df = \int_{-\infty}^\infty S_{pp}(\omega)d\omega \\ &= 2\int_0^\infty S_{pp}(\omega)d\omega, \end{aligned} \tag{1.7.3}$$

where the last integral results from the fact that the double-sided power spectral density is an even function, $S_{pp}(\omega) = S_{pp}(-\omega)$. Since $\omega = 2\pi f$,

$$\begin{aligned} 2\int_0^\infty S_{pp}(\omega)d\omega &= 2\int_0^\infty S_{pp}(2\pi f)2\pi df \\ &= \int_0^\infty G_{pp}(f)df. \end{aligned} \tag{1.7.4}$$

The two-sided and single-sided power spectra are then related by

$$G_{pp}(f) = 4\pi S_{pp}(2\pi f). \tag{1.7.5}$$

The contribution to the mean square response from frequencies that lie in the one-third octave band centered at the frequency f_3 is then

$$< p^2 >_{f_3} = \int_{\frac{f_3}{\sqrt{2^{1/3}}}}^{f_3\sqrt{2^{1/3}}} 4\pi S_{pp}(2\pi f)df. \tag{1.7.6}$$

In the special case where the power spectrum is independent of frequency, the integral can be performed analytically. If $S_{pp}(2\pi f) = S_{pp}$, carrying out the integration in Eq. (1.7.6) then gives

$$< p^2 >_{f_3} = 4\pi S_{pp} \int_{\frac{f_3}{\sqrt{2^{1/3}}}}^{f_3\sqrt{2^{1/3}}} df$$

$$= 4\pi S_{pp} f_3 \left(\sqrt{2^{1/3}} - \frac{1}{\sqrt{2^{1/3}}} \right). \tag{1.7.7}$$

Equations (1.7.1) and (1.7.7) give

$$SPL_3(f_3) = 10 \log_{10} \left(\frac{4\pi S_{pp} f_3 \times (2^{1/6} - 2^{-1/6})}{p_{ref}^2} \right). \tag{1.7.8}$$

1.8 The "Slow" Fourier Transform: Least Squares Extraction of a Harmonic Signal

The methods described above are used to examine the frequency content of broadband signals. There are many situations where we are interested in determining the amplitude and phase of a signal at a known frequency that is superimposed on other uncorrelated signals such as noise. To accomplish this we might be tempted to employ a FFT algorithm which would provide the amplitude and phase at numerous frequencies. This, however, provides a substantial amount of information on the contributions due to frequencies that are irrelevant to what is sought; i.e. one may care only about the contribution at one particular frequency, and that frequency may not match up well with the time step used in the analysis. Fortunately, because the frequency of interest is known, we can easily construct a linear least squares approach to obtain both the magnitude and phase of the desired signal.

Suppose that the measured signal is $x(t)$, which is sampled at N discrete instants of time, t_n, for $n = 1, \ldots, N$. Unlike the use of the FFT algorithm, in the present discussion we do not require that the sample times t_n be evenly spaced. We merely need to know what they are. The measured data are then stored in an array, $x_n = x(t_n)$, $n = 1, \ldots, N$. Our task is to estimate the amplitude and phase of a component of the measured signal having the known frequency, ω. This problem may then be expressed as one of estimating the coefficients a, b, and c in

$$x_n = a \cos(\omega t_n) + b \sin(\omega t_n) + c + \epsilon_n \quad n = 1, \ldots, N, \tag{1.8.1}$$

where the constant c accounts for the fact that in real signals there is often a DC shift causing the average of the signal over time to not be equal to zero. ϵ_n is the error at time t_n,

$$\epsilon_n = x_n - a \cos(\omega t_n) - b \sin(\omega t_n) - c. \tag{1.8.2}$$

We would like to find the coefficients a, b, and c so that the error is minimized in the least squares sense. That is, we'd like to minimize the total squared error, $E(a, b, c)$,

$$E(a, b, c) = \sum_{n=1}^{N} \epsilon_n^2. \tag{1.8.3}$$

Minimizing the total squared error with respect to the unknown coefficients in this linear model is the typical least squares problem. The solution is obtained by finding the extremum of the error,

$$\frac{\partial E}{\partial a} = \frac{\partial E}{\partial b} = \frac{\partial E}{\partial c} = 0. \tag{1.8.4}$$

Equations (1.8.2) through (1.8.4) give

$$2 \sum_{n=1}^{N} \epsilon_n \frac{\partial \epsilon_n}{\partial a} = -2 \sum_{n=1}^{N} \epsilon_n \cos(\omega t_n) = 0, \tag{1.8.5}$$

$$2 \sum_{n=1}^{N} \epsilon_n \frac{\partial \epsilon_n}{\partial b} = -2 \sum_{n=1}^{N} \epsilon_n \sin(\omega t_n) = 0, \tag{1.8.6}$$

and

$$2 \sum_{n=1}^{N} \epsilon_n \frac{\partial \epsilon_n}{\partial c} = -2 \sum_{n=1}^{N} \epsilon_n = 0. \tag{1.8.7}$$

Equations (1.8.2), (1.8.5), (1.8.6), and (1.8.7) give

$$\sum_{n=1}^{N} (x_n - a \cos(\omega t_n) - b \sin(\omega t_n) - c) \cos(\omega t_n) = 0, \tag{1.8.8}$$

$$\sum_{n=1}^{N} (x_n - a \cos(\omega t_n) - b \sin(\omega t_n) - c) \sin(\omega t_n) = 0, \tag{1.8.9}$$

and

$$\sum_{n=1}^{N} (x_n - a \cos(\omega t_n) - b \sin(\omega t_n) - c) = 0. \tag{1.8.10}$$

Equation (1.8.8) through (1.8.10) may be rearranged as a set of equations that may easily be solved for the unknowns a, b, and c,

$$\begin{bmatrix} \sum_{n=1}^{N} \cos^2(\omega t_n) & \sum_{n=1}^{N} \cos(\omega t_n)\sin(\omega t_n) & \sum_{n=1}^{N} \cos(\omega t_n) \\ \sum_{n=1}^{N} \cos(\omega t_n)\sin(\omega t_n) & \sum_{n=1}^{N} \sin^2(\omega t_n) & \sum_{n=1}^{N} \sin(\omega t_n) \\ \sum_{n=1}^{N} \cos(\omega t_n) & \sum_{n=1}^{N} \sin(\omega t_n) & \sum_{n=1}^{N} 1 \end{bmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} \sum_{n=1}^{N} x_n \cos(\omega t_n) \\ \sum_{n=1}^{N} x_n \sin(\omega t_n) \\ \sum_{n=1}^{N} x_n \end{pmatrix}. \tag{1.8.11}$$

Knowing a, b, and c, the least squares estimate, $x_{ls}(t)$, of the signal at the frequency ω may be computed from

$$x_{ls}(t) = a\cos(\omega t) + b\sin(\omega t) + c. \tag{1.8.12}$$

The amplitude, X, and phase, ϕ, of the harmonic component of the signal can be determined by

$$a\cos(\omega t) + b\sin(\omega t) = X\sin(\omega t + \phi) = X\sin(\omega t)\cos(\phi) + X\cos(\omega t)\sin(\phi), \tag{1.8.13}$$

so that

$$X = \sqrt{a^2 + b^2}, \quad \tan(\phi) = \frac{a}{b}. \tag{1.8.14}$$

The calculations in Eq. (1.8.11) may be easily accomplished in Matlab by expressing Eq. (1.8.1) as an overdetermined system of equations and neglecting the error. Define a matrix Φ having N rows and three columns,

$$\Phi = \begin{bmatrix} \cos(\omega t_n) & \sin(\omega t_n) & 1 \end{bmatrix}, \tag{1.8.15}$$

where each row corresponds to a value of the index n. The measured data is stored in the array X, where the nth row contains the value x_n. The solution for the unknown coefficients is then easily accomplished by the backslash operator in Matlab,

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} = \Phi \backslash X. \tag{1.8.16}$$

The least squares estimate of the signal at the frequency ω is then computed by

$$X_{ls} = \Phi \begin{pmatrix} a \\ b \\ c \end{pmatrix}. \tag{1.8.17}$$

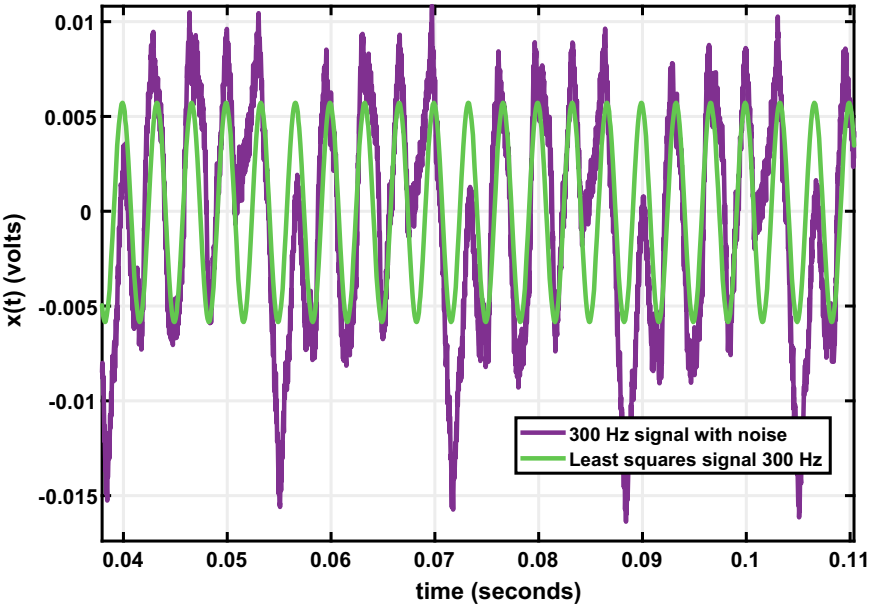


Fig. 1.2 Least squares method is able to extract the desired signal when it is significantly influenced by unwanted noise. The original data consisted of a 300 Hz tone corrupted by noise. The estimated component at the desired frequency was obtained using Eqs. (1.8.16) and (1.8.17)

To illustrate, measured data are shown in Fig. 1.2 in which the input to the system consisted of a pure tone at 300 Hz. The measured signal was influenced by noise as shown. The component of the measured signal at the frequency of interest, 300 Hz, was obtained by applying the least squares solution of Eq. (1.8.17). This approach can easily be extended to obtain any number of frequency components in a signal.

1.9 Frequency Response of Linear Systems with Random Input

The narrowband analysis of acoustic signals discussed above has been centered on the use of the auto-power spectral density, $S_{pp}(\omega)$. An even more powerful statistic available for the characterization of random signals is the cross power spectral density of two signals, say $p_1(t)$ and $p_2(t)$, $S_{p_1p_2}(\omega)$. Unlike the auto-power spectral density, this quantity will generally be complex. To illustrate one important application of the cross-spectral density, in the following we will derive an expression for the frequency response of a linear system in terms of the cross power spectral density of the input and output signals and the auto-power spectral density of the input.

We will assume that a linear, time-invariant system has a random input, $f(t)$ and an output, $p(t)$. The system is characterized by its impulse response, $h(t)$, which is zero for negative values of t. If the system is initially at rest, the impulse response enables the determination of the system output through the convolution integral,

$$p(t) = \int_0^t h(t-\tau')f(\tau')d\tau'. \tag{1.9.1}$$

The complex frequency response of the system is defined to be the Fourier transform of the impulse response,

$$H(\omega) = \int_0^\infty h(\tau)e^{-i\omega\tau}d\tau. \tag{1.9.2}$$

The impulse response may be obtained from the frequency response function through the use of the inverse transform,

$$h(\tau) = \frac{1}{2\pi}\int_{-\infty}^\infty H(\omega)e^{i\omega\tau}d\omega. \tag{1.9.3}$$

Note that Eqs. (1.9.2) and (1.9.3) form a Fourier transform pair where the factor of $1/(2\pi)$ has been interchanged from the usual convention.

We wish to show that given the cross power spectral density, $S_{fp}(\omega)$ and the auto-power spectral density, $S_{ff}(\omega)$, the frequency response function may be determined from

$$H(\omega) = \frac{S_{fp}(\omega)}{S_{ff}(\omega)}. \tag{1.9.4}$$

We will assume that the input signal, $f(t)$ is a weakly stationary random process.

$S_{fp}(\omega)$ is defined to be the Fourier transform of the cross- correlation function, $R_{fp}(\tau) = \mathrm{E}[f(t)p(t+\tau)]$, where $\mathrm{E}[\cdot]$ denotes the expected value. The weak stationarity of the processes results in the cross-correlation being independent of time, t. Equation (1.9.1) enables us to write the cross-correlation function as

$$R_{fp}(\tau) = \mathrm{E}[f(t)p(t+\tau)] = \mathrm{E}\left[f(t)\int_0^{t+\tau} h(t+\tau-\tau')f(\tau')d\tau'\right]. \tag{1.9.5}$$

The cross power spectral density is then obtained by taking the Fourier transform,

$$\begin{aligned} S_{fp}(\omega) &= \frac{1}{2\pi}\int_{-\infty}^\infty R_{fp}(\tau)e^{-i\omega\tau}d\tau \\ &= \frac{1}{2\pi}\int_{-\infty}^\infty \mathrm{E}\left[f(t)\int_0^{t+\tau} h(t+\tau-\tau')f(\tau')d\tau'\right]e^{-i\omega\tau}d\tau. \end{aligned} \tag{1.9.6}$$

The upper limit in the integration over τ' can be changed to ∞ since $h(t) = 0$ for $t < 0$. The lower limit can be changed to $-\infty$ since the force starts at time $t = 0$, $f(t) = 0$ for $t < 0$.

Changing the limits of integration enables us to also change the order of the integrations and rearrange Eq. (1.9.6) to the form

$$S_{fp}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(t+\tau-\tau')\mathrm{E}[f(t)f(\tau')]e^{-i\omega\tau}d\tau d\tau', \tag{1.9.7}$$

where we have also used the fact that $h(t)$ is deterministic so it can be taken outside the expected value operator. The expected value in Eq. (1.9.7) consists of the autocorrelation function,

$$R_{ff}(t-\tau') = R_{ff}(\tau'-t) = \mathrm{E}[f(t)f(\tau')], \tag{1.9.8}$$

since $R_{ff}(\tau)$ is an even function of τ. It is now helpful to make a change of variables by letting $\tau'' = t - \tau'$ so that $d\tau' = -d\tau''$ which enables us to write Eq. (1.9.7) as

$$S_{fp}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(\tau''+\tau)R_{ff}(\tau'')e^{-i\omega\tau}e^{-i\omega\tau''}e^{i\omega\tau''}d\tau d\tau'', \tag{1.9.9}$$

where we have also multiplied by unity,

$$1 = e^{-i\omega\tau''}e^{i\omega\tau''}. \tag{1.9.10}$$

Make one more change of variables, $\tau_1 = \tau'' + \tau$ so that Eq. (1.9.9) becomes

$$S_{fp}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{0}^{\infty} h(\tau_1)R_{ff}(\tau'')e^{-i\omega\tau_1}e^{i\omega\tau''}d\tau_1 d\tau'', \tag{1.9.11}$$

where we have replaced the lower limit of integration over τ_1 with zero since $h(\tau_1) = 0$ for $\tau_1 \leq 0$. The two integrations may now be carried out separately. Since the power spectral density is defined to be the Fourier transform of the autocorrelation function,

$$S_{ff}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{ff}(\tau'')e^{-i\omega\tau''}d\tau''. \tag{1.9.12}$$

Equations (1.9.2), (1.9.11), and (1.9.12) give

$$S_{fp}(\omega) = H(\omega)S_{ff}(\omega) \tag{1.9.13}$$

in agreement with Eq. (1.9.4) which is what we wished to show.

It is important to note that the results presented in previous sections to estimate the auto-power spectral density from discrete time data can be extended to the estimation

of the cross-spectral density. The estimation of the frequency response of linear systems through Eq. (1.9.4) or (1.9.13) provides a very powerful tool for characterizing systems.

1.10 Input–Output Relationships for a Linear System

In the following, we examine the calculation of the response of dynamic systems through the use of the FFT algorithm. Knowing the complex transfer function, $H(\omega)$, which relates the input $f(t)$, to the output signal, $x(t)$, of a linear system in the frequency domain, one can obtain the response of the system to any input signal, $f(t)$. This algorithm can also be extremely useful for obtaining the numerical derivative or integral of arbitrary time domain signals.

Consider a general linear system which has an input, $f(t)$ and an output response $x(t)$. This system may be composed of an electronic circuit or filter having any number of poles and zeros or a mechanical structure having any number of resonant modes. It is convenient to represent the response of such systems by use of the Fourier Transform. The Fourier Transform of the input and output signals are

$$\begin{aligned} F(\omega) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\omega t} f(t)dt \\ X(\omega) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\omega t} x(t)dt. \end{aligned} \tag{1.10.1}$$

The transfer function of the system is the ratio of these transforms,

$$H(\omega) = \frac{X(\omega)}{F(\omega)}. \tag{1.10.2}$$

If one knows $F(\omega)$ or $X(\omega)$, the corresponding time domain signals may be obtained by the inverse Fourier Transform,

$$\begin{aligned} f(t) &= \int_{-\infty}^{\infty} e^{i\omega t} F(\omega)d\omega \\ x(t) &= \int_{-\infty}^{\infty} e^{i\omega t} X(\omega)d\omega. \end{aligned} \tag{1.10.3}$$

If $H(\omega)$ and $F(\omega)$ are known, the second of Eq. (1.10.3) gives

$$x(t) = \int_{-\infty}^{\infty} e^{i\omega t} H(\omega)F(\omega)d\omega. \tag{1.10.4}$$

In some problems, it can be helpful to express the response $x(t)$ using the convolution integral,

$$x(t) = \int_0^t h(t-\tau) f(\tau) d\tau, \tag{1.10.5}$$

where $h(t)$ is the impulse response which is defined to be the inverse Fourier Transform of the transfer function,

$$h(t) = \int_{-\infty}^{\infty} e^{i\omega t} H(\omega) d\omega. \tag{1.10.6}$$

The impulse response is also defined such that

$$h(t) = 0, \quad \text{for} \quad t < 0. \tag{1.10.7}$$

Equation (1.10.5) may be shown to represent the system response by substituting it into Eq. (1.10.1) and using Eq. (1.10.2),

$$H(\omega)F(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \int_0^t h(t-\tau) f(\tau) d\tau dt. \tag{1.10.8}$$

We will assume that the force does not act for negative values of time so that $f(\tau) = 0$ for $\tau < 0$. In this case, the lower limit of the integral over τ may be changed from zero to $-\infty$ without changing the result,

$$H(\omega)F(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \int_{-\infty}^t h(t-\tau) f(\tau) d\tau dt. \tag{1.10.9}$$

In addition, because of Eq. (1.10.7), the upper limit of the integral over τ may also be changed to ∞ without changing the result,

$$H(\omega)F(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \int_{-\infty}^{\infty} h(t-\tau) f(\tau) d\tau dt. \tag{1.10.10}$$

Now we will change the order of the integration and multiply by $e^{-i\omega\tau} e^{i\omega\tau} = 1$,

$$\begin{aligned} &H(\omega)F(\omega) = \\ &\frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-i\omega t} h(t-\tau) f(\tau) e^{-i\omega\tau} e^{i\omega\tau} dt d\tau \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-i\omega(t-\tau)} h(t-\tau) dt f(\tau) e^{-i\omega\tau} d\tau. \end{aligned} \tag{1.10.11}$$

Let $t' = t - \tau$ so we can carry out the inner integral,

$$H(\omega)F(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-i\omega t'}h(t')dt' f(\tau)e^{-i\omega\tau}d\tau$$
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} e^{-i\omega t'}h(t')dt' \int_{-\infty}^{\infty} f(\tau)e^{-i\omega\tau}d\tau. \tag{1.10.12}$$

Using the first of Eqs. (1.10.1), (1.10.12) (and hence, Eq. (1.10.5)) will be true if we define the impulse response such that

$$H(\omega) = \int_{-\infty}^{\infty} e^{-i\omega t'}h(t')dt'. \tag{1.10.13}$$

Of course, the integration variable t' can be replaced by t. Note that Eq. (1.10.13) resembles the forward Fourier Transform of Eq. (1.10.1) but it lacks the factor of $1/(2\pi)$.

1.11 Spectral Approach to Evaluating the Convolution Integral

When the impulse response, $h(t)$, and the input signal $f(t)$ are given as specific functions, there are a few situations where the integral in Eq. (1.10.5) can be expressed in terms of a reasonably simple closed form expression. More often, however, the functions $f(t)$ and $h(t)$ do not permit the integral to be readily evaluated. We must often resort to numerical methods. Numerous techniques are available. In the following, we focus on the use of the Fast Fourier Transform (FFT) which can be particularly efficient for the case of transient excitations.

Suppose that rather than seeking the response at an arbitrary instant of time t, we limit our attention to evenly spaced values over an interval from 0 to T,

$$t_n = (n-1)\Delta t = (n-1)\frac{T}{N}, \tag{1.11.1}$$

where N is the number of time points. In using the FFT algorithm, the computations are greatly simplified if we limit the number of points such that N is a power of 2.

We will assume that while $f(t)$ may be a rather complicated function, it will be zero outside the interval for $0 < t < T$. The first integral in Eq. (1.10.1) may then be approximated by a summation over our finite set of values of t_n,

$$F(\omega) = \frac{1}{2\pi}\int_0^T f(t)e^{-i\omega t}dt$$
$$\approx \frac{1}{2\pi}\sum_{n=1}^{N} f_n e^{-i\omega t_n}\Delta t, \tag{1.11.2}$$

where $f_n = f(t_n)$, $\Delta t = T/N$, and $t_n = (n-1)\Delta t$.

To satisfy the Nyquist sampling criterion, the time series, $f(t)$ must be sampled at a high enough rate to obtain at least two samples during one period of the highest frequency component in the signal. If ω_{max} is the highest frequency component in the signal, the Nyquist criterion is

$$\Delta t \leq \frac{2\pi}{\omega_{max}} \frac{1}{2}. \tag{1.11.3}$$

It is helpful to note that $F(\omega)$ is periodic if it is obtained from Eq. (1.11.2), $F(\omega) = F(\omega + 2\pi/\Delta t)$. This can be seen from Eqs. (1.11.1) and (1.11.2),

$$\begin{aligned} F(\omega + 2\pi/\Delta t) &\approx \frac{1}{2\pi} \sum_{n=1}^{N} f_n e^{-i(\omega + 2\pi/\Delta t)(n-1)\Delta t} \Delta t \\ &= \frac{1}{2\pi} \sum_{n=1}^{N} f_n e^{-i2\pi(n-1)} e^{-i\omega(n-1)\Delta t} \Delta t \\ &= \frac{1}{2\pi} \sum_{n=1}^{N} f_n e^{-i\omega(n-1)\Delta t} \Delta t = F(\omega). \end{aligned} \tag{1.11.4}$$

Rather than evaluating Eq. (1.11.2) over all possible values of ω, we will again limit our attention to discrete values given by

$$\omega_j = (j-1)\frac{2\pi}{T}, \quad j = 1 \ldots N. \tag{1.11.5}$$

Note that $\omega_1 = 0$. Also, $\omega_2 = 2\pi/T$ is the lowest full cycle corresponding to the time interval of duration T. Using these values in Eq. (1.11.2) gives

$$\begin{aligned} F(\omega_j) &\approx \frac{1}{2\pi} \sum_{n=1}^{N} f_n e^{-i\omega_n t_n} \Delta t \\ &= \frac{1}{2\pi} \sum_{n=1}^{N} f_n e^{-i(n-1)(j-1)2\pi/n} \frac{T}{N}. \end{aligned} \tag{1.11.6}$$

The Fast Fourier Transform algorithm provides an extremely efficient means of evaluating the summation in Eq. (1.11.6),

$$FFT(f_n) = \sum_{n=1}^{N} f_n e^{-i(n-1)(j-1)2\pi/N}, \tag{1.11.7}$$

so that Eq. (1.11.6) becomes

$$F(\omega_j) \approx \frac{T}{2\pi N} FFT(f_n). \tag{1.11.8}$$

Note that the argument of the FFT algorithm is the entire array of values of f_n. The output of the algorithm is an array of values (generally complex) of $F(\omega_j) = F_j$.

One can also use the inverse FFT algorithm to approximate the inverse Fourier Transform as in Eq. (1.10.4). We will assume that the signals are filtered so that the only significant contributions to the integral are for $-\omega_{max} < \omega < \omega_{max}$, where $\omega_{max} = N\pi/T$ is the highest frequency component in the signal, as required by the Nyquist criterion in Eq. (1.11.3). Equation (1.10.4) then may be approximated by

$$\begin{aligned} x(t) &= \int_{-\infty}^{\infty} e^{i\omega t} H(\omega) F(\omega) d\omega \\ &\approx \int_{-\omega_{max}}^{\omega_{max}} e^{i\omega t} H(\omega) F(\omega) d\omega = 2\Re\left[\int_0^{\omega_{max}} e^{i\omega t} H(\omega) F(\omega) d\omega\right] \\ &= 2\Re\left[\sum_{j=1}^{N/2} e^{i\omega_j t} H(\omega_j) F(\omega_j)\right] \Delta\omega, \end{aligned} \tag{1.11.9}$$

where $\Delta\omega = 2\pi/T$, $\Re[\cdot]$ denotes the real part, and we have used the fact that evaluating the transfer function with negative values of ω yields the conjugate.

If we again limit our attention to discrete values of t given in Eqs. (1.11.1), (1.11.9) become

$$\begin{aligned} x(t_n) = x_n &= 2\Re\left[\sum_{j=1}^{N/2} e^{i\omega_j t_n} H(\omega_j) F(\omega_j) \Delta\omega\right] \\ &= 2\Re\left[\sum_{j=1}^{N/2} e^{i(j-1)(n-1)2\pi/N} H_j F_j \frac{2\pi}{T}\right]. \end{aligned} \tag{1.11.10}$$

Since the summation goes to $N/2$ rather than N, it is convenient to set $H_j = 0$, for $j = N/2 + 1, \ldots, N$. This enables Eq. (1.11.10) to be evaluated very efficiently by the use of the inverse FFT algorithm, which is equivalent to

$$iFFT(H_j F_j) = \frac{1}{N} \sum_{j=1}^{N} e^{i(j-1)(n-1)2\pi/N} H_j F_j. \tag{1.11.11}$$

The time domain response in Eq. (1.11.10) may then be computed by

$$x(t_n) = \frac{4\pi N}{T} \Re[iFFT(H_j F_j)]. \tag{1.11.12}$$

1.12 Example: FFT for Response of a Spring/Mass/Damper

As an example of using the FFT to find the response of a linear system, suppose we have an input signal consisting of a force input to a single degree of freedom spring/mass/damper. The spring/mass/damper will act like a second-order low-pass filter and the displacement of the mass will be the output signal. The response of this simple linear system may be obtained for any complicated forcing simply by directly solving the governing differential equation. Here we use the FFT approach described above.

In this example, consider a fairly complicated input force given by

$$\begin{aligned} f(t) &= F\sin(\pi t/t_1) \quad \text{for} \quad t < t_1 \\ &= 0 \quad \text{for} \quad t_1 \le t \le t_2 \\ &= -F \quad \text{for} \quad t_2 \le t \le t_3 \\ &= 0 \quad \text{for} \quad t_3 \le t. \end{aligned} \tag{1.12.1}$$

This force is also periodic with period T so that $f(t) = f(t+T)$, where $T > t_3$. Let $t_1 = 0.3\,\text{s}, t_2 = 0.6\,\text{s}, t_3 = 0.8\,\text{s}, T = 2\,\text{s}$, and $F = 1\,\text{N}$. The governing differential equation of the system is

$$\ddot{x} + \omega_0^2 x + 2\omega_0\zeta\dot{x} = \frac{f(t)}{m}, \tag{1.12.2}$$

where $\omega_0 = 10$ rad/s, $\zeta = 0.5$ and $m = 3$ kg. The transfer function of the system of Eq. (1.10.2) may be determined by taking the Fourier transform of Eq. (1.12.2),

$$H(\omega) = \frac{1/m}{\omega_0^2 - \omega^2 + 2i\zeta\omega_0\omega}. \tag{1.12.3}$$

The response obtained using the process above is shown in Fig. 1.3 as calculated using a Matlab script (see "fftFilterForArbitrarySignal120718.m").

1.13 Example: Numerical Differentiation and Integration Using the FFT

As another example, suppose we wish to numerically differentiate or integrate time domain data. Suppose that one is able to measure the velocity, $v(t)$, of a system using, for example, a laser vibrometer. The derivative may be written as

$$\dot{v}(t) = \frac{d}{dt}\int_{-\infty}^{\infty} e^{i\omega t}V(\omega)d\omega = \int_{-\infty}^{\infty} e^{i\omega t}i\omega V(\omega)d\omega. \tag{1.13.1}$$

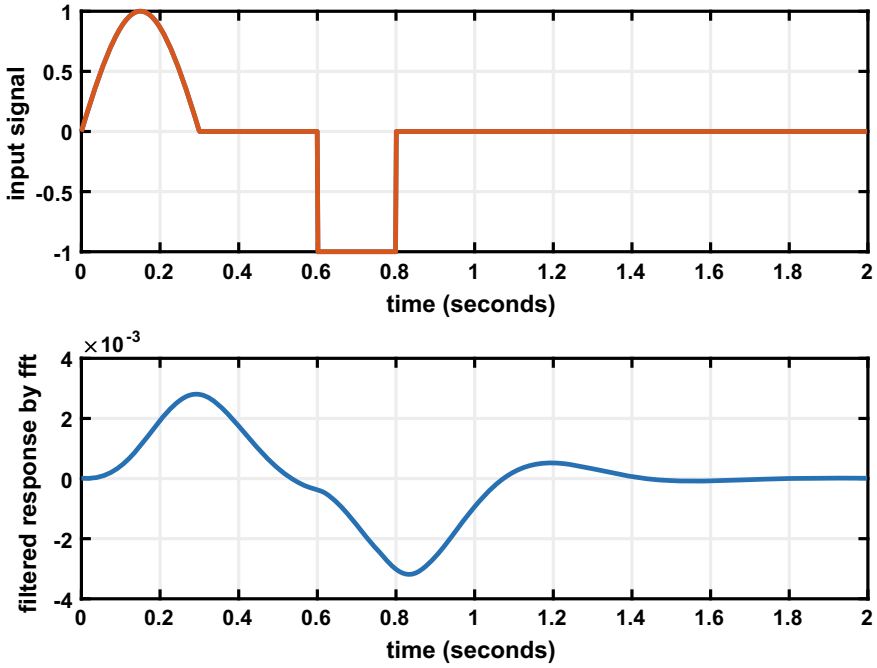


Fig. 1.3 FFT algorithm can be used to calculate the output of a linear system due to complex input signals. The system may be considered as a filter with a transfer function $H(\omega)$ given by Eq. (1.12.3)

Here we have expressed $v(t)$ in terms of the inverse Fourier transform as in Eq. (1.10.3). In the case of differentiation the filter then simply takes the form

$$H(\omega) = i\omega. \tag{1.13.2}$$

The time derivative of the velocity may then be computed at discrete times t_n by adapting Eq. (1.11.10),

$$\dot{v}(t_n) = a(t_n) = \frac{4\pi N}{T}\Re[iFFT(H_j V_j)], \tag{1.13.3}$$

where we have replaced the input $f(t)$ by $v(t)$ and V_j is obtained by applying the FFT algorithm on $v(t_n) = v_n$ as in Eq. (1.11.8),

$$V(\omega_j) = V_j \approx \frac{T}{2\pi N} FFT(v_n). \tag{1.13.4}$$

The use of the FFT algorithm to perform numerical integration of a velocity signal can be developed by expressing the displacement $x(t)$ as the integral of the velocity,

$$x(t) = \int_0^t \dot{x}(t)dt = \int_0^t \int_{-\infty}^{\infty} e^{i\omega t} V(\omega) d\omega dt, \tag{1.13.5}$$

where we have expressed $\dot{x}(t)$ using the inverse Fourier transform of $V(\omega)$, which is the Fourier transform of the velocity $v(t)$. Changing the order of the integration and carrying out the integration over time, t give

$$x(t) = \int_0^t \dot{x}(t)dt = \int_{-\infty}^{\infty} V(\omega) \frac{(e^{i\omega t} - 1)}{i\omega} d\omega = \int_{-\infty}^{\infty} V(\omega) \frac{e^{i\omega t}}{i\omega} d\omega - \int_{-\infty}^{\infty} \frac{V(\omega)}{i\omega} d\omega. \tag{1.13.6}$$

The first integral on the right-hand side can be evaluated using the FFT in a similar manner to Eq. (1.11.9),

$$\int_{-\infty}^{\infty} V(\omega) \frac{e^{i\omega t}}{i\omega} d\omega \approx 2\Re \left[\sum_{j=1}^{N/2} e^{i\omega_j t} \frac{V(\omega_j)}{i\omega_j} \right] \Delta\omega. \tag{1.13.7}$$

In this case, however, since the first frequency component is $\omega_1 = 0$, $V(\omega_1)$ corresponds to a constant velocity which will not occur in typical vibrating systems. We will then neglect the constant term by setting $V(\omega_1) = 0$. Evaluating Eq. (1.13.7) at discrete times t_n as in Eq. (1.11.12) gives

$$\int_{-\infty}^{\infty} V(\omega) \frac{e^{i\omega t_n}}{i\omega} d\omega \approx 2\Re \left[\sum_{j=1}^{N/2} e^{i\omega_j t_n} \frac{V(\omega_j)}{i\omega_j} \right] \Delta\omega \approx \frac{4\pi N}{T} \Re \left[iFFT \left(\frac{V(\omega_j)}{i\omega_j} \right) \right], \tag{1.13.8}$$

where, again, since there is no constant velocity, we set $\frac{V(\omega_1)}{i\omega_1} = 0$ before computing the inverse FFT.

We must now evaluate the second integral on the right side of Eq. (1.13.6). This can be accomplished as in Eq. (1.13.7),

$$\int_{-\infty}^{\infty} \frac{V(\omega)}{i\omega} d\omega \approx 2\Re \left[\sum_{j=1}^{N/2} \frac{V(\omega_j)}{i\omega_j} \right] \Delta\omega \approx 2\Re \left[\sum_{j=2}^{N/2} \frac{V(\omega_j)}{i\omega_j} \right] \frac{2\pi}{T}, \tag{1.13.9}$$

where we have again neglected the constant velocity term, $V(\omega_1)$ and have used the fact that $\Delta\omega = \frac{2\pi}{T}$. Note that the summation on the right hand side begins at $j = 2$. Equations (1.13.6), (1.13.8), and (1.13.9) give the integral of the velocity $v(t)$ (i.e. the displacement) as

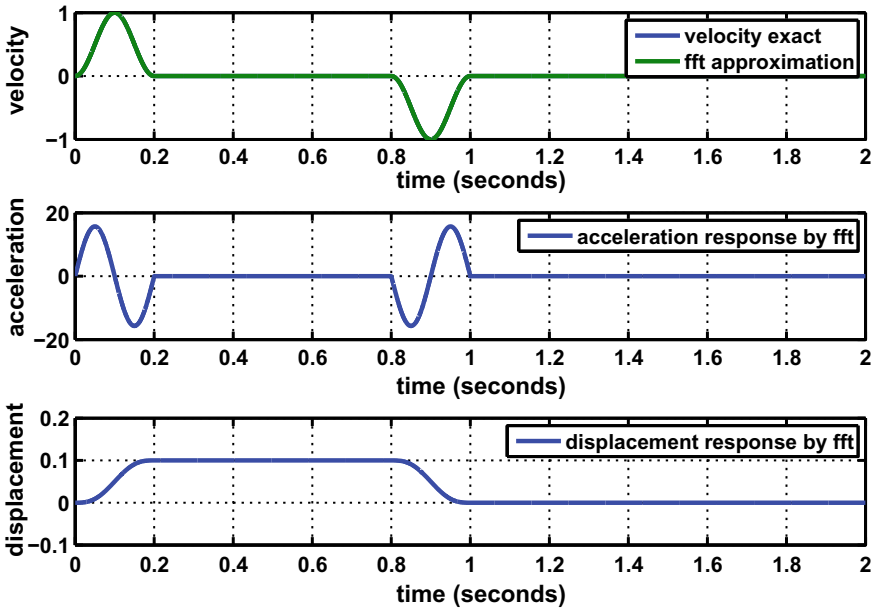


Fig. 1.4 FFT algorithm can be used to numerically differentiate and integrate a complex input signal. The differentiation may be considered as a filter with a transfer function $H(\omega)$ given by Eq. (1.13.2). The integration to calculate displacement was obtained using Eq. (1.13.10)

$$x(t_n) = \frac{4\pi N}{T}\Re\left[iFFT\left(\frac{V(\omega_j)}{i\omega_j}\right)\right] + \frac{4\pi}{T}\Re\left[\sum_{j=2}^{N/2}\frac{V(\omega_j)}{i\omega_j}\right]. \tag{1.13.10}$$

Here, we will apply this process to a velocity signal having smooth derivatives as shown in Fig. 1.4. The acceleration and displacements obtained using Eqs. (1.13.3) and (1.13.10), respectively, are as expected. These results were obtained using "fftFilterDifferentiationandIntegrationForArbitrarySignal120118.m".

1.14 FFT and iFFT to Estimate the Fourier Transform for Complex Functions

In Eq. (1.11.9) we developed a method of using the inverse FFT, iFFT, to estimate the time domain signal $x(t)$ from the system's frequency response $H(\omega)$. In this case, we assumed that the frequency response function was obtained from a typical dynamic system so that when it is evaluated for negative values of ω, one obtains

the complex conjugate. While this typically occurs for dynamic systems, there are situations where this is not the case. Consider again Eq. (1.11.9),

$$x(t) = \int_{-\infty}^{\infty} e^{i\omega t} H(\omega) F(\omega) d\omega \approx \int_{-\omega_{max}}^{\omega_{max}} e^{i\omega t} H(\omega) F(\omega) d\omega$$
$$= \int_{-\omega_{max}}^{0} e^{i\omega t} H(\omega) F(\omega) d\omega + \int_{0}^{\omega_{max}} e^{i\omega t} H(\omega) F(\omega) d\omega$$
$$= \int_{0}^{\omega_{max}} e^{-i\omega t} H(-\omega) F(-\omega) d\omega + \int_{0}^{\omega_{max}} e^{i\omega t} H(\omega) F(\omega) d\omega. \tag{1.14.1}$$

Approximating the integrals by finite summations at discrete values of ω gives

$$x(t) \approx \sum_{j=2}^{N/2} e^{-i\omega_j t} H(-\omega_j) F(-\omega_j) \Delta\omega + \sum_{j=1}^{N/2} e^{i\omega_j t} H(\omega_j) F(\omega_j) \Delta\omega, \tag{1.14.2}$$

where $\Delta\omega = 2\pi/T$ and $\omega_j = (j-1)\Delta\omega$. Note that we have started the first summation at $j = 2$ to avoid the contribution to the integral at ω_1 from being counted twice.

Again limiting our attention to discrete values of t given in Eqs. (1.11.1), (1.14.2) becomes

$$x(t_n) = x_n = \sum_{j=2}^{N/2} e^{-i\omega_j t_n} H(-\omega_j) F(-\omega_j) \Delta\omega + \sum_{j=1}^{N/2} e^{i\omega_j t_n} H(\omega_j) F(\omega_j) \Delta\omega. \tag{1.14.3}$$

Since the summation goes to $N/2$ rather than N, we will again set $H(\omega_j) = H(-\omega_j) = 0$, for $j = N/2 + 1, \ldots, N$. The second summation in Eq. (1.14.3) may be evaluated very efficiently by the use of the inverse FFT algorithm, which is equivalent to

$$iFFT(H(\omega_j)F(\omega_j)) = \frac{1}{N} \sum_{j=1}^{N} e^{i(j-1)(n-1)2\pi/N} H(\omega_j) F(\omega_j). \tag{1.14.4}$$

The first summation in Eq. (1.14.3) may be evaluated using the forward FFT algorithm, where the complex functions F and H are evaluated for negative values of ω_j. In order to start the summation at $j = 2$, we will set $H(-\omega_1) = 0$,

$$FFT(H(-\omega_j)F(-\omega_j)) = \sum_{j=1}^{N} e^{-i(j-1)(n-1)2\pi/N} H(-\omega_j) F(-\omega_j). \tag{1.14.5}$$

The time domain response in Eq. (1.14.3) may then be computed by

$$x(t_n) = x_n = \frac{2\pi}{T} FFT(H(-\omega_j)F(-\omega_j)) + \frac{2\pi N}{T} iFFT(H(\omega_j)F(\omega_j)). \tag{1.14.6}$$

1.15 Problems

1. A plane sound wave in air has a frequency of 1034 Hz and a peak acoustic pressure amplitude of 0.3 N/m^2. (a) What is its mean square pressure? (b) What is its sound pressure level re 20 μPa (i.e. $p_{ref} = 20 \times 10^{-6}$ Pa)?
2. A simple Matlab file, "SPLcalchw.m" is available that will read the file "timedatam.mat" to create two arrays, pressure and time, and plots pressure as a function of time. The file "timedatam.mat" contains simulated sound pressure data (pascals) as a function of time (seconds). Determine the sound pressure level (SPL) for the pressure data.
3. One period of a periodic sound pressure is shown in Fig. 1.5. Obtain exact values for the mean square pressure and the overall sound pressure level in decibels

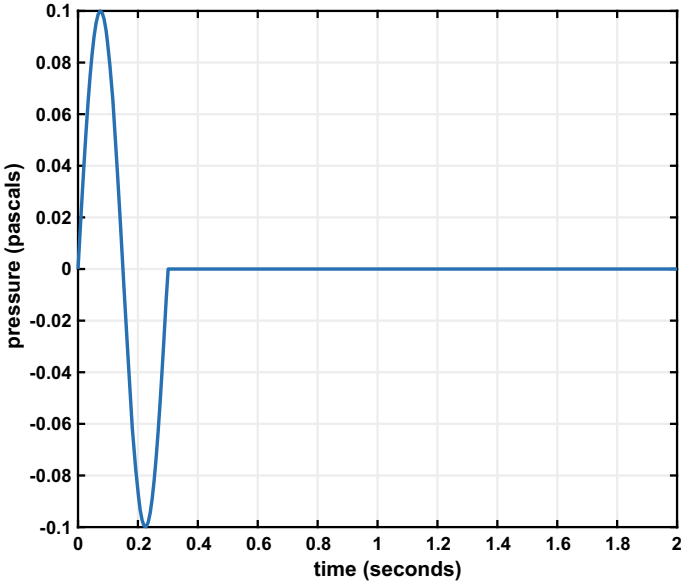


Fig. 1.5 One period of a periodic sound pressure signal

using a reference pressure of $p_{ref} = 20 \times 10^{-6}$ Pa. Note that the period of this signal is $T = 2$ s. While you may check your result using numerical or symbolic computations to evaluate the integral, you must evaluate the integral "by hand" and show your work.

4. Suppose that the two sound pressures for Problems 2 and 3 are detected simultaneously. Based on your estimates of the SPL for each one acting alone, use Eq. (1.1.6) to estimate the SPL that one would obtain if both were acting at the same time. This is a simple calculation. Do not do this problem by adding the time domain signals and integrating numerically. That is done in Problem 5.
5. Modify your Matlab program used to estimate the mean square pressure and SPL in Problem 2 to calculate the mean square pressure of the summation of the two pressure time histories examined in Problems 2 and 3. Calculate the SPL based on this time domain calculation and compare with the results obtained using the simple formula in Problem 4.
6. A harmonic sound pressure is given by $p(t) = 0.8\cos(\omega t + \phi)$ where $\phi = 0.4$ radians and ω is the frequency in radians/second. If this time varying pressure is expressed in the form $p(t) = \Re[Pe^{i\omega t}]$, where $\Re[\cdot]$ denotes the real part and $P = a + ib$, determine the real and imaginary parts (a and b) of the complex constant P.
7. For the sound pressure given in Problem 6, **derive** an expression for the mean square pressure $< p^2(t) >$ in terms of the real and imaginary parts of the complex constant $P = a + ib$.
8. A sound wave has a mean square pressure given by $< p^2 >= 0.15\,\text{Pa}^2$. If the standard reference pressure is 20×10^{-6} Pa, calculate the sound pressure level in decibels.
9. A room contains two noise sources, an air conditioner, and a computer cooling fan. The sound pressure level measured when the air conditioner operates alone is 72 dB. When the computer fan operates alone the measured sound pressure level is 74 dB. Determine the sound pressure level that would be measured if both noise sources are operated simultaneously.
10. According to ANSI S1.42-2001, Design Response of Weighting Networks for Acoustical Measurements, the A-weighting filter shape as a function of frequency, f in Hertz is

$$H(f) = \frac{\sqrt{3.5041384 \times 10^{16}}(if)^4}{(if + 20.598997)^2(if + 107.65265)(if + 737.86223)(if + 12194.217)^2}, \tag{1.15.1}$$

where $i = \sqrt{-1}$. The filter shape may be plotted versus frequency on a decibel scale by plotting

$$dBAWeight(f) = 10\log_{10}(H(f)H^*(f)), \tag{1.15.2}$$

Table 1.2 Octave band sound levels

Frequency (Hz)	Sound pressure level
63	67
125	77
250	90
500	68
1000	78
2000	62
4000	45
8000	39

where * denotes the complex conjugate. Sound pressure levels are usually quantified using either 1/3 octave band frequencies or full octave band frequencies. The 1/3 octave band frequencies in Hz are computed from $f = 10^{n/10}$ where $n = 0, 1, 2, \ldots, 50$ are the band numbers. The full octave band center frequencies may be computed from $f = 10^{3m/10}$ where $m = 0, 1, 2, \ldots, 16$. Measured octave band sound levels are shown in Table 1.2. (a) Use Eq. (1.15.2) to compute the A-weighting in decibels for each octave band frequency in Table 1.2, (b) compute the A-weighted octave band sound pressure levels at each of these frequencies, (c) compute the overall A-weighted sound pressure level in dBA and (d) the overall sound pressure level in dB.

11. If one has a pressure power spectral density, $G_{pp}(f)$ with units of Pa^2/Hz, the corresponding dBA sound pressure level is calculated from

$$SPL_A = 10\log_{10}\left(\int_0^\infty \frac{G_{pp}(f)}{P_{ref}^2} H(f)H^*(f)df\right), \tag{1.15.3}$$

where $P_{ref} = 20 \times 10^{-6}$ Pa is the mean square reference pressure and $H(f)$ is given in Eq. (1.15.1).
In some cases, the pressure power spectral density is independent of frequency so that it can be taken outside the integral,

$$SPL_A = 10\log_{10}\left(G_{pp}\int_0^\infty \frac{1}{P_{ref}^2} H(f)H^*(f)df\right) = 10\log_{10}(G_{pp}) + W_c. \tag{1.15.4}$$

(a) Carry out the integration to determine the constant W_c to at least four significant digits. You may use a numerical integration method or obtain an exact analytical expression. In other words, verify Eq. (1.4.9). (b) Suppose that a microphone is found to produce a noise signal having a constant power spectral density

given by $G_{pp} = 10^{-10}$ Pa^2/Hz. Use Eq. (1.15.4) to find the corresponding A-weighted sound pressure level in dBA.

12. The program, "soundcardspectrumanalyzer14.m", has been made available. Modify the program so that it analyzes a known, deterministic signal that you can use to verify that the program is correct. You may employ whatever checks are appropriate to verify the program. Do not assume that the program is correct. Turn in your modified program (with comments inserted) along with a written description of your method (in English sentences).
13. Use the program "soundcardspectrumanalyzer14.m" to acquire a sound signal. Modify the program to compute 1/3 octave band sound pressure levels from the narrowband data that the program acquires. Calculate the overall sound pressure level from your 1/3 octave levels and compare with the overall calculated from the time history and the spectrum.
14. Suppose we wish to analyze a simple periodic pulse pressure signal with period $T = 0.37152$ s. During the first period it is equal to $p(t) = 0.25$ Pa for $t < T/10$ and $p(t) = 0$ for $T/10 \leq t \leq T$. Again, it is periodic so that for all subsequent periods, $p(t) = p(t + T)$. Use an exact approach to obtain the power spectral density of the signal. You may use the finite Fourier transform as in Eqs. (1.6.6) and (1.6.7).

Chapter 2
One Dimensional Sound Fields



Sound in air that we hear with our ears consists of minute fluctuations in the ambient atmospheric air pressure. If we limit our attention to one dimensional spaces (like inside a narrow tube), we can write down a model for the fluctuating sound pressure. It helps to think of the total air pressure, $P(t)$ as the sum of the static, ambient pressure P_0 and the minute acoustic fluctuation, $p(t)$, $P(t) = P_0 + p(t)$. Of course, if the pressure changes at a point in space, even by a tiny fluctuating amount, we would also expect the density (or total mass of the molecules of the gas for a given volume) to also change depending on the change in pressure.

We must always keep in mind that other parameters that are commonly used to describe a gas will also fluctuate with the pressure. If the density changes, we would expect the average kinetic energy of the molecules, or the temperature to also vary according to the pressure fluctuations. Finally, if the pressure is not uniform in space as it fluctuates in time, the forces that act on a small region of air will not be in equilibrium. As a result, according to Newton's second law, the net force will cause an acceleration of the mass within that volume.

In the following, we will consider fundamental properties of a gas to construct a simple mathematical model for the propagation of sound. The primary coordinates we will consider are the pressure $P(x, t)$, the mean particle velocity, $U(x, t)$, and the density $\rho(x, t)$. The relationships between these coordinates will be derived through the use of three fundamental equations: Newton's second law, the conservation of mass, and the equation of state.

2.1 Newton's Second Law

Imagine a slab of air in a tube that has cross-sectional area A, and differential thickness dx where x is the spatial coordinate, as shown in Fig. 2.1. We will account for the spatial and temporal variation in the total pressure by expressing it as $P(x, t)$. The

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_2.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_2

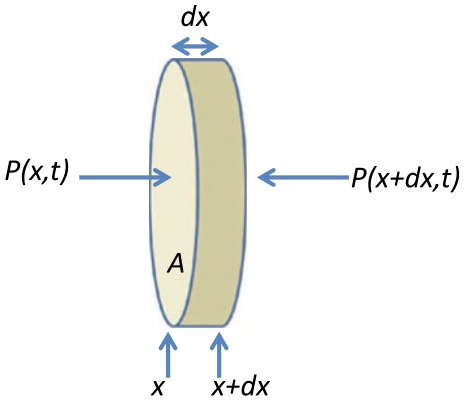


Fig. 2.1 An infinitesimal volume of air

left side of this infinitesimal volume of air is acted on by the pressure on its left face, which will apply a force to the right given by $P(x,t)A$. The right side of the volume of air having thickness dx is acted on by a force to the left given by $-P(x+dx,t)A$. It is very important to note that the pressure of the air (whether fluctuating as sound or not) always creates a force that is **normal** to any surface.

Air, of course, consists of a large number of molecules, or particles, of gas that are all traveling at very high velocities, depending on the temperature. If the gas is confined to a very small space, we would expect the average of the molecular velocities to be zero since the air, on average isn't going anywhere. If the volume of air is large enough or if the pressure varies with position the spatial average of the velocity of these molecules will be nonzero; the change in pressure with position causes the air to move. This is, of course, what causes wind on a large spatial scale. If we let $U(x,t)$ be the spatial average of the particle velocity (taken over each infinitesimal region of the air), then Newton's second law provides the relation between the net force, F_{net}, and the rate of change of the linear momentum, $\frac{\partial}{\partial t}(m(x,t)U(x,t))$, or the spatial variation in pressure and the movement of the air,

$$F_{net} = \frac{\partial}{\partial t}(m(x,t)U(x,t))$$
$$P(x,t)A - P(x+dx,t)A = \frac{\partial}{\partial t}(\rho(x,t)AdxU(x,t)), \tag{2.1.1}$$

where $m(x,t) = \rho(x,t)Adx$ is the mass of the air in the slab of thickness dx. Equation (2.1.1) may be rearranged to

$$-\frac{P(x+dx,t) - P(x,t)}{dx} = \frac{\partial}{\partial t}(\rho(x,t)U(x,t)). \tag{2.1.2}$$

Taking the limit as $dx \to 0$ gives

$$-\frac{\partial P(x,t)}{\partial x} = \frac{\partial}{\partial t}(\rho(x,t)U(x,t)). \tag{2.1.3}$$

Equation (2.1.3) is an extremely important equation in acoustics and is usually referred to as the Euler equation. It indicates the crucial relationship between the movement of air and the spatial gradient of the pressure. Note that we have written this in terms of the total pressure $P(x,t)$, which includes the ambient air pressure. Since the ambient pressure is spatially uniform we can also write it in terms of the acoustic fluctuating pressure,

$$-\frac{\partial p(x,t)}{\partial x} = \frac{\partial}{\partial t}(\rho(x,t)U(x,t)). \tag{2.1.4}$$

2.2 Conservation of Mass

In addition to requiring that Newton's second law be satisfied, it is also essential that the particles of air obey mass conservation. This will provide us a relationship between the average particle velocity and the fluctuation in density. Again consider the infinitesimal slab of air shown in Fig. 2.1. If $U(x,t)$ is the particle velocity averaged over the left face with area A, then the rate at which mass crosses into the volume through this left face will be $\rho(x,t)U(x,t)A$, where $\rho(x,t)$ is the mass per unit volume at x and t. Mass will pass out of the face on the right side of the infinitesimal volume at $x+dx$ with a rate of $\rho(x+dx,t)U(x+dx,t)A$. Since dx is infinitesimal, the rate of change of the mass contained within the volume Adx will equal the net rate at which mass enters,

$$\begin{aligned}&\frac{d}{dt}(\rho(x+dx/2,t)Adx) = \frac{d}{dt}(\rho(x,t)Adx)\\ &= \rho(x,t)U(x,t)A - \rho(x+dx,t)U(x+dx,t)A.\end{aligned} \tag{2.2.1}$$

Again, using the fact that dx is infinitesimal, Eq. (2.2.1) gives

$$-\frac{\partial(\rho U)}{\partial x} = \dot{\rho}. \tag{2.2.2}$$

The density ρ in Eqs. (2.2.1) and (2.2.2) consists of a constant ambient air density ρ_0 and a miniscule fluctuating part due to acoustic sound, ρ_a, $\rho = \rho_0 + \rho_a$. Because ρ_0 is constant in time and space, the right side of Eq. (2.2.2) simplifies to

$$-\frac{\partial(\rho U)}{\partial x} = \dot{\rho}_a. \tag{2.2.3}$$

2.3 Equation of State

Equations (2.1.4) and (2.2.3) provide two equations relating our three coordinates, pressure, velocity and density. We need an additional equation to fully characterize the sound field. This is given by the equation of state which expresses the total pressure as a function of the fluctuating density,

$$P(x,t) = P(\rho) = P_0 + \rho_a \frac{\partial P}{\partial \rho}|_{\rho=\rho_0}, \tag{2.3.1}$$

where the right side of Eq. (2.3.1) consists of a two-term Taylor's series of the total pressure as a function of the density. We limit our attention to small fluctuations in density that are much smaller than the ambient static air density ρ_0 so that two terms are sufficient. Let

$$\frac{\partial P}{\partial \rho}|_{\rho=\rho_0} = c^2, \tag{2.3.2}$$

where c^2 is a constant.

Since the total pressure $P(x,t)$ consists of the sum of the static pressure P_0 and the fluctuating acoustic pressure, $p(x,t)$, Eqs. (2.3.1) and (2.3.2) give a simple relation between the fluctuating pressure and the fluctuating density,

$$p(x,t) = \rho_a c^2. \tag{2.3.3}$$

Equations (2.1.4), (2.2.3), and (2.3.3) comprise three equations in the three unknowns $p(x,t)$, $U(x,t)$, and $\rho_a(x,t)$. These can be used to obtain a single differential equation for the pressure, $p(x,t)$. Differentiating equation (2.1.4) by x, Eq. (2.2.3) by t and subtracting give

$$\frac{\partial^2 p(x,t)}{\partial x^2} = \ddot{\rho}_a. \tag{2.3.4}$$

Equations (2.3.3), and (2.3.4) give

$$\frac{\partial^2 p(x,t)}{\partial x^2} = \frac{1}{c^2}\ddot{p}(x,t). \tag{2.3.5}$$

2.4 Solutions for Some Simple Fields

If we limit our attention to harmonic sound waves at the frequency ω, the solution to Eq. (2.3.5) can be expressed as

$$p(x,t) = p_1 e^{i(\omega t - kx)} + p_2 e^{i(\omega t + kx)} = e^{i\omega t}(p_1 e^{-ikx} + p_2 e^{ikx}), \tag{2.4.1}$$

where $k = \omega/c, i = \sqrt{-1}$, p_1 is the amplitude of the wave propagating in the positive x direction and p_2 is the amplitude of the wave traveling in the opposite direction.

One can use the form of solution given in Eq. (2.4.1) to solve for the natural frequencies of tubes with various boundary conditions. For example, if the tube is open on each end, we could have $p(0,t) = p(l,t) = 0$, where l is the length of the tube. The fact that $p(0,t) = 0$ leads to $p_1 = -p_2$. Using this along with the fact that $p(l,t) = 0$ gives

$$p_1(e^{-ikl} - e^{ikl}) = -2ip_1 \sin(kl) = 0, \tag{2.4.2}$$

so that the allowed solutions for kl are $kl = n\pi$ where n is any nonzero integer. The two waves thus combine to that the resulting standing wave satisfies the two boundary conditions.

We can also easily examine cases where other boundary conditions must be satisfied. Suppose, for now, that there is no boundary at $x = l$ so that the tube continues on to infinity. Let the left end (at $x = 0$) have a piston that moves with a velocity given by $u = Ue^{i\omega t}$. In order to relate the velocity boundary condition to the pressure (so we can determine the required constant), we need to use the linearized Euler equation, Eq. (2.1.3), which expresses the force or momentum balance in the sound field,

$$-\frac{\partial p(x,t)}{\partial x} = \rho_0 \dot{u}(x,t), \tag{2.4.3}$$

where we have used $\rho = \rho_0 + \rho_a$ and neglected the product $\rho_a U(x,t)$, which consists of small fluctuating terms and ρ_0 is again, the nominal air density. Note that this equation applies anywhere in the domain and must be satisfied in any one-dimensional sound field. Because there is no boundary on the right end of our domain, the reflected wave will not exist, $p_2 = 0$. Evaluating equation (2.4.3) at $x = 0$ then gives

$$ikp_1 e^{i\omega t} = i\rho_0 \omega U e^{i\omega t} \tag{2.4.4}$$

or, since $k = \omega/c$,

$$p_1 = \rho_0 c U \tag{2.4.5}$$

The pressure at any point in the tube will then be

$$p(x,t) = \rho_0 c U e^{i(\omega t - kx)}. \tag{2.4.6}$$

We have assumed that the piston motion is harmonic with frequency ω. In Sect. 2.7 we will obtain the pressure field in the more general case where the velocity of the piston is arbitrary, $u(t)$.

Now suppose that the tube is terminated at $x = l$ by a spring/mass/damper system governed by

$$M\ddot{x} + Kx + C\dot{x} = f = pA, \tag{2.4.7}$$

where A is the cross-sectional area of the tube. If we again assume harmonic time dependence so that $p = Pe^{i\omega t}$ then the velocity of the mass will be related to the pressure by

$$u(l,t) = \frac{i\omega PAe^{i\omega t}}{K - M\omega^2 + i\omega C}. \tag{2.4.8}$$

Evaluating equations (2.4.1) and (2.4.3) at $x = l$ and using Eq. (2.4.8) give

$$ik(p_1 e^{-ikl} - p_2 e^{ikl}) = -\omega^2 \rho_0 \frac{A(p_1 e^{-ikl} + p_2 e^{ikl})}{K - M\omega^2 + i\omega C}. \tag{2.4.9}$$

If, as before, we have a piston at $x = 0$, then Eq. (2.4.3) may be evaluated at $x = 0$ to give,

$$-ik(p_1 - p_2) = \rho_0 i\omega U. \tag{2.4.10}$$

Equations (2.4.9) and (2.4.10) may be used to solve for p_1 and p_2. The pressure at any point in the tube may then be determined using Eq. (2.4.1).

If the tube is terminated by a general impedance $Z = P/U$ where P and U are the complex amplitudes of the harmonic pressure and velocity, then the Euler equation at $x = l$ becomes

$$ik(p_1 e^{-ikl} - p_2 e^{ikl}) = i\omega\rho_0 \frac{(p_1 e^{-ikl} + p_2 e^{ikl})}{Z}. \tag{2.4.11}$$

Since $k = \omega/c$, this equation can be used to determine the ratio p_2/p_1, which is usually referred to as the reflection coefficient,

$$\frac{p_2}{p_1} = e^{-2ikl} \frac{(\frac{Z}{\rho_0 c} - 1)}{\frac{Z}{\rho_0 c} + 1}. \tag{2.4.12}$$

Note that if we redefine the origin of our coordinate system to be at the location of the impedance Z, then the exponential term in Eq. (2.4.12) will not appear.

If the system is driven with a piston at the left end (at $x = 0$), as before, with a velocity $u = Ue^{i\omega t}$, then Eqs. (2.4.10) and (2.4.12) may be solved for p_1 and p_2,

$$p_1 = \frac{\rho_0 c U}{1-\beta}, \quad p_2 = \frac{\rho_0 c U}{1/\beta - 1}, \tag{2.4.13}$$

where

$$\beta = e^{-2ikl} \frac{(\frac{Z}{\rho_0 c} - 1)}{\frac{Z}{\rho_0 c} + 1}. \tag{2.4.14}$$

2.5 Sound Intensity and the Sound Absorption Coefficient

Having a description of a one-dimensional sound field in the form of incident and reflected waves (p_1 and p_2), it is instructive to examine sound intensity and the sound absorption coefficient in terms of these two waves. The instantaneous sound intensity, $I(t)$ is defined to be the product of the pressure, $p(t)$ and the acoustic particle velocity, $u(t)$, $I(t) = p(t)u(t)$. This has the units of acoustic power per unit area. We typically are interested in the time average of the intensity,

$$< I(t) >= \frac{1}{T} \int_0^T I(t) dt, \tag{2.5.1}$$

where the integration time, T is typically equal to the period of the signal if it is periodic. Suppose the pressure and velocity are harmonic signals given by

$$\begin{aligned} p(t) &= p \cos(\omega t + \phi) \\ u(t) &= u \cos(\omega t + \psi). \end{aligned} \tag{2.5.2}$$

The time-averaged intensity is

$$\begin{aligned} &< I(t) >= \frac{1}{T} \int_0^T p \cos(\omega t + \phi) u \cos(\omega t + \psi) dt \\ &= \frac{1}{T} \int_0^T p \cos(\omega t + \phi) u \cos(\omega t + \phi + \beta) dt \\ &= \frac{1}{T} \int_0^T p \cos(\omega t + \phi) \\ &u(\cos(\omega t + \phi)\cos(\beta) - \sin(\omega t + \phi)\sin(\beta)) dt \\ &= \frac{1}{T} \int_0^T pu \cos^2(\omega t + \phi)\cos(\beta) dt \\ &= \frac{pu\cos(\beta)}{2} = \frac{pu\cos(\psi - \phi)}{2} = \frac{pu\cos(\phi - \psi)}{2}, \end{aligned} \tag{2.5.3}$$

where the period of the oscillation is $T = 2\pi/\omega$. This result can also be calculated using our more typical complex notation for the pressure and velocity, $Pe^{i\omega t}$ and $Ue^{i\omega t}$, respectively. Let the pressure and velocity be the real parts of these complex quantities,

$$\begin{aligned} p(t) &= \Re[Pe^{i\omega t}] = \Re[pe^{i(\omega t+\phi)}] = p\cos(\omega t + \phi) \\ u(t) &= \Re[Ue^{i\omega t}] = \Re[ue^{i(\omega t+\psi)}] = u\cos(\omega t + \psi), \end{aligned} \tag{2.5.4}$$

where $\Re[\cdot]$ denotes the real part. From Eq. (2.5.4) we can see that $P = pe^{i\phi} = p(\cos(\phi) + i\sin(\phi))$ and $U = ue^{i\psi} = u(\cos(\psi) + i\sin(\psi))$. The time-averaged intensity given in Eq. (2.5.3) may then be computed from our complex representation by

$$< I >= \frac{\Re[PU^*]}{2} = \frac{\Re[pe^{i\phi}ue^{-i\psi}]}{2} = \frac{\Re[P^*U]}{2} = \frac{\Re[pe^{-i\phi}ue^{i\psi}]}{2} = \frac{pu\cos(\phi-\psi)}{2}, \tag{2.5.5}$$

where the superscript * denotes the complex conjugate.

For a single plane wave traveling in the positive direction we know that $p(t)/u(t) = \rho_0 c = P/U$ where ρ_0 is the nominal air density and c is the wave propagation speed. In this case, Eq. (2.5.5) gives the sound intensity of a single wave to be

$$< I >= \frac{PP^*}{2\rho_0 c} = \frac{p^2}{2\rho_0 c}. \tag{2.5.6}$$

The sound intensity **level** is defined by

$$L_i = 10\log_{10}(< I > /I_0), \tag{2.5.7}$$

where $I_0 = 10^{-12}$ W/m^2 is the reference intensity.

It should be noted that this result can also show the relation between our complex representation of sound waves, as in our solution to the wave equation in Eq. (2.4.1) and the sound pressure level, which requires us to calculate the time average of the mean square pressure as in Eq. (1.1.1). If we consider the sound pressure to be expressed as

$$p(t) = \Re[Pe^{i\omega t}], \tag{2.5.8}$$

the mean square pressure may be determined merely by replacing U with P in Eq. (2.5.5). The mean square pressure is then

$$< p^2 >= \frac{PP^*}{2} = \frac{|P|^2}{2}. \tag{2.5.9}$$

Having defined sound intensity, we can now define the sound absorption coefficient, α. The absorption coefficient of a surface is given by the ratio of the sound intensity absorbed by the surface to the sound intensity of an incident wave.

Now consider the somewhat more general case where a sound field consists of the combination of an incident and reflected wave. The pressure can be expressed as

$$p(x,t) = e^{i\omega t}(p_1 e^{-ikx} + p_2 e^{ikx}) = P(x)e^{i\omega t}, \tag{2.5.10}$$

where p_1 and p_2 are the complex amplitudes of the incident and reflected waves, respectively and the wave number is $k = \omega/c$.

Suppose there is an absorbing surface at $x = 0$ and we would like to express the absorption coefficient α in terms of p_1 and p_2. We first need to express the absorbed intensity of the wave at $x = 0$. As in Eq. (2.5.5), this requires us to write both the pressure and the velocity. The velocity can be expressed in terms of p_1 and p_2 through the use of the Euler equation,

$$-\frac{\partial p}{\partial x} = \rho_0 \dot{u}. \tag{2.5.11}$$

Having expressed the pressure in the complex form of Eq. (2.5.10), we can write the velocity in the form $u(x,t) = U(x)e^{i\omega t}$. Substituting this and Eq. (2.5.10) into Eq. (2.5.11) and evaluating at $x = 0^-$ gives,

$$U(0) = \frac{p_1 - p_2}{\rho_0 c}. \tag{2.5.12}$$

Equations (2.5.5), (2.5.10), and (2.5.12) allow us to write the time-averaged sound intensity at $x = 0$ as

$$\begin{aligned} < I > &= \frac{\Re[(p_1 + p_2)(p_1^* - p_2^*)]}{2\rho_0 c} \\ &= |p_1|^2 \frac{\Re[(1 + p_2/p_1)(1 - p_2^*/p_1^*)]}{2\rho_0 c} \\ &= |p_1|^2 \frac{(1 + \Re[p_2/p_1 - p_2^*/p_1^*] - |p_2/p_1|^2)}{2\rho_0 c} \\ &= |p_1|^2 \frac{(1 - |p_2/p_1|^2)}{2\rho_0 c}. \end{aligned} \tag{2.5.13}$$

The ratio of the complex amplitudes of the reflected wave to that of the incident wave is referred to as the reflection coefficient,

$$r = \frac{p_2}{p_1}. \tag{2.5.14}$$

Also, the intensity of the incident wave is

$$< I_{inc} >= \frac{|p_1|^2}{2\rho_0 c}. \tag{2.5.15}$$

The sound absorption coefficient may then be written as

$$\alpha = \frac{< I >}{< I_{inc} >} = 1 - |p_2/p_1|^2 = 1 - |r|^2. \tag{2.5.16}$$

Now suppose there is an absorbing material at $x = 0$ having a known acoustic impedance Z. This material determines the relative amplitudes of the incident and reflected waves, p_1 and p_2. The impedance at $x = 0$ fixes the relationship between the complex amplitudes of the pressure and velocity at $x = 0^-$,

$$Z = \frac{P(0)}{U(0)}. \tag{2.5.17}$$

Since for our sound field, $P(0) = p_1 + p_2$, Eqs. (2.5.12), (2.5.14), and (2.5.17) may be rearranged to give

$$\frac{p_2}{p_1} = \frac{Z/(\rho_0 c) - 1}{Z/(\rho_0 c) + 1} = r. \tag{2.5.18}$$

Having the reflection coefficient r in terms of the impedance Z we can also express the absorption coefficient by combining Eqs. (2.5.16) and (2.5.18),

$$\begin{aligned} \alpha &= 1 - |r|^2 = 1 - \frac{Z/(\rho_0 c) - 1}{Z/(\rho_0 c) + 1} \times \frac{Z^*/(\rho_0 c) - 1}{Z^*/(\rho_0 c) + 1} \\ &= \frac{4\Re[Z/(\rho_0 c)]}{|Z/(\rho_0 c)|^2 + 2\Re[Z/(\rho_0 c)] + 1}. \end{aligned} \tag{2.5.19}$$

2.6 d'Alembert's Solution

The analysis above for one-dimensional sound fields was limited to harmonic fields with oscillating pressures at the frequency ω. In the following, we will examine the more general case where the time variation of the pressure can take a nearly arbitrary form. This can be useful when the field is due to a transient pulse. We will also obtain a

very general relationship between the pressure and the acoustic particle velocity that doesn't depend on the field being harmonic. Again, the partial differential equation governing the acoustic pressure, $p(x,t)$, in a one-dimensional sound field is given by

$$\frac{\partial^2 p}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}, \tag{2.6.1}$$

where c is the speed of propagation of the sound wave. This differential equation can be solved by introducing new independent variables,

$$\zeta = x - ct \quad \text{and} \quad \eta = x + ct. \tag{2.6.2}$$

The derivatives in Eq. (2.6.1) may be expressed in terms of these new variables,

$$\begin{aligned}\frac{\partial}{\partial x} &= \frac{\partial \zeta}{\partial x}\frac{\partial}{\partial \zeta} + \frac{\partial \eta}{\partial x}\frac{\partial}{\partial \eta} = \frac{\partial}{\partial \zeta} + \frac{\partial}{\partial \eta} \\ \frac{\partial}{\partial t} &= \frac{\partial \zeta}{\partial t}\frac{\partial}{\partial \zeta} + \frac{\partial \eta}{\partial t}\frac{\partial}{\partial \eta} = -c\frac{\partial}{\partial \zeta} + c\frac{\partial}{\partial \eta}.\end{aligned} \tag{2.6.3}$$

Equation (2.6.3) may be used to compute the derivatives in Eq. (2.6.1),

$$\begin{aligned}\frac{\partial^2 p}{\partial x^2} &= \left(\frac{\partial}{\partial \zeta} + \frac{\partial}{\partial \eta}\right)\left(\frac{\partial p}{\partial \zeta} + \frac{\partial p}{\partial \eta}\right) \\ &= \frac{\partial^2 p}{\partial \zeta^2} + 2\frac{\partial^2 p}{\partial \zeta \partial \eta} + \frac{\partial^2 p}{\partial \eta^2} \\ \frac{\partial^2 p}{\partial t^2} &= \left(-c\frac{\partial}{\partial \zeta} + c\frac{\partial}{\partial \eta}\right)\left(-c\frac{\partial p}{\partial \zeta} + c\frac{\partial p}{\partial \eta}\right) \\ &= c^2\frac{\partial^2 p}{\partial \zeta^2} - 2c^2\frac{\partial^2 p}{\partial \zeta \partial \eta} + c^2\frac{\partial^2 p}{\partial \eta^2}.\end{aligned} \tag{2.6.4}$$

Equations (2.6.1) and (2.6.4) give

$$\frac{\partial^2 p}{\partial \zeta \partial \eta} = 0. \tag{2.6.5}$$

This gives

$$p(x,t) = f(\zeta) + g(\eta) = f(x - ct) + g(x + ct), \tag{2.6.6}$$

where f and g are arbitrary. f is a wave traveling to the right, in the positive x direction with speed c and g is a wave traveling to the left. Note that the only restriction on the functions f and g is that they be differentiable.

While Eq. (2.6.6) does solve the differential equation (2.6.1), it does not yet satisfy any initial conditions or boundary conditions. We will first examine how to satisfy the initial conditions. Because Eq. (2.6.1) is second order in time, two initial conditions are required. These two conditions enable us to uniquely determine the functions f and g. Suppose that the initial conditions are

$$p(x,0) = p_0(x), \quad \dot{p}(x,0) = v_0(x). \tag{2.6.7}$$

Equation (2.6.6) at $t = 0$ is

$$p(x,0) = p_0(x) = f(x) + g(x). \tag{2.6.8}$$

The time derivative of $p(x,t)$, may be obtained using Eq. (2.6.3),

$$\frac{\partial p}{\partial t} = \frac{\partial \zeta}{\partial t}\frac{\partial f}{\partial \zeta} + \frac{\partial \eta}{\partial t}\frac{\partial g}{\partial \eta} = -c\frac{\partial f}{\partial \zeta} + c\frac{\partial g}{\partial \eta}. \tag{2.6.9}$$

But, since $\frac{\partial f}{\partial \zeta} = \frac{\partial f}{\partial x}$ and $\frac{\partial g}{\partial \eta} = \frac{\partial g}{\partial x}$, Eq. (2.6.9) becomes

$$\frac{\partial p}{\partial t} = -c\frac{\partial f}{\partial x} + c\frac{\partial g}{\partial x}, \tag{2.6.10}$$

which may be evaluated at $t = 0$ to give

$$\dot{p}(x,0) = v_0(x) = -c\frac{\partial f(x)}{\partial x} + c\frac{\partial g(x)}{\partial x}. \tag{2.6.11}$$

This equation may be integrated over x to give

$$g(x) - f(x) = \frac{1}{c}\int_0^x v_0(z)dz. \tag{2.6.12}$$

Equations (2.6.8) and (2.6.12) give

$$f(x) = \frac{p_0(x)}{2} - \frac{1}{2c}\int_0^x v_0(z)dz, \tag{2.6.13}$$

and

$$g(x) = \frac{p_0(x)}{2} + \frac{1}{2c}\int_0^x v_0(z)dz. \tag{2.6.14}$$

We may now replace $f(x)$ with $f(x - ct)$ and $g(x)$ with $g(x + ct)$. Equations (2.6.13) and (2.6.14) then lead to

$$p(x,t) = \frac{p_0(x-ct)}{2} + \frac{p_0(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} v_0(z)dz. \tag{2.6.15}$$

Equation (2.6.15) now contains no unknown or arbitrary quantities and satisfies the initial conditions. We have, however, not specified the boundary conditions. To simplify the discussion, let $v_0(x) = 0$. We will first suppose that the boundary condition at the left end of the domain is $p(0,t) = 0$. Evaluating equation (2.6.15) at $x = 0$ gives

$$p_0(-ct) = -p_0(ct), \tag{2.6.16}$$

which shows that to satisfy this boundary condition, the initial pressure distribution, $p_0(x)$, must be an odd function of x. So, to satisfy this boundary condition, we must specify the initial pressure distribution outside the domain of interest.

We may now consider a boundary condition at another location, $x = L$. Let $p(L,t) = 0$. Equation (2.6.15) then becomes (again while letting $v_0(x) = 0$),

$$p(L,t) = 0 = \frac{p_0(L-ct)}{2} + \frac{p_0(L+ct)}{2}. \tag{2.6.17}$$

We may also make use of the fact that $p_0(x)$ must be an odd function as in Eq. (2.6.16) so that $p_0(L+ct) = -p_0(-L-ct)$. If we then let $\zeta = -L - ct$, Eq. (2.6.17) becomes

$$p_0(2L+\zeta) = p_0(\zeta) \tag{2.6.18}$$

or,

$$p_0(2L+x) = p_0(x). \tag{2.6.19}$$

So, to satisfy the boundary condition $p(L,t) = 0$, we must have the initial pressure distribution be periodic with period $2L$.

2.7 Sound in an Infinite Tube

Given our ability to obtain general solutions to the wave Eq. (2.6.1), we can derive other general results for certain simple geometries. In the following, we will show that for an arbitrary sound wave propagating in one dimension, the fluctuating pressure and the acoustic particle velocity are related by the product of the air density ρ_0 and the speed of wave propagation, c.

Suppose we have an infinite tube with a piston at $x = 0$ having an imposed velocity $u(t)$. In order to relate the velocity boundary condition to the pressure, we need to use the Euler equation, which expresses the force or momentum balance in the sound

field,

$$-\frac{\partial p(x,t)}{\partial x} = \rho_0 \dot{u}(x,t), \tag{2.7.1}$$

where ρ_0 is the nominal air density. Note that this equation applies anywhere in the domain and must be satisfied in any one-dimensional sound field. Because there is no boundary on the right end of our domain, the wave that travels to the left will not exist, $g(x+ct) = 0$. The left side of Eq. (2.7.1) becomes

$$-\frac{\partial p(x,t)}{\partial x} = -\frac{\partial f(x,t)}{\partial x} = \frac{1}{c}\frac{\partial f(x,t)}{\partial t}. \tag{2.7.2}$$

Equations (2.7.1) and (2.7.2) give

$$\frac{\partial p(x,t)}{\partial t} = \rho_0 c \dot{u}(x,t). \tag{2.7.3}$$

Integrating equation (2.7.3) with respect to t gives

$$p(x,t) = \rho_0 c(u(x,t) + U_0), \tag{2.7.4}$$

where U_0 is a constant velocity which we will assume is zero since we are interested in fluctuating pressure fields.

Equation (2.7.4) shows that for any sound field consisting of a single, one-dimensional wave traveling in the positive x direction, the pressure and acoustic particle velocity are related by

$$\frac{p(x,t)}{u(x,t)} = \rho_0 c. \tag{2.7.5}$$

If the field consisted of a wave traveling to the left, $g(x,t)$ so that $f(x,t) = 0$, we would then obtain

$$\frac{p(x,t)}{u(x,t)} = -\rho_0 c. \tag{2.7.6}$$

The product $\rho_0 c$ is often referred to as the specific acoustic impedance of a sound wave propagating in one dimension. In air, $\rho_0 c \approx 415$ MKS rayls (pascal-seconds/meter). The impedance is normally defined only for time harmonic fluctuating signals but the remarkably simple solution in Eq. (2.6.6) enables us to obtain this result for arbitrary time fluctuations. It is also very important to note that for sound fields that fluctuate in multiple spatial dimensions, such as near a sound source in an open room, the relationship between pressure and particle velocity will be much more complicated than that shown in Eq. (2.7.5).

Evaluating equation (2.7.5) at $x = 0$ then gives the pressure at the piston surface,

$$p(0, t) = f(-ct) = \rho_0 c u(t). \tag{2.7.7}$$

The pressure at any location is then

$$p(x, t) = \rho_0 c u(t - x/c). \tag{2.7.8}$$

2.8 Problems

1. A plane sound wave in air has a frequency of 400 Hz and a peak acoustic pressure amplitude of 5 $\mathrm{N/m^2}$.

   (a) What is its wavelength?

   (b) What is its time-averaged intensity?

   (c) What is its peak particle velocity amplitude?

   (d) What is its peak particle displacement amplitude?

   (e) What is its rms pressure?

   (f) What is its sound pressure level using a reference pressure of 20×10^{-6} Pa?

2. We have examined a model for sound in one-dimension that depended on three fluctuating quantities, $P(x, t)$, the (total) pressure, $\rho(x, t)$, the density, and $U(x, t)$ the velocity of the gas molecules. These were shown to be related to each other by the momentum equation (2.8.1), the conservation of mass (2.8.2), and the equation of state (2.8.3) as given below.

$$-\frac{\partial P(x, t)}{\partial x} = \frac{\partial}{\partial t}(\rho U), \tag{2.8.1}$$

$$-\frac{\partial(\rho U)}{\partial x} = \dot{\rho}_a, \tag{2.8.2}$$

$$p(x, t) = \rho_a c^2, \tag{2.8.3}$$

where $\rho = \rho_0 + \rho_a(t)$ with ρ_0 being the static air density and ρ_a is the acoustic fluctuation in density. These equations can be manipulated to obtain a single equation for the acoustic fluctuating pressure $p(x,t)$,

$$\frac{\partial^2 p(x,t)}{\partial x^2} = \frac{1}{c^2}\ddot{p}(x,t). \tag{2.8.4}$$

Use Eqs. (2.8.1), (2.8.2), and (2.8.3) to obtain a single partial differential equation in terms of velocity, U rather than pressure p as in Eq. (2.8.4).

3. If we are interested in steady-state harmonic sound fields, we can express a solution to Eq. (2.8.4) in the form

$$p(x,t) = p_1 \sin(\omega t - kx + \phi_1) + p_2 \sin(\omega t + kx + \phi_2). \tag{2.8.5}$$

Equation (2.8.5) describes a wave to the right (proportional to p_1) and a wave to the left (proportional to p_2). Determine the constants p_1, p_2, ϕ_1, ϕ_2, and k to satisfy zero velocity at $x = 0$ and $x = L$, i.e. $U(0,t) = 0$, and $U(L,t)$=0. Hint: Use Eq. (2.8.1) to relate pressure and velocity. Also note that there are more unknowns than boundary conditions. This means that the solutions will not be unique; there are many possible solutions. You may modify the program "pressurewave.m" to check your answer.

4. Assume that Eq. (2.8.5) is the solution to Eq. (2.8.4). If p_1 in Eq. (2.8.5) is given as $p_1 = 0.2$ Pa, determine values for p_2, ϕ_1 and ϕ_2 and obtain an expression for all of the values of ω such that the pressure is zero at $x = 0$ and the derivative of the pressure is zero at $x = 3$ m, i.e. $p(0,t) = 0$, and $\frac{\partial p(x,t)}{\partial x}|_{x=3} = 0$. Provide plots of $p(x,t)$ for $0 \le x \le 3$ for any convenient instant of time, t for three of the values you obtained for ω.

5. The solution to the acoustic wave equation may be written as in Eq. (2.8.6),

$$p(x,t) = e^{i\omega t}(p_1 e^{-ikx} + p_2 e^{ikx}), \tag{2.8.6}$$

where p_1 and p_2 are the complex amplitudes of the incident and reflected waves, respectively and the wave number is $k = \omega/c$. We showed that the sound absorption coefficient is equal to $\alpha = 1 - |\frac{p_2}{p_1}|^2$. Suppose that there is a boundary at $x = L$ having a complex acoustic impedance Z_L. Derive an expression for the absorption coefficient as a function of Z_L.

6. We showed that the general solution to the acoustic wave equation, Eq. (2.6.1)

$$\frac{\partial^2 p}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} \tag{2.8.7}$$

can be written as in Eq. (2.6.6),

$$p(x,t) = f(\zeta) + g(\eta) = f(x - ct) + g(x + ct), \tag{2.8.8}$$

where f and g are arbitrary. f is a wave traveling to the right, in the positive x direction with speed c and g is a wave traveling to the left. We showed that if there is only a wave to the right, i.e. $g(x + ct) = 0$, then the ratio of pressure to velocity (i.e. the impedance) is a constant, $\frac{p(x,t)}{u(x,t)} = \rho_0 c$, where ρ_0 is the ambient air density and c is the speed of wave propagation. Derive an expression for the impedance if there is only a wave to the left so that $p(x, t) = g(x + ct)$.

7. For the general solution of the wave equation given in problem 2.8, for the case where both waves are present so that $p(x, t) = f(x - ct) + g(x + ct)$, show that the impedance is

$$\frac{p(x,t)}{u(x,t)} = \rho_0 c \frac{f(x-ct) + g(x+ct)}{f(x-ct) - g(x+ct)}. \tag{2.8.9}$$

Hint: Use Eq. (2.4.3) and integrate over time to find $u(x, t)$. Then consider the time and space derivatives of f and g.

8. A tube of length $l = 1.5$ m contains a sound field that consists of two waves traveling in opposite directions, each having a frequency of 1500 Hz. The sound pressure in the tube is given by $p(x, t) = \Re[e^{i\omega t}(p_1 e^{-ikx} + p_2 e^{ikx})]$, where $\Re[\cdot]$ denotes the real part. The amplitude of the wave traveling in the positive direction is $p_1 = 1$ Pa and the complex amplitude of the wave traveling in the negative direction is $p_2 = 0.2 + i0.8$. Write down an expression for the mean square pressure as a function of position x in the tube, $< p^2(x) >$. Plot your result versus x.
9. A tube 2.6 m long is closed at one end and open at the other. The tube is filled with air that has a sound speed of 344 m/s. (a) Determine the first three resonant frequencies in Hertz. (b) Write down expressions for and plot the mode shapes (eigenfunctions) corresponding to each of these resonant frequencies as a function of the position x along the length of the tube.
10. Determine the first four acoustic natural frequencies (in Hertz) and eigenfunctions for a tube of length $L = 0.9$ m with one end closed and the other open. Assume that the closed end has infinite impedance and that the impedance of the open end is zero.
11. A tube of area A and length l has a piston at $x = 0$ that moves with a velocity given by $u(t) = Ue^{i\omega t}$. The tube is terminated at $x = l$ with a spring/mass/dashpot. (a) Obtain an expression for the sound pressure as a function of x in the tube. (b) Determine possible values of the stiffness, mass, and damping relative to the area, A, [k/A, m/A, and c/A] of the termination at $x = l$ such that there is no reflected sound wave in the tube.
12. A tube of length 1 m and diameter 0.25 m is terminated at its right end by a material that has an acoustic impedance of $z = 0.4 + 0.2i$ at a frequency of 1000 Hz. Assume that an incident wave travels to the right with a frequency of $\omega = 2\pi * 1000$ radians/s and a pressure amplitude of 1 Pa. Determine (a) the amplitude of the reflected wave and the complex reflection factor, r, of this material, (b) the absorption coefficient, α, and (c) the acoustic intensity, I, at the surface of the material.

13. A tube of length L has a spring/mass/dashpot at $x = L$. The impedance of the termination at $x = L$ is

$$Z_L = i\omega M_L + K_L/(i\omega) + C_L, \tag{2.8.10}$$

where $M_L = .02$, kg/m^2 $K_L = 8.15 \times 10^5$ N/m^3, and $C_L = 207.5$ m/s/m^2. Determine and plot the absorption coefficient, α as a function of frequency in Hertz for the frequency range from $20 < f < 20,000$ Hz. Use a logarithmic frequency axis.

Chapter 3
Sound Transmission Loss



The transmission of sound through walls and barriers is an extremely important topic for noise control engineers. In this chapter we examine the basic principles of how sound passes through walls. The primary physical characteristics that influence sound transmission through walls are studied, including the importance of mass and how the bending rigidity degrades the transmission loss at higher frequencies. Because walls can be fairly complicated constructions, the use of transfer matrices is introduced to enable the prediction of sound transmission loss in walls having multiple layers.

3.1 The Mass Law

In the following, a model is developed for predicting the sound transmission through a simple wall. The wall is assumed to consist of a limp mass, having no stiffness or damping. The ability of sound to transmit through walls is usually characterized by determining the amplitude of a transmitted sound wave relative to that of an incident sound wave. Assuming the wall is at $x = 0$, we will assume that an incident wave is traveling to the right from $x = -\infty$ so that the "incident" side of the wall is where $x < 0$. The "transmitted" side of the wall is where $x > 0$. Because of the interaction of sound with the wall, in addition to the incident wave, there will also be a reflected wave on the incident side that travels in the negative x direction as shown in Fig. 3.1. On the transmitted side, it will be assumed that there are no boundaries for $x < \infty$ so that there is no reflected wave there.

Assuming a harmonic sound field at the frequency ω radians/s, the sound pressure on the incident side of the wall ($x < 0$) may be expressed as

$$p(x,t) = e^{i\omega t}(p_1 e^{-ikx} + p_2 e^{ikx}), \tag{3.1.1}$$

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_3.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_3

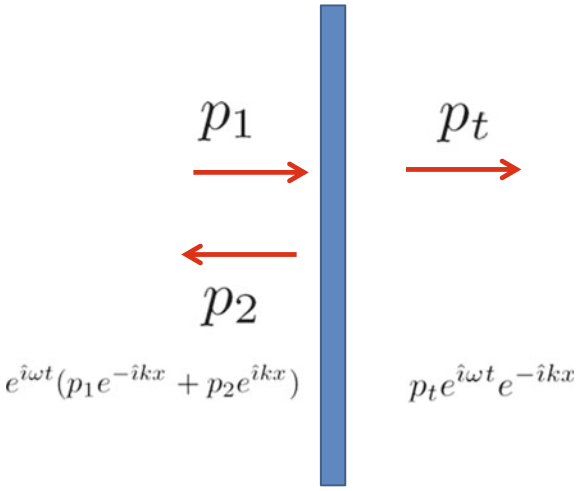


Fig. 3.1 A limp wall with a sound wave incident from the left with amplitude p_1, a reflected wave with amplitude p_2, and a transmitted wave with amplitude p_t

where p_1 and p_2 are the amplitudes of the incident and reflected waves, respectively, $k = \omega/c$ is the wave number and c is the sound speed.

The sound pressure on the transmitted side ($x > 0$) may be written as

$$p(x,t) = p_t e^{i\omega t} e^{-ikx}, \tag{3.1.2}$$

where p_t is the amplitude of the transmitted wave. The ability of sound to be transmitted through the wall is generally described by the ratio of the transmitted to incident sound pressures, p_t/p_1.

The relationship between the sound fields on the incident and transmitted sides of the wall is obtained by writing Newton's second law for the wall. Assuming that inertia effects dominate the restoring forces due to stiffness or damping, the equation of motion of the wall is given by

$$(p(0^-,t) - p(0^+,t))A = \rho_w h A \dot{u}_w, \tag{3.1.3}$$

where A is the area, ρ_w is the density of the wall, h is the thickness, and $u_w(t)$ is the wall's velocity in the x direction. Since the sound field is assumed to be harmonic at the frequency ω, the wall's velocity may be written as $u_w(t) = U_w e^{i\omega t}$ where U_w is the complex amplitude. Equations (3.1.1)–(3.1.3) then give

$$p_1 + p_2 - p_t = \rho_w h i \omega U_w, \tag{3.1.4}$$

In order to determine p_t/p_1 it is necessary to eliminate both p_2 and U_w from Eq. (3.1.4). This can be accomplished by writing the Euler equation relating the pressure and acoustic particle velocity on each side of the wall. Since the wall is assumed to not be porous, the velocity of the air on each side of the wall will be

equal to the velocity of the wall. The Euler equation evaluated at a point adjacent to the wall is then

$$-\frac{\partial p}{\partial x} = \rho_0 \dot{u}_w, \tag{3.1.5}$$

where ρ_0 is the nominal air density. Substituting Eq. (3.1.1) for the pressure on the incident side ($x = 0^-$) into Eq. (3.1.5) and simplifying give

$$p_1 - p_2 = \rho_0 c U_w. \tag{3.1.6}$$

Equation (3.1.5) may be evaluated on the transmitted side ($x = 0^+$) by substituting Eq. (3.1.2) for the pressure and simplifying,

$$p_t = \rho_0 c U_w. \tag{3.1.7}$$

Equations (3.1.4), (3.1.6) and (3.1.7) may be used to eliminate p_2 and U_w to obtain

$$\frac{p_t}{p_1} = \frac{1}{1 + \frac{i\rho_w h\omega}{2\rho_0 c}}. \tag{3.1.8}$$

The sound transmission coefficient is defined to be the ratio of the time averaged intensity of the transmitted wave to that of the incident wave. This is given by

$$\tau = |\frac{p_t}{p_1}|^2 = \frac{1}{1 + \left(\frac{\rho_w h\omega}{2\rho_0 c}\right)^2}. \tag{3.1.9}$$

The sound transmission loss in decibels is defined to be

$$\begin{aligned} TL =& 10\log_{10}(1/\tau) = 10\log_{10}\left(1 + \left(\frac{\rho_w h\omega}{2\rho_0 c}\right)^2\right) \\ \approx& 20\log_{10}\left(\frac{\rho_w h\omega}{2\rho_0 c}\right). \end{aligned} \tag{3.1.10}$$

Equation (3.1.10) is often referred to as the mass law. Evaluating Eq. (3.1.10) shows that if the surface density $\rho_w h$ (mass/area) is doubled, or if the frequency ω is doubled, the transmission loss will increase by approximately 6 decibels. In Eq. (3.1.9), the transmission coefficient depends on a single nondimensional parameter which we will refer to as β, $\beta = \frac{\rho_w h\omega}{\rho_0 c}$. The ratio of transmitted to incident pressures becomes

$$\frac{p_t}{p_1} = \frac{1}{1 + \frac{i\beta}{2}}. \tag{3.1.11}$$

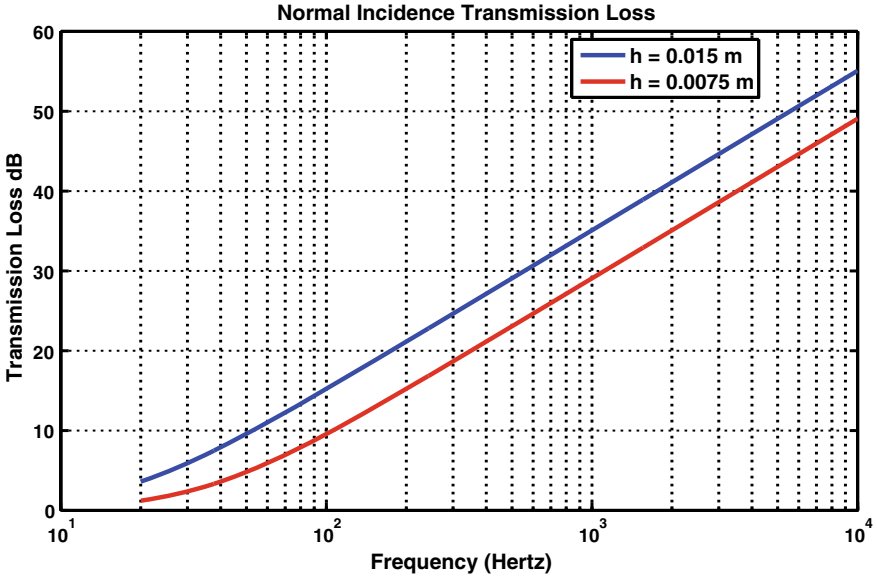


Fig. 3.2 Normal incidence transmission loss for walls having two different thicknesses and, hence, different surface densities. A doubling of the surface density increases the transmission loss by approximately 6 decibels over most of the frequency range

One can see that an increase in the surface density $\rho_w h$ has the same effect as an increase in frequency ω; a doubling of either of these quantities will produce the same increase in transmission loss of approximately 6 decibels.

As an example, Fig. 3.2 shows the predicted sound transmission loss versus frequency for two different wall densities. As the density is doubled, the transmission loss increases by the expected 6 decibels. The slope of each curve is approximately six decibels per octave.

3.2 Random Incidence Sound Transmission Loss

In most situations where we wish to estimate the sound transmission loss of a wall, it is necessary to account for the fact that the sound is incident from many possible directions. Figure 3.3 shows the wall with an incident wave with amplitude p_1 oriented at an angle relative to the wall with a reflected wave p_2 that is assumed to reflect according to Snell's law. The random incidence transmission loss is calculated by averaging the transmission coefficient corresponding to sound that is incident from all possible directions. Waves that drive the wall from directions other than the direction normal to the wall's surface cause it to bend, which must be accounted for in the model. This is accomplished by treating the wall as a plate with bending stiffness that depends on its thickness and elastic modulus. It is found that when the stiffness of the wall is included, it is possible to have what could be thought of as a "wave

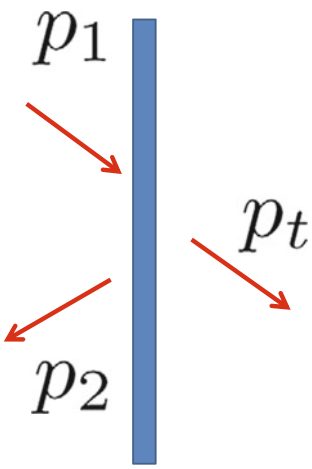


Fig. 3.3 A wall with a sound wave incident at an angle from the left with amplitude p_1, a reflected wave with amplitude p_2 and a transmitted wave with amplitude p_t

resonance" where waves at certain frequencies travel along the wall with very little resistance. This condition is known as coincidence and causes the wall to be nearly transparent to sound at a certain combination of the angle of incidence and frequency.

As before, we will characterize the ability of sound to transmit through walls by determining the amplitude of a transmitted sound wave relative to that of an incident sound wave. Assuming the wall is at $z = 0$, we will assume that an incident wave is traveling to the right from $z = -\infty$ so that the "incident" side of the wall is where $z < 0$. The "transmitted" side of the wall is where $z > 0$. Because of the interaction of sound with the wall, in addition to the incident wave, there will also be a reflected wave on the incident side that travels in the negative z direction. On the transmitted side, it will be assumed that there are no boundaries for $z < \infty$ so that there is no reflected wave there.

Assuming a harmonic sound field at the frequency ω radians/s, the sound pressure on the incident side of the wall ($z < 0$) may be expressed as

$$p(x, y, z, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 e^{-ik_z z} + p_2 e^{ik_z z}), \tag{3.2.1}$$

where p_1 and p_2 are the amplitudes of the incident and reflected waves, respectively. In order for Eq. (3.2.1) to satisfy the wave equation, the wave vector components, k_x, k_y, and k_z must satisfy $k_x^2 + k_y^2 + k_z^2 = k^2$, where $k = \omega/c$ is the wave number and c is the sound speed.

The sound pressure on the transmitted side ($z > 0$) may be written as

$$p(x, y, z, t) = p_t e^{i\omega t} e^{-i(k_x x + k_y y)} e^{-ik_z z}, \tag{3.2.2}$$

where p_t is the amplitude of the transmitted wave. The ability of sound to be transmitted through the wall is generally described by the ratio of the transmitted to incident sound pressures, p_t/p_1.

The relationship between the sound fields on the incident and transmitted sides of the wall is obtained by writing Newton's second law for the wall. Waves that are incident at angles other than normal incidence result in sinusoidal traveling pressure waves on the surface of the wall. As a result, the wall is forced to respond in the form of traveling bending waves. This means that it is necessary to include the bending stiffness of the wall in the wall's governing equation. Assuming the wall bends like an infinite plate, it's equation of motion is given by

$$D\nabla^4 w + \rho_w h\ddot{w} = (p(x, y, 0^-, t) - p(x, y, 0^+, t)) \\ = e^{i\omega t} e^{-i(k_x x + k_y y)}(p_1 + p_2 - p_t), \tag{3.2.3}$$

where ρ_w is the density of the wall, h is the thickness, and $w(x, y, t)$ is the wall's displacement in the z direction. The flexural rigidity is $D = Eh^3/12/(1 - \sigma^2)$, where E is Young's modulus of elasticity, and σ is Poisson's ratio. The particular solution to Eq. (3.2.3) may be shown to be

$$w(x, y, t) = We^{i\omega t} e^{-i(k_x x + k_y y)}, \tag{3.2.4}$$

where,

$$W = \frac{p_1 + p_2 - p_t}{D(k_x^2 + k_y^2)^2 - \omega^2 \rho_w h}. \tag{3.2.5}$$

In order to determine p_t/p_1 it is necessary to eliminate both p_2 and W from Eq. (3.2.5). This can be accomplished by writing the Euler equation relating the pressure and acoustic particle velocity on each side of the wall. Since the wall is assumed to not be porous, the velocity of the air on each side of the wall will be equal to the velocity of the wall. The Euler equation is

$$-\overrightarrow{\nabla p} = \rho_0 \dot{\overrightarrow{u}}, \tag{3.2.6}$$

where ρ_0 is the nominal air density and $\overrightarrow{u}(x, y, z, t)$ is the acoustic particle velocity vector. Since we wish to eliminate the z component of the wall's motion (W in Eqs. (3.2.4) and (3.2.5)) it is necessary to consider only the z component of the vector equation (3.2.6),

$$-\frac{\partial p}{\partial z} = \rho_0 \dot{u}_z. \tag{3.2.7}$$

This equation will be evaluated on each side of the wall (i.e., at $z = 0^-$ and at $z = 0^+$). Since the velocity and displacement of the wall must equal that of the air on both the incident and transmitted sides,

$$u_z(x, y, 0^-, t) = u_z(x, y, 0^+, t) = i\omega W e^{i\omega t} e^{-i(k_x x + k_y y)}, \tag{3.2.8}$$

where W is given in Eq. (3.2.5).

Substituting Eq. (3.2.1) for the pressure on the incident side ($z = 0^-$) into Eq. (3.2.7) and using Eq. (3.2.8) give

$$ik_z(p_1 - p_2) = -\omega^2 \rho_0 W \tag{3.2.9}$$

Equation (3.2.7) may be evaluated on the transmitted side ($z = 0^+$) by substituting Eq. (3.2.2) for the pressure and again using Eq. (3.2.8),

$$ik_z p_t = -\omega^2 \rho_0 W. \tag{3.2.10}$$

Equations (3.2.9) and (3.2.10) give

$$p_t = p_1 - p_2 \tag{3.2.11}$$

Equations (3.2.5), (3.2.10) and (3.2.11) may be used to eliminate p_2 and W to obtain

$$\frac{p_t}{p_1} = \frac{1}{1 + \frac{k_z(D(k_x^2+k_y^2)^2 - \rho_w h\omega^2)}{2\rho_0 ckI\omega}} \tag{3.2.12}$$

Equation (3.2.12) can be simplified by expressing the components of the wave vector, $\overrightarrow{k} = k_x i + k_y \hat{j} + k_z \hat{k}$, as shown in Fig. 3.4 in terms of the angles that describe the orientation of the incident wave. Let the angle relative to the vertical axis be θ and the angle in the xy plane be ϕ. The wave vector components may then be written as

$$\begin{aligned} &k_x = k\sin(\theta)\sin(\phi), \quad k_y = k\sin(\theta)\cos(\phi), \\ &k_z = k\cos(\theta) \end{aligned} \tag{3.2.13}$$

As before, $k = \omega/c$ is the wave number. Substituting k_x and k_y from Eqs. (3.2.13) into Eq. (3.2.12) and simplifying give,

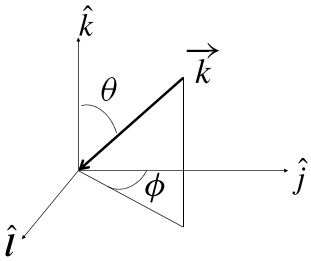


Fig. 3.4 Orientation of the incident sound wave

$$\frac{p_t}{p_1} = \frac{1}{1 + \frac{\cos(\theta)(Dk^4 \sin^4(\theta) - \rho_w h\omega^2)}{2\rho_0 ci\omega}}. \tag{3.2.14}$$

It is important to note that Eq. (3.2.14) can yield unity if

$$Dk^4 \sin^4(\theta) = \rho_w h\omega^2. \tag{3.2.15}$$

Since $k = \omega/c$, Eq. (3.2.15) can be solved for the frequency at which the transmitted pressure, p_t is equal to the incident pressure, p_1,

$$\omega_c = \left(\frac{c}{\sin(\theta)}\right)^2 \sqrt{\frac{\rho_w h}{D}}. \tag{3.2.16}$$

At the frequency, ω_c given by Eq. (3.2.16), the wall becomes transparent to sound. The lowest frequency at which this happens is usually called the "coincidence" frequency, which can be seen to correspond to the angle of incidence at which $\sin(\theta) = 1$, or $\theta = \pi/2$, or grazing incidence.

In practice, walls never become acoustically transparent because there is always some sort of energy dissipation within the structure. This can be conveniently accounted for in our model of the wall by including structural damping. This is accomplished by replacing Young's modulus, E, and the resulting flexural rigidity in Eq. (3.2.3), by a complex modulus, $E(1 + i\eta)$, where η is the loss modulus of the material. It can be shown that the loss modulus accounts for energy dissipation in the wall. The inclusion of a complex modulus and corresponding flexural rigidity, D, causes Eq. (3.2.14) to no longer produce unity at the coincidence frequency, ω_c.

In many situations, walls are driven by sound that arrives from many directions simultaneously. It is therefore common to calculate a transmission coefficient that is obtained by averaging over the angles of sound incidence that correspond to a "diffuse" sound field. A diffuse sound field contains waves that travel in all possible directions such that each wave vector component has equal probability subject to the limitation that $k_x^2 + k_y^2 + k_z^2 = k^2 = (\omega/c)^2$. This constraint could be used to eliminate k_z in Eq. (3.2.12) and express the ratio of p_t to p_1 in terms of k_x and k_y only. The result would then be a function of $k_x^2 + k_y^2$.

The averaged transmission coefficient for a diffuse sound field could then be calculated from

$$< \tau >= \frac{\int\int \tau(k_x, k_y) dk_x dk_y}{\int\int dk_x dk_y}. \tag{3.2.17}$$

The domain of integration is over all possible values of k_x and k_y, which is equal to a circle of radius $k = \omega/c$. It is then helpful to convert the integrations in Eq. (3.2.17) to polar coordinates by letting $dk_x dk_y = r dr d\psi$, where $r = \omega \sin(\theta)/c$. Equation (3.2.17) then becomes

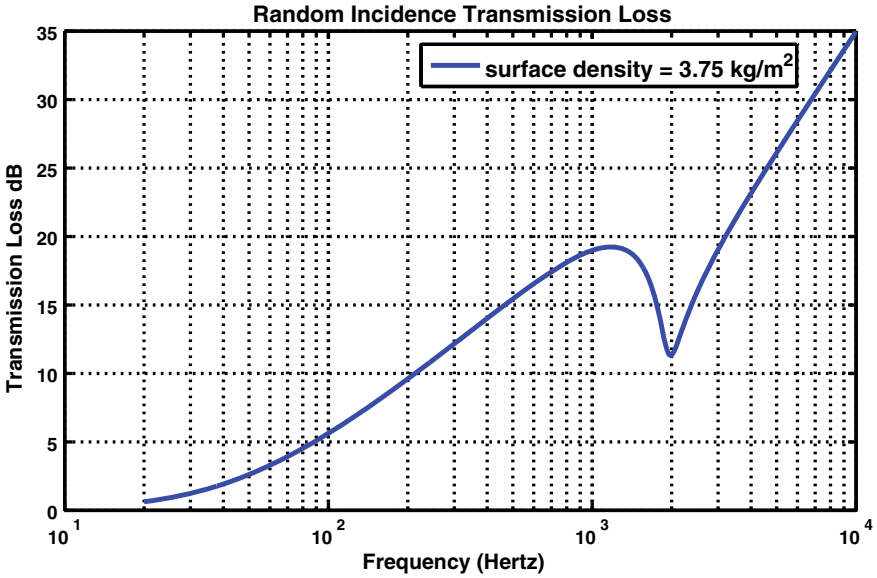


Fig. 3.5 Random incidence transmission loss for a wall. This wall has a coincidence dip at approximately 2000 Hz as predicted by Eq. 3.2.16

$$< \tau > = \frac{\int_0^{\pi/2} \sin(\theta)\cos(\theta)\tau(\theta)d\theta}{\int_0^{\pi/2} \sin(\theta)\cos(\theta)d\theta}$$
$$= 2\int_0^{\pi/2} \sin(\theta)\cos(\theta)\tau(\theta)d\theta, \tag{3.2.18}$$

where the integration over ψ is simply equal to 2π since the integrand does not depend on ψ (Fig. 3.5). To illustrate the influence of coincidence on the random incidence transmission loss, results are shown in Fig. 3.5 for a wall having a surface density of $\rho_w = 3.75$ kg/m^2 and flexural rigidity, $D = 386$ Nm. The loss modulus was taken to be $\eta = 0.01$. This wall has a coincidence frequency of approximately 2 kHz. The predicted transmission loss has a significant dip at this frequency.

The random incidence sound transmission loss in decibels is defined to be

$$TL = 10\log_{10}(1/ < \tau >). \tag{3.2.19}$$

In addition to random incidence transmission loss where the integration over θ is carried out from zero to 90°, it is also common to compute the "field" incidence transmission loss by integrating from zero to 78°. We thus have three types of transmission loss: normal incidence, random incidence, and field incidence. These are compared in Fig. 3.6 where they are plotted for a limp wall (i.e., having no bending stiffness, D) versus the nondimensional parameter, $\beta = \frac{\rho_w h\omega}{\rho_0 c}$ as in Eq. (3.1.11).

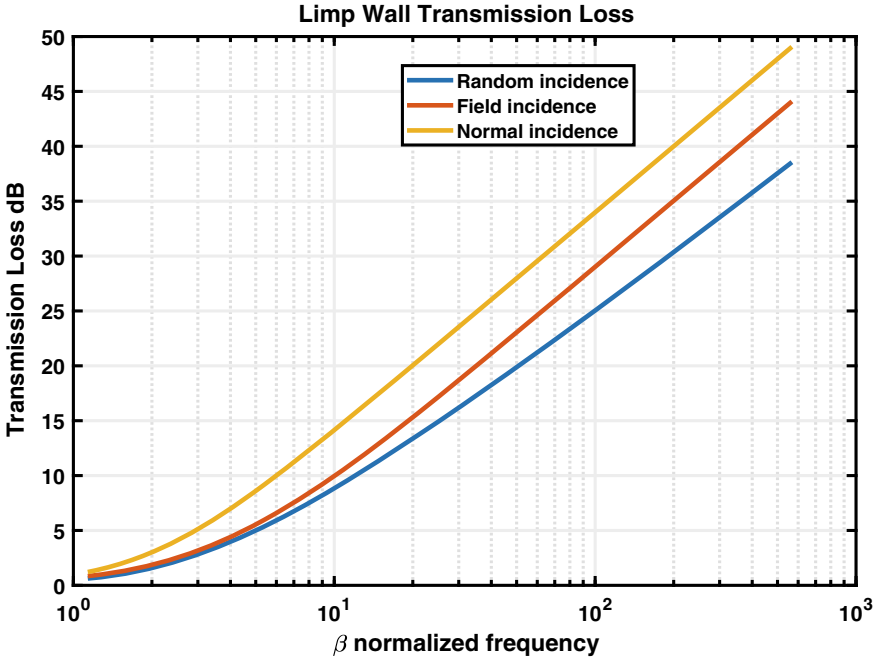


Fig. 3.6 Sound transmission loss for a limp wall. The transmission loss for random incidence, field incidence, and normal incidence sound is shown versus the nondimensional parameter $\beta = \frac{\rho_w h \omega}{\rho_0 c}$

Again, an increase in the surface density $\rho_w h$ has the same effect as an increase in frequency ω; a doubling of either of these quantities will produce the same increase in transmission loss of approximately 6 decibels.

3.3 Transmission Loss Calculations Using Transfer Matrices

A transfer matrix representation can be an extremely powerful means of modeling very complex coupled acoustical systems. We will develop transfer matrix models of multilayered walls in order to calculate the sound transmission loss and we will also use this approach to model the sound transmission through ducts and mufflers.

A transfer matrix can be used to relate the total acoustic pressure and velocity on each end of an acoustical system. Having the composite transfer matrix model one can compute both the sound transmission through, and the sound reflected by the system.

Because the thickness of the walls or lengths of the ducts to be examined will vary, it is convenient to define local coordinate systems to describe propagation on each side of the system. Let x be the spatial coordinate on the left, or incident side. The

incident side of the wall will then be located at $x = 0$. Let y be the spatial coordinate on the transmitted side and the right end of the wall will be located at $y = 0$.

To relate the complex reflection coefficient, r, and the transmission coefficient, τ, to the elements of the system's transfer matrix, suppose that the elements, T_{11}, T_{12}, T_{21}, and T_{22}, of the transfer matrix $[T]$ are known,

$$[T] = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix}. \tag{3.3.1}$$

As before, assume that the sound field to the left (the incident side) consists of a combination of a given incident wave with complex amplitude p_1 and a reflected wave with complex amplitude p_2 so that the sound pressure on the incident side of the system may be written as

$$p(x,t) = (p_1 e^{-ikx} + p_2 e^{ikx}) e^{i\omega t}. \tag{3.3.2}$$

On the right side of the system (the transmitted side) it will be assumed that there are no surfaces for the sound to reflect from so that the sound field consists of only a transmitted wave with complex amplitude p_t. On the transmitted side, the sound pressure is then given by

$$p(y,t) = p_t e^{-iky} e^{i\omega t}. \tag{3.3.3}$$

As in most situations, it will be assumed that the amplitude of the incident wave, p_1, is given and we would like to determine the amplitudes of the reflected wave, p_2, and the transmitted wave, p_t, in terms of the known elements in the transfer matrix $[T]$ given in Eq. (3.3.1).

Taking P_1 and U_1 to be the complex amplitudes of the total pressure and velocity on the incident side and P_2 and U_2 to be the corresponding amplitudes on the transmitted side, then the transfer matrix in Eq. (3.3.1) gives the relationships between them as

$$\begin{pmatrix} P_2 \\ U_2 \end{pmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix}. \tag{3.3.4}$$

The complex amplitude of the pressure on the incident side, P_1 may be related to p_1 and p_2 by evaluating Eq. (3.3.2) at $x = 0$ and dropping the factor $e^{i\omega t}$ since we are interested in only the complex amplitude of the pressure. This gives

$$P_1 = p_1 + p_2. \tag{3.3.5}$$

The amplitude of the pressure on the transmitted side, P_2 is determined by evaluating Eq. (3.3.3) at the interface between the wall and the air at $y = 0$,

$$P_2 = p_t. \tag{3.3.6}$$

Equations (3.3.5) and (3.3.6) can be used in Eq. (3.3.4) to eliminate P_1 and P_2. However, Eq. (3.3.4) still contains the unknown velocities U_1 and U_2. Two additional equations are needed to eliminate these velocities. The velocities may be related to the pressures p_1, p_2, and p_t by evaluating the Euler equation on both the incident and transmitted sides,

$$-\frac{\partial p}{\partial x} = \rho_0 \dot{u} \quad \text{and} \quad -\frac{\partial p}{\partial y} = \rho_0 \dot{u} \tag{3.3.7}$$

where ρ_0 is the nominal air density. This can be evaluated on the incident side by substituting Eq. (3.3.2) into the first of Eqs. (3.3.7) and setting $x = 0$. Rearranging the result gives

$$p_1 - p_2 = \rho_0 c U_1, \tag{3.3.8}$$

where we have used the fact that the wave number is $k = \omega/c$. The Euler equation (3.3.7) may be evaluated on the transmitted side by substituting Eq. (3.3.3) for the transmitted pressure into the second of Eqs. (3.3.7) and setting $y = 0$. This leads to

$$p_t = \rho_0 c U_2. \tag{3.3.9}$$

The matrix equation (3.3.4) and the scalar equations (3.3.8) and (3.3.9) comprise a total of four equations in the unknowns p_2/p_1, p_t/p_1, U_1 and U_2. To simplify the writing, let $r = p_2/p_1$. Substituting this and Eqs. (3.3.8) and (3.3.9) into Eq. (3.3.4) gives two equations for r and p_t/p_1. Rearranging this result slightly gives

$$\begin{aligned} \frac{p_t}{p_1} &= T_{11}(1+r) + \frac{T_{12}}{\rho_0 c}(1-r), \\ \frac{p_t}{p_1} &= \rho_0 c T_{21}(1+r) + T_{22}(1-r). \end{aligned} \tag{3.3.10}$$

Subtracting Eqs. (3.3.10) to eliminate $\frac{p_t}{p_1}$ gives

$$(1+r)(T_{11} - \rho_0 c T_{21}) + (1-r)\left(\frac{T_{12}}{\rho_0 c} - T_{22}\right) = 0. \tag{3.3.11}$$

Solving Eq. (3.3.11) for r gives

$$r = \frac{\rho_0 c T_{21} - T_{11} + T_{22} - \frac{T_{12}}{\rho_0 c}}{T_{11} - \rho_0 c T_{21} + T_{22} - \frac{T_{12}}{\rho_0 c}}. \tag{3.3.12}$$

Equation (3.3.12) may be substituted into either of Eqs. (3.3.10) to eliminate r. Simplifying this result gives

$$\frac{p_t}{p_1} = 2\frac{(T_{11}T_{22} - T_{12}T_{21})}{T_{11} - \rho_0 c T_{21} + T_{22} - \frac{T_{12}}{\rho_0 c}}. \tag{3.3.13}$$

3.4 Transfer Matrix for a Solid Element

Having Eqs. (3.3.12) and (3.3.13) to calculate the reflection factor and transmitted pressure through general walls modeled with a composite transfer matrix, we now need to develop transfer matrix models for the types of walls that are commonly encountered. The first, and most important element to consider is a solid panel as examined in Sect. 3.3. Our task will be to find the relationship between the pressure and velocity on the left side of an element, P_1 and U_1 to those on the right side, P_2 and U_2 in the form of Eq. (3.3.4). In the case of a solid, i.e., nonporous, element, the relationship between the velocities is simply $U_1 = U_2$ since we assume there is no deformation in the thickness of the element. We wrote a relation between the net pressure on a solid wall and the transverse deflection, $w(x, y, t)$ in Eq. (3.2.3). Assuming harmonic time dependence and assuming that the x and y dependence is of the form $e^{-i(k_x x + k_y y)}$, enables us to relate the net pressure $P_1 - P_2$ to the deflection in a manner similar to Eq. (3.2.5),

$$W = \frac{P_1 - P_2}{D(k_x^2 + k_y^2)^2 - \omega^2 \rho_w h}. \tag{3.4.1}$$

Because the transfer matrix relationship depends on velocities rather than displacement, it is more convenient to express this result in terms of the wall velocity, $U_w = i\omega W$. Equation (3.4.1) may then be written as

$$P_1 - P_2 = \frac{D(k_x^2 + k_y^2)^2 - \omega^2 \rho_w h}{i\omega} U_w = \frac{D \sin^4(\theta)(\omega/c)^4 - \omega^2 \rho_w h}{i\omega} U_w = Z_w(\theta) U_w, \tag{3.4.2}$$

where

$$Z_w(\theta) = \frac{D \sin^4(\theta)(\omega/c)^4 - \omega^2 \rho_w h}{i\omega}, \tag{3.4.3}$$

is the impedance of the wall and we have used Eqs. (3.2.13) to express k_x and k_y in terms of $\sin(\theta)$.

The continuity of velocity across the solid wall along with Eqs. (3.4.1) through (3.4.3) enable us to write the transfer matrix relation for this wall as

$$\begin{pmatrix} P_2 \\ U_2 \end{pmatrix} = \begin{bmatrix} 1 & -Z(\theta) \\ 0 & 1 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix}. \tag{3.4.4}$$

3.5 Transfer Matrix for an Air Gap

Since many acoustical systems contain regions where sound can be assumed to simply propagate as a plane wave, it is very important to have a transfer matrix for a simple air gap of length l. The sound field can be described by

$$p(x,t) = (p_1 e^{-ikx} + p_2 e^{ikx}) e^{i\omega t} = p(x) e^{i\omega t}. \tag{3.5.1}$$

where p_1 and p_2 denote the complex amplitude of waves traveling to the right and left in the gap, respectively. The velocity, $u(x,t)$, can be determined using the Euler equation,

$$-\frac{\partial p}{\partial x} = \rho_0 \frac{\partial u}{\partial t} = \rho_0 i\omega u(x) e^{i\omega t}. \tag{3.5.2}$$

The spatial dependence of the velocity is then

$$u(x) = \frac{p_1}{\rho_0 c} e^{-ikx} - \frac{p_2}{\rho_0 c} e^{ikx}. \tag{3.5.3}$$

To obtain a transfer matrix which relates the pressures and velocities at $x = 0$ and $x = l$ we will evaluate Eqs. (3.5.1) and (3.5.3) at these two locations and eliminate p_1 and p_2. At $x = 0$,

$$\begin{aligned} p(0) &= p_1 + p_2 \\ u(0) &= \frac{p_1}{\rho_0 c} - \frac{p_2}{\rho_0 c}. \end{aligned} \tag{3.5.4}$$

At $x = l$,

$$\begin{aligned} p(l) &= p_1 e^{-ikl} + p_2 e^{ikl} \\ u(l) &= \frac{p_1}{\rho_0 c} e^{-ikl} - \frac{p_2}{\rho_0 c} e^{ikl}. \end{aligned} \tag{3.5.5}$$

In matrix form, Eqs. (3.5.4) and (3.5.5) become

$$\begin{pmatrix} p(0) \\ u(0) \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ 1/(\rho_0 c) & -1/(\rho_0 c) \end{bmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}, \tag{3.5.6}$$

and,

$$\begin{pmatrix} p(l) \\ u(l) \end{pmatrix} = \begin{bmatrix} e^{-ikl} & e^{ikl} \\ e^{-ikl}/(\rho_0 c) & -e^{ikl}/(\rho_0 c) \end{bmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}. \tag{3.5.7}$$

Equation (3.5.6) may be easily solved for p_1 and p_2,

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \frac{\rho_0 c}{-2} \begin{bmatrix} -1/(\rho_0 c) & -1 \\ -1/(\rho_0 c) & 1 \end{bmatrix} \begin{pmatrix} p(0) \\ u(0) \end{pmatrix}. \tag{3.5.8}$$

Substituting Eq. (3.5.8) into Eq. (3.5.7) gives

$$\begin{pmatrix} p(l) \\ u(l) \end{pmatrix} = \frac{\rho_0 c}{-2} \begin{bmatrix} e^{-ikl} & e^{ikl} \\ e^{-ikl}/(\rho_0 c) & -e^{ikl}/(\rho_0 c) \end{bmatrix} \begin{bmatrix} -1/(\rho_0 c) & -1 \\ -1/(\rho_0 c) & 1 \end{bmatrix} \begin{pmatrix} p(0) \\ u(0) \end{pmatrix} \tag{3.5.9}$$

$$= \frac{\rho_0 c}{-2} \begin{bmatrix} (-e^{-ikl} - e^{ikl})/(\rho_0 c) & e^{ikl} - e^{-ikl} \\ (e^{ikl} - e^{-ikl})/(\rho_0 c)^2 & (-e^{-ikl} - e^{ikl})/(\rho_0 c) \end{bmatrix} \times \begin{pmatrix} p(0) \\ u(0) \end{pmatrix}. \tag{3.5.10}$$

Equation (3.5.10) can be simplified using

$$\cos(kl) = (e^{ikl} + e^{-ikl})/2, \tag{3.5.11}$$

and,

$$\sin(kl) = (e^{ikl} - e^{-ikl})/(2i). \tag{3.5.12}$$

Substituting Eqs. (3.5.11) and (3.5.12) into Eq. (3.5.10) gives

$$\begin{pmatrix} p(l) \\ u(l) \end{pmatrix} = \begin{bmatrix} \cos(kl) & -i\rho_0 c \sin(kl) \\ -\frac{i \sin(kl)}{\rho_0 c} & \cos(kl) \end{bmatrix} \begin{pmatrix} p(0) \\ u(0) \end{pmatrix}. \tag{3.5.13}$$

3.6 Air Gap with Non-normal Incident Sound

Suppose we have a simple air gap of length l. The sound field within the gap can be described by

$$p(x, y, z, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 e^{-ik_z z} + p_2 e^{ik_z z}) = p(z) e^{i\omega t} e^{-i(k_x x + k_y y)}. \tag{3.6.1}$$

Having the pressure, the velocity vector, $\overrightarrow{u}(x, y, z, t)$, can be determined using the Euler equation,

$$-\overrightarrow{\nabla} p = \rho_0 \dot{\overrightarrow{u}}, \tag{3.6.2}$$

where ρ_0 is the nominal air density. We will assume that the air gap lies in the plane normal to the z axis so that our primary interest is in the z component of the Euler equation (3.6.2).

Using Eq. (3.6.1) for the pressure in Eq. (3.6.2) enables us to determine the z component of the velocity,

$$u_z(x, y, z, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 e^{-ik_z z} - p_2 e^{ik_z z}) k_z / (\rho_0 c k), \tag{3.6.3}$$

where $k = \omega / c$.

Evaluating Eqs. (3.6.1) and (3.6.3) at the left end (i.e., at $z = 0$) of the air gap gives

$$p(x, y, 0, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 + p_2) = P_1 e^{i\omega t} e^{-i(k_x x + k_y y)}, \tag{3.6.4}$$

where,

$$P_1 = (p_1 + p_2), \tag{3.6.5}$$

and,

$$u_z(x, y, 0, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 - p_2) k_z / (\rho_0 c k) = U_1 e^{i\omega t} e^{-i(k_x x + k_y y)}, \tag{3.6.6}$$

where,

$$U_1 = (p_1 - p_2) k_z / (\rho_0 c k). \tag{3.6.7}$$

Evaluating Eqs. (3.6.1) and (3.6.3) at the right end (at $z = l$) of the air gap gives

$$p(x, y, l, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 e^{-ik_z l} + p_2 e^{ik_z l}) = P_2 e^{i\omega t} e^{-i(k_x x + k_y y)}, \tag{3.6.8}$$

where,

$$P_2 = (p_1 e^{-ik_z l} + p_2 e^{ik_z l}), \tag{3.6.9}$$

and

$$u_z(x, y, l, t) = e^{i\omega t} e^{-i(k_x x + k_y y)} (p_1 e^{-ik_z l} - p_2 e^{ik_z l}) k_z/(\rho_0 ck) = U_2 e^{i\omega t} e^{-i(k_x x + k_y y)}, \tag{3.6.10}$$

where,

$$U_2 = (p_1 e^{-ik_z l} - p_2 e^{ik_z l}) k_z/(\rho_0 ck). \tag{3.6.11}$$

The transfer matrix we seek will relate the pressures on either side of the air gap, P_1 and P_2 and the components of the velocity normal to the surface of the air gap, U_1 and U_2,

$$\begin{pmatrix} P_2 \\ U_2 \end{pmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix}. \tag{3.6.12}$$

In matrix form, Eqs. (3.6.5) and (3.6.7) become

$$\begin{pmatrix} P_1 \\ U_1 \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ k_z/(\rho_0 ck) & -k_z/(\rho_0 ck) \end{bmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}. \tag{3.6.13}$$

Equations (3.6.9) and (3.6.11) may be written as

$$\begin{pmatrix} P_2 \\ U_2 \end{pmatrix} = \begin{bmatrix} e^{-ikl} & e^{ikl} \\ e^{-ikl} k_z/(\rho_0 ck) & -e^{ikl} k_z/(\rho_0 ck) \end{bmatrix} \begin{pmatrix} p_1 \\ p_2 \end{pmatrix}. \tag{3.6.14}$$

We need to eliminate p_1 and p_2 from Eqs. (3.6.13) and (3.6.14). Equation (3.6.13) may be solved for p_1 and p_2,

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \frac{\rho_0 ck}{-2k_z} \begin{bmatrix} -k_z/(\rho_0 ck) & -1 \\ -k_z/(\rho_0 ck) & 1 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix}. \tag{3.6.15}$$

Substituting Eq. (3.6.15) into Eq. (3.6.14) gives

$$\begin{aligned} \begin{pmatrix} P_2 \\ U_2 \end{pmatrix} &= \frac{\rho_0 ck}{-2k_z} \begin{bmatrix} e^{-ikl} & e^{ikl} \\ e^{-ikl} k_z/(\rho_0 ck) & -e^{ikl} k_z/(\rho_0 ck) \end{bmatrix} \begin{bmatrix} -k_z/(\rho_0 ck) & -1 \\ -k_z/(\rho_0 ck) & 1 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix} \\ &= \frac{\rho_0 ck}{-2k_z} \begin{bmatrix} (-e^{-ikl} - e^{ikl}) k_z/(\rho_0 ck) & e^{ikl} - e^{-ikl} \\ (e^{ikl} - e^{-ikl})(k_z/(\rho_0 ck))^2 & (-e^{-ikl} - e^{ikl}) k_z/(\rho_0 ck) \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix} \end{aligned} \tag{3.6.16}$$

Equation (3.6.16) can be simplified using

$$\cos(kl) = (e^{ikl} + e^{-ikl})/2, \tag{3.6.17}$$

and,

$$\sin(kl) = (e^{ikl} - e^{-ikl})/(2i). \tag{3.6.18}$$

Substituting Eqs. (3.6.17) and (3.6.18) into Eq. (3.6.16) gives

$$\begin{pmatrix} P_2 \\ U_2 \end{pmatrix} = \begin{bmatrix} \cos(kl) & -\frac{ik\rho_0 c \sin(kl)}{k_z} \\ -\frac{ik_z \sin(kl)}{\rho_0 ck} & \cos(kl) \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix}. \tag{3.6.19}$$

As a check on these results, the transmission loss of the air gap can be calculated by using our result for the ratio of the transmitted to incident pressures,

$$\frac{p_t}{p_1} = \frac{2(T_{11}T_{22} - T_{12}T_{21})}{T_{11} - T_{21}\frac{k\rho_0 c}{k_z} + T_{22} - \frac{T_{12}k_z}{k\rho_0 c}}. \tag{3.6.20}$$

Substituting the elements of the transfer matrix of Eq. (3.6.19) into Eq. (3.6.20) gives

$$\frac{p_t}{p_1} = \frac{2}{2(\cos(kl) + i\sin(kl))} = e^{-ikl}. \tag{3.6.21}$$

The effect of the air gap is thus to create only a phase change in the transmitted signal relative to the incident sound of kl radians. The amplitude of the transmitted pressure will be identical to that of the incident pressure, giving a transmission loss of zero decibels. While this air gap does not attenuate the sound in the wall, in the following we will see that it does play a significant role in the interaction between the elements on either side of it.

3.7 Double Walls

As a first example, consider a double wall with normally incident sound where the outer elements consist of solid materials having a surface density of $\rho_w h = 10\,\mathrm{kg/m^2}$. The air gap between the outer elements has a thickness of $l = 2$ cm. The composite transfer matrix may be computed by multiplying the three element matrices,

$$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} = \begin{bmatrix} 1 & -Z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(kl) & -i\rho_0 c \sin(kl) \\ -\frac{i\sin(kl)}{\rho_0 c} & \cos(kl) \end{bmatrix} \begin{bmatrix} 1 & -Z_1 \\ 0 & 1 \end{bmatrix}, \tag{3.7.1}$$

The impedances of the outer layers are $Z_1 = i\rho_w \omega h_1$ and $Z_2 = i\rho_w \omega h_2$, where ρ_w is the density of the outer layers and h_1 and h_2 are the thicknesses. In this example we

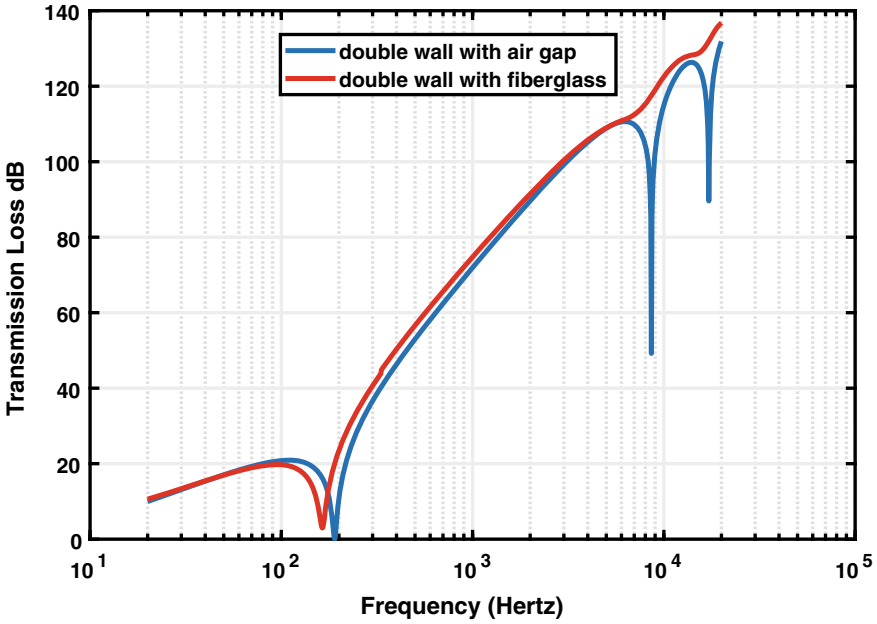


Fig. 3.7 Predicted sound transmission loss for a double wall. The sound is normally incident. The surface densities of the outer elements are each $10\,\mathrm{kg/m^2}$ and the thickness of the air gap between them is 2 cm. The transmission loss is computed with and without fiberglass included in the air gap. The fiberglass clearly damps out acoustic resonances within the wall.

will simply set $\rho_w h_1 = \rho_w h_2 = 10\,\mathrm{kg/m^2}$. We also have, as usual, $k = \omega/c$ where c is the sound speed. The predicted sound transmission loss of this double wall is shown in Fig. 3.7.

The figure shows that at low frequencies, the transmission loss increases at roughly 6 dB per octave, as expected for a single wall. At approximately 200 Hz, there is a deep dip in the transmission loss with the double wall becoming nearly acoustically transparent in a narrow range of frequencies. As we will see in Sect. 3.7.3 this corresponds to the double wall resonance where the two outer panels oscillate with the air space between them acting as a spring.

At frequencies above 200 Hz, the transmission loss increases at approximately 12 dB/octave, twice the slope of the low frequency, single wall response that occurs below the double wall resonance. Finally, as frequency is increased to nearly 9 kHz, we again see sharp dips which are associated with standing wave acoustic resonances within the wall.

We now consider a double wall consisting of two outer layers which are solid and are separated by a porous layer with thickness l that is filled with a sound absorbing material such as fiberglass insulation. This absorbing material will increase the transmission loss, especially at high frequencies, and eliminate standing wave resonances within the air gap. The presence of the absorbent material may be accounted for by

Table 3.1 Coefficients of the Delaney and Bazley fiberglass model [2]

Coefficient	$e_{norm} \leq 0.025$	$e_{norm} > 0.025$
a_1	0.396	0.179
α_1	0.458	0.674
a_{11}	0.135	0.102
α_{11}	0.646	0.705
b_1	0.0668	0.0235
β_1	0.707	0.887
b_{11}	0.196	0.0875
β_{11}	0.549	0.770

replacing the wave number $k = \omega/c$ with a complex wave number (or propagation constant, ik^g) and using a complex material impedance, z^g instead of $\rho_0 c$. There are many empirical models for the acoustical properties of fiberglass and other porous materials available in the literature. The model used here is just one example [1–3]. In this model, the complex propagation constant is given by

$$\gamma = ik^g = a + ib = \frac{\omega}{c}(a_1 e_{norm}^{-\alpha_1} + i(1 + a_{11} e_{norm}^{-\alpha_{11}})). \tag{3.7.2}$$

The complex impedance is given by

$$z^g = \rho c(1 + b_1 e_{norm}^{-\beta_1} - ib_{11} e_{norm}^{-\beta_{11}}), \tag{3.7.3}$$

where the constants are given in Table 3.1 for two ranges of values of the dimensionless constant $e_{norm} = \rho_0 \frac{\omega}{2\pi r_1}$. r_1 is the flow resistivity which we will take to be $r_1 = 1.6 \times 10^4$.

The composite transfer matrix of this wall is obtained by multiplying the three element matrices:

$$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} = \begin{bmatrix} 1 & -z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(k_z^g l) & -iz^g \frac{k^g}{k_z^g} \sin(k_z^g l) \\ -i\frac{k_z^g}{z^g k^g} \sin(k_z^g l) & \cos(k_z^g l) \end{bmatrix} \begin{bmatrix} 1 & -z_1 \\ 0 & 1 \end{bmatrix}, \tag{3.7.4}$$

where ik^g is the complex propagation constant in the porous material, $ik^g = a + ib$, z^g is the complex impedance of the material and $k_z^g = \sqrt{k^{g^2} - k_x^2 - k_y^2}$. k_x and k_y are the x and y components of the wave vector of the incident wave. Carrying out the multiplication gives

$$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} = \begin{bmatrix} 1 & -z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(k_z^g l) & -z_1 \cos(k_z^g l) - i z^g \frac{k^g}{k_z^g} \sin(k_z^g l) \\ -i \frac{k_z^g}{z^g k^g} \sin(k_z^g l) & {}_1 i \frac{k_z^g}{z^g k^g} \sin(k_z^g l) + \cos(k_z^g l) \end{bmatrix}$$
$$= \begin{bmatrix} \cos(k_z^g l) + \frac{z_2 i k_z^g}{z^g k^g} \sin(k_z^g l) & -(z_1 + z_2) \cos(k_z^g l) - \frac{i z^g k^g}{k_z^g} \sin(k_z^g l) - \frac{z_1 z_2 i k_z^g}{z^g k^g} \sin(k_z^g l) \\ -\frac{i k_z^g}{z^g k^g} \sin(k_z^g l) & \frac{z_1 i k_z^g}{z^g k^g} \sin(k_z^g l) + \cos(k_z^g l) \end{bmatrix}. \tag{3.7.5}$$

The ratio p_t/p_1 may be computed from the elements of the composite transfer matrix by

$$\frac{p_t}{p_1} = 2/\left(T_{11} + T_{22} - \frac{T_{12} k_z}{k \rho_0 c} - T_{21} \frac{k \rho_0 c}{k_z} \right), \tag{3.7.6}$$

where $k_z = \sqrt{k^2 - k_x^2 - k_y^2}$ and $k = \omega/c$.

It is helpful to look at some special cases. Consider normally incident sound in which $k = k_z$, $k_z^g = k^g$, and $k_x = k_y = 0$. Equation (3.7.6) simplifies to

$$\frac{p_t}{p_1} = 2/\left(T_{11} + T_{22} - \frac{T_{12}}{\rho_0 c} - T_{21} \rho_0 c \right). \tag{3.7.7}$$

Substituting the elements of the transfer matrix, T_{11}, T_{12}, T_{21}, and T_{22}, in Eq. (3.7.5) into Eq. (3.7.7) gives

$$\frac{p_t}{p_1} = \frac{2}{\cos(k^g l)\left(2 + \frac{z_1 + z_2}{\rho_0 c}\right) + i \sin(k^g l)\left(\frac{z_1 + z_2}{z^g} + \frac{z^g}{\rho_0 c} + \frac{z_1 z_2}{z^g \rho_0 c} + \frac{\rho_0 c}{z^g}\right)}. \tag{3.7.8}$$

3.7.1 Simple Air Gap

Suppose that the gap contains only air so that $z^g = \rho_0 c$ and $k^g = \omega/c = k_z = k$. The impedances of the outer layers are $z_1 = i \rho_w \omega h_1$ and $z_2 = i \rho_w \omega h_2$, where ρ_w is the density of the outer layers and h_1 and h_2 are the thicknesses. Equation (3.7.8) then becomes

$$\frac{p_t}{p_1} = \frac{2}{\cos(kl)\left(2 + \frac{i \rho \omega (h_1 + h_2)}{\rho_0 c}\right) + \sin(kl)\left(\frac{-\rho \omega (h_1 + h_2)}{\rho_0 c} + 2i - \frac{i (\rho \omega)^2 h_1 h_2}{(\rho_0 c)^2}\right)}. \tag{3.7.9}$$

This simplifies to

$$\frac{p_t}{p_1} = \frac{2}{e^{ikl}\left(2 + \frac{i\rho\omega(h_1+h_2)}{\rho_0 c}\right) - i\sin(kl)\frac{(\rho\omega)^2 h_1 h_2}{(\rho_0 c)^2}}. \tag{3.7.10}$$

Now consider the case where the thickness of the gap in the wall, l, is much less than the wavelength of sound, $l \ll \lambda$, or, since $\lambda = 2\pi/k$, $kl \ll 2\pi$. In this case $e^{ikl} \approx 1$ and $\sin(kl) \approx kl = \frac{\omega}{c} l$. Equation (3.7.10) then simplifies to

$$\frac{p_t}{p_1} = \frac{2}{2 + \frac{i\rho\omega(h_1+h_2)}{\rho_0 c} - i\frac{\omega}{c} l \frac{(\rho\omega)^2 h_1 h_2}{(\rho_0 c)^2}}. \tag{3.7.11}$$

3.7.2 *Low Frequencies*

At very low frequencies we can neglect terms that are proportional to ω^3 so that

$$\frac{p_t}{p_1} \approx \frac{1}{1 + \frac{i\rho\omega(h_1+h_2)}{2\rho_0 c}}. \tag{3.7.12}$$

Note that this is just the single wall mass law for a panel of thickness $h_1 + h_2$.

3.7.3 *Slightly Higher Frequencies—Double Wall Resonance*

From Eq. (3.7.11) we see that p_t/p_1 will be unity (and the TL will be zero) if

$$\frac{i\rho\omega(h_1 + h_2)}{\rho_0 c} = i\frac{\omega}{c} l(\rho\omega)^2 \frac{h_1 h_2}{(\rho_0 c)^2}. \tag{3.7.13}$$

Solving for ω gives

$$\omega_0 = \sqrt{\frac{\rho_0 c^2 (1/h_1 + 1/h_2)}{\rho l}}, \tag{3.7.14}$$

which is the double wall resonance frequency.

3.7.4 Very High Frequencies

In the above results, we have simplified things by neglecting the presence of the absorbent, porous material within the wall. At frequencies above ω_0 it is not valid to assume that $kl \ll 2\pi$ and as a result, there will be "standing" waves within the wall. If we include absorbent material in the model, however, the resonant standing waves will be damped out. Let's go back to Eq. (3.7.8) and set $z_1 = i\rho h_1 \omega$ and $z_2 = i\rho h_2 \omega$. Equation (3.7.8) is then

$$\frac{p_t}{p_1} = \frac{2}{\cos(k^g l)\left(2 + \frac{i\omega\rho(h_1+h_2)}{\rho_0 c}\right) + i\sin(k^g l)\left(\frac{i\omega\rho(h_1+h_2)}{z^g} + \frac{z^g}{\rho_0 c} - \frac{(\omega\rho)^2 h_1 h_2}{z^g \rho_0 c} + \frac{\rho_0 c}{z^g}\right)}. \tag{3.7.15}$$

For a porous material, the wave number and corresponding propagation constant are complex, $ik^g = a + ib$. Since we are still considering a normally incident sound field with $k_x = k_y = 0$ and $k^g = k_z^g$, then the sine and cosine functions in Eq. (3.7.15) may be written,

$$\cos(k^g l) = \frac{1}{2}(e^{al}e^{ibl} + e^{-al}e^{-ibl}), \tag{3.7.16}$$

and,

$$\sin(k^g l) = \frac{1}{2i}(e^{al}e^{ibl} - e^{-al}e^{-ibl}). \tag{3.7.17}$$

If al is large (the case where waves are highly attenuated by the porous material) then

$$\cos(k^g l) \approx \frac{1}{2}e^{al}e^{ibl}, \tag{3.7.18}$$

and,

$$\sin(k^g l) \approx \frac{1}{2i}e^{al}e^{ibl}. \tag{3.7.19}$$

Substituting Eqs. (3.7.18) and (3.7.19) into Eq. (3.7.15) gives

$$\frac{p_t}{p_1} \approx \frac{4e^{-al}e^{-ibl}}{2 + \frac{i\omega\rho(h_1+h_2)}{\rho_0 c} + \frac{i\omega\rho(h_1+h_2)}{z^g} + \frac{z^g}{\rho_0 c} - \frac{(\omega\rho)^2 h_1 h_2}{z^g \rho_0 c} + \frac{\rho_0 c}{z^g}}. \tag{3.7.20}$$

At high frequencies the term that is proportional to ω^2 dominates in the denominator so that

$$\frac{p_t}{p_1} \approx \frac{4e^{-al}e^{-ibl}}{-\frac{(\omega\rho)^2 h_1 h_2}{z^g \rho_0 c}}. \tag{3.7.21}$$

The transmission coefficient, τ, is $\tau = |p_t/p_1|^2$ and the transmission loss is $TL = 10\log_{10}(1/\tau)$,

$$TL \approx 20\log_{10}\left(\frac{(\omega\rho)^2 h_1 h_2 e^{al}}{4|z^g|\rho_0 c}\right). \tag{3.7.22}$$

Equation (3.7.22) may be written as

$$\begin{aligned} TL &= 20\log_{10}\left(\frac{\omega\rho h_1}{2\rho_0 c}\right) + 20\log_{10}\left(\frac{\omega\rho h_2}{2\rho_0 c}\right) + 20\log_{10}\left(\frac{\rho_0 c}{|z^g|}e^{al}\right) \\ &= TL_{\text{wall}_1} + TL_{\text{wall}_2} + TL_{\text{porous material}}. \end{aligned} \tag{3.7.23}$$

This approximate formula shows that the total transmission loss is roughly the sum of the transmission losses of each outer wall plus that of the porous material. This formula neglects any coupling between the elements.

3.8 Analysis of Double Walls with Mechanical Coupling

We discussed a model of a double wall consisting of two outer layers which are solid and are separated by a porous layer with thickness l. The composite transfer matrix of this wall was obtained by multiplying the three element matrices:

$$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} = \begin{bmatrix} 1 & -z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \cos(k_z^g l) & -iz^g \frac{k^g}{k_z^g}\sin(k_z^g l) \\ -i\frac{k_z^g}{z^g k^g}\sin(k_z^g l) & \cos(k_z^g l) \end{bmatrix} \begin{bmatrix} 1 & -z_1 \\ 0 & 1 \end{bmatrix}, \tag{3.8.1}$$

where ik^g is the complex propagation constant in the porous material, $ik^g = a + ib$, z^g is the complex impedance of the material and $k_z^g = \sqrt{k^{g^2} - k_x^2 - k_y^2}$. k_x and k_y are the x and y components of the wave vector of the incident wave.

There are many situations where there is some mechanical connection between the two outer walls which acts in parallel with the fiberglass. This connection can consist of wooden studs as in housing construction or mechanical isolation mounts. The mechanical connection between the outer walls tends to "short circuit" the effectiveness of the fiberglass layer. In the following, we will work out a procedure for accounting for this.

We will assume that the mechanical connection between the outer layers can be represented by a spring having some uniform stiffness K per unit area. This is, of

course, a rough approximation since in practice the coupling occurs at discrete points. Although this is a rather crude model, it should give us a qualitative understanding of how K influences the sound transmission through the wall.

Let the transfer matrix for the fiberglass layer be

$$\begin{bmatrix} T_{11}^g & T_{12}^g \\ T_{21}^g & T_{22}^g \end{bmatrix} = \begin{bmatrix} \cos(k_z^g l) & -iz^g \frac{k^g}{k_z^g} \sin(k_z^g l) \\ -i\frac{k_z^g}{z^g k^g} \sin(k_z^g l) & \cos(k_z^g l) \end{bmatrix}. \tag{3.8.2}$$

The transfer matrix for a spring connecting the two outer layers follows from noting that the forces (or pressures in this case) will be equal on two ends of a spring so that $p_1 = p_2$. This pressure is related to the spring constant K by $K(w_1 - w_2) = p$, where w_1 and w_2 are the displacements of the ends of the spring. These displacements are related to the velocities (assuming harmonic time dependence) by $w_1 = u_1/(i\omega)$ and $w_2 = u_2/(i\omega)$. The relation between pressure and velocity is then $K/(i\omega)(u_1 - u_2) = p$. The transfer matrix for the mechanical spring is then

$$\begin{bmatrix} T_{11}^s & T_{12}^s \\ T_{21}^s & T_{22}^s \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -i\omega/K & 1 \end{bmatrix}. \tag{3.8.3}$$

We can think of the transfer matrices defined in Eqs. (3.8.2) and (3.8.3) as providing relations between the pressures and velocities associated with the fiberglass and the spring. That is, if each element acted alone,

$$\begin{pmatrix} p_2^g \\ u_2^g \end{pmatrix} = \begin{bmatrix} T_{11}^g & T_{12}^g \\ T_{21}^g & T_{22}^g \end{bmatrix} \begin{pmatrix} p_1^g \\ u_1^g \end{pmatrix}, \tag{3.8.4}$$

and

$$\begin{pmatrix} p_2^s \\ u_2^s \end{pmatrix} = \begin{bmatrix} T_{11}^s & T_{12}^s \\ T_{21}^s & T_{22}^s \end{bmatrix} \begin{pmatrix} p_1^s \\ u_1^s \end{pmatrix}, \tag{3.8.5}$$

where the superscripts g or s denote the fiberglass or spring, respectively.

When these elements act in parallel, the velocities will be equal, $u_1^g = u_1^s$, and $u_2^g = u_2^s$. The pressures will add so that the total pressure on side #1 is $p_1 = p_1^g + p_1^s$ and the total pressure on side #2 is $p_2 = p_2^g + p_2^s$. We need to find a composite matrix which relates p_2, u_2, p_1, and u_1. To accomplish this it is helpful to rearrange equations (3.8.4) and (3.8.5) into the form of impedance matrices,

$$\begin{pmatrix} p_1^g \\ p_2^g \end{pmatrix} = \begin{bmatrix} Z_{11}^g & Z_{12}^g \\ Z_{21}^g & Z_{22}^g \end{bmatrix} \begin{pmatrix} u_1^g \\ u_2^g \end{pmatrix}, \tag{3.8.6}$$

and

$$\begin{pmatrix} p_1^s \\ p_2^s \end{pmatrix} = \begin{bmatrix} Z_{11}^s & Z_{12}^s \\ Z_{21}^s & Z_{22}^s \end{bmatrix} \begin{pmatrix} u_1^s \\ u_2^s \end{pmatrix}. \tag{3.8.7}$$

Since the pressures add and the velocities are equal as discussed above,

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{pmatrix} p_1^g + p_1^s \\ p_2^g + p_2^s \end{pmatrix} = \left\{ \begin{bmatrix} Z_{11}^g & Z_{12}^g \\ Z_{21}^g & Z_{22}^g \end{bmatrix} + \begin{bmatrix} Z_{11}^s & Z_{12}^s \\ Z_{21}^s & Z_{22}^s \end{bmatrix} \right\} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}. \tag{3.8.8}$$

Equation (3.8.8) can be rearranged to give the composite transfer matrix relation for the spring and the fiberglass in the form

$$\begin{pmatrix} p_2 \\ u_2 \end{pmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{pmatrix} p_1 \\ u_1 \end{pmatrix}. \tag{3.8.9}$$

Note that if one has a transfer matrix model such as that of Eqs. (3.8.4) or (3.8.5), one can rearrange the equation to obtain the impedance matrix as

$$\begin{pmatrix} p_1 \\ p_2 \end{pmatrix} = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} = \begin{bmatrix} -\frac{T_{22}}{T_{21}} & \frac{1}{T_{21}} \\ -\frac{T_{22}T_{11}}{T_{21}} & \frac{T_{11}}{T_{21}} \end{bmatrix} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}. \tag{3.8.10}$$

In addition, if one has the elements of the impedance matrix, as in Eqs. (3.8.6) or (3.8.7) the equations may be rearranged to give the transfer matrix,

$$\begin{pmatrix} p_2 \\ u_2 \end{pmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{pmatrix} p_1 \\ u_1 \end{pmatrix} = \begin{bmatrix} -\frac{Z_{22}}{Z_{12}} & Z_{21} - \frac{Z_{21}Z_{11}}{Z_{12}} \\ -\frac{1}{T_{12}} & -\frac{Z_{11}}{Z_{12}} \end{bmatrix} \begin{pmatrix} p_1 \\ u_1 \end{pmatrix}. \tag{3.8.11}$$

The elements of the composite transfer matrix in Eq. (3.8.9) may be shown to be related to the transfer matrices of the fiberglass and the spring by

$$T_{11} = \frac{T_{11}^g T_{21}^s + T_{11}^s T_{21}^g}{T_{21}^g T_{21}^s}, \tag{3.8.12}$$

$$T_{12} = T_{12}^g + T_{12}^s + \frac{T_{11}^s T_{22}^g + T_{11}^g T_{22}^s - T_{11}^g T_{22}^g - T_{11}^s T_{22}^s}{T_{21}^g T_{21}^s}, \tag{3.8.13}$$

$$T_{21} = \frac{T_{21}^g T_{21}^s}{T_{21}^g + T_{21}^s}, \tag{3.8.14}$$

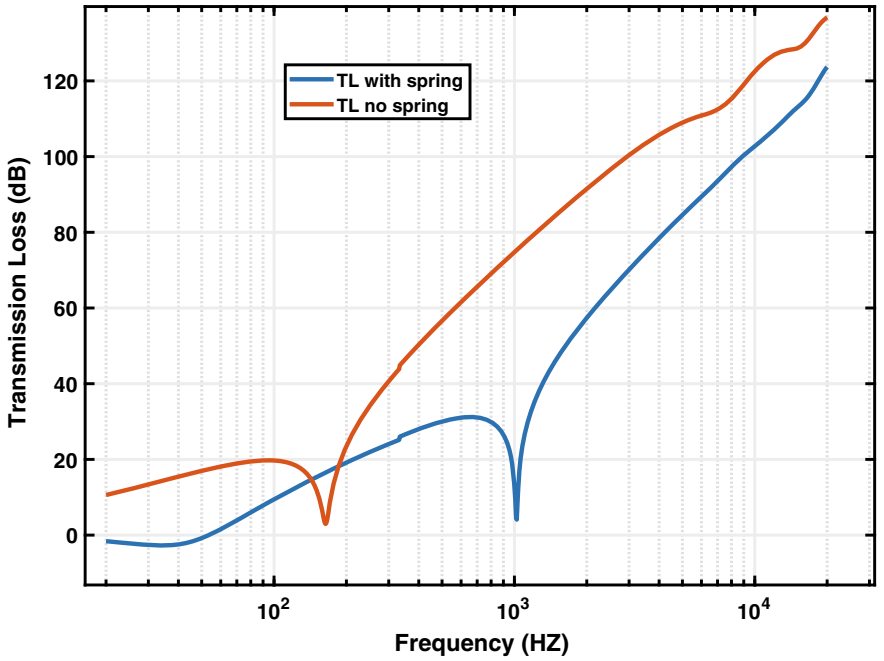


Fig. 3.8 Predicted sound transmission loss for the double wall examined in Fig. 3.7 with the addition of a spring with equivalent stiffness per unit area of $K = 2 \times 10^8\,\mathrm{N/m^3}$ attached between the two solid outer elements. The sound is normally incident. As in Fig. 3.7, the surface densities of the outer elements are each $10\,\mathrm{kg/m^2}$ and the thickness of the air gap between them is 2 cm. The spring increases the double wall resonance frequency and subsequently decreases the transmission loss

$$T_{22} = \frac{T_{22}^g T_{21}^s + T_{22}^s T_{21}^g}{T_{21}^g T_{21}^s}. \tag{3.8.15}$$

As an example, suppose that a spring with stiffness $K = 2 \times 10^8\,\mathrm{N/m^3}$ is attached between the solid elements in the wall examined in Fig. 3.7. This spring is stiff enough to increase the double wall resonance to approximately 1 kHz. The predicted results are shown in Fig. 3.8. The spring increases the double wall resonance frequency which causes a decrease in the high-frequency transmission loss.

The results shown in Fig. 3.8 show that a mechanical connection between the outer solid elements in the wall can have the effect of essentially short-circuiting the sound attenuation that is provided by the absorbent material within the wall. This illustrates the importance of minimizing these mechanical connections as much as possible to maximize the transmission loss.

3.9 Problems

1. Determine the sound transmission loss, TL, of a limp wall having a density of $\rho = 500\,\mathrm{kg/m^3}$ and a thickness of $h = 0.06\,\mathrm{m}$. Assume the density of air is $\rho = 1.18\,\mathrm{kg/m^3}$ and that the speed of sound is $c = 344\,\mathrm{m/s}$. Plot your results as a function of frequency using a logarithmic frequency axis over a range from 63 Hz to 16 kHz. Include an annotated listing of any computer programs written.
2. For the wall of Problem 1, plot the absorption coefficient α as a function of frequency using a logarithmic frequency axis over a range from 63 Hz to 16 kHz and a logarithmic vertical axis. Include an annotated listing of any computer programs written.
3. Derive Eqs. (3.2.5), (3.2.12) and (3.2.14).
4. Determine the random incidence transmission loss of the wall of Problem 1. Note that this wall is "limp" so that the flexural rigidity is $D = 0$. Plot your result along with that of Problem 1. You may adapt the program "singlewallrandomtl.m" to calculate the random incidence transmission loss in these problems.
5. A single wall consists of a layer of 1/2 in. thick gypsum wallboard having a surface density $\rho \times h = 6\,\mathrm{kg/m^2}$. The equivalent Young's modulus is approximately $E = 2 \times 10^8\ \mathrm{N/m^2}$, Poisson's ratio is $\sigma = 0.3$, the material loss factor is $\eta = 0.07$. Determine the **random incidence** transmission loss of this wall. Include (1) a write-up describing your program, (2) an annotated listing of your m file and (3) a properly labeled plot of the transmission loss for a frequency range of 20 Hz to 20 kHz. Plot your data using a logarithmic frequency axis (i.e., using the "semilogx" command in Matlab).

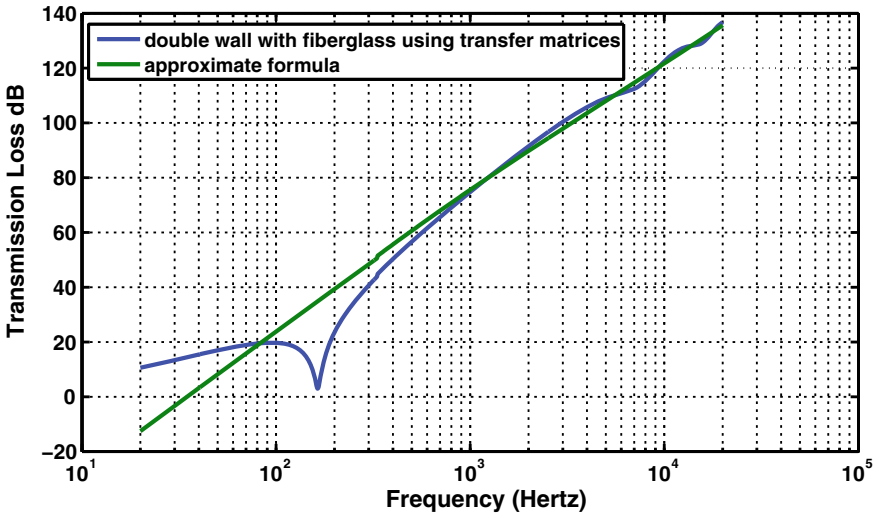


Fig. 3.9 Predicted sound transmission loss for the double wall examined in Fig. 3.7 using the approximate formula Eq. (3.7.22). The sound is normally incident. As in Fig. 3.7, the surface densities of the outer elements are each $10\,\mathrm{kg/m^2}$ and the thickness of the air gap between them is 2 cm

6. Use Eq. (3.2.16) to estimate the "coincidence frequency" for the wall of problem 3.9).
7. Suppose the wall of Problem 5 has no stiffness so $E = 0$. Plot the random incidence transmission loss in this case on the same plot as the results of Problem 5. Discuss why they are different (or the same).
8. Derive Eqs. (3.3.12) and (3.3.13).
9. Write a program to use transfer matrices to predict the normal incidence transmission loss of a double wall containing fiberglass having the properties indicated in Fig. 3.7. Compare your results with those shown in the figure. You may use the Matlab script, "fiberglassTmatrix.m", to create the transfer matrix for the fiberglass.
10. Use Eq. (3.7.22) to obtain an approximation to the transmission loss you obtained in Problem 9. Plot your approximate results on the same set of axes as used in Problem 9. Hint: Your solution should look something like Fig. 3.9.
11. A double wall has a $10\,\mathrm{kg/m^2}$ solid element on one side and $10\,\mathrm{kg/m^2}$ solid element on the other. The air gap (with no porous material) between the two outer walls has a thickness of 3 cm. Write a Matlab program to estimate the normal incidence transmission loss of this wall and plot your results over a frequency range of 20 Hz to 20 kHz. Use a logarithmic frequency axis. As always, include a description of your approach and an annotated listing of your Matlab program.

12. Modify your analysis of the wall of Problem 11 to incorporate fiberglass insulation within it rather than an air gap. You may use the Matlab script, "fiberglassTmatrix.m", to create the transfer matrix for the fiberglass. Plot your results on the same set of axes for the cases with and without the fiberglass.
13. Use Eq. (3.7.22) to obtain an approximation to the transmission loss you obtained in Problem 12. Plot your approximate results on the same set of axes as used in Problem 12.
14. Re-derive Eq. (3.6.19) and verify that there are no typographical errors in the derivation.
15. Design a multielement wall having as large an absorption coefficient α as possible over the frequency range that is most important for human speech, 500 Hz to 5 kHz. The wall must incorporate at least one solid element having a surface density of $10\,\mathrm{kg/m^2}$. It may also incorporate fiberglass with a total fiberglass thickness no greater than 3 cm. Plot your predicted absorption coefficient versus frequency using a logarithmic frequency axis.
16. Write a Matlab script to account for the spring attached between the two outer solid elements of the wall examined in Fig. 3.8. Plot the predicted transmission loss as shown in the figure.
17. Sound levels have been measured in a space that is intended to house a new acoustics testing facility. A goal of this facility is that the maximum sound levels will be below those indicated as "maximum acceptable noise level" in Fig. 3.10. Measured noise levels were obtained using a sound level meter in the space without any acoustic treatment. These data are shown in the figure along with the estimated amount of noise reduction needed to achieve the desired noise

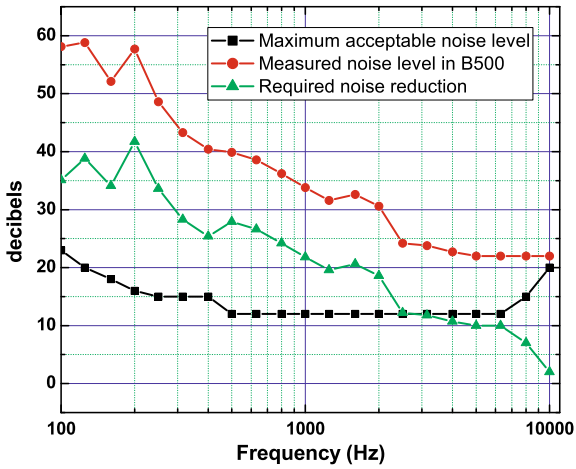


Fig. 3.10 Estimate of required noise reduction for the new acoustics testing facility

levels shown. We will assume that the noise reduction is approximately equal to the random incidence sound transmission loss of the walls that will enclose the testing facility. Design a wall using Gypsum wallboard and fiberglass insulation that meets or exceeds the required transmission loss over the entire range of frequencies shown. The thickness of the wall must not exceed 20 cm. You may use any number of layers of wallboard in your design as long as the total thickness is less than 20 cm. Plot the predicted normal incidence transmission loss of your design over the range of frequencies shown. Use a logarithmic frequency axis.

References for Chapter 3

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2. Beranek LL, Ver IL (1992) Noise and vibration control engineering-principles and applications. Wiley, New York, p 814
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Chapter 4
Analysis of Mufflers and Ducts



The transfer matrix analysis of the transmission through walls as discussed in Chap. 3 is readily adaptable to the analysis of mufflers. This approach will enable us to construct models of muffler and/or duct systems having nearly arbitrary complexity. In the following, we will examine several key muffler components using transfer matrices. A very common muffler construction is the expansion muffler, which consists of a section of a pipe or duct that has an increased cross-sectional area. To develop a transfer matrix model of this we will use three transfer matrices, one for each of the junctions at the ends of the muffler and one for the section between each junction.

Here we attempt only a brief introduction to the acoustics of mufflers and ducts. There are many excellent sources that treat this problem with the rigor it deserves. An excellent example is the text by Munjal [1].

4.1 The Junction of Two Pipes

The transfer matrix for the junction between a pipe of cross-sectional area s_1 with a pipe having cross-sectional area s_2 can be written by noting that the pressures at the two sides of the junction must be equal, $p_1 = p_2$ and the acoustic velocities are related by $u_1 s_1 = u_2 s_2$. The transfer matrix relation between each side of the junction is then,

$$\begin{pmatrix} p_2 \\ u_2 \end{pmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & s_1/s_2 \end{bmatrix} \begin{pmatrix} p_1 \\ u_1 \end{pmatrix} \tag{4.1.1}$$

Note that the determinant of this transfer matrix is not equal to 1. The reflection factor, r, may be calculated from

$$r = \frac{T_{21}\rho_0 c + T_{22} - T_{11} - T_{12}/(\rho_0 c)}{T_{11} + T_{22} - T_{12}/(\rho_0 c) - T_{21}\rho_0 c}. \tag{4.1.2}$$

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_4

Substituting the elements of the transfer matrix into Eq. (4.1.2) gives

$$r = \frac{s_1/s_2 - 1}{s_1/s_2 + 1}. \tag{4.1.3}$$

The ratio, p_t/p_1, becomes

$$\frac{p_t}{p_1} = \frac{2(T_{11}T_{22} - T_{21}T_{12})}{T_{11} + T_{22} - T_{12}/(\rho_0 c) - T_{21}\rho_0 c} = \frac{2s_1/s_2}{s_1/s_2 + 1}. \tag{4.1.4}$$

The transmission coefficient is

$$\tau = |p_t/p_1|^2 = \frac{4s_1^2}{(s_1 + s_2)^2}, \tag{4.1.5}$$

and the transmission loss is

$$TL = 20\log_{10}\left(\frac{s_1 + s_2}{2s_1}\right). \tag{4.1.6}$$

τ is equal to the ratio of the transmitted to incident intensity. In muffler design we often are interested in the ratio of transmitted to incident **power**,

$$\frac{\Pi_t}{\Pi_i} = \frac{I_t \times s_2}{I_i \times s_1} = \tau s_2/s_1 = \frac{4s_1 s_2}{(s_1 + s_2)^2}. \tag{4.1.7}$$

4.2 The Expansion Muffler

Consider the simple expansion muffler shown in Fig. 4.1. This muffler may be modeled by using transfer matrices to model the junctions and the middle section of length l. The composite transfer matrix is

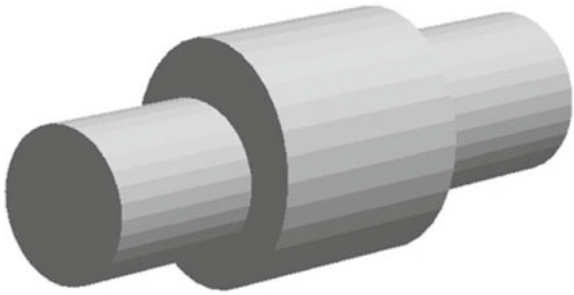


Fig. 4.1 An expansion chamber muffler

$$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} =$$
$$\begin{bmatrix} 1 & 0 \\ 0 & s_2/s_1 \end{bmatrix} \begin{bmatrix} \cos(kl) & -i\rho_0 c \sin(kl) \\ -\frac{i \sin(kl)}{\rho_0 c} & \cos(kl) \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & s_1/s_2 \end{bmatrix}$$
$$= \begin{bmatrix} \cos(kl) & -i\frac{s_1}{s_2}\rho_0 c \sin(kl) \\ -\frac{i s_2 \sin(kl)}{s_1 \rho_0 c} & \cos(kl) \end{bmatrix} \tag{4.2.1}$$

Then

$$\begin{pmatrix} p_2 \\ u_2 \end{pmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{pmatrix} p_1 \\ u_1 \end{pmatrix}. \tag{4.2.2}$$

The ratio of transmitted to incident pressure becomes

$$\begin{aligned} \frac{p_t}{p_1} &= \frac{2(T_{11}T_{22} - T_{21}T_{12})}{T_{11} + T_{22} - T_{12}/(\rho_0 c) - T_{21}\rho_0 c} \\ &= \frac{2}{2\cos(kl) + i\sin(kl)(s_1/s_2 + s_2/s_1)}. \end{aligned} \tag{4.2.3}$$

The transmission coefficient is

$$\begin{aligned} \tau &= |p_t/p_1|^2 = \frac{I_t}{I_i} \\ &= \frac{1}{\cos^2(kl) + \frac{\sin^2(kl)}{4}(s_1/s_2 + s_2/s_1)^2}. \end{aligned} \tag{4.2.4}$$

The transmission loss is

$$\begin{aligned} TL &= 10\log_{10}(1/\tau) \\ &= 10\log_{10}(\cos^2(kl) + \frac{\sin^2(kl)}{4}(s_1/s_2 + s_2/s_1)^2). \end{aligned} \tag{4.2.5}$$

The transmission loss thus depends only on the area ratio, $s_2/s_1 = m$ and frequency through $kl = \omega l/c$. The transmission loss is periodic in kl since the squared sine and cosine functions in Eq. (4.2.4) are periodic with period π. The transmission loss of the expansion chamber muffler is plotted in Fig. 4.2 as a function of kl for several values of the area ratio $m = s_2/s_1$.

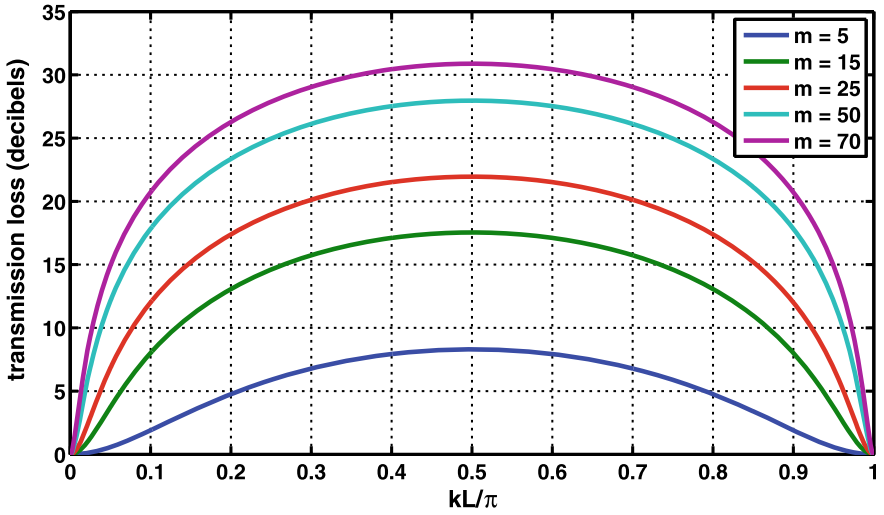


Fig. 4.2 Transmission loss (dB) of an expansion muffler of length l for various values of the ratio of the expansion area to the duct area, $m = s_2/s_1$ as in Eq. (4.2.5). The horizontal axis is the non-dimensional frequency, kL/π, where k is the wave number, the frequency divided by the sound speed, $k = \omega/c$

4.3 The Helmholtz Resonator

Another type of muffler consists of a Helmholtz resonator attached to the duct wall. This provides significant attenuation by changing the effective impedance of the wall of the duct. To analyze this we need to find the impedance of the opening of the resonator.

Figure 4.3 shows a schematic of a Helmholtz resonator. Resonance in this system occurs because the air in the neck of length l and area s acts like a mass and the air in the volume, V, acts like a spring. We will assume that the motion of the air in the neck is $w(t)$, where positive motion is in the direction of the opening, away from the volume, V. A positive pressure, p, on the exterior side of the neck will then result in a force in the negative direction. The equation of motion for the air in the neck is

$$-ps + p_{in}s = \rho_0 s l \ddot{w}, \tag{4.3.1}$$

where p is the pressure on the exterior of the opening, p_{in} is the pressure within the volume, s is the cross-sectional area of the neck, l is the length, and w is the deflection of the air in the neck. The internal pressure, p_{in} is due to the compression of air in the volume. This pressure can be expressed in terms of the deflection w using the equation of state:

$$p_{in} = c^2\rho_1 = \rho_0 c^2 \frac{\rho_1}{\rho_0} = -\rho_0 c^2 \frac{\Delta V}{V}, \tag{4.3.2}$$

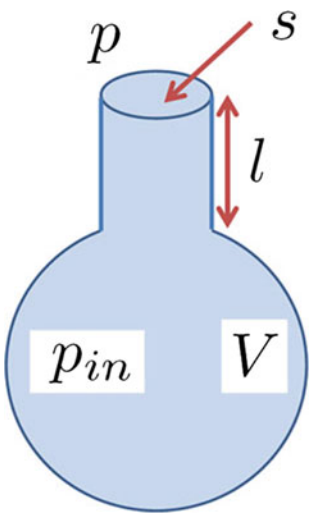


Fig. 4.3 The Helmholtz resonator consists of a volume of air, V with an opening through a duct or neck of length l

where ρ_1 is the fluctuating air density within the volume. The negative sign on the right in Eq. (4.3.2) is due to the fact that the density decreases as the volume increases. The increase in density relative to the ambient density, ρ_1/ρ_0, is equal to the negative of the ratio of the volume of gas flowing in through the opening, ΔV, relative to the total volume, V. The change in volume is equal to the displacement of the air in the opening multiplied by the area, $\Delta V = sw$, where again, we consider the positive direction to be toward the neck opening to the exterior. Equation (4.3.1) then becomes

$$-p = \rho_0 c^2 sw/V + \rho_0 l\ddot{w}. \tag{4.3.3}$$

Assuming harmonic time dependence, the velocity of the air at the opening is $u = i\omega w$ so that Eq. (4.3.3) may be written as

$$-p = \rho_0 c^2 \frac{su}{i\omega V} + \rho_0 li\omega u. \tag{4.3.4}$$

The impedance at the opening is then

$$z = p/u = \frac{i\rho_0 c^2 s}{\omega V} - \rho_0 li\omega. \tag{4.3.5}$$

Again note that we have taken the positive direction for velocity to be in the direction out of the neck, away from the volume, V. In some problems it may be more convenient to use a different sign convention which will require Eq. (4.3.5) to be modified.

The impedance will be zero when

$$\frac{\rho_0 c^2 s}{i\omega V} = -\rho_0 l i\omega. \tag{4.3.6}$$

Solving for ω gives the resonance frequency

$$\omega = \sqrt{\frac{c^2 s}{Vl}}. \tag{4.3.7}$$

4.4 Side Branches

Many mufflers (such as the Helmholtz resonator) consist of an acoustic impedance that is attached to a duct as a branch as illustrated in Fig. 4.4. The branch muffler creates an impedance, z_b, over the area of the junction, s_b. We would like to know the transmission loss due to the branch muffler. This can be calculated if we have a transfer matrix that relates the pressures and velocities across the branch.

The pressures on each side of the branch will be equal, $p_1 = p_2$. The velocities will be related by $su_1 = s_b u_b + su_2$, where s is the cross-sectional area of the duct, s_b is the cross-sectional area of the branch opening, u_1 is the acoustic velocity upstream of the branch and u_2 is the downstream acoustic velocity. The velocity into the branch, u_b is related to the pressure at the branch opening by

$$u_b = \frac{p_b}{z_b} = \frac{p_1}{z_b}, \tag{4.4.1}$$

where p_b is the pressure at the branch inlet and z_b is the acoustic impedance of the branch opening. Equation (4.4.1) permits us to relate the velocities by

$$u_1 = u_2 + \frac{s_b p_1}{s z_b}. \tag{4.4.2}$$

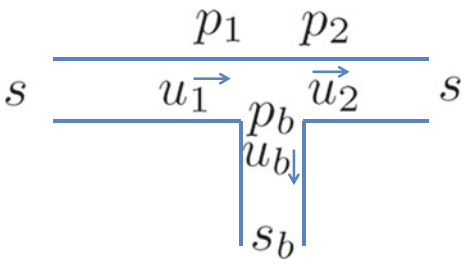


Fig. 4.4 A side branch in a duct

The transfer matrix relation across the branch opening is then

$$\begin{pmatrix} p_2 \\ u_2 \end{pmatrix} = \begin{bmatrix} 1 & 0 \\ -s_b/(sz_b) & 1 \end{bmatrix} \begin{pmatrix} p_1 \\ u_1 \end{pmatrix}. \tag{4.4.3}$$

The ratio of transmitted to incident pressures is

$$\begin{aligned} \frac{p_t}{p_1} &= \frac{2}{T_{11} + T_{22} - T_{12}/(\rho_0 c) - T_{21}\rho_0 c} \\ &= 1/\left(1 + \frac{s_b \rho_0 c}{2sz_b}\right). \end{aligned} \tag{4.4.4}$$

The effectiveness of the muffler increases as the branch impedance, z_b, becomes smaller.

If we consider the branch to consist of a Helmholtz resonator, we can use our previous result in Eq. (4.3.5) to calculate the impedance at the branch opening. We must modify our result slightly in this case because in our derivation of the Helmholtz resonator impedance we assumed that the positive direction for the motion of the air in the neck was **toward** the opening rather than away from the opening as in Fig. 4.4. The branch impedance needed here will then be the negative of the result shown in Eq. 4.3.5.

For a Helmholtz resonator the branch impedance is then

$$z_b = \frac{\rho_0 c^2 s_b}{i\omega V} + \rho_0 L i\omega. \tag{4.4.5}$$

Substituting Eq. (4.4.5) into Eq. (4.4.4) gives

$$\begin{aligned} \frac{p_t}{p_1} &= 1/\left(1 + \frac{s_b \rho_0 c/(2s)}{\frac{\rho_0 c^2 s_b}{i\omega V} + \rho_0 L i\omega}\right) \\ &= 1/\left(1 + \frac{c}{i2s(\omega L/s_b - c^2/(\omega V))}\right). \end{aligned} \tag{4.4.6}$$

The transmission coefficient is

$$\begin{aligned} \tau &= |p_t/p_1|^2 \\ &= 1/\left(1 + \frac{c^2}{4s^2(\omega L/s_b - c^2/(\omega V))^2}\right). \end{aligned} \tag{4.4.7}$$

Note that τ approaches zero at the resonant frequency of the resonator. This causes the transmission loss to become large at that frequency.

4.5 General Side Branch

Suppose we have a more complicated branch. If we have a transfer matrix representation for the branch we can determine the input impedance, $z_b = p_b/u_b$. If p_e and u_e are the pressure and velocity at the far end of the branch the transfer matrix for the branch is

$$\begin{pmatrix} p_e \\ u_e \end{pmatrix} = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \begin{pmatrix} p_b \\ u_b \end{pmatrix}. \tag{4.5.1}$$

To find the input impedance it is convenient to invert equation (4.5.1),

$$\begin{pmatrix} p_b \\ u_b \end{pmatrix} = \begin{bmatrix} T_{22} & -T_{12} \\ -T_{21} & T_{11} \end{bmatrix} \begin{pmatrix} p_e \\ u_e \end{pmatrix}, \tag{4.5.2}$$

where we have assumed that the determinant is unity. Assume that the far end of the side branch is terminated by an impedance, z_t, so that

$$p_e = z_t u_e. \tag{4.5.3}$$

Equation (4.5.2) may be written as

$$\begin{aligned} p_b &= T_{22} p_e - T_{12} u_e, \\ u_b &= -T_{21} p_e + T_{11} u_e. \end{aligned} \tag{4.5.4}$$

Equations (4.5.3) and (4.5.4) give

$$\begin{aligned} z_b = p_b/u_b &= \frac{T_{22} p_e - T_{12} u_e}{-T_{21} p_e + T_{11} u_e} \\ &= \frac{T_{22} z_t - T_{12}}{-T_{21} z_t + T_{11}}. \end{aligned} \tag{4.5.5}$$

4.6 Simple Pipe Side Branch

As an example, consider a side branch that consists of a simple pipe of cross-sectional area, s_b, which is terminated by an impedance, z_t. If the side branch has a length, l, then its transfer matrix is

$$\begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} = \begin{bmatrix} \cos(kl) & -i\rho_0 c \sin(kl) \\ -\frac{i \sin(kl)}{\rho_0 c} & \cos(kl) \end{bmatrix}. \tag{4.6.1}$$

From Eqs. (4.5.5) and (4.6.1) the branch impedance is

$$z_b = \frac{z_t \cos(kl) + i\rho_0 c \sin(kl)}{\cos(kl) + i\frac{z_t}{\rho_0 c}\sin(kl)}. \tag{4.6.2}$$

Equation (4.4.4) may be used with Eq. (4.6.2) to compute p_t/p_1.

Suppose that the end of the side branch is terminated with a zero impedance, $z_t = 0$. Equation (4.6.2) becomes

$$z_b = \frac{i\rho_0 c \sin(kl)}{\cos(kl)}. \tag{4.6.3}$$

Equation (4.4.4) then gives

$$\begin{aligned} p_t/p_1 &= 1/\left(1 + \frac{\rho_0 c s_b \cos(kl)}{2si\rho_0 c \sin(kl)}\right) \\ &= 1/\left(1 - \frac{is_b \cos(kl)}{2s \sin(kl)}\right). \end{aligned} \tag{4.6.4}$$

For small kl, which corresponds to low frequencies and/or short pipes, Eq. (4.6.3) becomes

$$z_b \approx i\rho_0 ckl = i\rho_0 c\frac{\omega}{c}l = i\rho_0 \omega l. \tag{4.6.5}$$

This is just the impedance due to the mass of the air in the pipe. Using the small kl assumption in Eq. (4.6.4) gives

$$p_t/p_1 \approx 1/\left(1 - \frac{is_b c}{2s\omega l}\right). \tag{4.6.6}$$

The transmission coefficient is

$$\tau = |p_t/p_1|^2 = 1/\left(1 + \left(\frac{s_b c}{2s\omega l}\right)^2\right), \tag{4.6.7}$$

and the transmission loss is

$$TL = 10\log_{10}(1/\tau) = 10\log_{10}\left(1 + \left(\frac{s_b c}{2s\omega l}\right)^2\right). \tag{4.6.8}$$

This muffler essentially acts like a high-pass filter.

4.7 Sound in Tubes of Varying Cross-Sectional Area

There are a number of practical situations where sound propagates in a tube or duct that has a cross-sectional area that varies smoothly along its length. We have already considered the transfer matrix for an abrupt change in cross-sectional area but we now need to consider how to describe the sound field in ducts having smoothly varying cross-sectional area, $A(x)$. If the domain has uniform area, the partial differential equation governing the acoustic pressure, $p(x,t)$, in a one-dimensional sound field is given by

$$\frac{\partial^2 p}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} \tag{4.7.1}$$

where c is the speed of propagation of the sound wave. We will consider sound in tubes where the wavelength of the sound is much larger than the cross-sectional dimensions. If the cross-sectional area A varies with x, we must revisit the three fundamental equations used to derive the wave equation and include the spatial dependence on $A(x)$.

The first equation relates fluctuations in mass of a small volume to the flow of mass into and out of the volume. Considering a volume having length dx and area A, the rate of change of mass is then

$$\frac{d}{dt}(\rho A dx) = \\ u(x,t)\rho(x,t)A(x) - u(x+dx,t)\rho(x+dx,t)A(x+dx) \tag{4.7.2}$$

where, $u(x,t)$ is the acoustic particle velocity, and $\rho(x,t)$ is the fluctuating density. This may be rearranged to give

$$\dot{\rho}(x,t) = -\frac{\partial(u\rho A)}{A\partial x} \tag{4.7.3}$$

The general expression for the Euler equation (Newton's second law) in multiple dimensions is

$$\vec{\nabla} p = -\frac{d}{dt}(\rho\vec{u}) \tag{4.7.4}$$

The Euler equation must be satisfied at any point in space and at any surface. Multiplying Eq. (4.7.4) by A and noting that the pressure varies only with one spatial coordinate, x, give

$$-A\frac{\partial p}{\partial x} = \frac{d}{dt}(A\rho u) \tag{4.7.5}$$

The equation of state remains unchanged from what we've had previously,

$$p = \rho_a c^2 \tag{4.7.6}$$

where ρ_a is the acoustic fluctuation in density about the nominal value ρ_0.

Substituting Eq. (4.7.6) into Eq. (4.7.3) and rearranging give

$$\frac{A\dot{p}}{c^2} = -\frac{\partial(u\rho A)}{\partial x} \tag{4.7.7}$$

Differentiating Eq. (4.7.5) with respect to x, and differentiating Eq. (4.7.7) with respect to t give

$$\frac{\partial^2(u\rho A)}{\partial x \partial t} = -\frac{\partial}{\partial x}\left(A\frac{\partial p}{\partial x}\right) = -\frac{A\ddot{p}}{c^2} \tag{4.7.8}$$

This gives the Webster horn equation,

$$\frac{\partial}{\partial x}\left(A\frac{\partial p}{\partial x}\right) = \frac{A\ddot{p}}{c^2} \tag{4.7.9}$$

If we limit our attention to harmonic sound waves at the frequency ω, the solution can be expressed as

$$p(x,t) = P(x)e^{i\omega t} \tag{4.7.10}$$

where ω, is the frequency in radians/second and $i = \sqrt{-1}$. Equation (4.7.9) may then be written as

$$\frac{d}{dx}\left(A\frac{dP}{dx}\right) = A_x P_x + A P_{xx} = -k^2 P A \tag{4.7.11}$$

where the subscript x denotes differentiation with respect to x and $k = \omega/c$.

Equation (4.7.11) may be written as

$$\frac{A_x}{A} = -\frac{k^2 P + P_{xx}}{P_x} \tag{4.7.12}$$

From Eq. (4.7.12), note that if the cross sectional area is effectively constant, $k^2 P \approx -P_{xx}$. In addition, if the quantities on the right hand side of Eq. (4.7.12) can be measured with reasonable accuracy, they can be used to estimate A_x/A.

4.7.1 Conical Horn

Obtaining solutions to Eq. (4.7.9) can be difficult if $A(x)$ is a complicated function. For certain common special cases, however, it is not too difficult to obtain solutions. The most important case is where $A(x)$ describes a cone so that

$$A(x) = (a + bx)^2 \tag{4.7.13}$$

where a and b are constants. To obtain solutions to Eq. (4.7.9), it is helpful to make the substitution

$$P' = P\sqrt{A} \tag{4.7.14}$$

Substituting this into Eq. (4.7.11) or (4.7.12) gives

$$P'_{xx} + \frac{P'}{A^2}\left(\frac{A_x^2}{4} - \frac{AA_{xx}}{2}\right) = -k^2 P' \tag{4.7.15}$$

where we have again assumed harmonic time dependence. Substituting Eq. (4.7.13) into Eq. (4.7.15) gives

$$\frac{A_x^2}{4} = \frac{AA_{xx}}{2} \tag{4.7.16}$$

so that Eq. (4.7.15) simplifies to

$$P'_{xx} = -k^2 P' \tag{4.7.17}$$

which has the solution

$$P'(x) = P_1 e^{-ikx} + P_2 e^{ikx} \tag{4.7.18}$$

where P_1 and P_2 depend on the boundary conditions. Equations (4.7.14) and (4.7.18) may then be used to write the solution of Eq. (4.7.9) for the pressure as

$$p(x,t) = P(x)e^{i\omega t} = \frac{e^{i\omega t}}{\sqrt{A}}(P_1 e^{-ikx} + P_2 e^{ikx}) \tag{4.7.19}$$

We may now proceed to construct a transfer matrix for sound propagating in a conical duct. This can be accomplished by using our solution (4.7.19) along with the Euler equation to write the pressure and velocity at the two ends, say $x = 0$ and $x = l$. The Euler equation may be used to evaluate $u(x,t) = U(x)e^{i\omega t}$,

$$U(x) = \frac{i}{\rho_0 \omega}\frac{\partial P}{\partial x} \tag{4.7.20}$$

Using (4.7.19) for $P(x)$ gives

$$U(x) = \frac{i}{\rho_0 \omega}\left(\frac{-A_x A^{-3/2}}{2} - ikA^{-1/2}\right) P_1 e^{-ikx} + \frac{i}{\rho_0 \omega}\left(\frac{-A_x A^{-3/2}}{2} + ikA^{-1/2}\right) P_2 e^{ikx} \tag{4.7.21}$$

To simplify things, let

$$\alpha(x) = \frac{-A_x A^{-3/2}}{2} - ikA^{-1/2} \tag{4.7.22}$$

Evaluating equations (4.7.19) and (4.7.21) at $x = 0$ and at $x = l$ then gives

$$\begin{pmatrix} P(0) \\ U(0) \end{pmatrix} = \begin{bmatrix} \frac{1}{\sqrt{A(0)}} & \frac{1}{\sqrt{A(0)}} \\ \frac{i\alpha(0)}{\rho_0 \omega} & \frac{i\alpha^*(0)}{\rho_0 \omega} \end{bmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = [B_0] \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \tag{4.7.23}$$

where the superscript * denotes the complex conjugate, and

$$\begin{pmatrix} P(l) \\ U(l) \end{pmatrix} = \begin{bmatrix} \frac{e^{-ikl}}{\sqrt{A(l)}} & \frac{e^{ikl}}{\sqrt{A(l)}} \\ \frac{i\alpha(l)}{\rho_0 \omega} e^{-ikl} & \frac{i\alpha^*(l)}{\rho_0 \omega} e^{ikl} \end{bmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = [B_l] \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \tag{4.7.24}$$

Equations (4.7.23) and (4.7.24) may be used to eliminate P_1 and P_2 to obtain the desired transfer matrix, $[T] = [B_l][B_0]^{-1}$,

$$\begin{pmatrix} P(l) \\ U(l) \end{pmatrix} = [B_l][B_0]^{-1} \begin{pmatrix} P(0) \\ U(0) \end{pmatrix} = [T] \begin{pmatrix} P(0) \\ U(0) \end{pmatrix} \tag{4.7.25}$$

4.7.2 Assemble Conical Elements to Model a Tube with Arbitrary Cross-Sectional Area

Equation (4.7.25) provides a means of obtaining the transfer matrix for a conical tube spanning the distance from $x = 0$ to $x = l$. This matrix can be used to assemble short segments together to approximate a rather general duct having a nearly arbitrary variation in cross-sectional area along its length. Having the pressure and velocity at one end of the tube, the pressure and velocity at the other end can be calculated by simply multiplying the element transfer matrices together.

Let the total length of the duct be divided up into n segments along the x-axis with each segment occupying the domain $x_i \le x \le x_{i+1}$, for $i = 1, 2, \ldots, n$. For

convenience, we will assume that each segment has equal length, $dl = x_{i+1} - x_i$. The duct cross-sectional area at $x = x_i$ will be A_i, and its derivative with respect to x will be A_{xi}. We will similarly evaluate Eq. (4.7.22) at each x_i and store $\alpha(x_i) = \alpha_i$. Following Eqs. (4.7.22) through (4.7.25), we may construct the element transfer matrices, $[T]_i$ by defining

$$[B]_i = \begin{bmatrix} \frac{e^{-ikx_i}}{\sqrt{A_i}} & \frac{e^{ikx_i}}{\sqrt{A_i}} \\ \frac{i\alpha_i}{\rho_0\omega} e^{-ikx_i} & \frac{i\alpha_i^*}{\rho_0\omega} e^{ikx_i} \end{bmatrix} \tag{4.7.26}$$

and

$$[T]_i = [B]_i [B]_{i-1}^{-1} \tag{4.7.27}$$

for $i = 1, 2, \ldots, n$ and $[B_0]$ is a 2×2 identity matrix. The pressures and velocities at the end of each segment are related by

$$\begin{pmatrix} P(x_{i+1}) \\ U(x_{i+1}) \end{pmatrix} = \begin{pmatrix} P_{i+1} \\ U_{i+1} \end{pmatrix} = [B]_i [B]_{i-1}^{-1} \begin{pmatrix} P(x_i) \\ U(x_i) \end{pmatrix} = [T]_i \begin{pmatrix} P_i \\ U_i \end{pmatrix}. \tag{4.7.28}$$

The process described above can be used to assemble a composite transfer matrix for any number of conical sections, which could be employed to represent a duct with varying cross-sectional area. If our primary interest is in the relation between the pressures and velocities at the input and output, we can eliminate all of the intermediate pressures and velocities to obtain a single composite transfer matrix,

$$\begin{pmatrix} P(x_n) \\ U(x_n) \end{pmatrix} = \begin{pmatrix} P_n \\ U_n \end{pmatrix} = \left(\prod_{i=n}^{1} [T]_i\right) \begin{pmatrix} P(x_1) \\ U(x_1) \end{pmatrix} = [T]_c \begin{pmatrix} P_1 \\ U_1 \end{pmatrix}, \tag{4.7.29}$$

where $[T]_c$ is the composite transfer matrix for the tube.

4.7.3 *Solving for the Pressure and Velocity Within the Domain*

While the transfer matrix approach described above provides an efficient way to eliminate coordinates that are not needed, there are situations where those coordinates are desired. If each transfer matrix is known, we can assemble the collection of transfer matrices into a global system matrix and solve for all of the desired unknowns. To accomplish this, let the elements of the ith transfer matrix be given by

$$[T]_i = \begin{bmatrix} T_{11}^i & T_{12}^i \\ T_{21}^i & T_{22}^i \end{bmatrix}. \tag{4.7.30}$$

As an example, suppose we have a system that is described by two transfer matrices so that

$$\begin{pmatrix} P_2 \\ U_2 \end{pmatrix} = \begin{bmatrix} T_{11}^1 & T_{12}^1 \\ T_{21}^1 & T_{22}^1 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \end{pmatrix} \tag{4.7.31}$$

and

$$\begin{pmatrix} P_3 \\ U_3 \end{pmatrix} = \begin{bmatrix} T_{11}^2 & T_{12}^2 \\ T_{21}^2 & T_{22}^2 \end{bmatrix} \begin{pmatrix} P_2 \\ U_2 \end{pmatrix}. \tag{4.7.32}$$

Equations (4.7.31) and (4.7.32) may be written in matrix form as

$$\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} T_{11}^1 & T_{12}^1 & -1 & 0 & 0 & 0 \\ T_{21}^1 & T_{22}^1 & 0 & -1 & 0 & 0 \\ 0 & 0 & T_{11}^2 & T_{12}^2 & -1 & 0 \\ 0 & 0 & T_{21}^2 & T_{22}^2 & 0 & -1 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \\ P_2 \\ U_2 \\ P_3 \\ U_3 \end{pmatrix}. \tag{4.7.33}$$

The four Eqs. (4.7.31) and (4.7.32) contain a total of six unknowns, three pressures, and three velocities. Unique solutions may be obtained only if we have two additional equations. These additional equations may specify the conditions at the input of the system (P_1 and U_1) and the output ((P_3 and U_3)). We may specify that the input consists of the velocity at location 1 being prescribed, $U_1 = U_{in}$ and the impedance at the output to be $Z_3 = P_3/U_3$. These two additional conditions along with Eq. (4.7.33) may be expressed as

$$\begin{bmatrix} T_{11}^1 & T_{12}^1 & -1 & 0 & 0 & 0 \\ T_{21}^1 & T_{22}^1 & 0 & -1 & 0 & 0 \\ 0 & 0 & T_{11}^2 & T_{12}^2 & -1 & 0 \\ 0 & 0 & T_{21}^2 & T_{22}^2 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & -Z_3 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \\ P_2 \\ U_2 \\ P_3 \\ U_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ u_1 \end{pmatrix}. \tag{4.7.34}$$

This may be solved for a unique, non-trivial solution for all six of the desired unknowns. The approach described above can easily be extended to systems having any number of transfer matrices and intermediate coordinates.

Alternatively, we may have the pressure at the output prescribed so that $P_3 = p_3$ and the impedance at the input prescribed, $Z_1 = P_1/U_1$. In this case, we obtain the following equations which may be solved for the unknown pressures and velocities

$$\begin{bmatrix} T_{11}^1 & T_{12}^1 & -1 & 0 & 0 & 0 \\ T_{21}^1 & T_{22}^1 & 0 & -1 & 0 & 0 \\ 0 & 0 & T_{11}^2 & T_{12}^2 & -1 & 0 \\ 0 & 0 & T_{21}^2 & T_{22}^2 & 0 & -1 \\ 1 & -Z_1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \\ P_2 \\ U_2 \\ P_3 \\ U_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ p_3 \end{pmatrix}. \tag{4.7.35}$$

Another useful case is where the pressures are imposed at each end so that $P_1 = p_1$ and $P_3 = p_3$ are known. Equation (4.7.35) then becomes

$$\begin{bmatrix} T_{11}^1 & T_{12}^1 & -1 & 0 & 0 & 0 \\ T_{21}^1 & T_{22}^1 & 0 & -1 & 0 & 0 \\ 0 & 0 & T_{11}^2 & T_{12}^2 & -1 & 0 \\ 0 & 0 & T_{21}^2 & T_{22}^2 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} P_1 \\ U_1 \\ P_2 \\ U_2 \\ P_3 \\ U_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ p_1 \\ p_3 \end{pmatrix}. \tag{4.7.36}$$

4.7.4 The Exponential Horn

Along with the cone shape examined above, another very important case to consider, and which permits closed-form solutions, is where the area of the duct increases exponentially with the axial coordinate, x. This is often used in horns. Let the spatial dependence of the area be expressed as

$$A(x) = A_0 e^{mx}, \tag{4.7.37}$$

where A_0 and m are given constants. Substituting Eq. (4.7.37) into Eq. (4.7.4) gives

$$A_x P_x + A P_{xx} = mAP_x + AP_{xx} = -k^2 PA \tag{4.7.38}$$

which gives a simple second-order ordinary differential equation for $P(x)$,

$$mP_x + P_{xx} + k^2 P = 0. \tag{4.7.39}$$

A solution for $P(x)$ is $P(x) = P_0 e^{\beta x}$, where P_0 and β are constants. Substituting this into Eq. (4.7.39) gives

$$m\beta + \beta^2 + k^2 = 0. \tag{4.7.40}$$

The two solutions for β are

$$\beta = -\frac{m}{2} \pm i\sqrt{k^2 - (m/2)^2}. \tag{4.7.41}$$

Let

$$\beta_1 = -\frac{m}{2} - i\sqrt{k^2 - (m/2)^2} \tag{4.7.42}$$

and

$$\beta_2 = -\frac{m}{2} + i\sqrt{k^2 - (m/2)^2}. \tag{4.7.43}$$

The solution to the wave equation in the horn is then

$$p(x,t) = e^{i\omega t}P(x) = e^{i\omega t}(P_1 e^{\beta_1 x} + P_2 e^{\beta_2 x}). \tag{4.7.44}$$

Note that if $k = \omega/c < m/2$, there is no wave propagation in the horn. The minimum frequency for waves to propagate

$$\omega_c = cm/2 \tag{4.7.45}$$

is called the horn cut-off frequency.

Having our solution for the pressure in Eq. (4.7.44), we may again construct a transfer matrix for the exponential horn. Proceeding as before, the Euler equation and Eq. (4.7.44) give

$$U(x) = \frac{i}{\rho_0 \omega}(\beta_1 P_1 e^{\beta_1 x} + \beta_2 P_2 e^{\beta_2 x}). \tag{4.7.46}$$

We may again evaluate $P(x)$ and $U(x)$ at $x = 0$ and $x = l$ to eliminate P_1 and P_2

$$\begin{pmatrix} P(0) \\ U(0) \end{pmatrix} = \begin{bmatrix} 1 & 1 \\ \frac{i}{\rho_0\omega}\beta_1 & \frac{i}{\rho_0\omega}\beta_2 \end{bmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = [C_0] \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} \tag{4.7.47}$$

and

$$\begin{pmatrix} P(l) \\ U(l) \end{pmatrix} = \begin{bmatrix} e^{\beta_1 l} & e^{\beta_2 l} \\ \frac{i}{\rho_0\omega}\beta_1 e^{\beta_1 l} & \frac{i}{\rho_0\omega}\beta_2 e^{\beta_2 l} \end{bmatrix} \begin{pmatrix} P_1 \\ P_2 \end{pmatrix} = [C_l] \begin{pmatrix} P_1 \\ P_2 \end{pmatrix}. \tag{4.7.48}$$

Eliminating P_1 and P_2 gives the desired transfer matrix, $[T] = [C_l][C_0]^{-1}$

$$\begin{pmatrix} P(l) \\ U(l) \end{pmatrix} = [C_l][C_0]^{-1} \begin{pmatrix} P(0) \\ U(0) \end{pmatrix} = [T] \begin{pmatrix} P(0) \\ U(0) \end{pmatrix}. \tag{4.7.49}$$

It is often important to determine the acoustic impedance at the input of the horn. This is a useful measure of the horn's efficiency. If the impedance at the output end is known, Z_l, then $P(l)/U(l) = Z_l$. Inverting Eq. (4.7.49) gives

$$\begin{pmatrix} P(0) \\ U(0) \end{pmatrix} = [T]^{-1} \begin{pmatrix} P(l) \\ U(l) \end{pmatrix}. \tag{4.7.50}$$

Denote the elements of $[T]^{-1}$ as

$$[T]^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}. \tag{4.7.51}$$

Equations (4.7.50) and (4.7.51) then give the impedance at the input

$$\frac{P(0)}{U(0)} = \frac{aP(l) + bU(l)}{cP(l) + dU(l)} = \frac{aZ_l + b}{cZ_l + d} = Z_0. \tag{4.7.52}$$

It can also be convenient to normalize the pressures and velocities in Eq. (4.7.49) by the velocity at the input, $U(0)$

$$\begin{pmatrix} P(l)/U(0) \\ U(l)/U(0) \end{pmatrix} = [T] \begin{pmatrix} P(0)/U(0) \\ U(0)/U(0) \end{pmatrix} = [T] \begin{pmatrix} Z_0 \\ 1 \end{pmatrix}. \tag{4.7.53}$$

4.7.5 Numerical Example for a Non-uniform Duct

The results given above for the sound field in non-uniform ducts may be used to examine sound propagation in a wide range of ducts having nearly arbitrary variation in cross-sectional area. As an example, suppose we have a duct having a length of $l = 0.2$ meters and radius at $x = 0$ of 1 cm, giving a cross-sectional area $A(0) = A_0 = \pi(0.01)^2$ square meter. We will assume the cross-sectional area varies as an exponential horn such that the area at $x = l$ is twice the initial area, $A(l) = 2\pi(0.01)^2$ square meter. Equation (4.7.26) then gives

$$m = \log_e(A(l))/\log_e(A_0)/l, \tag{4.7.54}$$

where $\log_e$ is the natural logarithm. For the parameters given above, Eq. (4.7.54) gives $m = 3.4657$/meter. A transfer matrix model can be constructed for this duct using the solution given in Eq. (4.7.49), which we could consider an "exact" solution obtained by solving the differential equation Eq. (4.7.39). An approximate model can also be constructed by cascading a series of cones having cross-sectional areas that match the distribution of areas of the actual exponential horn. This "approximate" model is constructed based on Eq. (4.7.29) where a total of 100 cones are used in the calculations. Models for the sound field in the duct based on these two approaches may be compared by examining the input impedance, $P(0)/U(0) = Z_0$, as given in Eq. (4.7.52). Figure 4.5 shows the impedance at the entrance of the duct obtained using these two methods. The figure shows that the methods yield very similar results. The numerical approach may be used for ducts for which closed-form solutions are not feasible (Fig. 4.5).

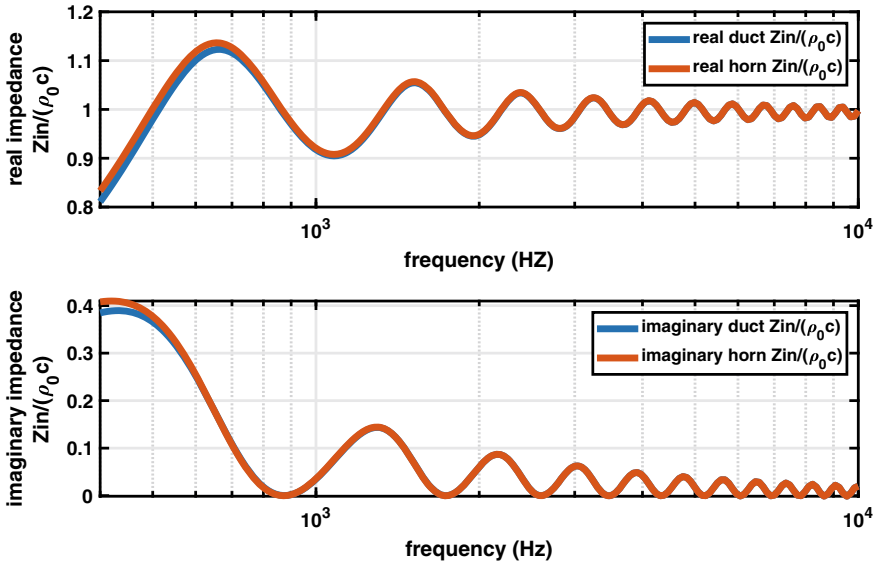


Fig. 4.5 Impedance at the entrance of a duct having a cross-sectional area that varies exponentially with position along the duct length. The impedance predicted using the "exact" solution of Eq. (4.7.39) is in close agreement with the approximate method of Eq. (4.7.29)

4.8 Problems

1. A duct has a cross-sectional area of 0.5 m^2. Design a simple expansion muffler that has the highest possible transmission loss over a range of frequencies from 1000 to 1100 Hz and that fits in a space less than or equal to 2 m^2 in cross-sectional area and is no more than 1 m long. Plot the predicted transmission loss of your design over a range of frequencies from 20 to 10000 Hz. Use a logarithmic frequency axis. In this and all of the following problems describe all equations **and** include an annotated listing of any computer programs used.
2. A duct has a cross-sectional area of 0.5 m^2. Design a side-branch muffler that attenuates sound at 600 Hz. The side branch can consist of either a straight section of pipe or a Helmholtz resonator. Plot the predicted transmission loss of your design over a range of frequencies from 20 to 2000 Hz. Use a logarithmic frequency axis.
3. A noise source creates a sound wave that travels in a circular ventilation duct having a diameter of 0.5 m. Design a muffler that fits in a space that is 1 m long and 1 m wide that can provide as much attenuation as possible over a frequency range of 50–1000 Hz. You must use at least one expansion chamber muffler and at least one side branch muffler in your design. The entrance and exit diameters of your muffler must each be 0.5 m. Write a program to create a composite transfer matrix model of the system and use that composite transfer matrix to compute

the transmission loss. Provide a plot of the transmission loss in decibels versus frequency in Hertz for your design using a logarithmic frequency axis. Include a detailed write-up describing your design and the equations used to carry out your design along with an annotated listing of any computer programs written.

4. For the duct of problem 3, design a muffler that incorporates at least five expansion chamber mufflers that can provide as much attenuation as possible over a frequency range of 50–1000 Hz. The expansion chamber mufflers should be tuned to different frequencies to improve the attenuation across frequency. Write a program to create a composite transfer matrix model of the system and use that composite transfer matrix to compute the transmission loss. Provide a plot of the transmission loss in decibels versus frequency in Hertz for your design using a logarithmic frequency axis. Include a detailed write-up describing your design and the equations used to carry out your design along with an annotated listing of any computer programs written.
5. A duct has a cross-sectional area of $0.5\,\mathrm{m}^2$. The duct is joined to a cone having area $0.5\,\mathrm{m}^2$ at $x = 0$ and area $2\,\mathrm{m}^2$ at $x = l = 0.3$ m. The duct then is joined to a section of pipe having the original area of $0.5\,\mathrm{m}^2$. This joint is completely sealed as in the junctions in an expansion chamber muffler. Find the transmission loss of this system. Plot your results over a range of frequencies from 20 to 10000 Hz. Use a logarithmic frequency axis.
6. A cone has a length $L = 0.2$ m and area $A(0) = 0.5\ \mathrm{m}^2$ at $x = 0$ and $A(L) = 2.0\ \mathrm{m}^2$ at $x = L$. If the large end terminates with an impedance $\rho_0 c = 415 = P(L)/U(L)$ mks rayls, find the input impedance of the cone, i.e. $P(0)/U(0)$. Plot the real and imaginary parts of this input impedance versus frequency for a range of frequencies from 20 Hz to 20,000 Hz. Use a logarithmic frequency axis. To do this calculation, you should first construct the transfer matrix that relates pressure and velocity at $x = 0$ to that at $x = L$.
7. A duct of length L has a cross-sectional area that varies with the position x according to

$$S(x) = a_0 + a_1 \frac{x}{l} + a_2 \left(\frac{x}{l}\right)^2 + a_3 \left(\frac{x}{l}\right)^3, \tag{4.8.1}$$

where a_0, a_1, a_2, and a_3 are constants. The areas at $x = 0$ and $x = l$ are given as $S(0) = S_0 = S(l) = S_l = 0.5\ \mathrm{m}^2$. (a) Use these two conditions on the area to eliminate a_0 and a_3 so that the only unknowns are a_1 and a_2. (b) Determine values (i.e., design) of a_1 and a_2 to maximize the input impedance of the duct at $x = 0$. Assume that the impedance at $x = l$ is $Z = \rho c$.

Reference for Chapter 4

1. Munjal ML (2014) Acoustics of ducts and mufflers. Wiley, New York

Chapter 5
Sound Radiation in Three Dimensions



In the following we will examine the radiation of sound in three dimensions. We will first write the basic equations for the sound field in three dimensions and then consider radiation from a simple point source of sound. We will then extend our study of the radiation from a simple point source to explore sound radiation and reflection in a wide range of acoustic spaces.

5.1 Wave Equation in Three Dimensions

The acoustic wave equation for pressure in three dimensions is

$$\nabla^2 p = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}, \tag{5.1.1}$$

where $p(x, y, z, t)$ is the sound pressure and c is the sound speed. The Euler equation in three dimensions is

$$-\overrightarrow{\nabla p} = \rho_0 \dot{\overrightarrow{u}}, \tag{5.1.2}$$

where ρ_0 is the nominal air density and $\overrightarrow{u}(x, y, z, t)$ is the acoustic particle velocity vector.

In a Cartesian (x, y, z) coordinate system having unit vectors $\overrightarrow{i}$, $\overrightarrow{j}$, and $\overrightarrow{k}$, the gradient of a function, $f(x, y, z)$ is

$$\overrightarrow{\nabla f} = \frac{\partial f}{\partial x}\overrightarrow{i} + \frac{\partial f}{\partial y}\overrightarrow{j} + \frac{\partial f}{\partial z}\overrightarrow{k}, \tag{5.1.3}$$

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_5.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_5

and consequently, $\nabla^2 f = \nabla \cdot \nabla f$, or,

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}. \tag{5.1.4}$$

In this coordinate system it is straightforward to represent **plane** waves in the form

$$p(x, y, z, t) = P_0 e^{i\omega t - i(k_x x + k_y y + k_z z)}. \tag{5.1.5}$$

Another very important type of acoustic pressure fluctuation is the spherical wave. To examine spherical waves it is helpful to define a spherical coordinate system having unit vectors $\vec{r}$, $\vec{\theta}$, and $\vec{\phi}$. In terms of $\vec{r}$, $\vec{\theta}$, and $\vec{\phi}$, the ∇ operator acting on a function $f(r, \theta, \phi)$ becomes

$$\vec{\nabla f} = \frac{\partial f}{\partial r}\vec{r} + \frac{1}{r}\frac{\partial f}{\partial \theta}\vec{\theta} + \frac{1}{r\sin(\theta)}\frac{\partial f}{\partial \phi}\vec{\phi}, \tag{5.1.6}$$

and

$$\nabla^2 f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2\sin(\theta)}\frac{\partial}{\partial \theta}\left(\sin(\theta)\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2(\theta)}\frac{\partial^2 f}{\partial \phi^2}. \tag{5.1.7}$$

If we are interested in spherically symmetric waves then p is independent of θ and ϕ. $\nabla^2 p$ is then

$$\nabla^2 p = \frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r}, \tag{5.1.8}$$

and

$$\vec{\nabla p} = \frac{\partial p}{\partial r}\vec{r}. \tag{5.1.9}$$

Equations (5.1.1) and (5.1.2) then become

$$\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2}, \tag{5.1.10}$$

and

$$-\frac{\partial p}{\partial r} = \rho_0 \dot{u}_r, \tag{5.1.11}$$

where u_r is the $\overrightarrow{r}$ component of the acoustic particle velocity vector $\overrightarrow{u}$. Equation (5.1.10) may be written as

$$\frac{\partial^2 p}{\partial r^2} + \frac{2}{r}\frac{\partial p}{\partial r} = \frac{1}{r}\frac{\partial^2 (rp)}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2 p}{\partial t^2} \tag{5.1.12}$$

or, after multiplying by r,

$$\frac{\partial^2 (rp)}{\partial r^2} = \frac{1}{c^2}\frac{\partial^2 (rp)}{\partial t^2} \tag{5.1.13}$$

Equation (5.1.13) has the same form as our wave equation in one dimension in Eq. (2.6.6) with p replaced by rp. The general solution to Eq. (5.1.13) is then

$$rp(r,t) = f(r-ct) + g(r+ct), \tag{5.1.14}$$

where f and g are arbitrary functions. Dividing Eq. (5.1.14) by r gives

$$p(r,t) = \frac{f(r-ct)}{r} + \frac{g(r+ct)}{r}, \tag{5.1.15}$$

Acoustic Impedance of Harmonic Waves

Consider a harmonic wave traveling in the positive $\overrightarrow{r}$ direction,

$$p(r,t) = \frac{D_0}{r} e^{i(\omega t - kr)}, \tag{5.1.16}$$

where D_0 is a constant that determines the magnitude of the pressure at a particular location r. Equation (5.1.16) can be shown to be a solution by substituting into Eq. (5.1.13). It is found to be a solution if $k = \omega/c$, as before.

To determine the impedance we need the acoustic particle velocity, which may be obtained from the Euler equation, Eq. (5.1.11),

$$-\frac{\partial}{\partial r}\frac{D_0}{r} e^{i(\omega t - kr)} = \rho_0 i\omega u_r, \tag{5.1.17}$$

where we have assumed that the time dependence of $u_r(r,t)$ is $e^{i\omega t}$. Differentiating the left side of Eq. (5.1.17) gives

$$\begin{aligned} -\frac{\partial}{\partial r}\frac{D_0}{r} e^{i(\omega t - kr)} &= -D_0(-ikr - 1)\frac{e^{i(\omega t - kr)}}{r^2} \\ &= \frac{D_0(1+ikr)e^{i(\omega t - kr)}}{r^2} = \rho_0 i\omega u_r, \end{aligned} \tag{5.1.18}$$

Equation (5.1.18) gives

$$u_r(r,t) = \frac{D_0(1+ikr)}{\rho_0 i\omega r^2} e^{i(\omega t - kr)}, \tag{5.1.19}$$

The acoustic impedance is

$$\begin{aligned} Z &= \frac{p(r,t)}{u_r(r,t)} = \frac{\frac{D_0}{r}e^{i(\omega t-kr)}}{\frac{D_0(1+ikr)}{\rho_0 i\omega r^2}e^{i(\omega t-kr)}} \\ &= \frac{\rho_0 cikr}{1+ikr} = \frac{\rho_0 c((kr)^2+ikr)}{1+(kr)^2}. \end{aligned} \tag{5.1.20}$$

5.2 Sound Intensity of Spherical Waves

The time-averaged acoustic intensity is given by

$$< I >= \frac{1}{T}\int_0^T P(r,t)U(r,t)dt, \tag{5.2.1}$$

where T is the period of one cycle, $T = 2\pi/\omega$, and $P(r,t)$ and $U(r,t)$ are real functions representing the pressure and velocity. Let

$$\begin{aligned} P(r,t) &= \Re\left[\frac{D_0}{r}e^{i(\omega t-kr)}\right] = \Re[\overline{P}e^{i\omega t}] \\ U(r,t) &= \Re\left[\frac{D_0(1+ikr)}{\rho_0 i\omega r^2}e^{i(\omega t-kr)}\right] = \Re[\overline{U}e^{i\omega t}]. \end{aligned} \tag{5.2.2}$$

where $\Re$ denotes the real part. In arriving at Eq. (2.5.5) we showed that

$$< I >= \frac{1}{2}\Re[\overline{P}\overline{U}^*] = \frac{1}{2}\Re[\overline{P}^*\overline{U}]. \tag{5.2.3}$$

Since

$$Z = \frac{\overline{P}}{\overline{U}} \tag{5.2.4}$$

the intensity may also be written as

$$< I >= \frac{1}{2}|\overline{P}|^2\Re\left[\frac{1}{Z^*}\right] = \frac{1}{2}|\overline{U}|^2\Re[Z^*]. \tag{5.2.5}$$

The squared magnitude of the pressure is

$$|\overline{P}|^2 = \frac{D_0}{r}e^{i(\omega t-kr)} \times \frac{D_0^*}{r}e^{-i(\omega t-kr)} = \frac{|D_0|^2}{r^2}, \tag{5.2.6}$$

and

$$\Re\left[\frac{1}{Z^*}\right] = \Re\left[\frac{1-ikr}{-\rho_0 cikr}\right] = \Re\left[\frac{1+i/(kr)}{\rho_0 c}\right] = \frac{1}{\rho_0 c}. \tag{5.2.7}$$

Using the first of Eq. (5.2.5) and Eqs. (5.2.6) and (5.2.7), the sound intensity of a spherical wave is

$$< I >= \frac{1}{2}|\overline{P}|^2\Re\left[\frac{1}{Z^*}\right] = \frac{1}{2\rho_0 c}\frac{|D_0|^2}{r^2}. \tag{5.2.8}$$

Note that the mean square pressure is

$$P_{rms}^2 = \frac{1}{2}|\overline{P}|^2 = \frac{1}{2}\frac{|D_0|^2}{r^2}, \tag{5.2.9}$$

so that the intensity is

$$< I >= \frac{P_{rms}^2}{\rho_0 c}. \tag{5.2.10}$$

5.3 Sound Radiation by a Simple Source

The simplest source of spherical sound waves is the pulsating sphere. Consider a sphere of radius a with a surface velocity in the radial direction given by

$$u(a,t) = u_s e^{i\omega t}. \tag{5.3.1}$$

We wrote the sound pressure for a spherical wave in Eq. (5.1.16),

$$p(r,t) = \frac{D_0}{r}e^{i(\omega t-kr)}. \tag{5.3.2}$$

Equation (5.1.19) gives the acoustic particle velocity of a spherical wave,

$$u_r(r,t) = \frac{D_0(1+ikr)}{\rho_0 i\omega r^2}e^{i(\omega t-kr)}. \tag{5.3.3}$$

Evaluating Eq. (5.3.3) at $r = a$ along with Eq. (5.3.1) enable us to determine D_0 for this pulsating sphere,

$$D_0 = u_s \frac{\rho_0 i\omega a^2}{(1+ika)} e^{ika}. \tag{5.3.4}$$

The pressure at a point r is then

$$p(r,t) = \frac{D_0}{r} e^{i(\omega t - kr)} = \frac{\rho_0 cika^2}{(1+ika)} \frac{u_s}{r} e^{i(\omega t - k(r-a))}. \tag{5.3.5}$$

It should be noted that this expression assumes that we are sufficiently far from the sphere's surface that the sound field is indeed described by a spherical wave. If the radius a is not small relative to the wavelength and if the distance r is of similar magnitude to a, the field will be much more complicated than that considered here. In this case, the sound radiated from different locations on the sphere may arrive at a point out of phase. This will result in dips in the pressure at certain frequencies.

The acoustic impedance is the ratio of the pressure to the velocity,

$$\begin{aligned} Z &= \frac{p(r,t)}{u_r(r,t)} = \frac{\frac{D_0}{r} e^{i(\omega t - kr)}}{\frac{D_0(1+ikr)}{\rho_0 i\omega r^2} e^{i(\omega t - kr)}} \\ &= \frac{\rho_0 cikr}{1+ikr} = \frac{\rho_0 c((kr)^2 + ikr)}{1+(kr)^2}. \end{aligned} \tag{5.3.6}$$

We would like to determine the acoustic power radiated by the sphere and the resulting acoustic pressure. The radiated power is easily determined if we know the sound intensity at the surface of the sphere. If $< I(a) >$ is the time averaged intensity at the surface at $r = a$ then the radiated power is simply

$$\begin{aligned} \Pi &= 4\pi a^2 < I(a) >= \frac{4\pi a^2}{2} |\overline{U}(a)|^2 \Re[Z]_a \\ &= 2\pi a^2 |u_s|^2 \Re\left[\frac{\rho_0 c((ka)^2 + ika)}{1+(ka)^2}\right] \\ &= 2\pi a^2 |u_s|^2 \frac{\rho_0 c(ka)^2}{1+(ka)^2}, \end{aligned} \tag{5.3.7}$$

where, again, $\Re[\cdot]$ denotes the real part.

5.4 Sound Field Inside a Pulsating Sphere

In addition to creating a sound field on the exterior as examined above, the inside of a pulsating sphere with surface velocity, $u_s e^{i\omega t}$ as in Eq. (5.3.1) and radius a, will also have sound pressure that can be assumed to be spherically symmetric. Within this domain, we can take the solution to Eq. (5.1.1) to consist of an outgoing and an incoming wave of the form

$$p(r,t) = e^{i\omega t}\left(\frac{P_1 e^{-ikr}}{r} + \frac{P_2 e^{ikr}}{r}\right). \tag{5.4.1}$$

The constants P_1 and P_2 must be determined by satisfying the boundary conditions. One boundary condition is that at $r = a$, the velocity of the air must equal that of the sphere. The Euler equation then gives the relation between the velocity and pressure as in Eq. (5.3.3),

$$u_a = P_1 \frac{1 + ika}{-\rho_0 i\omega a^2} e^{-ika} + P_2 \frac{1 - ika}{-\rho_0 i\omega a^2} e^{ika}. \tag{5.4.2}$$

The other boundary condition is that, owing to spherical symmetry, the velocity at $r = 0$ must vanish. This boundary condition is a bit more difficult to use since the pressure has a singularity at $r = 0$. It is helpful to consider the total rate at which air moves through a sphere of radius r_0 centered at $r = 0$. This must be zero as the radius $r_0 \to 0$. This can be accomplished by evaluating Eq. (5.4.2) at $a = r_0$ and multiplying by the area $4\pi r_0^2$. This gives the volume velocity through this sphere with radius r_0 as

$$u_{r_0} 4\pi r_0^2 = (P_1 \frac{1 + ikr_0}{-\rho_0 i\omega {r_0}^2} e^{-ikr_0} + P_2 \frac{1 - ikr_0}{-\rho_0 i\omega {r_0}^2} e^{ikr_0}) 4\pi r_0^2. \tag{5.4.3}$$

Taking the limit as $r_0 \to 0$ gives

$$P_1 = -P_2. \tag{5.4.4}$$

Equations (5.4.2) and (5.4.4) give

$$P_2 = u_a \frac{\rho_0 cka^2}{2ka\cos(ka) - 2\sin(ka)}. \tag{5.4.5}$$

Equations (5.4.1), (5.4.4) and (5.4.5) give the pressure at any location in the sphere

$$p(r,t) = e^{i\omega t} \frac{u_a \sin(kr)\rho_0 cika^2/r}{ka\cos(ka) - \sin(ka)}. \tag{5.4.6}$$

5.5 Simple Model of Sound Radiation from a Loudspeaker

In the following, we'd like to construct a highly simplified model of sound radiation from an electrodynamic loudspeaker based on a very approximate representation as a simple equivalent spherical sound source. We will examine this problem in more detail later but for now we want the simplest possible model.

The loudspeaker consists of a diaphragm that is mounted on one side of a sealed box. It is common to approximate the sound radiated by the diaphragm as if it were a

piston mounted in an infinite baffle which radiates into a half-space. A more detailed examination of this system will be presented in Sect. 5.11. The diaphragm is driven by a force, $F(t) = Fe^{i\omega t}$, that is proportional to the current in the voice coil which is placed within a magnetic field. Suppose that the mechanical properties of the diaphragm are such that it has an effective impedance of

$$Z_d = i\omega M + \frac{K}{i\omega} + C. \tag{5.5.1}$$

In order to have consistent units, we can divide the applied force by the diaphragm area, S_d so that it has the units of pressure, $P_d(t) = F(t)/S_d = P_d e^{i\omega t}$. For a first approximation, we will assume that the impedance of the air is negligible relative to Z so that the velocity of the diaphragm is not influenced by the radiation pressure at the external surface of the diaphragm. The velocity of the diaphragm will then be

$$U_d(t) = \frac{P_d(t)}{Z_d} = \frac{P_d e^{i\omega t}}{i\omega M + \frac{K}{i\omega} + C} = P_d e^{i\omega t} \frac{i\omega/M}{\omega_0^2 - \omega^2 + 2\omega_0 \zeta i\omega}, \tag{5.5.2}$$

where $\omega_0^2 = K/M$ is the square of the natural frequency and $\zeta = C/(2\sqrt{KM})$ is the damping ratio.

Knowing the velocity of the diaphragm, $U_d(t)$, we'd like to examine the sound field that results, again, using the simplest possible model. If our interest is in the sound field at significant distances from the loudspeaker, we can hope that the specific geometric details of the system are not important so it can be represented by an equivalent spherical source as studied in the previous section. We'd like to employ Eq. (5.3.5) where the velocity of the sphere, u_s is replaced by an equivalent velocity to represent the motion of the diaphragm.

An equivalent spherical source can be constructed such that the source "strength" is equivalent. This is accomplished by ensuring that the total volume of air displaced by both the diaphragm and the spherical source is identical. Since the sphere is assumed to be radiating into a half-space on one side of an infinite baffle, the radiating surface area will be $2\pi a^2$, half that of the full sphere. Multiplying each velocity by the area of the moving element gives the source strength, Q,

$$Q = u_s 2\pi a^2 = U_d \pi a_d^2, \tag{5.5.3}$$

where a_d is the radius of the diaphragm. The equivalent velocity of the spherical source may then be approximated by

$$u_s = \frac{U_d a_d^2}{2a^2}. \tag{5.5.4}$$

Substituting Eq. (5.5.4) into Eq. (5.3.5) gives

$$p(r,t) \approx \frac{\rho_0 c i k a_d^2}{1+ika}\frac{U_d}{2r}e^{i(\omega t - k(r-a))}. \tag{5.5.5}$$

This result still depends on the radius, a, of our equivalent simple source. Since this term is multiplied by k in Eq. (5.5.5), it becomes important only when ka is not negligible relative to unity, which occurs at high frequencies. The radiation from the loudspeaker at these frequencies cannot be accurately represented by our highly simplified model. But, we can obtain qualitative results by assuming that the radius of our equivalent spherical source is the same as that of the loudspeaker diaphragm, $a = a_d$ so that Eq. (5.5.5) becomes

$$p(r,t) \approx \frac{\rho_0 c i k a_d^2}{1+ika_d}\frac{U_d}{2r}e^{i(\omega t - k(r-a_d))}. \tag{5.5.6}$$

We will find in Sect. 5.11 that this very approximate result agrees with a more detailed analysis of a piston radiator in an infinite baffle for listening locations that are sufficiently far from the source.

We can also estimate the total acoustic power radiated by the loudspeaker. Since we consider radiation only into a half-space, the power will be half that given in Eq. (5.3.7) obtained for the simple spherical source,

$$\Pi = 2\pi a^2 < I(a) >= \pi a^2 |u_s|^2 \frac{\rho_0 c (ka)^2}{1+(ka)^2} = \pi a^2 |\frac{U_d a_d^2}{2a^2}|^2 \frac{\rho_0 c (ka)^2}{1+(ka)^2} = \frac{\pi a_d^2}{4}|U_d|^2 \frac{\rho_0 c (ka_d)^2}{1+(ka)^2}, \tag{5.5.7}$$

where we have used Eq. (5.5.4) to replace the spherical source velocity with an equivalent diaphragm velocity.

Substituting Eq. (5.5.2) for U_d gives the total radiated power as

$$\Pi = \frac{\pi a_d^2}{4} \times \left|\frac{P_d}{M}\right|^2 \times \frac{\omega^2}{(\omega_0^2-\omega^2)^2 + (2\omega_0\zeta\omega)^2} \times \frac{\rho_0 c (ka_d)^2}{1+(ka_d)^2}, \tag{5.5.8}$$

where, as we did in Eq. (5.5.6), we have replaced $1+(ka)^2$ with $1+(ka_d)^2$. Since $k = \omega/c$, Eq. (5.5.8) provides a rather simple expression for the frequency dependence of the power emitted by our loudspeaker as a function of key system parameters.

As an example, suppose we have a very small diaphragm with radius $a_d = 0.01$ m. The mass per unit area is $M = 0.4$ kg/m^2, the stiffness is $K = 100000$ N/m^3, and the dashpot constant is $C = 200$ Ns/m^3. The natural frequency is approximately 80 Hz and the damping ratio is $\zeta = 0.5$. The equivalent force per unit area applied by the current is assumed to be $P_d = 1$ N/m^2. The squared magnitude of the velocity of the diaphragm, $|U_d|^2$, is shown in Fig. 5.1. At frequencies below resonance, $|U_d|^2$ increases in proportion to ω^2 while it decreases inversely proportionally to ω^2 at frequencies above resonance.

The real part of the acoustic radiation impedance, $\Re[Z]_{a_d}$, is shown in Fig. 5.2. The real part of the radiation impedance is seen to increase in proportion to ω^2 for

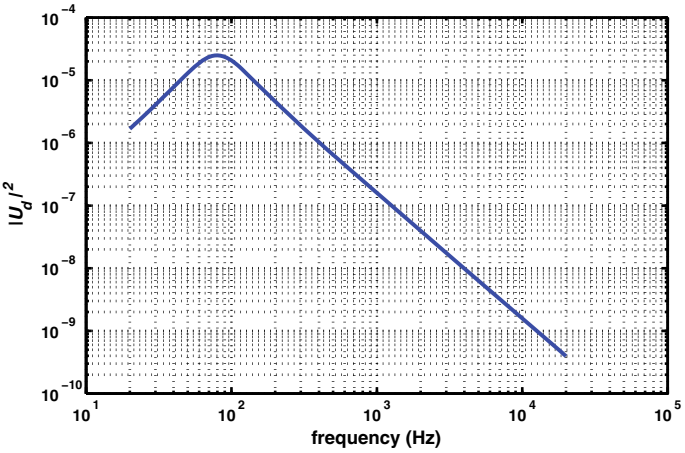


Fig. 5.1 The squared magnitude of the velocity of the diaphragm, $|U_d|^2$. At frequencies below resonance, $|U_d|^2$ increases in proportion to ω^2 while it decreases inversely proportionally to ω^2 at frequencies above resonance

frequencies below where $ka_d = \omega a_d/c << 1$. Above this frequency, the real part of the impedance becomes roughly constant. The product of the results shown in Figs. 5.1 and 5.2 is shown in Fig. 5.3. For frequencies below the resonant frequency of the diaphragm, both $|U_d|^2$ and $\Re[Z]_{a_d}$ increase in proportion to ω^2 so that the product increases in proportion to ω^4. At the resonant frequency of the diaphragm, $\omega = \omega_0$, there is a peak in response that depends on the damping ratio, ζ.

At frequencies above the resonant frequency of the diaphragm, $|U_d|^2$ decreases in proportion to ω^2. As a result, the product of $|U_d|^2$ and $\Re[Z]_{a_d}$ leads to a response that is approximately constant in frequency since the decrease of $|U_d|^2$ with ω^2 cancels the increase of $\Re[Z]_{a_d}$ with ω^2.

Finally, at frequencies above where $ka_d = \omega a_d/c \approx 1$, the fact that the radiation impedance becomes roughly constant causes the decrease of $|U_d|^2$ with ω^2 to result in a similar frequency dependence in the radiated acoustic power.

The approximate model for the radiation from a loudspeaker described above was based on the assumption that the acoustic pressure radiated at the diaphragm's surface doesn't influence the diaphragm's velocity so that the diaphragm velocity could be expressed in terms of only the internal pressure in Eq. (5.5.2). If the impedance of the diaphragm is not dominant over the acoustic radiation impedance, then the radiated pressure must be accounted for when determining the diaphragm velocity. The net pressure acting on the diaphragm will equal the difference between the effective pressure applied by the current, $P_d(t)$ and the pressure applied by the radiated sound $P(a,t)$. The velocity of the diaphragm may then be written as

$$U_d(t) = \frac{P_d(t) - P(a,t)}{Z_d} = \frac{P(a,t)}{Z(a)}, \tag{5.5.9}$$

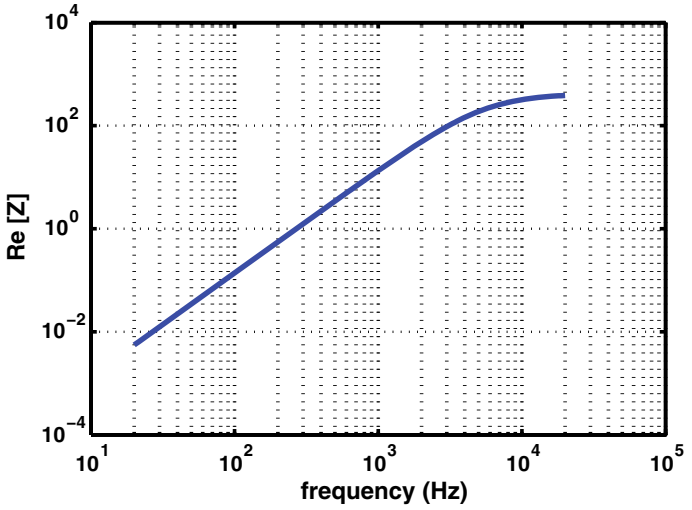


Fig. 5.2 The real part of the acoustic radiation impedance, $\Re[Z]_{a_d}$. The real part of the radiation impedance is seen to increase in proportion to ω^2 for frequencies below where $ka_d = \omega a_d/c << 1$. Above this frequency, the real part of the impedance becomes roughly constant

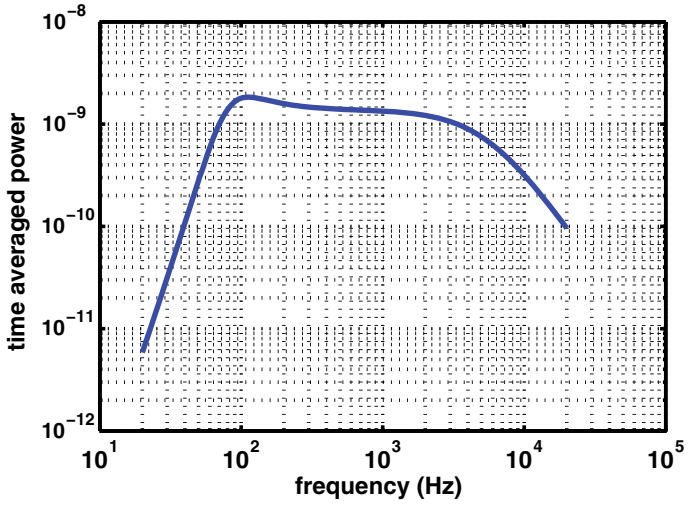


Fig. 5.3 The radiated acoustic power from Eq. (5.5.8). At frequencies below the resonant frequency of the diaphragm, the power increases in proportion to ω^4. At frequencies between the diaphragm resonant frequency and where $ka_d = \omega a_d/c \approx 1$, the radiated power is roughly constant. At higher frequencies where the radiation impedance is roughly constant, the fact that $|U_d|^2$ decreases inversely with ω^2 causes the power to also fall inversely with ω^2

where the right equation results from the fact that the diaphragm velocity must equal the effective acoustic particle velocity at a. Equations (5.5.9) may be used to eliminate the radiated pressure $P(a,t)$ to get

$$U_d(t) = \frac{P_d(t)}{Z(a) + Z_d}. \tag{5.5.10}$$

The impedances $Z(a)$ and Z_d clearly behave as if they are connected in series. The total effective impedance is then

$$Z(a) + Z_d = i\omega M + \frac{K}{i\omega} + C + \frac{\rho_0 c((ka)^2 + ika)}{1 + (ka)^2}, \tag{5.5.11}$$

where we have used Eqs. (5.5.1) and (5.1.20). The effective damping constant per unit area is increased to

$$C_{eff} = C + \frac{\rho_0 c(ka)^2}{1 + (ka)^2}, \tag{5.5.12}$$

and the effective mass per unit area is increased to

$$M_{eff} = M + \frac{\rho_0 a}{1 + (ka)^2}. \tag{5.5.13}$$

We may then proceed as before where Eq. (5.5.9) is used to calculate the diaphragm velocity instead of Eq. (5.5.2).

5.6 The Image Source

Having the model for sound radiation from a point source, it is possible to construct models for sound radiation from more complicated structures. In the following we will extend our previous results to determine the sound field due to a point source that is adjacent to a rigid boundary, which will be extended to analyze radiation from an acoustic dipole. We will then adapt these results to model sound radiation from a line source, or a radially oscillating cylinder. This will then lead us to an important goal: the calculation of sound radiation from general vibrating surfaces.

To examine the sound field created by a point sound source near a rigid wall, it is convenient to introduce the use of an image source as shown in Fig. 5.4. The actual sound source is denoted by S and is a distance d from the infinite, rigid wall. The boundary condition at the wall is that the x component (normal to the wall) of the acoustic particle velocity must be zero. The Euler equation then tells us that $\frac{\partial p}{\partial x} = 0$ at $x = 0$. To satisfy this boundary condition we can introduce an image source at the point B that is identical to the source at S. Assuming harmonic fluctuations at the

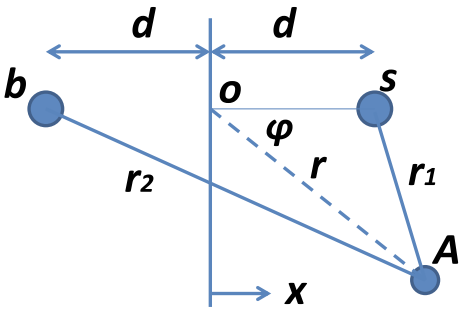


Fig. 5.4 Image source a distance d from a rigid, infinite wall at O. A listener at the point A receives sound from the original source S along with its image at the point b

frequency ω rad/s, the sound field at the point A will be the sum of the pressures due to both sources,

$$p(A,t) = D_0 e^{i\omega t}\left(\frac{e^{-ikr_1}}{r_1} + \frac{e^{-ikr_2}}{r_2}\right), \tag{5.6.1}$$

where D_0 is a constant, $k = \omega/c$, and c is the sound propagation speed.

The distance r_1 can be expressed as

$$\begin{aligned} r_1 &= \sqrt{(d\sin(\phi))^2 + (r_0 - d\cos(\phi))^2} \\ &= \sqrt{d^2 + r_0^2 - 2r_0 d\cos(\phi)} \\ &= r_0\sqrt{1 + \left(\frac{d}{r_0}\right)^2 - 2\frac{d}{r_0}\cos(\phi)}. \end{aligned} \tag{5.6.2}$$

If the sound source is fairly close to the wall so that $d/r_0 << 1$, Eq. (5.6.2) becomes

$$r_1 \approx r_0\left(1 - \frac{d}{r_0}\cos(\phi)\right) = r_0 - d\cos(\phi). \tag{5.6.3}$$

We also find that

$$r_2 \approx r_0 + d\cos(\phi). \tag{5.6.4}$$

Substituting Eqs. (5.6.3) and (5.6.4) into Eq. (5.6.1) gives

$$p(A,t) \approx D_0 e^{i\omega t}\left(\frac{e^{-ikr_0}e^{ikd\cos(\phi)}}{r_0 - d\cos(\phi)} + \frac{e^{-ikr_0}e^{-ikd\cos(\phi)}}{r_0 + d\cos(\phi)}\right), \tag{5.6.5}$$

If d/r_0 is small, we can neglect $d\cos(\phi)$ in the denominators of this expression, but it cannot be neglected in the arguments of the exponential. Equation (5.6.5) may then be approximated by

$$p(A,t) \approx D_0 e^{i\omega t}\frac{e^{-ikr_0}}{r_0}(e^{ikd\cos(\phi)} + e^{-ikd\cos(\phi)}). \tag{5.6.6}$$

This may be written as

$$p(A,t) \approx D_0 e^{i\omega t}\frac{e^{-ikr_0}}{r_0}2\cos(kd\cos(\phi)). \tag{5.6.7}$$

Note that if

$$kd\cos(\phi) = (2n+1)\pi/2, \tag{5.6.8}$$

where n is any integer, then $p(A,t) = 0$. This null occurs because the pressures due to the actual source and the image source cancel at these frequencies. This cancelation effect can be very important when making sound measurements near a rigid surface. We will also generally see maxima in $P(A,t)$ when

$$kd\cos(\phi) = m\pi, \tag{5.6.9}$$

where m is any integer. Note that since $k = \omega/c$, the nulls and maxima will occur at periodic values of ω. If the sound source is very close to the surface, taking the limit as d approaches zero in Eq. (5.6.7) gives

$$p(A,t) \approx 2D_0 e^{i\omega t}\frac{e^{-ikr_0}}{r_0}. \tag{5.6.10}$$

This shows that for sound sources that are very close to a rigid boundary, the observed pressure is twice what it would be if the boundary wasn't there.

To examine the effect acoustic reflections from a boundary can have on the sound field, suppose we have a point source, a distance $d = 0.5$ meter from a rigid boundary. Let the point of sound detection be at a distance $r_0 = 5$ m from the origin and at an angle $\phi = \pi/5$ rad as shown in Fig. 5.4. The results are shown in Fig. 5.5. The figure shows substantial dips that occur at periodic frequencies.

To illustrate the effect of the sound receiving point position, Fig. 5.6 shows the detected sound pressure at the much closer distance of $r_0 = 0.5$ m from the origin with all other dimensions, the same as used in Fig. 5.5. In this case, the range of sound pressure levels is considerably smaller than those of Fig. 5.5.

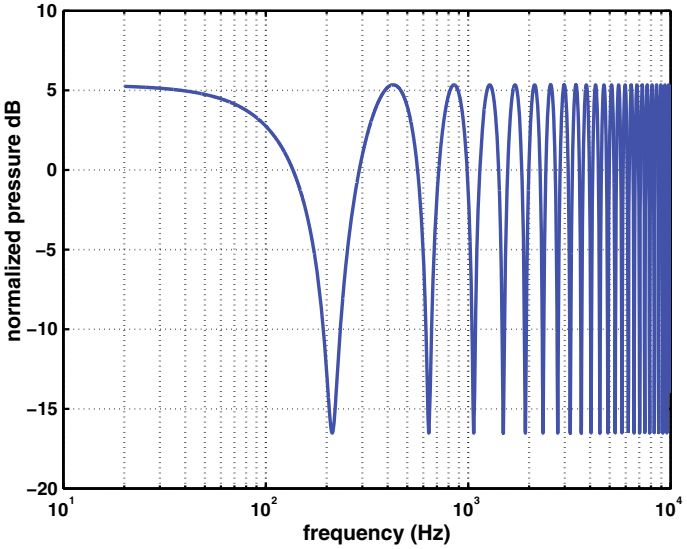


Fig. 5.5 Sound reflection from the boundary causes significant variations in the sound pressure across frequency. Even though the source strength is independent of frequency, the interference between the image source and the direct source causes the total pressure to vary by over 20 dB, depending on frequency. The pressure at the receiving point A is shown where $r_0 = 5$ m from the origin and at an angle $\phi = \pi/5$ radian. The point source is a distance $d = 0.5$ m from the rigid boundary

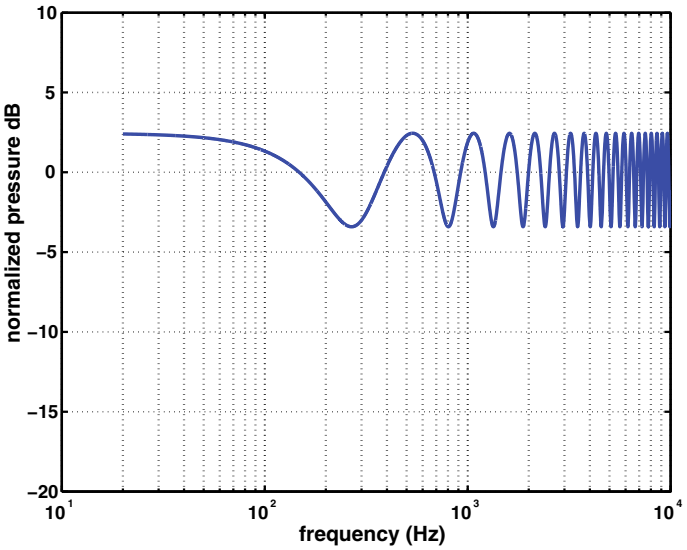


Fig. 5.6 Pressure at the receiving point A where $r = 0.5$ m from the origin and at an angle $\phi = \pi/5$ rad. The point source is a distance $d = 0.5$ m from the rigid boundary. Note that since d/r_0 is not small in this case, Eq. (5.6.1) has been used rather than the approximate result in Eq. (5.6.7)

5.6.1 Periodic Frequency Response Due to Reflections

The periodic frequency dependence of the sound pressure due to acoustic reflections from a rigid boundary is an important effect to keep in mind, especially when interpreting experimental data; when this is observed in experimental data one should be on the lookout for sources of sound reflections which may adversely impact the results. While this effect is to be expected when a rigid reflecting surface is near the source or receiver of sound, it can also influence the pressure even when the surface is considered to be highly absorbing (i.e., even in an anechoic chamber).

It is worthwhile to briefly consider a highly simplified system in which we can readily account for reflections from absorbing surfaces to see how the variations in pressure across frequency depend on the absorption of the surface. This can be examined by revisiting the simple one- dimensional wave of Sect. 2.5. We considered the effect of an absorbing surface at the location $x = 0$, which had an acoustic impedance, Z, and an absorption coefficient α as in Eq. (2.5.19). Equation (2.5.18) gives the ratio of the reflected to incident pressure due to the reflection of a normally incident plane wave where the reflecting surface has an acoustic impedance Z,

$$\frac{p_2}{p_1} = \frac{Z/(\rho_0 c) - 1}{Z/(\rho_0 c) + 1} = r, \tag{5.6.11}$$

where we have denoted this ratio as the reflection factor r.

The sound pressure at a distance l to the left of the reflecting surface, may be computed from Eq. (2.5.10), at $x = -l$,

$$p(-l, t) = e^{i\omega t}(p_1 e^{ikl} + p_2 e^{-ikl}) = e^{i\omega t} p_1 e^{ikl}(1 + re^{-2ikl}). \tag{5.6.12}$$

The squared magnitude of the pressure will then be

$$|p(-l, t)|^2 = |p_1|^2(1 + re^{-2ikl}) \times (1 + r^* e^{2ikl}) = |p_1|^2(1 + 2\Re[re^{-2ikl}] + |r|^2). \tag{5.6.13}$$

The frequency dependence of the pressure can be a bit complicated when the impedance Z in Eq. (5.6.11) is complex and, as often occurs, frequency dependent. To simplify matters, let r be real and assume that the absorption coefficient of the surface is near unity, say $\alpha \approx 0.99$, which would correspond to a highly absorbent surface. The reflection factor corresponding to this absorption coefficient will then be, from Eq. (2.5.19),

$$r = \sqrt{1 - \alpha} \approx 0.1. \tag{5.6.14}$$

Neglecting $|r|^2$ in Eq. (5.6.13) gives

$$|p(-l, t)|^2 \approx |p_1|^2(1 + 2\Re[re^{-2ikl}]) = |p_1|^2(1 + 2r\cos(2kl)). \tag{5.6.15}$$

The sound pressure level may then be computed using Eqs. (1.1.1) and (2.5.9),

$$SPL = 10\log_{10}\left(\frac{|p_1|^2}{2P_{ref}^2}(1+2r\cos(2kl))\right) = 10\log_{10}\left(\frac{|p_1|^2}{2P_{ref}^2}\right) \\ +10\log_{10}((1+2r\cos(2kl)), \tag{5.6.16}$$

where $P_{ref} = 20 \times 10^{-6}$ pascal is the standard reference pressure. The second term on the right of this result gives the fluctuation in the sound pressure level due to the reflection. Expanding this in a Taylor's series for small r gives

$$10\log_{10}((1+2r\cos(2kl)) \approx \frac{10}{\ln(10)}(2r\cos(2kl) + \left(\frac{2r\cos(2kl))^2}{2}\right) \approx \frac{20r\cos(2kl)}{\ln(10)}. \tag{5.6.17}$$

Since $\cos(2kl)$ varies between ± 1, the sound pressure level will vary across frequency by $\pm\frac{20r}{\ln(10)}$ decibels. If $r = 0.1$, corresponding to a rather absorbent surface with $\alpha = 0.99$, the variation will be approximately $\pm\frac{2}{\ln(10)} \approx \pm 0.87$ decibels. If the absorption coefficient is $\alpha = 0.9$, the fluctuation is found to be approximately ± 2.75 decibels.

5.7 The Acoustic Dipole

An acoustic dipole consists of two point sources adjacent to each other where the sources vibrate 180° out of phase. The situation is similar to our model of a sound source near a rigid wall as shown in Fig. 5.4. In this case, however, since the sources are out of phase, we subtract the image source instead of add it to the pressure from the actual source,

$$p(A,t) = D_0 e^{i\omega t}\left(\frac{e^{-ikr_1}}{r_1} - \frac{e^{-ikr_2}}{r_2}\right). \tag{5.7.1}$$

Again, for $d/r << 1$, when the sources are sufficiently close together, we can use Eqs. (5.6.3) and (5.6.4) for r_1 and r_2 so that Eq. (5.7.1) becomes

$$p(A,t) \approx D_0 e^{i\omega t}\left(\frac{e^{-ikr}e^{ikd\cos(\phi)}}{r - d\cos(\phi)} - \frac{e^{-ikr}e^{-ikd\cos(\phi)}}{r + d\cos(\phi)}\right). \tag{5.7.2}$$

This may again be approximated by

$$p(A,t) \approx D_0 e^{i\omega t}\frac{e^{-ikr}}{r}(e^{ikd\cos(\phi)} - e^{-ikd\cos(\phi)}). \tag{5.7.3}$$

This may be written as

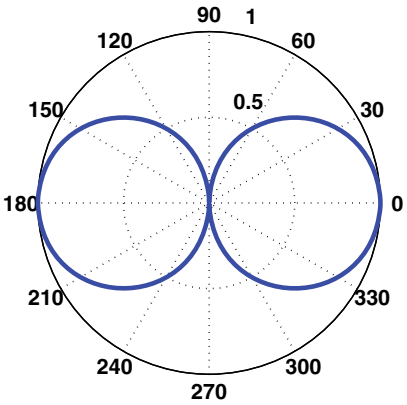


Fig. 5.7 Directivity pattern of an ideal dipole radiator

$$p(A,t) \approx D_0 e^{i\omega t} \frac{e^{-ikr}}{r} 2i \sin(kd\cos(\phi)). \tag{5.7.4}$$

Note that at very low frequencies or if d is sufficiently small so that $kd\cos(\phi) << 1$, then $\sin(kd\cos(\phi)) \approx kd\cos(\phi)$. Equation (5.7.4) then becomes

$$p(A,t) \approx D_0 e^{i\omega t} \frac{e^{-ikr}}{r} 2ikd\cos(\phi). \tag{5.7.5}$$

The surface that is equidistant between the two sources satisfies a zero pressure boundary condition rather than a zero velocity boundary condition as in the rigid wall.

The angular dependence of the radiation is usually displayed on a polar plot where the distance from the origin is a measure of the amplitude of the pressure. As shown in Eq. (5.7.5), at low frequencies, the angular dependence is given by $\cos(\phi)$. This is shown in Fig. 5.7. In this case the directivity plot consists of a curve where each point is a distance $\cos(\phi)$ from the origin where ϕ is the angle between the line to the point from the origin and the x axis. From Fig. 5.7, this curve appears as circles centered at $x = 1/2$ and having a radius equal to 1/2. This circle is described by

$$(x - 1/2)^2 + y^2 = 1/4. \tag{5.7.6}$$

In this case, $x = \cos(\phi)^2$ and $y = \cos(\phi)\sin(\phi)$. Substituting these into Eq. (5.7.6) and simplifying gives

$$\cos(\phi)^2 + \sin(\phi)^2 = 1 \tag{5.7.7}$$

so that the directivity pattern does describe the circle of Eq. (5.7.6).

It is also important to note that if one has the pressure given in Eq. (5.7.1), one can apply the Euler equation to calculate the velocity vector at any location. For harmonic waves, the Euler equation gives the velocity vector as

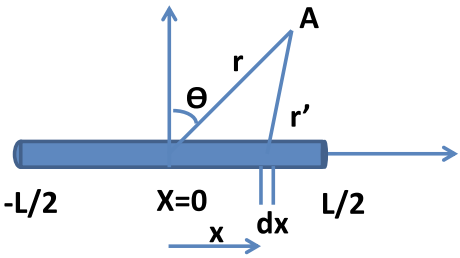


Fig. 5.8 Radially oscillating cylinder

$$\vec{u} = \frac{i\vec{\nabla p}}{k\rho_0 c}. \tag{5.7.8}$$

The gradient in Eq. (5.7.8) may be easiest to calculate by expressing r_1 and r_2 in terms of x and y in Eq. (5.7.1) based on Fig. 5.4,

$$r_1 = \sqrt{(x-d)^2 + y^2}, \quad r_2 = \sqrt{(x+d)^2 + y^2}. \tag{5.7.9}$$

5.8 The Line Source

As our next step in calculating the sound radiation from vibrating bodies, consider a radially oscillating cylinder of length L as shown in Fig. 5.8. We'd like to calculate the sound pressure at the point A due to the radiation from the cylinder which has a radial velocity given by

$$u(x,t) = U_0 e^{i\omega t}, \quad -L/2 \le x \le L/2. \tag{5.8.1}$$

Consider a differential element of the cylinder of length dx and radius a. Since the dimensions of this element are small compared to the sound wavelength, we can model its sound radiation by constructing an "equivalent" point source (or "simple source") which displaces a volume of fluid at the same rate as the differential element. For a pulsating sphere with radial velocity U_s, the rate of fluid displacement is

$$Q = 4\pi a^2 U_s. \tag{5.8.2}$$

Since our differential cylinder oscillates only in the direction perpendicular to its axis of symmetry, its rate of fluid displacement is

$$dQ = 2\pi a U_0 dx. \tag{5.8.3}$$

The pressure at a distance r' from a pulsating sphere is given in Eq. (5.3.5) and repeated here for convenience,

$$p(r',t) = \frac{\rho_0 cika^2}{1+ika}\frac{U_s}{r'}e^{i(\omega t - k(r'-a))}. \tag{5.8.4}$$

By using Eq. (5.8.2), Eq. (5.8.4) can be expressed in terms of the rate of fluid displacement. If we also assume that $ka << 1$, Eq. (5.8.4) becomes

$$p(r',t) \approx \frac{\rho_0 cikQ}{4\pi r'}e^{i(\omega t - kr')}. \tag{5.8.5}$$

To find the pressure due to the differential element of our line source, replace Q in Eq. (5.8.5) with dQ from Eq. (5.8.3),

$$dp(r',t) \approx \frac{\rho_0 cikdQ}{4\pi r'}e^{i(\omega t - kr')} = \frac{\rho_0 cik2\pi a U_0 dx}{4\pi r'}e^{i(\omega t - kr')} \tag{5.8.6}$$

We now need to determine $r'(x)$ and add up the contributions to the pressure for $-L/2 \le x \le L/2$. From Fig. 5.8,

$$\begin{aligned} r' &= \sqrt{(r\sin(\theta) - x)^2 + (r\cos(\theta))^2} \\ &= \sqrt{r^2 + x^2 - 2rx\sin(\theta)} \\ &= r\sqrt{1 + \left(\frac{x}{r}\right)^2 - 2\frac{x}{r}\sin(\theta)}. \end{aligned} \tag{5.8.7}$$

The pressure at the point A may then be calculated from

$$p(r,\theta,t) \approx \int_{-L/2}^{L/2} \frac{\rho_0 cik2\pi a U_0 dx}{4\pi r'}e^{i(\omega t - kr')}, \tag{5.8.8}$$

where we have again assumed that the radius of the cylinder is such that $ka << 1$.

When Eq. (5.8.7) is substituted for r', obtaining an analytical expression for the integral in Eq. (5.8.8) can be challenging. Fortunately, we can evaluate the integral in closed form by considering limiting cases. If $r >> L/2$, so that the listening point is sufficiently far from the source, the square root can be expanded in Eq. (5.8.7) and higher powers of x/r can be neglected,

$$r' \approx r\left(1 - \frac{x}{r}\sin(\theta)\right) = r - x\sin(\theta). \tag{5.8.9}$$

The pressure at the point A is then

$$p(r,\theta,t) \approx e^{i\omega t}\int_{-L/2}^{L/2}\frac{\rho_0 cik2\pi aU_0}{4\pi(r-x\sin(\theta))}e^{-ik(r-x\sin(\theta))}dx$$
$$= \frac{i\rho_0 cke^{i\omega t}aU_0e^{-ikr}}{2}\int_{-L/2}^{L/2}\frac{e^{ikx\sin(\theta)}}{r-x\sin(\theta)}dx. \tag{5.8.10}$$

If we again use the assumption that $r >> L/2$, we may neglect $x\sin(\theta)$ in the denominator of Eq. (5.8.10), which greatly simplifies the task of carrying out the integration. This gives

$$p(r,\theta,t) \approx \frac{i\rho_0 cke^{i\omega t}aU_0e^{-ikr}}{2r}\int_{-L/2}^{L/2}e^{ikx\sin(\theta)}dx$$
$$= \frac{i\rho_0 cke^{i\omega t}aU_0e^{-ikr}}{2r}\left[\frac{e^{ikL/2\sin(\theta)}-e^{-ikL/2\sin(\theta)}}{ik\sin(\theta)}\right]$$
$$= \frac{i\rho_0 cke^{i\omega t}aU_0e^{-ikr}}{2rik\sin(\theta)}\sin(kL/2\sin(\theta))2i$$
$$= \frac{L}{2}i\rho_0 cke^{i\omega t}aU_0\frac{e^{-ikr}}{r}\frac{\sin(kL/2\sin(\theta))}{kL/2\sin(\theta)}. \tag{5.8.11}$$

5.9 Surface Source

The idea of summing the contributions to the pressure due to a distribution of infinitesimal surfaces can be extended to compute the radiation from many complex vibrating surfaces. As our next example, consider a vibrating surface that is part of an infinite rigid wall. As before, we need to determine the source strength, or rate of fluid displacement from each differential area, but in this case we assume that the radiation is into a half space. Recall that the pressure radiated from a pulsating sphere having a surface velocity U_s is given in Eq. (5.8.5). We must replace U_s in this expression with an equivalent source strength for radiation into a half space. The source strength, $Q_{1/2}$ of a pulsating sphere radiating into a half space will be half that of Eq. (5.8.2),

$$Q_{1/2} = 2\pi a^2 U_s. \tag{5.9.1}$$

The pressure at r' from this source is, from Eq. (5.8.5),

$$p(r',t) = \frac{Q_{1/2}\rho_0 cik}{2\pi r'}e^{i(\omega t-kr')}, \tag{5.9.2}$$

where we have again assumed that $ka << 1$.

If the radiation is from the velocity $\vec{U}$ of a differential surface of area $\vec{ds}$ where the direction of the vector is normal to the surface, then the differential source strength will be $dQ_{1/2} = \vec{U}\cdot\vec{ds}$. For a simple flat surface, $dQ_{1/2} = Uds$. The corresponding differential contribution to the pressure at r' is obtained by replacing $Q_{1/2}$ in

Eq. (5.9.2) with Uds,

$$dp(r', t) = \frac{Uds\rho_0 cik}{2\pi r'} e^{i(\omega t - kr')} \tag{5.9.3}$$

so that the pressure due to the entire radiating surface of area S is

$$p(r, t) = \int_S \frac{U\rho_0 cik}{2\pi r'} e^{i(\omega t - kr')} ds, \tag{5.9.4}$$

where r indicates the point at which the pressure is calculated and r' is the distance between this point and a differential area on the vibrating surface. This expression is often referred to as the Rayleigh integral.

5.10 Sound Radiation from a Piston in a Baffle

The Rayleigh integral in Eq. (5.9.4) is a very powerful tool for determining the sound field resulting from the vibration of a surface. In the following, we will apply this method to construct a model for the sound radiation from a piston in an infinite wall. This is a convenient approximate model for the sound radiated by a circular loudspeaker driver. The Rayleigh integral relates the surface velocity, U, of a region with area S to the pressure,

$$p(r, t) = \int_S \frac{U\rho_0 cik}{2\pi r'} e^{i(\omega t - kr')} ds, \tag{5.10.1}$$

where $k = \omega/c$ with ω being the frequency in rad/s, ρ_0 is the nominal air density and c is the sound propagation speed. We must express r' and r in coordinates that are convenient for carrying out the surface integral. It may be easiest to begin by expressing them in terms of cartesian coordinates and then convert to polar coordinates to evaluate the integral.

5.11 Single Piston

Our first task is to determine the sound field radiating from a single piston centered at the origin of our coordinate system. We will then extend these results to account for the radiation from any number of circular pistons. From Fig. 5.9, let the coordinates of the point A be (X, Y, Z) and let the coordinates of a differential surface on the radiating piston be (x, y, z). Since the piston is on the (Y, Z) plane, $x = 0$. The distance between the differential surface and A is

$$r' = \sqrt{(X - x)^2 + (Y - y)^2 + (Z - z)^2}. \tag{5.11.1}$$

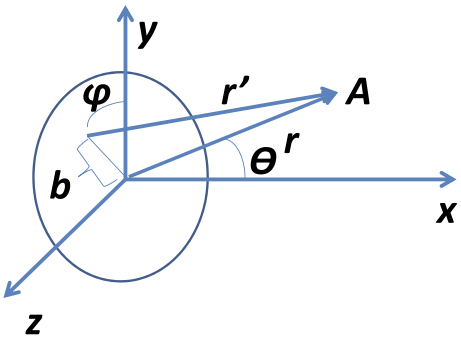


Fig. 5.9 Piston in an infinite baffle. The piston is centered at the origin of the (x, y, z) coordinate system and is placed in the (y, z) plane. A listener is at the location A

Let b be the distance between the differential radiating point and the origin and ϕ be the angle between that radius and the Y axis. So, $(x, y, z) = (0, b\cos(\phi), b\sin(\phi))$. The location of the point A is $(X, Y, Z) = (r\cos(\theta), r\sin(\theta), 0)$. Equation (5.11.1) then becomes

$$\begin{aligned} r' &= \sqrt{(r\cos(\theta))^2 + (r\sin(\theta) - b\cos(\phi))^2 + (b\sin(\phi))^2} \\ &= \sqrt{r^2 + b^2 - 2rb\sin(\theta)\cos(\phi)}. \end{aligned} \tag{5.11.2}$$

If we limit our attention to the case where the listening point A is sufficiently far from the piston, $r >> a$, where a is the radius of the piston, then Eq. (5.11.2) may be approximated by

$$r' \approx r - b\sin(\theta)\cos(\phi). \tag{5.11.3}$$

This must be substituted into Eq. (5.10.1) and the integral must be evaluated. Note that the r' terms in the denominator can be approximated by r if $r >> a$ so that Eq. (5.10.1) becomes

$$p(r, t) = e^{i(\omega t)} \frac{U\rho_0 cik}{2\pi r} e^{-ikr} \int_S e^{ikb\sin(\theta)\cos(\phi)} ds. \tag{5.11.4}$$

Since we are integrating in polar coordinates, $ds = bd\phi db$, so that Eq. (5.11.4) becomes

$$p(r, t) = e^{i(\omega t)} \frac{U\rho_0 cik}{2\pi r} e^{-ikr} \int_0^{2\pi} \int_0^a e^{ikb\sin(\theta)\cos(\phi)} bdbd\phi. \tag{5.11.5}$$

The integral may be evaluated by using the zeroth order Bessel function of the first kind,

$$J_0(kb\sin(\theta)) = \frac{1}{2\pi} \int_0^{2\pi} e^{ikb\sin(\theta)\cos(\phi)} d\phi. \tag{5.11.6}$$

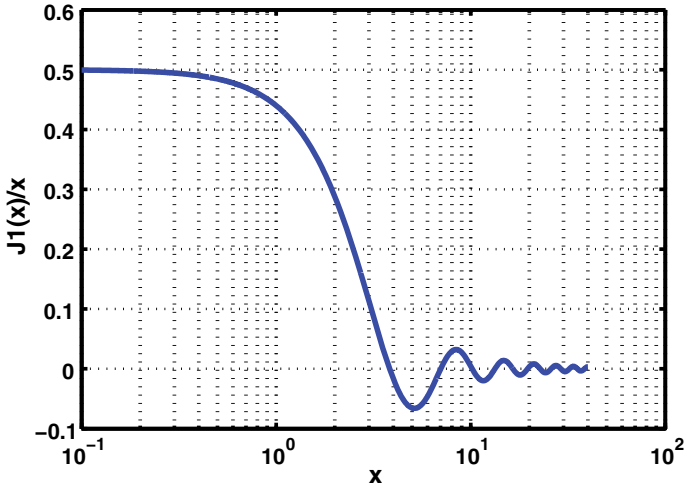


Fig. 5.10 $J_1(x)/x$ versus x. For values of x greater than unity, the function varies rapidly with x

Combining Eqs. (5.11.5) and (5.11.6) gives

$$p(r,t) = e^{i(\omega t)} \frac{U\rho_0 cik}{r} e^{-ikr} \int_0^a J_0(kb\sin(\theta))bdb. \tag{5.11.7}$$

An important property of these Bessel functions is that

$$\int x J_0(x)dx = x J_1(x). \tag{5.11.8}$$

Carrying out the remaining integral in Eq. (5.11.7) then gives

$$p(r,t) = e^{i(\omega t)} \frac{U\rho_0 cik}{r} e^{-ikr} \frac{a^2 J_1(ka\sin(\theta))}{ka\sin(\theta)}. \tag{5.11.9}$$

The frequency response of the piston radiator is strongly affected by the term $\frac{J_1(ka\sin(\theta))}{ka\sin(\theta)}$ in Eq. (5.11.9). At small frequencies or small angles θ, this function tends to be constant. However, when the argument $ka\sin(\theta)$ is on the order of unity or above, the function can change rapidly with the argument. This is illustrated in Fig. 5.10 which shows $J_1(x)/x$ versus x. For small x this approaches 1/2. The reasonably detailed analysis of the acoustic radiation of a piston radiator leading up to Eq. (5.11.9) agrees with our very approximate result in Eq. (5.5.6) when ka and $\sin(\theta)$ are small. Equation (5.11.9) may be used to examine the sound pressure as a function of the angle θ. This is shown in a polar plot in Fig. 5.11 for values of ka varying from 1 to 5. The figure shows that as ka increases, the loudspeaker driver becomes increasingly directional.

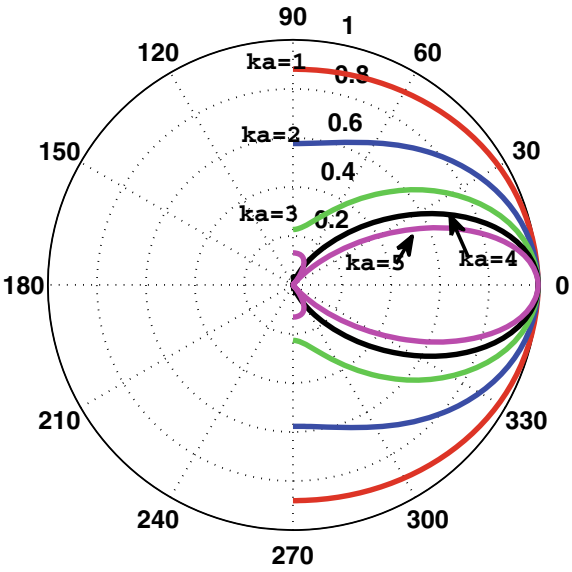


Fig. 5.11 Directivity patterns for a loudspeaker driver in a rigid wall based on Eq. (5.11.9) for values of ka from 1 to 5. The results are normalized relative to the sensitivity at $\theta = 0$

5.12 Piston Vibration

In addition to the argument of the Bessel function in Eq. (5.11.9), the dependence on frequency is also strongly determined by the frequency dependence of the product Uik since $k = \omega/c$ and the velocity U can vary with frequency. In the case of a typical loudspeaker, however, the actual input to the system is a voltage, v_{in} provided by a power amplifier. In an electrodynamic loudspeaker, the driving force applied to the diaphragm is due to an electromagnetic force,

$$f = i_c BL, \tag{5.12.1}$$

where i_c is the current in the voice coil, B is the flux density of the magnetic field (teslas), and L is length (m) of wire in the coil. The current is related to the input voltage through Ohm's law, $v_{in} = iZ_{in}$, where Z_{in} is the electrical impedance of the loudspeaker.

The displacement, $w(t)$, of the loudspeaker diaphragm can be modeled as a single degree of freedom spring/mass/damper,

$$\ddot{w} + \omega_0^2 w + 2\omega_0 \zeta \dot{w} = f/m, \tag{5.12.2}$$

where ω_0 is the undamped natural frequency in rad/s, ζ is the damping ratio, and m is the mass. If we assume that the input voltage provided by the power amplifier is harmonic with frequency ω, $v(t) = Ve^{i\omega t}$, where V is the complex amplitude. The governing equation for the diaphragm displacement can be expressed as

$$\ddot{w} + \omega_0^2 w + 2\omega_0 \zeta \dot{w} = V e^{i\omega t} \frac{BL}{Z_{in} m}. \tag{5.12.3}$$

The steady-state solution for $w(t)$ is $We^{i\omega t}$ where

$$W = \frac{VBL}{Z_{in} m(\omega_0^2 - \omega^2 + 2i\omega\omega_0\zeta)}. \tag{5.12.4}$$

The diaphragm velocity is $u(t) = \dot{w}(t) = i\omega W e^{i\omega t} = U e^{i\omega t}$. The complex amplitude of the diaphragm's velocity may then be written as

$$U = \frac{i\omega VBL}{Z_{in} m(\omega_0^2 - \omega^2 + 2i\omega\omega_0\zeta)}. \tag{5.12.5}$$

Equation (5.12.5) shows that the velocity relative to the input voltage varies strongly with frequency according to

$$\frac{U}{V} = \frac{i\omega BL}{Z_{in} m(\omega_0^2 - \omega^2 + 2i\omega\omega_0\zeta)}. \tag{5.12.6}$$

Substituting this result into Eq. (5.11.9) gives

$$\frac{p(r,t)}{V} = e^{i(\omega t)} \frac{i\omega BL}{Z_{in} m(\omega_0^2 - \omega^2 + 2i\omega\omega_0\zeta)} \frac{\rho_0 cik}{r} e^{-ikr} \frac{a^2 J_1(ka\sin(\theta))}{ka\sin(\theta)}. \tag{5.12.7}$$

Since $k = \omega/c$, the numerator of this expression is proportional to ω^2 which multiplies the Bessel function term which is plotted in Fig. 5.10. At frequencies where $\omega << \omega_0$, since the velocity increases in proportion to frequency, the result of Eq. (5.12.7) will increase in proportion to ω^2. It reaches a maximum near $\omega \approx \omega_0$. At frequencies where $\omega >> \omega_0$, the velocity becomes inversely proportional to frequency so that the result of Eq. (5.12.7) is approximately constant and may be approximated by

$$\frac{p(r,t)}{V} \approx e^{i(\omega t)} \frac{BL\rho_0}{rZ_{in} m} e^{-ikr} \frac{a^2 J_1(ka\sin(\theta))}{ka\sin(\theta)}. \tag{5.12.8}$$

Based on the results shown in Fig. 5.10, at frequencies below where $ka\sin(\theta) \approx 1$, the pressure due to a given voltage becomes approximately independent of frequency. Above that frequency range, we can expect the response to decrease with increasing frequency.

As an example, Fig. 5.12 shows the predicted response due to a diaphragm with radius $a = 0.05\,\text{m}$ at the location $r = 5\,\text{m}$ and $\theta = \pi/3$. The resonant frequency is 100 Hz, the damping ratio is $\zeta = 0.707$ (which may be shown to give flat frequency

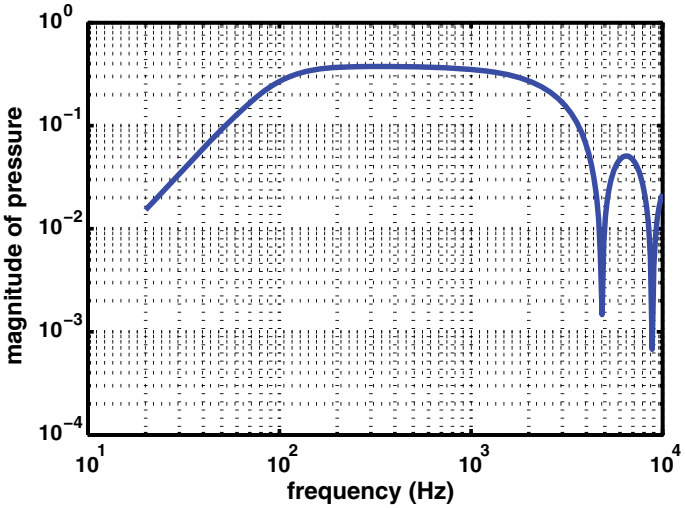


Fig. 5.12 Predicted response due to a diaphragm with radius $a = 0.05\,\text{m}$ at the location $r = 5\,\text{m}$ and $\theta = \pi/3$. The resonant frequency is 100 Hz, the damping ratio is $\zeta = 0.707$ and $BL = 100$. The response is seen to rise at low frequencies until about 100 Hz and is fairly flat up to frequencies above about 2 kHz

response) and $BL = 100$. The response is seen to rise at low frequencies until about 100 Hz and is fairly flat up to frequencies above about 2 kHz.

5.13 Multiple Piston Radiators

Since loudspeaker systems typically consist of multiple drivers (sometimes large numbers of them), it would be useful to be able to compute the pressure from any number of piston radiators by extending the approximate expression in Eq. (5.11.9). Suppose that the nth radiator is displaced a distance $(\Delta x_n, \Delta y_n, \Delta z_n)$ from the origin of the (x, y, z) system shown in Fig. 5.9. The distance between the nth radiator's differential surface and the listening point A is obtained by modifying Eq. (5.11.1),

$$r_n' = \sqrt{(X - \Delta x_n - x)^2 + (Y - \Delta y_n - y)^2 + (Z - \Delta z_n - z)^2}. \tag{5.13.1}$$

To simplify our notation, let

$$X_n = X - \Delta x_n, \quad Y_n = Y - \Delta y_n, \quad Z_n = Z - \Delta z_n. \tag{5.13.2}$$

Again substituting $(x, y, z) = (0, b\cos(\phi), b\sin(\phi))$, Eqs. (5.13.1) and (5.13.2) give

$$r_n' = \sqrt{X_n^2 + (Y_n - b\cos(\phi))^2 + (Z_n - b\sin(\phi))^2}$$
$$= \sqrt{r_n^2 - 2Y_n b\cos(\phi) - 2Z_n b\sin(\phi) + b^2}, \tag{5.13.3}$$

where,

$$r_n^2 = X_n^2 + Y_n^2 + Z_n^2. \tag{5.13.4}$$

As before, it is reasonable to assume that the listening point is far away from each radiator so that $r_n >> b$. Expanding Eq. (5.13.3) in a Taylor's series and neglecting small terms give

$$r_n' \approx r_n - Y_n b\cos(\phi)/r_n - Z_n b\sin(\phi)/r_n. \tag{5.13.5}$$

The contribution to the pressure due to the nth radiator may now be obtained by substituting Eq. (5.13.5) into Eq. (5.10.1),

$$p_n(r_n, t) = \frac{e^{i(\omega t)}}{2\pi}\int_S \frac{U\rho_0 cike^{-ik(r_n - Y_n b\cos(\phi)/r_n - Z_n b\sin(\phi)/r_n)}}{r_n - Y_n b\cos(\phi)/r_n - Z_n b\sin(\phi)/r_n} ds. \tag{5.13.6}$$

As before, the evaluation of the integral is greatly simplified if we neglect small terms in the denominator of the integrand. Since the listening point is far from any radiator,

$$p_n(r_n, t) \approx \frac{e^{i(\omega t - kr_n)}}{2\pi r_n}\int_S U\rho_0 cike^{ik(Y_n b\cos(\phi)/r_n + Z_n b\sin(\phi)/r_n)} ds. \tag{5.13.7}$$

Since we will integrate using polar coordinates, $ds = bdbd\phi$. The integral to be evaluated is then

$$\int_0^{2\pi}\int_0^a e^{ik(Y_n b\cos(\phi)/r_n + Z_n b\sin(\phi)/r_n)} bdbd\phi. \tag{5.13.8}$$

To cast this in a more familiar form, it is helpful to combine the sine and cosine terms using

$$D_n\cos(\phi + \alpha_n) = Y_n\cos(\phi)/r_n + Z_n\sin(\phi)/r_n, \tag{5.13.9}$$

where

$$D_n = \frac{1}{r_n}\sqrt{Y_n^2 + Z_n^2}, \quad \tan(\alpha_n) = -\frac{Z_n}{Y_n}. \tag{5.13.10}$$

The integral in (5.13.8) then becomes

$$\int_0^{2\pi}\int_0^a e^{ikbD_n\cos(\phi+\alpha_n)}bdbd\phi. \tag{5.13.11}$$

Since the integrand is periodic in ϕ, the result can be expressed using the Bessel function of the first kind as in Eq. (5.11.6),

$$J_0(kbD_n) = \frac{1}{2\pi}\int_0^{2\pi} e^{ikbD_n\cos(\phi+\alpha_n)}d\phi. \tag{5.13.12}$$

Equation (5.13.7) becomes

$$p_n(r_n,t) \approx \frac{e^{i(\omega t-kr_n)}}{r_n}\int_0^a U\rho_0 cikbJ_0(kbD_n)db. \tag{5.13.13}$$

Again using Eq. (5.11.8) and inserting the limits of integration give

$$p_n(r_n,t) \approx \frac{e^{i(\omega t-kr_n)}}{r_n}U\rho_0 cika^2\frac{J_1(kaD_n)}{kaD_n}. \tag{5.13.14}$$

If there are n radiators the total pressure will be the sum of all of these contributions,

$$P(A,t) \approx \sum_{i=1}^{N}\frac{e^{i(\omega t-kr_n)}}{r_n}U_n\rho_0 cika^2\frac{J_1(kaD_n)}{kaD_n}, \tag{5.13.15}$$

where we have replaced U with U_n to allow for the possibility that each radiator may vibrate differently. It is also a simple matter to allow the radiators to have dissimilar radii by replacing a with a_n.

5.14 Analysis and Design of Loudspeakers in Vented Boxes

Many loudspeakers incorporate a vent in the box to enhance the sound output at low frequencies. In Eq. (5.5.8) we developed a highly simplified model of the sound power radiated by a loudspeaker having a single diaphragm at low frequencies and at distances that are sufficiently far from the radiator relative to the sound wavelength. In the following we will examine the essential design parameters of the system to determine how to obtain improved response at low frequencies.

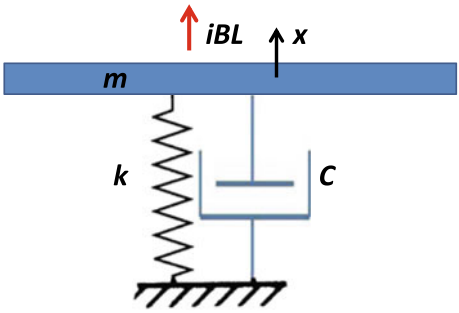


Fig. 5.13 Second order oscillator model of a loudspeaker diaphragm. This simple model neglects the effect of the air in the closed space behind the diaphragm

5.14.1 Loudspeaker Diaphragm in a Closed Box

Many loudspeaker systems consist of a diaphragm (or driver) that is mounted in a box to prevent sound from the back of the diaphragm from canceling that radiated by the front, causing it to radiate like a dipole. When the diaphragm is mounted in a very large box where the compression of the internal air doesn't create a noticeable force on the diaphragm, the response may be modeled as a linear second order oscillator as depicted in Fig. 5.13.

This system may be examined using the following equation of motion

$$m\ddot{x} + kx + \mathcal{C}\dot{x} = i_c BL, \tag{5.14.1}$$

where m is the diaphragm mass (kg), k is the mechanical stiffness (N/m), $\mathcal{C}$ is the viscous damping coefficient (N-s/m), and $i_c BL$ is the force (N) due to the current i_c in the coil of length L (m). B is the magnetic flux density (teslas), a measure of the strength of the magnetic field.

5.14.2 Diaphragm Force Due to Back Volume of Air

As the loudspeaker diaphragm moves, it changes the volume of the enclosed air in the box. In most cases, the box holding the loudspeaker diaphragm is not large enough for the effects of the resulting compression of the internal air to be negligible. If the dimensions of the air chamber behind the diaphragm are much smaller than the wavelength of sound, we can assume that the air pressure in the volume is independent of location. The air in this volume will then act like a linear spring; it exerts a restoring force on whatever acts to change the volume of the air. The assembly is shown schematically in Fig. 5.14. If the diaphragm moves in such a way that the volume is reduced, an internal pressure, P_d will be applied to the interior side of the diaphragm due to the compression of air in the volume. This pressure can be expressed in terms of the deflection x using the equation of state, Eq. (2.3.3):

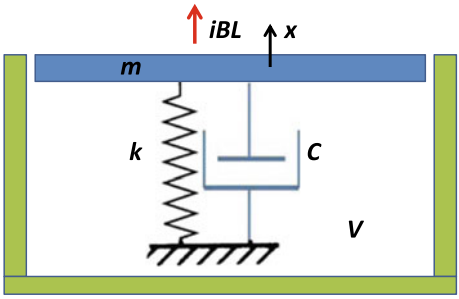


Fig. 5.14 Schematic representation of a loudspeaker diaphragm with an enclosed back volume of air

$$P_d = \rho_a c^2 = \rho_0 c^2 \frac{\rho_a}{\rho_0} = -\rho_0 c^2 \frac{\Delta V}{V}, \tag{5.14.2}$$

where ρ_a is, as before, the fluctuating air density within the volume. The negative sign on the right in Eq. (5.14.2) is due to the fact that the density decreases as the volume increases. The increase in density relative to the ambient density, ρ_a/ρ_0, is equal to the negative of the ratio of the change in volume, ΔV, relative to the total volume, V; a reduction in volume increases the density and increases the pressure. The change in volume is equal to the displacement of the air in the opening multiplied by the area, $\Delta V = S_d x$, where again, x is taken to be positive in the upward direction shown in Fig. 5.14. S_d is the area of the diaphragm. The fluctuating pressure in the volume, V, due to a fluctuation in the volume, ΔV, resulting from the outward motion of the diaphragm, x, is then

$$P_d = -\rho_o c^2 \Delta V / V = -\rho_o c^2 S_d x / V, \tag{5.14.3}$$

where ρ_o is the density of air and c is the sound speed. This pressure in the volume exerts a force on the diaphragm given by

$$F_d = P_d \times S_d = -\rho_o c^2 S_d^2 x / V = -K_d x, \tag{5.14.4}$$

where

$$K_d = \rho_o c^2 S_d^2 / V \tag{5.14.5}$$

is the equivalent spring constant of the air in N/m.

The force due to the air in the back volume adds to the restoring force due to the mechanical stiffness of the diaphragm. Including the air in the back volume, Eq. (5.14.1) becomes

$$m\ddot{x} + kx + K_d x + C\dot{x} = i_c BL, \tag{5.14.6}$$

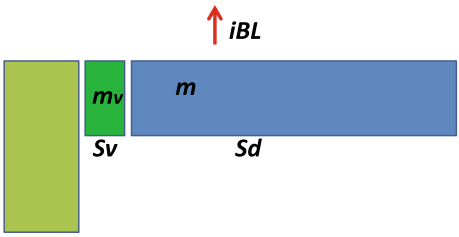


Fig. 5.15 Schematic representation of the air in a slit or vent that equalizes static pressure differences between the back volume and the exterior

5.14.3 Effect of the Air in the Vent

As will be examined in detail in the following, many loudspeaker systems incorporate a vent or port in the box to improve the sound radiation at low frequencies. This greatly complicates the analysis and design. In the following, we will construct a dynamical model of this system based on the forces acting on each element.

The air in the vent will be forced to move due to the fluctuating pressures both within the space behind the diaphragm and in the external sound field. We can again assume that the dimensions of this volume of moving air are much smaller than the wavelength of sound so that it can be represented by a single lumped mass of mass m_v. Figure 5.15 shows a rough representation of the air in the vent. An outward displacement of the air in the vent, x_v, causes a change in volume of the air in the back volume given by $-S_v x_v$ and a corresponding pressure given by

$$P_{vv} = -\rho_o c^2 S_v x_v / V, \tag{5.14.7}$$

where S_v is the area of the vent on which the pressure acts.

The pressure due to the motion of the air in the vent applies a restoring force on the mass of air in the vent given by

$$F_{vv} = -\rho_o c^2 S_v^2 x_v / V = -K_{vv} x_v, \tag{5.14.8}$$

where

$$K_{vv} = \rho_o c^2 S_v^2 / V. \tag{5.14.9}$$

The pressure due to the motion of the air in the vent also exerts a force on the diaphragm given by

$$F_{vd} = P_{vv} * S_d = -\rho_o c^2 S_v S_d x_v / V = -K_{vd} x_v, \tag{5.14.10}$$

where

$$K_{vd} = \rho_o c^2 S_v S_d / V. \tag{5.14.11}$$

Likewise, the pressure due to the motion of the diaphragm in Eq. (5.14.3) produces a force on the air in the vent that is given by

$$F_{dv} = P_d S_v = -\rho_o c^2 S_d S_v x / V = -K_{dv} x, \tag{5.14.12}$$

where $K_{dv} = K_{vd}$ as given in Eq. (5.14.11).

Because the air in the vent is squeezed through an opening, we may account for the viscous effects which result in a velocity-dependent restoring force on the air in the vent,

$$F_{viscous} = -c_v \dot{x}_v, \tag{5.14.13}$$

where c_v is a viscous damping coefficient that depends on the details of the flow.

Summing the forces on the moving elements of the system gives the following pair of governing equations

$$\begin{aligned} m\ddot{x} + (k + K_d)x + K_{vd} x_v + \mathcal{C}\dot{x} &= i_c BL, \\ m_v \ddot{x}_v + k_{vv} x_v + K_{dv} x + c_v \dot{x}_v &= 0. \end{aligned} \tag{5.14.14}$$

5.14.4 Response Due to Harmonic Sound Fields

If we assume the current, i_c varies harmonically with frequency ω then let $i_c = I_c e^{\hat{\imath}\omega t}$. The diaphragm and vent displacements may then be expressed as $x(t) = Xe^{\hat{\imath}\omega t}$ and $x_v(t) = X_v e^{\hat{\imath}\omega t}$. Equation (5.14.14) may be solved to give the steady-state response relative to the amplitude of the current, which may then be written as

$$\begin{pmatrix} X/I_c \\ X_v/I_c \end{pmatrix} = \begin{bmatrix} k + K_d - \omega^2 m + i\omega\mathcal{C} & k_{vd} \\ k_{dv} & k_{vv} - \omega^2 m_v + i\omega c_v \end{bmatrix}^{-1} \begin{pmatrix} BL \\ 0 \end{pmatrix}. \tag{5.14.15}$$

The response of the loudspeaker diaphragm is then

$$X/I_c = \frac{(k_{vv} - \omega^2 m_v + i\omega c_v) BL}{[(k + K_d - \omega^2 m + i\omega\mathcal{C}) * (k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv}]} \tag{5.14.16}$$

The response of the air in the vent is

$$X_v/I_c = \frac{-k_{vd} BL}{[(k + K_d - \omega^2 m + i\omega\mathcal{C}) * (k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv}]} \tag{5.14.17}$$

5.14.5 Sound Radiation from Loudspeakers in Vented Boxes

Equations (5.14.16) and (5.14.17) enable us to calculate the amplitudes of the displacements of the diaphragm and vent due to a harmonically varying current, $i(t)$. To construct an approximate model for the sound radiation due to both the diaphragm and vent, we will modify Eq. (5.5.8) to account for radiation from two pistons, consisting of the diaphragm with velocity $U_d(t) = i\omega X e^{\hat{i}\omega t}$ and the vent with velocity $U_v(t) = i\omega X_v e^{\hat{i}\omega t}$. We will again employ Eq. (5.3.5) where the velocity of the sphere, u_s is replaced by an equivalent velocity to account for the total volume of air moved by the diaphragm and the vent.

It is helpful to think of both the diaphragm and the vent as equivalent point sources where we wish to compute the far field sound power a low frequencies. We will find the net acoustic particle velocity and sound pressure at a distance r_0 from the loudspeaker.

In our analysis of the sound radiation from a simple spherical sound source, we considered a pulsating sphere with radius a having velocity u_s. Equations (5.3.2) through (5.1.20) give the pressure, velocity field and impedance of the pulsating sphere (repeated here for convenience),

$$p(r,t) = \frac{D_0}{r} e^{i(\omega t - kr)}, \tag{5.14.18}$$

$$u_r(r,t) = \frac{D_0(1+ikr)}{\rho_0 i\omega r^2} e^{i(\omega t - kr)}, \tag{5.14.19}$$

$$D_0 = u_s \frac{\rho_0 i\omega a^2}{(1+ika)} e^{ika}, \tag{5.14.20}$$

and,

$$\begin{aligned} Z = \frac{p(r,t)}{u_r(r,t)} &= \frac{\frac{D_0}{r} e^{i(\omega t - kr)}}{\frac{D_0(1+ikr)}{\rho_0 i\omega r^2} e^{i(\omega t - kr)}} \\ &= \frac{\rho_0 cikr}{1+ikr} = \frac{\rho_0 c((kr)^2 + ikr)}{1+(kr)^2}. \end{aligned} \tag{5.14.21}$$

When examining the low frequency response of a loudspeaker, we will assume that $ka << 1$ so that Eq. (5.14.20) becomes

$$D_0 \approx u_s \rho_0 i\omega a^2. \tag{5.14.22}$$

It is common to assume that the loudspeaker is mounted in an infinite baffle so that the radiation is into a half-space as was done in Sect. 5.5. The simple source then radiates into a hemisphere having area $S = 2\pi a^2$, half that considered for a simple spherical source. Recall that this is what was done in calculating the sound radiation from a surface source in Eq. (5.9.1). In this case, we may write D_0 as

$$D_0 \approx u_s \rho_0 i\omega a^2 = u_s \rho_0 i\omega \frac{S}{2\pi}. \tag{5.14.23}$$

To construct our simplified low-frequency model of sound radiation from a vented loudspeaker, we will compute the pressure and velocity at a position r_0 due to the pressures and velocities from both the diaphragm and the vent. Let P_d be the amplitude due to the diaphragm, which may be obtained by replacing the sphere velocity, u_s with U_d and the sphere area, S with S_d in Eq. (5.14.23),

$$P_d = U_d \rho_0 i\omega \frac{S_d}{2\pi}. \tag{5.14.24}$$

The amplitude due to the vent motion will be

$$P_v = U_v \rho_0 i\omega \frac{S_v}{2\pi}. \tag{5.14.25}$$

With the appropriate substitutions, Eqs. (5.14.24), (5.14.25), (5.14.18) and (5.14.19) enable us to write the pressures and velocities at the position r_0 due to the motions of the diaphragm and the vent, U_d and U_v,

$$p(r_0, t) = \frac{P_d + P_v}{r_0} e^{i(\omega t - kr_0)}, \tag{5.14.26}$$

$$u_r(r_0, t) = \frac{(P_d + P_v)(1 + ikr_0)}{\rho_0 i\omega r_0^2} e^{i(\omega t - kr_0)}. \tag{5.14.27}$$

The time averaged sound intensity at r_0 can be computed in a manner similar to that of our analysis of a spherical wave in Eq. (5.2.3),

$$< I >= \frac{1}{2}\Re[\overline{PU}^*] = \frac{1}{2}\Re[\overline{P}^*\overline{U}] = \frac{1}{2}|U_d S_d + U_v S_v|^2 \left(\frac{\rho_0 \omega}{2\pi}\right)^2 \times \frac{1}{r_0^2 \rho_0 c}. \tag{5.14.28}$$

If we consider the loudspeaker to be radiating into a half-space (where it is assumed to be mounted in an infinite baffle), the total power radiated will be the intensity multiplied by half the area of the sphere with radius r_0,

$$\Pi =< I > \times 2\pi r_0^2 = |U_d S_d + U_v S_v|^2 \frac{\omega^2 \rho_0}{4\pi c}. \tag{5.14.29}$$

Equations (5.14.24) through (5.14.24) give the total power radiated by the vented loudspeaker in terms of the velocities of the diaphragm and vent U_d and U_v.

5.15 Problems

1. A small sphere with radius $a = 5$ mm is driven with an internal fluctuating pressure with amplitude 1 pascal at all frequencies as was examined in Sect. 5.5. Suppose that the mechanical properties of the sphere are such that it has an effective impedance of

$$Z = i\omega M + \frac{K}{i\omega} + C, \tag{5.15.1}$$

   where the effective mass is $M = 0.4$ kg/m^2, the effective stiffness is $K = 10^5$ N/m^3 and the effective dashpot constant is $C = 150$ Ns/m^3. Plot the total sound power in watts produced by the sphere over the frequency range of $20 \leq f \leq 20{,}000$ Hz. Use a log-log plot. Interpret your results. In other words, explain why the frequency response looks the way it does based on your physical analysis of the system.
2. Suppose the sphere of Problem 1 has a radius that is reduced to $a = 4$ mm with the other properties remaining the same. Plot the total sound power in watts produced by the sphere over the frequency range of $20 \leq f \leq 20,000$ Hz. Compare your results with those of Problem 1.
3. Determine the sound pressure in decibels on the interior surface of a radially oscillating sphere as a function of frequency for a sphere of nominal radius $r = 0.25$ m with a surface velocity of $u_a = 1$ m/s. Plot your result versus frequency in Hertz using a logarithmic frequency axis.
4. A harmonic point sound source is a distance of $d = 0.4$ meters from a rigid wall as shown in Fig. 5.4. Use an image source to find the sound pressure at a point A where $r = 5$ meters and $\phi = \pi/3$ rad. Plot the magnitude of the pressure at A as a function of frequency in Hertz over the frequency range of $20 \leq f \leq 20{,}000$ Hz. Use a log-log plot.
5. A thin cylinder vibrates with a radial velocity $U(x, t) = U_0 e^{i\omega t}$ where $U_0 = 1$ m/s. It has a radius $a = 0.001$ m and length $L = 0.1$ m. We would like to calculate the sound field at the point A where $r = 1$ m and $\theta = \pi/3$ as shown in Fig. 5.8. (a) calculate the frequencies (in Hertz) where the amplitude of the pressure at A will be a minimum and (b) plot the amplitude of the pressure at A as a function of frequency for a range of frequencies from 20 to 20,000 Hertz. Use a log-log plot.

6. Suppose we wish to calculate the amplitude of the sound pressure as in Problem 5 but at the much closer location where $r = 0.1$ m with $\theta = \pi/3$ as before. In this case it is no longer true that $r >> L/2$. The pressure should be estimated from the more accurate expression, Eq. (5.8.8). In this expression, as in Eq. (5.8.11), we have assumed that the radius of the cylinder is such that $ka << 1$. (a) Integrate Eq. (5.8.8) numerically and use this result to plot the amplitude of the pressure at A as a function of frequency for a range of frequencies from 20 to 20,000 Hz. (b) Use Eq. (5.8.11) to calculate the pressure and plot this on the same plot as in (a). Use a log-log plot.
7. Suppose that the velocity of the cylinder of Problem 5 is not uniform along the length but is instead given by

$$U(x,t) = U_0 \cos(\pi(x + L/2)/L)e^{i\omega t}. \tag{5.15.2}$$

Modify Eq. (5.8.8) for the case where the cylinder velocity varies with x and integrate this numerically to determine the pressure at A where $r = 0.1$ m and $\theta = \pi/3$ as before.
8. In high fidelity audio and sound reinforcement systems it is common to use a linear array of loudspeaker drivers. The sound field radiated from a column of loudspeakers could be examined approximately by representing it as a cylindrical radiator as shown in Fig. 5.8. We will assume that all of the drivers vibrate with the same velocity due to a force applied by the current in their voice coils. The drivers are all identical and have resonant frequencies of 500 Hz so that $\omega_0 = 2\pi * 500$ and identical damping ratios, $\zeta = 0.707$. The velocity of the equivalent cylinder radiator is then proportional to

$$U(\omega) = G\frac{i\omega}{\omega_0^2 - \omega^2 + 2\omega_0\zeta i\omega}, \tag{5.15.3}$$

where G is a proportionality constant which we will assume to be $G = 1$. The height (or length) of the array is $L = 2$ m. Let the effective radius of the cylinder be $a = 0.01$ meter. We would like to examine the uniformity of the sound field for listeners at a distance $Y = 20$ m from the center of the cylinder, where Y is in the direction orthogonal to the cylinder's axis of symmetry. If the center of the cylinder is at $(x, y) = (0, 0)$, calculate the sound pressure at the following locations: (0, 20), (0.5, 20), (1.0, 20), (1.5, 20), and (2.0, 20), where the distances are measured in meters. plot the amplitude of the pressure for each location as a function of frequency for a range of frequencies from 20 to 20,000 Hz. Use a log-log plot. It is preferable to display all of the results on one plot.
9. For the cylinder in Problem (8), determine the length of the cylindrical radiator so that all of the listening locations listed in that problem will receive pressure amplitudes that differ by no more than a factor of 2 at 8 kHz. Plot the results of this design as a function of frequency as you did in the previous problem.
10. A Matlab file, "multipleloudspeakerwithdrivervib102614.m" has been made available. This program may contain errors. Examine the program for consis-

tency with the section titled "Multiple Piston Radiators." Develop a set of test cases to determine if the results from the program are reliable.

11. We wish to analyze the sound field radiated by an array of loudspeaker drivers that is two meters long such as was modeled in question (8) above. Assume that all of the drivers have identical properties and their resonant frequencies are 500 Hz with damping ratios of $\zeta = 0.707$. The radius of each driver and the number of drivers is up to you. Use the program (or a modified version) "multipleloudspeakerwithdrivervib102614.m" to calculate the field at various positions, including those of Problem (8). Compare your results qualitatively with those obtained for the line source of Problems (8) and (9).

Chapter 6
Computer-Aided Acoustics



In the following we will outline the use of computer-aided design models for the analysis of complicated acoustic spaces. The overall goal is to solve the acoustic wave equation for the domain and satisfy the boundary conditions when the geometry is considerably more complicated than can be properly accommodated in analytical solutions. In many studies of sound fields that rely on computational approaches such as the finite element or boundary element methods, the most cumbersome task is in properly describing the geometric details of the domain and boundary conditions. Having the geometry described in a computationally convenient way, the formulation and solution of the equations can be relatively straightforward. The main goal of the following is to provide a brief introduction to the process for formulating and numerically solving the equations based on a geometric description that is readily available through the use of common software used in computer-aided design.

The approach used in the following is based on the use of the Green's function for an unbounded acoustical domain. This is used to first calculate the sound field due to radiation from a surface that vibrates in an infinite baffle. The formulation is then generalized to construct an integral equation that can be solved to find the sound field created by an arbitrary vibrating surface in a domain that also contains rigid surfaces. Finally, the numerical solution of the integral equation is then examined. The method described here is essentially the boundary element method, adapted to the use of computer-aided design tools.

There is a long and rich literature concerning computational methods for predicting sound fields and the interested reader is encouraged to examine these more in-depth contributions. For a fairly comprehensive discussion the thesis by Peter Juhl [1] should be consulted. This work has been continuously extended and has led to publicly available Matlab software available at http://www.openbem.dk/. Also see [2–4].

While the present discussion focuses on the boundary element method rather than the finite element method (or others), no attempt to convince the reader to prefer one method over another is intended. Ease of program development, computational

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_6

efficiency, and accuracy depend strongly on specific applications, computer technology, and personal preference. Regardless of one's choice of the preferred method, it is essential that multiple approaches be used to verify one's results.

6.1 The Green's Function

The calculation of the sound field due to a radiator in a multidimensional domain is greatly facilitated knowing the appropriate Green's function. The task of finding the Green's function is similar to that of finding the sound field in a uniform, three-dimensional space due to a point radiator. In the case of spherical symmetry, the solution to the acoustic wave equation,

$$\nabla^2 p = \frac{\ddot{p}}{c^2} \tag{6.1.1}$$

is

$$p(r,t) = p_o \frac{e^{i(\omega t - kr)}}{r} \tag{6.1.2}$$

The Green's function in the infinite domain (in the case of spherical symmetry) is the solution to

$$\nabla^2 G + k^2 G = -\delta(\bar{r} - \bar{r}_s), \tag{6.1.3}$$

where $G(\bar{r}, \bar{r}_s)$ is the Green's function, and $\bar{r}$ and $\bar{r}_s$ are vectors giving the positions of two points in space. In cartesian coordinates these vectors are

$$\begin{aligned} \bar{r} &= x\hat{i} + y\hat{j} + z\hat{k} \\ \bar{r}_s &= x_s\hat{i} + y_s\hat{j} + z_s\hat{k}. \end{aligned} \tag{6.1.4}$$

The Dirac delta function of the vector argument on the right side of Eq. (6.1.3) is equivalent to

$$\delta(\bar{r} - \bar{r}_s) = \delta(x - x_s) \times \delta(y - y_s) \times \delta(z - z_s). \tag{6.1.5}$$

The solution to Eq. (6.1.3) is

$$G(\bar{r}, \bar{r}_s) = \frac{e^{\pm ik(|\bar{r} - \bar{r}_s|)}}{4\pi|\bar{r} - \bar{r}_s|}, \tag{6.1.6}$$

where the negative sign corresponds to an outgoing spherical wave and the positive sign corresponds to an incoming wave. In calculations of the sound field in an infinite

domain, the incoming wave is assumed to not exist since there are no boundaries for the outgoing wave to reflect from. We will therefore neglect the incoming wave.

In the case where $\bar{r} \neq \bar{r}_s$, Eq. (6.1.6) is the solution of Eq. (6.1.3). This can be shown by following the same process used to examine Eq. (5.1.16). In this case, the δ function is zero and it will still be a solution if the factor of 4π is replaced by an arbitrary constant. In order to verify that it is also a solution at the point where $\bar{r} = \bar{r}_s$, we will integrate the equation over a spherical volume centered at the point where $|\bar{r} - \bar{r}_s| = 0$ and then reduce the radius of this sphere to zero. The normalization factor of 4π will be found to be essential for Eq. (6.1.6) to also be a solution at this point of singularity of the δ function.

To simplify things, let $R = |\bar{r} - \bar{r}_s|$. We will integrate Eq. (6.1.3) over a sphere with radius R_0. The volume integral of Eq. (6.1.3) is

$$\begin{aligned}
&\int\int\int (\nabla^2 G + k^2 G) dv = -\int\int\int \delta(\bar{r} - \bar{r}_s) dv \\
&= \int\int\int (\bar{\nabla} \cdot \bar{\nabla} G + k^2 G) dv = -1 \\
&= \int\int \hat{n} \cdot \bar{\nabla} G ds + k^2 \int\int\int G dv = -1,
\end{aligned} \tag{6.1.7}$$

where we have used Gauss' theorem to transform the first volume integral into a surface integral and $\hat{n}$ is the unit vector on the surface pointing outward from the enclosed volume. Since we have radial symmetry,

$$\begin{aligned}
&n \cdot \bar{\nabla} G = \frac{\partial}{\partial R}\left(\frac{e^{-ikR}}{4\pi R}\right) \\
&= -ik\frac{e^{-ikR}}{4\pi R} - \frac{e^{-ikR}}{4\pi R^2}
\end{aligned} \tag{6.1.8}$$

The surface integral in Eq. (6.1.7) is performed over the sphere with radius R_0,

$$\begin{aligned}
&\int\int \hat{n} \cdot \bar{\nabla} G ds = 4\pi R_0^2 \frac{\partial G}{\partial R}|_{R=R_0} \\
&= 4\pi R_0^2 \left(-\frac{ike^{-ikR_0}}{4\pi R_0} - \frac{e^{-ikR_0}}{4\pi R_0^2}\right)
\end{aligned} \tag{6.1.9}$$

Taking the limit as $R_0 \Rightarrow 0$ gives

$$\lim_{R_0 \to 0} \int\int \hat{n} \cdot \bar{\nabla} G ds = -1. \tag{6.1.10}$$

The volume integral in Eq. (6.1.7) may be written as

$$k^2 \int\int\int G dv = k^2 \int_0^{R_0} \frac{4\pi R^2 e^{-ikR}}{4\pi R} dR$$
$$= -ik(e^{-ikR_0}(1 - R_0) - 1) \tag{6.1.11}$$

Taking the limit of this volume integral as $R_0 \Rightarrow 0$ gives

$$\lim_{R_0 \to 0} k^2 \int\int\int G dv$$
$$= \lim_{R_0 \to 0} -ik(e^{-ikR_0}(1 - R_0) - 1) = 0 \tag{6.1.12}$$

Equations (6.1.10) and (6.1.12) show that Eq. (6.1.7) is satisfied for an infinitesimal volume at the location where the delta function in Eq. (6.1.3) becomes singular. Equation (6.1.6) then provides a solution to Eq. (6.1.3) in the infinite, unbounded domain.

6.2 Green's Function for Infinite Half-Space Bounded by an Infinite Rigid Plane

To calculate the sound field radiated by a vibrating surface that is surrounded by an infinite baffle (i.e., a rigid plane), it is very helpful to first construct the Green's function for the domain in which no surface is moving. This function is the solution to Eq. (6.1.3) that satisfies the boundary condition for a rigid plane. We will also neglect the negative solution in Eq. (6.1.6) since the domain is infinite and no waves will exist that are propagating toward the radiator.

Let the rigid plane be at $z = 0$. Euler's equation gives the boundary condition on this plane as

$$\hat{n} \cdot \bar{\nabla} G|_{z=0} = 0. \tag{6.2.1}$$

This boundary condition can be satisfied through the use of an image source. Let

$$G = \frac{e^{-ik(|\bar{r} - \bar{r}'|)}}{4\pi|\bar{r} - \bar{r}'|} + \frac{e^{-ik(|\bar{r} - \bar{r}''|)}}{4\pi|\bar{r} - \bar{r}''|}, \tag{6.2.2}$$

where

$$\bar{r} - \bar{r}' = (x - x')\hat{i} + (y - y')\hat{j} + (z - z')\hat{k}$$
$$\bar{r} - \bar{r}'' = (x - x')\hat{i} + (y - y')\hat{j} + (z + z')\hat{k}, \tag{6.2.3}$$

G is essentially the sound field at $\bar{r}$ resulting from normalized sound sources at $\bar{r}'$ and at $\bar{r}''$ (the image source on the left side of the boundary). The Green's function G in Eq. (6.2.2) is then a function of (x, y, z) and (x', y', z'), $G(x, y, z; x', y', z')$.

For our plane boundary at $z = 0$, Eq. (6.2.1) becomes

$$\hat{n} \cdot \nabla \bar{G}|_{z=0} = \frac{\partial G}{\partial z}|_{z=0} = 0. \tag{6.2.4}$$

To show that this is satisfied through the use of our image source, it is convenient to let

$$\begin{aligned} R' &= |\bar{r} - \bar{r}'| = \sqrt{(x - x')^2 + (y - y')^2 + (z - z')^2} \\ R'' &= |\bar{r} - \bar{r}''| = \sqrt{(x - x')^2 + (y - y')^2 + (z + z')^2}. \end{aligned} \tag{6.2.5}$$

Equation (6.2.4) then may be written as

$$\frac{\partial G}{\partial z}|_{z=0} = \frac{\partial G}{\partial R'}\frac{\partial R'}{\partial z}|_{z=0} + \frac{\partial G}{\partial R''}\frac{\partial R''}{\partial z}|_{z=0}. \tag{6.2.6}$$

Note that from Eqs. (6.2.5) at $z = 0$, $R' = R''$ so that $\frac{\partial G}{\partial R'} = \frac{\partial G}{\partial R''}$ there. Also,

$$\frac{\partial R'}{\partial z} = \frac{(z - z')}{R'^2}, \quad \frac{\partial R''}{\partial z} = \frac{(z + z')}{R''^2}. \tag{6.2.7}$$

Evaluating Eqs. (6.2.7) at $z = 0$ and substituting into Eq. (6.2.6) shows that the boundary condition Eq. (6.2.4) is satisfied.

Equation (6.2.2) is the solution to

$$\nabla^2 G + k^2 G = -\delta(\bar{r} - \bar{r}') - \delta(\bar{r} - \bar{r}''). \tag{6.2.8}$$

Keep in mind that $\bar{r}''$ is the position of the image source, which is on the opposite side of the infinite boundary from our listening position at $\bar{r}$.

We are now ready to set up the solution to the problem of calculating the sound pressure due to vibration on a portion of the infinite boundary. The sound pressure must satisfy

$$\nabla^2 p + k^2 p = 0 \tag{6.2.9}$$

subject to the following boundary condition

$$\begin{aligned} -\hat{n} \cdot \nabla p &= i\omega\rho_0 U \quad \text{on} \quad S_1 \\ &= 0 \quad \text{otherwise}, \end{aligned} \tag{6.2.10}$$

where $\hat{n}$ is a unit normal vector on the boundary that points in the positive z direction and U is the velocity of the area S_1.

Multiplying Eq. (6.2.8) by $p(\bar{r})$, Eq. (6.2.9) by G, subtracting and integrating over the entire volume on the side of the boundary for which $z \geq 0$ gives

$$\begin{aligned} &\int\int\int (G\nabla^2 p - p\nabla^2 G + k^2(Gp - pG))dv \\ &= \int\int\int (p(\bar{r})(\delta(\bar{r} - \bar{r}') + \delta(\bar{r} - \bar{r}''))dv \end{aligned} \tag{6.2.11}$$

Note that the integration over second delta function on the right side of Eq. (6.2.11) will be zero since the integration is carried out only on the side of the plane where $z > 0$. Equation (6.2.11) then becomes

$$\int\int\int (G\nabla^2 p - p\nabla^2 G)dv = \int\int\int (G\bar{\nabla} \cdot \bar{\nabla} p - p\bar{\nabla} \cdot \bar{\nabla} G)dv = p(\bar{r}'). \tag{6.2.12}$$

These integrals may be converted to surface integrals through the use of Gauss' theorem,

$$\int\int (G\hat{N} \cdot \bar{\nabla} p - p\hat{N} \cdot \bar{\nabla} G)ds = p(\bar{r}'), \tag{6.2.13}$$

where the integration covers the entire boundary of the domain on the side of the plane where $z \geq 0$. It is important to note that in this case, because of Gauss' theorem, $\hat{N}$ is a unit normal vector pointing in the outward direction from the volume that is integrated in Eq. (6.2.12). This vector will then point into the boundary at $z = 0$ and will thus point in the negative z direction. As a result, $\hat{n} = -\hat{N}$, where $\hat{n}$ is the unit normal vector pointing in the positive z direction as in the boundary condition in Eq. (6.2.10).

From Eq. (6.2.10), the first integral then becomes

$$\int\int G\hat{N} \cdot \bar{\nabla} p ds = \int\int_{S_1} Gi\omega\rho_0 U ds, \tag{6.2.14}$$

where the integration is taken only over the portion of the area S_1 where the surface velocity U is nonzero.

The second term in the integrand of Eq. (6.2.13) is zero because the Green's function satisfies the boundary condition in Eq. (6.2.1). The sound pressure at a point $\bar{r}'$ is then

$$p(\bar{r}') = \int\int_{S_1} Gi\omega\rho_0 U ds, \tag{6.2.15}$$

which can usually be evaluated numerically. This result is essentially identical to the Rayleigh integral in Eq. (5.9.4).

6.3 Integral Equation for the General Sound Radiation Problem

Equation (6.2.15) provides a convenient method of calculating the sound radiated by a portion of a surface on an infinite plane. Unfortunately, of course, there are few practical situations where the domain can be accurately described by this model. We would like to construct a more general approach that enables the calculation of the sound field due to a finite vibrating surface in any orientation and accompanied by rigid surfaces, again in any orientation. This situation occurs when a sound source such as a loudspeaker is placed in an anechoic chamber. The loudspeaker driver radiates sound but the enclosure that contains the driver substantially affects the sound radiation and cannot be accounted for by using Eq. (6.2.15). One may also have a radiator that does not contain an enclosure such as a simple unbaffled loudspeaker driver. There may also be rigid surfaces in the domain that reflect sound which have significant affects on the field.

To set up the analysis of these more general situations, we will use an approach that extends that of the previous section. In this case, let's assume we use the Green's function for the infinite domain as given in Eq. (6.1.6), which again, is the solution to Eq. (6.1.3) in the unbounded domain. This Green's function will not satisfy any of the boundary conditions of the present problem unlike what was used in the case of the infinite baffle in the previous section.

Let this more general domain contain a total surface area S that contains a portion S_1 that moves with velocity U. As above, the sound pressure p must satisfy

$$\nabla^2 p + k^2 p = 0 \tag{6.3.1}$$

subject to the following boundary condition

$$\begin{aligned} -\hat{n} \cdot \nabla p &= i\omega\rho_0 U \quad \text{on} \quad S_1 \\ &= 0 \quad \text{otherwise} \end{aligned} \tag{6.3.2}$$

where $\hat{n}$ is a unit outward normal vector on the boundary pointing toward the acoustic domain and U is the velocity of the surface.

Again proceeding as above, multiplying Eq. (6.1.3) by $p(\bar{r})$, Eq. (6.3.1) by G, subtracting and integrating over the entire volume of the domain gives

$$\begin{aligned} &\int\int\int (G\nabla^2 p - p\nabla^2 G + k^2(Gp - pG))dv \\ &= \int\int\int p(\bar{r})(\delta(\bar{r} - \bar{r}'))dv \end{aligned} \tag{6.3.3}$$

Equation (6.3.3) then becomes

$$\int\int\int (G\nabla^2 p - p\nabla^2 G)dv = \int\int\int (G\bar{\nabla}\cdot\bar{\nabla}p - p\bar{\nabla}\cdot\bar{\nabla}G)dv = p(\bar{r}'). \tag{6.3.4}$$

As in Eqs. (6.2.12) and (6.2.13), these integrals may be converted to surface integrals through the use of Gauss' theorem,

$$\int\int (G\hat{N}\cdot\bar{\nabla}p - p\hat{N}\cdot\bar{\nabla}G)ds = p(\bar{r}'), \tag{6.3.5}$$

where the integration covers the entire boundary of the domain and the point $\bar{r}'$ is any point in the interior. We will need to modify this result slightly to account for the situation where $\bar{r}'$ is on the surface. Note that, as before, because of Gauss' theorem, $\hat{N}$ is a unit normal vector pointing toward the surface, in the outward direction from the volume that is integrated in Eq. (6.3.5). Recall that $\hat{n} = -\hat{N}$, where $\hat{n}$ is the unit normal vector pointing away from the surface, toward the interior of the volume as in the boundary condition in Eq. (6.3.2).

Using the boundary condition Eq. (6.3.2) in Eq. (6.3.5), the first integral becomes

$$\int\int G\hat{N}\cdot\bar{\nabla}p\,ds = \int\int_{S_1} Gi\omega\rho_0 U\,ds, \tag{6.3.6}$$

where the integration is taken only over the portion of the area S_1 where the surface velocity U is nonzero.

Equation (6.3.5) may then be written as

$$p(\bar{r}') + \int\int (p\hat{N}\cdot\bar{\nabla}G)ds = \int\int_{S_1} Gi\omega\rho_0 U\,ds, \tag{6.3.7}$$

where, again, $\bar{r}'$ indicates any point within the interior. To evaluate Eq. (6.3.7), one must first determine p on all surfaces in order to evaluate the integral on the left side of the equation. This can be accomplished by confining the evaluation point $\bar{r}'$ to points on the surface S. As mentioned above, however, Eq. (6.3.7) follows from Eqs. (6.3.3) and (6.3.4) where the volume integral over the Dirac delta function gave

$$\int\int\int p(\bar{r})(\delta(\bar{r} - \bar{r}'))dv = p(\bar{r}'). \tag{6.3.8}$$

When carrying out this integral for a point $\bar{r}'$ within the volume, the domain of integration could be a small sphere enclosing the point $\bar{r}'$. To avoid confusion between interior points and locations on the surface, let the surface locations be denoted by $\bar{r}_s'$. For interior points, the volume integral is as before,

$$\int\int\int p(\bar{r})(\delta(\bar{r}-\bar{r}'))dv = p(\bar{r}'). \tag{6.3.9}$$

If, however, we wish to compute the pressure on a point on the surface rather than an interior point, the integration domain must include only that volume that is within the enclosure. Let locations on the surface be denoted by $\bar{r}_s'$. For surface points we must carry out the integration over the delta function over a hemisphere rather than a sphere. Note also that if the point $\bar{r}_s'$ is on an edge or a corner, the integration domain must be modified accordingly so that it is over only locations that are in the interior of the acoustic space. When evaluating the integral at a point on a smooth surface, the result will then be one half that for an interior point in Eq. (6.3.9),

$$\int\int\int p(\bar{r})(\delta(\bar{r}-\bar{r}_s'))dv = \frac{1}{2}p(\bar{r}_s'). \tag{6.3.10}$$

The pressure in the interior is then calculated using

$$p(\bar{r}') + \int\int(p\hat{N}\cdot\bar{\nabla}G)ds = \int\int_{S_1} Gi\omega\rho_0 U ds. \tag{6.3.11}$$

Advance knowledge of the pressure on the surface is required to evaluate this. Equations (6.3.3) and (6.3.10) lead to a Fredholm integral equation that may be solved for the surface pressure, $p(\bar{r}_s')$

$$\frac{p(\bar{r}_s')}{2} + \int\int(p\hat{N}\cdot\bar{\nabla}G)ds = \int\int_{S_1} Gi\omega\rho_0 U ds. \tag{6.3.12}$$

6.4 Numerical Solution

The determination of $p(\bar{r}_s')$ in Eq. (6.3.12) can be a formidable task. We will attempt to construct a computational approach that can be applied to a wide range of spatial configurations.

The Green's function G in Eq. (6.3.12) is a function of both $\bar{r}$ and $\bar{r}'$,

$$G(\bar{r}',\bar{r}) = \frac{e^{\pm ik(|\bar{r}'-\bar{r}|)}}{4\pi|\bar{r}'-\bar{r}|}. \tag{6.4.1}$$

The Green's function is clearly symmetric so that $G(\bar{r}',\bar{r}) = G(\bar{r},\bar{r}')$.

Suppose that the spatial domain is discretized into n reasonably small areas, S_j, where $j = 1,\ldots,n$. The velocity of surface j will be U_j. Let $\bar{r}_j$ be the position vector describing the location of $\bar{r}$, the midpoint of S_j. The location of $\bar{r}'$ will be denoted by $\bar{r}_l$, for $l = 1,\ldots,n$. The integrals in Eq. (6.3.12) may then be expressed as summations over the areas S_j, which gives

$$\frac{p(\bar{r}_s')}{2} = -\sum_j \int\int_{S_j} (p(\bar{r}_s)\hat{N}\cdot\nabla\bar{G})ds + \sum_{j=1}^{n} i\omega\rho_0 \int\int_{S_j} GUds. \tag{6.4.2}$$

Equation (6.4.2) may be evaluated at any $\bar{r}_s'$. Let p_l be the spatial average of $p(\bar{r}_s')$ over the area S_l. Expressing the area average of Eq. (6.4.2) gives

$$\frac{p_l}{2} = \frac{< p(\bar{r}_s') >}{2} = -\frac{1}{S_l}\int\int_{S_l}\sum_{j=1}^{n}\int\int_{S_j}(p(\bar{r}_s)\hat{N}\cdot\nabla\bar{G})dsds' + \frac{1}{S_l}\int\int_{S_l}\sum_{j=1}^{n} i\omega\rho_0 U_j \int\int_{S_j} Gdsds' \tag{6.4.3}$$

If the areas S_l are sufficiently small and if $p(\bar{r})$ is nearly constant over each area, Eq. (6.4.3) may be approximated by pulling the average, p_j outside the inner integrals. This gives

$$\begin{aligned}\frac{p_l}{2} &\approx -\frac{1}{S_l}\int\int_{S_l}\sum_{j=1}^{n} p_j \int\int_{S_j}(\hat{N}\cdot\nabla\bar{G})dsds' + \frac{1}{S_l}\int\int_{S_l}\sum_{j=1}^{n} i\omega\rho_0 U_j\int\int_{S_j} Gdsds' \\ &= -\sum_{j=1}^{n} p_j \frac{1}{S_l}\int\int_{S_l}\int\int_{S_j}(\hat{N}\cdot\nabla\bar{G})dsds' + \sum_{j=1}^{n} i\omega\rho_0 U_j \frac{1}{S_l}\int\int_{S_l}\int\int_{S_j} Gdsds'\end{aligned} \tag{6.4.4}$$

Let

$$B_{lj} = \frac{1}{S_l}\int\int_{S_l}\int\int_{S_j}(\hat{N}\cdot\nabla\bar{G})dsds', \tag{6.4.5}$$

and

$$C_{lj} = i\omega\rho_0\frac{1}{S_l}\int\int_{S_l}\int\int_{S_j} Gdsds'. \tag{6.4.6}$$

The estimation of the double surface integrals in Eqs. (6.4.5) and (6.4.6) comprises the most difficult numerical task in this approach to analyzing sound fields. This problem will be addressed in Sect. 6.8 below. Assuming that B_{lj} and C_{lj} can be computed and stored, Eq. (6.4.4) then becomes

$$\frac{p_l}{2} \approx -\sum_{j=1}^{n} p_j B_{lj} + \sum_{j=1}^{n} U_j C_{lj}. \tag{6.4.7}$$

Equation (6.4.7) may be expressed as a linear system of equations for p_j,

$$\sum_{j=1}^{n} K_{lj} p_j = \sum_{j=1}^{n} U_j C_{lj} \quad \text{for} \quad l = 1, \ldots, n. \tag{6.4.8}$$

This may be expressed in matrix form as

$$[K]P = F \tag{6.4.9}$$

where the elements of $[K]$ are

$$K_{lj} = \frac{\delta_{lj}}{2} + B_{lj}, \tag{6.4.10}$$

where δ_{lj} is the Kroniker delta function, and the elements of the vector F are

$$F_l = \sum_{j=1}^{n} U_j C_{lj}. \tag{6.4.11}$$

6.5 Effect of Boundary Impedance

In the above discussion, we limited our attention to the case where the boundary conditions were cast in terms of an imposed velocity on the surface (either specified to have amplitude U or equal to zero). Fortunately, the method can easily be extended to the case where rather than imposing a zero velocity, we specify a frequency dependent and location dependent boundary impedance, Z. To accomplish this, we must modify Eq. (6.3.2). Again, the partial differential equation we wish to solve for the pressure is

$$\nabla^2 p + k^2 p = 0. \tag{6.5.1}$$

To account for a general impedance boundary, the boundary condition will be written as

$$\begin{aligned} -\hat{n} \cdot \nabla p &= i\omega\rho_0 U \quad \text{on} \quad S_1 \\ &= i\omega\rho_0 \frac{p}{Z} \quad \text{on} \quad S_2 \end{aligned} \tag{6.5.2}$$

where S_2 is the entire surface area other than S_1. As before, $\hat{n}$ is a unit outward normal vector on the boundary pointing toward the interior of the acoustic domain and U is the velocity of the surface S_1. Our previous results correspond to the special case where the impedance of the boundary is infinite.

The two possible boundary conditions of Eq. (6.5.2) may be easily accommodated by adding them together and defining Z to be infinite on S_1 and U to vanish on S_2,

$$-\hat{n} \cdot \nabla p = i\omega\rho_0 U + i\omega\rho_0 \frac{p}{Z}, \tag{6.5.3}$$

where $U = 0$ on S_2 and $Z = \infty$ on S_1.

Proceeding exactly as in Eqs. (6.4.2) through (6.4.7), we find that if a portion of the surface has a finite boundary impedance so that the boundary condition is as in Eq. (6.5.3), we may account for this by modifying Eq. (6.4.7) to be

$$\frac{p_l}{2} \approx -\sum_{j=1}^{n} p_j B_{lj} + \sum_{j=1}^{n} \left(U_j + \frac{p_j}{Z_j} \right) C_{lj}. \tag{6.5.4}$$

Equation (6.4.8) becomes

$$\sum_{j=1}^{n} \left(K_{lj} + \frac{C_{lj}}{Z_j} \right) p_j = \sum_{j=1}^{n} K'_{lj} p_j = \sum_{j=1}^{n} U_j C_{lj} \quad \text{for} \quad l = 1, \ldots, n, \tag{6.5.5}$$

where

$$K'_{lj} = K_{lj} + \frac{C_{lj}}{Z_j}. \tag{6.5.6}$$

Equation (6.5.5) is again a linear system of equations for the unknown surface pressures p_j.

6.6 Sound Field Due to a Point Source in an Enclosure

We would now like to extend the results presented above to the situation where there is a point sound source at a location $\bar{r}_p$. This point source will enable us to estimate the sound pressure on complex surfaces that results from some externally imposed sound source rather than the velocity of the surface itself. To account for this, we must modify the Helmholtz equation (6.5.1) to account for the point source of sound. When we first considered spherical waves we assumed that a sound field existed in the form of Eq. (5.1.16) without examining the partial differential equation that it solves. Having examined the solution to Eq. (6.1.3) throughout the domain, we can write the solution to

$$\nabla^2 P + k^2 P = -4\pi D_0 \delta(\bar{r} - \bar{r}_p), \tag{6.6.1}$$

where $\bar{r}$ is a vector giving the listener location and $\bar{r}_p$ is the location of the point source. The solution is

$$p(\bar{r}, \bar{r}_p, t) = \frac{D_0}{|\bar{r} - \bar{r}_p|} e^{i(\omega t - k|\bar{r} - \bar{r}_p|)}, \tag{6.6.2}$$

where D_0 is a constant that accounts for the equivalent strength of the source. As before, if the source is a small pulsating sphere with radius a,

$$D_0 = \frac{\rho_0 c i k a^2}{1 + ika} u_s e^{ika}. \tag{6.6.3}$$

If the domain contains a point source at the location $\bar{r}_p$, we may then extend Eq. (6.5.1) to be

$$\nabla^2 p + k^2 p = -4\pi D_0 \delta(\bar{r} - \bar{r}_p). \tag{6.6.4}$$

To account for a general impedance boundary, the boundary condition is again

$$\begin{aligned} -\hat{n} \cdot \nabla p &= i\omega\rho_0 U \quad \text{on} \quad S_1 \\ &= i\omega\rho_0 \frac{p}{Z} \quad \text{on} \quad S_2 \end{aligned} \tag{6.6.5}$$

where S_2 is the entire surface area other than S_1. As before, $\hat{n}$ is a unit outward normal vector on the boundary pointing toward the acoustic domain and U is the velocity of the surface S_1.

Multiplying Eq. (6.1.3) by $p(\bar{r})$, where $\bar{r}'$ is a point in the volume (i.e., not on the surface), multiplying Eq. (6.6.4) by $G(\bar{r}, \bar{r}')$, subtracting and integrating throughout the domain give

$$\begin{aligned} &\int\int\int (G\nabla^2 p - p\nabla^2 G + k^2(Gp - pG))dv \\ &= \int\int\int (p(\bar{r})(\delta(\bar{r} - \bar{r}')) - G(\bar{r}, \bar{r}')(4\pi D_0 \delta(\bar{r} - \bar{r}_p)))dv \\ &= p(\bar{r}') + 4\pi D_0 G(\bar{r}_p, \bar{r}') \end{aligned} \tag{6.6.6}$$

As in Eq. (6.3.10), if the location at which we would like to calculate the pressure is on the surface, $\bar{r}_s$, the integration over the Dirac delta function yields half the value for points within the volume. For points on the surface (which must be determined before evaluating the integral in Eq. (6.6.6)), we then obtain

$$\begin{aligned} &\int\int\int (G\nabla^2 p - p\nabla^2 G + k^2(Gp - pG))dv \\ &= \int\int\int (p(\bar{r})(\delta(\bar{r} - \bar{r}')) - G(\bar{r}, \bar{r}')(4\pi D_0 \delta(\bar{r} - \bar{r}_p)))dv \\ &= \frac{p(\bar{r}_s')}{2} + 4\pi D_0 G(\bar{r}_p, \bar{r}') \end{aligned} \tag{6.6.7}$$

Following the same steps used to obtain Eq. (6.6.7) gives,

$$\frac{p(\bar{r}_s')}{2} + \int\int (p\hat{N}\cdot\nabla\bar{G})ds - \int\int_{S_2} Gi\omega\rho_0\frac{p}{Z}ds = \int\int_{S_1} Gi\omega\rho_0 U ds + 4\pi D_0 G(\bar{r}_p,\bar{r}'), \tag{6.6.8}$$

where again, the first integral on the left side of this equation is taken over the entire boundary.

To construct a numerical solution to Eq. (6.6.8) we must evaluate $G(\bar{r}_p,\bar{r}')$ in the last term on the right hand side. We will again assume that the surface area is discretized into a large number of small areas S_l for $l = 1,\ldots,n$ and that the midpoint of each of these areas is $\bar{r}_l$. Similar to Eq. (6.1.4), let

$$\bar{r}_l' = R_{l1}\hat{i} + R_{l1}\hat{j} + R_{l1}\hat{k}, \tag{6.6.9}$$

where R_{l1}, R_{l2}, and R_{l3} are the cartesian coordinates of $\bar{r}_l'$. The position of the point source will be given by

$$\bar{r}_p = R_{p1}\hat{i} + R_{p2}\hat{j} + R_{p3}\hat{k}. \tag{6.6.10}$$

We may then approximate $G(\bar{r}_p,\bar{r}')$ by

$$G_{lp} = \frac{e^{-ikD_{lp}}}{4\pi D_{lp}}, \tag{6.6.11}$$

where

$$D_{lp} = |\bar{r}_l - \bar{r}_p| = \sqrt{(R_{l1}-R_{p1})^2 + (R_{l2}-R_{p2})^2 + (R_{l3}-R_{p3})^2}. \tag{6.6.12}$$

We may now proceed as in Eqs. (6.5.4) through (6.5.6) to obtain a modification of Eq. (6.5.5) that accounts for a point source in the domain,

$$\sum_{j=1}^{n}\left(K_{lj} + \frac{C_{lj}}{Z_j}\right)p_j = \sum_{j=1}^{n} K_{lj}' p_j = \sum_{j=1}^{n} U_j C_{lj} + 4\pi D_0 G_{lp} \quad \text{for} \quad l = 1,\ldots,n. \tag{6.6.13}$$

6.7 Calculation of the Pressure at an Arbitrary Point in Space

We would now like to construct a convenient computational approach to calculate the sound pressure at an arbitrary location within the domain. In the previous discussion, the vector $\bar{r}_s$ denotes the location of a point on the surface over which the integrals

are carried out. The vector $\bar{r}_s'$ is a point on the surface at which the pressure is to be calculated and $\bar{r}_p$ denotes the location of a point source within the domain. Since we often would like to calculate the pressure at points other than on the surface, we will define an additional position vector, $\bar{r}_L$, which denotes the location of a listener at any point in the domain.

To calculate the pressure at this arbitrary location, one must first solve Eq. (6.6.8) for $p(\bar{r}')$ to determine the pressure at all points on the surfaces. Once this is accomplished, the pressure at any location can be computed by replacing $\bar{r}'$ with $\bar{r}_L$ everywhere in Eq. (6.6.8). The calculations can be facilitated by storing the components of the listener locations as R_{L1}, R_{L2}, and R_{L3}, so that

$$\bar{r}_L = R_{L1}\hat{\imath} + R_{L2}\hat{\jmath} + R_{L3}\hat{k}. \tag{6.7.1}$$

A matrix containing the distance between each listener location and each surface can then be constructed as was done in Eq. (6.6.12),

$$D_{Lj} = |\bar{r}_j - \bar{r}_L| = \sqrt{(R_{j1} - R_{L1})^2 + (R_{j2} - R_{L2})^2 + (R_{j3} - R_{L3})^2}. \tag{6.7.2}$$

Let

$$G_{Lj} = \frac{e^{-ikD_{Lj}}}{4\pi D_{Lj}}. \tag{6.7.3}$$

We must also compute a new version of Eqs. (6.4.1) through (6.4.11) where we replace the surface location i with the listener location L. In addition, because we are here computing the pressure at a point in space rather than on the surface, the left hand term of Eq. (6.6.8) must be modified from $\frac{p(\bar{r}_s')}{2}$ to $p(\bar{r}')$. The pressure within the domain is then determined by evaluating

$$p(\bar{r}') = -\int\int (p\hat{N}\cdot\nabla\bar{G})ds + \int\int_{S_2} Gi\omega\rho_0 \frac{p}{Z} ds + \int\int_{S_1} Gi\omega\rho_0 U ds + 4\pi D_0 G(\bar{r}_p, \bar{r}'). \tag{6.7.4}$$

To account for the doubling of the first term in Eq. (6.6.8), and replacing a surface location i with a listener location L Eq. (6.6.13) becomes

$$p_2 = \sum_{j=1}^{n} \left(\left(-B_{Lj} p_j + \frac{C_{Lj}}{Z_j} p_j \right) + U_j C_{Lj} \right) + 4\pi D_0 G_{Lp}. \tag{6.7.5}$$

Also note that for small areas S_l and for listener locations that are sufficiently far from any surface, the double surface integrals that define B_{Lj}, C_{Lj}, in Eqs. (6.4.5) and (6.4.6) and subsequently K_{Lj} may be approximated by

$$B_{Lj} \approx \hat{N}_j \cdot \nabla \bar{G}_{Lj} S_j, \tag{6.7.6}$$

and

$$C_{Lj} = i\omega\rho_0 G_{Lj} S_j, \tag{6.7.7}$$

where

$$\nabla \bar{G}_{Lj} = -\frac{e^{-ikD_{Lj}}}{4\pi D_{Lj}^2}\left(ik + \frac{1}{D_{Lj}}\right) \times ((R_{L1} - R_{j1})i + (R_{L2} - R_{j2})\hat{j} + (R_{L3} - R_{j3})\hat{k}) \tag{6.7.8}$$

and

$$B_{Lj} = \hat{N}_j \cdot \nabla \bar{G}_{Lj} = -\frac{e^{-ikD_{Lj}}}{4\pi D_{Lj}^2}\left(ik + \frac{1}{D_{Lj}}\right) \times (N_{j1}(R_{L1} - R_{j1}) + N_{j2}(R_{L2} - R_{j2}) + N_{j3}(R_{L3} - R_{j3})) \tag{6.7.9}$$

where N_{j1}, N_{j2}, and N_{j3} are the three cartesian components of the unit normal vector $\hat{N}_j$ for surface S_j. The determination of this unit normal vector for a given geometry will be discussed below.

The pressure at the location $\bar{r}_L$ may now be computed by modifying Eq. (6.6.13)

$$p_2 = -\sum_{j=1}^{n} p_j B_{Lj} S_j + \sum_{j=1}^{n} i\omega\rho_0 \frac{p_j}{Z_j} G_{Lj} S_j + \sum_{j=1}^{n} i\omega\rho_0 U_j G_{Lj} S_j + 4\pi D_0 G_{Lp} \quad L = 1, \ldots, n_L \tag{6.7.10}$$

where n_L is the number of listener locations and p_j is the pressure at surface location j obtained by solving Eq. (6.6.13).

6.8 Approximate Evaluation of Integrals

Our next task is to construct a convenient means of calculating and storing, B_{lj}, C_{lj} (in Eq. (6.4.5)), S_j, G_{lj}, and specifying the surface velocities U_j. We must also determine the vector quantities $\hat{N}_j$ and $\nabla \bar{G}_{lj}$ that correspond to each surface j or pair of surfaces lj.

The methods described in this chapter can require substantial computational effort. As a result, once the essential ideas for computing rather general sound fields are understood, numerical efficiency must be attended to. Our present purpose, however,

is to lay out the essential ideas without addressing too many computational details. As a result, minimal effort is made in this introductory discussion to construct a calculation that is "optimized" for computational efficiency. In addition, a rigorous examination of numerical errors is best examined elsewhere.

6.8.1 *Elements that Are Sufficiently Separated: Estimation Using Midpoints*

The difficulty of accurately estimating the integrals in Eq. (6.4.5) depends strongly on the size of each pair of elements and the distance between them. If the elements are sufficiently small relative to their separation distance, we can expect the integrand to vary slowly so that the integral can be estimated with a minimum of function evaluations. Actually, for the vast majority of pairs of elements we may assume that the integrals may be estimated simply using the function evaluations at the element midpoints. In this case, Eq. (6.4.5) may be approximated by

$$B_{lj} = \frac{1}{S_l} \int \int_{S_l} \int \int_{S_j} (\hat{N} \cdot \nabla \bar{G}) ds ds' \approx \hat{N}_j \cdot \nabla \bar{G}_{lj} S_j, \tag{6.8.1}$$

and

$$C_{lj} = i\omega\rho_0 \frac{1}{S_l} \int \int_{S_l} \int \int_{S_j} G(\bar{r}', \bar{r}) ds ds' \approx i\omega\rho_0 G_{lj} S_j. \tag{6.8.2}$$

The intermediate vector quantities such as $\hat{N}_j$, $\bar{r}_l$, and $\nabla \bar{G}_{lj}$ in Eq. (6.8.1) have components corresponding to each dimension of the domain. Again, recall that $\bar{r}_l$ is the position vector describing the location of the middle of area S_l. These may be stored in arrays with additional indices that correspond to each dimension. For example, the components of the vector $\bar{r}_l$ may be stored in the matrix $[R]$ having elements given by R_{lI} where $I = 1, 2, 3$ corresponds to each coordinate in the x, y, or z dimension. If we use the usual unit vectors in the x, y, or z directions, $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$, then the vector may be expressed as

$$\bar{r}_l = R_{l1}\hat{\imath} + R_{l2}\hat{\jmath} + R_{l3}\hat{k}. \tag{6.8.3}$$

We can also store the components of $\hat{N}_j$ in a matrix $[N]$ with elements $N_{j,I}$ where $j = 1, \ldots, n$ with n being the total number of elements and $I = 1, 2, 3$ so that

$$\hat{N}_j = N_{j1}i + N_{j2}\hat{\jmath} + N_{j3}\hat{k}. \tag{6.8.4}$$

The evaluation of G_{lj} and $\nabla \bar{G}_{lj}$ may be facilitated by first storing the matrix $[D]$ which stores $|\bar{r}_l - \bar{r}_j|$ so that its elements are given by

$$
\begin{aligned}
D_{lj} &= |\bar{r}_l - \bar{r}_j| \\
&= \sqrt{(R_{l1} - R_{j1})^2 + (R_{l2} - R_{j2})^2 + (R_{l3} - R_{j3})^2} \\
&= \sqrt{X^2 + Y^2 + Z^2}
\end{aligned} \tag{6.8.5}
$$

where $X^2 = (R_{l1} - R_{j1})^2$, $Y^2 = (R_{l2} - R_{j2})^2$, and $Z^2 = (R_{l3} - R_{j3})^2$.

This result may be substituted directly into Eq. (6.7.3),

$$
G_{lj} = \frac{e^{-ikD_{lj}}}{4\pi D_{lj}}. \tag{6.8.6}
$$

The gradient may then be written as

$$
\nabla \bar{G}_{lj} = \frac{\partial G_{lj}}{\partial D_{lj}} \left(\frac{dD_{lj}}{dX}\hat{\imath} + \frac{dD_{lj}}{dY}\hat{\jmath} + \frac{dD_{lj}}{dZ}\hat{k} \right). \tag{6.8.7}
$$

Differentiating Eq. (6.8.6) gives

$$
\frac{\partial G_{lj}}{\partial D_{lj}} = -\frac{e^{-ikD_{lj}}}{4\pi D_{lj}} \left(ik + \frac{1}{D_{lj}} \right). \tag{6.8.8}
$$

Equations (6.8.5) through (6.8.8) give

$$
\begin{aligned}
\nabla \bar{G}_{lj} &= -\frac{e^{-ikD_{lj}}}{4\pi D_{lj}^2} \left(ik + \frac{1}{D_{lj}} \right) (X\hat{\imath} + Y\hat{\jmath} + Z\hat{k}) \\
&= -\frac{e^{-ikD_{lj}}}{4\pi D_{lj}^2} \left(ik + \frac{1}{D_{lj}} \right) \\
&\times ((R_{l1} - R_{j1})\hat{\imath} + (R_{l2} - R_{j2})\hat{\jmath} + (R_{l3} - R_{j3})\hat{k}).
\end{aligned} \tag{6.8.9}
$$

If the unit normal vector is expressed as in Eq. (6.8.4), Eq. (6.8.1) becomes

$$
\begin{aligned}
B_{lj} &\approx \hat{N}_j \cdot \nabla \bar{G}_{lj} \\
&= -\frac{e^{-ikD_{lj}}}{4\pi D_{lj}^2} \left(ik + \frac{1}{D_{lj}} \right) \\
&\times (N_{j1}(R_{l1} - R_{j1}) + N_{j2}(R_{l2} - R_{j2}) + N_{j3}(R_{l3} - R_{j3}))
\end{aligned} \tag{6.8.10}
$$

As mentioned above, the results presented in Eqs. (6.8.1) through (6.8.10) assume that the pair of elements corresponding to lj are far enough apart relative to their dimensions that the integrals may be approximated by only a single function evaluation over each area. When the elements are adjacent to each other this assumption will clearly lead to errors. Unfortunately, performing a more rigorous evaluation of the integrals for adjacent or proximate areas adds enormous computational effort. We

will leave this task for more specialized studies and hope that the errors introduced by our approximations will not invalidate the results.

While it may be permissible to neglect errors introduced when elements l and j are near each other, there are considerable troubles introduced when $l = j$. In this case, the integrals in Eqs. (6.8.1) and (6.8.2) contain singularities which must be accounted for with care as described in the following section to avoid substantial errors.

6.8.2 *Estimation of B_{ll}*

We will first consider an estimation of the integrals in Eq. (6.8.1) when $l = j$,

$$B_{ll} = \frac{1}{S_l} \int \int_{S_l} \int \int_{S_l} (\hat{N} \cdot \nabla \bar{G}) ds ds'. \tag{6.8.11}$$

Of course, to evaluate this integral exactly one must know the detailed geometry of the element. If, again, there are a sufficient number of small elements in the model we can hope that the contribution of this diagonal term in B_{lj} will not dramatically influence the overall solution so that small errors in its evaluation will likely have minimal impact.

We will attempt to estimate the integrals in Eq. (6.8.11) by first considering the form of the Green's function, G and by examining some properties of Eq. (6.3.12). Recall the form of the Green's function in Eq. (6.4.1),

$$G(\bar{r}', \bar{r}) = \frac{e^{\pm ik(|\bar{r}' - \bar{r}|)}}{4\pi |\bar{r}' - \bar{r}|}. \tag{6.8.12}$$

If the integration domain is sufficiently small relative to the sound wavelength, note that when carrying out the integration in Eq. (6.8.11) we can expect the argument of the exponential term in Eq. (6.8.12) to be significantly less than unity. If so, the wave number k will not influence the result and we can safely set it to zero. If so, the task of evaluating Eq. (6.8.11) becomes identical to that of analyzing a static pressure field, i.e., the case where $k = \omega/c = 0$. We may obtain an estimate of B_{ii} in Eq. (6.8.11) by examining the solution to Eq. (6.3.12) in this special case. Our approach here is essentially a recasting of the argument made by Seybert et al. [5]. When $\omega = 0$, Eq. (6.3.12) becomes

$$\frac{p(\bar{r}_s')}{2} + \int \int (p \hat{N} \cdot \nabla \bar{G}) ds = 0. \tag{6.8.13}$$

This corresponds to the problem of finding the static pressure distribution when there is no surface velocity or source of pressure. The Helmholtz equation, Eq. (6.3.1) may be solved in this case with p equal to a constant in both time and space. If the pressure $p(\bar{r}_s)$ is a constant, Eq. (6.8.13) leads to

$$\int\int (\hat{N} \cdot \nabla \bar{G}) ds = -\frac{1}{2}, \tag{6.8.14}$$

where the integration is over the entire surface area of the domain. Note that the integrand of Eq. (6.8.14) depends on the vector describing the primed coordinate location $\bar{r}_s'$ on ds'. Since the result is equal to a constant, $-1/2$, performing the surface average over the primed coordinate as in Eq. (6.8.11) will have no effect. Our discrete version of Eq. (6.8.14) then follows from Eq. (6.4.7) by letting p_l be a constant for all l and using the fact that $C_{lj} = 0$ when $\omega = 0$ (from Eq. (6.4.6)). This gives

$$\sum_{j=1}^{n} B_{lj} \approx -\frac{1}{2}. \tag{6.8.15}$$

Once B_{lj} has been estimated for $l \neq j$ through the use of Eqs. (6.8.10), (6.8.15) may then be used to estimate B_{ll},

$$B_{ll} \approx -\sum_{j=1, j\neq l}^{n} B_{lj} - \frac{1}{2}. \tag{6.8.16}$$

6.8.3 *Estimation of C_{ll}*

In estimating the values of B_{ll} we resorted to an approximation involving "mathematical trickery" in which we avoided extensive computational difficulties by making use of the properties of Eq. (6.8.10) and thereby avoided the need to specify the geometric details of each element. While a similar approach might be effective in estimating C_{ll}, it has yet to reveal itself to me and will doubtless provide amusement to some future reader.

In the following, C_{ll} will be estimated starting with the same approximation as in B_{ll} above, namely that the dimensions of each element are sufficiently small relative to the wavelength of sound. This, again, reduces our Green's function in Eq. (6.8.12) to not depend on the argument of the exponential so that we can safely assume that the wavenumber $k = 0$. Our Green's function then reduces to that of Laplace's equation,

$$\nabla^2 P = 0. \tag{6.8.17}$$

Fortunately, boundary element-based solutions to Laplace's equation are sought in a great many fields, in particular, electrostatics, where integrals of the type we are interested in have been worked out.

It is quite common to represent complex surfaces as a large collection of triangular shaped area elements. We will then limit our examination to those and leave

more general areas to other more specialized studies. The description of complicated surfaces as an assembly of small triangles is used in common software tools that fall into the category of Computer-Aided Design.

6.8.4 Computer-Aided Design Model to Specify Geometry

To construct a model of an acoustic domain, it is necessary to specify a large number of geometric details. Since computer-aided design (CAD) software is widely available for describing complex geometries, in the following we will examine the translation of data provided by typical CAD programs to provide the required parameters. Having specified the geometry in the desired CAD program, the user must then export the data in an STL file which may then be processed to determine all of the needed quantities.

An STL file contains a description of triangles used to define the surfaces in a solid model. The two-dimensional array V contains the XYZ coordinates of each vertex, or nodes, used to define the triangles. The XYZ coordinates of vertex l will be (V_{l1}, V_{l2}, V_{l3}) so that the row number indicates the number of the vertex and the number of the column (1, 2, or 3) indicates the X, Y, or Z coordinate.

The F array (face) indicates which vertices define the corners of each face. The row number indicates the face number and the three columns indicate the vertex numbers (which correspond to the rows of the V array. For face number l, the coordinates of the three vertices will then be $(V_{F(l,1)1}, V_{F(l,1)2}, V_{F(l,1)3})$, for vertex 1, $(V_{F(l,2)1}, V_{F(l,2)2}, V_{F(l,2)3})$, for vertex 2, $(V_{F(l,3)1}, V_{F(l,3)2}, V_{F(l,3)3})$, for vertex 3.

The midpoint of face number i, will define the vector $\bar{r}_l$, as in Eq. (6.8.3). The coordinates of $\bar{r}_l$ will be given by

$$\begin{aligned} R_{l1} &= \frac{1}{3}(V_{F(l,1)1} + V_{F(l,2)1} + V_{F(l,3)1}) \\ R_{l2} &= \frac{1}{3}(V_{F(l,1)2} + V_{F(l,2)2} + V_{F(l,3)2}) \\ R_{l3} &= \frac{1}{3}(V_{F(l,1)3} + V_{F(l,2)3} + V_{F(l,3)3}) \end{aligned} \tag{6.8.18}$$

We also need to determine the area S_l of each triangular area. To calculate the area, we will define two vectors, one that begins at the first vertex and terminates at the second $\bar{L}_{12l}$ and one that begins at the first vertex and ends at the third, $\bar{L}_{13l}$. Given the coordinates of the vertices, these two vectors are given by

$$\begin{aligned} \bar{L}_{12l} &= (V_{F(l,2)1} - V_{F(l,1)1})l + (V_{F(l,2)2} - V_{F(l,1)2})\hat{j} \\ &\quad + (V_{F(l,2)3} - V_{F(l,1)3})\hat{k} \\ \bar{L}_{13l} &= (V_{F(l,3)1} - V_{F(l,1)1})l + (V_{F(l,3)2} - V_{F(l,1)2})\hat{j} \\ &\quad + (V_{F(l,3)3} - V_{F(l,1)3})\hat{k} \end{aligned} \tag{6.8.19}$$

The area of the triangle is given by

$$S_l = \frac{1}{2}|\bar{L}_{12l} \times \bar{L}_{13l}|, \tag{6.8.20}$$

where $\times$ denotes the vector cross product.

The unit normal vector is in the direction of the cross product. Normalizing the cross product so that it has unit length gives

$$\hat{N}_l = \frac{\bar{L}_{12l} \times \bar{L}_{13l}}{|\bar{L}_{12l} \times \bar{L}_{13l}|}. \tag{6.8.21}$$

The unit normal of each surface element is included in an STL file so it is convenient to simply read it rather than calculate it.

6.8.5 *Evaluation of Singular Integrals for C_{ll}*

Once a given surface geometry has been represented as a collection of triangles as above, we may carry out the integrals required to estimate C_{ll} in Sect. 6.8.3. As mentioned above, the integrand in Eq. (6.4.6) becomes singular when $l = j$. This has, fortunately, been carried out analytically [6]. For each l let

$$\begin{aligned} a &= \sum_{j=1}^{3} (V_{F(l,3)j} - V_{F(l,1)j}) \times (V_{F(l,3)j} - V_{F(l,1)j}) \\ b &= \sum_{j=1}^{3} (V_{F(l,3)j} - V_{F(l,1)j}) \times (V_{F(l,3)j} - V_{F(l,2)j}) \\ c &= \sum_{j=1}^{3} (V_{F(l,3)j} - V_{F(l,2)j}) \times (V_{F(l,3)j} - V_{F(l,2)j}) \end{aligned} \tag{6.8.22}$$

$$\begin{aligned} I_l = &\frac{\log\left(\frac{(a-b+\sqrt{a}\sqrt{a-2b+c})(b+\sqrt{ac})}{(-b+\sqrt{ac})(-a+b+\sqrt{a}\sqrt{a-2b+c})}\right)}{6\sqrt{a}} \\ &+ \frac{\log\left(\frac{(-b+c+\sqrt{c}\sqrt{a-2b+c})(b+\sqrt{ac})}{(-b+\sqrt{ac})(b-c+\sqrt{c}\sqrt{a-2b+c})}\right)}{6\sqrt{c}} \\ &+ \frac{\log\left(\frac{(a-b+\sqrt{a}\sqrt{a-2b+c})(-b+c+\sqrt{c}\sqrt{a-2b+c})}{(b-c+\sqrt{c}\sqrt{a-2b+c})(-a+b+\sqrt{a}\sqrt{a-2b+c})}\right)}{6\sqrt{a-2b+c}} \end{aligned} \tag{6.8.23}$$

The diagonal elements of the matrix $[G]$ may then be computed from

$$G_{ll} \approx \frac{S_l I_l}{\pi}. \tag{6.8.24}$$

This gives an approximation of the diagonal elements of $[C]$ in Eq. (6.8.2) as

$$C_{ll} = i\omega\rho_0 \frac{1}{S_l} \int\int_{S_l} \int\int_{S_l} G(\bar{r}', \bar{r}) ds ds' \approx i\omega\rho_0 S_l^2 \frac{I_l}{\pi}. \tag{6.8.25}$$

6.9 Numerical Results

In the following, we show results obtained from the numerical approach described above along with those that are calculated from simple closed-form solutions for the sound field in various situations. While a detailed examination of sources of error in the numerical method is highly desirable, this will be left for others. The results shown in the following can provide some guidance on which situation can be handled with reasonable accuracy.

6.9.1 Sound Inside a Rectangular Tube

In this example, we examine the sound field in a rectangular tube as shown in Fig. 6.1. The tube is 1 m long with width and depth equal to 0.05 m. It will be assumed, for now, that all interior surfaces are rigid boundaries, except the surface at $z = 0.5$, which will have a somewhat arbitrary complex impedance of $Z = \rho_0 c \times (1 + 0.7i)/2$. We will examine the sound field within the tube due to an imposed velocity $U_0 e^{i\omega t}$ of the surface located at $z = -0.5$ m where $U_0 = 1.0$ m/s. This simple domain was modeled using the CREO (ProEngineer) computer-aided design program to create an STL file. A total of 668 areas were defined to represent the domain.

The sound field in a tube having a moving piston at one end and an impedance termination at the other has been examined in Eqs. (2.4.13) and (2.4.14). The present situation differs only in the choice of a coordinate system since our current domain is $-0.5 \le z \le 0.5$ rather than $0 \le x \le l$ as considered previously. A similar analysis to that leading to Eqs. (2.4.13) and (2.4.14) gives the pressure field in the tube to be

$$p(z, t) = e^{i\omega t}(p_1 e^{-ikz} + p_2 e^{ikz}), \tag{6.9.1}$$

where

$$p_1 = \frac{\rho_0 c U_0}{e^{ikl/2} - Re^{-ikl/2}}, \tag{6.9.2}$$

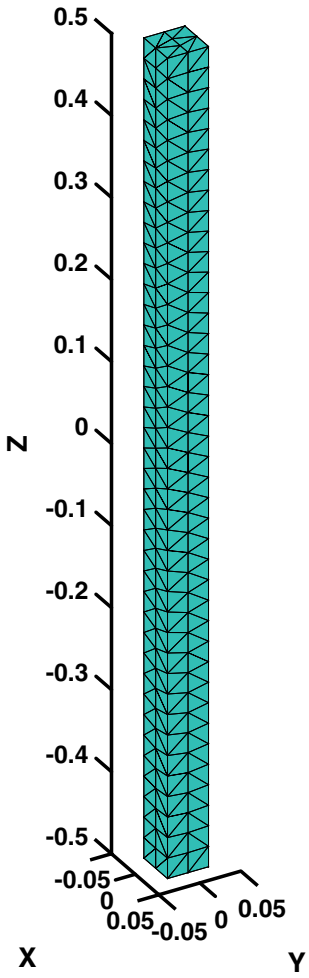


Fig. 6.1 Computer-aided design model of a rectangular tube

$$p_2 = r p_1 \tag{6.9.3}$$

$$r = \frac{1 - \rho_0 c/Z}{1 + \rho_0 c/Z} e^{-ikl}, \tag{6.9.4}$$

and $l = 1$.

Figure 6.2 shows a comparison of the predicted pressure amplitude at $z = -0.5$ with that obtained using the solution based on the assumption that the domain is

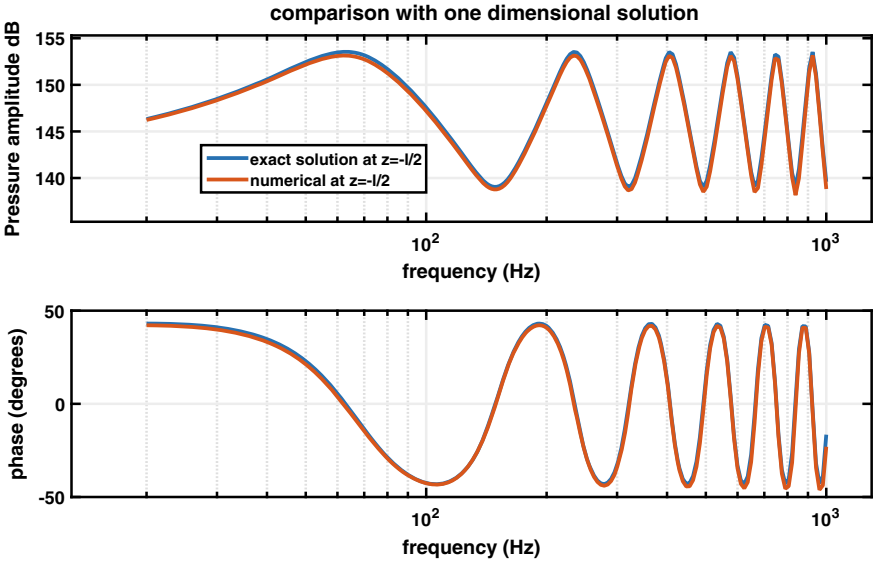


Fig. 6.2 Comparison of the amplitude and phase of the pressure within the tube at $z = -0.5$ predicted using the one-dimensional solution with that obtained using the numerical solution based on the CAD model. All interior surfaces are rigid boundaries, except the surface at $z = 0.5$, which will have a somewhat arbitrary complex impedance of $Z = \rho_0 c \times (1 + 0.7i)/2$. The pressure results from an imposed velocity $U_0 e^{i\omega t}$ of the surface located at $z = -0.5\,\text{m}$ where $U_0 = 1.0\,\text{m/s}$. The approximate numerical results are in very close agreement with those obtained using the one-dimensional solution

one-dimensional as in Eqs. (6.9.1) through (6.9.4). The results are in very close agreement.

Figure 6.3 shows the distribution of pressure on the interior of the tube at a frequency of 1000 Hz. A quantitative comparison of the pressure distribution within the tube based on our numerical method with that obtained using Eqs. (6.9.1) through (6.9.4) is shown in Fig. 6.4.

Predicted results will be presented in Sect. 7.3 for the sound field within the enclosure shown in Fig. 6.1 due to a point source within the enclosure.

The pressure distribution obtained for a circular tube having a substantially larger number of surface elements is shown in Fig. 6.5. The boundary conditions and overall dimensions are similar to the rectangular tube examined above. In this case, a total of 13248 elements are used to describe the surface. Figure 6.6 shows the amplitude and phase of the pressure within the tube compared to those predicted by Eqs. (6.9.1) through (6.9.4). The numerical results for this tube having a finer mesh show better agreement with the "exact" result than is seen in Fig. 6.4.

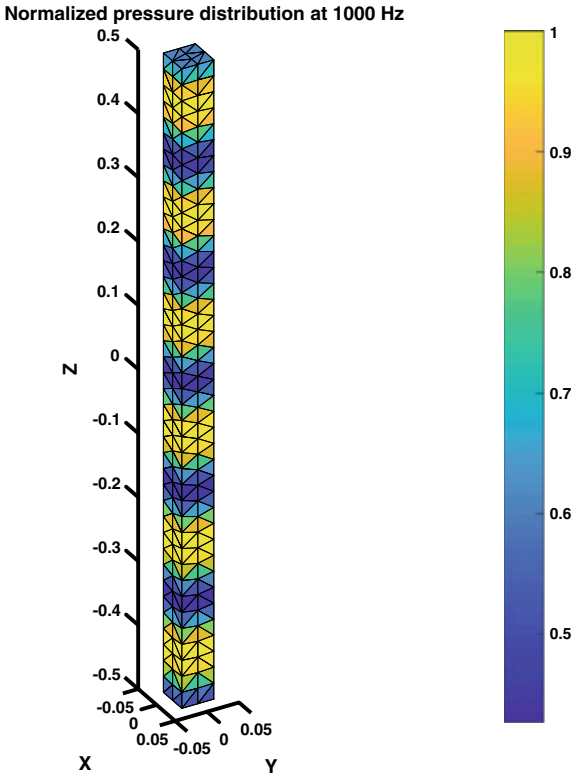


Fig. 6.3 The distribution of predicted pressure on the interior of the surfaces on the tube of Figs. 6.1 and 6.2 at a frequency of 1000 Hz

6.9.2 Sound Inside a Radially Oscillating Sphere

As another comparison, consider the problem of finding the sound field within a sphere having a radially oscillating surface as examined in Eq. (5.4.6). Figure 6.7 shows the predicted sound pressure on the interior surface as a function of frequency for a sphere of nominal radius $r = 0.25\,\text{m}$ with a surface velocity of $u_a = 1\,\text{m/s}$. The results obtained using the numerical boundary integral approach are in close agreement with those of Eq. (5.4.6) for a range of frequencies up to about 10 kHz. It should be noted that there is no energy dissipation in this system so that the levels at resonances and antiresonances will not be accurately represented.

The distribution of the sound field on the interior surface of the sphere is shown in Fig. 6.8. These results correspond to a frequency of 987 Hz, which is the frequency of the first dominant resonance peak shown in Fig. 6.7. Note that the color scale indicates that the pressure on the surface is nearly uniform.

The numerical results shown in Fig. 6.7 also show a small peak at a frequency of about 733.5 Hz which does not appear in the field predicted by Eq. (5.4.6). The pressure distribution corresponding to this frequency is shown in Fig. 6.9. It appears

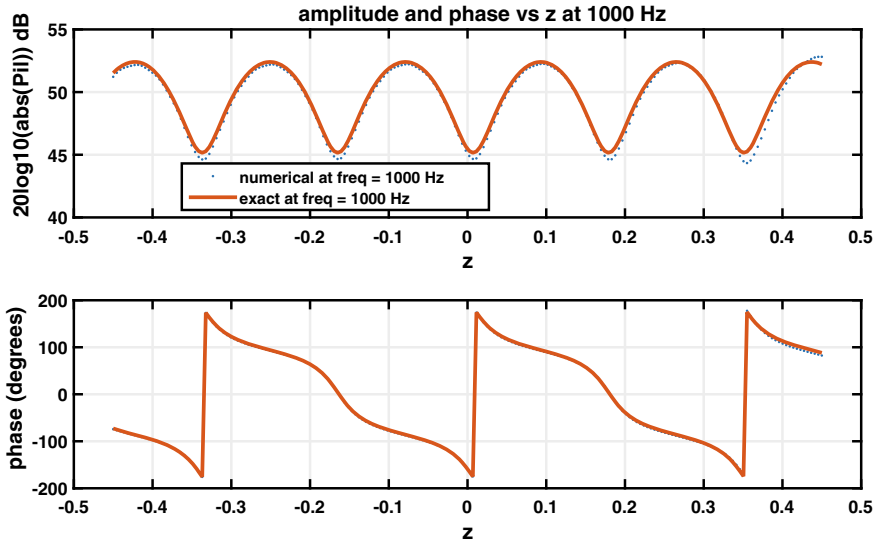


Fig. 6.4 Comparison of exact and numerical distribution of predicted pressure within the tube of Figs. 6.1 and 6.2 at a frequency of 1000 Hz. The "exact" results were obtained using Eqs. (6.9.1) through (6.9.4). The results are found to be in fairly close agreement

to be a standing wave resonance that is not predicted by the spherically symmetric analysis.

6.9.3 Sound Radiation from a Radially Oscillating Cylinder

We previously examined the sound field radiated by a "simple source", or a radially oscillating sphere with radial surface velocity u_s and radius a. It was assumed that the sound field radiated by the sphere can be properly represented by the equivalent point source approach of Eq. (5.3.5),

$$p(r,t) = \frac{D_0}{r} e^{i(\omega t - kr)} = \frac{\rho_0 c i k a^2}{1 + ika} \frac{u_s}{r} e^{i(\omega t - k(r-a))}. \tag{6.9.5}$$

In deriving this result, we started with the assumption that the field was described by an outgoing wave with constant amplitude D_0 of the form shown on the left of Eq. (6.9.5). If the radius of the sphere is on the order of the wavelength of the radiated sound, however, we can expect sound radiated by portions of its surface to cancel that radiated by other portions causing cancelations and summations at certain frequencies. This will cause the pressure to vary significantly with frequency as the frequency is increased.

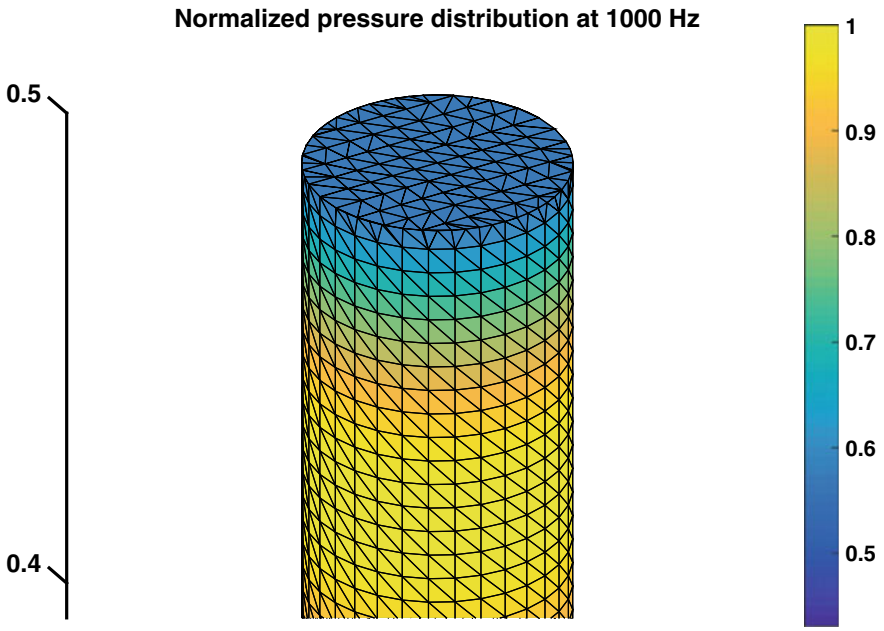


Fig. 6.5 The distribution of predicted pressure on the interior of the surfaces of a circular tube. The surface is represented by a total of 13248 triangular elements. The boundary conditions are the same as those of the rectangular tube considered above. The figure shows only the portion of the tube near $z = 0.5$ to better show the element distribution

Figure 6.10 shows the predicted sound pressure level due to a radially oscillating sphere having a radius of $a = 0.125\,\text{m}$ at two different distances from the center of the sphere. The radial velocity of the sphere is $u_s = 1\,\text{m/s}$. Results obtained using Eq. (6.9.5) are shown along with those obtained using the boundary element approach described above. The surface of the sphere is discretized into 2400 triangular areas as shown in Fig. 6.9. The size of this sphere becomes on the order of the wavelength of sound in the frequency range shown. As a result, portions of the sphere emit sound that arrives with time delays that are sufficient to result in interference causing dips in the numerically predicted sound pressure level. This effect is not accounted for in the analytical expression (6.9.5) which assumes the sphere is much smaller than the sound wavelength.

Figure 6.11 shows the predicted sound pressure level at the same distances from the center of a sphere as used in Fig. 6.10 but in this case, the radius of the sphere is reduced by a factor of 10 to $a = 0.0125\,\text{m}$. In this case, the wave interference observed in the numerical result of Fig. 6.10 has much less effect on the results; the numerical prediction agrees well with the approximate analytical formula (6.9.5) except at the highest frequencies shown.

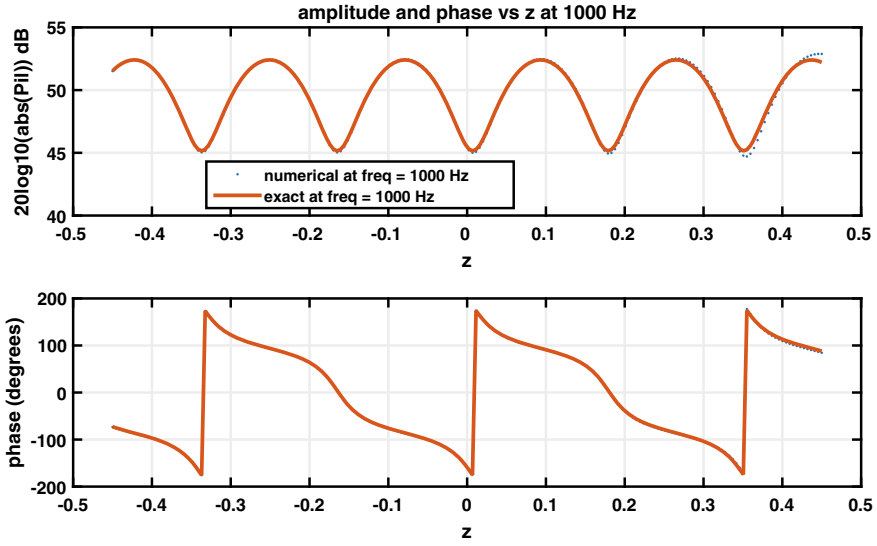


Fig. 6.6 Comparison of exact and numerical distribution of predicted pressure within the circular tube of Fig. 6.5 at a frequency of 1000 Hz. The "exact" results were obtained using Eqs. (6.9.1) through (6.9.4). The results are found to be in closer agreement than those shown in Fig. 6.4

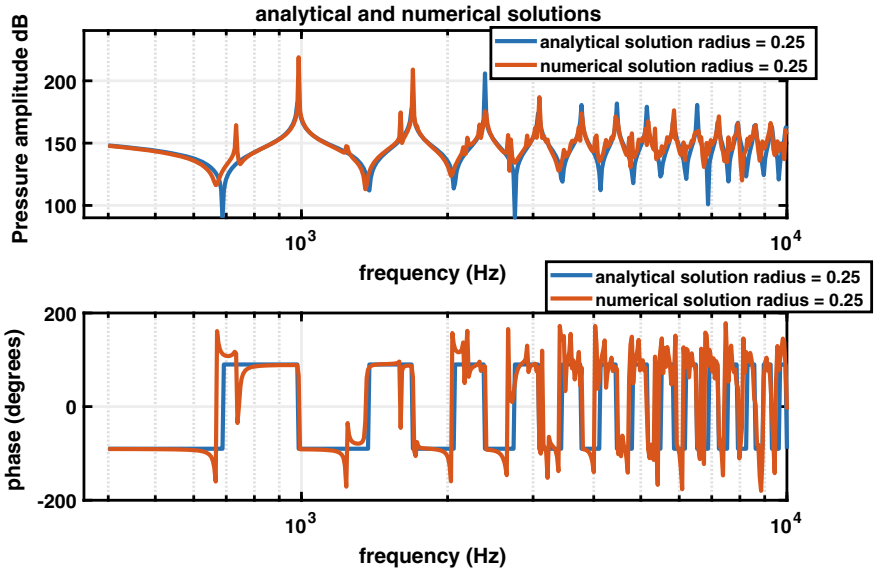


Fig. 6.7 Comparison of the predicted pressure amplitude within a sphere with a radially oscillating surface obtained using Eq. (5.4.6) and using the numerical method described here. The sphere has a radius of 0.25 m. The approximate numerical results are in very close agreement with those obtained using the solution given in Eq. (5.4.6). Note that the sound pressure levels are unreasonably high owing to the 1 m/s surface velocity

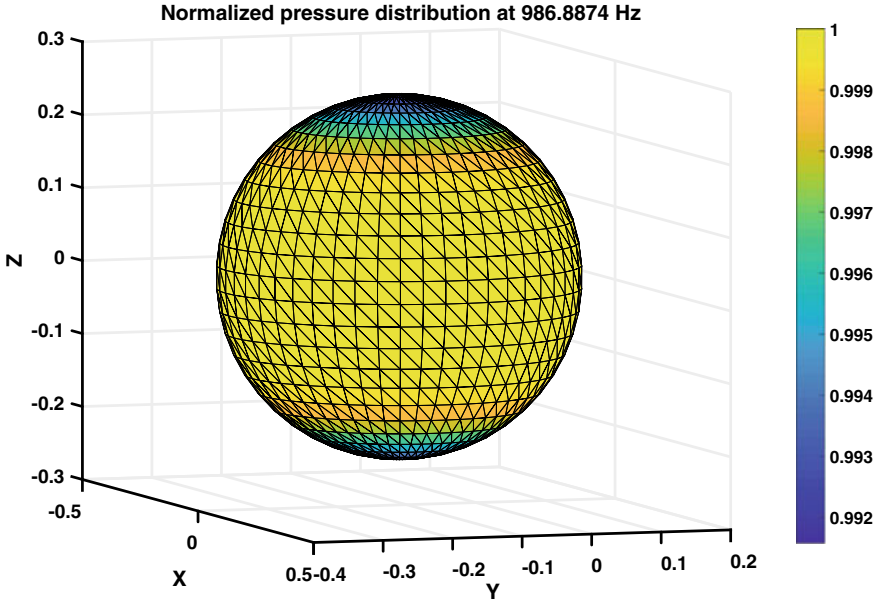


Fig. 6.8 The pressure amplitude on the interior surface of a radially oscillating sphere. The sphere has a nominal radius of 0.25 m. Note that the color scale indicates that the pressure on the surface is nearly uniform

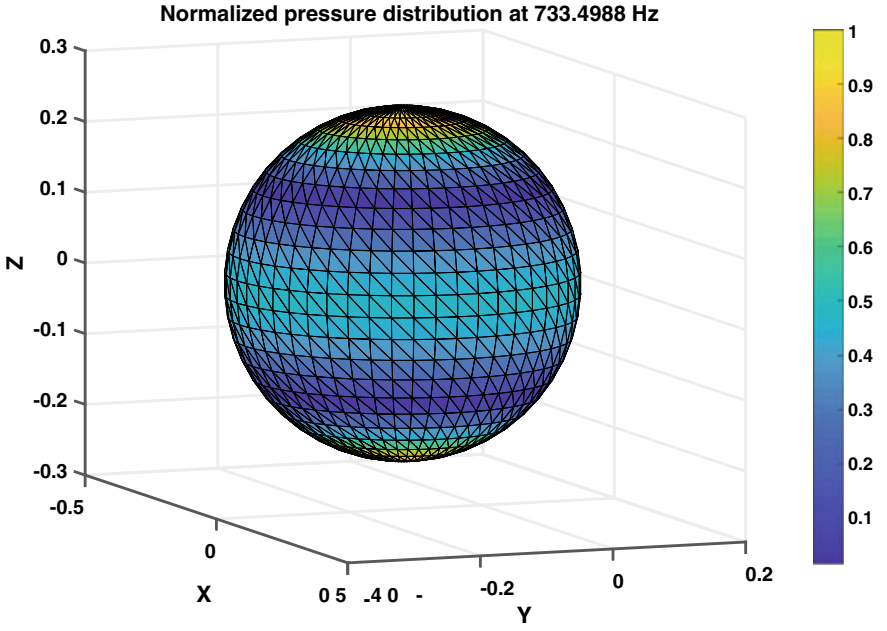


Fig. 6.9 The pressure amplitude on the interior surface of a radially oscillating surface at a frequency of 733 Hz. The sphere has a radius of 0.25 m

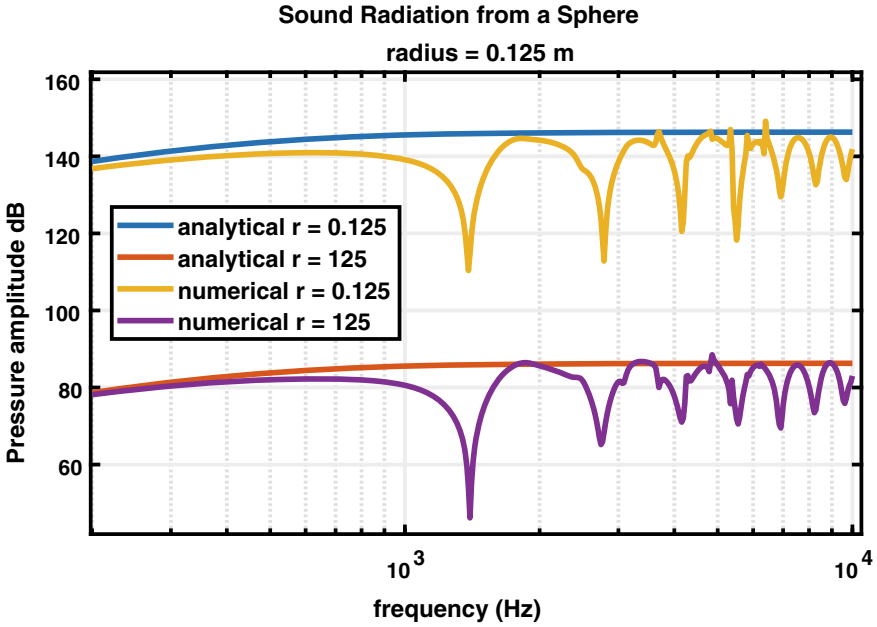


Fig. 6.10 Predicted sound pressure level due to a radially oscillating sphere of radius $a = 0.125$ at distances $r = 0.125\,\text{m}$ and $r = 125\,\text{m}$

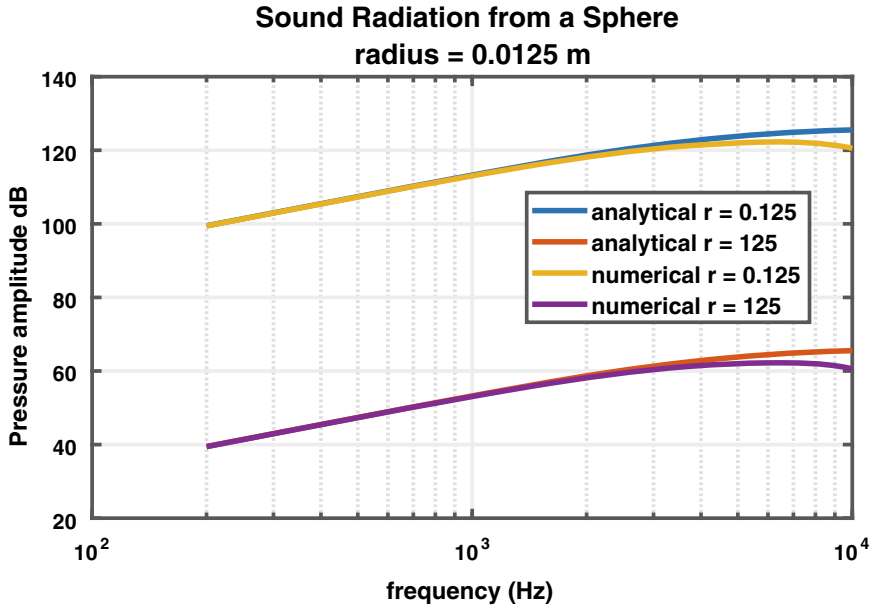


Fig. 6.11 Predicted sound pressure level due to a radially oscillating sphere of radius $a = 0.0125\,\text{m}$ at distances $r = 0.125\,\text{m}$ and $r = 125\,\text{m}$

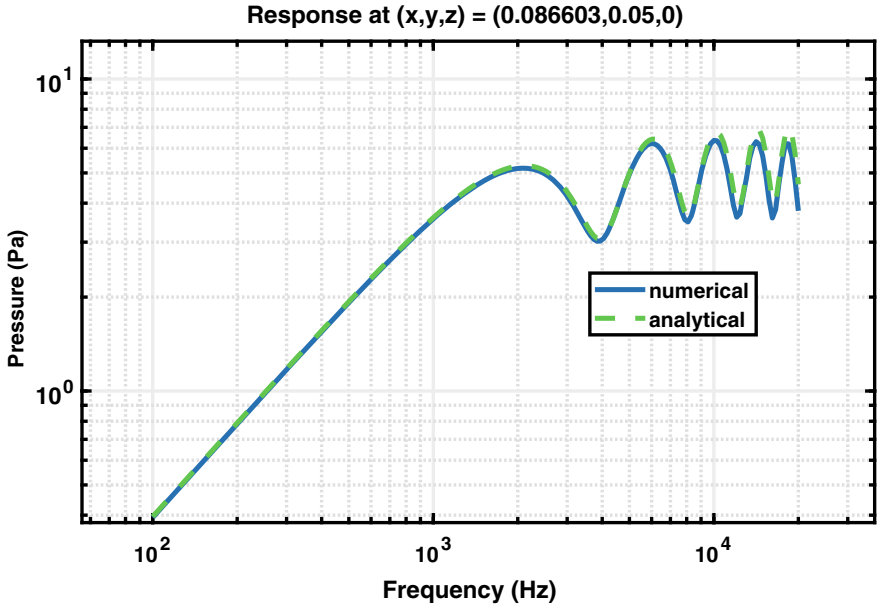


Fig. 6.12 Sound field at the location $(x, y, z) = (0.0866, 0.05, 0)$ due to a radially oscillating cylinder of length $L = 0.1$ m, radius $a = 0.001$ m having a radial surface velocity of $U_s = 1$ m/s. The cylinder is oriented parallel to the x axis and is centered at the origin. Numerical results are obtained using the methods described above and the analytical results were obtained using the method described in Sect. 5.8

6.9.4 Sound Radiation from a Radially Oscillating Cylinder

In Sect. 5.8 we considered the sound radiation from a line source with the aim of estimating the radiation from a radially oscillating cylinder. This problem amounts to finding the sound radiated from a structure similar to that shown in Fig. 6.5. Predicted sound pressure amplitudes obtained using the numerical approach described above are compared to those obtained using the approximate method of Sect. 5.8. The surface of the cylinder was described using 2016 triangles in the numerical solution. The results are shown in Fig. 6.12 which shows that the numerical and analytical results are in reasonable agreement over the frequency range shown.

Results obtained at a location much closer to the center of the cylinder $(x, y, z) = (0.02, 0.02, 0)$ are shown in Fig. 6.13. Again, the numerical results are in very close agreement with those obtained using the analytical method of Sect. 5.8.

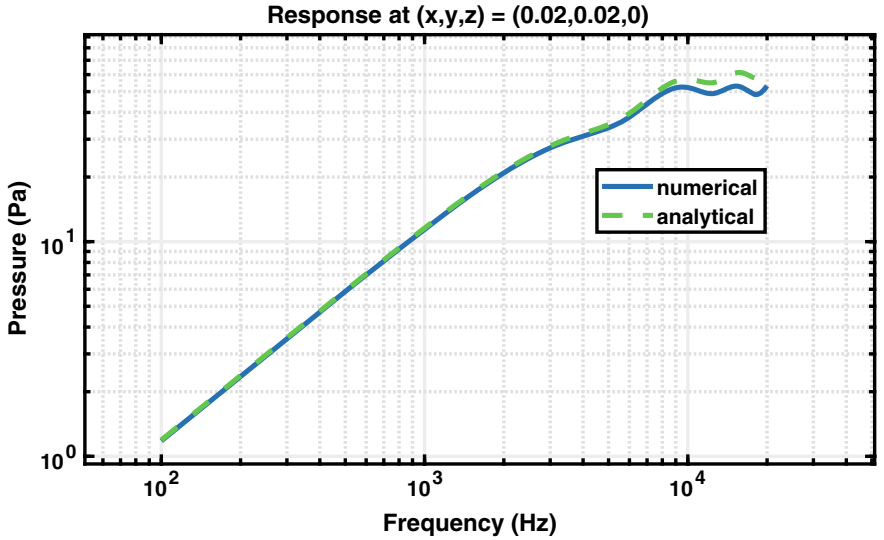


Fig. 6.13 Sound field at the location $(x, y, z) = (0.0866, 0.05, 0)$ due to a radially oscillating cylinder of length $L = 0.1$ m, radius $a = 0.001$ m having a radial surface velocity of $U_s = 1$ m/s. The cylinder is oriented parallel to the x axis and is centered at the origin. Numerical results are obtained using the methods described above and the analytical results were obtained using the method described in Sect. 5.8

6.10 Problems

1. Use the method of the present chapter to re-do Problem 3 of Chap. 5.
2. Use the method of the present chapter to re-do Problem 4 of Chap. 5.
3. Use the method of the present chapter to re-do Problem 5 of Chap. 5.

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Chapter 7
Modal Solutions for the Sound in Enclosures



While the integral equation approach of determining the sound field described in the previous sections is very general and applicable to a wide range of domain shapes, it rarely leads to equations that are simple enough to provide insight into the nature of the sound field; the results must be obtained numerically. For harmonic sound fields, when the amplitude of the pressure is examined as a function of frequency, one often finds sharp peaks and dips that are very similar to those due to resonances in vibrating structures. We then expect that the wave equation may often be solved using the method of separation of variables as is common in solving vibration problems; the sound field within enclosures can be efficiently described in terms of the superposition of the eigenfunctions of the system.

The simplest three dimensional enclosure to study is the rectangular box. We will assume that all of the walls have rigid boundaries so that the acoustic particle velocity normal to the walls is equal to zero on all surfaces. We would like to compute the sound field due to radiation from a point sound source in the interior. We have written the Helmholtz equation accounting for a point source in Eq. (6.6.1),

$$\nabla^2 P + k^2 P = -4\pi D_0 \delta(\bar{r} - \bar{r}_p), \tag{7.0.1}$$

where $\bar{r}$ is a vector giving the listener location and $\bar{r}_p$ is the location of the point source. The pressure from the point source is assumed to vary harmonically at the frequency ω. The time varying pressure field is then, $p(x, y, z, t) = P(x, y, z)e^{i\omega t}$ so that the wave equation we need to solve is

$$\nabla^2 p - \ddot{p}/c^2 = -4\pi D_0 e^{i\omega t} \delta(\bar{r} - \bar{r}_p). \tag{7.0.2}$$

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_7.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_7

7.1 Natural Frequencies and Eigenfunctions for the Rectangular Enclosure

To solve Eq. (7.0.2), we will first solve

$$\nabla^2 p - \ddot{p}/c^2 = 0 \tag{7.1.1}$$

to determine the eigenfrequencies and eigenfunctions. The domain will be assumed to be a rectangular enclosure with dimensions (L_x, L_y, L_z). To simplify comparisons with solutions obtained using the boundary integral formulation described in the previous sections, we will assume that the domain is described in terms of the maximum and minimum values of each coordinate,

$$\begin{aligned} L_{xmin} \le x \le L_{xmax} \\ L_{ymin} \le y \le L_{ymax} \\ L_{zmin} \le z \le L_{zmax} \end{aligned} \tag{7.1.2}$$

so that $(L_x, L_y, L_z) = (L_{xmax} - L_{xmin}, L_{ymax} - L_{ymin}, L_{zmax} - L_{zmin})$.

To use separation of variables, we will first separate the spatial variables from the time dependence by letting

$$p(x, y, z, t) = P(x, y, z)\eta(t). \tag{7.1.3}$$

Substituting this into Eq. (7.1.1) gives

$$\nabla^2 P\eta - P\ddot{\eta}/c^2 = 0. \tag{7.1.4}$$

Rearranging this gives

$$c^2 \frac{\nabla^2 P}{P} = \frac{\ddot{\eta}}{\eta}. \tag{7.1.5}$$

Since the left side of this equation depends only on spatial coordinates and the right side depends only on time, both must equal a constant (to be determined later), which we will set to $-\omega^2$,

$$\frac{\ddot{\eta}}{\eta} = -\omega^2 \tag{7.1.6}$$

which is simply the familiar ordinary differential equation for the harmonic oscillator,

$$\ddot{\eta} + \omega^2 \eta = 0. \tag{7.1.7}$$

The solution of Eq. (7.1.7) is

$$\eta(t) = a_0 \cos(\omega t) + b_0 \sin(\omega t) = A_0 e^{i\omega t}, \tag{7.1.8}$$

where a_0, b_0 and A_0 are as yet unknown constants. Note that A_0 may be complex. Equations (7.1.5) and (7.1.6) give

$$\nabla^2 P + Pk^2 = 0, \tag{7.1.9}$$

where $k = \omega/c$ is the wave number.

We may also attempt to obtain a solution with separable spatial variables,

$$P(x, y, z) = \alpha(x)\gamma(y)\beta(z). \tag{7.1.10}$$

Substituting this into Eq. (7.1.9) and rearranging gives

$$\frac{\alpha_{xx}}{\alpha} + \frac{\gamma_{yy}}{\gamma} + \frac{\beta_{zz}}{\beta} = -k^2. \tag{7.1.11}$$

As in Eq. (7.1.5), each term on the left side of this equation depends on a different spatial variable (either x, y, or z). Since the sum of these must be a constant, each of them must be a constant so that we may write

$$\frac{\alpha_{xx}}{\alpha} = -k_x^2, \quad \frac{\gamma_{yy}}{\gamma} = -k_y^2, \quad \text{and} \quad \frac{\beta_{zz}}{\beta} = -k_z^2. \tag{7.1.12}$$

We thus obtain three ordinary differential equations for the spatial dependence, each having the same familiar form as that for $\eta(t)$ in Eq. (7.1.7). The unknown constants, k_x, k_y, and k_z will be determined using the boundary conditions.

The solutions to Eq. (7.1.12) are

$$\begin{aligned} \alpha(x) &= a_1 \cos(k_x x) + b_1 \sin(k_x x) \\ \gamma(y) &= a_2 \cos(k_y y) + b_2 \sin(k_y y) \\ \beta(x) &= a_3 \cos(k_z z) + b_3 \sin(k_z z), \end{aligned} \tag{7.1.13}$$

where a_i and b_i are unknown constants for $i = 1, 2, 3$. Since the walls are rigid, the Euler equations say that the normal derivative of the pressure must be zero at all boundaries,

$$\left.\frac{\partial P}{\partial x}\right|_{x=L_{xmin}} = \left.\frac{\partial P}{\partial x}\right|_{x=L_{xmax}} = \left.\frac{\partial P}{\partial y}\right|_{y=L_{ymin}} = \left.\frac{\partial P}{\partial y}\right|_{y=L_{ymax}} = \left.\frac{\partial P}{\partial z}\right|_{z=L_{zmin}} = \left.\frac{\partial P}{\partial z}\right|_{z=L_{zmax}} = 0. \tag{7.1.14}$$

The derivatives of Eq. (7.1.13) are

$$\alpha_x(x) = k_x(-a_1 \sin(k_x x) + b_1 \cos(k_x x))$$
$$\gamma_y(y) = k_y(-a_2 \sin(k_y y) + b_2 \cos(k_y y))$$
$$\beta_z(x) = k_z(-a_3 \sin(k_z z) + b_3 \cos(k_z z)). \tag{7.1.15}$$

The first two equations in (7.1.14) and the first of Eq. (7.1.15) give

$$\begin{bmatrix} \sin(k_x L_{xmin}) & -\cos(k_x L_{xmin}) \\ \sin(k_x L_{xmax}) & -\cos(k_x L_{xmax}) \end{bmatrix} \begin{pmatrix} a_1 \\ b_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{7.1.16}$$

For nontrivial solutions for a_1 and b_1, the determinant of the matrix must equal zero. This gives

$$\sin(k_x(L_{xmax} - L_{xmin})) = \sin(k_x L_x) = 0. \tag{7.1.17}$$

Equations (7.1.14) and (7.1.15) then give

$$k_x = \frac{i_x \pi}{L_x} \quad k_y = \frac{i_y \pi}{L_y} \quad \text{and} \quad k_z = \frac{i_z \pi}{L_z} \quad , \tag{7.1.18}$$

where i_x, i_y, and i_z can take on any integer values from zero to infinity. The coefficients in Eq. (7.1.15) must satisfy

$$\frac{b_1}{a_1} = \tan(k_x L_{xmin}), \quad \frac{b_2}{a_2} = \tan(k_y L_{ymin}), \quad \frac{b_3}{a_3} = \tan(k_z L_{zmin}). \tag{7.1.19}$$

The functions in Eq. (7.1.13) become

$$\alpha(x) = a_1(\cos(k_x x) + \tan(k_x L_{xmin}) \sin(k_x x))$$
$$\gamma(y) = a_2(\cos(k_y y) + \tan(k_y L_{ymin}) \sin(k_y y))$$
$$\beta(x) = a_3(\cos(k_z z) + \tan(k_z L_{zmin}) \sin(k_z z)). \tag{7.1.20}$$

Our solution to (7.1.1) may then be written using Eqs. (7.1.8), (7.1.10), and (7.1.20),

$$\begin{aligned} p(x, y, z, t) &= A_0 e^{i\omega t} \frac{a_1 a_2 a_3}{\cos\left(\frac{i_x \pi L_{xmin}}{L_x}\right) \cos\left(\frac{i_y \pi L_{ymin}}{L_y}\right) \cos\left(\frac{i_z \pi L_{zmin}}{L_z}\right)} \\ &\times \cos\left(\frac{i_x \pi (x - L_{xmin})}{L_x}\right) \cos\left(\frac{i_y \pi (y - L_{ymin})}{L_y}\right) \cos\left(\frac{i_z \pi (z - L_{zmin})}{L_z}\right) \\ &= B_{i_x, i_y, i_z} e^{i\omega t} \cos\left(\frac{i_x \pi (x - L_{xmin})}{L_x}\right) \cos\left(\frac{i_y \pi (y - L_{ymin})}{L_y}\right) \cos\left(\frac{i_z \pi (z - L_{zmin})}{L_z}\right). \end{aligned} \tag{7.1.21}$$

Since the coefficients, $A_0 a_1 a_2 a_3$ are unknown, they have been combined into one unknown constant that will depend on the three indices, B_{i_x, i_y, i_z}. We have not yet

identified the value of ω in this equation. This may be determined from Eqs. (7.1.11), (7.1.12), and (7.1.18) and the fact that $k = \omega/c$,

$$\omega = \omega_{i_x,i_y,i_z} = c\sqrt{\left(\frac{i_x\pi}{L_x}\right)^2 + \left(\frac{i_y\pi}{L_y}\right)^2 + \left(\frac{i_z\pi}{L_z}\right)^2}. \tag{7.1.22}$$

We thus have a discrete spectrum of possible values of ω that depend on the three indices i_x, i_y, and i_z.

The fact that we have an infinite number of solutions depending on three integer values, i_x, i_y, and i_z, can result in very awkward notation when using these to solve for the pressure due to a sound source. To simplify the writing, we will combine the indices into one by assuming that we will use only a finite number of each so that $i_x = 0, 1, 2, \ldots, n_x - 1$, $i_y = 0, 1, 2, \ldots, n_y - 1$, and $i_z = 0, 1, 2, \ldots, n_z - 1$. We may then store our natural frequencies and wave vector components (given in Eq. (7.1.18)) according to the following algorithm, expressed roughly in the syntax used in Matlab,

$$\begin{aligned}
&I = 0\\
&for \quad i_x = 0 : n_x\\
&for \quad i_y = 0 : n_y\\
&for \quad i_z = 0 : n_z\\
&I = I + 1\\
&k_{xI} = \frac{i_x\pi}{L_x}\\
&k_{yI} = \frac{i_y\pi}{L_y}\\
&k_{zI} = \frac{i_z\pi}{L_z}\\
&\omega_I = c\sqrt{k_{xI}^2 + k_{yI}^2 + k_{zI}^2}\\
&k_I = \omega_I/c\\
&end\\
&end\\
&end,
\end{aligned} \tag{7.1.23}$$

I will then vary from $I = 1$ to $I = n_x n_y n_z$. We may then define our multi-dimensional eigenfunction to be

$$\Phi_I(x, y, z) = \Phi_I = \cos(k_{xI}(x - L_{xmin}))\cos(k_{yI}(y - L_{ymin}))\cos(k_{zI}(z - L_{zmin})). \tag{7.1.24}$$

7.2 Solution for the Pressure Field

Having the natural frequencies in Eq. (7.1.22) and the corresponding eigenfunctions in Eq. (7.1.24), we may now obtain solutions to Eq. (7.0.2). To use a modal approach, we will seek solutions in the form of a series of the eigenfunctions,

$$p(x, y, z, t) = \sum_{I=1}^{N} a_I(t)\Phi_I(x, y, z), \tag{7.2.1}$$

where $a_I(t)$ are unknown functions of time to be determined and $N = n_x n_y n_z$ is the total number of terms in the sum. Since we are interested in steady-state solutions and the time dependence of the sound source is of the form $e^{i\omega t}$, we may assume that $a_I(t) = A_I e^{i\omega t}$, where A_I are complex constants. Substituting Eq. (7.2.1) into Eq. (7.0.2) and rearranging give

$$\sum_{I=1}^{N} A_I(\nabla^2\Phi_I + k^2\Phi_I) = -4\pi D_0\delta(\bar{r} - \bar{r}_p). \tag{7.2.2}$$

From Eqs. (7.1.23) and (7.1.24), it can be seen that

$$\nabla^2\Phi_I = -k_I^2\Phi_I \tag{7.2.3}$$

so that Eq. (7.2.2) becomes

$$\sum_{I=1}^{N} A_I(k^2 - k_I^2)\Phi_I = -4\pi D_0\delta(\bar{r} - \bar{r}_p). \tag{7.2.4}$$

The unknown A_I may be determined by multiplying Eq. (7.2.4) by Φ_J and integrating over the entire domain, V. After some rearranging this gives

$$\sum_{I=1}^{N} A_I(k^2 - k_I^2)\int_V \Phi_J\Phi_I dV = -\int_V \Phi_J 4\pi D_0\delta(\bar{r} - \bar{r}_p)dV. \tag{7.2.5}$$

The volume integral on the left side of Eq. (7.2.5) consists of the product of three integrals, one for each spatial coordinate. From Eq. (7.1.21), the integral over x has the form

$$\begin{aligned}\int_{L_{xmin}}^{L_{xmax}} \cos(k_{xI}(x - L_{xmin})) \times \cos(k_{xJ}(x - L_{xmin}))dx &= norm_x(I, J)\\ &= \delta_{k_{xI}k_{xJ}} L_x/2 \quad \text{if} \quad k_{xI} \neq 0\\ &= \delta_{k_{xI}k_{xJ}} L_x \quad \text{if} \quad k_{xI} = 0,\end{aligned} \tag{7.2.6}$$

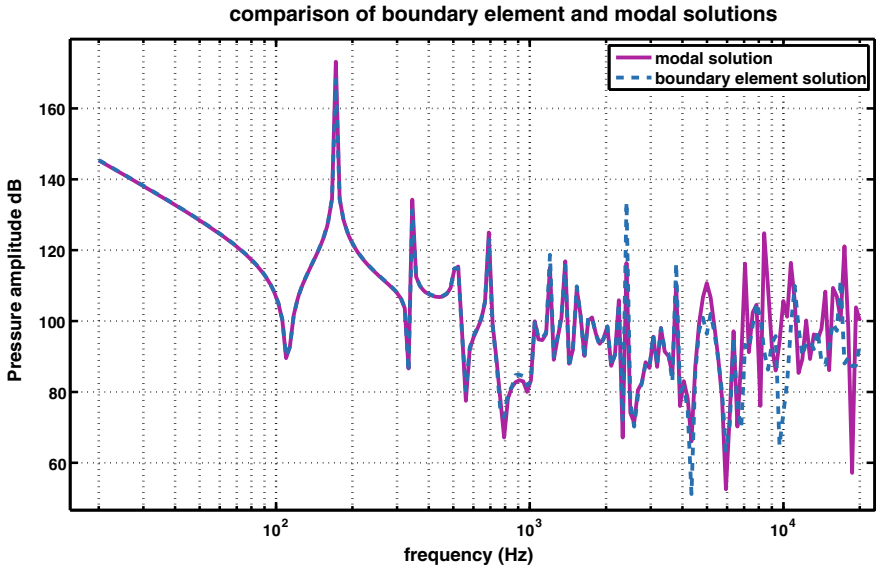


Fig. 7.1 Sound pressure at $z = -0.4$ due to a point source at $z = -0.273$. Results were computed using the modal solution and using the boundary integral approach described in Chap. 6. The results are in excellent agreement

where $\delta_{k_{xI}k_{xJ}}$ is the Kroniker delta function,

$$\begin{aligned} \delta_{k_{xI}k_{xJ}} &= 1 \quad \text{if} \quad k_{xI} = k_{xJ} \\ &= 0 \quad \text{if} \quad k_{xI} \neq k_{xJ}. \end{aligned} \tag{7.2.7}$$

Integrations over y and z may be performed similarly and will be denoted by $norm_y(I, J)$ and $norm_z(I, J)$.

The integral on the left side of Eq. (7.2.5) becomes

$$\int_V \Phi_J \Phi_I dV = \delta_{IJ} \int_V \Phi_I^2 dV = norm_x(I, J) \times norm_y(I, J) \times norm_z(I, J) = Norm_I. \tag{7.2.8}$$

This constant may be stored for each index I. The sifting property of the Dirac delta function enables us to evaluate the integral on the right side of Eq. (7.2.5),

$$\int_V \Phi_J 4\pi D_0 \delta(\bar{r} - \bar{r}_p) dV = 4\pi D_0 \Phi_J(R_{p1}, R_{p2}, R_{p3}), \tag{7.2.9}$$

where R_{p1}, R_{p2}, and R_{p3}, are the coordinates of the point source as in Eq. (6.6.10).

Equation (7.2.5) may then be solved for A_J,

$$A_J = \frac{4\pi D_0 \Phi_J(R_{p1}, R_{p2}, R_{p3})}{(k_J^2 - k^2) Norm_J}. \tag{7.2.10}$$

Equation (7.2.1) then gives the pressure anywhere in the domain,

$$p(x, y, z, t) = e^{i\omega t} \sum_{I=1}^{N} \frac{4\pi D_0 \Phi_I(R_{p1}, R_{p2}, R_{p3})}{(k_I^2 - k^2) Norm_I} \Phi_I(x, y, z), \tag{7.2.11}$$

where we have replaced J with I in Eq. (7.2.10) since the summation is expressed in terms of I.

7.3 Numerical Results

To examine the accuracy of results obtained using the modal approach described above, a simple domain was studied having dimensions identical to that of Fig. 6.1. This model included a point source with amplitude $P0 = 0.01$ at the location $(R_{p1}, R_{p2}, R_{p3}) = (0, 0, -0.273)$. The calculations were performed using $n_x = n_y = 8$ and $n_z = 100$ for a total of $N = 6400$ terms in the series of Eq. (7.2.11). The results are shown in Fig. 7.1 along with those obtained using the boundary element solution as described in Chap. 6.

A Matlab file have been made available to obtain the modal solution (see "AmplitudeComparisonWithModalFromrigidboxwithpointsource042217.m").

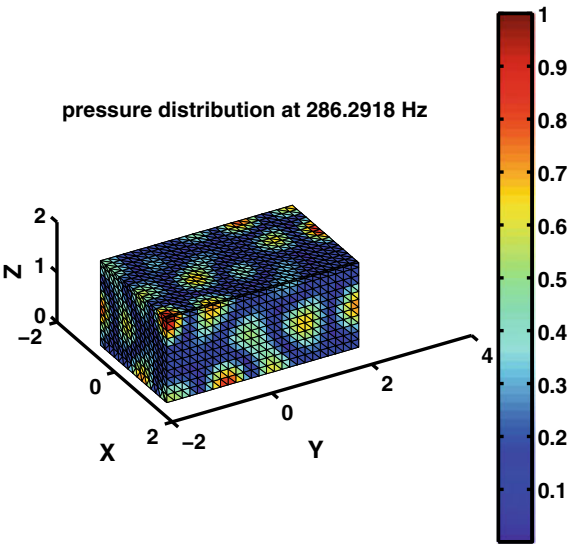


Fig. 7.2 Distribution of sound pressure at the frequency 286.3 Hz due to a point source at $(R_{p1}, R_{p2}, R_{p3}) = (0, 0, 0.5467)$. Results were computed using the boundary integral approach

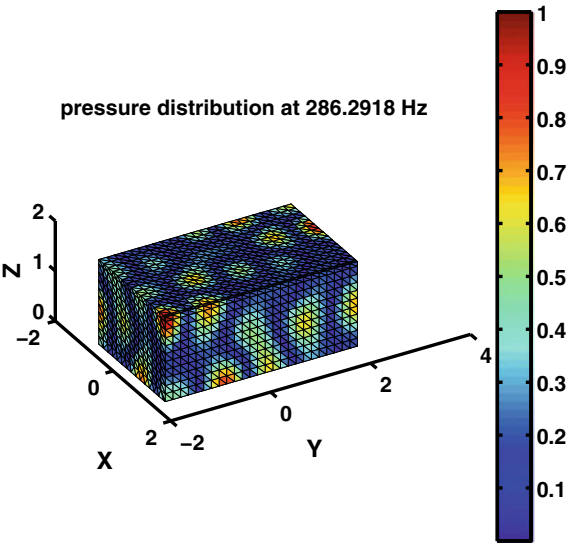


Fig. 7.3 Distribution of sound pressure at the frequency 286.3 Hz due to a point source at $(R_{p1}, R_{p2}, R_{p3}) = (0, 0, 0.5467)$. Results were computed using the modal solution. The results are in excellent agreement with those shown in Fig. 7.2

As another verification, the sound field has been computed in a rectangular enclosure having dimensions $(Lx, Ly, Lz) = (2.3011, 3.8452, 1.6907)$ meters with a point sound source located at $(R_{p1}, R_{p2}, R_{p3}) = (0, 0, 0.5467)$ with amplitude $P0 = 0.01$. In this enclosure, $(L_{xmin}, L_{ymin}, L_{zmin}) = (-0.928, -1.7478, 0)$. The number of terms in the modal expansion has been taken to be $n_x \times n_y \times n_z = 50 \times 50 \times 100 = 250{,}000$. A total of 3260 triangles were used to describe the surface pressure.

The distribution of pressure on the surface as calculated using the boundary integral approach is shown in Fig. 7.2. These results are essentially identical to those obtained using the modal solution shown in Fig. 7.3.

Chapter 8
Geometrical Room Acoustics



The analytical methods of the previous chapters enable predictions of the sound field in an enclosure accounting for numerous geometric and material details of the boundary conditions. In practice however, many acoustical spaces of interest have far too many geometric complexities to permit meaningful predictions of a sound field. In this case, rather than perform painstaking, detailed estimates of the field, it often suffices to estimate the spatial average of the field due to changes in some global, or average acoustical features of the space. For example, it is often extremely important to be able to estimate how the average sound levels in a space will be affected by changes in the average sound absorption on the surfaces. The ability to perform this sort of rough estimate can provide valuable guidance in designs and gives much needed insight without the burden of complicated mathematics. This approach is often referred to as "geometrical" acoustics. The primary quantities used to characterize the sound field will be the sound energy density, $\mathcal{E}$, and the sound intensity, I, that is incident on all of the surfaces. The calculations are greatly simplified by seeking the spatial average of the energy density and by assuming that the contribution due to reflections within the enclosure create a spatially uniform, or diffuse, field. Once the sound energy density is estimated, the mean square pressure may be calculated from

$$< p^2(t) >= \rho_0 c^2 \mathcal{E} \tag{8.0.1}$$

The sound pressure level, SPL, is then calculated from Eq. (1.1.1)

$$SPL = 10 \log_{10} \left(\frac{< p^2(t) >}{P_{ref}^2} \right). \tag{8.0.2}$$

We will need a relationship between the sound energy density and the sound power per unit area, or intensity I, that is incident on all of the absorbing surfaces as was

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_8

discussed in Sect. 2.5. This can be expressed in detail for the simple case of a plane sound wave propagating in the positive x direction. A general solution to the wave equation in this case was examined in Sect. 2.7.

8.1 Relation Between Energy Density and Intensity

The sound energy per unit volume in a plane sound wave may be expressed by considering the rate at which energy enters and leaves a differential volume as shown in Fig. 8.1. Let the intensity at the left end of this volume be $I(x,t)$ and the intensity on the right surface be $I(x+dx,t)$. The net rate at which energy (i.e., the power) enters per unit volume is

$$\lim_{dx\to 0} \frac{I(x,t) - I(x+dx,t)}{dx} = -\frac{\partial I}{\partial x}. \tag{8.1.1}$$

The energy density will then be equal to the integral of this expression over time,

$$\mathcal{E} = -\int \frac{\partial I}{\partial x} dt. \tag{8.1.2}$$

In the case of a simple plane sound wave the spatial derivative in the integrand can be expressed as a time derivative because of the form of d'Alembert's solution in Eq. (2.6.6). In Sect. 2.7 we showed that the pressure in this one dimensional plane wave may be written as in Eq. (2.6.6) with $g(x+ct) = 0$. Fortunately, both the pressure, $p(x,t)$ and the acoustic particle velocity, $u(x,t)$ and hence their product, the intensity, $I(x,t)$, may be expressed as functions of $\eta = x - ct$, where c is the sound speed,

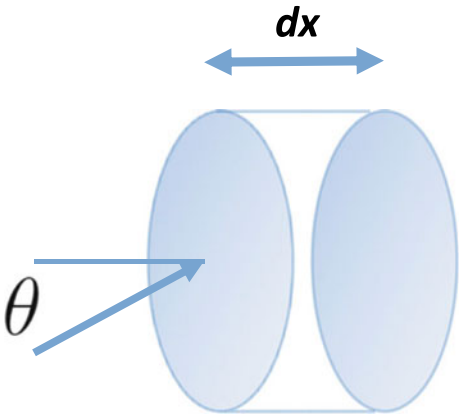


Fig. 8.1 Differential volume in a diffuse sound field

$$I(x,t) = p(x-ct)u(x-ct) = I(x-ct) = I(\eta). \tag{8.1.3}$$

The derivative in the integrand of Eq. (8.1.2) may then be written as

$$\frac{\partial I}{\partial x} = \frac{\partial I}{\partial \eta}\frac{\partial \eta}{\partial x} = \frac{\partial I}{\partial \eta} \tag{8.1.4}$$

since $\frac{\partial \eta}{\partial x} = 1$. Now consider the derivative of I with respect to t,

$$\frac{\partial I}{\partial t} = \frac{\partial I}{\partial \eta}\frac{\partial \eta}{\partial t} = \frac{\partial I}{\partial \eta}(-c) \tag{8.1.5}$$

since, again, $\eta = x - ct$. Equations (8.1.4) and (8.1.5) then give

$$\frac{\partial I}{\partial x} = \frac{-1}{c}\frac{\partial I}{\partial t}\frac{-1}{c}. \tag{8.1.6}$$

Equation (8.1.2) then becomes simply

$$\mathcal{E} = -\int \frac{\partial I}{\partial x}dt = -\frac{-1}{c}\int \frac{\partial I}{\partial t}dt = \frac{I}{c}. \tag{8.1.7}$$

Again, this simple result holds for a plane wave traveling in one direction. Fortunately, a similar result is also obtained for radiation from a spherical source at sufficient distances from the source.

Suppose now that instead of a plane sound wave we have waves incident from all possible directions with intensity I. A plane wave incident at an angle θ on the differential volume shown in Fig. 8.1 adds power per unit area equal to $I\cos(\theta)$. Adding up all of the energy density due to all possible waves incident on the left surface of this differential volume gives

$$\int_{-\pi/2}^{\pi/2} \frac{I}{c}\cos(\theta)d\theta = \frac{2I}{c}. \tag{8.1.8}$$

An equal amount of energy will be added to the differential volume due to waves incident on the right surface so that the relation between the total energy density and the intensity of the sound waves is

$$\mathcal{E} = \frac{4I}{c}. \tag{8.1.9}$$

This assumes that all waves in the space have equal sound intensity, which implies that the field is “diffuse”.

8.2 Effect of Sound Absorption Coefficient on Steady-State Sound Fields

If one knows the sound intensity that is incident on all of the surfaces of an enclosure, the absorption coefficient, α, of all of those surfaces along with the amount of power that is input to the sound field, then this information must enable one to employ the principle of conservation of energy to construct useful relationships between these quantities. In the following, we will make use of this power balance to estimate the average sound pressure level in an enclosure due to the sound power input from a source of sound.

Suppose a sound source emits acoustic power Π as a spherically symmetric, nondirectional source. The sound intensity a distance R from this source due to direct radiation (i.e., that has not been reflected by any surface) will be

$$I_d = \frac{\Pi}{4\pi R^2}. \tag{8.2.1}$$

The acoustic power input to the enclosure must equal the total power absorbed by all of the surfaces,

$$\Pi = \int I\alpha ds. \tag{8.2.2}$$

The total sound intensity, I, incident on any surface of the enclosure will be the sum of the direct intensity radiated by the source and that intensity that is due to reflections, the reverberant intensity I_{rev}

$$I = I_d + I_{rev}. \tag{8.2.3}$$

Substituting Eq. (8.2.3) into Eq. (8.2.2) gives

$$\Pi = \int I_d\alpha ds + \int I_{rev}\alpha ds. \tag{8.2.4}$$

We will approximate the first integral as

$$\int I_d\alpha ds \approx \frac{\Pi}{S}\int \alpha ds = \Pi < \alpha >, \tag{8.2.5}$$

where $< \alpha >$ is the surface average of the absorption coefficient and S is the total surface area. We will also assume that the reverberant component of the intensity, I_{ref} is uniform on the surface. Equation (8.2.4) then may be approximated by

$$\Pi = \Pi < \alpha > + I_{rev}\int \alpha ds = \Pi < \alpha > + I_{rev}S < \alpha > . \tag{8.2.6}$$

The reverberant part of the intensity then becomes

$$I_{rev} = \frac{\Pi(1- < \alpha >)}{S < \alpha >}. \tag{8.2.7}$$

Equations (8.2.1), (8.2.3) and (8.2.7) give the total intensity incident on the surfaces

$$I = \frac{\Pi}{4\pi R^2} + \frac{\Pi(1- < \alpha >)}{S < \alpha >}. \tag{8.2.8}$$

The sound energy density within the enclosure is then

$$\mathcal{E} = \frac{\Pi}{c4\pi R^2} + 4\frac{\Pi(1- < \alpha >)}{cS < \alpha >}, \tag{8.2.9}$$

where we have used the fact that the first term is due to the direct radiation of a spherical source and the second term corresponds to the diffuse reverberant field as in Eq. (8.1.9). The mean square pressure is obtained from Eqs. (8.0.1) and (8.2.9),

$$< p^2(t) >= \rho_0 c^2 \mathcal{E} = \rho_0 c \left(\frac{\Pi}{4\pi R^2} + 4\frac{\Pi(1- < \alpha >)}{S < \alpha >} \right). \tag{8.2.10}$$

Another important situation occurs when the source of sound is such that the direct component of the intensity is fairly uniform on the surfaces of the enclosure. This happens when the source of sound is transmission through the walls from the exterior, as occurs in transportation vehicles. We may then take the direct component of the sound intensity to simply be $I_d = \Pi/S$. The total sound intensity then becomes

$$I = \frac{\Pi}{S} + \frac{\Pi(1- < \alpha >)}{S < \alpha >} = \frac{\Pi}{S < \alpha >}. \tag{8.2.11}$$

Because the direct intensity is uniform on the surface, it may be considered to be diffuse along with the reverberant component so that the energy density and mean square pressure become

$$\mathcal{E} = \frac{4I}{c} = \frac{4\Pi}{Sc < \alpha >} \quad < p^2(t) >= \frac{4\rho_0 c\Pi}{S < \alpha >}. \tag{8.2.12}$$

The sound pressure level in the enclosure may be written as

$$SPL = 10\log_{10}\left(\frac{4\rho_0 c\Pi}{S < \alpha > p_{ref}^2} \right) = 10\log_{10}\left(\frac{4\rho_0 c\Pi}{Sp_{ref}^2} \right) - 10\log_{10}(< \alpha >). \tag{8.2.13}$$

8.3 Effect of Sound Absorption Coefficient on Decaying Sound Fields

The approach employed above to estimate the steady-state sound in an enclosure may be utilized to obtain a simple relationship between the rate at which sound decays in an enclosure after a sound source has been removed and the sound absorption coefficient of the surfaces. This again, may be constructed by considering the power balance between the input power and the absorbed power. The rate of change of sound energy in the enclosure with volume V and energy $\mathcal{E}$ may be written as the difference between the power input, Π, and the power absorbed by the walls,

$$V\frac{d\mathcal{E}}{dt} = \Pi - I < \alpha > S. \tag{8.3.1}$$

If we assume that the sound field is diffuse so that $\mathcal{E} = 4I/c$ as in Eq. (8.1.9), we may eliminate the intensity to obtain a first order ordinary differential equation for the sound energy density,

$$V\frac{d\mathcal{E}}{dt} + \frac{c\mathcal{E}}{4} < \alpha > S = \Pi. \tag{8.3.2}$$

If the power Π is constant in time for $t < 0$ and shut off at $t = 0$, Eq. (8.3.2) becomes

$$\frac{d\mathcal{E}}{dt} + \frac{c\mathcal{E}}{4V} < \alpha > S = 0 \tag{8.3.3}$$

for $t > 0$, which has the solution

$$\mathcal{E} = \mathcal{E}_0 e^{-\frac{c\mathcal{E}}{4V} < \alpha > St} = \mathcal{E}_0 e^{-t/\tau}, \tag{8.3.4}$$

where $\mathcal{E}_0$ is the initial energy density at $t = 0$ and the decay rate is

$$\tau = \frac{4V}{c\mathcal{E} < \alpha > S}. \tag{8.3.5}$$

The "reverberation time" of an enclosure is taken to be the time it takes for the sound pressure level to drop by 60 dB from the sound level when the sound source is turned off. The sound pressure level will be

$$\begin{aligned} SPL &= 10\log_{10}\left(\frac{\rho_0 c^2 \mathcal{E}}{p_{ref}^2}\right) = 10\log_{10}\left(\frac{\rho_0 c^2 \mathcal{E}_0 e^{-t/\tau}}{p_{ref}^2}\right) \\ &= 10\log_{10}\left(\frac{\rho_0 c^2 \mathcal{E}_0}{p_{ref}^2}\right) + 10\log_{10}(e^{-t/\tau}). \end{aligned} \tag{8.3.6}$$

The reverberation time will be the time t when

$$10\log_{10}(e^{-t/\tau}) = -60. \tag{8.3.7}$$

The solution is

$$t = \frac{6\tau}{\log_{10} e} \approx \frac{13.8 \times 4V}{Sc < \alpha >} \approx \frac{0.16V}{S < \alpha >}. \tag{8.3.8}$$

8.4 Problems

1. An enclosure has dimensions $3 \times 6.5 \times 9.1$ m. A calibrated sound source emits sound power $\Pi = 0.01$ acoustic watts. The sound pressure level is measured to be 94 dB and is dominated by the reverberant component. Determine the average sound absorption coefficient of the room, α.
2. A portion of the surface of the room of Problem 1 is covered with an absorbent material having an area of $2\,\text{m}^2$. The measured sound pressure level is reduced by 2.5 dB. Determine the absorption coefficient of the added material.
3. For the room of Problem 2, plot the sound pressure level as a function of the distance from the sound source including both the direct and reverberant components. At what distance is the reverberant component equal to the direct component?
4. The floor and ceiling of a rectangular room have dimensions 6.5×9.1 m and the ceiling is 3 m above the floor. The floor is carpeted with a carpet having an absorption coefficient of $\alpha = 0.4$, and the floors and ceiling are made of hardwood plywood paneling having an absorption coefficient of $\alpha = 0.02$. If a listener is 4 m from a sound source, determine the change in sound level at 1000 Hz due to replacing the carpet with glazed tile having an absorption coefficient of $\alpha = 0.01$.
5. The absorption coefficient of a certain material was measured in a reverberation chamber of volume $1300\,\text{m}^3$ and surface area of $720\,\text{m}^2$. The area of the test sample is $30\,\text{m}^2$. The measured average reverberation times (RT) in seconds at several frequencies are shown in the table below. Find the absorption coefficients of the material at the frequencies given.

Frequency (Hertz)	RT empty	RT with sample
125	16.8	10.2
250	20.1	10.4
500	18.5	9.4
1000	14.5	8.0
2000	9.1	6.1

6. A room 16 m long, 10 m wide and 5 m high which was previously used as a laboratory is to be converted to use as a lecture room for 200 people. The original wall and floor surfaces are hard plaster and concrete whose average absorption coefficient is 0.05. Acoustical tiles having an absorption coefficient of 0.75 are available for application to the walls and ceiling. Calculate the area of tile to

be applied to achieve a reverberation time of 0.5 s. Assume that the effective absorption coefficient multiplied by the effective surface area of a seated audience (per person) is $0.4\,m^2$ metric Sabines.

7. An auditorium is observed to have a reverberation time of 2.0 s. Its dimensions are $7 \times 15 \times 30$ m. (a) What acoustic power is required to cause the spatial average of the steady-state sound pressure level to be 60 dB relative to $20\,\mu$ Pa? (b) What is the average sound absorption coefficient of the interior surface of the auditorium?

Chapter 9
Effects of Viscosity



There are many situations where we are interested in the sound field in the vicinity of small objects. This often requires us to account for the viscosity of the medium. This greatly complicates the relationship between fluid velocity near solid obstacles and the fluctuating pressure. In the following, the differential equations for acoustic fluctuations in a viscous fluid are presented. Obtaining solutions to these equations can provide endless challenges. Here we solve them for some specialized situations that are relevant for acoustic sensing.

9.1 Basic Equations for Sound in a Viscous Fluid

Let the velocity of the acoustic medium be described by a vector having components u_j, for $j = 1, 2, 3$. It is fortunate that for most sound pressure fluctuations that are normally encountered, the fluid velocity is typically small so that it may be represented by the linearized Navier–Stokes equations. The constitutive equation for a newtonian fluid may then be expressed as

$$\sigma_{ij} = -\mathcal{P}\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) - \frac{2}{3}\mu\delta_{ij}\frac{\partial u_k}{\partial x_k} \tag{9.1.1}$$

where σ_{ij} is the stress tensor, u_i is the velocity, $\mathcal{P}$ is the thermodynamic pressure, δ_{ij} is the Kroniker delta function, and μ is the *dynamic viscosity*. In air, $\mu \approx 1.846 \times 10^{-5}$ kg/m/s. As usual when using tensor notation, the repeated index in the last term implies summation. The dynamic viscosity is defined to be the ratio of shear stress to the velocity gradient in simple shear flow. For example, if there is a velocity gradient $\frac{\partial u_1}{\partial x_3}$, and all other velocity gradients are zero, the resulting stress will be

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_9.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_9

$$\sigma_{13} = \mu \frac{\partial u_1}{\partial x_3} \tag{9.1.2}$$

It is sometimes convenient to define a *kinematic viscosity* by $\nu = \mu/\rho$ where ρ is the density. The conservation of mass in any differential volume may be written as

$$\dot{\rho} + \frac{\partial \rho u_k}{\partial x_k} = \dot{\rho} + \frac{\partial \rho u_1}{\partial x_1} + \frac{\partial \rho u_2}{\partial x_2} + \frac{\partial \rho u_3}{\partial x_3} = 0. \tag{9.1.3}$$

We will again consider the density to consist of the combination of the static air density, ρ_0 and an acoustic fluctuation ρ_a, so that $\rho = \rho_0 + \rho_a$. The pressure is also a combination of the static ambient pressure, P_0 and the acoustic fluctuating pressure, p, $\mathcal{P} = P_0 + p$. The acoustic fluctuation in the pressure is related to the fluctuation in the density, ρ_a through the equation of state, Eq. (2.3.3),

$$p = \rho_a c^2 \tag{9.1.4}$$

If we consider an arbitrary volume V having area S, the conservation of momentum may be written as

$$\int_s \sigma_{ij} n_i dS + \int_V \rho f_j dV = \frac{D}{Dt} \int_V \rho u_j dV \tag{9.1.5}$$

where n_i is the unit normal vector, f_j is the body force and $\frac{D}{Dt}$ is the "total derivative." The total derivative of some quantity, α is the change in α as seen by an observer who is moving with the fluid. In tensor notation, the total derivative may be written as

$$\frac{D\alpha}{Dt} = \frac{\partial \alpha}{\partial t} + u_k \frac{\partial \alpha}{\partial x_k} \tag{9.1.6}$$

In acoustics, since we are usually concerned only with very small motions of the fluid, the total derivative is usually replaced by the eulerian derivative $\partial\alpha/\partial t$. Body forces f_j in Eq. (9.1.5) are also normally of negligible importance in acoustics so that the momentum conservation equation becomes

$$\int_s \sigma_{ij} n_i dS = \frac{d}{dt} \int_V \rho u_j dV \tag{9.1.7}$$

The surface integral on the left side of this equation may be written as a volume integral through the use of Gauss' theorem,

$$\int_V \frac{\partial \sigma_{ij}}{\partial x_i} dV = \frac{d}{dt} \int_V \rho u_j dV \tag{9.1.8}$$

Since this must be true for an arbitrary volume, V,

$$\frac{\partial \sigma_{ij}}{\partial x_i} = \frac{d}{dt}(\rho u_j) \tag{9.1.9}$$

We can again use the fact that the density consists of the combination of the static air density ρ_0 plus an acoustic fluctuation ρ_a,

$$\frac{\partial \sigma_{ij}}{\partial x_i} = \frac{d}{dt}((\rho_0 + \rho_a)u_j) \approx \rho_0 \frac{du_j}{dt}, \tag{9.1.10}$$

where we have neglected products of small fluctuating terms, $\rho_a u_j$.

Substituting the constitutive equation (9.1.1) into (9.1.10) gives the equation of motion in terms of small fluctuating quantities,

$$\begin{aligned}\rho_0 \frac{du_j}{dt} &= -\frac{\partial p}{\partial x_i}\delta_{ij} + \mu \frac{\partial}{\partial x_j}\left(\frac{\partial u_i}{\partial x_i}\right) + \mu \frac{\partial^2 u_j}{\partial x_i^2} - \frac{2}{3}\mu \frac{\partial}{\partial x_j}\left(\frac{\partial u_k}{\partial x_k}\right) \\ &= -\frac{\partial p}{\partial x_j} + \frac{\mu}{3}\frac{\partial}{\partial x_j}\left(\frac{\partial u_k}{\partial x_k}\right) + \mu \frac{\partial^2 u_j}{\partial x_i^2}.\end{aligned} \tag{9.1.11}$$

This result can be simplified by using the mass conservation for small fluctuating quantities along with the equation of state. For small fluctuations, Eq. (9.1.3) becomes

$$\dot{\rho}_a + \rho_0 \frac{\partial u_k}{\partial x_k} = 0. \tag{9.1.12}$$

Combining this with Eq. (9.1.4) gives

$$\frac{\partial u_k}{\partial x_k} = -\frac{\dot{p}}{\rho_0 c^2}. \tag{9.1.13}$$

Since we are primarily interested in harmonic acoustic signals having frequency ω, all time varying quantities will be denoted as a complex magnitude multiplied by $e^{i\omega t}$. The pressure is then $p = Pe^{i\omega t}$ where P is a function of location only. Equations (9.1.11) and (9.1.13) then give

$$\rho_0 \frac{du_j}{dt} = -\frac{\partial^2 P}{\partial x_j}\left(1 + \frac{i\mu\omega}{3\rho_0 c^2}\right) + \mu \frac{\partial^2 u_j}{\partial x_i^2}. \tag{9.1.14}$$

Let

$$\frac{i\mu\omega}{3\rho_0 c^2} = \frac{i\omega}{\omega_C} \tag{9.1.15}$$

where $\omega_C = 3\rho_0 c^2/\mu$ indicates the frequency where this term becomes comparable to unity. For typical values of ρ, c, and μ, $\omega_C \approx 2.3 \times 10^{10}$ or 3.7×10^9 Hz. For typical frequencies of interest, our governing equation may then be approximated by

$$\rho_0 \frac{du_j}{dt} \approx -\frac{\partial p}{\partial x_j} + \mu \frac{\partial^2 u_j}{\partial x_i^2}. \tag{9.1.16}$$

If we let the unit vectors in the 1, 2, and 3 directions be $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$, respectively, then the ∇ operator is defined as

$$\nabla = \frac{\partial}{\partial x_1}\hat{\imath} + \frac{\partial}{\partial x_2}\hat{\jmath} + \frac{\partial}{\partial x_3}\hat{k} \tag{9.1.17}$$

The ∇ operator dotted with itself is

$$\nabla \cdot \nabla = \frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial x_2^2} + \frac{\partial^2}{\partial x_3^2} \tag{9.1.18}$$

which may be written in tensor notation as

$$\nabla \cdot \nabla = \frac{\partial^2}{\partial x_i \partial x_i} \tag{9.1.19}$$

The velocity vector is

$$\bar{u} = u_1\hat{\imath} + u_2\hat{\jmath} + u_3\hat{k} \tag{9.1.20}$$

Equations (9.1.17), (9.1.19), and (9.1.20) enable us to write the momentum conservation equation (9.1.16) in vector form as

$$\rho_0 \frac{d\bar{u}}{dt} = -\bar{\nabla} p + \mu \nabla^2 \bar{u}. \tag{9.1.21}$$

In addition to momentum conservation, we may also write the linearized mass conservation equation (9.1.13),

$$\bar{\nabla} \cdot \bar{u} = -\frac{\dot{p}}{\rho_0 c^2}. \tag{9.1.22}$$

The three vector components of Eq. (9.1.21) along with the scalar equation (9.1.22) comprise four partial differential equations in the four unknowns consisting of the three components of the vector velocity, $\bar{u}$ and the scalar pressure p. These equations are solvable (in principle) once the boundary and initial conditions are properly specified. As an illustration, in the following we will limit our attention to a

two-dimensional problem that submits to known methods of obtaining closed-form solutions of partial differential equations.

9.2 Viscous Flow in One Dimension

There are a number of situations, particularly in the design of miniature acoustic sensors, where the effects of viscosity can have a marked influence. In Sect. 10.4 we will find that viscosity plays an important role in determining the response of pressure microphones at low frequencies. These microphones use a pressure-sensing diaphragm having one side (the front side) that is exposed to the incident sound and having its back side enclosed by the microphone package. The diaphragm responds to the difference in pressure on its external surface and the pressure within the closed volume of air. In order to eliminate the influence of atmospheric pressure changes, pressure microphones incorporate a small vent, which enables very low frequency pressure fluctuations to equalize on either side of the pressure-sensing diaphragm. A very rough model of the effect of this vent can be developed using Eqs. (9.1.21) and (9.1.22) where we greatly simplify the flow by considering it to be dominated by velocity in only one dimension.

Let the momentum conservation equation (9.1.21) be written as

$$-\nabla P + \mu \nabla^2 u = \rho_0 \dot{u} \tag{9.2.1}$$

The domain will consist of a slit having depth (in the direction, x, of the flow) L and width $2h$. The slit will be assumed to be sufficiently small that the pressure gradient does not vary across the width and will be equal to the difference in pressure on either side divided by the depth, L, $\nabla P = \partial P/\partial x = P_x = (p_1 - p_2)/L$, where p_1 is the external pressure and p_2 is the pressure on the interior of the back volume. We will assume that the velocity of the air in the slit is a constant through its depth (in the x direction) so that $u = u(y)$. In order to obtain an approximate expression for the equivalent damping of the air in the slit, we will first assume that the flow of the air is steady so that $\dot{u} = 0$.

The Eq. (9.2.1) then becomes

$$-P_x + \mu \frac{\partial^2 u}{\partial y^2} = 0 \tag{9.2.2}$$

Integrating this equation twice over the depth, y gives

$$u(y) = \frac{P_x}{2\mu L} y^2 + ay + b \tag{9.2.3}$$

where a and b are constants of integration. If the velocity of the wall of the slit at $y = -h$ is $u(-h) = 0$ and $u(h) = U_0$ then Eq. (9.2.3) becomes

$$u(y) = -P_x \frac{h^2}{2\mu}(1 - (y/h)^2) + \frac{U_0}{2}(1 + y/h) \tag{9.2.4}$$

Equation (9.2.4) reduces to Couette flow when $P_x = 0$ and Poiseuille flow when $U_0 = 0$.

If we assume the the slit (or vent) has stationary walls, Eq. (9.2.4) gives the velocity of the viscous flow in the slit in the form of a parabolic profile given by

$$u(y) = -P_x \frac{h^2}{2\mu}(1 - (y/h)^2), \tag{9.2.5}$$

where $\mu = 1.846 \times 10^{-5}$ kg/m/s, and h is half the distance across the gap. To compute the equivalent viscous damping coefficient we need to relate the net force acting on the air in the gap to the average velocity, $< u(y) >$, where

$$< u(y) >= \frac{1}{2h}\int_{-h}^{h} u(y)dy = -P_x \frac{h^2}{2\mu}\frac{1}{2h}\int_{-h}^{h}(1 - (y/h)^2)dy = -P_x \frac{h^2}{3\mu}. \tag{9.2.6}$$

The gradient of the pressure in the slit may be taken as the net pressure on its exterior divided by the depth, $L \approx 2\,\mu$m, $\frac{\partial P}{\partial x} = (p_1 - p_2)/L = P_{net}/L$, so that

$$P_{net} = -\frac{3\mu L}{h^2} < u(y) > . \tag{9.2.7}$$

If the total length of the slit is w then the net force is

$$f_{net} = P_{net} 2hw = -\frac{6\mu L w}{h} < u(y) > . \tag{9.2.8}$$

The equivalent dash-pot constant is then

$$c_v = \frac{6\mu L w}{h}. \tag{9.2.9}$$

If $L = 2\,\mu$m, $h = 1\,\mu$m, $w = 4$ mm, and $\mu = 1.846 \times 10^{-5}$ kg/m/s, the dashpot constant due to the slits is $c_v = 8.861 \times 10^{-7}$ N-s/m. This result assumes that the walls of the slit are stationary.

Now consider the damping of a diaphragm that has a narrow slit around its perimeter. In this case, one of the sides of the slit will move as the diaphragm. The ratio of shear stress on the face of the slit that is on the diaphragm to the rate of shear strain of the air in the slit is the coefficient of dynamic viscosity, μ, so that

$$\mu = \frac{\sigma}{U_0/(2h)} \tag{9.2.10}$$

where, again, the width of the slit is $2h$. The force on the diaphragm is $f = \sigma L w$ where w is the length of the slit and L is the depth. The equivalent dashpot is then

$$c = \frac{f}{U_0} = \frac{\mu L w}{2h} \tag{9.2.11}$$

9.3 Squeeze Film Damping in a Compressible Fluid in Two Dimensions

As will be discussed in the following chapters, most microphones rely on the use of parallel plate capacitive sensing to transduce the motion of a pressure-sensing diaphragm into an electronic signal. This is accomplished through the use of a fixed electrode placed parallel to the moving diaphragm. Changes in capacitance between these parallel plates provide a convenient means of obtaining an electronic output. The electrical sensitivity tends to increase as the gap between the diaphragm and the back plate electrode is reduced. An unfortunate design challenge with this type of capacitive microphone is that for small gaps, the air between these parallel electrodes tends to act as a viscous fluid for gaps that are used in typical designs (on the order of 10 microns).

The air between the diaphragm and back plate can impose forces on the diaphragm that substantially influence its ability to respond to sound. At low frequencies, the transverse motion of the diaphragm causes the air to flow in the direction perpendicular to the diaphragm motion, which pushes the air toward the edges of the diaphragm where it may be vented to the surrounding air. The horizontal motion of the air is resisted by viscosity, resulting in a velocity-dependent force imposed on the diaphragm (i.e. damping). At sufficiently high frequencies, the viscous force on the fluid can greatly restrict its ability to flow at all, causing this small air space to act as a closed volume. The diaphragm motion then results in compression of the air which imposes a force on the diaphragm that resembles that of a linear restoring spring. We thus obtain a frequency-dependent restoring force on the diaphragm that acts as a damper or a spring, or a combination of the two.

The importance of this squeeze-film effect in capacitive sensors has motivated much effort to investigate it [1–7]. Here, we will limit our attention to a two-dimensional application of Eqs. (9.1.21) and (9.1.22) to introduce the essential ideas without becoming overwhelmed by mathematical troubles. The system is shown in Fig. 9.1. The motion of the viscous, compressible fluid results from the vertical motion of the upper solid surface shown in the figure. Let the horizontal component of the fluid velocity between the two solids be $u(x, z, t)$ and the vertical component be $v(x, z, t)$. Because the fluid film is thin, $d_0 \ll L$, we will take the usual lubrication approximation that the pressure does not vary in the z direction so that the pressure is $p(x, t)$.

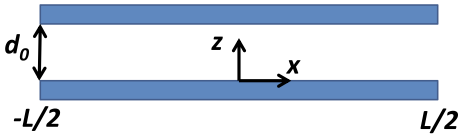


Fig. 9.1 A compressible, viscous fluid (air) is sandwiched between two solid surfaces separated in the z direction by a distance d_0. The width of the domain is L in the x direction. The system is assumed to be uniform in the direction out of the plane (y). The lower solid surface may be assumed to be stationary and the upper surface moves with a velocity $\mathcal{V}_0$. The origin is taken to be the center of the top of the lower solid surface

We will assume no slip between the fluid and solid interfaces so that the velocity boundary conditions are

$$\begin{aligned} &u(x,0,t) = u(x,d_0,t) = 0, \quad -L/2 \le x \le L/2 \\ &v(x,0,t) = 0, \quad v(x,d_0,t) = \mathcal{V} = \dot{d}_0, \quad -L/2 \le x \le L/2 \end{aligned} \tag{9.3.1}$$

We will also assume that the fluctuating pressure is zero at the openings so that

$$p(-L/2,t) = p(L/2,t) = 0 \tag{9.3.2}$$

Because $d_0 \ll L$ and the horizontal velocity must equal zero on either side of the gap, we will expect $\frac{\partial^2 u}{\partial z^2} \gg \frac{\partial^2 u}{\partial x^2}$. In this case, the two-dimensional version of Eq. (9.1.21) becomes

$$\begin{aligned} \frac{\partial p}{\partial x} &= \mu \frac{\partial^2 u}{\partial z^2} - \rho_0 \dot{u} \\ \frac{\partial p}{\partial z} &= 0 \end{aligned} \tag{9.3.3}$$

In this system, the linearized mass conservation equation (9.1.22) becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} = -\frac{\dot{p}}{\rho_0 c^2}. \tag{9.3.4}$$

We will assume harmonic time dependence so that

$$\begin{aligned} u(x,z,t) &= U(x,z)e^{i\omega t} \\ v(x,z,t) &= V(x,z)e^{i\omega t} \\ p(x,t) &= P(x)e^{i\omega t} \end{aligned} \tag{9.3.5}$$

The first of Eq. (9.3.3) then becomes

$$\frac{\partial^2 U}{\partial z^2} - \frac{i\omega\rho_0}{\mu}U = \frac{1}{\mu}\frac{\partial P}{\partial x}$$
$$\frac{\partial^2 U}{\partial z^2} + m^2 U = \frac{1}{\mu}\frac{\partial P}{\partial x} \tag{9.3.6}$$

where

$$m^2 = -\frac{i\omega\rho_0}{\mu} \tag{9.3.7}$$

Equation (9.3.6) is not difficult to solve since the right hand side is independent of z. The solution is

$$U(x,z) = a(x)\cos(mz) + b(x)\sin(mz) + \frac{1}{m^2\mu}\frac{\partial P}{\partial x} \tag{9.3.8}$$

Satisfying the first boundary condition Eq. (9.3.1) enables us to determine $a(x)$ and $b(x)$. This gives

$$U(x,z) = -\frac{1}{m^2\mu}\frac{\partial P}{\partial x}\left(1 - \cos(mz) - \frac{1-\cos(md_0)}{\sin(md_0)}\sin(mz)\right) \tag{9.3.9}$$

We need to differentiate this result with respect to x in order to use Eq. (9.3.4) to determine v,

$$\frac{\partial U}{\partial x} = -\frac{1}{m^2\mu}\frac{\partial^2 P}{\partial x^2}\left(1 - \cos(mz) - \frac{1-\cos(md_0)}{\sin(md_0)}\sin(mz)\right) \tag{9.3.10}$$

Equations (9.3.5), (9.3.4), and (9.3.10) give

$$\frac{\partial V}{\partial z} = -\frac{i\omega P}{\rho_0 c^2} - \frac{\partial U}{\partial x} = -\frac{i\omega P}{\rho_0 c^2} + \frac{1}{m^2\mu}\frac{\partial^2 P}{\partial x^2}\left(1 - \cos(mz) - \frac{1-\cos(md_0)}{\sin(md_0)}\sin(mz)\right) \tag{9.3.11}$$

Integrating this result over z gives

$$V(x,z) = -\frac{i\omega P z}{\rho_0 c^2} + \frac{1}{m^2\mu}\frac{\partial^2 P}{\partial x^2}\left(z - \frac{\sin(mz)}{m} - \frac{1-\cos(md_0)}{m\sin(md_0)}\cos(mz)\right) + \mathcal{C} \tag{9.3.12}$$

The boundary condition $v(x,0) = 0$ in Eq. (9.3.1) enables us to determine $\mathcal{C}$,

$$\mathcal{C} = -\frac{1}{m^2\mu}\frac{\partial^2 P}{\partial x^2}\left(\frac{1-\cos(md_0)}{m\sin(md_0)}\right) \tag{9.3.13}$$

The vertical velocity of the upper, moving plate in Fig. 9.1 is $V(x, d_0) = \mathcal{V} = \dot{d}_0$ (which doesn't depend on x) so that evaluating Eqs. (9.3.12) and (9.3.13) at $z = d_0$ gives

$$V(d_0) = \mathcal{V} = -\frac{i\omega P d_0}{\rho_0 c^2} + \frac{d_0}{m^2\mu}\frac{\partial^2 P}{\partial x^2}\left(1 - \frac{\sin(md_0)}{md_0} - \frac{(1-\cos(md_0))^2}{md_0\sin(md_0)}\right) \tag{9.3.14}$$

To simplify things, let

$$\beta = 1 - \frac{\sin(md_0)}{md_0} - \frac{(1-\cos(md_0))^2}{md_0\sin(md_0)} \tag{9.3.15}$$

Equation (9.3.14) may then be written as a simple second order differential equation for the pressure $P(x)$,

$$\frac{\partial^2 P}{\partial x^2} - \frac{i\omega m^2\mu}{\beta\rho_0 c^2}P = \mathcal{V}\frac{m^2\mu}{d_0\beta} \tag{9.3.16}$$

We are again fortunate that the right side of this equation is independent of x so that the solution may be written as

$$P(x) = A\cos(\gamma x) + B\sin(\gamma x) + \mathcal{V}\frac{i\rho_0 c^2}{\omega d_0} \tag{9.3.17}$$

where

$$\gamma^2 = -\frac{i\omega m^2\mu}{\beta\rho_0 c^2} = \left(\frac{\omega}{c}\right)^2\frac{1}{\beta} \tag{9.3.18}$$

and we have used Eq. (9.3.7) to eliminate m^2. Using the boundary condition in Eq. (9.3.2) enables us to determine A and B so that the solution for the pressure in the film becomes

$$P(x) = \mathcal{V}\frac{i\rho_0 c^2}{\omega d_0}\left(1 - \frac{\cos(\gamma x)}{\cos(\gamma L/2)}\right) \tag{9.3.19}$$

Since the motion of the upper moving solid element in Fig. 9.1 is typically assumed to be independent of x, we are often interested in the average pressure on $-L/2 < x < L/2$,

$$< P(x) >= \frac{1}{L}\int_{-L/2}^{L/2} P(x)dx = \mathcal{V}\frac{i\rho_0 c^2}{\omega d_0}\left(1 - \frac{2\sin(\gamma L/2)}{L\gamma\cos(\gamma L/2)}\right) \tag{9.3.20}$$

The ratio of the average pressure to the velocity, or the equivalent impedance, Z, is then

$$\frac{< P(x) >}{\mathcal{V}} = Z = \frac{i\rho_0 c^2}{\omega d_0}\left(1 - \frac{2\sin(\gamma L/2)}{L\gamma\cos(\gamma L/2)}\right) \tag{9.3.21}$$

9.3.1 Limiting Cases at Low and High Frequencies

It is also helpful to consider some limiting cases. When the moving element oscillates at a very low frequency, it is reasonable to neglect the time derivatives in Eqs. (9.3.3) and (9.3.4) so that

$$\begin{aligned} \frac{\partial p}{\partial x} &\approx \mu\frac{\partial^2 u}{\partial z^2} \\ \frac{\partial p}{\partial z} &= 0 \end{aligned} \tag{9.3.22}$$

In this system, the linearized mass conservation equation (9.1.22) becomes

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} \approx 0 \tag{9.3.23}$$

Integrating the first of Eqs. (9.3.22) over z twice and satisfying the boundary conditions $u(x, 0, t) = u(x, d_0, t) = 0$ in Eq. (9.3.1) give

$$U(x, z) = \frac{1}{2\mu}\frac{\partial P}{\partial x}(z^2 - zd_0) \tag{9.3.24}$$

where we have, again, assumed harmonic time dependence. Equation (9.3.23) then becomes

$$\frac{\partial V}{\partial z} = -\frac{\partial U}{\partial x} = -\frac{1}{2\mu}\frac{\partial^2 P}{\partial x^2}(z^2 - zd_0) \tag{9.3.25}$$

so that

$$V(x, z) = -\frac{1}{2\mu}\frac{\partial^2 P}{\partial x^2}\left(\frac{z^3}{3} - \frac{z^2 d_0}{2} + D\right) \tag{9.3.26}$$

where D is a constant of integration. Since $V(x, 0) = 0$, we have $D = 0$. The other boundary condition on V is $V(x, d_0) = \mathcal{V}$, using Eq. (9.3.1). Equation (9.3.26) becomes

$$\frac{\partial^2 P}{\partial x^2} = \mathcal{V}\frac{12\mu}{d_0^3} \tag{9.3.27}$$

Integrating this result twice over x and satisfying the boundary conditions in Eq. (9.3.1) give

$$P(x) = \mathcal{V}\frac{12\mu}{d_0^3}\left(\frac{x^2}{2} - \frac{L^2}{8}\right) \tag{9.3.28}$$

Finally, averaging this over the domain and simplifying give a simple approximate expression for the impedance that is valid at low frequencies,

$$\frac{< P(x) >}{\mathcal{V}} = \frac{\frac{1}{L}\int_{-L/2}^{L/2} P(x)dx}{\mathcal{V}} = -\frac{\mu L^2}{d_0^3} \tag{9.3.29}$$

As mentioned above, at sufficiently high frequencies, the viscous force can become large enough that it resists the fluid from flowing at all. To obtain an approximate expression in this case, we may assume that the horizontal velocity of the fluid, $u(x, z, t)$ is zero there. Equation (9.3.3) then indicates that the boundary condition on the pressure at $x = \pm L/2$ is

$$\frac{\partial P}{\partial x} = 0, \quad \text{at} \quad x = \pm L/2 \tag{9.3.30}$$

We may now return to Eq. (9.3.17) to determine the unknown constants, A and B to satisfy Eq. (9.3.30). Differentiating equation (9.3.17) with respect to x gives

$$\frac{\partial P}{\partial x} = -A\frac{\sin(\gamma x)}{\gamma} + B\frac{\cos(\gamma x)}{\gamma} \tag{9.3.31}$$

Equations (9.3.30) and (9.3.31) give $A = B = 0$. Equation (9.3.17) then leads to the impedance corresponding to when the compressible film is contained within a closed volume,

$$\frac{P(x)}{\mathcal{V}} = Z = \frac{i\rho_0 c^2}{\omega d_0} \tag{9.3.32}$$

This result is independent of x and of the viscosity μ. It is also helpful to note that because the velocity of the moving element is related to the displacement, $\mathcal{X}$, through $\mathcal{V} = i\omega\mathcal{X}$, Eq. (9.3.32) takes the form of a restoring spring, with spring constant K, as we expect for a closed volume of air

$$\frac{P(x)}{\mathcal{X}} = -\frac{\rho_0 c^2}{d_0} = -K \tag{9.3.33}$$

9.3.2 *Numerical Results*

Numerical results are presented in Fig. 9.2. The figure shows that, as expected, the effective damping and stiffness imposed on the moving element varies with frequency. It tends to act as a pure dashpot as estimated by Eq. (9.3.29) at low frequencies and as a compressible volume of air as estimated by Eq. (9.3.33) at high frequencies.

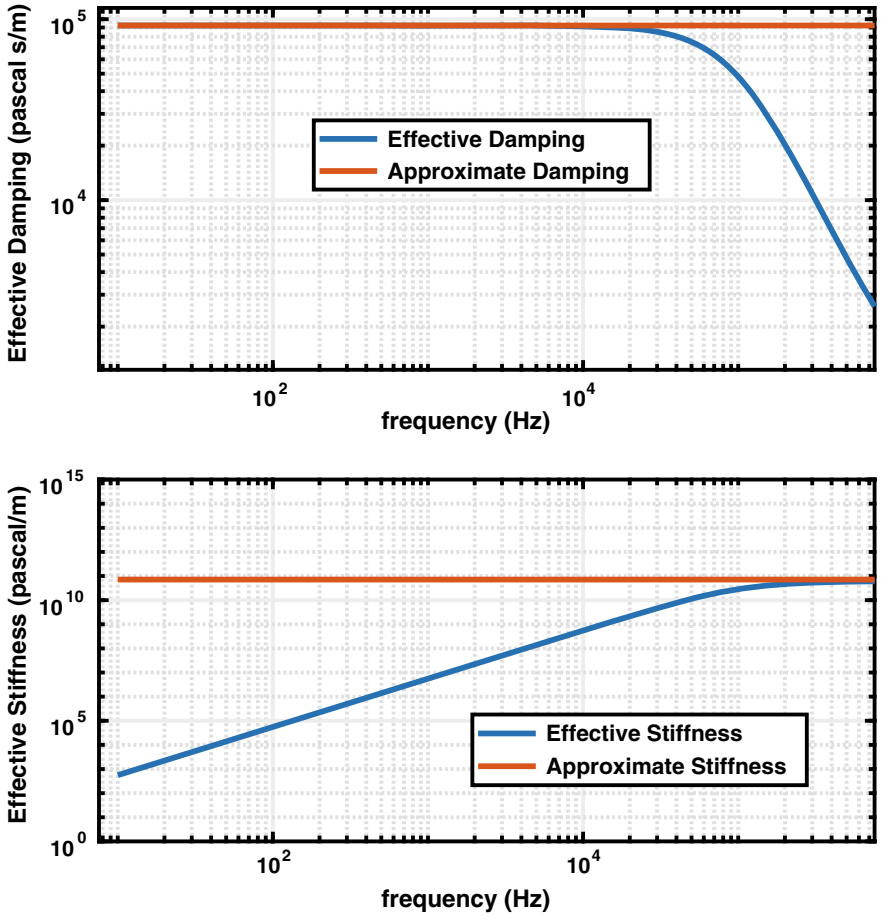


Fig. 9.2 Results obtained by evaluating Eqs. (9.3.21), (9.3.29) and (9.3.33) for $L = 200\,\mu\text{m}$, $d_0 = 2$ μm, and $\mu = 1.846 \times 10^{-5}$ Pa s. The sound speed is taken to be $c = 344$ m/s and the density is $\rho_0 = 1.206$ kg/m^3. The film acts as a viscous dashpot at low frequencies and as a compressible volume of air at high frequencies

9.4 Viscous Force on an Oscillating Cylinder

To illustrate the pronounced effect viscosity can have on the interaction of solids with an acoustic fluid, in the following we review a solution for the fluid forces on a circular cylinder that is oscillating in one direction. This is based on a remarkable solution presented by Stokes [8].

Since our problem involves fluid motion in only two directions, we will revert to a notation in which the spatial coordinates are denoted by (x, y, z) rather than the indices $(1, 2, 3)$, respectively. Let z denote the long axis of our circular cylinder and let its motion be in the x direction. The x and y components of the velocity of the fluid will be u and v, respectively so that $\bar{u} = ui + v\hat{j}$. Equation (9.1.21) may then be written as

$$\begin{aligned} \frac{\partial p}{\partial x} &= \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) - \rho_0 \dot{u} \\ \frac{\partial p}{\partial y} &= \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) - \rho_0 \dot{v} \end{aligned} \tag{9.4.1}$$

The pressure p can be eliminated from these equations by differentiating the first of Eqs. (9.4.1) by y, differentiating the second of Eqs. (9.4.1) by x and subtracting,

$$\mu\frac{\partial}{\partial y}(u_{xx} + u_{yy}) - \rho_0 \dot{u}_y - \mu\frac{\partial}{\partial x}(v_{xx} + v_{yy}) + \rho_0 \dot{v}_x = 0 \tag{9.4.2}$$

where subscripts denote partial differentiation.

If we limit our attention to steady-state harmonic motions, let $u(x, y, t) = U(x, y)e^{i\omega t}$ and $v(x, y, t) = V(x, y)e^{i\omega t}$ where $U(x, y)$ and $V(x, y)$ are complex amplitudes. Equation (9.4.2) becomes

$$\mu\frac{\partial}{\partial y}(U_{xx} + U_{yy}) - i\omega\rho_0 U_y - \mu\frac{\partial}{\partial x}(V_{xx} + V_{yy}) + \rho_0 i\omega V_x = 0 \tag{9.4.3}$$

Following Stokes [8], we will assume that compressibility can be neglected in the expression for conservation of mass, Eq. (9.1.22) which gives

$$u_x + v_y = U_x + V_y = 0 \tag{9.4.4}$$

Again following Stokes [8], define a stream function ψ such that

$$U = \psi_y \quad \text{and} \quad V = -\psi_x \tag{9.4.5}$$

We then have

$$U_y = \psi_{yy} \quad \text{and} \quad V_x = -\psi_{xx} \tag{9.4.6}$$

Equation (9.4.3) then becomes

$$\left(\mu\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial x^2}\right)-i\omega\rho_0\right)\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}\right)\psi=0 \tag{9.4.7}$$

or,

$$(\mu\nabla^2-i\omega\rho_0)\nabla^2\psi=0 \tag{9.4.8}$$

The solution of this factored partial differential equation may be written as

$$\psi=\psi_1+\psi_2 \tag{9.4.9}$$

where

$$\nabla^2\psi_1=0 \quad \text{and} \quad \mu\nabla^2\psi_2-i\omega\rho_0\psi_2=0 \tag{9.4.10}$$

Our problem is now reduced to one of solving two second order partial differential equations. The first is Laplace's equation for ψ_1 and the second is a Helmholtz equation for ψ_2.

For a cylinder, it is of course beneficial to express the motion using cylindrical coordinates, so that Eqs. (9.4.10) become

$$\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\frac{\partial}{\partial r}+\frac{1}{r^2}\frac{\partial^2}{\partial\theta^2}\right)\psi_1=0 \tag{9.4.11}$$

and

$$\left(\frac{\partial^2}{\partial r^2}+\frac{1}{r}\frac{\partial}{\partial r}+\frac{1}{r^2}\frac{\partial^2}{\partial\theta^2}-m^2\right)\psi_2=0 \tag{9.4.12}$$

where $m^2=\frac{i\omega\rho_0}{\mu}$. Fortunately, the independent variables are separable so we may express the solutions as

$$\psi_1=\sin(\theta)F_1(r) \quad \text{and} \quad \psi_2=\sin(\theta)F_2(r) \tag{9.4.13}$$

Equations (9.4.11)–(9.4.13) give ordinary differential equations for $F_1(r)$ and $F_2(r)$,

$$\left(\frac{d^2}{dr^2}+\frac{1}{r}\frac{d}{dr}-\frac{1}{r^2}\right)F_1=0 \tag{9.4.14}$$

and

$$\left(\frac{d^2}{dr^2}+\frac{1}{r}\frac{d}{dr}-\left(\frac{1}{r^2}+m^2\right)\right)F_2=0 \tag{9.4.15}$$

The solution to Eq. (9.4.14) is

$$F_1(r)=\frac{A_{11}}{r}+A_{12}r \tag{9.4.16}$$

Since we are interested only in solutions that are finite as $r\to\infty$, $A_{12}=0$.

The solution to Eq. (9.4.15) is

$$F_2(r)=A_{21}I_1(mr)+A_{22}K_1(mr) \tag{9.4.17}$$

where $I_1(mr)$ is the modified Bessel function of the first kind of order 1 and $K_1(mr)$ is the modified Bessel function of the second kind of order 1. We again must ensure finite solutions for infinite values of r so we set $A_{21}=0$. b remains a constant to be determined. The solution of Eq. (9.4.8) is then

$$\psi(r,\theta)=\left(\frac{A_{11}}{r}+A_{22}K_1(mr)\right)\sin(\theta) \tag{9.4.18}$$

We must now find the constants A_{11} and A_{22} so that the fluid velocity is equal to that of the solid cylinder of radius a at its surface, $r=a$.

In polar coordinates, the continuity equation (9.4.4) may be written as

$$\frac{\partial U_r}{\partial r}+\frac{U_r}{r}+\frac{\partial U_\theta}{r\partial\theta}=0 \tag{9.4.19}$$

where the fluid velocity is given by

$$\bar{U}=U_r\bar{e_r}+U_\theta\bar{e_\theta} \tag{9.4.20}$$

and $\bar{e_r}$ and $\bar{e_\theta}$ are radial and rotational unit vectors, respectively in polar coordinates. The stream function may also be related to the components of the velocity vector in polar coordinates,

$$U_r=\frac{\partial\psi}{r\partial\theta}\quad\text{and}\quad U_\theta=-\frac{\partial\psi}{\partial r} \tag{9.4.21}$$

Let the circular cylinder of radius a have a velocity only in the x direction, $\dot{w}(z,t)i=U_c(z)e^{i\omega t}i=i\omega W(z)e^{i\omega t}i$ where, again, z measures the distance along the cylinder's axis, perpendicular to the (x,y) plane which has its origin at the center of the cylinder. $W(z)$ is the amplitude of the cylinder's deflection in the x direction. The motion of the fluid may be assumed to be equal to that of the solid cylinder at the cylinder's surface. Let U_a and U_θ be the radial and angular components of the velocity of the fluid at the surface of the cylinder. If the radial unit vector is oriented

at an angle θ relative to the x axis, the radial and angular components of the fluid velocity at the cylinder's surface are related to the cylinder's velocity by

$$U_a = U_c(z)\cos(\theta) \quad \text{and} \quad U_\theta = -U_c(z)\sin(\theta) \tag{9.4.22}$$

Evaluating Eq. (9.4.21) at $r = a$ and using (9.4.22) provide two boundary conditions for our stream function $\psi(r, \theta)$.

Equations (9.4.18), (9.4.21) (with $r = a$), and (9.4.22) then give two equations for the unknowns A_{11} and A_{22},

$$\begin{aligned} \frac{A_{11}}{a} + A_{22}K_1(ma) &= aU_c(z) \\ -\frac{A_{11}}{a^2} + A_{22}\frac{d}{dr}K_1(mr)|_{r=a} &= U_c(z) \end{aligned} \tag{9.4.23}$$

The derivative of the Bessel function in Eq. (9.4.23) may be written as

$$\frac{d}{dr}K_1(mr)|_{r=a} = -mK_0(ma) - \frac{K_1(ma)}{a} \tag{9.4.24}$$

Equations (9.4.23) and (9.4.24) give

$$\begin{aligned} A_{11} &= \frac{(2aK_1(ma) + ma^2K_0(ma))}{mK_0(ma)}U_c(z) \\ A_{22} &= -\frac{2}{mK_0(ma)}U_c(z) \end{aligned} \tag{9.4.25}$$

Having satisfied the boundary conditions, we may now turn to the problem of determining the force on the cylinder. The force may be obtained from the pressure, p, which we promptly eliminated from Eq. (9.4.1). Following Stokes [8], the force in the direction of motion (i.e. the x direction) on a length of the cylinder dl may be written as

$$\begin{aligned} F &= \rho a dl i\omega \int_0^{2\pi} \left(a\frac{\partial \psi_1}{\partial r}|_{r=a} + \psi_2(a) \right) \sin(\theta) d\theta \\ &= \rho a dl i\omega \int_0^{2\pi} \left(-\frac{A_{11}}{a} + A_{22}K_1(ma) \right) \sin^2(\theta) d\theta \\ &= \rho a dl i\omega\pi \left(-\frac{A_{11}}{a} + A_{22}K_1(ma) \right) \end{aligned} \tag{9.4.26}$$

The force per unit length in the x direction is then

$$f = \rho a i \omega \pi \left(-\frac{A_{11}}{a} + A_{22} K_1(ma) \right) \tag{9.4.27}$$

Substituting Eq. (9.4.25) into (9.4.27) leads to

$$f = -U_c(z) \frac{\rho a i \omega \pi}{m} \left(\frac{4K_1(ma)}{K_0(ma)} + ma \right) \tag{9.4.28}$$

where again, $m = \sqrt{\frac{i\omega\rho}{\mu}}$. This well-known result is essentially the same as that presented in numerous texts on viscous flow [9].

9.5 Viscous Acoustic Excitation of a Thin Beam Having Circular Cross Section

We may now use the results above for the viscous acoustic forces on a thin cylinder to construct an approximate analytical model to examine the dominant forces and response of a nanofiber in a sound field. The fiber is modeled as a beam including simple Euler–Bernoulli bending and axial tension and is subjected to fluid forces by the surrounding air. This analysis shows that for sufficiently small diameter fibers, the motion is entirely dominated by forces applied by the fluid (i.e. air); the mechanical forces associated with the fiber's elasticity and mass become negligible. This simple result is entirely inline with any observations of thin fibers in air; the thinner they are, the more easily they move with subtle air currents. The dominance of fluid forces on thin fibers makes them ideal for sensing sound [10–13].

The analysis presented here is based on the assumption that the motion of the fiber and of the surrounding fluid can be adequately represented by considering both to be continua. The fluid is taken to be a rarefied gas. A continuum model is considered to be valid when the Knudsen number Kn, given by ratio of the mean free path λ, of the molecules relative to some characteristic dimension of the system is less than about $Kn \approx 10^{-2}$ [14]. The mean free path for air is approximately $\lambda \approx 65 \times 10^{-9}$ meters [14]. If we take the characteristic dimension to be the fiber diameter, the continuum model is then considered reliable for diameters greater than about 6.5 microns, greater than those of interest here.

In spite of the limitations of the simplified continuum model presented here, experimental results presented in [10–13] indicate that the flow-induced motion of sub-micron diameter fibers closely resembles that of the spatial average of the velocity of the molecules comprising the fluid that are in close proximity to the fiber.

The long axis of the nanofiber is assumed to be orthogonal to the direction of propagation of a harmonic plane wave. Let the z direction be parallel to the nanofiber axis and the x direction be the direction of sound propagation. The harmonic plane sound

wave at the frequency ω (radians/second) creates a pressure field identical to that in a tube as in Eq. (2.4.6), $p(x,t) = Pe^{i(\omega t - kx)}$, where $k = \omega/c$ is the wave number, P is the complex wave amplitude, $i = \sqrt{-1}$, and c is the speed of wave propagation. The plane sound wave also creates a fluctuating acoustic particle velocity field, $u(x,t)$, in the x direction,

$$u(x,t) = Ue^{i(\omega t - kx)} = \frac{p(x,t)}{\rho_0 c} = \frac{Pe^{i(\omega t - kx)}}{\rho_0 c} \tag{9.5.1}$$

where ρ_0 is the nominal air density and $U = P/(\rho_0 c)$ is the complex amplitude of the acoustic particle velocity. We will consider the fiber to be located at $x = 0$.

Let the transverse deflection in the x direction (orthogonal to the long axis) of the nanofiber be $w(z,t) = W(z)e^{i\omega t}$. The fluid motion in the immediate vicinity of the fiber will be strongly influenced by the presence of the fiber. Our current goal is to obtain an analytical model for the fiber motion relative to the fluid motion that would occur if the fiber were not present (i.e. that given by Eq. (9.5.1)).

The fluid forces on the fiber may be determined by considering the problem of a straight cylinder that is moving with some velocity $v(t) = Ve^{i\omega t}$ within a viscous fluid that is at rest at locations far from the fiber. The forces on this moving cylinder along with the flow field near the cylinder were worked out in the previous section, following Stokes [8]. Stokes' series solution to the governing differential equations may be written in terms of Bessel functions as in Eq. (9.4.28) [9–11]

$$f_v(t) = F_v e^{i\omega t} = \frac{\rho_0 ckr\pi i}{m}\left(4\frac{K_1(mr)}{K_0(mr)} + mr\right)Ve^{i\omega t} = Z(\omega)Ve^{i\omega t} \tag{9.5.2}$$

where $K_0(mr)$ and $K_1(mr)$ are the modified Bessel functions of the second kind, of order 0 and 1, respectively, $m = \sqrt{i\omega\rho/\mu}$, and μ is the dynamic viscosity. We may define $Z(\omega)$ to be the impedance of the fiber,

$$Z(\omega) = \frac{F_v}{V} = \frac{\rho_0 ckr\pi i}{m}\left(4\frac{K_1(mr)}{K_0(mr)} + mr\right) \tag{9.5.3}$$

The real and imaginary parts of the impedance may be interpreted as an equivalent frequency-dependent dashpot $R(\omega)$ and co-vibrating mass (i.e. the equivalent mass of fluid that moves with the fiber), $M(\omega)$,

$$Z(\omega) = R(\omega) + i\omega M(\omega) \tag{9.5.4}$$

where $R(\omega)$ is the real part of $Z(\omega)$ and $\omega M(\omega)$ is the imaginary part.

Since we are interested in the fluid force and subsequent fiber motion due to a sound-induced fluid velocity, $u(0,t)$, we will take $v(t) = Ve^{i\omega t} = u(0,t) - \dot{w}(z,t) = (U - i\omega W(z))e^{i\omega t}$ to be the relative velocity between the fiber and the fluid, where again, $U = P/(\rho_0 c)$ is the complex amplitude of the acoustic particle velocity and P is the complex amplitude of the sound pressure.

We will assume that the fiber is connected at each of its two ends to a rigid substrate. It is, of course, not difficult to account for any other boundary conditions on the fiber so we could also consider the fiber to be supported at only one end, as in a cantilever beam. If the fiber has a circular cross section of radius r and moves as an Euler-Bernoulli beam of length L, governing differential equation of motion may be written as [15],

$$EIw_{zzzz} - EAw_{zz}\left(\frac{Q(L)}{L} + \frac{1}{2L}\int_0^L w_z^2 dz\right) + \rho_m A\ddot{w} = f_v(t) \tag{9.5.5}$$

where E is Young's modulus of elasticity, $I = \pi r^4/4$ is the area moment of inertia, $A = \pi r^2$ is the cross sectional area, ρ_m is the volume density of the material and again, r is the radius. Subscripts denote partial differentiation with respect to the spatial variable z. The axial displacement of the fiber is taken to be zero at $z = 0$ and $Q(L)$ is the axial displacement of the end at $z = L$. The integral in Eq. (9.5.5) accounts for stretching of the fiber as it undergoes displacements that are on the order of its diameter [15].

Note that all of the terms on the left side of Eq. (9.5.5) depend strongly on the radius of the fiber. It is helpful to express each term in terms of the radius,

$$E\frac{\pi r^4}{4}w_{zzzz} - E\pi r^2 w_{zz}\left(\frac{Q(L)}{L} + \frac{1}{2L}\int_0^L w_z^2 dz\right) + \rho_m \pi r^2 \ddot{w} = f_v(t) \tag{9.5.6}$$

It is evident that all terms on the left side of this equation are proportional to either r^4 or r^2. The dependence on the radius r on the right side of Eqs. (9.5.5) or (9.5.6) is, unfortunately, not easy to see in this equation.

Figure 9.3 shows the result of evaluating Eq. (9.5.2) at frequency of $\omega = 2\pi \times 1000$ for a range of values of radius from 50 nanometers to 10 microns. The viscous force is found to be a very weak function of the radius for values of r of interest here.

For sufficiently small values of the radius, r, the governing equation of motion of the fiber (Eqs. (9.5.5) or (9.5.6)) becomes simply

$$0 \approx f_v(t) = \frac{\rho_0 ckr\pi i}{m}\left(4\frac{K_1(mr)}{K_0(mr)} + mr\right)(u(0,t) - \dot{w}(x,t)) \tag{9.5.7}$$

which has the solution $\dot{w}(z,t) = u(0,t)$, where, again, $u(0,t) = u(x,t)_{x=0}$. Of course, this shows that the fiber moves with the fluid when the fiber is sufficiently thin.

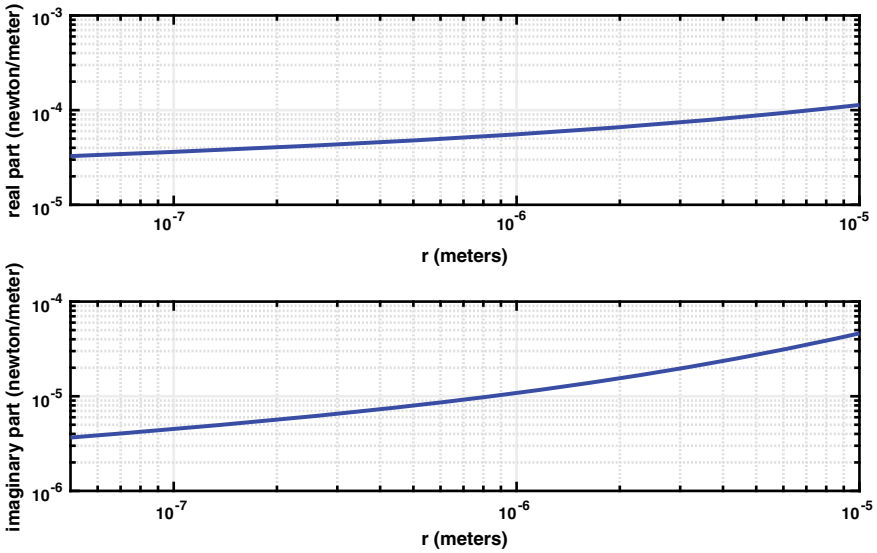


Fig. 9.3 The viscous force is not a strong function of the fiber radius r. The result of evaluating Eq. (9.5.2) is shown for a wide range of values of the radius r, assuming the frequency is 1 kHz. The fiber is assumed to undergo a velocity of 1 m/s at each frequency. The fluid is assumed to be stationary at large distances from the fiber. The force varies by roughly a factor of 10 as the radius varies by a factor of 100 from. 1 μm to 10 μm. As a result, as the fiber radius becomes small, fluid forces dominate over the forces on the left side of Eqs. (9.5.5) and (9.5.6)

9.6 Solutions for the Acoustic Response

In the following, we will obtain solutions of Eq. (9.5.6) in order to examine the range of values of the radius r in which viscous forces dominate the response of the fiber in a harmonic plane-wave sound field. In the simplest case, we first consider the response of a fiber that is infinitely long so that no waves are reflected by its boundaries.

9.6.1 Response of an Infinitely Long Uniform Fiber

In the absence of boundaries, the solution of Eq. (9.5.6), the displacement, $w(z, t)$, will be a constant in z. If we denote the response of this infinitely long fiber by $w_I(t)$, the governing equation becomes,

$$\rho_m \pi r^2 \ddot{w}_I = f_v(t) \tag{9.6.1}$$

For our harmonic sound field having frequency ω, let $w_I(t) = W_I e^{i\omega t}$. In this case, the sound-induced velocity of the fiber (rather than the displacement) relative to the acoustic particle velocity is simply [11]

$$v_I(t) = V_I e^{i\omega t} = \dot{w}_I(t) = i\omega w_I(t) = i\omega W_I e^{i\omega t} \tag{9.6.2}$$

Equations (9.5.2), (9.5.3) and (9.6.1) then give

$$i\omega\rho_m \pi r^2 V_I = Z(\omega)(U - V_I) \tag{9.6.3}$$

Equations (9.6.3) and (9.5.4) give

$$\frac{V_I}{U} = \frac{Z(\omega)}{Z(\omega) + i\omega\rho_m \pi r^2} = \frac{R(\omega) + i\omega M(\omega)}{R(\omega) + i\omega(M(\omega) + \rho_m \pi r^2)} \tag{9.6.4}$$

This simple result along with Eq. (9.6.3) shows that the fiber velocity is a sufficient approximation to the air velocity(i.e. $V_I/U \approx 1$) if the mass of the fiber is negligible relative to the co-vibrating mass of fluid, i.e. $M(\omega) \ll \rho_m \pi r^2$.

9.6.2 *Response of a Fiber of Finite Length*

In the case of a fiber of finite length, L, the ratio of the fiber velocity at the location z to the acoustic particle velocity due to a plane harmonic wave with frequency ω may then be calculated by expressing the solution of Eq. (9.5.6) in orthogonal eigenfunctions. The solution may be shown to be [11]

$$\frac{V_F(z)}{U} = i\omega \sum_{j=1}^{\infty} \frac{Z(\omega)\phi_j(z)\frac{2}{L}\int_0^L \phi_j(z)dz}{(EIp_j^4 + EAp_j^2 Q(L)/L + i\omega(Z(\omega) + i\omega\rho_m \pi r^2))} \tag{9.6.5}$$

In the calculations, the eigenfunctions were taken to be those of a beam with simple hinge supports on each end. Comparisons of the above analysis with experimental results are presented in [11]. The predicted results are shown to be in very reasonable agreement those of the experiment. Predicted and measured data indicate that for fibers that are on the order of 0.5 μm in diameter, the fiber motion is very similar to that of the acoustic particle velocity over a frequency range of 1–50 kHz [10].

An alternative approach to obtaining solutions to Eq. (9.5.6) is to apply a more direct approach rather than the expansion in eigenfunctions. To simplify the calculations, let the terms associated with axial tension that are proportional to w_{xx} be neglected. Equation (9.5.6) then becomes

$$E\frac{\pi r^4}{4} w_{zzzz} + \rho_m \pi r^2 \ddot{w} = f_v(t) \tag{9.6.6}$$

In this case, for our harmonic sound field having frequency ω, let $w_(z, t) = W(z)e^{i\omega t}$. Equation (9.6.6) may then be written as

$$E\frac{\pi r^4}{4} W(z)_{zzzz} + \frac{Z(\omega)}{i\omega} W(z) - \omega^2 \rho_m \pi r^2 W(z) = Z(\omega)U \tag{9.6.7}$$

where we have divided out the time dependence, $e^{i\omega t}$. Since the right side of this equation is independent of z, it is not difficult to write the solution in terms of the homogenous and particular solutions., $W(z) = W_h(z) + W_p(z)$. The particular solution is simply

$$W_p(z) = W_p = \frac{Z(\omega)U}{\frac{Z(\omega)}{i\omega} - \omega^2\rho_m\pi r^2} \tag{9.6.8}$$

which is clearly independent of the spatial coordinate z. $W_h(z)$ is the solution to the homogeneous equation

$$W(z)_{zzzz} + p^4 W(z) = 0 \tag{9.6.9}$$

where

$$p^4 = \frac{\frac{Z(\omega)}{i\omega} - \omega^2\rho_m\pi r^2}{E\frac{\pi r^4}{4}} \tag{9.6.10}$$

p is computed at each frequency ω. The solution of Eq. (9.6.9) is

$$W_h(z) = A_1 e^{pz} + A_2 e^{-pz} + A_3 e^{ipz} + A_4 e^{-ipz} \tag{9.6.11}$$

The four constants A_1, A_2, A_3, and A_4 are determined so that the complete solution, $W(z) = W_h(z) + W_p(z)$ satisfies the four boundary conditions.

If the beam (or fiber) is simply supported at $x = 0$ and $x = L$, the boundary conditions are

$$W(0) = 0, \quad W_{xx}(0) = 0, \quad W(L) = 0, \quad W_{xx}(L) = 0 \tag{9.6.12}$$

Equations (9.6.9), (9.6.11) and (9.6.12) then give four equations that may readily be solved for the unknown constants, A_1, A_2, A_3, and A_4, at each frequency ω

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ p^2 & p^2 & -p^2 & -p^2 \\ e^{pL} & e^{-pL} & e^{ipL} & e^{-ipL} \\ p^2e^{pL} & p^2e^{-pL} & -p^2e^{ipL} & -p^2e^{-ipL} \end{bmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix} = \begin{pmatrix} -W_p \\ 0 \\ -W_p \\ 0 \end{pmatrix} \tag{9.6.13}$$

Knowing these constants, the response at any location may be computed from

$$W(z) = A_1 e^{pz} + A_2 e^{-pz} + A_3 e^{ipz} + A_4 e^{-ipz} + W_p \tag{9.6.14}$$

The velocity of the fiber relative to the acoustic particle velocity is

$$\frac{V_f(z)}{U} = i\omega\frac{W(z)}{U} \tag{9.6.15}$$

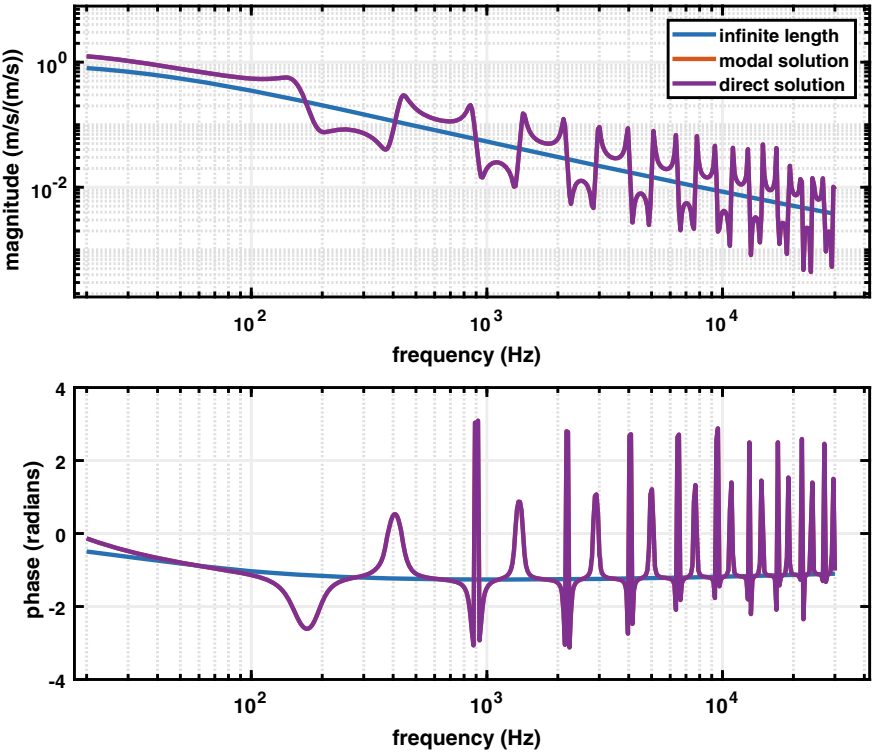


Fig. 9.4 Results of the modal solution computed from Eq. (9.6.5) are in excellent agreement with those obtained using the direct solution approach of Eqs. (9.6.7) through (9.6.15). The predicted fiber velocity is shown relative to the acoustic particle velocity for a simply supported fiber of length $L = 3$ cm and diameter $d = 2r = 6$ μm at the center $z = L/2$. The fiber is assumed to have Young's modulus $E = 180 \times 10^9$ N/m^2 and density $\rho_m = 7900$ kg/m^3. The dynamic air viscosity is taken to be $\mu = 1.8462 \times 10^{-5}$ Ns/m^2

Results obtained from Eqs. (9.6.4), (9.6.5) and (9.6.15) are shown in Fig. 9.4. The results obtained using Eqs. (9.6.5) and (9.6.15) are indistinguishable.

As another example, suppose we have a fiber that is suspended by a hinge at $x = 0$ and free at $x = L$, so that the boundary conditions are

$$W(0) = 0, \quad W_{xx}(0) = 0, \quad W_{xx}(L) = 0, \quad W_{xxx}(L) = 0 \tag{9.6.16}$$

These four boundary conditions lead to

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ p^2 & p^2 & -p^2 & -p^2 \\ p^2 e^{pL} & p^2 e^{-pL} & -p^2 e^{ipL} & -p^2 e^{-ipL} \\ p^3 e^{pL} & -p^3 e^{-pL} & -ip^3 e^{ipL} & ip^3 e^{-ipL} \end{bmatrix} \begin{pmatrix} A_1 \\ A_2 \\ A_3 \\ A_4 \end{pmatrix} = \begin{pmatrix} -W_p \\ 0 \\ 0 \\ 0 \end{pmatrix} \tag{9.6.17}$$

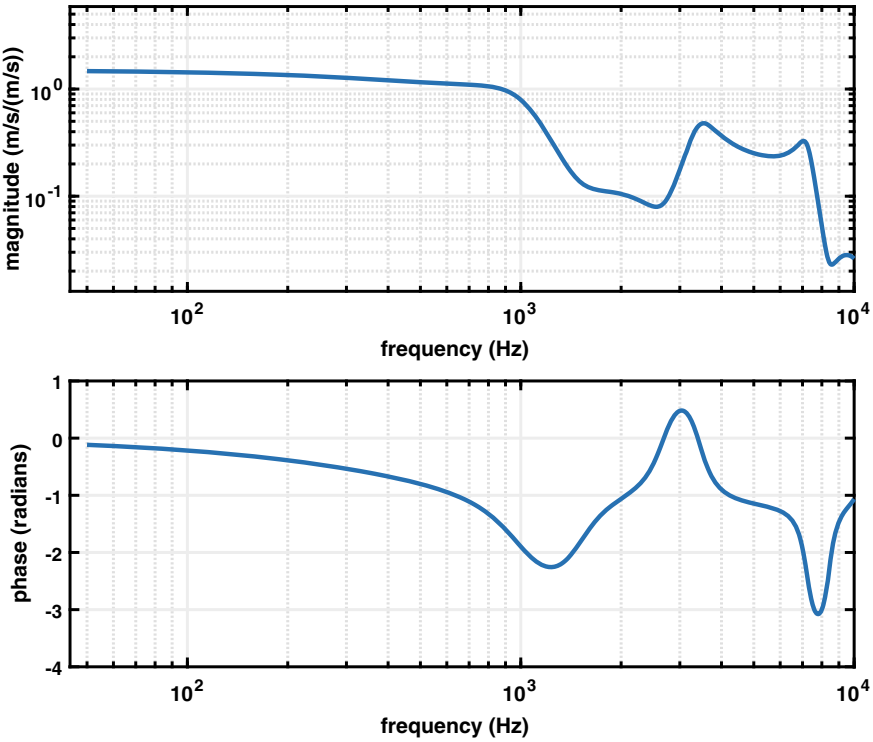


Fig. 9.5 Predicted fiber velocity relative to the acoustic particle velocity for a fiber supported by a hinge at $x = 0$ and free at $x = L$, of length $L = 1.5$ cm and diameter $d = 2r = 3$ μm. Results are shown for the free end $z = L$. The fiber is assumed to have a Young's modulus $E = 2 \times 10^9$ N/m^2, and density $\rho_m = 2267$ kg/m^3. The dynamic air viscosity is taken to be $\mu = 1.8462 \times 10^{-5}$ Ns/m^2

The predicted velocity of the free end of the fiber relative to the acoustic particle velocity is shown in Fig. 9.5. The approach above could readily be modified to account for a wide range of boundary conditions, including those that result in energy dissipation.

9.7 Pressure and Flow Around a Periodic Array of Thin Beams Including Viscosity and Compressibility

As another example of the use of Eqs. (9.1.21) and (9.1.22), suppose we have a surface in the (x, y) plane in which the normal component of velocity at the frequency ω, $v(x, y, z, \omega)$ at $z = 0$ prescribed and is uniform in y and periodic in x with period L so that $v(x, y, 0, \omega) = v(x \pm L, y, 0, \omega)$ for any x. We'll assume that L is small enough

that viscosity plays a role in determining the flow field near $z \approx 0$ and influences the pressure everywhere. The domain of interest is an infinite half-space with velocity boundary conditions imposed at $z = 0$ and finite solutions at $z = \infty$. The equations governing the fluid velocity and the pressure are Eqs. (9.1.21) and (9.1.22), repeated here for convenience,

$$\rho_0 \frac{d\bar{u}}{dt} = -\bar{\nabla} p + \mu \nabla^2 \bar{u}. \tag{9.7.1}$$

and

$$\bar{\nabla} \cdot \bar{u} = -\frac{\dot{p}}{\rho_0 c^2}. \tag{9.7.2}$$

Let $u(x, y, z, \omega)$ be the flow velocity in the x direction, parallel to the vibrating surface. Our three unknowns are then $u(x, z, \omega)$, $v(x, z, \omega)$, and $p(x, z, \omega)$, where we have dropped the dependence on y since we assume the motion is uniform in the y direction. The equations to be solved are then

$$\begin{aligned} i\rho_0 \omega u &= -\frac{\partial p}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ i\rho_0 \omega v &= -\frac{\partial p}{\partial z} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial z^2} \right) \\ -\frac{i\omega p}{\rho_0 c^2} &= \frac{\partial u}{\partial x} + \frac{\partial v}{\partial z} \end{aligned} \tag{9.7.3}$$

The third equation may be used to eliminate p from the first two equations (momentum conservation),

$$\begin{aligned} i\rho_0 \omega u &= \frac{\rho_0 c^2}{i\omega} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x \partial z} \right) + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ i\rho_0 \omega v &= \frac{\rho_0 c^2}{i\omega} \left(\frac{\partial^2 u}{\partial x \partial z} + \frac{\partial^2 v}{\partial z^2} \right) + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial z^2} \right) \end{aligned} \tag{9.7.4}$$

Multiplying by $\frac{i\omega}{\rho_0 c^2}$ gives

$$\begin{aligned} -\left(\frac{\omega}{c}\right)^2 u &= \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 v}{\partial x \partial z} + \frac{i\omega\mu}{\rho_0 c^2} \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial z^2} \right) \\ -\left(\frac{\omega}{c}\right)^2 v &= \frac{\partial^2 u}{\partial x \partial z} + \frac{\partial^2 v}{\partial z^2} + \frac{i\omega\mu}{\rho_0 c^2} \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial z^2} \right) \end{aligned} \tag{9.7.5}$$

9.7.1 Fourier Transform Solution for the Velocities and Pressure

The pair of partial differential equations (9.7.5) can be turned into a pair of ordinary differential equations by using the Fourier transform to convert the x domain to the spatial frequency k_x. For example, the Fourier transform of $u(x, y, z, \omega)$ is

$$U(k_x, z, \omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} u(x, y, z, \omega)e^{-ik_x x}dx \tag{9.7.6}$$

Applying the Fourier transform to Eq. (9.7.5) gives

$$-\left(\frac{\omega}{c}\right)^2 U = -k_x^2 U + ik_x\frac{\partial V}{\partial z} + \frac{i\omega\mu}{\rho_0 c^2}\left(-k_x^2 U + \frac{\partial^2 U}{\partial z^2}\right) \tag{9.7.7}$$

and

$$-\left(\frac{\omega}{c}\right)^2 V = ik_x\frac{\partial U}{\partial z} + \frac{\partial^2 V}{\partial z^2} + \frac{i\omega\mu}{\rho_0 c^2}\left(-k_x^2 V + \frac{\partial^2 V}{\partial z^2}\right) \tag{9.7.8}$$

Equation (9.7.7) may be solved for $\frac{\partial V}{\partial z}$,

$$\begin{aligned}\frac{\partial V}{\partial z} &= \left(\frac{\omega}{c}\right)^2\frac{i}{k_x}U + \frac{k_x^2}{ik_x}U - \frac{i\omega\mu}{\rho_0 c^2 ik_x}\left(-k_x^2 U + \frac{\partial^2 U}{\partial z^2}\right)\\ &= \left(\left(\frac{\omega}{c}\right)^2\frac{i}{k_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right)U - \frac{\omega\mu}{\rho_0 c^2 k_x}\frac{\partial^2 U}{\partial z^2} = \alpha U + \beta\frac{\partial^2 U}{\partial z^2}\end{aligned} \tag{9.7.9}$$

where

$$\alpha = \left(\frac{\omega}{c}\right)^2\frac{i}{k_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2} \quad \beta = -\frac{\omega\mu}{\rho_0 c^2 k_x} \tag{9.7.10}$$

Differentiating equation (9.7.8) with respect to z gives

$$-\left(\frac{\omega}{c}\right)^2\frac{\partial V}{\partial z} = ik_x\frac{\partial^2 U}{\partial z^2} + \frac{\partial^3 V}{\partial z^3} + \frac{i\omega\mu}{\rho_0 c^2}\left(-k_x^2\frac{\partial V}{\partial z} + \frac{\partial^3 V}{\partial z^3}\right) \tag{9.7.11}$$

Differentiating equation (9.7.9) twice with respect to z gives

$$\frac{\partial^3 V}{\partial z^3} = \alpha\frac{\partial^2 U}{\partial z^2} + \beta\frac{\partial^4 U}{\partial z^4} \tag{9.7.12}$$

Rewriting Eq. (9.7.11) to group terms,

$$ik_x \frac{\partial^2 U}{\partial z^2} + \frac{\partial^3 V}{\partial z^3}\left(1 + \frac{i\omega\mu}{\rho_0 c^2}\right) + \frac{\partial V}{\partial z}\left(\left(\frac{\omega}{c}\right)^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2\right) = 0 \tag{9.7.13}$$

Equations (9.7.9) and (9.7.12) may be used to eliminate V from Eq. (9.7.13),

$$\begin{aligned} &ik_x \frac{\partial^2 U}{\partial z^2} + \left(1 + \frac{i\omega\mu}{\rho_0 c^2}\right)\alpha \frac{\partial^2 U}{\partial z^2} - \frac{\omega\mu}{\rho_0 c^2 k_x}\frac{\partial^4 U}{\partial z^4} \\ &+ \left(\left(\frac{\omega}{c}\right)^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2\right)\left[\left(-\left(\frac{\omega}{c}\right)^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right)U - \frac{\omega\mu}{\rho_0 c^2 k_x}\frac{\partial^2 U}{\partial z^2}\right] = 0 \end{aligned} \tag{9.7.14}$$

Grouping terms and simplifying give

$$\begin{aligned} &\frac{\partial^4 U}{\partial z^4}\left[-\left(1 + \frac{i\omega\mu}{\rho_0 c^2}\right)\frac{\omega\mu}{\rho_0 c^2 k_x}\right] \\ &+ \frac{\partial^2 U}{\partial z^2}\left[\left(1 + \frac{i\omega\mu}{\rho_0 c^2}\right)\left(-\left(\frac{\omega}{c}\right)^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right) - \left(\left(\frac{\omega}{c}\right)^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2\right)\frac{\omega\mu}{\rho_0 c^2 k_x} + ik_x\right] \\ &+ U\left[\left(\left(\frac{\omega}{c}\right)^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2\right)\left(-\left(\frac{\omega}{c}\right)^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right)\right] = 0 \end{aligned} \tag{9.7.15}$$

We finally have a fourth order ordinary differential equation of the form

$$\frac{\partial^4 U}{\partial z^4} + b\frac{\partial^2 U}{\partial z^2} + aU = 0 \tag{9.7.16}$$

where

$$a = \frac{((\frac{\omega}{c})^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2)(-(\frac{\omega}{c})^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2})}{-(1 + \frac{i\omega\mu}{\rho_0 c^2})\frac{\omega\mu}{\rho_0 c^2 k_x}} \tag{9.7.17}$$

and

$$b = \frac{(1 + \frac{i\omega\mu}{\rho_0 c^2})(-(\frac{\omega}{c})^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}) - ((\frac{\omega}{c})^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2)\frac{\omega\mu}{\rho_0 c^2 k_x} + ik_x}{-(1 + \frac{i\omega\mu}{\rho_0 c^2})\frac{\omega\mu}{\rho_0 c^2 k_x}} \tag{9.7.18}$$

The solution of Eq. (9.7.16) is

$$U = Ae^{pz} \tag{9.7.19}$$

Substituting (9.7.19) into Eq. (9.7.16) leads to four values of p,

$$p = \pm\sqrt{\frac{-b \pm \sqrt{b^2 - 4a}}{2}} \tag{9.7.20}$$

Two of these values will have positive real parts and will therefore be neglected because we seek finite solutions for U as z tends to ∞.

Let p_1 and p_2 be the two values of p from Eq. (9.7.20) having negative real parts. The solution for $U(k_x, z, \omega)$ may then be written

$$U(k_x, z, \omega) = Ae^{p_1 z} + Be^{p_2 z} \tag{9.7.21}$$

Because the fluid is assumed to stick to the surface at $z = 0$ and the surface motion is assumed to be only in the z direction,

$$U(k_x, 0, \omega) = A + B = 0 \tag{9.7.22}$$

so that

$$U(k_x, z, \omega) = A(e^{p_1 z} - e^{p_2 z}) \tag{9.7.23}$$

The value of the remaining unknown constant A may be determined by satisfying the boundary condition for v (or V) at $z = 0$. Substituting Eq. (9.7.23) into (9.7.9) and integrating give

$$V(k_x, z, \omega) = \left(-\left(\frac{\omega}{c}\right)^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right) A\left(\frac{e^{p_1 z}}{p_1} - \frac{e^{p_2 z}}{p_2}\right) - \frac{\omega\mu}{\rho_0 c^2 k_x} A(p_1 e^{p_1 z} - p_2 e^{p_2 z}) + V_0 \tag{9.7.24}$$

where V_0 is a constant of integration. Since we assume the fluid to be at rest at $z = \infty$, we will set $V_0 = 0$.

The velocity of the surface is identical to that of the fluid at $z = 0$ so we will take $v(x, 0, \omega)$ to be a given boundary condition. The Fourier transform of $v(x, 0, \omega)$ is

$$V(k_x, 0, \omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} v(x, y, 0, \omega) e^{-ik_x x} dx \tag{9.7.25}$$

so that $V(k_x, 0, \omega)$ is the given boundary condition in the k_x domain. Evaluating Eq. (9.7.24) at $z = 0$ then enables us to determine A,

$$A = \frac{V(k_x, 0, \omega) ik_x}{(-(\frac{\omega}{c})^2 + k_x^2 + \frac{i\omega\mu k_x^2}{\rho_0 c^2})(\frac{1}{p_1} - \frac{1}{p_2}) - \frac{i\omega\mu}{\rho_0 c^2}(p_1 - p_2)} \tag{9.7.26}$$

Once A is known, we can calculate $U(k_x, z, \omega)$ using Eq. (9.7.23) and $V(k_x, z, \omega)$ using Eq. (9.7.24).

The Fourier transform of the pressure may be obtained by transforming the third of Eqs. (9.7.3),

$$P(k_x, z, \omega) = -\frac{\rho_0 c^2}{i\omega}\left(ik_x U(k_x, z, \omega) + \frac{\partial V}{\partial z}\right) \tag{9.7.27}$$

Equations (9.7.23), (9.7.24), and (9.7.27) then give

$$P(k_x, z, \omega) = -A\frac{\rho_0 c^2}{i\omega}\left[ik_x(e^{p_1 z} - e^{p_2 z}) + \left(-\left(\frac{\omega}{c}\right)^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right) \times (e^{p_1 z} - e^{p_2 z}) - \frac{\omega\mu}{\rho_0 c^2 k_x}(p_1^2 e^{p_1 z} - p_2^2 e^{p_2 z})\right] \tag{9.7.28}$$

9.7.2 *Numerical Evaluation*

Having obtained solutions for the velocities and pressure in the Fourier transform domain, k_x, our next task is to convert the transform solution into the spatial domain through the use of the inverse Fourier transform. For example, having $U(k_x, z, \omega)$, the velocity in the physical domain, $u(x, z, \omega)$ is

$$u(x, z, \omega) = \int_{-\infty}^{\infty} U(k_x, y, z, \omega) e^{ik_x x} dx \tag{9.7.29}$$

It is, unfortunately, not feasible to carry out the necessary integrations in the inverse Fourier transforms of U, V, and P due to the complicated dependence on the integration variable k_x. We will, instead, rely on the fast Fourier transform algorithm to estimate the inverse transform. To evaluate the inverse Fourier transforms numerically, we must confine the range of integration to a finite domain, $-k_{xmax} < k_x < k_{xmax}$. In the case of $U(k_x, y, z, \omega)$, the integral then takes the form

$$u(x, z, \omega) = \int_{-k_{xmax}}^{k_{xmax}} U(k_x, y, z, \omega) e^{ik_x x} dx \tag{9.7.30}$$

It is convenient to split this integral into two domains,

$$\begin{aligned} u(x, z, \omega) &= \int_{-k_{xmax}}^{0} U(k_x, y, z, \omega) e^{ik_x x} dx + \int_{0}^{k_{xmax}} U(k_x, y, z, \omega) e^{ik_x x} dx \\ &= \int_{0}^{k_{xmax}} U(-k_x, y, z, \omega) e^{-ik_x x} dx + \int_{0}^{k_{xmax}} U(k_x, y, z, \omega) e^{ik_x x} dx \end{aligned} \tag{9.7.31}$$

Equation (9.7.31) is strongly reminiscent of Eq. (1.14.1), where we developed a reasonably general approach to estimate the inverse Fourier transform.

To employ the fast Fourier transform, as in Eq. (1.6.1), we will discretize the wave vector domain into a set of components k_{x_j} for $j = 1, \ldots, N$ and the spatial domain x as x_n for $n = 1, \ldots, N$. If the domain of interest is $0 \le x \le L$, let $x_n = (n-1) * L/N = (n-1)\Delta x$. To satisfy the Nyquist criterion, the spatial domain

must be sampled with a sufficiently small spatial separation Δx so that two samples are obtained within one wavelength of the highest spatial frequency component in the signal, k_{xmax}. Let $k_{x_j} = (j-1)\Delta k_x$ where $\Delta k_x = 2\pi/L$. As in Eq. (1.6.2), the maximum spatial frequency component will be $k_{xmax} \leq \pi/\Delta x = \pi N/L$. To satisfy the Nyquist criterion we will then ensure there are no frequency components above k_{x_j} for $j = N/2$. As in Eq. (1.14.2), approximating the integrals by finite summations at discrete values $k_x = k_{x_j}$ gives

$$u(x,z,\omega) \approx \sum_{j=2}^{N/2} e^{-ik_{x_j}x} U(-k_{x_j},z,\omega)\Delta k_x + \sum_{j=1}^{N/2} e^{ik_{x_j}t} U(k_{x_j},z,\omega)\Delta k_x \quad (9.7.32)$$

Note that we have again started the first summation at $j = 2$ to avoid the contribution to the integral at k_{x_1} from being counted twice.

Again limiting our attention to a total of N discrete values of x by letting $x_n = (n-1) * L/N$, for $n = 1, \ldots, N$, in Eq. (9.7.32) gives

$$u(x_n,z,\omega) = u_n = \sum_{j=2}^{N/2} e^{-ik_{x_j}t_n} U(-k_{x_j},z,\omega)\Delta k_x + \sum_{j=1}^{N/2} e^{ik_{x_j}t_n} U(k_{x_j},z,\omega)\Delta k_x \quad (9.7.33)$$

Since the summation goes to $N/2$ rather than N, we will again set $U(k_{x_j},z,\omega) = U(-k_{x_j},z,\omega) = 0$, for $j = N/2+1, \ldots, N$. The second summation in Eq. (9.7.33) may be evaluated very efficiently by the use of the inverse FFT algorithm, which is equivalent to

$$iFFT(U(k_{x_j},z,\omega)) = \frac{1}{N}\sum_{j=1}^{N} e^{i(j-1)(n-1)2\pi/N} U(k_{x_j},z,\omega) \quad (9.7.34)$$

The first summation in equation (9.7.33) may be evaluated using the forward FFT algorithm, where the complex function U is evaluated for negative values of k_{x_j}.

$$FFT(U(-k_{x_j},z,\omega)) = \sum_{j=1}^{N} e^{-i(j-1)(n-1)2\pi/N} U(-k_{x_j},z,\omega) \quad (9.7.35)$$

In order to start the summation at $j = 2$, we will set $U(-k_{x_1},z,\omega) = 0$. The response in equation (1.14.3) may then be computed by

$$u(x_n,z,\omega) = u_n = \frac{2\pi}{L} FFT(U(-k_{x_j},z,\omega)) + \frac{2\pi N}{L} iFFT(U(k_{x_j},z,\omega)) \quad (9.7.36)$$

The corresponding inverse transforms for $V(k_{x_j}, z, \omega)$ and $P(k_{x_j}, z, \omega))$ are

$$v(x_n, z, \omega) = u_n = \frac{2\pi}{L} FFT(V(-k_{x_j}, z, \omega)) + \frac{2\pi N}{L} iFFT(V(k_{x_j}, z, \omega)) \tag{9.7.37}$$

and

$$p(x_n, z, \omega) = u_n = \frac{2\pi}{L} FFT(P(-k_{x_j}, z, \omega)) + \frac{2\pi N}{L} iFFT(P(k_{x_j}, z, \omega)) \tag{9.7.38}$$

Having obtained solutions for the two velocity components and the pressure in the spatial frequency domain, U, V, and P, we may then easily compute the solutions in the spatial domain.

Our calculations depend on knowing the velocity boundary condition $v(x, 0, \omega)$ and its spatial frequency domain representation, $V(k_x, 0, \omega)$ in order to determine the constant A in Eq. (9.7.26). This can be accomplished either by evaluating Eq. (9.7.25)

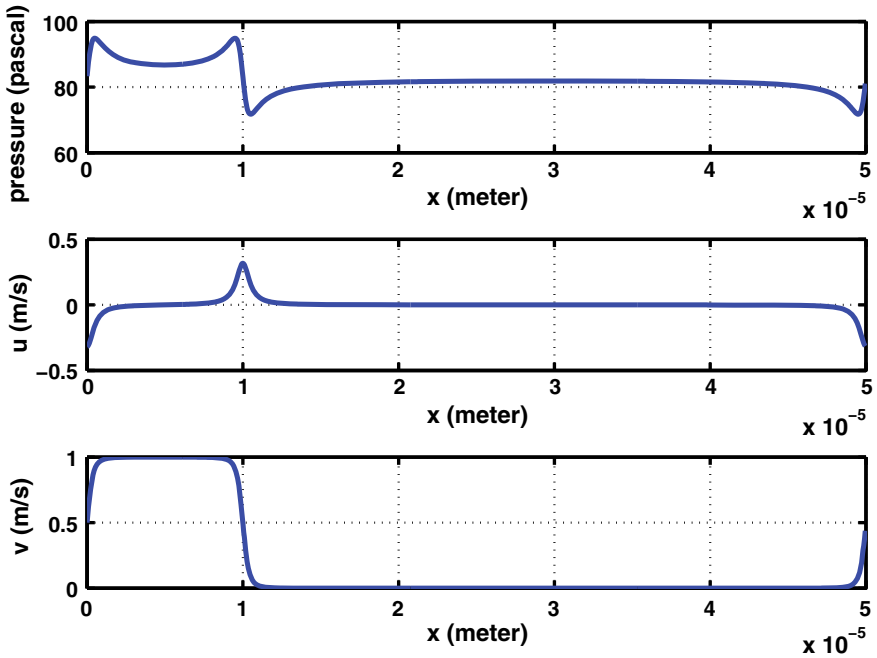


Fig. 9.6 The real part of the pressure and velocity at a distance $z = 0.5\,\mu$m from a spatially periodic vibrating surface at 1000 Hz. The imposed transverse velocity is $v(x, 0, \omega) = 1$ m/s for $0 \le x \le D$ where $D = 10\,\mu$m and $v(x, 0, \omega) = 0$ for $D < x \le L$, with $L = 50\,\mu$m

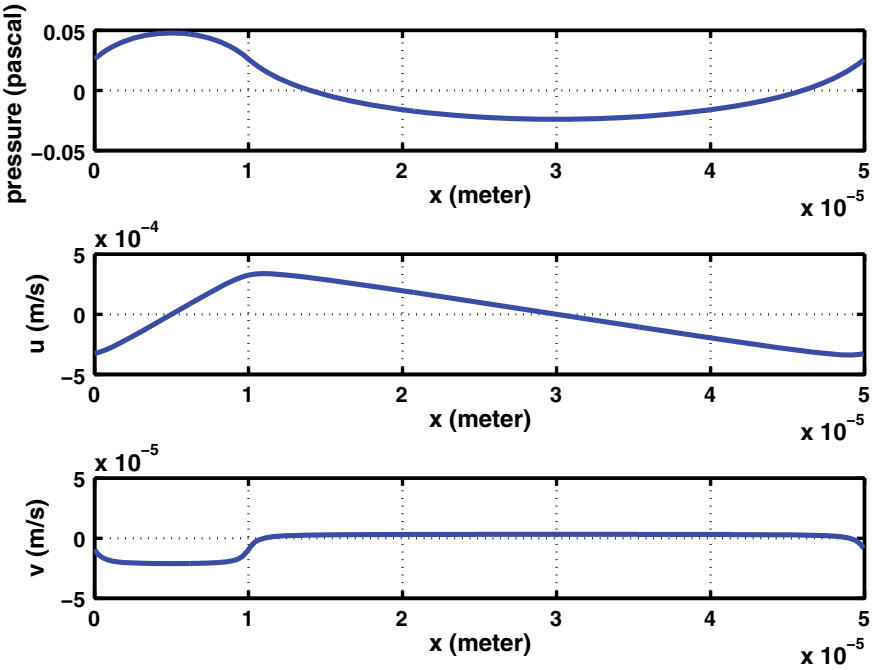


Fig. 9.7 The imaginary part of the pressure and velocity at a distance $z = 0.5$ μm from a spatially periodic vibrating surface at 1000 Hz. The imposed transverse velocity is $v(x, 0, \omega) = 1$ m/s for $0 \leq x \leq D$ where $D = 10$ μm and $v(x, 0, \omega) = 0$ for $D < x \leq L$, with $L = 50$ μm

analytically (when this is feasible) for a given $v(x, 0, \omega)$, or by using the FFT algorithm to evaluate it numerically, in a manner that is directly analogous to equation 1.11.8),

$$V(k_x, 0, \omega) \approx \frac{L}{2\pi N} FFT(v(x_n, 0, \omega)) \tag{9.7.39}$$

9.7.3 Numerical Example

As an example, suppose the surface at $z = 0$ has an imposed transverse velocity of $v(x, 0, \omega) = 1$ m/s for $0 \leq x \leq D$ where $D = 10$ μm and $v(x, 0, \omega) = 0$ for $D < x \leq L$, with $L = 50$ μm. Let the motion be periodic in x so that $v(x, 0, \omega) = v(x \pm L, 0, \omega)$. The surface has no velocity in the plane so that $u(x, 0, \omega) = 0$ for all x. The frequency is $\omega = 2\pi 1000$ rad/s, the sound speed is $c = 343$ m/s, the air density is $\rho_0 = 1.2$ kg/m^3, and the viscosity is $\mu = 1.814 \times 10^{-5}$ kg/(ms). In this case, the velocity boundary condition in the spatial frequency domain may be obtained by evaluating the integral in equation (9.7.25) which gives

$$V(k_x, 0, \omega) = \frac{1 - e^{-ik_x D}}{2\pi i k_x} \tag{9.7.40}$$

Note that when $k_x = 0$ this becomes $V(0, 0, \omega) = \frac{D}{2\pi}$.

Calculated results are shown in Figs. 9.6 and 9.7.

9.8 Sound Pressure and Viscous Flow Around a Periodic Array of Thin Beams

In the following, we will extend the approach of Sect. 9.7 to consider the effects of viscosity and fluid compressibility on the sound field due to a periodic array of vibrating beams having infinitesimal thickness. When the beams are at rest at $z = 0$, we assume that they are separated by some small distance in the planar direction (normal to the z axis). The space around the beams is assumed to be filled with air and the domain contains no rigid boundary at $z = 0$ as in Sect. 9.7.

We will again utilize the Fourier transform to express the Stokes equations, (9.7.5) to ultimately be expressed as ordinary differential equations (9.7.9) and (9.7.16), repeated here for convenience,

$$\begin{aligned}\frac{\partial V}{\partial z} &= \left(\frac{\omega}{c}\right)^2 \frac{i}{k_x} U + \frac{k_x^2}{ik_x} U - \frac{i\omega\mu}{\rho_0 c^2 i k_x}\left(-k_x^2 U + \frac{\partial^2 U}{\partial z^2}\right) \\ &= \left(\left(\frac{\omega}{c}\right)^2 \frac{i}{k_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}\right) U - \frac{\omega\mu}{\rho_0 c^2 k_x}\frac{\partial^2 U}{\partial z^2} = \frac{\alpha}{k_x} U + \frac{\beta}{k_x}\frac{\partial^2 U}{\partial z^2}\end{aligned} \tag{9.8.1}$$

where

$$\alpha = \left(i\frac{\omega}{c}\right)^2 - ik_x^2 + \frac{\omega\mu k_x^2}{\rho_0 c^2}, \quad \beta = -\frac{\omega\mu}{\rho_0 c^2} \tag{9.8.2}$$

$$\frac{\partial^4 U}{\partial z^4} + b\frac{\partial^2 U}{\partial z^2} + aU = 0 \tag{9.8.3}$$

where

$$a = \frac{((\frac{\omega}{c})^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2)(-(\frac{\omega}{c})^2 \frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2})}{-(1 + \frac{i\omega\mu}{\rho_0 c^2})\frac{\omega\mu}{\rho_0 c^2 k_x}} = \frac{((\frac{\omega}{c})^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2)(i(\frac{\omega}{c})^2 - ik_x^2 + \frac{\omega\mu k_x^2}{\rho_0 c^2})}{-(1 + \frac{i\omega\mu}{\rho_0 c^2})\frac{\omega\mu}{\rho_0 c^2}} \tag{9.8.4}$$

and

$$b = \frac{(1+\frac{i\omega\mu}{\rho_0 c^2})(-(\frac{\omega}{c})^2\frac{1}{ik_x} - ik_x + \frac{\omega\mu k_x}{\rho_0 c^2}) - ((\frac{\omega}{c})^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2)\frac{\omega\mu}{\rho_0 c^2 k_x} + ik_x}{-(1+\frac{i\omega\mu}{\rho_0 c^2})\frac{\omega\mu}{\rho_0 c^2 k_x}}$$
$$= \frac{(1+\frac{i\omega\mu}{\rho_0 c^2})(i(\frac{\omega}{c})^2 - ik_x^2 + \frac{\omega\mu k_x^2}{\rho_0 c^2}) - ((\frac{\omega}{c})^2 - \frac{i\omega\mu}{\rho_0 c^2}k_x^2)\frac{\omega\mu}{\rho_0 c^2} + ik_x^2}{-(1+\frac{i\omega\mu}{\rho_0 c^2})\frac{\omega\mu}{\rho_0 c^2}} \tag{9.8.5}$$

As in the solution form given in Eq. (9.7.19), there will be four solutions as given in Eq. (9.7.20),

$$p = \pm\sqrt{\frac{-b \pm \sqrt{b^2 - 4a}}{2}} \tag{9.8.6}$$

As before, we will let p_1 and p_2 denote the roots of p having negative real parts. Because our domain now spans $-\infty < z < \infty$, we will use the remaining roots, p_3 and p_4 to express the solution for $z < 0$ since those roots have positive real parts causing the solution to be bounded for all z. We will therefore let the solution be expressed separately for the two domains,

$$U_1(k_x, z, \omega) = A_1 e^{p_1 z} + A_2 e^{p_2 z} = A_1(e^{p_1 z} - e^{p_2 z}) \tag{9.8.7}$$

for $z > 0$, and

$$U_2(k_x, z, \omega) = A_3 e^{p_3 z} + A_4 e^{p_4 z} = A_3(e^{p_3 z} - e^{p_4 z}) \tag{9.8.8}$$

for $z < 0$. We have used the requirement that $u(x, z, \omega) \to 0$ as $z \to 0$, which causes the transform of u to also approach zero as $z \to 0$ giving $A_2 = -A_1$ and $A_4 = -A_3$. Our task is then to determine A_1 and A_3, which are the unknowns in the transform domain and are functions of k_x, $A_1(k_x)$ and $A_3(k_x)$. The velocity in the z direction will also be separated as $V_1(k_x, z, \omega)$ for $z > 0$ and $V_2(k_x, z, \omega)$ for $z < 0$.

Knowing U_1 and U_2 from Eqs. (9.8.7) and (9.8.8) we may determine $V_1(k_x, z, \omega)$ and $V_2(k_x, z, \omega)$ using Eq. (9.8.1). This gives

$$V_1(k_x, z, \omega) = A_1\left(\frac{\alpha}{k_x}\left(\frac{e^{p_1 z}}{p_1} - \frac{e^{p_2 z}}{p_2}\right) + \frac{\beta}{k_x}(p_1 e^{p_1 z} - p_2 e^{p_2 z})\right) \quad z > 0 \tag{9.8.9}$$

and

$$V_2(k_x, z, \omega) = A_3\left(\frac{\alpha}{k_x}\left(\frac{e^{p_3 z}}{p_3} - \frac{e^{p_4 z}}{p_4}\right) + \frac{\beta}{k_x}(p_3 e^{p_3 z} - p_4 e^{p_4 z})\right) \quad z < 0 \tag{9.8.10}$$

Since we require continuity of velocity between the two domains $z > 0$ and $z < 0$, we must have $V_1(k_x, 0, \omega) = V_2(k_x, 0, \omega)$. Equations (9.8.9) and (9.8.10) require that

$$\frac{A_1}{k_x}\left(\alpha\left(\frac{1}{p_1} - \frac{1}{p_2}\right) + \beta(p_1 - p_2)\right) = \frac{A_3}{k_x}\left(\alpha\left(\frac{1}{p_3} - \frac{1}{p_4}\right) + \beta(p_3 - p_4)\right) \tag{9.8.11}$$

However, since $p_1 = -p_3$ and $p_2 = -p_4$, this gives simply $A_1 = -A_3$.

We will assume that there is a moving beam with velocity $\mathcal{V}$ at $z = 0$ for $-D/2 < x < D/2$ and that the entire domain is given by $-L/2 < x < L/2$ with $D < L$. To simplify expressing the boundary condition for v, define a switch function, $S_e(x)$ to be equal to unity for $-D/2 < x < D/2$ (i.e. the domain of the moving beam) and $S_e(x) = 0$ in the region occupied by only air. We could then consider a switch function that is unity in the region occupied by air, $S_a(x) = 1 - S_e(x)$ for all $-L/2 < x < L/2$.

In the region outside that of the solid beam (and at $z = 0$), we expect the pressure to be zero since this region receives sound equally and in opposite phase from both sides of the moving beam. In addition, since the horizontal velocity is zero at $z = 0$, the continuity equation, the last of equations (9.7.3) leads to

$$\frac{\partial v}{\partial z}|_{z=0} = 0 \tag{9.8.12}$$

for the region $-L/2 < x < -D/2$ and $D/2 < x < L/2$, which contains only air.

We must now utilize the boundary conditions expressed in the physical domain, along the x axis, in order to determine values of $A_1(k_x)$ and $A_3(k_x)$ in the transform domain, k_x. This can be accomplished using our switch functions $S_e(x)$ and $S_a(x)$ along with the inverse Fourier transform,

$$v(x, 0, \omega) = \mathcal{V}S_e(x) = \int_{-k_{xmax}}^{k_{xmax}} e^{ik_x x}(S_e(x)V_1(k_x, 0, \omega) + S_a(x)\frac{\partial V_1}{\partial z}|_{z=0})dk_x \tag{9.8.13}$$

$V_1(k_x, 0, \omega)$ and $\frac{\partial V_1}{\partial z}|_{z=0}$ may be expressed in terms of $A_1(k_x)$ through Eqs. (9.8.1) and (9.8.7). Equation (9.8.13) may then be written as an integral equation for the unknown $A_1(k_x)$,

$$\mathcal{V}S_e(x) = \int_{-k_{xmax}}^{k_{xmax}} e^{ik_x x}\frac{A_1(k_x)}{k_x}(S_e(x)\alpha_e(k_x) + S_a(x)\alpha_a(k_x))dk_x \tag{9.8.14}$$

where

$$\alpha_e(k_x) = \left(i\left(\frac{\omega}{c}\right)^2 - ik_x^2 + \frac{\omega\mu k_x^2}{\rho_0 c^2}\right)\left(\frac{1}{p_1} - \frac{1}{p_2}\right) - \frac{\omega\mu}{\rho_0 c^2}(p_1 - p_2) = \alpha\left(\frac{1}{p_1} - \frac{1}{p_2}\right) + \beta(p_1 - p_2) \tag{9.8.15}$$

and

$$\alpha_a(k_x) = -\frac{\omega\mu}{\rho_0 c^2}(p_1^2 - p_2^2) = \beta(p_1^2 - p_2^2) \tag{9.8.16}$$

Note that the integrand in equation (9.8.14) has a singularity at $k_x = 0$. This may be avoided by letting

$$A_1'(k_x) = \frac{A_1(k_x)}{k_x} \tag{9.8.17}$$

so that Eq. (9.8.14) may be solved for our new unknown $A_1'(k_x)$,

$$\mathcal{V}S_e(x) = \int_{-k_{xmax}}^{k_{xmax}} e^{ik_x x} A_1'(k_x)(S_e(x)\alpha_e(k_x) + S_a(x)\alpha_a(k_x))dk_x \tag{9.8.18}$$

Having determined $A_1'(k_x)$, our desired solution is simply $A_1(k_x) = A_1'(k_x) \times k_x$.

As in Eq. (9.7.27), the Fourier transform of the pressure may be obtained by transforming the third of equations (9.7.3),

$$P(k_x, z, \omega) = -\frac{\rho_0 c^2}{i\omega}(ik_x U(k_x, z, \omega) + \frac{\partial V}{\partial z}) \tag{9.8.19}$$

Equations (9.8.7), (9.8.17), and (9.8.1) enable us to express the pressure for $z \geq 0$ as

$$P(k_x, z, \omega) = A_1'(k_x)\frac{\rho_0 c^2}{i\omega}(ik_x(e^{p_1 z} - e^{p_2 z}) + (\alpha(e^{p_1 z} - e^{p_2 z}) + \beta(p_1^2 e^{p_1 z} - p_2^2 e^{p_2 z}))) \tag{9.8.20}$$

For $z \leq 0$ the pressure may be calculated by

$$P(k_x, z, \omega) = A_3'(k_x)\frac{\rho_0 c^2}{i\omega}(ik_x(e^{p_3 z} - e^{p_4 z}) + (\alpha(e^{p_3 z} - e^{p_4 z}) + \beta(p_3^2 e^{p_3 z} - p_4^2 e^{p_4 z}))) \tag{9.8.21}$$

Having determined the velocities and pressure in the transform domain, we may now approximate the inverse Fourier transform of these quantities to obtain results in the physical domain. The inverse transforms have the same form as in Eq. (9.7.29),

$$u(x,z,\omega)=\int_{-\infty}^{\infty}U(k_x,z,\omega)e^{ik_x x}dk_x\approx\int_{-k_{xmax}}^{k_{xmax}}U(k_x,z,\omega)e^{ik_x x}dk_x, \tag{9.8.22}$$

$$v(x,z,\omega)=\int_{-\infty}^{\infty}V(k_x,z,\omega)e^{ik_x x}dk_x\approx\int_{-k_{xmax}}^{k_{xmax}}V(k_x,z,\omega)e^{ik_x x}dk_x, \tag{9.8.23}$$

and

$$p(x,z,\omega)=\int_{-\infty}^{\infty}P(k_x,z,\omega)e^{ik_x x}dk_x\approx\int_{-k_{xmax}}^{k_{xmax}}P(k_x,z,\omega)e^{ik_x x}dk_x \tag{9.8.24}$$

9.8.1 Numerical Results

To solve Eq. (9.8.14) for $A_1(k_x)$, we will discretize x by letting $x_n=-L/2+\frac{n-1}{N-1}L$, for $n=1,\ldots,N$. Let k_x also be discretized according to $k_{x_j}=-k_{xmax}+\frac{j-1}{N-1}2k_{xmax}$, where $k_{xmax}=\pi N/L$. Here, we will use an alternate approach to the FFT algorithm of Sect. 9.7.2 and select N to be an odd integer so that x_n and k_{x_j} can take on the value zero at the center of the domain. The integral in equation (9.8.14) may then be written as a summation,

$$\mathcal{V}_n S_e(x_n)=\sum_{j=1}^{N}e^{ik_{x_j}x_n}A'_{1_j}(S_e(x_n)\alpha_{e_j}+S_a(x_n)\alpha_{a_j})\Delta k_x \tag{9.8.25}$$

where

$$\mathcal{V}_n=\mathcal{V}(x_n),\quad A'_{1_j}=A'_1(k_{x_j}),\quad A_{1_j}=A'_{1_j}k_{x_j}\quad \alpha_{a_j}=\alpha_a(k_{x_j}),\quad \text{and}\quad \Delta k_x=2k_{xmax}/(N-1) \tag{9.8.26}$$

Equation (9.8.25) is a linear system of equations that may be readily solved numerically for the unknown A'_{1_j} and hence, A_{1_j} for $j=1,\ldots,N$.

The integrals in equations (9.8.22), (9.8.23), and (9.8.24) may be approximated as summations as in Eq. (9.8.25),

$$u(x_n,z,\omega)=\sum_{j=1}^{N}e^{ik_{x_j}x_n}U(k_{x_j},z,\omega)\Delta k_x, \tag{9.8.27}$$

$$v(x_n,z,\omega)=\sum_{j=1}^{N}e^{ik_{x_j}x_n}V(k_{x_j},z,\omega)\Delta k_x, \tag{9.8.28}$$

$$p(x_n, z, \omega) = \sum_{j=1}^{N} e^{ik_{x_j} x_n} P(k_{x_j}, z, \omega) \Delta k_x, \tag{9.8.29}$$

We are often interested in calculating the surface average of the pressure applied to the moving element. This can be accomplished by integrating over its width,

$$p_{ave}(\omega) = \frac{1}{D} \int_{-D/2}^{D/2} p(x, 0, \omega) dx \approx \frac{1}{D} \sum_{n=1}^{N} p_n(\omega) S_e(x_n) \Delta x \tag{9.8.30}$$

where $p_n(\omega) = p(x_n, 0, \omega)$ and Δx is the constant spacing between the points along the x axis, $\Delta x = x_{n+1} - x_n$.

In our formulation above, we have found the fluid velocity around the moving element with the fluid assumed to be stationary at an infinite distance from the moving element. Because we are interested in the flow and forces on a surface due to sound, it is also instructive to imagine that the fluid is moving with velocity $\mathcal{V}$ at x and z equal to infinity with the sensing element being stationary. This can be accomplished simply by replacing our calculated $v(x_n, z, \omega)$ with $v(x_n, z, \omega) - \mathcal{V}$.

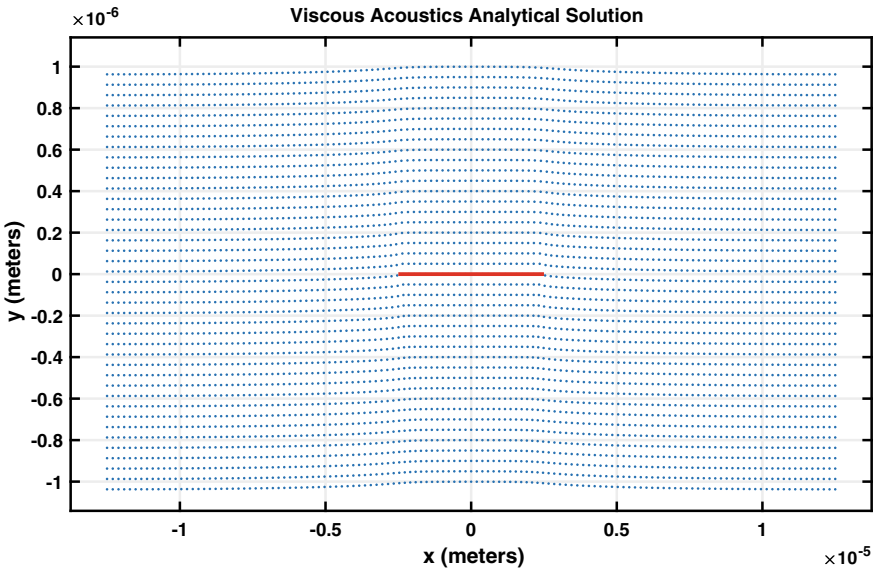


Fig. 9.8 Predicted viscous flow field around a fixed sensing element due to imposed oscillatory flow in the z direction. The flow oscillates at a frequency of 2 kHz with amplitude $\mathcal{V} = \frac{1}{\rho_0 c}$. The width (in the x direction) of each infinitesimally thin moving electrode is $D = 5\,\mu$m. The electrodes are arranged in a planar, periodic array with a gap $g = 10\,\mu$m between each pair. A video version of this figure is available here (see "2kHzinviscous5micronwidthMovingElectrode042619.avi" in supplementary material)

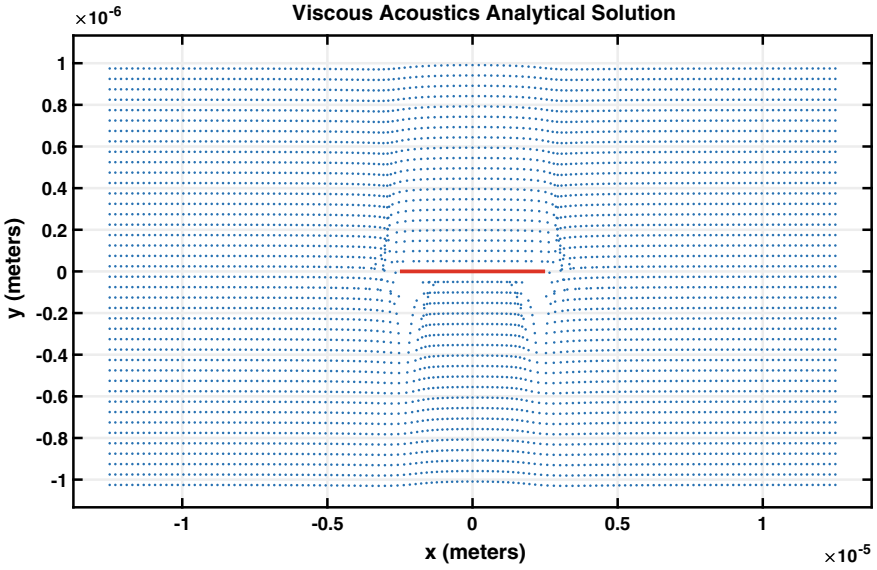


Fig. 9.9 Predicted inviscid flow field around a fixed sensing element due to imposed oscillatory flow in the z direction. The flow oscillates at a frequency of 2 kHz with amplitude $\mathcal{V} = \frac{1}{\rho_0 c}$. The width (in the x direction) of each infinitesimally thin moving electrode is $D = 5$ μm. The electrodes are arranged in a planar, periodic array with a gap $g = 10$ μm between each pair. The parameters in this case are identical to those used in Fig. 9.8 except that the viscosity has been reduced by a factor of 10^{10}. A video version of this figure is available here (see "2kHzinviscid5micronwidthMovingElectrode042619.avi" in supplementary material)

Let the fluid velocity at distances far from the sensing element be $\mathcal{V} = \frac{1}{\rho_0 c}$, where $\rho_0 c$ is the acoustic impedance of the fluid. In this case, if the sensor at $z = 0$ is an impervious, massless, infinitesimally thin membrane, in a plane acoustic wave, the pressure applied to it would be 1 Pa. We may then compare the spatial average of the pressure applied to our periodic array of sensing elements to this unit plane wave pressure.

To examine a specific example, suppose that the width (in the x direction) of each infinitesimally thin moving electrode is $D = 5$ μm. The electrodes are arranged in a planar, periodic array with a gap $g = 10$ μm between each pair. The width of each periodic unit is then $L = D + 2g$. Let the number of terms in our series approximation of the Fourier transforms be $N = 1025$. Numerical results are shown in Fig. 9.8. The figure shows a cross section in the (x, z) plane of the stationary element (shown in red) and the displaced positions of fluid particles around the element due to an imposed fluid motion in the z direction (the upward direction in the figure). In this figure, the motions have been exaggerated to make the flow field easier to see.

Figure 9.8 shows that the fluid particles are displaced in a fairly smooth pattern with relatively little motion relative to that of the sensing element. To illustrate the effect viscosity has on the fluid motion, Fig. 9.9 shows the predicted flow field with

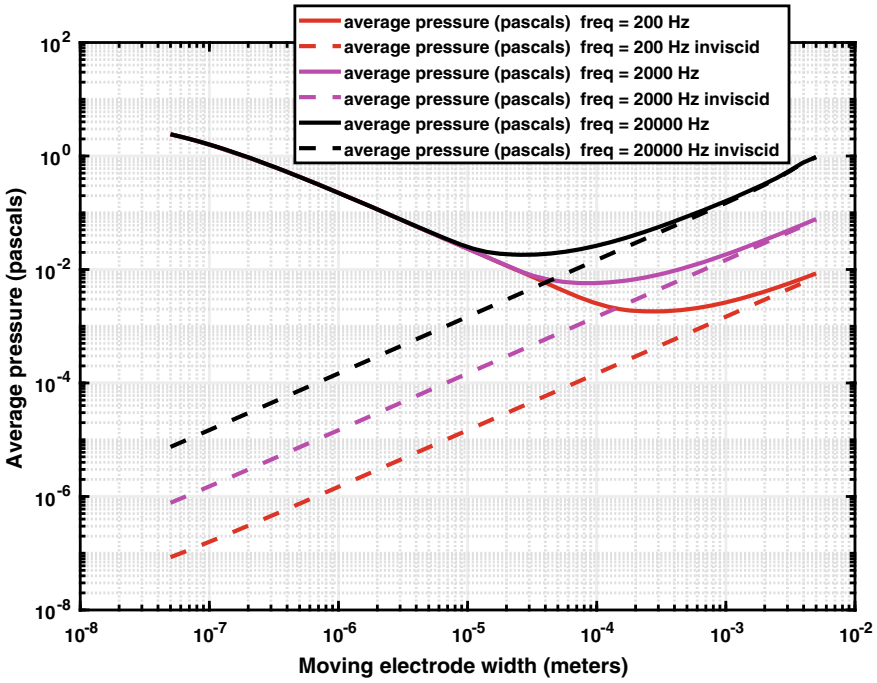


Fig. 9.10 Predicted average pressure applied to the sensing element as a function of viscosity, frequency and element width, D. As the element width, D, is changed, the gap between the electrodes is changed in proportion so that the ratio of $D/g = 5/10$ is kept the same. In the inviscid case (shown in dashed lines), as the width of the element is reduced, the average pressure reduces significantly. When viscosity is included, when the width of the element is approximately 10 microns, the average pressure increases with decreasing D and becomes essentially independent of frequency

all parameters kept the same except that the viscosity has been reduced by a factor of 10^{10}, i.e. essentially inviscid. The flow field in this case is significantly different. The fluid particles near the solid element are now free to translate in the x direction with the particles on the leading surface spreading apart and those on the trailing surface bunching together, leaving regions of diminished density at the edges at $x = \pm D/2$.

In addition to influencing the flow pattern, viscosity has a marked influence on the average pressure applied to the sensing element as its size is decreased. Figure 9.10 shows the predicted average pressure applied to the sensing element as a function of viscosity, frequency and element width, D. As the element width, D, is changed, the gap between the electrodes is changed in proportion so that the ratio of $D/g = 5/10$ is kept the same. These results are, again, for the case where the sensing element moves in the z direction with velocity $\mathcal{V} = \frac{1}{\rho_0 c}$ relative to the velocity of the fluid at an infinite distance away. In the inviscid case (shown in dashed lines), as the width of the element is reduced, the average pressure reduces significantly. When

viscosity is included, when the width of the element is approximately 10 microns, the average pressure increases with decreasing D and becomes essentially independent of frequency.

9.9 Problems

1. Carry out the integration of $P(x)$ to obtain the result given in Eq. (9.3.29) where $P(x)$ is given in Eq. (9.3.28).
2. Write a Matlab script to verify the results shown in Fig. 9.4.
3. Write a Matlab script to verify the results shown in Fig. 9.5.
4. A fiber has a length $L = 3$ cm. Its ends are fully fixed so that the displacement and slope is zero on its ends. Write a Matlab script to calculate the response due to sound to compare with the simply-supported fiber of Fig. 9.4. Compare your results using both the direct and modal solutions.
5. For the fiber of Fig. 9.4, use your Matlab script to determine a fiber diameter such that the fiber response is no less than 80% of the air velocity over the frequency range shown in the figure. Plot your result using the same axes ranges of Fig. 9.4.
6. For the fiber of Fig. 9.5, calculate and plot the fiber response when the fiber is fully fixed (i.e. zero displacement and slope at the supported end) and compare your result with that shown in the figure.

References for Chapter 9

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Chapter 10
Acoustic Sensing



Acoustic sensors normally consist of two essential elements: (1) a mechanical structure that is intended to respond to the minute fluctuations in the medium in a sound field and (2) a means of transducing that response into an electronic signal. In this chapter, we will mainly concentrate on the analysis and design of a suitable structure to respond to these small motions of the medium. The problem of transducing that motion into an electronic signal will be considered in Chap. 11.

10.1 Mechanical Sensitivity of a Microphone Diaphragm

In the following an approximate model is developed for the effects due to the air in the back volume behind an omnidirectional microphone diaphragm and due to the narrow vent used to equalize the static pressure difference between the back volume and the exterior. The air in the back volume is assumed to act like a spring and, due to the narrowness of the vent, viscous forces control the flow of air through it. It is found that the vent and back volume have a pronounced effect on the diaphragm response. The natural frequency of the diaphragm is increased significantly due to the stiffness of the air in the volume and the low-frequency response is reduced.

10.2 Diaphragm with No Vent

When only one side of the microphone diaphragm is exposed to the incident sound, the diaphragm response may be modeled as a linear second-order oscillator as depicted in Fig. 10.1.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_10

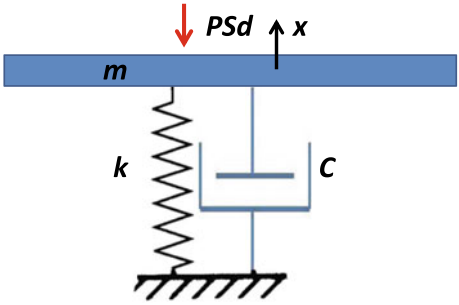


Fig. 10.1 Second-order oscillator model of a microphone diaphragm. This simple model neglects the effect of the air in the closed space behind the diaphragm

This system may be examined using the following equation of motion

$$m\ddot{x} + kx + \mathcal{C}\dot{x} = -PS_d, \tag{10.2.1}$$

where m is the diaphragm mass, k is the mechanical stiffness, $\mathcal{C}$ is the viscous damping coefficient, S_d is the diaphragm area and P is the pressure due to the applied sound field. It is assumed that a positive pressure at the diaphragm's exterior will result in a force in the negative direction. If the resonant frequency, $\omega_0 = \sqrt{k/m}$ is above the highest frequency of interest then the mechanical sensitivity will be $S_m \approx S_d/k$.

10.3 Diaphragm Force Due to Back Volume of Air

Microphones that are intended to respond to sound pressure typically incorporate an enclosed volume of air that surrounds the back side of the diaphragm. If the dimensions of the air chamber behind the diaphragm are much smaller than the wavelength of sound, we can assume that the air pressure in the volume is independent of location. The air in this volume will then act like a linear spring; it exerts a restoring force on whatever acts to change the volume of the air. The assembly is shown schematically in Fig. 10.2. If the diaphragm moves in such a way that the volume is reduced, an internal pressure, P_d will be applied to the interior side of the diaphragm due to the compression of air in the volume. This pressure can be expressed in terms of the deflection x using the equation of state, Eq. (2.3.3):

$$P_d = \rho_a c^2 = \rho_0 c^2 \frac{\rho_a}{\rho_0} = -\rho_0 c^2 \frac{\Delta V}{V}, \tag{10.3.1}$$

where ρ_a is, as before, the fluctuating air density within the volume. The negative sign on the right in Eq. (10.3.1) is due to the fact that the density decreases as the

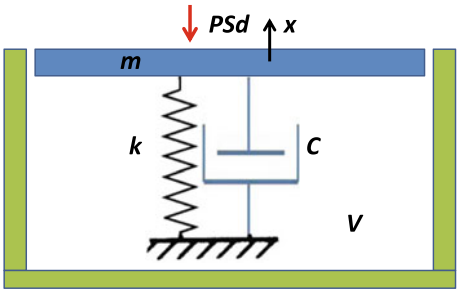


Fig. 10.2 Schematic representation of a microphone diaphragm with an enclosed back volume of air

volume increases. The increase in density relative to the ambient density, ρ_a/ρ_0, is equal to the negative of the ratio of the change in volume, ΔV, relative to the total volume, V; a reduction in volume increases the density and increases the pressure. The change in volume is equal to the displacement of the air in the opening multiplied by the area, $\Delta V = S_d x$, where again, x is taken to be positive in the upward direction shown in Fig. 10.2. The fluctuating pressure in the volume, V, due to a fluctuation in the volume, ΔV, resulting from the outward motion of the diaphragm, x, is then

$$P_d = -\rho_o c^2 \Delta V/V = -\rho_o c^2 S_d x/V, \tag{10.3.2}$$

where ρ_o is the density of air and c is the sound speed. This pressure in the volume exerts a force on the diaphragm given by

$$F_d = P_d \times S_d = -\rho_o c^2 S_d^2 x/V = -K_d x, \tag{10.3.3}$$

where

$$K_d = \rho_o c^2 S_d^2/V, \tag{10.3.4}$$

is the equivalent spring constant of the air in N/m.

The force due to the air in the back volume adds to the restoring force due to the mechanical stiffness of the diaphragm. Including the air in the back volume, Eq. (10.2.1) becomes

$$m\ddot{x} + kx + K_d x + \mathcal{C}\dot{x} = -PS_d, \tag{10.3.5}$$

so that the mechanical sensitivity now becomes $S_m \approx S_d/(k + K_d)$.

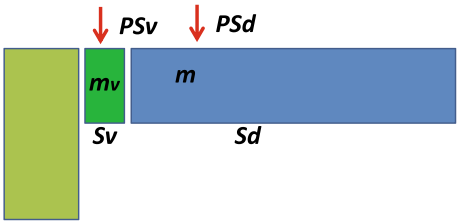


Fig. 10.3 Schematic representation of the air in a slit or vent that equalizes static pressure differences between the back volume and the exterior

10.4 Effect of the Air in the Vent

Because any microphone diaphragm must respond to minute acoustic pressure fluctuations, it is necessary to have the mechanical sensitivity as high as possible, which means we must have the effective stiffness, $k + K_d$, be as small as possible. There are, of course, numerous practical difficulties with this requirement. One is that if the microphone can be represented by the system in Eq. (10.3.5), we must consider the fact that the tiny acoustic pressure fluctuation, P, is also accompanied by comparatively huge fluctuations in atmospheric pressure. These very low-frequency pressure changes (associated with changes in the weather) are often large enough to either damage or at least impose nonlinear stresses on the diaphragm, significantly affecting the sensitivity.

A simple approach to eliminating this difficulty is to pierce the diaphragm or incorporate some vent in the structure that permits the flow of air between the back volume and the exterior. This essentially short circuits the low frequency, or quasi-static response due to atmospheric pressure changes. While the vent will obviously eliminate the static response of the diaphragm, it is far from obvious what effect the vent has as the frequencies are increased from zero. In the following, we attempt to construct a very simplified model to aid in understanding the essential mechanisms that determine the low-frequency response.

The air in the vent will be forced to move due to the fluctuating pressures both within the space behind the diaphragm and in the external sound field. We can again assume that the dimensions of this volume of moving air are much smaller than the wavelength of sound so that it can be represented by a single lumped mass of mass m_v. Figure 10.3 shows a rough representation of the air in the vent. An outward displacement of the air in the vent, x_v, causes a change in volume of the air in the back volume given by $-S_v x_v$ and a corresponding pressure given by

$$P_{vv} = -\rho_o c^2 S_v x_v / V, \tag{10.4.1}$$

where S_v is the area of the vent on which the pressure acts.

The pressure due to the motion of the air in the vent applies a restoring force on the mass of air in the vent given by

$$F_{vv} = -\rho_o c^2 S_v^2 x_v / V = -K_{vv} x_v, \tag{10.4.2}$$

where

$$K_{vv} = \rho_o c^2 S_v^2 / V. \tag{10.4.3}$$

The pressure due to the motion of the air in the vent also exerts a force on the diaphragm given by

$$F_{vd} = P_{vv} * S_d = -\rho_o c^2 S_v S_d x_v / V = -K_{vd} x_v, \tag{10.4.4}$$

where

$$K_{vd} = \rho_o c^2 S_v S_d / V. \tag{10.4.5}$$

Likewise, the pressure due to the motion of the diaphragm in Eq. (10.3.2) produces a force on the air in the vent that is given by

$$F_{dv} = P_d S_v = -\rho_o c^2 S_d S_v x / V = -K_{dv} x, \tag{10.4.6}$$

where $K_{dv} = K_{vd}$ as given in Eq. (10.4.5).

Because the air in the vent is squeezed through a relatively small opening, we must account for the viscous effects which result in a velocity dependent restoring force on the air in the vent,

$$F_{viscous} = -c_v \dot{x}_v, \tag{10.4.7}$$

where c_v is a viscous damping coefficient that depends on the details of the flow, as discussed below.

Finally, the externally applied force on the air in the vent due to the incident sound field is

$$F_v = -P S_v. \tag{10.4.8}$$

Summing the forces on the moving elements of the system gives the following pair of governing equations

$$\begin{aligned} m\ddot{x} + (k + K_d)x + K_{vd} x_v + \mathcal{C}\dot{x} &= -P S_d, \\ m_v \ddot{x}_v + k_{vv} x_v + K_{dv} x + c_v \dot{x}_v &= -P S_v. \end{aligned} \tag{10.4.9}$$

10.5 Response Due to Harmonic Sound Fields

If we assume the sound pressure is harmonic with frequency ω then let $P = \mathcal{P}e^{i\omega t}$. The response may also be written as $x = Xe^{i\omega t}$, and $x_v = X_v e^{i\omega t}$. Equations (10.4.9) may be solved to give the steady-state response relative to the amplitude of the pressure, which may then be written as

$$\begin{pmatrix} X/\mathcal{P} \\ X_v/\mathcal{P} \end{pmatrix} = \begin{bmatrix} k + K_d - \omega^2 m + i\omega\mathcal{C} & k_{vd} \\ k_{dv} & k_{vv} - \omega^2 m_v + i\omega c_v \end{bmatrix}^{-1} \begin{pmatrix} -S_d \\ -S_v \end{pmatrix}. \tag{10.5.1}$$

The response of the microphone diaphragm relative to the pressure, the mechanical sensitivity is then

$$S_m = X/\mathcal{P} = \frac{-S_d(k_{vv} - \omega^2 m_v + i\omega c_v) + S_v K_{vd}}{[(k + K_d - \omega^2 m + i\omega\mathcal{C}) * (k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv}]}. \tag{10.5.2}$$

Note that Eqs. (10.4.3) and (10.5.2) give $S_d K_{vv} = S_v K_{vd}$ so that Eq. (10.5.2) becomes

$$S_m = X/\mathcal{P} = \frac{-S_d(-\omega^2 m_v + i\omega c_v)}{[(k + K_d - \omega^2 m + i\omega\mathcal{C}) * (k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv}]}. \tag{10.5.3}$$

The ω dependence in the numerator of this expression clearly shows that the response has a high-pass filter characteristic. If we assume the damping of the vent dominates so that $c_v >> C$, the cut-off frequency of the high-pass response is given by

$$\omega_{cut} \approx \frac{k_{vv}k}{c_v(k + K_d)}, \tag{10.5.4}$$

where we have used the fact that $k_{vv}K_d = k_{vd}k_{dv}$.

Note that for a sufficiently large value of c_v, Eq. (10.5.3) becomes

$$S_m = X/\mathcal{P} \approx \frac{-S_d}{k + K_d - \omega^2 m + i\omega\mathcal{C}}, \tag{10.5.5}$$

in which case the response behaves as if the enclosure is sealed with an equivalent stiffness $k + K_d$.

Another important special case occurs if the diaphragm's mechanical stiffness is significantly lower than the stiffness due to the air behind the diaphragm, $k << K_d$

in Eq. (10.5.3). In this case, Eq. (10.5.3) becomes

$$S_m = X/\mathcal{P} \approx \frac{-S_d(-\omega^2 m_v + i\omega c_v)}{(K_d - \omega^2 m + i\omega\mathcal{C}) * (k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv}} = \frac{-S_d(-\omega^2 m_v + i\omega c_v)}{\left[(-\omega^2 m + i\omega\mathcal{C})(-\omega^2 m_v + i\omega c_v) + k_{vv}(-\omega^2 m + i\omega\mathcal{C}) + K_d(-\omega^2 m_v + i\omega c_v)\right]}. \tag{10.5.6}$$

If we limit our attention to the lower frequencies where terms that are proportional to ω^2 may be neglected, Eq. (10.5.6) becomes

$$S_m = X/\mathcal{P} = \frac{-S_d i\omega c_v}{k_{vv}(i\omega\mathcal{C}) + K_d i\omega c_v) = \frac{-S_d c_v}{k_{vv}\mathcal{C} + K_d c_v}}. \tag{10.5.7}$$

If $c_v >> \mathcal{C}$ and we use Eqs. (10.3.4) and (10.4.3), Eq. (10.5.7) leads to a simple expression for the pass-band sensitivity

$$S_m = X/\mathcal{P} = \frac{-S_d c_v}{K_d c_v} \approx \frac{-S_d}{K_d} = \frac{-S_d}{(\rho_o c^2 S_d^2/V)} = -\frac{V}{\rho_o c^2 S_d}. \tag{10.5.8}$$

Note that $\rho_o c^2 \approx 1.4 \times 10^5$ Pa. Also, if the depth of the back volume is d, so that $V = d \times S_d$, then Eq. (10.5.8) becomes

$$S_m = X/\mathcal{P} = -\frac{d}{\rho_o c^2}. \tag{10.5.9}$$

Suppose the depth of the backvolume is equal to the thickness of a commonly used silicon wafer, then $d = 500\,\mu$m. The magnitude of the mechanical sensitivity is then $|S_m = X/\mathcal{P}| \approx 3.5$ nm/Pa.

10.6 The Microphone Package as a Helmholtz Resonator

It can be very instructive to compare the performance of a given design to the performance of a highly idealized system where the properties are set to ideal (but perhaps impractical) values. In the following we consider the limiting case where the diaphragm of a pressure-sensing microphone is reduced to have no mass or stiffness. In this case, we consider only the movement of the air where the diaphragm would be. The microphone package may then be viewed as a simple Helmholtz resonator where the 'diaphragm' motion is identical to the motion of the air in the neck. A pressure-sensing microphone that is nondirectional nearly always consists

of a flexible diaphragm backed by a back volume of air. This back volume is created by including an acoustically opaque enclosure that protects the backside of the diaphragm from being acted on by the external sound pressure. Because the dimensions of the back volume are typically much smaller than the wavelength of the sound being detected, this structure looks much like a Helmholtz resonator, as considered in Sect. 4.3 and shown in Fig. 4.3.

We found that the relation between the pressure p and the displacement w of the air in the opening of the resonator may be written as Eq. (4.3.3),

$$-p = \rho_0 c^2 s w / V + \rho_0 l \ddot{w}. \tag{10.6.1}$$

where s is the cross-sectional area of the opening, ρ_0 is the nominal air density, c is the speed of sound propagation, and l is the length of the neck of the resonator. If we focus on only the low-frequency sensitivity, we may neglect $\ddot{w}$ so that the displacement of the air in the opening of the resonator relative to the fluctuating acoustic pressure becomes

$$\frac{w}{p} \approx \frac{V}{\rho_0 c^2 s}. \tag{10.6.2}$$

Equation (10.6.2) gives us a limiting microphone mechanical sensitivity in terms of only two system parameters, the diaphragm area s and the package volume V. In the numerical example presented in Sect. 10.10 below, Eq. (10.6.2) gives $\frac{w}{p} \approx 7 \times 10^{-9}$ m/Pa. This is reasonably close to the pass band sensitivity predicted for the diaphragm of 5.4×10^{-9} m/Pa shown in Fig. 10.4.

10.7 An Important Note on the Ideal Diaphragm Properties

Note that the sensitivity given in Eq. (10.5.8) is achieved when the diaphragm's mechanical stiffness is much less than that of the air spring so that $k << K_d$. This highly compliant diaphragm has a frequency response and sensitivity that no longer depends on the size of or the damping in the vent, c_v, or any properties of the diaphragm at all. The diaphragm's purpose is merely to provide a convenient means of detecting the sound pressure-induced motion of the air next to it. The ideal diaphragm is thus massless and should have infinite compliance. Unfortunately, one must also have a means of converting the diaphragm's motion into an electronic signal. This imposes additional difficulties. The ideal mechanical or purely acoustical requirements on the diaphragm are straight forward: no mass, no stiffness.

If the low-frequency cut-off as predicted by Eq. (10.5.4) is a key parameter in a given design, diaphragm stiffness can be determined to satisfy this requirement. Solving Eq. (10.5.4) for k gives

$$k = \frac{\omega_{cut} c_v K_d}{k_{vv} - \omega_{cut} c_v}. \tag{10.7.1}$$

10.8 Damping Due to the Vent

Estimating the equivalent viscous damping due to the flow through the vent can be rather difficult and strongly dependent on the vent geometry. As a very approximate example, we will assume that the vent is composed of a narrow slit only 2 μm wide, has a depth that is equal to the thickness of the diaphragm, approximately 1 μm and runs the entire perimeter of a 1 mm by 1 mm square diaphragm so that it has a total length of 4 mm. We will assume that the flow of air through it is primarily determined by viscous effects. Again, the approach here is extremely rough and is useful only to obtain an initial approximation. The velocity of the viscous flow in the vent may be approximated by a parabolic profile as we obtained in Eq. (9.2.5),

$$u(y) = -\frac{\partial P}{\partial x}\frac{h^2}{2\mu}(1-(y/h)^2), \tag{10.8.1}$$

where $\mu = 1.846 \times 10^{-5}$ kg/m/s is the dynamic viscosity, and $h = 1\,\mu$m is half the distance across the gap. Equation (10.8.1) is valid only in situations where the thickness, or depth of the vent is much greater than its width, which being only 1 μm relative to the width of 2 μm is certainly not the case here. To compute the equivalent viscous damping coefficient we need to relate the net force acting on the air in the gap to the average velocity, $< u(y) >$, where

$$< u(y) >= -\frac{\partial P}{\partial x}\frac{h^2}{2\mu}\frac{1}{2h}\int_{-h}^{h}(1-(y/h)^2)dy = -\frac{\partial P}{\partial x}\frac{h^2}{3\mu}. \tag{10.8.2}$$

The gradient of the pressure in the vent may be taken as the net pressure on its exterior divided by the depth, $L \approx 2\,\mu$m, $\frac{\partial P}{\partial x} = P_{net}/L$, so that

$$P_{net} = -\frac{3\mu L}{h^2} < u(y) > . \tag{10.8.3}$$

If the total length of the vent is $w = 4$ mm then the net force is

$$f_{net} = P_{net}2hw = -\frac{6\mu L w}{h} < u(y) > . \tag{10.8.4}$$

The equivalent dash-pot constant is then

$$c_v = \frac{6\mu L w}{h}. \tag{10.8.5}$$

Since $L = 2\,\mu$m, $h = 1\,\mu$m, $w = 4$ mm, and $\mu = 1.846 \times 10^{-5}$ kg/m/s, the dashpot constant due to the vent is $c_v \approx 8.861 \times 10^{-7}$ N-s/m.

10.9 Acoustic Radiation Loading on the Diaphragm

If we assume that the microphone diaphragm vibrates like an ideal piston, the results presented above can be modified to account for the radiation load on the piston. The total reaction force on a square piston has been calculated by Morse and Ingard [1]. The force relative to the complex amplitude of the displacement is given by

$$\frac{F_\omega}{X} \approx \rho c d^2[\omega^2 8d/(9\pi c) + i\omega(\omega d/c)^2/16], \tag{10.9.1}$$

where we have assumed the length of each side of the square piston is d. Equation (10.9.1) can be used to identify a "co-vibrating" mass due to the air near the diaphragm that is given by

$$m_{rad} \approx \rho c d^2 8d/(9\pi c). \tag{10.9.2}$$

One can also identify the frequency-dependent equivalent damping constant due to the radiation,

$$C_{rad} \approx \rho c d^2(\omega d/c)^2/16. \tag{10.9.3}$$

By including the mass and damping effects of Eqs. (10.9.2) and (10.9.3), Eqs. (10.4.9) become

$$\begin{aligned}
&(m + m_{rad})\ddot{x} + (k + K_d)x + K_{vd}x_v + (\mathcal{C} + C_{rad})\dot{x} \\
&= -PS_d, \\
&m_v\ddot{x}_v + k_{vv}x_v + K_{dv}x + c_v\dot{x}_v \\
&= -PS_v.
\end{aligned} \tag{10.9.4}$$

The response given by Eq. (10.5.7) then becomes

$$S_m = X/\mathcal{P} = \frac{-S_d(-\omega^2 m_v + i\omega c_v)}{\begin{array}{l}(k + K_d - \omega^2(m + m_{rad}) \\ \quad + i\omega(\mathcal{C} + C_{rad}))\times(k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} \times k_{dv}\end{array}}. \tag{10.9.5}$$

10.10 Numerical Examples

To illustrate the effects of key parameters on the response, we will examine a hypothetical design where the diaphragm has dimensions 1 mm by 1 mm and is fabricated out of a 1 μm thick film of polysilicon. We will assume that the diaphragm moves like a rigid piston and has an equivalent stiffness such that its natural frequency (without the effects of the back volume of air) is 20 kHz. Based on the

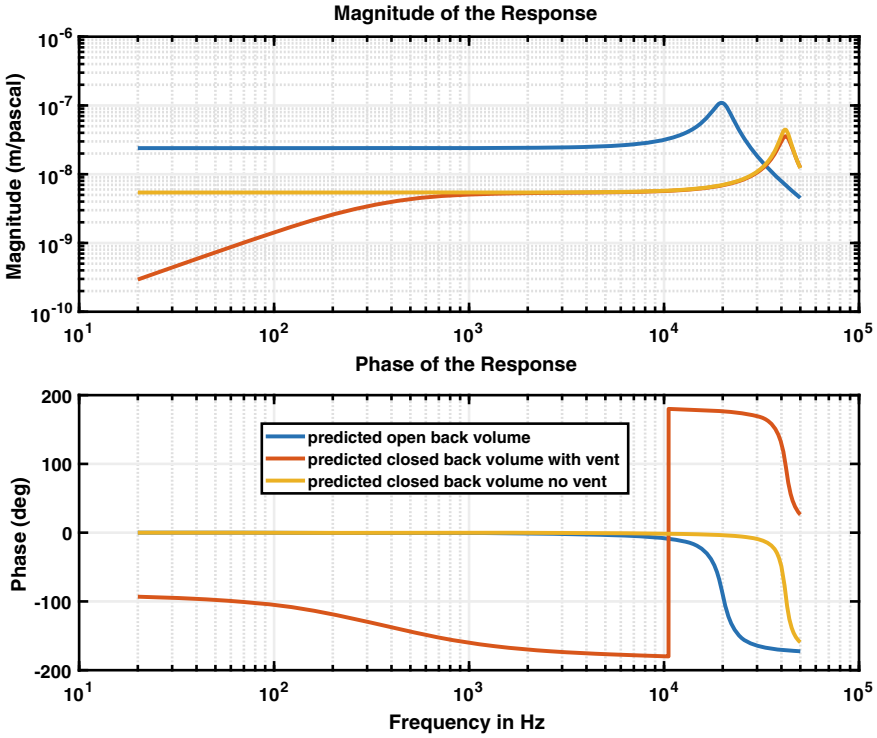


Fig. 10.4 Predicted mechanical response of a microphone diaphragm. The mechanical stiffness of the diaphragm is set so that the resonance frequency is 20 kHz without the effect of the back volume. Results are shown with and without the effects of the back volume. The results obtained with the back volume are obtained with and without the effects of the vent. Presence of the vent clearly attenuates the low-frequency response at frequencies below the cut-off frequency of 371 Hz estimated using Eq. (10.5.4)

density of polysilicon (approximately 2300 kg/m^3), the mass of the diaphragm is roughly 2.3×10^{-9} kg. The co-vibrating mass of air is obtained from Eq. (10.9.2), $m_{rad} = 3.4134 \times 10^{-10}$ kg. Assuming the mechanical stiffness to be such that the natural frequency is 20 kHz then gives

$$k = (m_o + m_{rad}) * (20000 * 2 * \pi)^2 \approx 41.71\,\text{N/m}. \tag{10.10.1}$$

Let the back volume be a cube with all dimensions equal to 1 mm. Equation (10.3.4) then gives $K_d = 142.76$ N/m.

If the dimensions of the vent are as described above, the mechanical sensitivity of the diaphragm is found to be as shown in Fig. 10.4.

For this design, the cut-off frequency predicted using Eq. (10.5.4) is found to be $\omega_{cut} = 2331.5$ rad/s, or 371 Hz. At this frequency, we estimate that the response should have half the power of that in the pass band. The response level in the pass

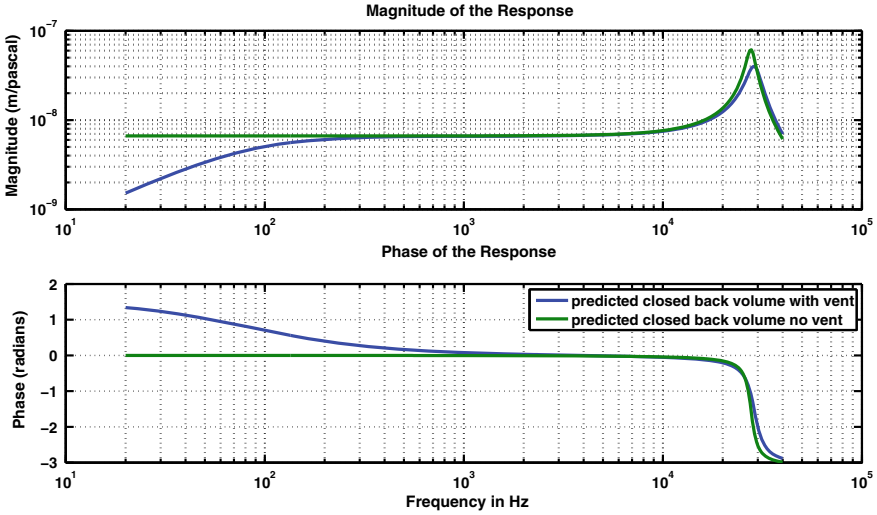


Fig. 10.5 Vent in the microphone back volume has a significant effect on the low-frequency response of a microphone. The parameters of this design are as in those of the results shown in Fig. 10.4 except that the mechanical stiffness of the diaphragm is reduced to $k = 7.803$ N/m. This reduces the low-frequency cut-off to approximately 85 Hz as predicted by Eq. (10.5.4)

band (the flat portion of the curves corresponding to the 'closed back volume' is approximately 5.4×10^{-9} m/Pa. As mentioned in Sect. 10.6, considering the microphone to consist merely of a Helmholtz resonator leads to a sensitivity estimate of $\frac{w}{p} \approx 7 \times 10^{-9}$ m/Pa as given by Eq. (10.6.2). At the cut-off frequency of 371 Hz, the response is found to be approximately 3.8×10^{-9} m/Pa, which is approximately $\sqrt{2}/2$ multiplied by 5.4×10^{-9} m/Pa, as expected.

Equation (10.6.2) gives us a limiting microphone mechanical sensitivity in terms of only two system parameters, the diaphragm area s and the package volume V. In the numerical example presented in Sect. 10.10 below, Eq. (10.6.2) gives $\frac{w}{p} \approx 7 \times 10^{-9}$ m/Pa.

The low-frequency response of our sample design, as shown in Fig. 10.4, is rather poor for audio applications since frequency components below 371 Hz are significantly attenuated. As discussed above, Eq. (10.5.4) indicates that the low-frequency cut-off can be reduced by reducing the mechanical stiffness of the diaphragm, k. If the mechanical stiffness of the diaphragm given in Eq. (10.10.1) is reduced to $k = 7.803$ N/m, the cut-off frequency estimated by Eq. (10.5.4) is reduced to approximately 85 Hz.

The data in Figs. 10.4 and 10.5 show that the phase of the diaphragm's motion is significantly affected by the vent along with the amplitude. This phase distortion thus occurs in the low frequencies.

10.11 Analysis of Thermal-Mechanical Noise in a Pressure-Sensing Microphone

Because the microphone resides in a gas, ballistic interactions between the air molecules with the diaphragm will result in random mechanical vibrations that will be detected as if they resulted from random acoustic fluctuating pressures. If one had an ideal, noiseless transduction scheme to convert the diaphragm motion into an electronic signal, the signal would still contain the noise due to the impinging air molecules when no sound was present.

The amount of vibrational energy imparted to the diaphragm due to these random molecular interactions will depend on their kinetic energy, which is characterized as the temperature of the gas. While the random motion of the diaphragm that results from this thermal interaction is quite complicated when viewed in the time domain, certain low-order statistics of the motion can be calculated through rather simple formulas. We will assume that the diaphragm and the surrounding gas are in thermal equilibrium.

Our interest is not only in how much the diaphragm moves due to the thermal excitation but in the equivalent sound pressure that would produce this random motion. The amount of fluctuating sound pressure that can incite the diaphragm to move the same as if it is driven by random bombardment by air molecules will correspond to the lowest sound pressure signal that the diaphragm can reliably sense. We usually want this equivalent sound pressure signal to be as small as possible, which requires the microphone diaphragm to be as sensitive to sound pressure as possible in order for it to respond to sound as well as it responds to the thermal excitation. In addition to seeking designs with high sensitivity to pressure, we would also like the diaphragm to effectively average the uncorrelated pressures applied by the randomly moving molecules in the gas. If one could estimate the average of a very large number of these interactions with the diaphragm, the average motion will converge to the mean motion due to acoustic pressure. This averaging can be accomplished by using a diaphragm of sufficient surface area. There are thus two key principles to keep in mind when designing a microphone having low thermal noise, the diaphragm should be as sensitive as possible to sound pressure while being of sufficient size to average out the thermal excitations by impinging molecules.

The amount of energy imparted to the diaphragm due to interactions with the thermally excited surrounding medium can be calculated using the equipartition theorem. The equipartition theorem states that if the system is in thermal equilibrium, the thermal energy, $K_B T/2$, where $K_B = 1.38064852 \times 10^{-23}$ (J/K) is the Boltzmann constant and T is absolute temperature (kelvin), imparted to the diaphragm is equal to the expected value of the randomly fluctuating kinetic or potential energy in the system

$$\frac{1}{2}K_B T = \frac{1}{2}(k + K_d)\mathrm{E}[X^2]_N, \tag{10.11.1}$$

where $\mathrm{E}[X^2]_N$ is the mean square of the response of the mode due to thermal excitation and k is the mechanical stiffness of the diaphragm. We will assume, for the purpose of calculating x_N, the response to thermal excitation, that the effects of the slit in the back volume of air can be neglected. The governing equation of motion for random thermal response is

$$(m + m_{rad})\ddot{x}_N + (k + K_d)x_N + \mathcal{C}\dot{x}_N = P_T(t)S_d, \tag{10.11.2}$$

where $P_T(t)$ is the random, net fluctuating pressure on the diaphragm due to the ballistic impacts with the air molecules. Because $P_T(t)$ is not deterministic, we can calculate only certain statistics of the random response $x_N(t)$. The two-sided power spectral density (with units of meters^2/(rad/s)) of the diaphragm's motion is related to that of the random pressure $P_T(t)$ by

$$S_{XX}(\omega) = \frac{S_{pp}(\omega)(S_d/(m + m_{rad}))^2}{(\omega_X^2 - \omega^2)^2 + (2\omega_X\zeta_X\omega)^2}, \tag{10.11.3}$$

where $\omega_X = \sqrt{\frac{k+K_d}{m+m_{rad}}}$ is the natural frequency in rad/s and $\zeta_X = \frac{\mathcal{C}}{2\sqrt{(k+K_d)(m+m_{rad})}}$ is the damping ratio. The mean square response in Eq. (10.11.1) may be determined by integrating Eq. (10.11.3) over all frequencies,

$$\mathrm{E}[X^2]_N = \int_{-\infty}^{\infty} \frac{S_{pp}(\omega)(S_d/(m + m_{rad}))^2}{(\omega_X^2 - \omega^2)^2 + (2\omega_X\zeta_X\omega)^2} d\omega. \tag{10.11.4}$$

Because the pressure $P_T(t)$ results from a huge number of impacts of molecules with the diaphragm surface, it can be considered to be equivalent to white noise, which has a power spectral density that is constant at all frequencies so that $S_{pp}(\omega) = S_{pp}$. The integration can then be carried out with help from the residue theorem, as in Eq. (C.19). The result is

$$\mathrm{E}[X^2]_N = \frac{S_{pp}\pi S_d^2}{2\omega_X^3\zeta_X(m + m_{rad})^2}. \tag{10.11.5}$$

Equations (10.11.1) and (10.11.5) give a simple result for the the power spectral density of the equivalent fluctuating pressure due to thermal noise

$$S_{pp} = \frac{K_B T\mathcal{C}}{\pi S_d^2} \tag{10.11.6}$$

This simple result tells us which design parameters ultimately influence the noise performance of a microphone. The only parameters in Eq. (10.11.6) that a designer has control over are the damping constant, C, and the diaphragm area, S_d. It is clearly important that the damping constant be minimized and that the area, being squared and in the denominator, should be maximized. Unfortunately, increasing the

area also increases cost, especially for miniature silicon microphones. This motivates designers to seek methods of reducing the damping in order to minimize thermal noise in pressure-sensing microphones. In Sect. 10.13, we will examine the thermal noise in a microphone that responds to the acoustic particle velocity rather than pressure. In this case, it will be shown that for this type of microphone it is beneficial to increase the damping constant rather than minimize it, enabling a radically different design approach.

Equation (10.11.3) then becomes

$$S_{XX}^{N}(\omega) = \frac{K_B T \mathcal{C}/(\pi(m + m_{rad})^2)}{(\omega_X^2 - \omega^2)^2 + (2\omega_X \omega \zeta_X)^2} \tag{10.11.7}$$

Equation (10.11.7) gives the power spectral density of the random thermal noise response with units of meters^2/(rad/s). We would like to compute the effective power spectral density of the sound pressure that would produce this same response.

If the back volume is effectively sealed, the sound pressure will produce the same diaphragm motion as the thermal excitation. In this case, the sound pressure-referred noise floor will be as predicted by Eq. (10.11.6). This is a remarkably simple result. Equation (10.11.6) clearly shows that to improve the noise floor of a microphone, one should seek a design that minimizes damping, $\mathcal{C}$, and maximizes area, S_d. No other design parameters appear in this equation; the equivalent input noise due to thermal excitation is not impacted by the stiffness or mass of the diaphragm.

It is a bit confusing and not intuitive that higher damping means more noise. It is important to remember that we are assuming the system is in thermal equilibrium which means that whatever energy flows to the diaphragm from the air molecules must be dissipated by the damping. This also means that the more damping the system has, the more energy will flow to it from the air molecules. This essentially means that the equivalent force (or pressure) applied by all those molecules will be higher when the damping is higher.

The kinetic energy imparted by the surrounding gas must equal the mean potential energy as in Eq. (10.11.1). Since the total energy is fixed, it is desirable to have the system be such that a very small equivalent input pressure will drive the system to have this same amount of energy. This will occur if the system has extremely light damping; it will require a very small equivalent input pressure power spectral density in Eq. (10.11.5) to create that same mean square response and corresponding potential energy.

If the back volume is vented to the exterior, the acoustic excitation will not elicit the same diaphragm response as the thermal excitation. The vent significantly attenuates the diaphragm response at low frequencies as shown in Fig. 10.4. In this case the equivalent input noise can be computed by dividing the result of Eq. (10.11.7) by $|X/\mathcal{P}|^2$ in Eq. (10.9.5),

$$S_{pp}^{N}(\omega) = \frac{K_B T \mathcal{C}/(\pi(m + m_{rad})^2)}{(\omega_X^2 - \omega^2)^2 + (2\omega_X \omega \zeta_X)^2} \times \frac{1}{|X/\mathcal{P}|^2}. \tag{10.11.8}$$

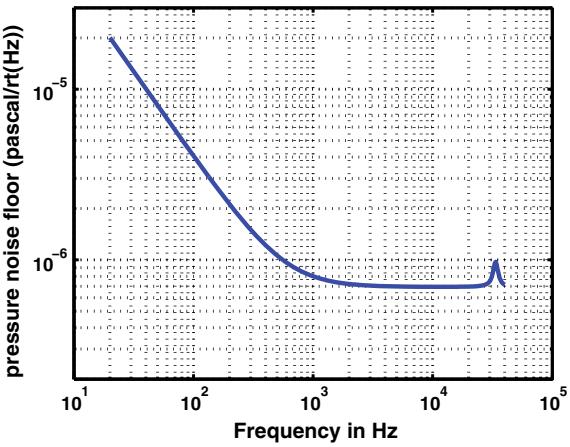


Fig. 10.6 Predicted thermal noise floor of the microphone diaphragm. The results are obtained using Eq. (10.11.8). The square root of the pressure power spectral density as a function of frequency, f (Hz), $G_{pp}(f) = 2\pi S^N_{pp}(\omega)$ are shown with units of pascals/$\sqrt{\text{Hz}}$

This gives the power spectral density of the thermal noise floor with units of pascal^2/(rad/s). This is often referred to as the equivalent input noise, EIN. The results are shown in Fig. 10.6. Since the acoustic response of the microphone falls with decreasing frequency due to the vent, dividing by $|X/\mathcal{P}|^2$ in Eq. (10.11.8) causes the equivalent input noise to increase significantly at low frequencies.

10.12 Limiting Case of Thermal-Mechanical Noise in an Ideal Pressure-Sensing Microphone

As another example of a calculation of the thermal response of an acoustic sensor, suppose we have a microphone diaphragm that responds to sound without any influence of the back volume of air. The diaphragm will then respond in the same manner as the limp wall shown in Fig. 3.1. In this case, we will assume that the pressure we would like to measure is the incident pressure traveling to the right with amplitude p_1. This is the pressure amplitude one would have if the diaphragm (i.e. the limp wall) were not present. If the diaphragm were not present, this plane wave would also have a fluctuating air velocity with amplitude $u_1 = \frac{p_1}{\rho_0 c}$, after Eq. (2.7.5). Using a process nearly identical to that of the discussion following Fig. 3.1, it can be shown that the velocity of the diaphragm relative to that of the incident sound wave is given by

$$u_w = \frac{u_1}{1 + \frac{\rho h i\omega}{2\rho_0 c}} = \frac{p_1/(\rho_0 c)}{1 + \frac{\rho h i\omega}{2\rho_0 c}}. \tag{10.12.1}$$

The equipartition theorem may now be applied to equate the energy imparted by the thermally excited gas to the time average of the kinetic energy,

$$\frac{1}{2}K_B T = \frac{1}{2}\mathrm{E}[MU_w^2]. \tag{10.12.2}$$

where M is the effective mass moving with velocity $U_w(t)$. The expected value of the squared velocity may be expressed as an integral over the power spectral density,

$$\mathrm{E}[U_w^2] = \frac{1}{(\rho_0 c)^2}\int_{-\infty}^{\infty} \frac{S_{pp}}{1+\left(\frac{\rho h\omega}{2\rho_0 c}\right)^2} d\omega. \tag{10.12.3}$$

Carrying out the integration and using Eq. (10.12.2) give an expression for the effective power spectral density of the pressure which is constant at all frequencies,

$$S_{pp} = \frac{K_B T \rho_0 c \rho h}{2M\pi}. \tag{10.12.4}$$

This result can be expressed in terms of the single-sided power spectral density, G_{pp}, with units of pascal^2/Hz using Eq. (1.7.5),

$$G_{pp} = 2\frac{K_B T \rho_0 c \rho h}{M}. \tag{10.12.5}$$

We now need to include an estimate of the mass M that moves with the diaphragm. If we assume that the diaphragm mass dominates over that of the air that moves with it,

$$M \approx \rho h S_d \tag{10.12.6}$$

where S_d is the surface area of the diaphragm. Equation (10.12.5) then becomes

$$G_{pp} = 2\frac{K_B T \rho_0 c}{S_d}. \tag{10.12.7}$$

In this simplified case, the input-referred noise density is clearly inversely proportional to the diaphragm area.

It can also be instructive to consider the mass per unit area of the co-vibrating air to be $\rho_0 h_0$, where h_0 is the effective thickness of the co-vibrating air. The total moving mass may then be expressed as

$$M = (\rho h + \rho_0 h_0) S_d. \tag{10.12.8}$$

Equation (10.12.5) then becomes

$$G_{pp} = 2\frac{K_B T \rho_0 c \rho h}{(\rho h + \rho_0 h_0) S_d}. \tag{10.12.9}$$

Equation (10.12.9) shows that to minimize the input-referred sound pressure, in addition to increasing the diaphragm area S_d, one should keep the mass per unit area of the diaphragm, ρh as small as possible.

These results can be expressed as the A-weighted sound pressure level of the thermal noise pressure using Eq. (1.4.9),

$$SPL_A \approx 10 \log_{10}(G_{pp}) + 135.2423. \tag{10.12.10}$$

Because these calculations assume the diaphragm motion is not impeded by the stiffness of the air in the back volume or by any mechanical stiffness of the diaphragm, one could consider this result to be a limiting case giving the lowest possible input-referred noise that can be achieved for a given diaphragm mass per unit area, ρh and area S_d. If we assume the mass is dominated by the diaphragm material, a diaphragm having area $S_d = 10^{-6}$ m^2, leads to an input-referred noise floor of

$$SPL_A \approx 20.5 \text{ dBA}. \tag{10.12.11}$$

10.13 Thermal Response of a Velocity Driven Spring/Mass

Many animals that hear, do so by using hairs that respond to air motion rather than tympana that respond to pressure. One example is the mosquito, as shown in Fig. 10.7, which hears through the sound-induced motion of its antennae. Flow sensitive hairs detect fluid motion because the relative motion between the hair and the mean flow results in a viscous force on the hair. The viscous force is proportional to the relative velocity between the fluid and the hair.

If we consider the hair to move as a one degree of freedom system, the essential features of the response can be accounted for by a spring/mass/damper that is driven



Fig. 10.7 A mosquito

by motion at the end of the dashpot. The motion, $x(t)$, of this system is governed by the following equation.

$$\begin{aligned} m\ddot{x} &= -kx - c\dot{x} + c\dot{y} \\ m\ddot{x} + kx + c\dot{x} &= c\dot{y} = f(t), \end{aligned} \tag{10.13.1}$$

where m is the mass, k is the stiffness, c is the equivalent viscous damping constant, and $y(t)$ is the displacement of the fluid. The effective driving force is $c\dot{y} = f(t)$. If we divide by m, the governing equation can be expressed in terms of the natural frequency, $\omega_0 = \sqrt{k/m}$, and the damping ratio, $\zeta = \frac{c}{2\sqrt{km}}$. This gives

$$\ddot{x} + \omega_0^2 x + 2\omega_0\zeta\dot{x} = 2\omega_0\zeta\dot{y}. \tag{10.13.2}$$

One can calculate the power spectral density of the equivalent random force (or that of the velocity, $\dot{y}$) due to thermal excitation through the equipartition theorem. Since the system motion is dominated by a single degree of freedom, the mean square response, $\mathrm{E}[x^2]$, due to thermal excitation is related to the absolute temperature, T by

$$\frac{1}{2}K_b T = \frac{1}{2}k\mathrm{E}[x^2], \tag{10.13.3}$$

where $K_b = 1.38 \times 10^{-23}\,\mathrm{m^2kg/(s^2K)}$ is Boltzmann's constant. If $f(t)$ is a weakly stationary random process with two-sided power spectral density S_{ff} with units of $\mathrm{N^2/(rad/s)}$, which is constant at all frequencies, it may be shown that the mean square response is given by

$$\mathrm{E}[x^2] = \frac{S_{ff}\pi}{kc}. \tag{10.13.4}$$

Equations (10.13.3) and (10.13.4) give the equivalent power spectral density of the force due to thermal excitation as

$$S_{ff} = \frac{cK_bT}{\pi}. \tag{10.13.5}$$

We must now relate this equivalent force power spectral density to the acoustic excitation. Since $f(t) = c\dot{y}$, the force power spectral density is simply related to the power spectral density of the acoustic velocity,

$$S_{ff} = c^2 S_{\dot{y}\dot{y}}. \tag{10.13.6}$$

Equations (10.13.5) and (10.13.6) may be used to determine the power spectral density of the equivalent acoustic particle velocity due to thermal excitation,

$$S_{\dot{y}\dot{y}} = \frac{K_b T}{c\pi}. \tag{10.13.7}$$

In an acoustic plane wave traveling in the positive direction, the acoustic pressure, $p(t)$, at any location is simply related to the fluctuating particle velocity through

$$p(t) = \rho_0 c_0 \dot{y}(t), \tag{10.13.8}$$

where ρ_0 is the nominal air density and c_0 is the speed of propagation of the wave. The power spectral densities of these quantities are then related by

$$S_{pp} = (\rho_0 c_0)^2 S_{\dot{y}\dot{y}}. \tag{10.13.9}$$

Equations (10.13.7) and (10.13.9) give the power spectral density of the equivalent acoustic pressure due to thermal excitation of the hair,

$$S_{pp} = (\rho_0 c_0)^2 \frac{K_b T}{c\pi}. \tag{10.13.10}$$

Note that the power spectral densities described above are two-sided (i.e. are defined for $-\infty < \omega < \infty$) and are expressed using radians/s. Since it is common to measure power spectral densities that are single-sided and are expressed using Hertz, it is helpful to recast these results in terms of single-sided spectral densities.

Given the double-sided power spectrum of a random process, $S_{pp}(\omega)$ and the single-sided spectrum $G_{pp}(f)$, it is useful to recall that the total mean square response, $E[p^2]$ is obtained by integrating the power spectrum over all frequencies,

$$\begin{aligned} \mathrm{E}[p^2] &= \int_0^\infty G_{pp}(f)df = \int_{-\infty}^\infty S_{pp}(\omega)d\omega \\ &= 2\int_0^\infty S_{pp}(\omega)d\omega, \end{aligned} \tag{10.13.11}$$

where the last integral results from the fact that the double-sided power spectrum is an even function, $S_{pp}(\omega) = S_{pp}(-\omega)$. Since $\omega = 2\pi f$,

$$\begin{aligned} 2\int_0^\infty S_{pp}(\omega)d\omega &= 2\int_0^\infty S_{pp}(2\pi f)2\pi df \\ &= \int_0^\infty G_{pp}(f)df. \end{aligned} \tag{10.13.12}$$

The two-sided and single-sided power spectra are then related by

$$G_{pp}(f) = 4\pi S_{pp}(2\pi f). \tag{10.13.13}$$

The single-sided power spectral density of the equivalent sound pressure (with units of pascal^2/Hz) due to thermal excitation is then

$$G_{pp} = (\rho_0 c_0)^2 \frac{4K_b T}{c}. \tag{10.13.14}$$

The corresponding single-sided power spectral density of the equivalent particle velocity (with units of $(\mathrm{m/s})^2/\mathrm{Hz}$) is

$$G_{\dot{y}\dot{y}} = \frac{4K_b T}{c}. \tag{10.13.15}$$

Equations (10.13.14) and (10.13.15) show that for an acoustic sensor that is driven by viscous forces, i.e., velocity rather than pressure, the equivalent excitation due to thermal noise is minimized by **maximizing** the viscous damping coefficient, c. This is the opposite of what we find for the pressure-sensing microphone in Eq. (10.11.6) where the equivalent pressure due to thermal noise is minimized by **minimizing** the viscous damping coefficient; the approach to design for these two types of acoustic sensors is quite different.

10.13.1 Application to a Hair for Sensing Sound

The results above apply to a simple spring/mass/damper system that is driven by motion through a dashpot. If we assume that a hair sensor rotates through the angle θ as a rigid bar in a compliant socket, for small motions the governing equation may be expressed as

$$I\ddot{\theta} + k_t\theta + c_t\dot{\theta} = M, \tag{10.13.16}$$

where I is the mass moment of inertia about the pivot, k_t is the effective torsional stiffness of the socket, c_t is the equivalent torsional dashpot constant, and M is the applied moment. We may take the horizontal displacement of a point x that is a distance l along the length of the hair from the point of rotation as a generalized coordinate. The applied moment will then be $M = fl$, and $x = l\theta$. Substituting these expressions into (10.13.16) and comparing with Eqs. (10.13.1) gives

$$m = I/l^2, \quad k = k_t/l^2, \quad c = c_t/l^2. \tag{10.13.17}$$

It has been found experimentally that the natural frequency of the mosquito antenna is approximately $\omega_0 \approx 500\,\mathrm{rad/s}$ and the damping ratio is $\zeta \approx 0.25$. It has also been estimated that the mass moment of inertia is $I \approx 2 \times 10^{-15}\,\mathrm{kg\,m^2}$. The total length of the antenna is approximately 1 mm so that $l \approx 0.5\,\mathrm{mm}$. These estimates lead to an estimate of the equivalent dashpot constant of $c \approx 2 \times 10^{-6}\,\mathrm{Ns/m}$.

For sound propagation in air, $\rho_0 c_0 \approx 415$ ralys. Taking the temperature to be $T = 294\,\mathrm{K}$ (room temperature), Eq. (10.13.14) gives $G_{pp} \approx 4.87 \times 10^{-9}\,\mathrm{Pa^2/Hz}$. Also, the square root of the power spectral density is $\sqrt{G_{pp}} \approx 6.98 \times 10^{-5}\,\mathrm{Pa}/\sqrt{\mathrm{Hz}}$.

Equation (10.13.15) gives the corresponding results in terms of acoustic velocity. This gives $G_{\dot{y}\dot{y}} \approx 2.83 \times 10^{-14}\,(\mathrm{m/s})^2/\mathrm{Hz}$, and $\sqrt{G_{\dot{y}\dot{y}}} \approx 1.68 \times 10^{-7}\,\mathrm{m/s}/\sqrt{\mathrm{Hz}}$.

The noise floor of microphones is typically quantified in terms of a dBA weighted equivalent sound pressure level. This may be calculated from the power spectral density of the equivalent sound pressure by

$$SPL_A \approx 10\log_{10}(G_{pp}) + 135.2 \approx 52.1\,\mathrm{dBA}. \tag{10.13.18}$$

Note that the input sound pressure-referred results assume the sound field consists of a plane wave. The input acoustic particle velocity-referred results are for an arbitrary sound field.

10.13.2 Thermal Noise Response Increases with the Number of Degrees of Freedom

The analysis of the effect of thermal noise given in Sect. 10.13 above is greatly simplified by the assumption that the sensor can be modeled as a single degree of freedom system. The number of degrees of freedom in structures has a marked influence on the response to thermal noise. To gain some insight on how the degrees of freedom can influence the thermal noise response, the thermal noise response of a two degree of freedom system is examined below.

If a system has **no** degrees of freedom, i.e., it is completely rigid, there will be no response to thermal excitation, but there would also be no response to sound. Of course, we are interested in sensing sound, so there must be at least one degree of freedom. In order to respond to the diminutive fluid forces in a sound field, the system should have minimal stiffness and mass. Typical structures for sensing sound consist of membranes, thin plates, or beams, and practical designs typically possess multiple resonant modes within the frequency range of interest. If one wishes to design an acoustic sensor having low thermal noise response and that is driven by viscous forces rather than the more typical acoustic pressure, it is desirable that the structure's response be dominated by that of only a single degree of freedom.

To see the effect of additional degrees of freedom on the thermal noise response of an acoustic sensor that uses viscous forces to sense sound, we will examine the simple case of a system with only two degrees of freedom as shown in Fig. 10.8. To keep things as simple as possible, let $k_1 = k_3 = k$ and $m_1 = m_2 = m$. To examine the noise response, we'd like to estimate the contribution to the responses at x_1 and x_2 due to the noise forces f_1 and f_2 compared to that due to the acoustic flow excitation $\dot{y}$. f_1 and f_2 represent forces due to the random impacts of air molecules, while $\dot{y}$ represents the spatial average of the velocity of those molecules as they move due to sound. The effect of the number of degrees of freedom can be explored by adjusting the value of k_2; $k_2 = 0$ corresponds to two uncoupled degrees of freedom, while $k_2 = \infty$ corresponds to a system with only one degree of freedom.

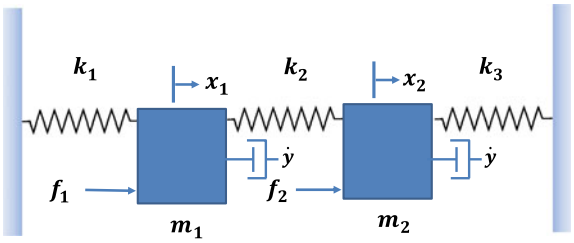


Fig. 10.8 Simplified two degree of freedom system. Direct forces f_1 and f_2 represent uncorrelated thermal noise. The acoustic excitation is represented by the acoustic velocity $\dot{y}$ which is applied to the masses m_1 and m_2 through viscosity represented by the two dashpots. The masses are attached to fixed points through the springs k_1 and k_3 and to each other through the spring k_2

First consider the response due to noise only with no contribution due to sound so that $\dot{y} = 0$. If $k_2 = 0$, the governing differential equation for the response of each mass becomes

$$m\ddot{x} + kx + C\dot{x} = f, \tag{10.13.19}$$

where f is the noise force and C is the equivalent damping constant for the dashpots shown in Fig. 10.8. If f is white noise with two-sided power spectral density K, it is shown in Eqs. (C.16) and (C.26) in Appendix C that the mean square response of $x(t)$ is

$$E[x^2(t)] = \frac{\pi K}{m^2 2\omega_0^3 \zeta}, \tag{10.13.20}$$

where $E[\cdot]$ denotes the expected value and $\zeta = C/(2\sqrt{km})$ is the damping ratio. Substituting this for the damping ratio in Eq. (10.13.20) gives

$$E[x^2(t)] = \frac{\pi K}{kC}. \tag{10.13.21}$$

Next, we consider the case where $k_2 \to \infty$, which causes the system to possess only one degree of freedom. The governing differential equation for the single mass $m_1 + m_2$ is

$$(m_1 + m_2)\ddot{x} + (k_1 + k_3)x + 2C\dot{x} = f_1 + f_2, \tag{10.13.22}$$

where again C is the equivalent constant for the dashpots shown in Fig. 10.8.

Since we are interested only in the response $x(t)$ long after initial transients have died out, we need to consider only the particular solution of Eq. (10.13.22). Because the differential equation is linear, we can find the response due to each of the uncorrelated forces, f_1 and f_2 separately and then add them together to obtain

the complete response. Let $x(t) = \chi_1(t) + \chi_2(t)$ where

$$\begin{aligned}(m_1 + m_2)\ddot{\chi}_1 + (k_1 + k_3)\chi_1 + 2C\dot{\chi}_1 &= f_1\\ (m_1 + m_2)\ddot{\chi}_2 + (k_1 + k_3)\chi_2 + 2C\dot{\chi}_2 &= f_2.\end{aligned} \tag{10.13.23}$$

Since f_1 and f_2 are uncorrelated, and the two governing equations in (10.13.23) are uncoupled, $\chi_1(t)$ and $\chi_2(t)$ will also be uncorrelated. This means that

$$\begin{aligned}E[(f_1 + f_2)^2] &= E[f_1^2] + 2E[f_1 f_2] + E[f_2^2] = E[f_1^2] + E[f_2^2] \quad \text{and}\\ E[(\chi_1 + \chi_2)^2] &= E[\chi_1^2] + 2E[\chi_1\chi_2] + E[\chi_2^2] = E[\chi_1^2] + E[\chi_2^2]\end{aligned} \tag{10.13.24}$$

since for uncorrelated signals $E[f_1 f_2] = E[\chi_1\chi_2] = 0$

The mean square of χ_1 can be written similarly to Eq. (10.13.21),

$$E[\chi_1^2(t)] = \frac{\pi K_{f_1}}{(k_1 + k_3)2C}, \tag{10.13.25}$$

where K_{f_1} is the two-sided power spectral density of f_1 and $2C$ is the sum of the two dashpots in Fig. 10.8. A similar result can be obtained for $E[\chi_2^2(t)]$,

$$E[\chi_2^2(t)] = \frac{\pi K_{f_2}}{(k_1 + k_3)2C}, \tag{10.13.26}$$

where K_{f_2} is the two-sided power spectral density of f_2. Equations (10.13.23) through (10.13.26) then give the total system mean square response as

$$E[x^2] = E[(\chi_1 + \chi_2)^2] = \frac{\pi(K_{f_1} + K_{f_2})}{(k_1 + k_3)2C}, \tag{10.13.27}$$

We may view this as one way to characterize the thermal noise response of the system when k_2 is large so that it reduces to a one degree of freedom system.

The result in Eq. (10.13.27) should be compared to what we obtain when the coupling spring is reduced to zero, $k_2 = 0$. In that case the total thermal noise response of the two independent oscillators becomes

$$E[x_1^2] + E[x_2^2] = \frac{\pi K_{f_1}}{k_1 C} + \frac{\pi K_{f_2}}{k_3 C}. \tag{10.13.28}$$

To further simplify our comparison, assume that the two noise spectral densities are the same, $K_{f_1} = K_{f_2} = K_f$, and that the spring constants are equal, $k_1 = k_3 = k$. Equation (10.13.27) for the coupled system simplifies to

$$E[x^2] = E[(\chi_1 + \chi_2)^2] = \frac{1}{2}\frac{\pi K_f}{kC}, \tag{10.13.29}$$

while Eq. (10.13.28) for the uncoupled systems becomes

$$E[x_1^2] + E[x_2^2] = 2\frac{\pi K_f}{kC}. \tag{10.13.30}$$

This shows that for the system shown in Fig. 10.8, the thermal noise in the coupled system given in Eq. (10.13.29) is **divided** by 2 while that of the uncoupled system in Eq. (10.13.30) is **multiplied** by 2, causing the uncoupled system to have 4 times as much response to thermal noise as the uncoupled system,

$$\frac{\text{uncoupled}}{\text{coupled}} = \frac{E[x_1^2] + E[x_2^2]}{E[(\chi_1 + \chi_2)^2]} = 4. \tag{10.13.31}$$

Of course, the approach taken here is overly simplified in an effort to avoid distracting effects. It is reasonable to conclude, however, that as the number of degrees of freedom increases the thermal noise response of systems driven by acoustic flow will increase substantially; keeping the number of degrees of freedom at a minimum can have a significant influence on the response to (uncorrelated) thermal noise.

10.14 Viscous Flow-Induced Response to Sound and Signal to Noise Ratio, SNR

In addition to determining the response to thermal excitation, it is essential that we compare this to the response to the sound we wish to measure. We'd like to calculate the equivalent acoustic signal that would correspond to the amount of response the system has to thermal noise. This means we need to determine the response of the masses in Fig. 10.8 to the acoustic input, $\dot{y}$, the acoustic flow velocity. To be consistent with our analysis of noise given above, we should determine the acoustic response for the coupled case ($k_2 \to \infty$) and for the uncoupled case ($k_2 = 0$). In the coupled case, this is accomplished by replacing each of the thermal noise forces f_1 and f_2 in Eq. (10.13.22) with the force due to viscous acoustic flow, $C\dot{y}$,

$$(m_1 + m_2)\ddot{x} + (k_1 + k_3)x + 2C\dot{x} = 2C\dot{y}. \tag{10.14.1}$$

In order to enable a convenient comparison with the response to white thermal noise, consider the acoustic excitation to also be white noise and that the sound field consists of a plane wave so that the acoustic velocity and the pressure are related by a constant, $\rho_0 c_0$, $p = \rho_0 c_0 \dot{y}$, where ρ_0 is the ambient air density and c_0 is the propagation speed of acoustic waves. Let the double-sided power spectral density of the acoustic velocity be $S_{\dot{y}\dot{y}}$, with units of $(\text{m/s})^2/(\text{radian/s})$. The double-sided power spectral density of the viscous acoustic force will be $C^2 S_{\dot{y}\dot{y}}$. To obtain the mean square response due to random acoustic flow, we can replace K_f in Eq. (10.13.29)

with $C^2 S_{\dot{y}\dot{y}}$. The mean square of the acoustic response of each of the coupled masses will then be

$$E[x_1^2] = E[x_2^2] = \frac{1}{2}\frac{\pi C^2 S_{\dot{y}\dot{y}}}{kC} = \frac{\pi C S_{\dot{y}\dot{y}}}{2k}. \tag{10.14.2}$$

An identical result for the mean square of the acoustic response is obtained when the masses are uncoupled.

The signal to noise ratio (SNR) for the uncoupled system can be calculated by taking the ratio of the result of Eq. (10.14.2) to that of Eq. (10.13.30). This gives

$$SNR_{uncoupled} = \frac{\pi C S_{\dot{y}\dot{y}}}{k} \times \frac{kC}{2\pi K_f} = \frac{C^2 S_{\dot{y}\dot{y}}}{2K_f}, \tag{10.14.3}$$

where again K_f is the two-sided power spectral density of the random thermal noise force. The signal to noise ratio (SNR) for the coupled system can be calculated by taking the ratio of the result of Eq. (10.14.2) to that of Eq. (10.13.29). This gives

$$SNR_{coupled} = \frac{\pi C S_{\dot{y}\dot{y}}}{k} \times 2\frac{kC}{\pi K_f} = 2\frac{C^2 S_{\dot{y}\dot{y}}}{K_f}. \tag{10.14.4}$$

This signal to noise ratio is, again, **four times greater** for the coupled system having only one degree of freedom than it is for the uncoupled, two degree of freedom system. In both cases, it increases with the square of the damping constant, C.

10.15 Effect of Sound Angle of Incidence on a Pressure Microphone

In the above discussion of the response of a pressure-sensing microphone diaphragm, it was assumed that the diaphragm is very small relative to the sound wavelength so that the pressure can be assumed to be constant over its area. For large diaphragms, such as those used in sound reinforcement and recording, this may no longer be true. In the following, we will examine the average pressure that acts on a diaphragm. If the diaphragm responds as a piston without higher order vibrational modes, the response should be proportional to the average pressure.

Most microphone diaphragms are circular so our determination of the average pressure will involve an integration over a circular area which will be assumed to have a radius a. The incident sound will be assumed to consist of a plane wave at the frequency ω. In cartesian coordinates this wave may be expressed as

$$p(x, y, z, t) = Pe^{i\omega t}e^{-ik_x x - ik_y y - ik_z z}, \tag{10.15.1}$$

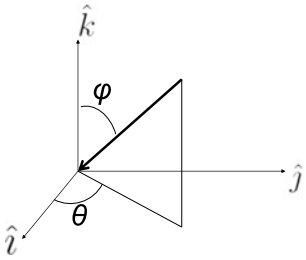


Fig. 10.9 Orientation of the incident sound wave

where k_x, k_y, and k_z are the components of the wave vector. We will assume that the microphone diaphragm is located at $z = 0$. The sound wave will be assumed to be incident on the diaphragm according to the geometry shown in Fig. 10.9. The components of the wave vector are related to the angles shown in Fig. 10.9,

$$k_x = \frac{\omega}{c}\cos(\theta)\sin(\phi), \quad k_y = \frac{\omega}{c}\sin(\theta)\sin(\phi), \quad k_z = \frac{\omega}{c}\cos(\phi), \tag{10.15.2}$$

where $\omega/c = k$ is the wave number. Since the diaphragm will be assumed to be circular, it is most convenient to use polar coordinates to describe each differential surface element. Let $x = b\cos(\psi)$ and $y = b\sin(\psi)$. The total force due to the incident pressure on the diaphragm of radius a is then given by

$$f(t) = e^{i\omega t}\int_0^a\int_0^{2\pi} Pe^{-i\frac{\omega}{c}\cos(\theta)\sin(\phi)b\cos(\psi)}e^{-i\frac{\omega}{c}\sin(\theta)\sin(\phi)b\sin(\psi)}bd\psi db. \tag{10.15.3}$$

This can be simplified to

$$f(t) = e^{i\omega t}\int_0^a\int_0^{2\pi} Pe^{-ikb\cos(\theta-\psi)\sin(\phi)}bd\psi db. \tag{10.15.4}$$

We have previously encountered a very similar integral in Eq. (5.11.5). The integration over ψ can be accomplished using the zeroth order Bessel function of the first kind,

$$J_0(kb\sin(\phi)) = \frac{1}{2\pi}\int_0^{2\pi} e^{ikb\sin(\phi)\cos(\psi)}d\psi. \tag{10.15.5}$$

Using the fact that the cosine is periodic with period 2π and the fact that J_0 is real, Eq. (10.15.4) becomes

$$f(t) = 2\pi e^{i\omega t}\int_0^a PJ_0(kb\sin(\phi))bdb. \tag{10.15.6}$$

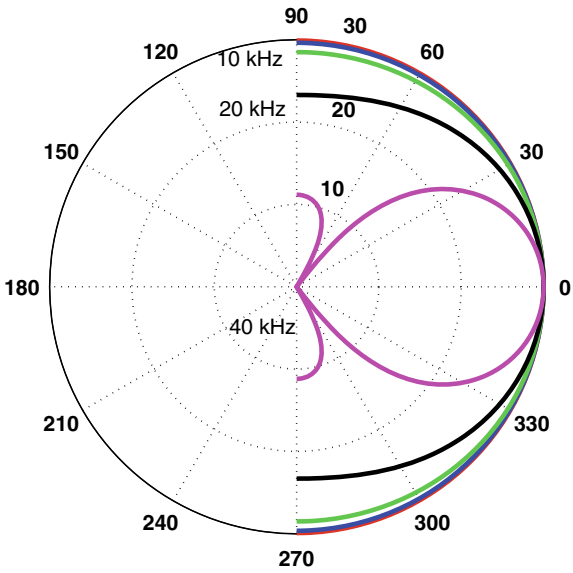


Fig. 10.10 A microphone with a 1/2 in. diameter diaphragm can be quite directional at frequencies above 20 kHz. The predicted directivity pattern is shown in decibels for a 1/2 in. B&K microphone. The data are normalized relative to the most sensitive direction

We may again make use of the property of these Bessel functions as in Eq. (5.11.8),

$$\int x J_0(x) dx = x J_1(x). \tag{10.15.7}$$

The net force then becomes

$$f(t) = 2\pi P e^{i\omega t} a \frac{J_1(ka \sin(\phi))}{k \sin(\phi)}. \tag{10.15.8}$$

The dependence of the net force on frequency, ω (since $k = \omega/c$) and ϕ can be deduced from Fig. 5.10. At frequencies where the wavelength of sound approaches the size of the diaphragm, the directivity pattern can differ significantly from the nondirectional response at low frequencies and it varies significantly with frequency, normally a very undesirable feature of microphones.

The directivity pattern for a 1/2 in. diameter microphone as predicted using Eq. (10.15.8) is shown in Fig. 10.10. The results are all normalized relative to the response at the most sensitive direction. The predicted and measured patterns are qualitatively similar. The microphone is clearly quite directional at 40 kHz.

The predicted directivity pattern presented here is based merely on an estimate of the average pressure incident on a circular diaphragm without taking into account any influence of the diaphragm itself on the pressure near it. In practice, one must always account for the influence of the microphone on the sound field, especially when using large microphones.

10.16 Acceleration Sensitivity

In the following, we examine the response of a microphone to acceleration of its case, or package. Unfortunately, the diaphragm of a microphone along with the air around it act exactly like a proof mass in an accelerometer. We hope that the design is much more sensitive to sound pressure than acceleration of the package. Otherwise, motion of the package will manifest itself as an electronic signal at the output which will be indistinguishable from an acoustic input pressure.

To account for the package acceleration, we need to modify the governing equation of motion (10.9.4). In our previous analysis of the response of a microphone diaphragm, x and x_v are taken to be the relative motion between the diaphragm or the air in the vent and the nominal position a distance d from the back plate. The stiffness k acts like a linear spring connecting the diaphragm to the package. If the base, or package, of the system undergoes a displacement y, we must be careful to properly express the accelerations so that they are expressed relative to an inertial reference frame. The governing equation needs to be modified to account for the forces due to motion relative to the package and those due to the accelerations relative to an inertial reference frame. The governing equation (10.9.4) in this case becomes

$$\begin{aligned}(m+m_{rad})(\ddot{x}+\ddot{y})+(k+K_d)x+K_{vd}x_v+\mathcal{C}\dot{x}+C_{rad}(\dot{x}+\dot{y})&=-PS_d,\\ m_v(\ddot{x}_v+\ddot{y})+k_{vv}x_v+K_{dv}x+c_v\dot{x}_v&=-PS_v.\end{aligned}\tag{10.16.1}$$

It should be noted that here we have assumed that whatever it is that causes the microphone package to accelerate, does not itself contribute to the sound pressure. If, for example, the microphone is placed on a moving piston, or the head of a shaker, the piston motion will create a sound pressure, which could contribute substantially to the output signal. Measurements of the vibration sensitivity of microphones, particularly small microphones, must consider this important effect [2]. To examine the sensitivity of the diaphragm to acceleration, consider the case where the acoustic pressure, P is zero. This gives,

$$\begin{aligned}(m+m_{rad})\ddot{x}+(k+K_d)x+K_{vd}x_v+\mathcal{C}\dot{x}+C_{rad}\dot{x}&=-(m+m_{rad})\ddot{y}-C_{rad}\dot{y},\\ m_v\ddot{x}_v+k_{vv}x_v+K_{dv}x+c_v\dot{x}_v&=-m_v\ddot{y}.\end{aligned}\tag{10.16.2}$$

Let the base acceleration be harmonic at the frequency ω with amplitude Y_a so that $\ddot{y}=Y_ae^{i\omega t}$. As in Eq. (10.5.1), the response may also be written as $x=X_ae^{i\omega t}$, and $x_v=X_{va}e^{i\omega t}$, where X_a and X_{va} are the diaphragm and vent displacements due to base acceleration only, i.e. with $P=0$. Equation (10.16.2) then gives

$$\begin{pmatrix} X_a \\ X_{va} \end{pmatrix} = \begin{bmatrix} k + K_d - \omega^2 m + i\omega(\mathcal{C} + C_{rad}) & k_{vd} \\ k_{dv} & k_{vv} - \omega^2 m_v + i\omega c_v \end{bmatrix}^{-1} \begin{pmatrix} -(m + m_{rad})Y_a - C_{rad}Y_a/(i\omega) \\ -m_v Y_a \end{pmatrix}. \tag{10.16.3}$$

While any microphone having a compliant moving element will no doubt respond to base acceleration, our interest is in minimizing the electronic output due to this acceleration. As when considering the influence of noise on the output signal, it is helpful to estimate the equivalent sound pressure level that would produce the same output as a 1 g base acceleration. We desire this equivalent sound level to be below that of the actual sound to be measured. This can be computed knowing the solution to Eq. (10.16.3) along with the solution due to pressure along in Eq. (10.9.5). The ratio of the acceleration response to the pressure response may then be written as

$$P_{accel} = \frac{X_a/Y_a}{X/P}. \tag{10.16.4}$$

This gives the equivalent pressure (pascals) due to a $1\,\mathrm{m/s^2}$ base acceleration. It is perhaps more convenient to express this as pascals/g, where g is the acceleration due to gravity. Figure 10.11 shows the results for the microphone example examined in Fig. 10.4.

10.17 Basics of Directional Microphones

Pressure-sensing microphones that have an output signal that depends on the direction of propagation of the incident sound wave generally detect the sound at multiple locations in space. The output signal is sensitive to differences in those detected signals along with, in many cases, the average of those signals. In many situations, the distance between the measurement locations is much smaller than the wavelength of sound. In this case, the microphones comprise a 'small aperture array.' When the pressure is detected at two spatial locations, the difference in output signals will be proportional to the first-order pressure gradient; hence we call this a 'first-order array.' When three microphones are used, one can obtain an output that approximates the difference between the gradients, providing a 'second-order array.'

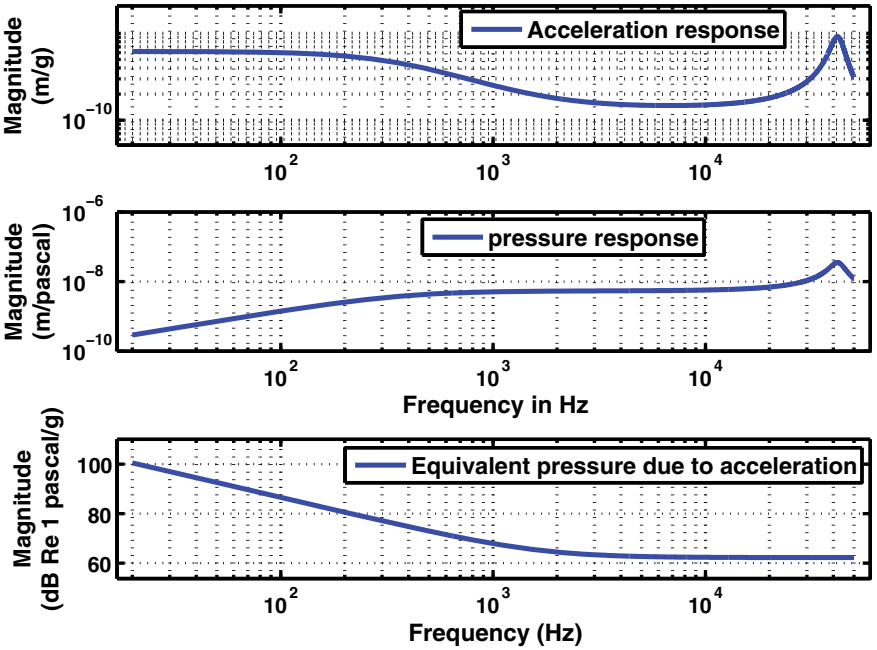


Fig. 10.11 Predicted microphone response due to base acceleration. The upper panel shows the predicted diaphragm displacement due to base acceleration as predicted by Eq. (10.16.3). The middle panel shows the predicted diaphragm displacement due to pressure only using Eq. (10.9.5). The lower panel shows the equivalent sound pressure in decibels detected by the microphone due to base excitation using Eq. (10.16.4). A 1 g acceleration results in an output that is equivalent to a sound level of approximately 60 dB at high frequencies

10.17.1 First-Order Directional Microphones

Suppose we have two nondirectional microphones separated by a distance d. Let the x axis be parallel to the line connecting the microphones with its origin halfway between the microphones. The sound pressure at $y = 0$ due to a plane harmonic sound wave with amplitude P and frequency ω can be described by

$$p(x,t) = Pe^{i(\omega t - kx\cos(\phi))}, \tag{10.17.1}$$

where $k = \omega/c$ is the wave number, c is the speed of propagation of the wave, $i = \sqrt{-1}$ and ϕ is the angle between the propagation direction and the x axis. The pressures at the two microphones may be written as

$$\begin{aligned} p(d/2,t) &= Pe^{i\omega t}e^{-ik(d/2)\cos(\phi)} \\ p(-d/2,t) &= Pe^{i\omega t}e^{ik(d/2)\cos(\phi)}. \end{aligned} \tag{10.17.2}$$

The difference of these pressures will be

$$\begin{aligned} & p(d/2,t) - p(-d/2,t) \\ & = Pe^{i\omega t}(e^{-ik(d/2)\cos(\phi)} - e^{ik(d/2)\cos(\phi)}) \\ & = Pe^{i\omega t}2i\sin(-k(d/2)\cos(\phi)) \\ & \approx -Pe^{i\omega t}2ik(d/2)\cos(\phi) = -Pe^{i\omega t}ikd\cos(\phi), \end{aligned} \tag{10.17.3}$$

where our approximation is valid for $kd << 1$. Note that if the incident angle is such that the difference response is maximum, i.e. $\phi = 0$, the difference signal will be less than the amplitude of the plane wave by the factor $kd = \frac{\omega d}{c}$.

Note that from Eq. (10.17.1), the pressure gradient is

$$\frac{\partial p(x,t)}{\partial x} = -iPk\cos(\phi)e^{i(\omega t - kx\cos(\phi))}. \tag{10.17.4}$$

Of course, from Eqs. (10.17.3) and (10.17.4),

$$\frac{\partial p(0,t)}{\partial x} \approx \frac{1}{d}(p(d/2,t) - p(-d/2,t)) \approx -iPk\cos(\phi)e^{i(\omega t)}. \tag{10.17.5}$$

The pressure gradient clearly depends on the orientation of the sound wave according to the factor $\cos(\phi)$. If we combine a measure of the pressure gradient with a measure of the pressure in the form

$$p(0,t) + \gamma\frac{\partial p(x,t)}{\partial x}, \tag{10.17.6}$$

where γ is some adjustable constant, then the result will depend on ϕ according to

$$p(0,t) + \gamma\frac{\partial p(x,t)}{\partial x} = Pe^{i(\omega t)}(1 - i\gamma k\cos(\phi)). \tag{10.17.7}$$

The directivity pattern is clearly described by (Fig. 10.12)

$$D(\phi) = 1 - i\gamma k\cos(\phi). \tag{10.17.8}$$

If we adjust γ so that $\gamma = \frac{1}{ik}$, then

$$D(\phi) = 1 - \cos(\phi), \tag{10.17.9}$$

which is simply the cardioid directivity pattern. The parameter γ may be adjusted to obtain any of the well-known first-order directivity patterns.

It is sometimes convenient to express the combination of a nondirectional sensor and a pressure gradient sensor in the form

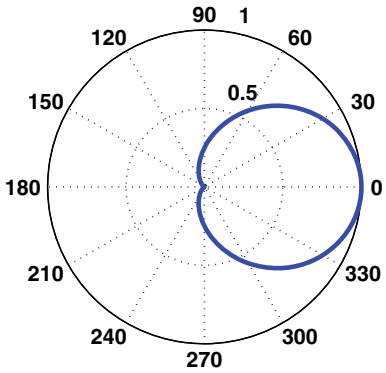


Fig. 10.12 The cardioid directivity pattern obtained when $A = 1$ and $B = -1$ in Eq. (10.17.10)

$$D(\phi) = A + B\cos(\phi), \tag{10.17.10}$$

where A and B are constants that can be adjusted to achieve a range of first-order directivity patterns. When $A = 1$ and $B = -1$, the familiar cardioid pattern is obtained as shown in Fig. 10.12.

10.17.2 Second-Order Directional Microphones

Suppose now that we have three nondirectional microphones oriented along a line and separated from each other by a distance $d/2$. The outer microphones will again be separated by the same distance d as in the two-microphone array. As above, let the x axis be parallel to the line connecting the microphones with its origin at the center microphone. The sound pressure at $y = 0$ due to a plane harmonic sound wave with amplitude P and frequency ω can be described by

$$p(x,t) = Pe^{i(\omega t - kx\cos(\phi))}, \tag{10.17.11}$$

where $k = \omega/c$ is the wave number, c is the speed of propagation of the wave, $i = \sqrt{-1}$ and ϕ is the angle between the propagation direction and the x axis. The pressures at the three microphones may be written as

$$\begin{aligned} p(d/2,t) &= Pe^{i\omega t}e^{-ik(d/2)\cos(\phi)} \\ p(0,t) &= Pe^{i\omega t} \\ p(-d/2,t) &= Pe^{i\omega t}e^{ik(d/2)\cos(\phi)}. \end{aligned} \tag{10.17.12}$$

We may now take the difference between the two outer microphones and the central microphone, and then subtract these two differences,

$$
\begin{aligned}
&(p(d/2,t) - p(0,t)) - (p(0,t) - p(-d/2,t)) = p(d/2,t) + p(-d/2,t) - 2p(0,t) \\
&= Pe^{i\omega t}(e^{-ik(d/2)\cos(\phi)} + e^{ik(d/2)\cos(\phi)} - 2) \\
&= 2Pe^{i\omega t}(\cos(-k(d/2)\cos(\phi)) - 1) \\
&\approx -Pe^{i\omega t}(kd/2)^2\cos^2(\phi),
\end{aligned} \tag{10.17.13}
$$

where, again, our approximation is valid for $kd << 1$. Note that if the incident angle is such that the difference response is maximum, i.e. $\phi = 0$, the second-order difference signal will be less than the amplitude of the plane wave by the factor $(kd/2)^2$. Unlike the first-order array, the dependence on the angle of sound incidence, ϕ is given by $\cos^2(\phi)$.

10.18 The Ribbon Microphone

An early embodiment of a directional microphone consists of a thin ribbon that is placed in a magnetic field [3, 4]. The ribbon is driven by the difference in pressure across its two planar surfaces. If the effective distance that a plane sound wave travels between the two planar surfaces of the ribbon is d, the net pressure can be expressed as in Eq. (10.17.3),

$$
\begin{aligned}
&p(d/2,t) - p(-d/2,t) \\
&= Pe^{i\omega t}(e^{-ik(d/2)\cos(\phi)} - e^{ik(d/2)\cos(\phi)}) \\
&= Pe^{i\omega t}2i\sin(-k(d/2)\cos(\phi)),
\end{aligned} \tag{10.18.1}
$$

where ϕ is the angle between the direction normal to the ribbon surface and the direction of sound propagation.

The displacement of the diaphragm, or ribbon, $x(t)$ may be calculated by accounting for the effective stiffness, K, mass, M, and viscous damping, C, in a single degree of freedom lumped parameter model,

$$
M\ddot{x} + Kx + C\dot{x} = -APe^{i\omega t}2i\sin(k(d/2)\cos(\phi)), \tag{10.18.2}
$$

where A is the area of one of the two planar faces. Assuming steady-state harmonic time dependence of the form $x(t) = Xe^{i\omega t}$ enables us to solve for the complex amplitude of the response,

$$
X = \frac{-AP2i\sin(k(d/2)\cos(\phi))}{K - \omega^2 M + i\omega C}. \tag{10.18.3}
$$

In ribbon microphones [3, 5], the effective stiffness K is typically taken to be very small so that the motion of the diaphragm is primarily controlled by its mass. Equation (10.18.3) becomes

$$X \approx \frac{-AP2i\sin(k(d/2)\cos(\phi))}{-\omega^2 M} \approx \frac{-APi\omega d\cos(\phi)/c}{-\omega^2 M}, \tag{10.18.4}$$

where our approximation in the right hand equation applies for $kd << 1$. The displacement then becomes inversely proportional to frequency ω. If the diaphragm motion is transduced to an electronic signal through the use of electrodynamic transduction, the output voltage will be approximately equal to $v = Ve^{i\omega t} = BL\dot{x} = BLi\omega Xe^{i\omega t}$ where B is the magnetic flux density and L is the length of the moving conductor. The amplitude of the output voltage will then be approximately independent of frequency,

$$V = -BL\frac{AP\omega^2 d\cos(\phi)/c}{\omega^2 M} = -BL\frac{APd\cos(\phi)/c}{M}. \tag{10.18.5}$$

It is also interesting to compare the velocity of the ribbon to the velocity of the acoustic particles in a plane wave sound field. The velocity in this field is related to the pressure as given in Eq. (2.7.5), $u_a(t) = \frac{p(t)}{\rho_0 c}$. Let $u_a(t) = U_a e^{i\omega t}$. The velocity of the ribbon relative to the acoustic particle velocity may be written using Eq. (10.18.4),

$$\frac{U}{U_a} = \frac{i\omega X}{U_a} \approx \frac{2A\omega\rho_0 c\sin(k(d/2)\cos(\phi))}{\omega^2 M}. \tag{10.18.6}$$

To examine the essential parameters that determine the mechanical sensitivity, consider normally incident sound, $\phi = 0$, and assume that $kd << 1$, with $k = \omega/c$. The mass of the ribbon will be $M = \rho Ah$, where ρ is the density of the ribbon material, and h is the thickness. Equation (10.18.6) then becomes

$$\frac{U}{U_a} \approx \frac{2A\omega\rho_0 c\omega/c(d/2))}{\omega^2 M} = \frac{A\rho_0 d}{M} = \frac{\rho_0 d}{\rho h}. \tag{10.18.7}$$

This simple approximate result indicates that one should use a ribbon material having low density, ρ, minimal thickness, h, and the system should be designed to increase the effective separation distance, d. It is important to consider, however, that the transduction method used must be considered in calculating the overall performance. When electrodynamic transduction is used, the noise performance will be influenced by the ribbon thickness and length, which is not considered in Eq. (10.18.7).

It is also important to note that increasing d can have adverse effects on the frequency response. To examine this, we must estimate the velocity ratio, $\frac{U}{U_a}$ without making the small kd assumption. As an example, suppose that the ribbon is constructed out of aluminum which has a density of $\rho \approx 2700\,\text{kg/m}^3$ with a thickness $h = 1\,\mu\text{m}$. Let the effective separation distance be $d = 2\,\text{cm}$. This gives

$$\frac{U}{U_a} \approx \frac{\rho_0 d}{\rho h} \approx 8.9, \tag{10.18.8}$$

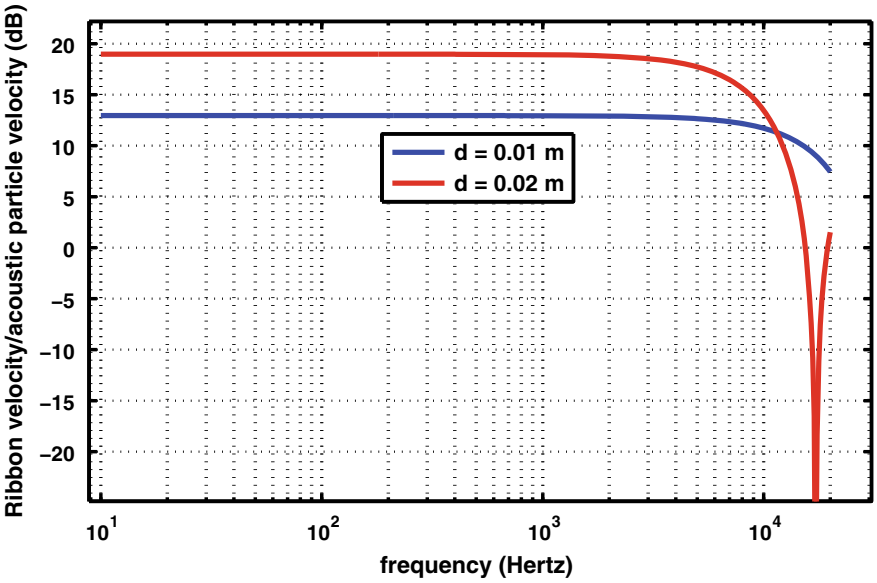


Fig. 10.13 Ribbon velocity relative to the velocity of the acoustic particle velocity as estimated by Eq. (10.18.6). The ribbon is assumed to be made of aluminum with a thickness $h = 1\,\mu\text{m}$. Increasing the effective separation distance d improves the low-frequency sensitivity but adversely affects the high frequency response

so that according to this approximation, the ribbon velocity is nearly nine times as much as the acoustic particle velocity. Figure 10.13 shows the frequency response predicted using Eq. (10.18.6) for two values of the separation distance d. Increasing d is found to adversely affect the response at high frequencies.

10.19 Effect of Gain Errors on Directivity Index

All directional sound sensing systems that rely on detection of sound pressure must use pressures sensed at multiple spatial locations. This is usually accomplished with multiple microphones where the signals from the microphones are combined electronically. The effectiveness of these systems is often strongly dependent on the accuracy of the pressure amplitude and phase measured by the microphones. In order to examine the effect of these sensitivity mismatches on the resulting directivity, Eq. (10.17.8) may be written in the form

$$D(\phi) = \beta + \cos(\phi), \tag{10.19.1}$$

where β measures the relative contribution of the pressure and pressure gradient in the combined signal.

One can calculate the directivity factor for the microphone by

$$Q = \frac{|D|^2_{\text{ref}}}{< |D|^2 >}, \tag{10.19.2}$$

where $|D|^2_{\text{ref}}$ is the amplitude of the response due to sound incident from some reference direction such as $\phi = 0$ and $< |D|^2 >$ is the diffuse field response obtained by averaging the responses over all possible angles of incidence.

Because the pair of nondirectional microphones are placed along a line on the x axis, the sensing system is clearly unchanged by a rotation about this axis. The geometry is actually nearly identical to that shown in Fig. 5.4 in our discussion of the acoustic dipole.

We may then view the spatial domain as being two dimensional (rather than three dimensional) and assume that an incident plane wave will arrive at an angle ϕ relative to the x axis and that the z axis is orthogonal to the x axis. Since the wave can arrive from any direction, the range of possible values of the x component of the wave vector will be $-\omega/c < k_x < \omega/c$. The diffuse field response is then the average response over all possible values of the wave vector component along the x direction,

$$< |D|^2 >= \frac{\int_{-\omega/c}^{\omega/c} |D(k_x)|^2 dk_x}{\int_{-\omega/c}^{\omega/c} dk_x}. \tag{10.19.3}$$

Since ϕ is the incidence angle relative to the x axis, $k_x = \frac{\omega}{c}\cos\phi$ and $dk_x = -\frac{\omega}{c}\sin\phi d\phi$. Equation (10.19.3) may then be transformed into an integration over ϕ. When changing the variable of integration from k_x to ϕ, care must be taken to note that k_x varies from $-\omega/c$ to ω/c as ϕ varies from π to zero as well as when ϕ varies from π to 2π. The integration over all possible values of ϕ then becomes

$$\begin{aligned} < |D|^2 >= & \frac{-\int_{\pi}^{0} |D(\phi)|^2 \frac{\omega}{c}\sin\phi d\phi - \int_{\pi}^{2\pi} |D(\phi)|^2 \frac{\omega}{c}\sin\phi d\phi}{-\int_{\pi}^{0} \frac{\omega}{c}\sin\phi d\phi - \int_{\pi}^{2\pi} \frac{\omega}{c}\sin\phi d\phi} \\ = & \frac{\int_0^{2\pi} |D(\phi)|^2 |\sin\phi| d\phi}{4}. \end{aligned} \tag{10.19.4}$$

This applies to any $D(\phi)$. In our special case $D(\phi) = D(-\phi)$ so the integration range can be simplified. Using Eqs. (10.19.1) and (10.19.4) and taking $\phi = 0$ to be our reference direction, the directivity factor becomes

$$Q = \frac{2(|\beta|^2 + 2\text{Re}[\beta] + 1)}{\int_0^{\pi} (|\beta|^2 + 2\text{Re}[\beta]\cos\phi + \cos^2\phi)\sin\phi d\phi}, \tag{10.19.5}$$

where Re denotes the real part. After carrying out the integration, Q becomes,

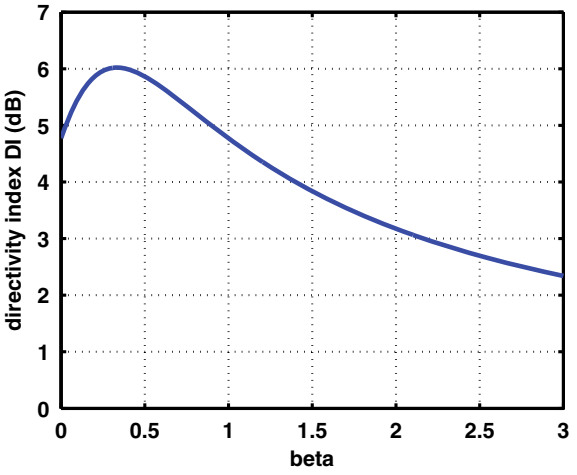


Fig. 10.14 Directivity index as a function of β in Eq. (10.19.1)

$$Q = \frac{|\beta|^2 + 2\text{Re}[\beta] + 1}{|\beta|^2 + 1/3}. \tag{10.19.6}$$

It can be shown that the maximum directivity factor of $Q = 4$ is obtained when $\beta = 1/3$. The directivity index is $DI = 10\log_{10}(Q)$.

If the pressure and pressure gradient are combined according to Eq. (10.19.1), it is important to examine the sensitivity of the directionality to changes in the relative combination of the two signals. It would, of course, be desirable for the directionality of the array to not be a strong function of β. The directivity index as a function of β is shown in Fig. 10.14 for a range of values for $0 < \beta < 3$. The figure shows that for this rather wide range of values, the directivity index varies only between about 4.8 and 6 dB. We can conclude that if one has a practical means of estimating the pressure gradient (which produces the factor of $\cos\phi$ in Eq. (10.19.1)), then the directional response of the system will not be strongly dependent on the absolute sensitivities of the microphones.

Of course, estimating the pressure gradient is the difficult part in constructing a directional sound sensor. Since the most common approach in creating a directional output is to utilize a pair of nondirectional microphones as in Eqs. (10.17.2), we should examine the sensitivity of the directivity index to errors in the gains between these two microphones. The sensitivity to gain errors can depend on the signal processing but for now, lets assume the directional output is taken to be the difference in the two signals where one is biased by a gain error δ. Let the difference signal be

$$\begin{aligned}\Delta &= \delta p(d/2, t) - p(-d/2, t) \\ &= Pe^{i\omega t}(\delta e^{-ik(d/2)\cos(\phi)} - e^{ik(d/2)\cos(\phi)})).\end{aligned} \tag{10.19.7}$$

The difference signal Δ depends on ϕ, the wave number $k = \omega/c$, and the distance between the microphones, d. To examine the sensitivity of δ and the other parameters on the directivity index, let

$$D_\Delta(\phi) = \delta e^{-ik(d/2)\cos(\phi)} - e^{ik(d/2)\cos(\phi)}. \tag{10.19.8}$$

Let

$$Q_\Delta = \frac{|D_\Delta|^2_{\text{ref}}}{< |D_\Delta|^2 >}, \tag{10.19.9}$$

where we will take the reference value, $|D_\Delta|^2_{\text{ref}}$ to be the maximum value of $|D_\Delta|^2$ as ϕ is varied. In this case is is not always obvious at which value of ϕ the maximum occurs. The average can be computed by integrating numerically,

$$< |D_\Delta|^2 >= \frac{1}{2}\int_0^\pi |D_\Delta|^2 \sin\phi d\phi. \tag{10.19.10}$$

The corresponding directivity index will be $DI_\Delta = 10\log_{10}(Q_\Delta)$.

To examine the sensitivity to gain error δ for a specific example, suppose that the two microphones are $d = 0.01$ m apart. Figure 10.15 shows the effect of δ on the directivity index of the difference of a pair of nondirectional microphones. The directivity index is shown for three frequencies. It is clear that as the frequency decreases, the sensitivity to changes in δ increases significantly.

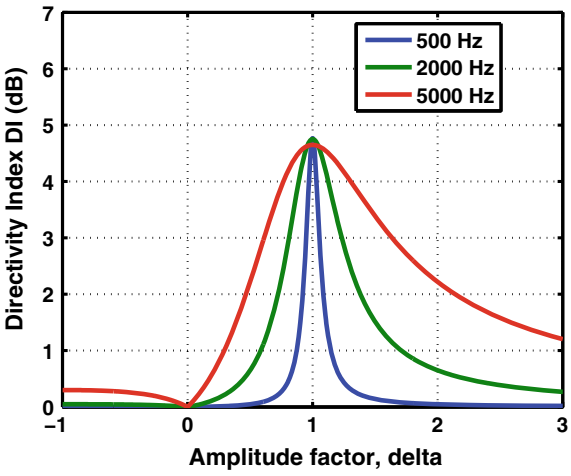


Fig. 10.15 Directivity index as a function of δ in Eq. (10.19.7)

10.20 Empirical Microphone Compensation for Desired Directivity

It is common to employ multiple microphones to achieve a desired directivity pattern. As discussed above, the directivity pattern can be significantly affected if the microphones used do not have ideal characteristics. It is helpful to be able to identify the compensation factor that should be applied to each microphone signal to achieve the desired directivity pattern. In the following, we will assume that an array of N microphones will be used to achieve a given pattern. We will assume that the response of each microphone in the array is measured as a function of the angle of incidence of a plane wave, θ for a large number of angles. The microphone responses are stored as complex transfer functions $H_l(\theta_j)$ for $l = 1, 2, \ldots, N$ and θ_j are the known angles of incidence for $j = 1, 2, \ldots, M$, and M is the number of angles. Let the output of one of the microphones in the array serve as the reference signal in the transfer functions. As a result, one of the transfer functions, say for $l = 1$, will be unity.

Let the desired directivity pattern of the array be

$$R(\theta) = A + B\cos(\theta) + C\cos^2(\theta), \tag{10.20.1}$$

where A, B, and C are constants that describe the desired pattern. The output of the array will be obtained by adding the signals from the N microphones with weighting filters a_l applied to each microphone signal. The output may then be expressed as

$$V(\theta_j) = \sum_{l=1}^{N} a_l H_l(\theta_j) \quad j = 1, \ldots, M, \tag{10.20.2}$$

where a_l are complex, frequency-dependent compensations to be applied to each microphone so that the θ dependence of the output closely approximates the desired pattern in Eq. (10.20.1). We will find the coefficients a_l by minimizing the norm of the difference between $R(\theta)$ and $V(\theta)$. Let the difference, or error in the measured directivity at each angle θ_j be

$$\epsilon_j = V(\theta_j) - R(\theta_j) = \sum_{l=1}^{N} a_l H_l(\theta_j) - A - B\cos(\theta_j) - C\cos^2(\theta_j) \quad j = 1, \ldots, M. \tag{10.20.3}$$

If we set $\epsilon_j = 0$, this becomes an over-determined system of equations for the a_l if $M > N$. The total error to be minimized in the least squares sense may be defined to be

$$E = \sum_{j=1}^{M} \epsilon_j \epsilon_j^*, \tag{10.20.4}$$

where * denotes the complex conjugate. As discussed in Appendix 2, the total error will be minimized if

$$\sum_{j=1}^{M} \frac{\partial \epsilon_j^*}{\partial a_l^*} \epsilon_j = \sum_{j=1}^{M} H_l^*(\theta_j) \sum_{l=1}^{N} (a_l H_l(\theta_j) - R_j) = 0, \quad l = 1, 2, \ldots, N, \tag{10.20.5}$$

where

$$R_j = A + B\cos(\theta_j) + C\cos^2(\theta_j) \quad j = 1, \ldots, M \tag{10.20.6}$$

Equation (10.20.5) is a linear system of equations for the unknown a_l,

$$\sum_{l=1}^{N} a_l \sum_{j=1}^{M} H_l^*(\theta_j) H_l(\theta_j) = \sum_{j=1}^{M} R_j H_l^*(\theta_j) \quad l = 1, 2, \ldots, N, \tag{10.20.7}$$

As an example, data were obtained using two miniature nondirectional microphones spaced $d = 12$ mm apart as shown in Fig. 10.16. Transfer functions, $H_l(\theta_j)$, between each miniature microphone (for $l = 1, 2$) and a Bruel and Kjaer reference microphone were stored for 300 equally spaced angles of sound incidence from zero to 360° (i.e. for $j = 1, 2, \ldots, 300$). Since only two microphones were used, it is pos-

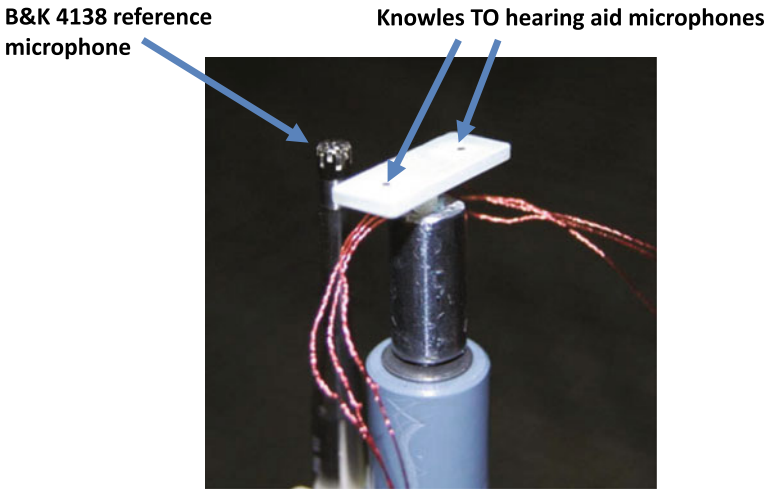


Fig. 10.16 Uncoupled Knowles omnidirectional TO38-30886-B21 microphone array. The microphones are assembled so that the distance between them is 12 mm. A Bruel and Kjaer 4138 1/8 in. microphone is also shown. This microphone was used as a calibrated reference microphone in the measurements

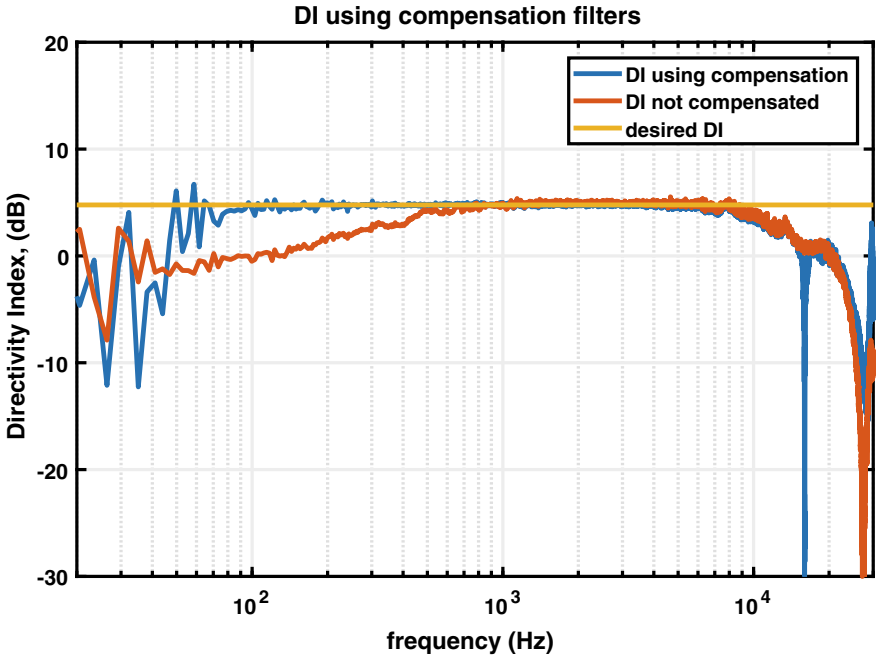


Fig. 10.17 The use of a compensation filter can provide significant improvement in the directivity index. Directivity index corresponding to $A = 0$, $B = 1$ and $C = 0$ in Eq. (10.20.1). The directivity index corresponding to the desired figure eight pattern is shown along with that obtained by using Eq. (10.20.2) in which the compensation coefficients, a_l resulted from solving Eq. (10.20.7) (shown as 'DI using compensation'). Also shown is the directivity index obtained simply by subtracting the outputs of the two nondirectional microphones (shown as 'DI not compensated')

sible to determine compensation coefficients, a_l only up to first order. Consequently, we consider directivity patterns with $C = 0$ in Eq. (10.20.1).

If we set $A = 0$, $B = 1$ and $C = 0$, the directivity pattern should correspond to figure eight pattern as discussed in Sect. 5.7. This also corresponds to $\beta = 0$ in Eq. (10.19.1), giving a directivity factor of $Q = 3$ in Eq. (10.19.6) and a directivity index of $DI = 10\log_{10}(Q) \approx 4.77\,\text{dB}$. Figure 10.17 shows this desired directivity index along with that obtained by using Eq. (10.20.2) in which the compensation coefficients, a_l resulted from solving Eq. (10.20.7) (shown as 'DI using compensation'). The figure also shows the directivity index obtained simply by subtracting the outputs of the two nondirectional microphones (shown as 'DI not compensated'). The use of the compensation coefficients improves the directivity index at low frequencies by correcting for amplitude and phase mismatches in the microphones. At high frequencies (above about 10 kHz), however, the assumption that the microphones are spaced closer than the wavelength of sound causes the compensated directivity index to depart from the desired value.

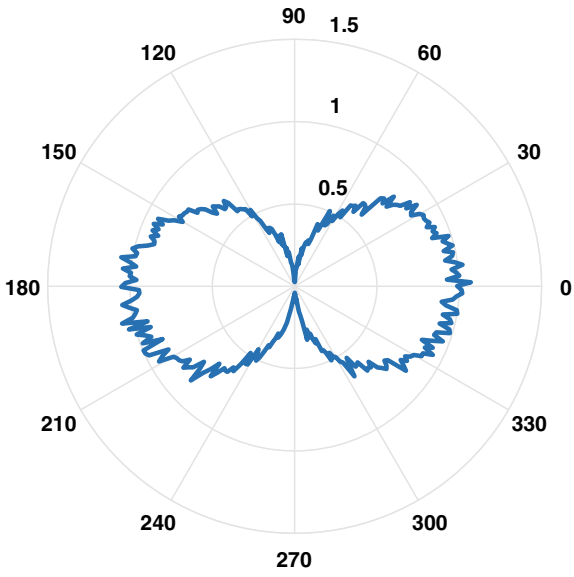


Fig. 10.18 Directivity pattern of the microphone array at 125 Hz obtained using the compensation coefficients as used in the Fig. 10.17. The pattern approximates the desired figure eight

The polar plot of the directivity of the microphone array at 125 Hz obtained using the compensation coefficients is shown in Fig. 10.18. This shows that the directivity of the output of Eq. (10.20.7) resembles a figure eight pattern. The compensation coefficients are shown as a function of frequency in (Fig. 10.19).

Note that compensation coefficients can be estimated using Eq. (10.20.7) for other combinations of A and B in Eq. (10.20.1) for this first-order array. Figure 10.20 shows a polar plot obtained at 125 Hz when the coefficients in Eq. (10.20.1) are taken to correspond to a cardioid pattern.

10.21 Noise in Directional Microphones

Because directional microphones must sense the difference in pressure at least two spatial locations, their output will diminish as the spacing between these locations is reduced. This presents a significant challenge in designing directional microphones that are small, which is the case in many applications. The effect of the separation distance, d, is shown in Eq. (10.17.3), which gives the pressure difference between two microphones in a plane sound wave with wave number $k = \omega/c$,

$$p(d/2,t) - p(-d/2,t) = \Delta p(t) = \Delta_p e^{i\omega t} \approx -Pe^{i\omega t} ikd\cos(\phi), \qquad (10.21.1)$$

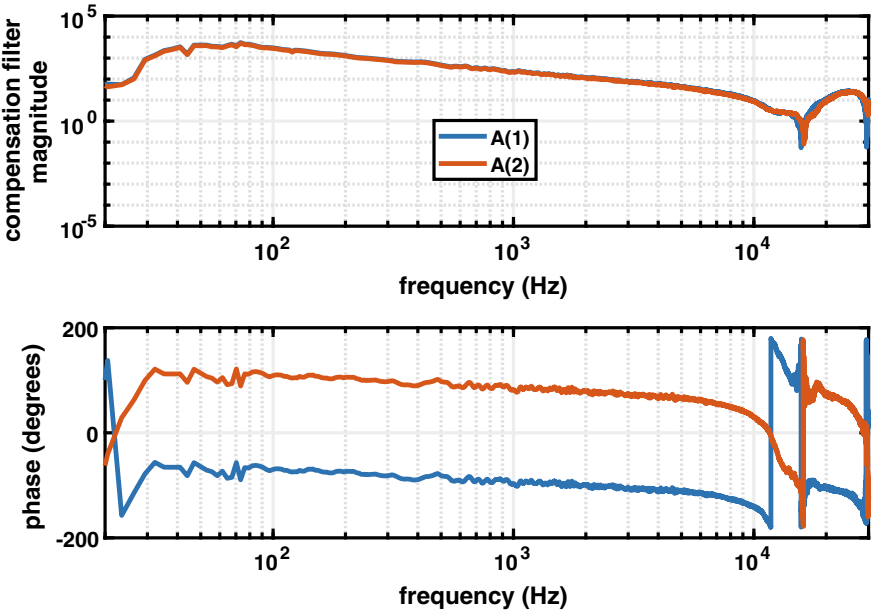


Fig. 10.19 Magnitude and phase of the compensation coefficients a_1 and a_2 used in Figs. 10.17 and 10.18

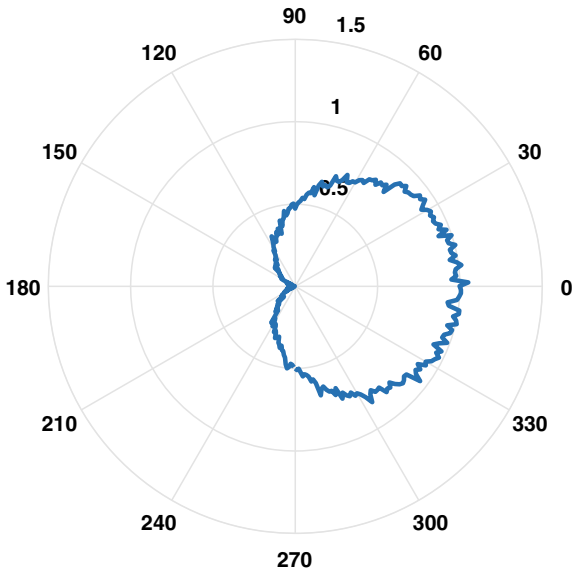


Fig. 10.20 Directivity pattern at 125 Hz using compensation coefficients a_1 and a_2 from Eq. (10.20.7) with $A = 0.5$ and $B = 0.5$ in Eq. (10.20.1). The directivity is a reasonable approximation to the desired cardioid pattern

where,

$$\Delta_p \approx -Pikd\cos(\phi), \tag{10.21.2}$$

is the complex amplitude of the difference signal. Again, our approximation is valid for plane sound waves incident at the angle ϕ relative to the line joining the two pressure measurement points and for $kd << 1$. Equation (10.21.2) shows that the sensitivity of the difference signal is less than that of the nondirectional microphone signal by a factor $kd\cos(\phi)$.

The equivalent input-referred noise of this directional system can be estimated knowing the power spectral densities of the noise floors of the two omnidirectional microphones, $S_{oo}^N(\omega)$, and by accounting for the loss of sensitivity to sound when the outputs of the microphones are subtracted to achieve a directional response. We will assume that the angle of incidence of the sound is $\phi = 0$, which is the direction where the signal is greatest. As shown in Eqs. (10.21.1) and (10.21.2) this loss of sensitivity can be accounted for by multiplying the output sensitivity of the omnidirectional microphones by $\omega d/c$ where ω is the frequency in radians/second, d is the distance between the microphones, and c is the sound speed. In addition, because the power of the uncorrelated noise of the microphones is added when the output signals are subtracted, the resultant noise power spectrum is effectively increased by a factor of two. By taking the difference of the two signals to achieve a directional output, the equivalent sound input-referred noise is then

$$S_{dd}^N(\omega) = 2\left(\frac{c}{\omega d_o}\right)^2 S_{oo}^N(\omega). \tag{10.21.3}$$

To compare the equivalent input pressure noise levels of a two omnidirectional microphone system with that of the present study, it can be more convenient to express the equivalent input noise levels in decibels. If $SPL_{oo}(\omega)$ is the equivalent input sound pressure level of the two omnidirectional microphones then the input-referred noise level of the directional system may be estimated from

$$SPL_{dd}(\omega) = 10\log_{10} S_{dd}^N(\omega) = SPL_{oo}(\omega) + 3 - 20\log_{10}\left(\frac{\omega d_o}{c}\right). \tag{10.21.4}$$

It is clear from Eqs. (10.21.3) and (10.21.4) that the loss of sensitivity of directional microphones at low frequencies results in dramatically higher sound input-referred noise.

The measured effect on sensitivity of subtracting the outputs of two miniature microphones is shown in Fig. 10.21. The distance between the microphones is 1.2 cm. It is clear that the sensitivity of the difference of the signals is significantly lower than that of the pressure itself.

The power spectral densities of the output noise signals from the individual microphones and of their difference signal were measured in addition to the sensitivities shown in Fig. 10.21. The power spectral density of the sound input-referred noise

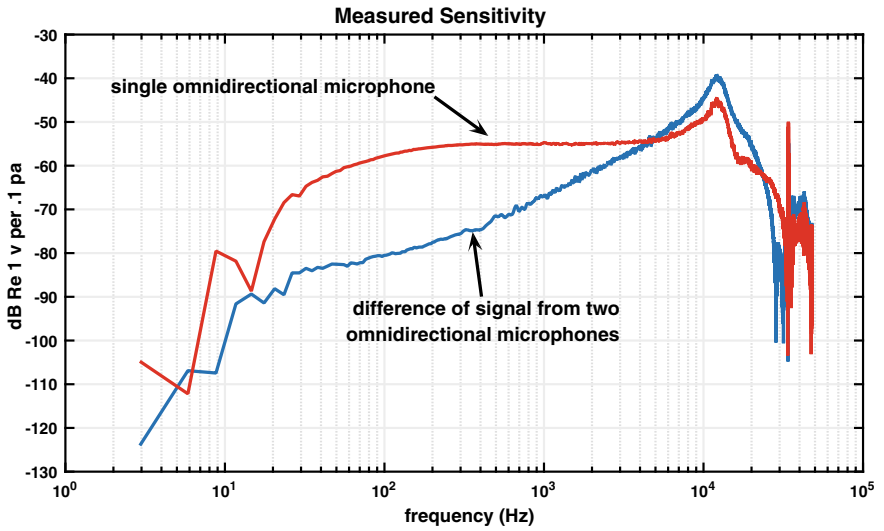


Fig. 10.21 Measured sensitivity of Knowles TO microphones configured as shown in Fig. 10.16. When measuring the difference in pressure, the sensitivity is reduced substantially over much of the frequency range of interest. The data are shown in dB relative to 1 V per 0.1 Pa

may then be determined through Eq. (11.4.1). This result may then be used to determine the equivalent sound pressure levels of the noise floor in one-third octave bands as shown in Fig. 10.22.

The noise floor of the difference signal was predicted knowing the noise floors of the individual omnidirectional microphones and using Eq. (10.21.4). Figure 10.22 shows that these predicted results are very close to the measured difference signal noise over a wide range of frequencies.

10.22 Directivity Index by Integration in the Wave Vector Domain

The discussion of the directivity of a pair of nondirectional microphones above was simplified by the fact that the system is symmetric about the line connecting the two microphones. In the following we will construct a more general calculation of the directivity index that is applicable for a more general acoustic sensor. This will comprise an extension of the calculation of the random incidence transmission coefficient, τ in Eq. (3.2.17), repeated here for convenience,

$$< \tau >= \frac{\int\int \tau(k_x, k_y) dk_x dk_y}{\int\int dk_x dk_y}. \tag{10.22.1}$$

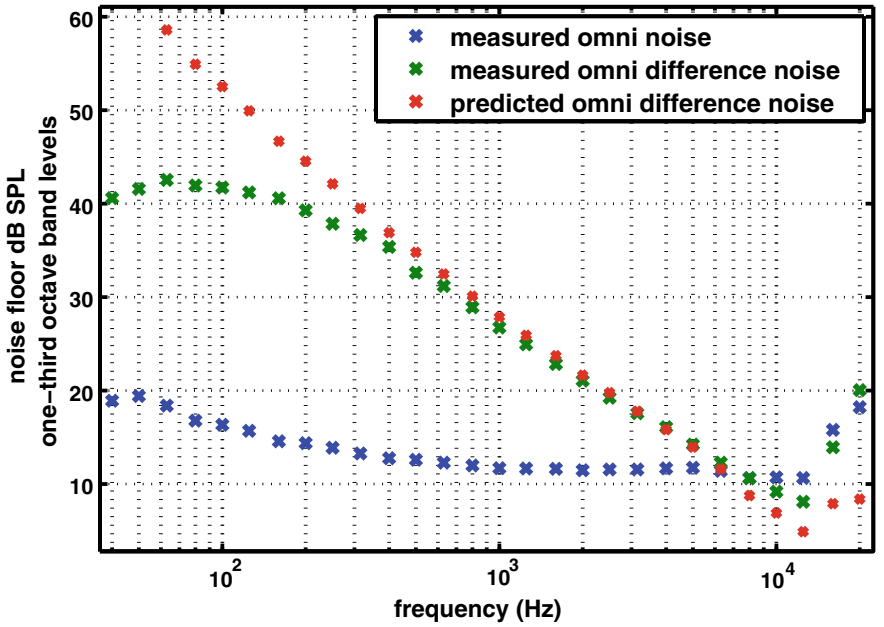


Fig. 10.22 Measured sound input-referred noise floors of the difference in output of two Knowles TO microphones configured as shown in Fig. 10.16 along with that of the two individual microphones. The difference signal has substantially higher input-referred noise as predicted from Eq. (10.21.4). The data are shown in dB sound pressure levels (SPL) in each one-third octave band

The averaged transmission coefficient for a diffuse sound field thus results from an integration over all possible acoustic wave vector components, k_x and k_y. The domain of integration in the k_x, k_y plane consists of a circle with radius equal to the acoustic wave number, $k = \omega/c$. This more general diffuse field response is essentially a two dimensional version of the calculation in Eq. (10.19.3) where the directivity, $|D|^2$, or directional sensitivity of the sensor is expressed as a function of both wave vector components, k_x and k_y,

$$< |D|^2 >= \frac{\int\int |D(k_x, k_y)|^2 dk_x dk_y}{\int\int dk_x dk_y}. \tag{10.22.2}$$

As an example, we will calculate the directivity index for the difference in output from a pair of microphones as examined above, placed along the x or i axis for sound incident at an angle θ relative to the z axis as shown in Fig. 10.9. The sensitivity of the difference output is

$$D = Pe^{i\omega t}(e^{-ik_x d/2} - e^{ik_x d/2}) = -Pe^{i\omega t} 2i \sin(k_x d/2) \approx -Pe^{i\omega t} ik_x d, \tag{10.22.3}$$

where we have assumed that $k_x d << 1$. Following the geometry of Fig. 10.9, Eq. (10.15.2) gives the x component of the wave vector as,

$$k_x = \frac{\omega}{c}\cos(\phi)\sin(\theta), \tag{10.22.4}$$

Since the domain of integration is over all acoustic wave numbers, $k = \omega/c = \sqrt{k_x^2 + k_y^2 + k_z^2}$, the differential area we are integrating over is a circle of radius k. We will then integrate using polar coordinates so that

$$dk_x dk_y = r dr d\phi = \left(\frac{\omega}{c}\sin(\theta)\right) d\left(\frac{\omega}{c}\sin(\theta)\right) d\phi = \left(\frac{\omega}{c}\right)^2 \sin(\theta)\cos(\theta) d\theta d\phi \tag{10.22.5}$$

where θ varies from 0 to $\pi/2$ and $0 \le \phi \le 2\pi$. The integral in the denominator of Eq. (10.22.2) then becomes

$$\int\int dk_x dk_y = \left(\frac{\omega}{c}\right)^2 \int_0^{2\pi}\int_0^{\pi/2} \sin(\theta)\cos(\theta) d\theta d\phi = \pi\left(\frac{\omega}{c}\right)^2, \tag{10.22.6}$$

The numerator of Eq. (10.22.2) is obtained from Eqs. (10.22.3) and (10.22.5),

$$\begin{aligned}&\int\int |D(k_x, k_y)|^2 dk_x dk_y \\ &= \int_0^{2\pi}\int_0^{\pi/2} |Pd|^2 \left(\frac{\omega}{c}\right)^4 \cos^2(\phi)\sin^3(\theta)\cos(\theta) d\theta d\phi = |Pd|^2 \left(\frac{\omega}{c}\right)^4 \frac{\pi}{4}.\end{aligned} \tag{10.22.7}$$

Equations (10.22.2), (10.22.6), and (10.22.7) then give

$$< |D|^2 >= \frac{|Pd|^2 (\frac{\omega}{c})^4 \frac{\pi}{4}}{\pi\left(\frac{\omega}{c}\right)^2} = |Pd|^2 \left(\frac{\omega}{c}\right)^2 \frac{1}{4}. \tag{10.22.8}$$

The directivity factor is again defined in Eq. (10.19.2),

$$Q = \frac{|D|^2_{\text{ref}}}{< |D|^2 >}, \tag{10.22.9}$$

where $|D|^2_{\text{ref}}$ is the sensitivity at some reference direction, usually the most sensitive direction. In the present case this will be

$$|D|^2_{\text{ref}} = |Pd|^2 \left(\frac{\omega}{c}\right)^2. \tag{10.22.10}$$

Equations (10.22.8)–(10.22.10) then give the directivity factor

$$Q = 4 \tag{10.22.11}$$

as expected for a pressure gradient microphone.

10.23 Acoustic Particle Velocity Estimation Using a Pressure Differential Microphone

Acoustical engineers seek accurate measurements of the acoustic particle velocity because it is essential in characterizing both sources of noise and acoustically absorbing materials. Knowledge of the acoustic particle velocity in the near field of a sound source, along with the pressure, permits the estimation of the sound intensity (power per unit area) emitted by the source. The current standard for measuring sound intensity, used throughout the noise control community, employs the use of two closely spaced microphones to estimate the pressure gradient, which is then processed to yield the acoustic particle velocity and the sound intensity (ANSI S1.9-1996 (R 2001) American National Standard Instruments for the Measurement of Sound Intensity). Sound intensity measuring systems based on this principle are available from a number of vendors. See for example, http://www.soundintensity.com/si_technical.htm. The following describes the technical basis for the measurement of acoustic particle velocity and then suggest methods for validating the data.

10.23.1 Measurement of Acoustic Particle Velocity

The relationship between the acoustic particle velocity and the gradient of the acoustic pressure is a consequence of Newton's second law ($F = ma$ for a particle of constant mass) as applied to an infinitesimal volume of air (or any Newtonian fluid). For the range of fluctuating pressures and acoustic particle velocities commonly experienced in sound fields, this relationship is usually stated in the following form,

$$\rho_0 \dot{\vec{u}} = -\vec{\nabla} p, \tag{10.23.1}$$

where the dot denotes the time derivative so that $\dot{\vec{u}}$ denotes the acceleration vector of the acoustic particles and $\vec{\nabla} p$ is the gradient of the sound pressure, which is also a vector. ρ_0 is the nominal mass density of the medium.

Equation (10.23.1) is the foundation of acoustic wave motion and can be found in most acoustics texts (see for example, Eq. (6.2.7) in [1]). It is important to note that this equation applies regardless of the nature of the sound source; it is equally applicable in the "near" or "far" fields of a sound source.

To simplify things, let's limit our attention to sound fields that travel in only one direction so that the vector quantities in Eq. (10.23.1) may be written as scalars. Equation (10.23.1) may then be written,

$$\rho_0 \dot{u} = -\frac{\partial p}{\partial r}, \tag{10.23.2}$$

where r is a spatial coordinate in the direction of $u(t)$.

Because of Eq. (10.23.2), the measurement of the particle velocity (or the particle acceleration, $\dot{u}$) can be accomplished by adequately estimating the pressure gradient, $\frac{\partial p}{\partial r}$. As mentioned above, the pressure gradient is normally estimated by measuring the pressures $p_1(t)$ and $p_2(t)$ at two points in space separated by a small distance r. The gradient of the pressure can then be estimated by

$$\frac{\partial p}{\partial r} \approx \frac{p_2(t) - p_1(t)}{r}. \tag{10.23.3}$$

Equations (10.23.2) and (10.23.3) then give

$$\dot{u} \approx -\frac{p_2(t) - p_1(t)}{\rho_0 r}. \tag{10.23.4}$$

One must integrate the results of the right hand side of Eq. (10.23.4) over time in order to estimate the acoustic particle velocity, u.

Widely used sound intensity measurement systems, as mentioned above, use Eq. (10.23.4) along with the fact that the pressure at a point half-way between the two microphones may be assumed to be

$$p(t) \approx \frac{p_1(t) + p_2(t)}{2}. \tag{10.23.5}$$

The sound intensity (sound power per unit area) at an instant of time t is

$$I(t) = p(t)u(t). \tag{10.23.6}$$

In general, since the acoustic particle velocity is a vector quantity, the intensity is a vector as well.

10.23.2 Pressure Gradient and Velocity Microphones

The NR-3158 microphone by Knowles Electronics contains a diaphragm that responds to the difference in pressure at its two sound inlet ports. Its electrical output is proportional to the pressure gradient taken along the line connecting these ports. As in Eq. (10.23.4), the output is thus proportional to the acceleration of the air motion,

not its velocity. Although the velocity can be determined with this microphone, it is more appropriate to call it a "pressure gradient" or a "pressure differential" microphone (http://www.emkayproducts.com/html/nr_series.htm).

Microphones that have been called "velocity" microphones, such as ribbon microphones, have highly compliant diaphragms such that a pressure gradient across them is proportional to the diaphragm's acceleration (owing to Newton's second law). The acceleration of the diaphragm is then proportional to the acceleration of the air (after Eq. (10.23.4)). Because of the electromagnetic transduction used in these microphones, the electrical voltage is proportional to the diaphragm velocity, which is also proportional to the air velocity. While this type of microphone responds to pressure gradients (just like the Knowles NR-3158) the conversion of the pressure gradient to an electrical signal is fundamentally different.

If one can adequately estimate the pressure gradient, then one can deduce the acoustic particle velocity. It must again be emphasized that this is true **anywhere** in a sound field. It applies with equal validity regardless of how close one is to a sound source.

The relationship between pressure and velocity (or impedance) is often rather complicated in the near field of a source. In the case of a harmonic source, we can visualize the sound field as having a component in which pressure and velocity are in phase (as in the far field) and another component where they are 90° out of phase. The spatial dependence of, and the relationship between, the pressure and the velocity depend strongly on the details of the source and the boundaries of the domain. However, as long as the pressure differential detected by the microphone serves as a reliable estimate of the pressure gradient, it still serves as a reliable acoustic particle velocity (or acceleration) sensor.

10.23.3 Measurement Error

There are four primary sources of error that can influence the measurement of the pressure gradient. The first is that the presence of the microphone itself may influence the pressure gradient near it. The next possible source of error is that the microphone sound inlet ports are too far apart (the distance r) in order for the sensed pressure differential to accurately estimate the pressure gradient. One must have the separation distance r be sufficiently small so that $\omega r/c << 1$, where ω is the frequency in radians/second and c is the sound propagation speed, for Eq. (10.23.3) to be valid. A third concern may be that the ports don't equally communicate the sound to the microphone diaphragm. It is possible that one of the ports introduces a greater time delay for the sound arriving at one side of the diaphragm than the other. This would have the effect of an error in the orientation of the microphone during measurements. One way to examine this is to measure the directivity pattern of the microphone, which should be a figure eight with nulls that are perpendicular to the line connecting the ports and equal (but opposite polarity) responses when the orientation is changed by 180°. The fourth concern I would raise is that the pressure gradient in the sound

field could vary strongly with position so that the orientation of the microphone and its position must be controlled very carefully. This could, of course, be examined by measuring the gradient at numerous locations and directions within the sound field.

Another method of verifying the use of the NR-3158 microphone for measuring pressure gradients is to employ a small probe microphone to accurately map the sound field in the region of interest. By accurately recording the positions of these pressure measurements, one can process these data to estimate the gradient of the pressure in any direction. These results could then be compared with those obtained using the NR-3158. Yet another way to check the results is to use the laser vibrometer to measure the velocity of the loudspeaker diaphragm used to create the low-frequency sound field. The velocity of the air must equal that of the diaphragm at points very close to the diaphragm. These data could also be compared with those obtained using the differential microphone.

10.24 Two-Microphone Intensity Measurement

The current method of measuring sound intensity relies on the use of two closely spaced microphones. The difference between the microphone signals is used to estimate the component of the gradient of the sound field in the direction parallel to the line between the microphones. This is then used to estimate the acoustic particle velocity. The average of the microphone signals is taken as the pressure. These signals can be conveniently analyzed in the frequency domain to obtain a spectrum of the sound intensity. The principles behind this measurement method are described below.

The sound intensity (sound power per unit area) at an instant of time t is

$$I(t) = p(t)u(t), \tag{10.24.1}$$

where $p(t)$ is the pressure and $u(t)$ is the velocity of the medium as it moves in the sound field. We will assume for now that the motion is in only one direction but it should be clear that since, in general, the acoustic particle velocity is a vector quantity, the intensity is a vector as well.

The time average of the intensity is $\mathrm{E}[p(t)u(t)]$ where $\mathrm{E}[\cdot]$ denotes the expected value. If we consider the sound field to be a weakly stationary random process then the intensity can be analyzed using spectral analysis. If $I(\omega)$ is the power spectral density of the intensity then it may be related to the time average intensity by integrating over all frequencies,

$$I = \mathrm{E}[p(t)u(t)] = \int_{-\infty}^{\infty} I(\omega)d\omega. \tag{10.24.2}$$

The intensity spectrum $I(\omega)$ is related to the cross spectral density of the pressure $p(t)$ and the velocity $u(t)$. To see this note that for a weakly stationary random

process the cross correlation function between pressure and velocity $R_{pu}(\tau)$ is equal to the inverse Fourier transform of the cross power spectral density $S_{pu}(\omega)$,

$$R_{pu}(\tau) = \mathrm{E}[p(t)u(t+\tau)] = \int_{-\infty}^{\infty} e^{i\omega\tau} S_{pu}(\omega) d\omega, \tag{10.24.3}$$

where $i = \sqrt{-1}$.

Evaluating Eq. (10.24.3) at $\tau = 0$ and comparing this result with equation (2) gives the relation between the intensity spectrum and the cross spectrum between pressure and velocity.

$$I(\omega) = S_{pu}(\omega). \tag{10.24.4}$$

Unfortunately, it is not a simple matter to measure the velocity of the medium as it moves due to the fluctuating sound pressure. In all existing acoustic intensity probes the velocity is estimated using the gradient of the sound field which is approximated from the pressures measured by two closely spaced microphones. The estimation of the acoustic particle velocity is made possible by the Euler equation which relates particle velocity to the gradient of the pressure,

$$\rho_0 \dot{u} = -\frac{\partial p}{\partial r}, \tag{10.24.5}$$

where ρ_0 is the density of the medium and r is a spatial coordinate in the direction of $u(t)$ (again we are assuming that the sound is propagating in one direction).

If one measures the pressures $p_1(t)$ and $p_2(t)$ with two microphones space apart by a small distance r then the gradient of the pressure can be estimated by

$$\frac{\partial p}{\partial r} \approx \frac{p_2(t) - p_1(t)}{r}. \tag{10.24.6}$$

Equations (10.24.5) and (10.24.6) then give

$$\dot{u} \approx -\frac{p_2(t) - p_1(t)}{\rho_0 r}. \tag{10.24.7}$$

The pressure at a point half-way between the two microphones may be assumed to be

$$p(t) \approx \frac{p_1(t) + p_2(t)}{2}. \tag{10.24.8}$$

We wish to compute the cross spectrum between pressure and velocity $S_{pu}(\omega)$ as in Eq. (10.24.4) in order to compute the power spectrum of the intensity. Fortunately, the cross spectrum $S_{pu}(\omega)$ can be related to the measured quantities $p_1(t)$ and $p_2(t)$. We will first compute $S_{p\dot{u}}(\omega)$ in terms of $p_1(t)$ and $p_2(t)$ since the time derivative

of u is available through Eq. (10.24.7). The resulting cross spectrum may then be shown to be related to our desired cross spectrum through

$$S_{pu}(\omega) = \frac{1}{i\omega} S_{p\dot{u}}(\omega). \tag{10.24.9}$$

The cross spectrum between the pressure and the derivative of the velocity is given by the Fourier transform of the corresponding cross correlation function

$$\begin{aligned} S_{p\dot{u}}(\omega) =& \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\tau} R_{p\dot{u}}(\tau) d\tau \\ & \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\tau} \mathrm{E}[p(t)\dot{u}(t+\tau)] d\tau. \end{aligned} \tag{10.24.10}$$

Substituting Eqs. (10.24.7) and (10.24.8) into Eq. (10.24.10) gives

$$\begin{aligned} S_{p\dot{u}}(\omega) = -\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega\tau} \mathrm{E} \Bigg[\left(\frac{p_1(t) + p_2(t)}{2} \right) \\ \times \left(\frac{p_2(t+\tau) - p_1(t+\tau)}{\rho_0 r} \right) \Bigg] d\tau. \end{aligned} \tag{10.24.11}$$

Expanding the expected value in Eq. (10.24.11) gives

$$\begin{aligned} & S_{p\dot{u}}(\omega) = \\ & \frac{-1}{4\pi\rho_0 r} \int_{-\infty}^{\infty} e^{-i\omega\tau} (R_{p_1 p_2}(\tau) - R_{p_1 p_1}(\tau) \\ & + R_{p_2 p_2}(\tau) - R_{p_2 p_1}(\tau)) d\tau \\ =& \frac{-1}{2\rho_0} (S_{p_1 p_2}(\omega) - S_{p_1 p_1}(\omega) + S_{p_2 p_2}(\omega) - S_{p_2 p_1}(\omega)), \end{aligned} \tag{10.24.12}$$

where R and S denote the corresponding correlation functions and cross spectra, respectively.

By using properties of the spectra in Eq. (10.24.12) and using Eqs. (10.24.4) and (10.24.9), the power spectrum of the acoustic intensity may be shown to be

$$I(\omega) = -\frac{\mathrm{Im}(S_{p_1 p_2}(\omega))}{\omega \rho_0 r}. \tag{10.24.13}$$

Note that if the sound field travels in only one direction so that it arrives at microphone 1 at time t and at microphone 2 at time $t + r/c$, where c is the speed of propagation of the sound, then

$$S_{p_1 p_2}(\omega) = S_{p_1 p_1}(\omega) e^{-i\omega r/c}. \tag{10.24.14}$$

Substituting Eq. (10.24.14) into Eq. (10.24.13) gives

$$I(\omega) = \frac{S_{p_1 p_1}(\omega)\sin(\omega r/c)}{\omega \rho_0 r}. \tag{10.24.15}$$

If the separation distance is sufficiently small so that $\omega r/c << 1$ then Eq. (10.24.15) reduces to

$$I(\omega) \approx \frac{S_{p_1 p_1}(\omega)}{\rho_0 c}. \tag{10.24.16}$$

In order for the sensor to operate well in a given frequency range one must have the separation distance r be sufficiently small so that $\omega r/c << 1$ for Eq. (10.24.16) to be valid. There are, however, practical limitations on the separation distance between the two microphones other than those due to their physical size. The primary limitation is that all of the spectral quantities employed above must be estimated from measured time domain data. As r becomes very small the imaginary part of $S_{p_1 p_2}(\omega)$ as needed in Eq. (10.24.13) becomes insignificant relative to the real part. This causes numerical difficulties because the cross spectrum is estimated numerically using the Fast Fourier Transform (FFT) algorithm from digitized time domain data. In effect, a very small separation distance causes $S_{p_1 p_2}(\omega) \approx S_{p_1 p_1}(\omega)$ which is entirely real. The cross spectrum is then dominated by its real part and the imaginary part is quite difficult to accurately detect.

10.25 Problems

1. A microphone has a circular diaphragm with diameter 0.6 mm. The diaphragm is made of polycrystalline silicon and has a thickness of 2 μ. The natural frequency is 10 kHz and the damping ratio is found to be $\zeta = 0.4$. We will assume that sound drives only one side of the diaphragm. Derive an equation for the displacement of the diaphragm due to sound that is incident normal to its surface.
2. For the microphone diaphragm of problem (1), plot the amplitude of the diaphragm displacement response relative to the amplitude of the pressure as a function of frequency in Hz (with units of meters/pascal). Use a log-log plot.
3. Use the results of Sect. 10.11 to determine the A-weighted thermal noise floor (in dBA) for the microphone of problem (1).
4. Suppose that the diaphragm of problem (1) is fabricated on a silicon chip that has a thickness of 500 μ which is then attached to a circuit board. This creates a back volume of air behind the diaphragm having a thickness of 500 μ. Assume that the area of the back volume is the same as that of the diaphragm so that the volume is equal to the diaphragm area multiplied by the depth of 500 μ. (a) Determine the natural frequency and damping ratio for this configuration. (b) Plot the response

and compare with that of problem (2). (c) Determine the noise floor in dBA for this configuration.

5. A microphone diaphragm has an effective mechanical stiffness $K = 9\,\text{N/m}$, the back volume is $V = 10^{-9}\,\text{m}^3$, the effective mass of the diaphragm is $M = 2.3 \times 10^{-9}\,\text{kg}$, and the diaphragm area is $S_d = 10^{-6}\,\text{m}^2$. Determine the dimensions of a vent (i.e. a slit) to achieve a low-frequency cut-off of $f_{cut} = 80\,\text{Hz}$, as predicted by Eq. (10.5.4) where $\omega_{cut} = 2\pi f_{cut}$. Assume that the dynamic viscosity of air is $\mu = 1.846 \times 10^{-5}\,\text{kg/m/s}$. Plot the magnitude and phase of the mechanical sensitivity of the diaphragm in m/Pa over a frequency range from 30 Hz to 50 kHz. Use a log-log plot.
6. Calculate and plot the directivity patterns for a microphone having a diameter of 1 in. for frequencies of 2500 Hz, 5 kHz, 10 kHz, 20 kHz, and 40 kHz. Discuss how the directivity of this 1 in. microphone compares to that of the 1/2 in. diameter microphone as shown in Fig. 10.10.
7. Two nondirectional microphones are to be used to create a directional array as examined in Eq. (10.19.7). If the microphones are separated by a distance of $d = 1\,\text{cm}$, calculate and plot the directivity index in dB function of frequency in Hz for a gain error $\delta = 1.5$ along with the directivity index that corresponds to $\delta = 1$ (i.e. no gain error). Use a log frequency axis.

References for Chapter 10

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Chapter 11
Electronic Transduction for Acoustic Sensors



Chapter 10 was concerned mainly with the mechanical response of a sensing element to sound. In the following, we focus on the problem of how to convert that mechanical motion into an electronic signal. Because the mechanical sensing element must normally be very compliant, the design of the electronic transduction can result in significant forces applied that have a marked influence on the sensor motion. As a result, great care must often be taken to ensure that the transduction system leads to the desired performance.

11.1 Electrical Sensitivity of a Microphone

In most microphones, the mechanical design cannot be performed properly without also considering the approach used to convert the diaphragm motion into an electronic signal. In the following we will assume that this transduction is accomplished through capacitive sensing, as is used in the vast majority of current designs.

To use capacitive sensing, it is normally necessary to place the diaphragm in an electric field created by some sort of electrode. The diaphragm and the adjacent electrode form a capacitor. The capacitance fluctuates with the relative distance between the diaphragm and the electrode. Since in early capacitive microphone designs this electrode was normally placed behind and parallel to the diaphragm, it is normally called a "backplate". We will assume that a bias voltage V_b is applied between the microphone diaphragm and the backplate. This bias could be the result of an applied voltage due to a battery or the charge, Q, incorporated in an electret.

The output voltage of the microphone will vary in proportion to the fluctuating acoustic pressure.

$$v_o(t) = v_o(p(t)) \approx V_0 + \left.\frac{\partial v_o}{\partial p}\right|_{p=0} \times p(t) = S_t \times p(t) \tag{11.1.1}$$

where S_t is the sensitivity of the microphone in volts/pascal.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_11

When examining the total sensitivity of the microphone, it is very helpful to consider it in terms of the product of the mechanical sensitivity of the diaphragm, S_m in meters/pascal and the sensitivity of the transduction of the diaphragm motion into an electronic signal, S_e with units of volts/meter. This follows from the fact that the electrical output of the circuit as a function of the time varying pressure can be expressed as a Taylor's series since the all of the time varying quantities are small fluctuations about their nominal values. The voltage depends on a change in the capacitance about its nominal value, C_n, the capacitance depends on the fluctuating displacement about its nominal value, x_n, and the displacement depends on the fluctuating pressure. The output voltage may then be expressed as

$$v_o(t) = v_o(C(x(p(t)))) \approx V_0 + \left.\frac{\partial v_o}{\partial C}\right|_{C=C_n} \times \left.\frac{\partial C}{\partial x}\right|_{x=x_n} \times \left.\frac{\partial x}{\partial p}\right|_{p=0} \times p(t) = V_0 + S_e \times S_m \times p(t) \tag{11.1.2}$$

where V_0 is a constant that depends on the details of the circuit. S_m is the mechanical sensitivity in meters/pascal,

$$S_m = \left.\frac{\partial x}{\partial p}\right|_{p=0} \tag{11.1.3}$$

The electrical sensitivity is

$$S_e = \left.\frac{\partial v_o}{\partial C}\right|_{C=C_n} \times \left.\frac{\partial C}{\partial x}\right|_{x=x_n} \tag{11.1.4}$$

The total sensitivity is thus the product of the electrical and mechanical sensitivities,

$$S_t = S_e \times S_m \tag{11.1.5}$$

11.1.1 Microphone Sensitivity Using a Constant Charge, High Impedance Voltage Amplifier

The electrical output is typically sensed with a high impedance circuit as shown in Fig. 11.1. If the input impedance of the circuit is large enough, we can assume no current flows and the charge on the capacitor is constant. The capacitance is defined to be the amount of charge for a given voltage. Assume, for now, that initially the charge is given by $Q = C_n V_b$, where C_n is the nominal capacitance of the microphone. Since the charge is assumed to stay constant, the voltage across the capacitor will vary as the capacitance varies. For a constant initial charge given by $Q = C_n V_b$, the fluctuating voltage across the capacitor will be

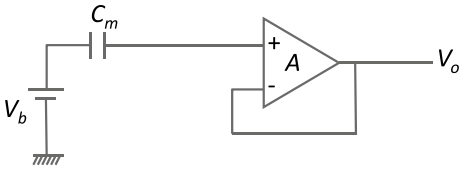


Fig. 11.1 A simple buffer amplifier to obtain an output voltage from the capacitive microphone C_m and the bias voltage supply V_b. Since the charge on the capacitor remains constant, changes in the capacitance will produce a fluctuation in the output voltage, v_o

$$V_m = \frac{q_m}{C_m} = \frac{C_n V_b}{C_m} \tag{11.1.6}$$

where V_b is the constant bias voltage and C_m is the fluctuating capacitance of the microphone of Fig. 11.1. The output voltage of the microphone will be the difference between the bias voltage and the voltage across the capacitor,

$$v_o = V_b - V_m = V_b - \frac{q_m}{C_m} = V_b \left(1 - \frac{C_n}{C_m}\right) \tag{11.1.7}$$

The change in output voltage with capacitance is then

$$\left.\frac{\partial v_o}{\partial C_m}\right|_{C_m=C_n} = V_b \left.\frac{C_n}{C_m^2}\right|_{C_m=C_n} = \frac{V_b}{C_n} \tag{11.1.8}$$

From Eqs. (11.1.4) and (11.1.8) the electrical sensitivity is

$$S_e = \frac{V_b}{C_n} \times \left.\frac{\partial C_m}{\partial x}\right|_{x=x_n} \tag{11.1.9}$$

The electrical sensitivity is then given by the product of the bias voltage, V_b and the change in capacitance with x relative to the nominal capacitance.

11.1.2 Effect of Buffer Amplifier Input Impedance—Parasitic Capacitance

It is often very important to account for the effects of parasitic capacitance due to the wiring or the input capacitance of the amplifier. This is illustrated in Fig. 11.2.

To analyze the system with parasitic capacitance, C_p, we will use Hamilton's principle, which will prove very convenient when we wish to examine the forces on the diaphragm due to the electronic interface circuit. To apply Hamilton's principle, it is necessary to carefully select coordinates that completely describe the system's

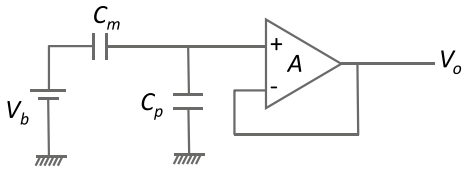


Fig. 11.2 A simple buffer amplifier including the bias supply, V_b and both the microphone capacitance, C_m and the parasitic capacitor C_p

energy state. We will use the diaphragm displacement x and the charge across the microphone diaphragm capacitor, q_m and the charge across the parasitic capacitor, q_p as generalized coordinates. The potential energy of the capacitors has the form

$$\mathcal{V}_m = \frac{1}{2} C_m V_m^2 \tag{11.1.10}$$

The voltage across the capacitor, V_m is related to the unknown charge, q_m, and the unknown displacement x since the capacitance is $C_m = q_m / V_m$. It is more convenient to express the potential energy of the capacitor in terms of the charge and the capacitance rather than in terms of the voltage,

$$\mathcal{V}_m = \frac{1}{2} \frac{q_m^2}{C_m} \tag{11.1.11}$$

The potential energy of the parasitic capacitor may be expressed similarly,

$$\mathcal{V}_p = \frac{1}{2} \frac{q_p^2}{C_p} \tag{11.1.12}$$

Including the potential energy in the spring k that represents the stiffness of the diaphragm, total potential energy in the system is then

$$\mathcal{V} = \frac{1}{2} k x^2 + \frac{1}{2} \frac{q_m^2}{C_m} + \frac{1}{2} \frac{q_p^2}{C_p} \tag{11.1.13}$$

The kinetic energy is

$$\mathcal{T} = \frac{1}{2} m \dot{x}^2 \tag{11.1.14}$$

The total virtual work done by the nonconservative elements in the system is that due to the applied sound pressure and due to viscous damping, C,

$$\delta W = -p S_d \delta x - C \dot{x} \delta x \tag{11.1.15}$$

When applying Hamilton's principle, we will need to compute the variation of the potential energy of the microphone capacitance, $\delta\mathcal{V}_m = \delta(\frac{1}{2}\frac{q_m^2}{C_m})$. Since this depends on two generalized coordinates, x and q_m,

$$\delta\mathcal{V}_m = \delta\left(\frac{1}{2}\frac{q_m^2}{C_m}\right) = \frac{q_m\delta q_m}{C_m} - \frac{q_m^2}{2C_m^2}\frac{\partial C_m}{\partial x}\delta x \tag{11.1.16}$$

Hamilton's principle then becomes

$$\int_{t_1}^{t_2}\left(-\left(m\ddot{x} + kx - \frac{q_m^2}{2C_m^2}\frac{\partial C}{\partial x} + c\dot{x} + pS_d\right)\delta x - \frac{q_m}{C_m}\delta q_m - \frac{q_p}{C_p}\delta q_p\right)dt = 0 \tag{11.1.17}$$

While we have described the energies and charges in the two capacitors with different coordinates δq_m and δq_p, we must account for the fact that the total charge in the system remains constant due to the infinite impedance of the amplifier,

$$q_0 = q_m + q_p \tag{11.1.18}$$

where q_0 is the constant initial system charge. Taking the variation of Eq. (11.1.18) gives

$$\delta q_m + \delta q_p = 0 \tag{11.1.19}$$

The terms that multiply the variation of the charge in Eq. (11.1.17) then give

$$-\frac{q_m}{C_m} + \frac{q_p}{C_p} = 0 \tag{11.1.20}$$

where we have used the fact that the variation δq_m is arbitrary. Equation (11.1.18) enables us to eliminate the charge in the parasitic capacitor, q_p, which gives

$$-\frac{q_m}{C_m} + \frac{q_0 - q_m}{C_p} = 0 \tag{11.1.21}$$

The charge across the microphone capacitor is

$$q_m = (V_b - v_o)C_m \tag{11.1.22}$$

so that Eq. (11.1.21) gives

$$v_o = V_b - \frac{q_0}{C_m + C_p} \tag{11.1.23}$$

Assuming that the initial output of the amplifier is at zero volts,

$$q_0 = V_b C_n \tag{11.1.24}$$

where, as before, C_n is the nominal capacitance of the microphone diaphragm. The output voltage of the microphone in Eq. (11.1.23) then becomes

$$v_o = V_b \left(1 - \frac{C_n}{C_m + C_p}\right) \tag{11.1.25}$$

This result should be compared to Eq. (11.1.7). As the microphone capacitance, C_m, changes due to impinging sound, the parasitic capacitance, C_p clearly will diminish the output signal. Equation (11.1.17) also gives the governing equation for the diaphragm displacement as

$$m\ddot{x} + kx - \frac{q_m^2}{2C_m^2}\frac{\partial C_m}{\partial x} + c\dot{x} + pS_d = 0 \tag{11.1.26}$$

Equations (11.1.21) and (11.1.26) may be solved numerically.

11.1.3 Microphone Sensitivity Using a Constant Voltage Bias

Another effective method of transducing the fluctuating capacitance into a voltage signal is to employ a charge amplifier as shown in Fig. 11.3. Because of the virtual short circuit between the positive and negative terminals of the operational amplifier, this circuit causes the constant battery voltage V_b to always be applied across the capacitor C_m. The total amount of charge on the two capacitors, q_m on the microphone capacitor and q_f, on the feedback capacitor remains constant, q_0, because no current can flow into the high input impedance inputs of the operational amplifier,

$$q_0 = q_m + q_f \tag{11.1.27}$$

Since the charge is equal to the product of the capacitance and the voltage across each capacitor, the virtual short circuit at the inputs of the operational amplifier allows us to write the charge on the microphone capacitor as

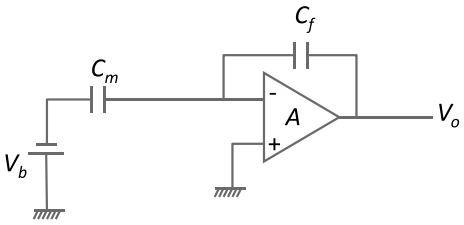


Fig. 11.3 Charge amplifier circuit. In this configuration, a constant voltage, V_b is applied across the microphone C_m. Note that in practice, one must incorporate a large value resistor in parallel with the feedback capacitor, C_f to reduce the DC gain

$$q_m = C_m V_b \tag{11.1.28}$$

The output voltage will be

$$v_o = \frac{q_f}{C_f} \tag{11.1.29}$$

Substituting Eqs. (11.1.27) and (11.1.28) into (11.1.29) gives

$$v_o = \frac{q_0 - q_m}{C_f} = \frac{q_0 - C_m V_b}{C_f} \tag{11.1.30}$$

If we take the initial constant charge, q_0 to be the charge on the microphone capacitor in its nominal, undeflected position, $V_b C_n$, where C_n is the microphone's nominal capacitance, then Eq. (11.1.30) becomes

$$v_o = \frac{q_0 - C_m V_b}{C_f} = V_b \frac{C_n - C_m}{C_f} \tag{11.1.31}$$

The output voltage is thus proportional to the difference between the nominal capacitance C_n and the fluctuating capacitance C_m.

Note that parasitic capacitance between the amplifier input and ground has no effect on the sensitivity in this case, unlike what was found for the constant charge circuit as shown in Eq. (11.1.25). Parasitic capacitance is shunted to ground by the virtual short circuit at the amplifier input.

11.2 Effect of the Electronic Circuit on the Diaphragm Mechanical Response

The electronic interface circuit will have a substantial effect on the mechanical response of the diaphragm due to the electrostatic force applied between the biased electrode and the diaphragm. This force can depend strongly on the details of the sensing circuit. In the following, we will extend the results presented above to examine the interactions between the circuit and the diaphragm response for both the constant charge circuit of Fig. 11.1 and the charge amplifier of Fig. 11.3, which maintains a constant voltage across the sensing capacitor rather than constant charge.

11.2.1 Electric Force Due to Constant Charge

For now we will assume that the microphone is placed in a package where the back-volume is completely sealed. The mechanical response of the microphone diaphragm may be examined by modeling it as a linear second order oscillator, as in Eq. (10.2.1). When a biasing electric field is applied, there will be an additional force, F_e, so that Eq. (10.2.1) becomes

$$m\ddot{x} + kx + \mathcal{C}\dot{x} = -PS_d + F_e, \tag{11.2.1}$$

where m is the diaphragm mass, k is the stiffness, $\mathcal{C}$ is the viscous damping coefficient, P is the pressure due to the applied sound field, S_d is the diaphragm area on which the pressure acts, and F_e is the force on the diaphragm due to the electric charge Q. It is assumed that a positive pressure at the diaphragm's exterior will result in a force in the negative direction. If the resonant frequency, $\omega_0 = \sqrt{k/m}$ is above the highest frequency of interest then the mechanical sensitivity will be $S_m \approx S_d/k$.

To examine the electric force, F_e, we will apply Hamilton's principle to derive the equation of motion of a simple system that contains electrostatic forces. The potential energy of the microphone capacitor is

$$\mathcal{V}_m = \frac{1}{2}C_m V_m^2 = \frac{1}{2}C_m(V_b - v_o)^2 \tag{11.2.2}$$

We have used the fact that in the circuit of Fig. 11.1, the voltage across the capacitor is again, the difference between the battery voltage, V_b and the output voltage, v_o, $V_C = V_b - v_o$. Because the total charge across the capacitor, Q, is constant in this high impedance circuit, from Eq. (11.1.6) we can also express the potential energy in the capacitor in terms of the constant bias voltage, V_b,

$$\mathcal{V}_m = \frac{1}{2}C_m V_m^2 = \frac{1}{2}V_b^2\frac{C_n^2}{C_m} \tag{11.2.3}$$

If the displacement of the diaphragm is x, the total potential energy which is the sum of that due to the spring and the capacitor may be written as

$$\mathcal{V} = \frac{1}{2}kx^2 + \frac{1}{2}V_b^2\frac{C_n^2}{C_m} \tag{11.2.4}$$

where we have assumed that the spring is unstretched when $x = 0$.

The kinetic energy of the diaphragm is

$$T = \frac{1}{2}m\dot{x}^2 \tag{11.2.5}$$

The virtual work done by the dashpot and the applied acoustic pressure, $p(t)$ is

$$\delta W_d + \delta W_p = -c\dot{x}\delta x - pS_d\delta x \tag{11.2.6}$$

where δ is the variational operator and S_d is the area of the diaphragm. We have assumed that the applied acoustic pressure acts in the negative direction since a positive pressure will push the diaphragm downward.

We will assume that the capacitance is some known function of the system's displacement, x, so that $\delta C_m = \frac{\partial C_m}{\partial x}\delta x$. We can now construct the governing equation for the mechanical system including the effects of the capacitance. Hamilton's principle states that

$$\int_{t_1}^{t_2} (\delta(T - \mathcal{V}) + \delta W)dt = 0 \tag{11.2.7}$$

Substituting Eqs. (11.2.4), and (11.2.5), into Eq. (11.2.7) and carrying out the usual integration by parts and grouping terms in applying Hamilton's principle give

$$-\int_{t_1}^{t_2} \left(m\ddot{x} + kx - \frac{1}{2}V_b^2 \frac{C_n^2}{C_m^2}\frac{\partial C_m}{\partial x} + c\dot{x} + pS_d \right) \delta x dt = 0 \tag{11.2.8}$$

The equation of motion is then

$$m\ddot{x} + kx + c\dot{x} = -pS_d + \frac{1}{2}V_b^2 \frac{C_n^2}{C_m^2}\frac{\partial C_m}{\partial x} \tag{11.2.9}$$

The force applied to the mass by the capacitor is

$$F_C = \frac{1}{2}V_b^2 \frac{C_n^2}{C_m^2}\frac{\partial C_m}{\partial x} \tag{11.2.10}$$

Suppose that the capacitor consists of a parallel electrode having a nominal distance d from the mass. If a positive displacement x causes the mass to move toward the electrode, then the capacitance is

$$C_m = \frac{\epsilon A}{d - x} \tag{11.2.11}$$

where $\epsilon \approx 8.86 \times 10^{-12}$ farad/m is the permittivity of the air in the gap and A is the surface area of the electrode, which is assumed to be the same as that of the mass. Equation (11.2.11) assumes that the gap, $d - x$ is much smaller than the width or length of the electrode. The nominal capacitance will be taken to be the capacitance when $x = 0$,

$$C_n = \frac{\epsilon A}{d} \tag{11.2.12}$$

The potential energy in Eq. (11.1.6) becomes

$$\mathcal{V}_m = \frac{1}{2}V_b^2\frac{C_n^2}{C_m} = \frac{1}{2}V_b^2\frac{(\epsilon A/d)^2}{\epsilon A/(d-x)} = \frac{1}{2}V_b^2\frac{\epsilon A}{d}\frac{d-x}{d} \tag{11.2.13}$$

In this special case of constant charge and parallel plate capacitance, the electric force on the diaphragm becomes independent of position. The governing equation becomes

$$m\ddot{x} + kx + c\dot{x} = -pA + \frac{1}{2}V_b^2\frac{\epsilon A}{d^2} \tag{11.2.14}$$

Note that the electric force on the right side of Eq. (11.2.14) is independent of the position x and is always positive, causing the diaphragm to be pushed toward the backplate as V_b is increased. The diaphragm will strike the backplate at $x = d$ when

$$V_b = \sqrt{\frac{2kd^3}{\epsilon A}}. \tag{11.2.15}$$

This result applies only when the input impedance of the amplifier is infinite, which is never realized in practice. In practice, a constant charge may be created by using an electret backplate rather than biasing the microphone with a voltage source.

11.2.2 Electric Force Due to Constant Voltage

We will now examine the force applied to the diaphragm when a charge amplifier as shown in Fig. 11.3 is used to obtain an electronic readout. In this case, the virtual short circuit at the input terminals of the operational amplifier A causes the constant voltage V_b to be applied across the sensor capacitance C_m. The amplifier effectively has zero input impedance. Since the circuit doesn't have infinite input impedance, it is necessary to account for the flow of charge out of the bias voltage supply, V_b. Since charge can flow from the voltage source, it effectively can do work and supply energy to the mechanical system. We may account for this by considering the virtual work done by the battery, $\delta W_b = V_b \delta q$, where δq is the virtual charge. Since the capacitance is given by the charge relative to the applied voltage, the virtual charge can be related to the variation of the microphone capacitance,

$$\delta q = V_b \delta C_m \tag{11.2.16}$$

where we again, take the voltage V_b to be constant.

The potential energy of the capacitor is again

$$\mathcal{V}_m = \frac{1}{2} C_m V_m^2 = \frac{1}{2} C_m V_b^2 \tag{11.2.17}$$

where we have used the fact that $V_m = V_b$ is applied across the capacitor in the charge amplifier. As above, the kinetic energy of the diaphragm is

$$T = \frac{1}{2} m \dot{x}^2 \tag{11.2.18}$$

and the total virtual work done by the dashpot, the applied acoustic pressure, $p(t)$, and the battery is

$$\delta W_d + \delta W_p + \delta W_b = -c\dot{x}\delta x - pA\delta x + V_b \delta q = -c\dot{x}\delta x - pA\delta x + V_b^2 \delta C_m \tag{11.2.19}$$

where δ is the variational operator and A is the area of the diaphragm.

Since the capacitance is a function of the system's displacement, x, $\delta C_m = \frac{\partial C_m}{\partial x}\delta x$. Hamilton's principle in Eq. (11.2.7) then gives

$$-\int_{t_1}^{t_2} \left(m\ddot{x} + kx + \frac{1}{2} V_b^2 \frac{\partial C_m}{\partial x} + c\dot{x} + pA - V_b^2 \frac{\partial C_m}{\partial x} \right) \delta x dt = 0 \tag{11.2.20}$$

The equation of motion is then

$$m\ddot{x} + kx + c\dot{x} = -pA + \frac{1}{2} V_b^2 \frac{\partial C_m}{\partial x} \tag{11.2.21}$$

The force applied to the mass by the capacitor is

$$F_C = \frac{1}{2} V_b^2 \frac{\partial C_m}{\partial x} \tag{11.2.22}$$

In the special case where a parallel plate capacitor is used, the capacitance is given in Eq. (11.2.11), $C_m = \epsilon A/(d - x)$. The force in Eq. (11.2.22) is then

$$F_C = \frac{V_b^2}{2} \frac{\mathrm{d}C_m}{\mathrm{d}x} = \frac{V_b^2 \epsilon A}{2(d-x)^2} \tag{11.2.23}$$

Note that as in the case of constant charge in Eq. (11.2.14), the force due to the electric field is always positive. The equivalent stiffness is

$$k_C = -\frac{dF_C}{dx} = \frac{-V_b^2 \epsilon A}{(d-x)^3}. \tag{11.2.24}$$

Since $x < d$, the equivalent stiffness will always be negative. The applied voltage V_b thus always acts to soften the system and will generally reduce the system's

natural frequency relative to the case when no bias voltage is applied. If the overall stiffness approaches zero, the system will become unstable. To examine the stability of the system, we need to determine the static solution of Eq. (11.2.21), $kx - F_C = 0$. Substituting our result for F_C given above yields

$$kx = \frac{V_b^2 \epsilon A}{2(d-x)^2} \tag{11.2.25}$$

Equation (11.2.25) may be solved for V_b^2,

$$V_b^2 = \frac{2kx(d-x)^2}{\epsilon A} \tag{11.2.26}$$

Substituting Eq. (11.2.26) into Eq. (11.2.24) gives

$$k_C = \frac{-2kx}{d-x}. \tag{11.2.27}$$

The total equivalent stiffness is then

$$k + k_v = k \times \left(1 - \frac{2x}{d-x}\right) \tag{11.2.28}$$

This stiffness goes to zero at $x = d/3$, which corresponds to the unstable "collapse" condition. Evaluating Eq. (11.2.26) at $x = d/3$ gives the collapse voltage, V_c,

$$V_b = V_{collapse} = \sqrt{\frac{8}{27}\frac{kd^3}{\epsilon A}}. \tag{11.2.29}$$

To determine the actual spring constant due to the electric field in a given design one must solve Eq. (11.2.25) for the equilibrium position $x = x_n$ and substitute this result into Eq. (11.2.27). Since Eq. (11.2.25) is cubic in x, the solution can't be written in a simple, convenient form. There will be three solutions to Eq. (11.2.25) and the physically realizable solution will be the one in which $0 < d - x < d$ and $k + k_v > 0$.

11.3 Optimum Capacitance for Sensitivity and Stability

The results in the previous sections enable us to examine the diaphragm motion as a function of the bias voltage, V_b, and the capacitance C_m. In nearly all sensors, the capacitance varies with the deflection x in a nonlinear way. The manner in which the capacitance varies with the deflection is a complicated function of the geometry. Nearly all current microphones utilize a parallel plate capacitor as examined above.

Future designs, however, may enable very different geometries that might permit one to tailor the dependence of the capacitance as a function of deflection. In the following, we show that if one can realize a linear dependence of capacitance with deflection, it can lead to substantial design advantages.

The dependence of capacitance with deflection, of course, has a marked influence on the electrical output. Our ultimate goal is to achieve maximal sensitivity of the electrical output as a function of the input pressure. The interface circuit is designed to detect capacitance which depends on the diaphragm displacement x. For the common high impedance buffer amplifier shown in Fig. 11.1, we have expressed the sensitivity of the microphone in Eq. (11.1.9),

$$S_e = \frac{V_b}{C_n} \times \left.\frac{\partial C_m}{\partial x}\right|_{x=x_n} \tag{11.3.1}$$

It is always desirable for the bias voltage to be high and the change in capacitance as a function of deflection to be large.

The equilibrium position is determined by solving Eq. (11.2.9) for x where the time derivatives are set to zero,

$$kx - \frac{1}{2}V_b^2 \frac{\partial C_m}{\partial x} = 0 \tag{11.3.2}$$

Small motions about the equilibrium, or nominal position, x_n, may be examined by expanding the derivative of the capacitance,

$$\frac{\partial C_m}{\partial x} \approx \left.\frac{\partial C_m}{\partial x}\right|_{x=x_n} + \left.\frac{\partial^2 C_m}{\partial x^2}\right|_{x=x_n} (x - x_n) \tag{11.3.3}$$

If we let the motion about the equilibrium position be $\delta = x - x_n$, Eq. (11.2.9) may then be approximated by

$$m\ddot{x} + kx - \frac{1}{2}V_b^2 \frac{\partial C_m}{\partial x} + c\dot{x} \approx m\ddot{\delta} + k_e\delta + c\dot{\delta} = -pA \tag{11.3.4}$$

where the equivalent stiffness, which includes the influence of the electric force, is

$$k_e = k - \frac{1}{2}V_b^2 \left.\frac{\partial^2 C_m}{\partial x^2}\right|_{x=x_n} \tag{11.3.5}$$

If the microphone is operated at frequencies well below its first resonant frequency (as is commonly the case), Eq. (11.3.4) enables us to calculate the mechanical sensitivity, or the change in deflection for a given pressure

$$S_m = \frac{\delta}{p} \approx \frac{-A}{k_e}, \quad \frac{\text{meter}}{\text{pascal}} \tag{11.3.6}$$

The maximum mechanical sensitivity is clearly obtained when the total equivalent stiffness, k_e, is as small as feasible. Of course, when it becomes zero (or negative) the system becomes unstable, which must be avoided.

From Eqs. (11.1.3), (11.1.4), (11.1.4), (11.3.5) and (11.3.6) the total sensitivity, S_t of the electrical output can be expressed as,

$$S_t = S_e \times S_m = V_b \frac{\frac{\partial C_m}{\partial x}|_{x=x_n}}{C_n} \times \frac{-A}{k_e} = \left(\frac{-A}{C_n}\right) \frac{V_b \frac{\partial C_m}{\partial x}|_{x=x_n}}{k - \frac{1}{2} V_b^2 \frac{\partial^2 C_m}{\partial x^2}|_{x=x_n}}, \quad \frac{\text{volts}}{\text{pascal}} \tag{11.3.7}$$

This result enables us to see what are the most desirable characteristics of the dependence of the capacitance on the deflection. It is very desirable to be able to use a large bias voltage, V_b to obtain a large electrical output. This can, however, lead to instability if $\frac{\partial^2 C_m}{\partial x^2}|_{x=x_n} > 0$. On the other hand, if $\frac{\partial^2 C_m}{\partial x^2}|_{x=x_n} < 0$, the system will always be stable but an increase in bias voltage V_b will decrease the sensitivity for large values of V_b. The ideal situation is then when the second derivative of the capacitance is negative and as small as possible and when the first derivative is large:

$$\left.\frac{\partial^2 C_m}{\partial x^2}\right|_{x=x_n} \approx 0 \quad \text{and} \quad \left.\frac{\partial C_m}{\partial x}\right|_{x=x_n} \rightarrow \text{large} \tag{11.3.8}$$

We thus want the capacitance to vary **linearly** with the deflection.

In the discussion above it has been assumed that the motion of the sensing element, the microphone diaphragm, is to be detected using a single biased electrode. In Chap. 12 we will examine in more detail how the use of multiple electrodes enhance the electronic output and produce a sensing scheme that is unconditionally stable. Our goal is to design a capacitive sensor that permits almost arbitrarily large bias voltages (and hence high electronic output sensitivity) without suffering the adverse effects of instability or collapse.

11.4 Sound Input-Referred Noise

In addition to sensitivity, another essential design parameter is the noise floor, or the quietest sound that can reliably be detected by the diaphragm. All microphones output a signal consisting of a combination of the signal that represents the desired sound pressure along with a signal due to noise sources created either through mechanical excitation of the microphone diaphragm or through the electronic interface. Our interest is always in how much noise there is in relation to the signal, not in the absolute level of either quantity.

The primary effect of this microphone self-noise is to place a limit on how quiet a sound pressure signal can be reliably detected before it becomes obscured by the

microphone noise. The noise floor of the microphone will be taken to be the sound pressure that produces an output signal that is of equal power to that of the noise. Because the microphone's sensitivity and noise are both frequency dependent, it is convenient to express the noise floor in the frequency domain through the use of the power spectral density. The power spectral density of the sound pressure to be detected may be expressed with the units of $\text{Pa}^2/(\text{rad/s})$. When characterizing the noise performance of electronic systems its common to express noise in terms of the square root of the power spectral density so that the units will be $\text{Pa}/\sqrt{(\text{rad/s})}$. The power spectral density may also be expressed using units of frequency in Hz rather than radians/s.

To determine the sound input-referred noise power spectral density, one must determine, by measurement or prediction, the power spectral density of the voltage noise signal produced by the microphone, $S_{vv}^N(\omega)$ with units of $\text{v}^2/(\text{rad/s})$. If one also knows the total sensitivity of the microphone, $S_t(\omega)$, with units of v/pa, one can determine the power spectral density of the input-referred noise floor, $S_{pp}^N(\omega)$ by

$$S_{pp}^N(\omega) = \frac{S_{vv}^N(\omega)}{S_e(\omega)^2} \tag{11.4.1}$$

where the units of $S_{pp}^N(\omega)$ will be $\text{Pa}^2/(\text{rad/s})$.

11.5 Effects of Electronic Noise and Stray Capacitance

The performance of a capacitive microphone can often be limited by the noise in the amplifier circuit. We have already examined the effects of thermal-mechanical noise in Sect. 10.11, which drives the sensing element before electronic transduction. Here we consider the additional noise due to the electronic circuit. Along with the addition of noise, the sensitivity can be significantly affected by stray capacitance of the signal path between the diaphragm and the amplifier. These two effects can be examined by analyzing the circuit shown in Fig. 11.4 below.

In Fig. 11.4, the parasitic capacitance, C_p must be determined from the geometry of the package. The noise source due to the input resistor is assumed to be stationary white noise with a power spectral density given by

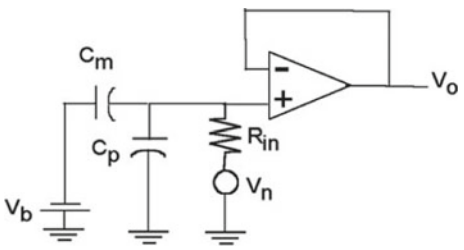


Fig. 11.4 Buffer amplifier including resistor noise, V_n and stray capacitance, C_p

$$S_{R_{in}} = 4K_bTR_{in} \tag{11.5.1}$$

where $K_b = 1.38 \times 10^{-23}$ Joules/^{oK} is Boltzmann's constant, and T is the absolute temperature. The noise voltage may be computed from

$$V_n = \sqrt{4K_bTR_{in} * Freq_{max}} * \text{randn} \tag{11.5.2}$$

where $Freq_{max}$ is the maximum frequency of interest and randn is a Gaussian random signal with zero mean and unit variance (as provided by Matlab).

In the following, it is assumed that the dominant electrical noise source is the thermal noise of the resistor. We neglect noise introduced by the bias source, V_b and the noise of the amplifier. This is probably not a safe assumption but these effects can be included using a similar approach.

The current, $\dot{q}$, in the microphone can be determined by analyzing the circuit of Fig. 11.4,

$$\dot{q} = \frac{V_b - V_n - q/C_m}{R_{in}} - \frac{C_p q\dot{x}}{\epsilon A(1 + C_p/C_m)} \tag{11.5.3}$$

and the output voltage is

$$V_0 = V_b - q/C_m. \tag{11.5.4}$$

Equation (11.5.3) may be solved numerically with Eq. (11.1.26). From Eq. (11.5.3) it is clear that as the input resistance, R_{in}, is increased the effect of the resistor noise, V_n, is reduced.

11.6 Noise Analysis of a Charge Amplifier

An alternative to the use of the high impedance buffer circuit described above is to use an inverting amplifier with capacitive feedback as shown in Fig. 11.5 below. A feedback resistor, R_f is required to prevent open loop operation at DC. The thermal noise voltage of the resistor is, as in Eq. (11.5.2),

$$V_f = \sqrt{4K_bTR_f * Freq_{max}} * \text{randn} \tag{11.6.1}$$

The charge amplifier circuit of Fig. 11.5 can be analyzed in terms of the charge in the feedback capacitor, Q_{cf} to give

$$\dot{Q}_{cf} = -\frac{V_b \epsilon A\dot{x}}{(d + x)^2} - (Q_{cf}/C_f - V_f)/R_f \tag{11.6.2}$$

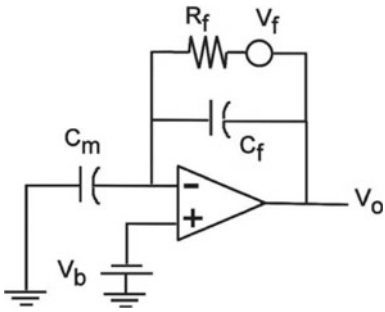


Fig. 11.5 Charge amplifier circuit including feedback resistance, R_f and noise, V_f

and the output voltage is

$$V_0 = V_b - Q_{cf}/C_f. \tag{11.6.3}$$

The charge in the microphone is

$$Q = C_m V_b \tag{11.6.4}$$

where, again, the capacitance of the microphone is $C_m = \epsilon A/(d + x)$. Equations (11.6.2) and (11.6.4) may be solved numerically with with Eq. (11.1.26). Again, Eq. (11.6.2) shows that the effect of the noise voltage, V_f is diminished by increasing R_f. For sufficiently large values of R_f, the charge in the feedback capacitor becomes roughly equal to that of the microphone so that $Q \approx Q_{cf}$ and the output voltage becomes

$$V_0 \approx V_b(1 - C_m/C_f) \tag{11.6.5}$$

Substituting the relation between C_m and the diaphragm displacement x gives

$$V_0 \approx V_b(1 - \frac{\epsilon A}{dC_f}) + V_b\frac{\epsilon A}{C_f}(x/d - (x/d)^2 + (x/d)^3). \tag{11.6.6}$$

Assuming the output voltage is ac coupled and neglecting nonlinear terms give

$$V_0 \approx V_b\frac{\epsilon A}{C_f}x/d \tag{11.6.7}$$

If the diaphragm displacement is $x \approx -PA/k$ then

$$V_0 \approx -P\frac{V_b \epsilon A}{dC_f} \tag{11.6.8}$$

so that the electrical sensitivity is

$$S_e = \frac{V_b \epsilon A}{dC_f} = \frac{V_b C_n}{C_f} \tag{11.6.9}$$

volts/Pascal, where C_n is the nominal microphone capacitance.

11.7 Microphone Equivalent Input Noise Including Electronics

The examination of the circuits in the previous sections facilitate the simulation of the coupled nonlinear differential equations that describe the mechanical diaphragm and the read-out electronic circuit used for capacitive sensing. In this section, we return to the problem of determining the equivalent input noise to include the effect of electronic noise in the circuit.

Since the electronic read-out circuit can be considered to be a linear system, its output signal, $V_o(t)$ can be considered to be made up of three statistically independent components that are added together: $V_p(t)$, the desired acoustic signal, $V_{TN}(t)$, the output voltage due to thermal-mechanical random vibrations of the microphone diaphragm, and $V_{EN}(t)$, the electronic noise created within the read-out circuit,

$$V_o(t) = V_p(t) + V_{TN}(t) + V_{EN}(t) \tag{11.7.1}$$

Again, because these three components arise from completely independent effects, they may be considered to be statistically independent and uncorrelated. The voltage level of the output noise is not a particularly useful quantity since it obviously depends strongly on the gain of the circuit, which can be set to nearly any value. What matters to any user of a microphone is the impact the noise signals have on detecting quiet sounds. We must then express these noise signals in terms of the input pressure signal being sensed as we did in our previous discussion of thermal noise.

It would be ideal if we could decompose an input pressure signal, $p_{in}(t)$ in terms of three corresponding input pressures,

$$p_{in}(t) = p_p(t) + p_{TN}(t) + p_{EN}(t) \tag{11.7.2}$$

where $p_p(t)$ is the desired acoustic pressure, $p_{TN}(t)$ is the equivalent input pressure due to the thermally-induced random vibrations of the diaphragm and $p_{EN}(t)$ is an equivalent pressure that causes the electronic output noise $V_{EN}(t)$.

Because of the extremely complicated nature of the three components on the right side of Eq. (11.7.2), they can be characterized only in a statistical sense. A convenient

way of characterizing complicated random signals such as these is the two-sided power spectral density, $S_{pp}(\omega)$, where ω is the frequency in radians/second. $S_{pp}(\omega)$ will have the units of pascal^2/(rad/s). This is referred to as "two-sided" because it may be evaluated for both positive and negative frequency with $S_{pp}(\omega) = S_{pp}(-\omega)$. Engineers often use the "single-sided" power spectral density, $G_{pp}(f)$ where f is the frequency in Hertz. $G_{pp}(f)$ will have the units of pascal^2/(Hz). These two spectral densities are related by

$$G_{pp}(f) = 4\pi S_{pp}(2\pi f) \tag{11.7.3}$$

As mentioned above in Eq. (1.4.7), the integrals over all frequencies will produce the mean square response,

$$< p^2(t) >= \int_{-\infty}^{\infty} S_{pp}(\omega)d\omega = \int_0^{\infty} G_{pp}(f)df \tag{11.7.4}$$

The power spectral density is defined to be the Fourier transform of the autocorrelation function,

$$S_{pp}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} R_{pp}(\tau)e^{-i\omega\tau}d\tau \tag{11.7.5}$$

where the autocorrelation function is

$$R_{pp}(\tau) = \mathrm{E}[p(t)p(t+\tau)] \tag{11.7.6}$$

and E[·] is the expected value. The autocorrelation function in Eq. (11.7.6) is not a function of time t if $p(t)$ is a weakly stationary random process. The autocorrelation function of $p_{in}(t)$ in Eq. (11.7.2) will be

$$\begin{aligned} &R_{ppin}(\tau) = \mathrm{E}[p_{in}(t)p_{in}(t+\tau)] = \\ &\mathrm{E}[p_p(t)p_p(t+\tau)] + \mathrm{E}[p_{TN}(t)p_{TN}(t+\tau)] \\ &+ \mathrm{E}[p_{EN}(t)p_{EN}(t+\tau)] \end{aligned} \tag{11.7.7}$$

where the cross correlation terms such as $\mathrm{E}[p_{TN}(t)p_{EN}(t+\tau)] = 0$ since the signals are uncorrelated. The power spectral density of $p_{in}(t)$ will then be equal to the sum of the power spectra of each of the uncorrelated component signals,

$$S_{ppin}(\omega) = S_{pp}(\omega) + S_{ppTN}(\omega) + S_{ppEN}(\omega) \tag{11.7.8}$$

Our current goal is to determine each of the three power spectra on the right side of Eq. (11.7.8) given the power spectra of the three output signals on the right side of Eq. (11.7.1). Let the power spectra of the three output voltage signals be $S_{vvp}(\omega)$,

$S_{vTN}(\omega)$, and $S_{vEN}(\omega)$, corresponding to $V_p(t)$, $V_{TN}(t)$, and $V_{EN}(t)$, respectively. Each of these power spectra will have the units of volts^2/(rad/s).

If we suppose that each power spectral density for the three output voltages are known (or measured), we can convert them to input power spectra if we know the transfer function of the system from an acoustic pressure input to a voltage output. This transfer function should account for any gain or filtering in the system. Let the system transfer function be $H_{pv}(\omega)$, with units of volts/pascal. To obtain the equivalent input-referred power spectral density of the pressure in pascals^2/(rad/s), we merely divide the known output spectral densities by $|H_{pv}(\omega)|^2$,

$$S_{ppTN}(\omega) = \frac{S_{vTN}(\omega)}{|H_{pv}(\omega)|^2} \quad S_{ppEN}(\omega) = \frac{S_{vEN}(\omega)}{|H_{pv}(\omega)|^2} \tag{11.7.9}$$

The power spectral density of the total equivalent input noise is simply the sum of these.

As an example, suppose that the microphone design examined in Fig. 10.6 incorporates capacitive sensing with a backplate a distance d from the diaphragm and a bias voltage V_b. As discussed above, the output voltage is related to the diaphragm displacement by

$$V_0 = -x\frac{V_b}{d} \tag{11.7.10}$$

Equation (11.7.10) along with Eq. (10.9.5) enables us to calculate the transfer function between the input pressure and the output voltage,

$$H_{pv}(\omega) = \frac{V_b}{d}\frac{X}{\mathcal{P}} = \frac{-A(-\omega^2 m_v + i\omega c_v)V_b/d}{k + K_d - \omega^2(m + m_{rad}) + i\omega(\mathcal{C} + C_{rad})*(k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv}} \tag{11.7.11}$$

We will assume that the buffer amplifier has an output noise power spectrum that is independent of frequency and equal to 20 nV/$\sqrt{(\text{Hz})}$ so that $S_{vEN}(\omega) = (20 \times 10^{-9})^2$ volts^2/Hz. Equation (11.7.9) then gives the results shown in Fig. 11.6.

11.8 Electrodynamic Microphones

The vast majority of microphones in use today rely on capacitive transduction to convert the diaphragm motion into an electronic signal. The next most common type of microphone uses the principles of electrodynamics in which a voltage is generated due to the motion of charge in a magnetic field. These are usually constructed in a similar way to the electrodynamic loudspeakers discussed above, where a diaphragm

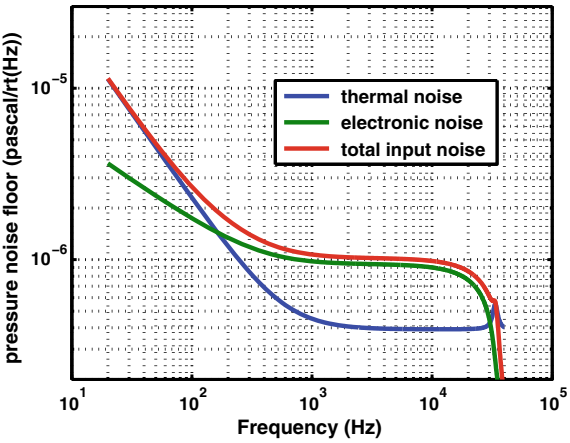


Fig. 11.6 The predicted pressure-referred noise increases significantly as the frequency is reduced and is dominated by the microphone thermal noise at low frequencies. The noise power spectra are shown in pascals $/\sqrt{\text{Hz}}$

incorporates a conducting coil which is moved due to sound while it is immersed in a magnetic field. The charges in the coil experience a force that is proportional to their velocity relative to the magnet. The force on those charges leads to a voltage that can be measured across the two free ends of the coil. The voltage is created due to the change in the potential energy of the charges that result from this force.

The basic principle of an electrodynamic microphone follows from the experimental observation that a charge q_0 (coulombs) in the conducting coil moving with a velocity $\vec{U}$ (m/s) experiences a force that is given by

$$\vec{F}_q = q_0 \vec{U} \times \vec{B} \tag{11.8.1}$$

where $\vec{B}$ is the magnetic flux density vector and $\times$ is the vector cross product. $\vec{B}$ is expressed in units of teslas. With proper magnetic materials, designs can be achieved where the magnitude of $\vec{B}$ is roughly one tesla.

It is important to note that the force $\vec{F}_q$ applied by the magnetic field on the charge is always orthogonal to the velocity $\vec{U}$. It is therefore a conservative force which does not work as the charge travels along any closed path.

As the charge travels along a path of differential length $\vec{dl}$ the force due to the magnetic field does differential work given by

$$dW = \vec{F}_q \cdot \vec{dl} = (q_0 \vec{U} \times \vec{B}) \cdot \vec{dl} = -d\mathcal{V} \tag{11.8.2}$$

where the differential change in potential energy, $d\mathcal{V}$ is the negative of the differential work done. Summing the differential potential energy of all of the charge in the

moving conductor of length L gives the total potential energy due to the movement of charges in the magnetic field,

$$\mathcal{V} = -\int_0^L (q_0 \vec{U} \times \vec{B}) \cdot \vec{dl} \tag{11.8.3}$$

Assuming all of the charge remains in the conductor, the open circuit voltage across its length is the potential energy per unit charge,

$$V = \frac{\mathcal{V}}{q_0} = -\int_0^L (\vec{U} \times \vec{B}) \cdot \vec{dl} \tag{11.8.4}$$

Equation (11.8.4) indicates that it is desirable for the three vectors $\vec{U}$, $\vec{B}$, and $\vec{dl}$ to be orthogonal to each other. The conductor typically consists of a coil having uniform velocity $\vec{U}$ along its length in a direction perpendicular to the coil radius. The coil surrounds one pole of the magnet with the other magnet pole outside the coil. This gives a flux density vector $\vec{B}$ that is parallel to the coil radius and orthogonal to both the differential length $\vec{dl}$ and the velocity $\vec{U}$. In this case the voltage across the coil is

$$V = -UBL \tag{11.8.5}$$

Where the minus sign depends on which end of the coil is taken to be ground and/or which pole of the magnet is chosen to be north or south.

Knowing the relationship between the diaphragm velocity and the output voltage, we may now adapt our model for the mechanical motion of a diaphragm in Eq. (10.9.5) to calculate the output voltage for a given pressure amplitude $\mathcal{P}$. For sound due to harmonic pressure at the frequency ω, the velocity amplitude U is related to the diaphragm displacement amplitude X by $U = i\omega X$. Equation (10.9.5) may then be used to compute the electrical sensitivity of output voltage as The response given by Eq. (10.5.7) then becomes

$$S_e = V/\mathcal{P} = \frac{-i\omega BLA(-\omega^2 m_v + i\omega c_v)}{\begin{array}{l} k + K_d - \omega^2(m + m_{rad}) \\ \quad + i\omega(\mathcal{C} + C_{rad}) * (k_{vv} - \omega^2 m_v + i\omega c_v) - k_{vd} * k_{dv} \end{array}}. \tag{11.8.6}$$

Chapter 12
Estimation of Capacitance



The vast majority of microphones that are produced each year utilize capacitive sensing either through the use of a charged electret material or an applied bias voltage. The design of these microphones, in essentially all cases, relies on simple formulas to estimate the capacitance of parallel plate electrodes. This greatly limits the designs to follow the ubiquitous parallel plate configuration, with all of its common design challenges. In an attempt to overcome this limitation, in the following we review the basic principles of sensing charge and examine a capacitive sensing geometry that overcomes several of the common limitations encountered in capacitive microphone design. Our approach follows that described in [1], which presents an electrostatic sensing scheme that results in a minimum of electrostatic force and stiffness on the moving electrode.

12.1 Coulomb's Law and the Electric Potential

The first concept to examine in electrostatics is the relationship between the electric potential or imposed voltage on the surfaces of conductors and the distribution of electric charge. In the following, we review Coulomb's law which is the source of this relationship. Coulomb's law follows from the empirical observation that the force applied to two point charges q_1 and q_2 is given by

$$f_{12} = \frac{q_1 q_2}{4\pi\epsilon r^2}, \tag{12.1.1}$$

where ϵ is an empirical constant, the permittivity of free space, and r is the charge separation distance. This force acts along the line separating the charges. The sign of the force is such that for q_1 and q_2 having the same sign, the force acts in the direction of increasing r, pushing them apart. When q_1 and q_2 opposite signs, the force acts to pull them together.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_12

Because the force in Eq. (12.1.1) is a function of position only and contains no time derivatives, it can be integrated over any path in space to determine the work done by it. Consider the work done by this force on the charge q_1 as q_1 moves from some reference distance r_0 to the distance r_1 from the charge q_2. Since the force always acts along the line joining the two charges and acts in the direction of increasing r, the work done by the force acting on q_1 is then

$$W_{12} = \int_{r_0}^{r_1} \frac{q_1 q_2}{4\pi\epsilon r^2} dr. \tag{12.1.2}$$

Carrying out the integration gives

$$W_{12} = \frac{q_1 q_2}{4\pi\epsilon}\left(\frac{1}{r_0} - \frac{1}{r_1}\right). \tag{12.1.3}$$

The potential energy, V_{12} (or the amount of work the force could do) is the negative of the work,

$$V_{12} = -W_{12} = \frac{q_1 q_2}{4\pi\epsilon}\left(\frac{1}{r_1} - \frac{1}{r_0}\right). \tag{12.1.4}$$

It is helpful to set the initial distance to infinity, $r_0 = \infty$, so that

$$V_{12} = \frac{q_1 q_2}{4\pi\epsilon r_1}. \tag{12.1.5}$$

In obtaining Eq. (12.1.5) we determined the work done by the force on the charge q_1. It is common to express this potential energy as the product of the charge q_1 and the electric voltage potential $\phi_1(r_1)$, which can be thought of as the source of the electric field experienced by the charge q_1,

$$V_{12} = \frac{q_1 q_2}{4\pi\epsilon r_1} = q_1\phi_1(r_1), \tag{12.1.6}$$

where

$$\phi_1(r_1) = \frac{q_2}{4\pi\epsilon r_1}. \tag{12.1.7}$$

The relationship between charge and electric potential is very important to examine when designing electrostatic sensors. In our simple system composed of two point charges, we can also examine this from the perspective of charge q_2 rather than considering the work done by the electrostatic force on charge q_1 as above. Consider now the work done by the coulomb force on the charge q_2 as q_2 moves from some reference distance r_0 to the distance r_1 from the charge q_1. As before, this force always acts along the line joining the two charges and acts in the direction of increasing r,

The work done by the force acting on q_2 is then

$$W_{21} = \int_{r_0}^{r_1} \frac{q_1 q_2}{4\pi\epsilon r^2} dr. \tag{12.1.8}$$

This work is, of course, identical to the work done by the force on q_1 as in Eq. (12.1.3), so $W_{21} = W_{12}$ and the potential energy of the system is also unchanged, $V_{21} = V_{12}$. The potential energy of the system due to the work by the electrostatic force on the charge q_2 may be expressed as the product of the charge q_2 multiplied by an electric potential $\phi_2(r_1)$,

$$V_{21} = \frac{q_1 q_2}{4\pi\epsilon r_1} = q_2 \phi_2(r_1), \tag{12.1.9}$$

where

$$\phi_2(r_1) = \frac{q_1}{4\pi\epsilon r_1}. \tag{12.1.10}$$

To facilitate the analysis of more complicated electrode configurations, it is helpful to express the potential energy of the system in the following form,

$$V_{12} = V_{21} = \frac{1}{2} \begin{pmatrix} \phi_1(r_1) & \phi_2(r_1) \end{pmatrix} \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \frac{q_1 q_2}{4\pi\epsilon r_1}. \tag{12.1.11}$$

The charges q_1 and q_2 may be related to the electric potential vector by

$$\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{pmatrix} \phi_1(r_1) \\ \phi_2(r_1) \end{pmatrix} \tag{12.1.12}$$

Equation (12.1.11) then becomes

$$\begin{aligned} V_{12} &= V_{21} \\ &= \frac{1}{2} \begin{pmatrix} \phi_1(r_1) & \phi_2(r_1) \end{pmatrix} \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} \begin{pmatrix} \phi_1(r_1) \\ \phi_2(r_1) \end{pmatrix} \\ &= \frac{1}{2} \Phi^t [C] \Phi = \frac{q_1 q_2}{4\pi\epsilon r_1} \end{aligned} \tag{12.1.13}$$

where Φ is a vector of electric potentials and $[C]$ is a capacitance matrix. In the case of two point charges, this capacitance matrix is given by

$$[C] = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} 0 & 4\pi\epsilon r_1 \\ 4\pi\epsilon r_1 & 0 \end{bmatrix}. \tag{12.1.14}$$

Our examination of two point charges provides a framework for studying complicated electrode configurations as discussed below.

12.2 Integral Equation Relating Charge and Potential

Suppose we have a system of N conducting electrodes which are each maintained at uniform electric potentials. When these potentials are maintained at the surface of a conductor, we will denote them as voltages $v_i, i = 1, \ldots, N$ relative to a given ground potential. These electrodes will create an electric field in the medium which we will assume has a uniform permittivity, ϵ. This will result in a geometry-dependent distribution of charge with surface density ρ coulomb/m^2. Let the position of any point on the surface of the electrodes be $\vec{r}$. The given surface voltage will then be $v(\vec{r})$ and the unknown surface charge density will be $\rho(\vec{r})$. The electric potential at a point may be expressed as the summation of the potential due to differential charges, $\rho(\vec{r})ds$ everywhere in the domain. In a three-dimensional domain these quantities are then related by the following integral equation,

$$v(\vec{r}) = \int \frac{\rho(\vec{r}')}{\epsilon 4\pi R(\vec{r}, \vec{r}')} ds', \tag{12.2.1}$$

where

$$R(\vec{r}, \vec{r}') = \sqrt{(\vec{r} - \vec{r}') \cdot (\vec{r} - \vec{r}')} \tag{12.2.2}$$

is the distance between $\vec{r}$ and $\vec{r}'$. When the domain is two-dimensional, Eq. (12.2.1) becomes

$$v(\vec{r}) = -\int \frac{\rho(\vec{r}') \log R(\vec{r}, \vec{r}')}{\epsilon 2\pi} ds'. \tag{12.2.3}$$

Since capacitance comprises the relationship between charge and voltage, one must solve Eq. (12.2.1) for the unknown surface charge density, $\rho(\vec{r})$, in order to calculate the total charge on each surface.

To evaluate Eq. (12.2.1) numerically, we will suppose that the entire surface area is divided into finite areas S_i which each have a midpoint $\vec{r}_i$. The integral may then be estimated by a summation. We may also express the average of the applied voltage over the area S_i as v_i. This average may then be written as

$$v_i = \frac{1}{S_i} \int_{S_i} \int_{S_j} \frac{\rho(\vec{r}')}{\epsilon 4\pi R(\vec{r}, \vec{r}')} ds' ds \approx \sum_{j=1}^{N} \frac{\rho_j}{\epsilon 4\pi R_{ij}} S_j, \tag{12.2.4}$$

and in the case of two-dimensional domains,

$$v_i = -\frac{1}{S_i}\int_{S_i}\int_{S_j}\frac{\rho(\vec{r}')\log R(\vec{r},\vec{r}')}{\epsilon 2\pi}ds'ds \approx -\sum_{j=1}^{N}\frac{\rho_j \log R_{ij}}{\epsilon 2\pi}S_j, \tag{12.2.5}$$

where $i \neq j$. R_{ij} is the distance between the midpoints of S_i and S_j. Since $R_{ii} = 0$, one must use care when evaluating the integrals when $j = i$ because the integrands contain singularities. Because two-dimensional domains are fraught with far fewer numerical difficulties than are three-dimensional domains, we will limit our current attention to two dimensions.

When the domain of integration consists of a single line segment, the integral in Eq. (12.2.5) becomes

$$v_i = -\frac{1}{S_i}\int_{S_i}\int_{S_i}\frac{\rho(\vec{r}')\log R(\vec{r},\vec{r}')}{\epsilon 2\pi}ds'ds. \tag{12.2.6}$$

Because the domain of integration is a single line, we can take the variables of integration to be x and x' and if y is in the direction normal to x, $ds = ydx$, and $ds' = ydx'$. Let the length of the line segment be L so that $S_i = Ly$. The integration along the length of the line can then be taken over $0 \leq x \leq L$ so that Eq. (12.2.6) becomes

$$v_i = -\frac{1}{Ly}\int_0^L\int_0^L\frac{\rho_i \log|x-x'|}{\epsilon 2\pi}y^2 dxdx'. \tag{12.2.7}$$

The integrals may be evaluated to obtain

$$v_i = -\frac{yL_i\rho_i}{\epsilon 2\pi}(\log(L_i) - 3/2), \tag{12.2.8}$$

where $L = L_i$ is taken to be the length of segment i.

Equations (12.2.4) and (12.2.5) lead to the following system of equations relating the unknown charge densities ρ_j to the voltages v_i,

$$v_i \approx \sum_{j=1}^{N} G_{ij}S_j\rho_j, \quad i = 1, 2, \ldots, N, \tag{12.2.9}$$

where for three-dimensional domains

$$G_{ij} \approx \frac{1}{\epsilon 4\pi R_{ij}}, \tag{12.2.10}$$

and in two-dimensional domains

$$G_{ij} \approx -\frac{\log R_{ij}}{\epsilon 2\pi}, \tag{12.2.11}$$

and, again, we require $i \neq j$. From Eq. (12.2.8), the case $i = j$ gives

$$G_{ii} = -\frac{1}{\epsilon 2\pi}(\log(L_i) - 3/2). \tag{12.2.12}$$

Having stored the elements G_{ij} of the matrix $[G]$ and the given elements of the applied voltages v_i in a vector $\underline{v}$, the vector of surface charges, $\underline{Q}$, where each element of $\underline{Q}$ is $Q_j = \rho_j S_j$, is easily computed from

$$\underline{Q} = [G]^{-1}\underline{v} = [C]\underline{v}. \tag{12.2.13}$$

The matrix $[G]^{-1} = [C]$ is the $N \times N$ capacitance matrix relating the charge on each surface element to the distribution of applied voltages, $\underline{v}$.

12.3 Numerical Evaluation for Three-Dimensional Domains

As discussed in Sect. 6.8.4, a number of numerical details must be addressed for the evaluation of the surface integrals for three-dimensional domains. When analyzing three-dimensional domains, it is common to discretize the surfaces into triangular areas. Since computer-aided design (CAD) software is widely available for describing complex geometries, we will utilize the analysis of data provided by typical CAD programs as worked out in Sect. 6.8.4 to obtain the required parameters. We will assume that the geometry has been accounted for by exporting an STL file using the desired CAD program.

An STL file contains a description of triangles used to define the surfaces in a solid model. The two-dimensional array V contains the XYZ coordinates of each vertex, or nodes, used to define the triangles. The XYZ coordinates of vertex i will be (V_{i1}, V_{i2}, V_{i3}) so that the row number indicates the number of the vertex and the number of the column (1, 2, or 3) indicates the X, Y, or Z coordinate.

The F array (face) indicates which vertices define the corners of each face. The row number indicates the face number and the three columns indicate the vertex numbers (which correspond to the rows of the V array. For face number i, the coordinates of the three vertices will then be $(V_{F(i,1)1}, V_{F(i,1)2}, V_{F(i,1)3})$, for vertex 1, $(V_{F(i,2)1}, V_{F(i,2)2}, V_{F(i,2)3})$, for vertex 2, $(V_{F(i,3)1}, V_{F(i,3)2}, V_{F(i,3)3})$, for vertex 3.

The midpoint of face number i, will define the vector $\vec{r}_i$, as in Eq. (12.2.2). The components of $\vec{r}_i$ will be given by Eq. (6.8.18). We also need to determine the area S_i of each triangular area. To calculate the area, we will define two vectors, one that begins at the first vertex and terminates at the second $\bar{L}_{12i}$ and one that begins at the first vertex and ends at the third, $\bar{L}_{13i}$. These two vectors are given by Eq. (6.8.19). The area of the triangle is given in Eq. (6.8.20).

It is sometimes necessary to determine the orientation of the triangular area, which is described by the unit normal vector. The unit normal vector is in the direction of the cross product of the vectors joining the vertices. The normalized cross product, which has unit length is given in Eq. (6.8.21). The unit normal of each surface element is normally included in an STL file so it is convenient to simply read it rather than calculate it.

As mentioned in Sect. 6.8.4, the integrand in Eq. (12.2.4) becomes singular when $i = j$. The integral in this case is evaluated in Eq. (6.8.23).

12.4 Potential Energy for a Distribution of Charge Densities

We may now write down the differential contribution to the total system potential energy due to the work by the electrostatic force on the differential charge $\rho(\vec{r}')ds'$ as this charge is moved from a point at infinity to $\vec{r}'$,

$$dV_e = \frac{dQ(\vec{r})dQ(\vec{r}')}{\epsilon 4\pi R(\vec{r},\vec{r}')} = \frac{\rho(\vec{r})ds\rho(\vec{r}')ds'}{\epsilon 4\pi R(\vec{r},\vec{r}')}. \tag{12.4.1}$$

This will also be equal to the differential contribution to the total system potential energy due to the work by the electrostatic force on the differential charge $\rho(\vec{r})ds$ as it is moved from a point at infinity to the point $\vec{r}$. To compute the system's total potential energy, when we sum the contributions over all surfaces ds and ds', we must divide by 2 in order to avoid counting energy twice. The total electric potential energy will then be

$$V_e = \frac{1}{2}\int\int \frac{dQ(\vec{r})dQ(\vec{r}')}{\epsilon 4\pi R(\vec{r},\vec{r}')} = \frac{1}{2}\int\int \frac{\rho(\vec{r})ds\rho(\vec{r}')ds'}{\epsilon 4\pi R(\vec{r},\vec{r}')}. \tag{12.4.2}$$

The integral over ds' may be replaced using Eq. (12.2.1),

$$V_e = \frac{1}{2}\int \rho(\vec{r})v(\vec{r})ds. \tag{12.4.3}$$

In most problems, we will divide the surface into N finite areas S_i as in Eqs. (12.2.4) and (12.2.5) so that the integral over the entire area in Eq. (12.4.3) may be written as

$$V_e \approx \frac{1}{2}\sum_{i=1}^{N}\rho_i S_i v_i = \frac{1}{2}\sum_{i=1}^{N} Q_i v_i = \frac{1}{2}\underline{Q}^t\underline{v} = \frac{1}{2}\underline{v}^t\underline{Q}. \tag{12.4.4}$$

From Eq. (12.2.13), the surface charges may be eliminated from this equation to give the total electric potential energy in terms of the capacitance matrix, $[C]$

$$V_e \approx \frac{1}{2}\sum_{i=1}^{N} Q_i v_i = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N} C_{ij} v_j v_i = \frac{1}{2}\underline{v}^t [C]\underline{v}, \tag{12.4.5}$$

where, again, $\underline{v}$ is a vector of voltages applied to each surface element.

12.5 Equation of Motion for a System Having a Single Degree of Freedom

One reason for developing expressions for the electrostatic potential energy is that it can be used to develop the governing differential equations of a dynamic system that incorporates electrostatic forces. To keep things simple, we will assume that the dynamic system has only a single generalized coordinate, α. This quantity could represent a simple translational displacement in cartesian coordinates, a rotation, or any convenient coordinate that describes the system motion. To write the governing differential equation of motion, we will rely on Hamilton's principle which states

$$\int_{t_1}^{t_2} (\delta T - \delta \mathcal{V} + \delta W) dt = 0, \tag{12.5.1}$$

where T is the system kinetic energy, $\mathcal{V}$ is the total system potential energy, δW is the virtual work done by all nonconservative forces, δ is the variational operator, and t_1 and t_2 are arbitrary instants of time. The governing equation of motion results from expressing each of the quantities in the integrand of Eq. (12.5.1) in terms of the generalized coordinate α and carrying out the integrations by parts necessary to rearrange Eq. (12.5.1) into the form

$$\int_{t_1}^{t_2} \left(-\frac{d}{dt}\frac{\partial}{\partial \dot{\alpha}}(T - \mathcal{V}) + \frac{\partial}{\partial \alpha}(T - \mathcal{V}) + F_{nc} \right) \delta\alpha dt = 0, \tag{12.5.2}$$

where we have expressed the virtual work as $\delta W = F_{nc}\delta\alpha$ with F_{nc} being the nonconservative force expressed in terms of the generalized coordinate α.

The electrode arrangement that often interests us is where voltages are applied to the various electrode surfaces and we seek to estimate the resulting charge distribution and the forces applied to the electrodes. When voltages are applied by voltage sources, these sources are always active elements, in that they provide electrostatic energy to the system. As a result, while a system having fixed charge may be regarded as conservative, one in which the electrodes have prescribed voltages is not conservative. In this case, we must account for the virtual work, δW done by nonconservative forces.

In any application of Hamilton's variational principle, it is essential that we recognize which variables may be varied in accord with the calculus of variations and which variables have no variation. Because the distribution of voltages is prescribed, we

must have $\delta \underline{v} = \underline{0}$. The charge distribution is not prescribed and hence its variation, $\delta \underline{Q}$ must be arbitrary. The vector of charges on our distributed collection of surfaces may be viewed as a spatially dependent intermediate coordinate which depends on the system's generalized coordinate α. The variation of the distributed charge may then be expressed as

$$\delta \underline{Q} = \frac{\partial \underline{Q}}{\partial \alpha} \delta \alpha. \tag{12.5.3}$$

The virtual work done by the voltage sources may then be expressed in terms of our generalize coordinate, α as

$$\delta W = \underline{v}^t \frac{\partial \underline{Q}}{\partial \alpha} \delta \alpha, \tag{12.5.4}$$

so that we may recognize the force due to the voltage sources as

$$F_v = \underline{v}^t \frac{\partial \underline{Q}}{\partial \alpha}. \tag{12.5.5}$$

This force is a scalar because we have considered the system to have only one generalized coordinate.

To apply Eq. (12.5.2) it is helpful to separate contributions to the potential energy that are due to mechanical forces like springs, $\mathcal{V}_m$ and those due to electrostatic forces, $\mathcal{V}_e$ so that $\mathcal{V} = \mathcal{V}_m + \mathcal{V}_e$. The kinetic energy T will depend only on mechanical inertia because electrostatic effects are not dependent on time. We may also consider the virtual work done by nonconservative forces to be decomposed into those associated due to mechanical forces, F_m and those due to the applied voltage, F_v, $F_{nc} = F_m + F_v$. Hamilton's principle then becomes

$$-\frac{d}{dt}\frac{\partial}{\partial \dot{\alpha}} T + \frac{\partial}{\partial \alpha} T + \frac{d}{dt}\frac{\partial}{\partial \dot{\alpha}} \mathcal{V}_m - \frac{\partial}{\partial \alpha} \mathcal{V}_m + F_m - \frac{\partial}{\partial \alpha} \mathcal{V}_e + F_v = 0. \tag{12.5.6}$$

The last two terms are our current concern since the remaining terms are identical to those encountered in typical problems in analytical dynamics. Using Eqs. (12.4.4) and (12.5.5) gives

$$-\frac{\partial}{\partial \alpha} \mathcal{V}_e + F_v = -\frac{1}{2}\frac{\partial}{\partial \alpha} \underline{v}^t \underline{Q} + \underline{v}^t \frac{\partial \underline{Q}}{\partial \alpha} = \frac{1}{2} \underline{v}^t \frac{\partial \underline{Q}}{\partial \alpha} = \frac{\partial}{\partial \alpha} \mathcal{V}_e. \tag{12.5.7}$$

Equation (12.5.6) may then be rearranged in the form

$$\frac{d}{dt}\frac{\partial}{\partial \dot{\alpha}} T - \frac{\partial}{\partial \alpha} T - \frac{d}{dt}\frac{\partial}{\partial \dot{\alpha}} \mathcal{V}_m + \frac{\partial}{\partial \alpha} \mathcal{V}_m = F_m + \frac{\partial}{\partial \alpha} \mathcal{V}_e. \tag{12.5.8}$$

The last term gives the force on the system due to the electrostatic field.

If we consider the conservative mechanical forces to be accounted for by the term $\frac{\partial}{\partial\alpha}\mathcal{V}_m$ and the electrostatic force to be represented by $\frac{\partial}{\partial\alpha}\mathcal{V}_e$, we arrive at the result that the conservative mechanical force, F_{mc} is obtained from

$$F_{mc} = -\frac{\partial}{\partial\alpha}\mathcal{V}_m, \tag{12.5.9}$$

which is the negative of the result we obtain for the electrostatic force

$$F_{ec} = \frac{\partial}{\partial\alpha}\mathcal{V}_e. \tag{12.5.10}$$

This apparent contradiction is due to the fact that energy (i.e., work) is provided by the voltage source to maintain the electric field.

In the simple case where the kinetic energy and mechanical potential energy are given by

$$T = \frac{1}{2}m\dot{\alpha}^2 \quad \text{and} \quad \mathcal{V}_m = \frac{1}{2}k\alpha^2, \tag{12.5.11}$$

where m and k are equivalent mass and stiffness, respectively, the equation of motion follows from Eq. (12.5.8),

$$m\ddot{\alpha} + k\alpha = F_m + \frac{\partial}{\partial\alpha}\mathcal{V}_e. \tag{12.5.12}$$

Because the electrostatic potential energy $\mathcal{V}_e$ can be a nonlinear function of the coordinate α, it is helpful to examine small motions about the system's static equilibrium point, α_0. The static equilibrium point is the solution to

$$k\alpha_0 = \frac{\partial\mathcal{V}_e}{\partial\alpha}|_{\alpha=\alpha_0}. \tag{12.5.13}$$

Consider small motions σ about the equilibrium point,

$$\alpha = \alpha_0 + \sigma. \tag{12.5.14}$$

For small σ, the derivative of the electrostatic potential energy in Eq. (12.5.12) becomes

$$\frac{\partial\mathcal{V}_e}{\partial\alpha} \approx \frac{\partial\mathcal{V}_e}{\partial\alpha}|_{\alpha=\alpha_0} + \sigma\frac{\partial^2\mathcal{V}_e}{\partial\alpha^2}|_{\alpha=\alpha_0}. \tag{12.5.15}$$

Substituting Eqs. (12.5.14) and (12.5.15) into Eq. (12.5.12) gives the governing equation for small motions about equilibrium, σ,

$$m\ddot{\sigma} + k\sigma = F_m + \sigma \frac{\partial^2 \mathcal{V}_e}{\partial \alpha^2}|_{\alpha=\alpha_0} \tag{12.5.16}$$

or

$$m\ddot{\sigma} + \left(k - \frac{\partial^2 \mathcal{V}_e}{\partial \alpha^2}|_{\alpha=\alpha_0}\right)\sigma = F_m. \tag{12.5.17}$$

The second derivative of the electrostatic potential energy evaluated at the equilibrium point may thus be viewed as the negative of the effective stiffness of an electrostatic spring.

12.6 Example of a Softening Electrode: Parallel Plate

Consider a highly simplified sensor composed of a mechanical spring/mass/damper and an electrode biased with a voltage V. Let α is the displacement of the sensing element. In this case, the charge may be expressed in terms of the capacitance through Eq. (12.2.13) where the electrode and the sensing element form a capacitor with capacitance $C_1(\alpha)$ (farads). The electrostatic potential energy may be expressed in terms of the capacitance as

$$\mathcal{V}_e = \frac{1}{2} C_1(\alpha) V_1^2. \tag{12.6.1}$$

The equation of motion given in Eq. (12.5.12) becomes

$$m\ddot{\alpha} + k\alpha + c\dot{\alpha} = f(t) + \frac{1}{2} V_1^2 \frac{\partial C_1}{\partial \alpha}, \tag{12.6.2}$$

where m is the equivalent mass, k is the equivalent stiffness, c is the equivalent damping constant, and $f(t)$ is an applied force. The term on the right accounts for the force applied to the moving element due to the electric field that results from the voltage V_1 applied between the electrode and the moving sensing element.

After solving Eq. (12.5.13) for α_0, we may examine small perturbations about this equilibrium point by expanding the derivative of $C_1(\alpha)$ in the first two terms of a Taylor's series,

$$\frac{\partial C_1(\alpha)}{\partial \alpha} \approx C_{10} + C_{11}\sigma, \tag{12.6.3}$$

where

$$C_{10} = \frac{\partial C_1(\alpha)}{\partial \alpha}|_{\alpha=\alpha_0} \quad \text{and} \quad C_{11} = \frac{\partial^2 C_1(\alpha)}{\partial \alpha^2}|_{\alpha=\alpha_0}, \tag{12.6.4}$$

and $\sigma = \alpha - \alpha_0$ is a small perturbation about α_0. Equation (12.6.2) then becomes

$$m\ddot{\sigma} + \left(k - \frac{1}{2}V_1^2 C_{11}\right)\sigma + c\dot{\sigma} = f(t). \tag{12.6.5}$$

Stability of small motions about $\sigma = 0$ is assured only when $k > \frac{1}{2}V_1^2 C_{11}$.

Using a typical charge amplifier to obtain an electronic readout provides an output voltage given by

$$V_o = -\frac{C_1(\alpha)}{C_f}V_1 \approx -\left(C_1(\alpha_0) + \sigma\frac{\partial C_1(\alpha_0)}{\partial \alpha}\right)\frac{V_1}{C_f}, \tag{12.6.6}$$

where the constant feedback capacitor C_f determines the gain of the amplifier. The electrical sensitivity to small changes in the position of the electrode is then

$$\frac{\partial V_o}{\partial \sigma} \approx -\frac{\partial C_1(\alpha_0)}{\partial \alpha}\frac{V_1}{C_f} = -C_{10}\frac{V_1}{C_f}. \tag{12.6.7}$$

Increasing the bias voltage, V_1, clearly provides an increase in the electrical sensitivity. Unfortunately, the selection of the bias voltage is typically constrained due to its effect on the system stiffness, as discussed below. An extremely important task in capacitive sensor design is to explore electrode configurations that permit the performance benefits of large bias voltages without the adverse effects on system stability or stiffness.

In the case of the common parallel plate electrode configuration, the capacitance may be approximated by

$$C_1(\alpha) = \frac{\epsilon A}{d - \alpha}, \tag{12.6.8}$$

where d is the nominal distance between the moving and fixed electrodes and A is the effective electrode area. The constants in Eq. (12.6.4) become

$$\begin{aligned} C_{10} &= \frac{\partial C_1}{\partial \alpha}|_{\alpha=\alpha_0} = \frac{\epsilon A}{(d-\alpha_0)^2}, \\ C_{11} &= \frac{\partial^2 C_1}{\partial \alpha^2}|_{\alpha=\alpha_0} = 2\frac{\epsilon A}{(d-\alpha_0)^3} \end{aligned} \tag{12.6.9}$$

Since C_{11} is always positive for physically realizable values of α_0, the parallel plate capacitor will always reduce the effective stiffness (i.e., it is softening) as can be seen in Eq. (12.6.5).

Equation (12.5.13) becomes

$$k\alpha_0 = \frac{1}{2}V_1^2\frac{\partial C_1(\alpha_0)}{\partial \alpha} = \frac{1}{2}V_1^2\frac{\epsilon A}{(d-\alpha_0)^2}. \tag{12.6.10}$$

Equation (12.6.10) may be solved for V_1^2,

$$V_1^2 = \frac{2k\alpha_0(d-\alpha_0)^2}{\epsilon A}. \tag{12.6.11}$$

Equations (12.6.5), (12.6.9), and (12.6.11) lead to the well-known result that stability will exist only for equilibrium positions in which $\alpha_0 < d/3$. Substituting $\alpha_0 = d/3$ into Eq. (12.6.11) gives the familiar value for the maximum voltage that may be used to avoid collapse of a parallel plate capacitor,

$$V_1 < V_{collapse} = \sqrt{\frac{8}{27}\frac{kd^3}{\epsilon A}}. \tag{12.6.12}$$

This collapse instability in parallel plate capacitive sensing excludes the use of moving electrodes having arbitrary compliance and the use of large bias voltages to improve sensitivity.

Another approach to the use of a parallel plate capacitive scheme is to use two electrodes which gives the possibility for the system total potential energy to be roughly independent of the moving electrode's position. To examine this case, assume the two capacitive sensing electrodes have applied voltages V_1 and V_2 and capacitances C_1 and C_2. The equation of motion then may then be expressed as an extension of Eq. (11.2.21),

$$m\ddot{x} + kx + c\dot{x} = -pA + \frac{1}{2}V_1^2\frac{\partial C_1}{\partial x} + \frac{1}{2}V_2^2\frac{\partial C_2}{\partial x}. \tag{12.6.13}$$

To examine the possible designs enabled by the use of this pair of capacitors, let each capacitance derivative be expanded in a Taylor's series about the equilibrium position, x_0. Keeping only the first two terms gives

$$\frac{\partial C_1}{\partial x} \approx C_{10} + C_{11}x, \tag{12.6.14}$$

and

$$\frac{\partial C_2}{\partial x} \approx C_{20} + C_{21}x. \tag{12.6.15}$$

Substituting Eqs. (12.6.14) and (12.6.15) into (12.6.13) and rearranging gives

$$\begin{aligned} &m\ddot{x} + \left(k - \left(\frac{1}{2}V_1^2C_{11} + \frac{1}{2}V_2^2C_{21}\right)\right)x + c\dot{x} \\ &= -pA + \frac{1}{2}V_1^2C_{10} + \frac{1}{2}V_2^2C_{20} \end{aligned} \tag{12.6.16}$$

For stability we must have

$$k - \left(\frac{1}{2}V_1^2 C_{11} + \frac{1}{2}V_2^2 C_{21}\right) > 0. \tag{12.6.17}$$

Capacitors having C_{11} or C_{21} that are positive will tend to reduce the effective stiffness and will be referred to as "softening" while when C_{11} or C_{21} are negative, we will refer to them as "hardening". If the passive system is stable (i.e., $k > 0$), stability can be assured if

$$\frac{1}{2}V_1^2 C_{11} + \frac{1}{2}V_2^2 C_{21} \approx 0. \tag{12.6.18}$$

Values of the bias voltages V_1 and V_2 can be determined to achieve this as long as C_{11} and C_{21} have opposite signs. That is, we would like one of the capacitors to be hardening while the other is softening. This enables us to tune the bias voltages to effectively cancel their effect on the overall system stiffness. Once the ratio of the bias voltages is chosen to satisfy Eq. (12.6.18), their magnitude can be set to large values without adversely affecting system stability.

A common capacitive sensing configuration involves the use of two parallel electrodes, one on either side of the moving electrode. In this case, as the moving electrode moves in the positive direction the gap between them becomes smaller while the gap between the moving electrode and the electrode on its opposite side becomes smaller. The two capacitors may be approximated by

$$C_1 = \frac{\epsilon A}{d + x} \quad \text{and} \quad C_2 = \frac{\epsilon A}{d - x}. \tag{12.6.19}$$

If the magnitudes of the bias voltages are taken to be identical, the static equilibrium position will be $x = 0$. In this case the capacitor constants in Eq. (12.6.16) are

$$\begin{aligned} C_{10} &= \frac{\partial C_1}{\partial x}|_{x=0} = \frac{-\epsilon A}{(d+x)^2}|_{x=0}, \quad C_{11} = \frac{\partial^2 C_1}{\partial x^2}|_{x=0} = \frac{\epsilon A}{(d+x)^3}|_{x=0} \\ C_{20} &= \frac{\partial C_2}{\partial x}|_{x=0} = \frac{\epsilon A}{(d-x)^2}|_{x=0}, \quad C_{21} = \frac{\partial^2 C_2}{\partial x^2}|_{x=0} = \frac{\epsilon A}{(d-x)^3}|_{x=0} \end{aligned} \tag{12.6.20}$$

In this case, since both C_{11} and C_{21} are positive, both capacitors are "softening". In this parallel plate sensing configuration, it is thus not possible to ensure stability unless k is sufficiently large.

12.7 Example of a Stiffening Electrode: Electrostatic Pendulum

The fact that the parallel plate capacitance can be approximated by a convenient analytical expression attracts designers to focus on that configuration rather than complex geometries that cannot be modeled with convenient expressions. Having the more

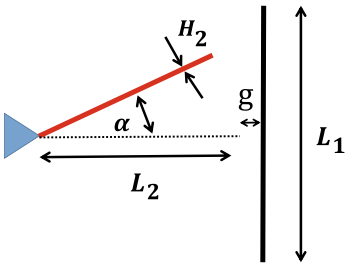


Fig. 12.1 Two electrode pendulum where the moving electrode of length L_2 rotates about a pivot through the angle α and the fixed electrode has a length L_1

general electrostatic analysis available through Eqs. (12.2.10) through (12.2.13), we are able to examine sensors having very complicated electrodes.

As an example, consider the electrode configuration shown in Fig. 12.1, which is similar to that described in [1]. The figure represents a two-dimensional cross-section of the system which is unchanged throughout the dimension that is perpendicular to this section. The moving electrode consists of a thin flexible element of length L_2 and thickness H_2 (shown in red) shown deflected relative to the horizontal orientation by the angle α. Here we show the moving electrode as a straight solid body that pivots about its attachment point. It could also consist of a flexible beam or string hung at one end having small enough bending stiffness that it is free to rotate in a manner similar to that shown in the figure.

A vertical, fixed electrode of length L_1 and thickness H_1 is shown to the right of the moving electrode in Fig. 12.1. When the moving electrode is in the horizontal position, perpendicular to the fixed electrode, the distance between the two electrodes is g. As the moving electrode rotates, it changes the charge distribution and net capacitance between this pair of electrodes. Let the voltage applied to the moving electrode be $v = 1$ volt and let the fixed electrode be set at a voltage of $v = 0$. In this example, to simplify our calculations we will assume that the moving electrode maintains the shape of a straight line so that it moves as a rigid pendulum that is free to pivot about its attachment point. Small deviations from this straight line shape due to bending will not significantly change the results. Let the moving electrode have a length $L_2 = 20$ μm and thickness $H_2 = 10$ nm. When the moving electrode is horizontal, with a pivot angle of $\alpha = 0$, its free end is a distance $g = 1.5$ μm from the fixed electrode. The fixed electrode has a length $L_1 = 22$ μm and thickness $H_1 = 1$ μm.

The charge distribution corresponding to a variety of rotations α has been computed by discretizing the electrode areas into 960 elements. Knowing the imposed voltages on each area and having calculated the charge distribution as a function of α, the force can be computed by numerical differentiation to estimate Eq. (12.5.5). The estimated first and second derivatives of the electrostatic potential energy as a function of rotation angle are shown in Fig. 12.2. It is clear that this force (or moment) is always attractive and stiffening since it always acts to return the pendulum to the equilibrium position at $\alpha = 0$.

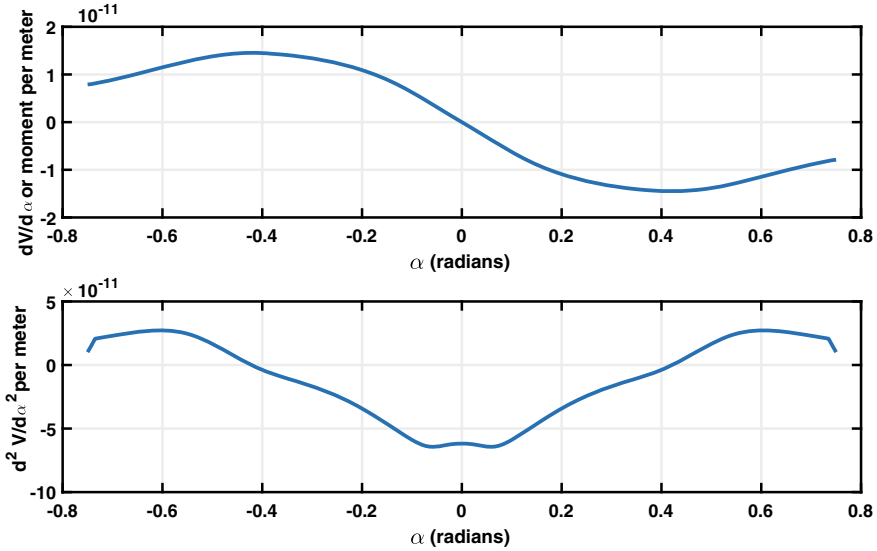


Fig. 12.2 First and second derivatives of the potential energy as a function of rotation angle α for the two electrode pendulum of Fig. 12.1. This force is always attractive and stiffening since it always acts to return the pendulum to the equilibrium position at $\alpha = 0$

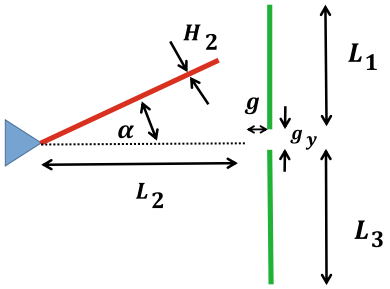


Fig. 12.3 Three electrode pendulum where the moving electrode of length L_2 rotates about a pivot through the angle α and the fixed electrode of Fig. 12.1 has been split into two fixed electrodes having lengths L_1 and L_3 and separated by a length g_y

It can be useful to note that the system of Fig. 12.1 can be modified by dividing the fixed electrode into two separated electrodes with a small gap g_y as shown in Fig. 12.3. In this case, if g_y is sufficiently small, and both fixed electrodes are held at the same potential, we can expect the energy and force on the moving electrode will be very similar to those of the system of Fig. 12.1. One could then detect the changes in charge on the two fixed electrodes as the moving electrode is displaced. This would result in a sensor that is globally stable with minimal force applied to the moving electrode.

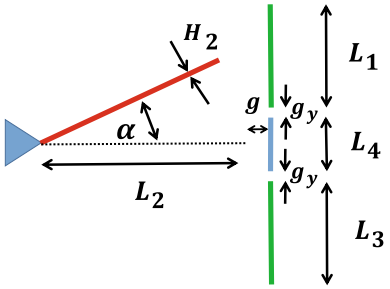


Fig. 12.4 Four electrode pendulum where the moving electrode of length L_2 rotates about a pivot through the angle α and the fixed electrode of Fig. 12.1 has been split into three fixed electrodes having lengths L_1, L_3 and L_4 and separated from each other by a length g_y

12.8 Example of an Adjustable Electrode: Electrostatic Pendulum with Reduced Stiffness at Equilibrium

Having constructed a model of the electrostatic force for the configuration of Fig. 12.1, it is not difficult to divide the fixed electrode into three separate electrodes that could be set to a variety of voltages as shown in Fig. 12.4. This provides the possibility of constructing a configuration inspired by the repulsive electrode arrangement studied in [2, 3].

We will consider this system to be comprised of four electrodes, where the vertical electrode of Fig. 12.1 has been divided into three isolated regions, denoted as electrodes 1, 3, and 4, with electrode 2 being the moving electrode as before. The thicknesses of electrodes 1, 3, and 4 will be set equal to that of electrode 1 of Fig. 12.1, i.e., $H_1 = H_3 = H_4 = 1$ μm. The lengths of these electrodes will be $L_1 = L_3 = 10$ μm and $L_4 = 2$ μm so that the total length of this set of three electrodes will be $L_1 + L_3 + L_4 = 22$ μm, equal to the length of the fixed electrode in Fig. 12.1. We will set the voltages of the two longer lengths of fixed electrode to be $V_1 = V_3 = 0$ volts. The voltage of the middle fixed electrode may be adjusted to achieve the desired force profile for the moving electrode. In order to enable these electrodes to have differing voltages, we introduce a gap between them given by $g_y = 1$ μm. As studied in [2, 3], the fixed electrode with adjustable voltage, V_4 can result in a repulsive force on the moving electrode with voltage V_1 with an appropriate choice of V_4.

The ability to set V_4 to any desired value enables the designer to adjust the electrostatic stiffness applied to the moving electrode to a wide range of values, either positive, negative, or nearly zero, without suffering adverse consequences such as instability or pull-in, which has always hampered the design of highly compliant electrostatic sensors. Note that setting $V_4 = 0$ causes the system to be nearly identical to the two electrode configuration of Fig. 12.1.

Increasing V_4 above the value of zero tends to reduce the attractive force that causes the moving electrode to return to $\alpha = 0$. As an example, if we set $V_4 = 0.20324$, the restoring moment becomes quite small for a range of values of α near the static equi-

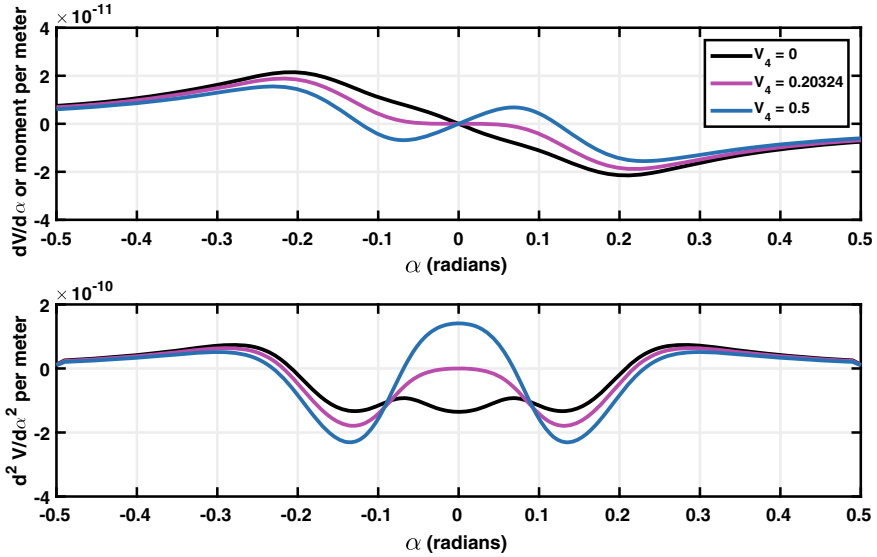


Fig. 12.5 Four electrode configuration of Fig. 12.4 makes it possible to minimize the electrostatic force on the moving electrode. In this example, setting the voltage of electrode 4 to $V_4 = 0.28947$ volts minimizes the electrostatic energy and, subsequently, the electrostatic force. The first and second derivatives of the electrostatic potential energy of the system are shown for three values of V_4. The dimensions of the system are taken to be $L_1 = 1\,\mu\text{m}$, $L_2 = 20\,\mu\text{m}$, $L_3 = 1\,\mu\text{m}$, $L_4 = 1\,\mu\text{m}$, $H_1 = 1\,\mu\text{m}$, $H_2 = 10$ nm, $H_3 = 1\,\mu\text{m}$, $H_4 = 1\,\mu\text{m}$, $g_y = 1\,\mu\text{m}$, and $g = 1.5\,\mu\text{m}$. The applied voltages (other than V_4 which is specified in the figure) are $V_1 = V_3 = 0$ and $V_2 = 1$ volt

librium point, $\alpha = 0$, as shown in Fig. 12.5. The effective torsional stiffness (being the derivative of the restoring moment) becomes very small near the equilibrium position. Figure 12.5 also shows the second derivative of the electrostatic potential energy with respect to α, which gives the torsional stiffness near the equilibrium point at $\alpha = 0$. The figure shows that for $V_4 = 0.20324$, this second derivative is very small indicating that the effective stiffness is also small. The ability to effectively null out the electrostatic stiffness near the equilibrium position, and without modifying the equilibrium position, opens up wide design space for electrostatic sensors. We may now utilize moving electrodes having essentially no mechanical stiffness in the direction of motion. The electrode may have significant stiffness in the x direction, i.e., along its length but is allowed to have negligible stiffness in the direction of motion, α. The mechanical stiffness of the electrode may be set without regard to the electrostatic forces or concerns about stability.

It should also be mentioned that one could choose a negative value of V_4, resulting in an increase in electrostatic stiffness beyond that obtained in the system of Fig. 12.1. The adjustment of V_4 provides a convenient way to either increase or decrease the system sensitivity.

12.9 Example of an Adjustable Electrode: A Bistable Electrostatic Pendulum

The ability to continuously adjust the stiffness at the equilibrium position of the moving electrode as discussed above also suggests that a further increase in the value of the middle fixed electrode voltage, V_4 could cause the effective stiffness to become negative. This is illustrated in Fig. 12.5 where we have set $V_4 = 0.5$ volts. In this case, the equilibrium position at $\alpha = 0$ becomes unstable and two new stable equilibrium points are created on either side of it. The system thus becomes bistable. This could prove to provide an extremely convenient way to create an oscillator or switch.

12.10 Electronic Sensitivity

To obtain an electronic voltage output from the sensor that is proportional to the motion α, we may detect the difference in charge on electrodes 3 and 1 using the circuits shown in Figs. 12.6 or 12.7. These circuits consist of either charge amplifiers or simple voltage buffers that produce output voltages that are then subtracted by a conventional differential amplifier. Note that the circuits shown here are idealized and additional components should be incorporated in practice. For example, one must normally use a large resistor in parallel with the feedback capacitor to ensure finite DC gain. In addition, it is helpful to have a parallel combination of a capacitor and resistor from the positive input to the ground that matches the impedance of the negative input rather than the zero impedance connection shown here. This helps to minimize the effects of the input bias currents.

It can be instructive to compare the sensor configuration of Fig. 12.4 with a commonly used design. Microphones are extremely common capacitive sensors and they clearly benefit from the use of a moving electrode (i.e., pressure-sensing diaphragm) having as little mass and stiffness as possible. Small mass and stiffness can be beneficial in this application because the forces on the diaphragm due to diminutive sound pressures are exceedingly small.

Nearly all current miniature microphones employ capacitive sensing where the moving diaphragm and backplate electrode form a parallel plate capacitor as discussed in Eqs. (12.6.8) through (12.6.12). Conventional microphones have enjoyed decades of design optimization and there are many effective designs available. Our present purpose is not yet to demonstrate superior performance relative to the current state of the art but to suggest a design approach that eliminates well-known barriers to performance, namely the need for significant diaphragm stiffness, a well-known adversary of microphone designers.

As a reasonably typical miniature microphone design, we will assume that the moving diaphragm has dimensions of 1 mm by 1 mm so that the area is 10^{-6} m^2. Assume the nominal gap between the diaphragm and backplate electrode is $d = 10$

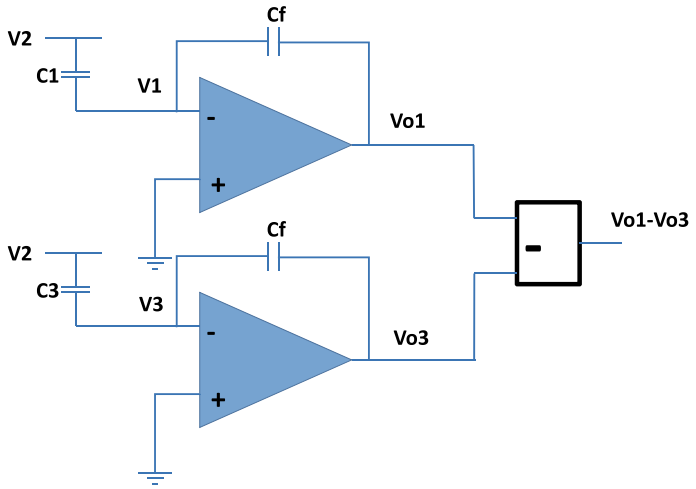


Fig. 12.6 Charge amplifier readout circuit to convert the difference in charge at electrodes 1 and 3 into an output voltage

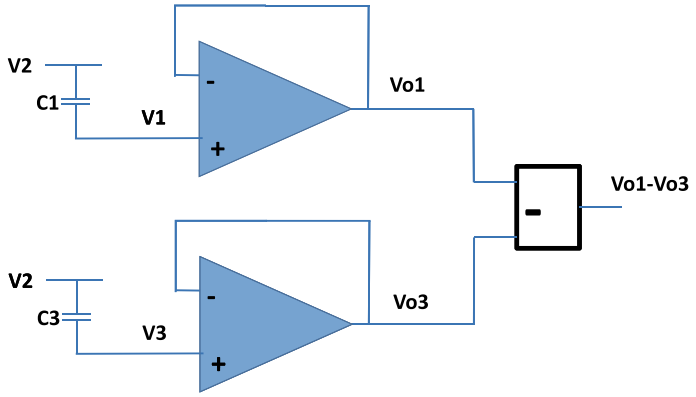


Fig. 12.7 Voltage amplifier readout circuit to convert the difference in charge at electrodes 1 and 3 into an output voltage

μm. For comparison's sake, we will take the bias voltage to be unity and assume the signal is read using a simple charge amplifier. This amplifier will cause the voltage across the sensing capacitor to be fixed and fluctuations in the microphone capacitance will convert the charge into a fluctuating voltage. In this comparison, we will take the deflection, x, of the diaphragm to be the coordinate, α and will assume that the equilibrium position is reasonably close to $x = 0$ so that d is the gap distance at equilibrium. For the parallel plate capacitor, this fluctuating charge will be given by

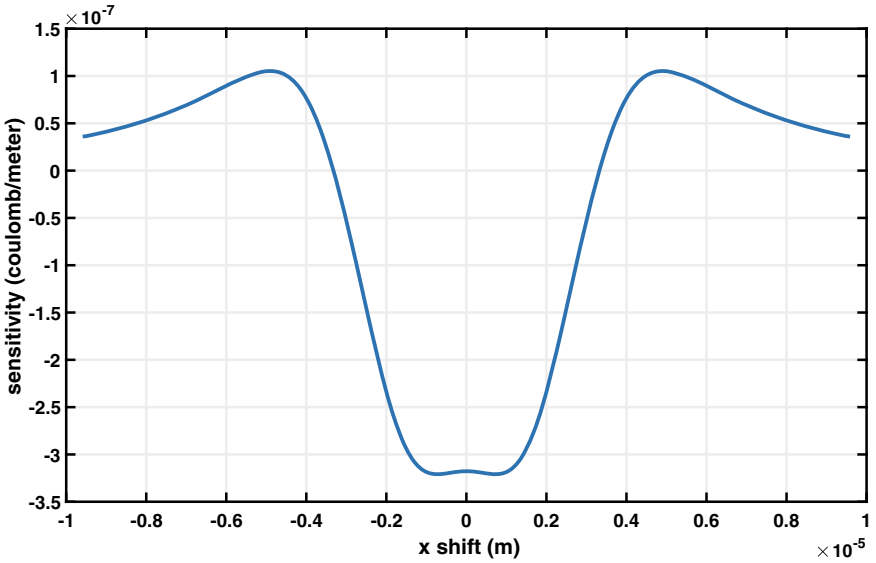


Fig. 12.8 Charge sensitivity in coulombs per meter as a function of the tip deflection of the moving electrode. The bias voltage of electrode 4 is $V_4 = 0.2032$ volt

$$\frac{dQ}{dx} = \frac{d}{dx}\frac{\epsilon A}{d-x}|_{x=0} = \frac{\epsilon A}{d^2} \approx 8.85 \times 10^{-8} \quad \text{C/m}, \tag{12.10.1}$$

where $\epsilon = 8.85 \times 10^{-12}$ F/m is the permittivity of free space.

It is, of course, not a simple matter to perform a completely valid comparison between sensing methods that differ significantly but, let us assume that the four electrode sensor of the present study is to be fabricated on a silicon chip having dimensions of 2 mm by 2 mm. If we assume that approximately half of this area can be devoted to the moving electrode area having length $L_2 = 20$ μm (as shown in Fig. 12.4), we may estimate that the total length of the moving electrode (i.e., in the direction normal to the section shown in Fig. 12.4) will be approximately $L_z \approx 4 \times 10^{-6}/(2 \times 20 \times 10^{-6}) = 0.1$ m. The charge calculated by our two-dimensional electrostatic calculations will be the total charge per unit length in the direction normal to the section shown in Fig. 12.4. We may then estimate the total charge by multiplying our calculated results by $L_z = 0.1$ m.

The calculated sensitivity of the sensor depicted in Fig. 12.4 is shown in Fig. 12.8. For deflections of the pendulum within approximately 1 μm from zero, the sensitivity is nearly constant and equal to approximately 32×10^{-8} coulomb/meter.

12.11 Compliant Repulsive Actuator

In addition to examining the electrode scheme of Fig. 12.4 as a sensor, we should also consider its use as an actuator. In this case, we may apply a small time-varying differential voltage, $Vi_{13} = V_1 - V_3$ to electrodes 1 and 3 which will effectively modulate the system's equilibrium position about $\alpha = 0$. The voltage applied to electrode 4, V_4 may be set to a value that adjusts the electrostatic stiffness to nearly any value desired, leaving the motion to be limited only by the mechanical stiffness and mass of the moving electrode. The use of an extremely compliant and lightweight moving electrode material, such as for example, graphene, would enable actuation with very small driving voltage, Vi_{13}.

As an example, we consider the special case where the mechanical stiffness of the moving electrode may be neglected and estimate its equilibrium position as a function of the driving voltage, Vi_{13}. The results are shown in Fig. 12.9 for two values of the compliance adjusting voltage, V_4. The figure shows the equilibrium position of the system as a function of Vi_{13}. When $V_4 = 0.21$ volt, i.e., close to the zero stiffness value of $V_4 = 0.2032$ volt, we can see that the moving electrode will

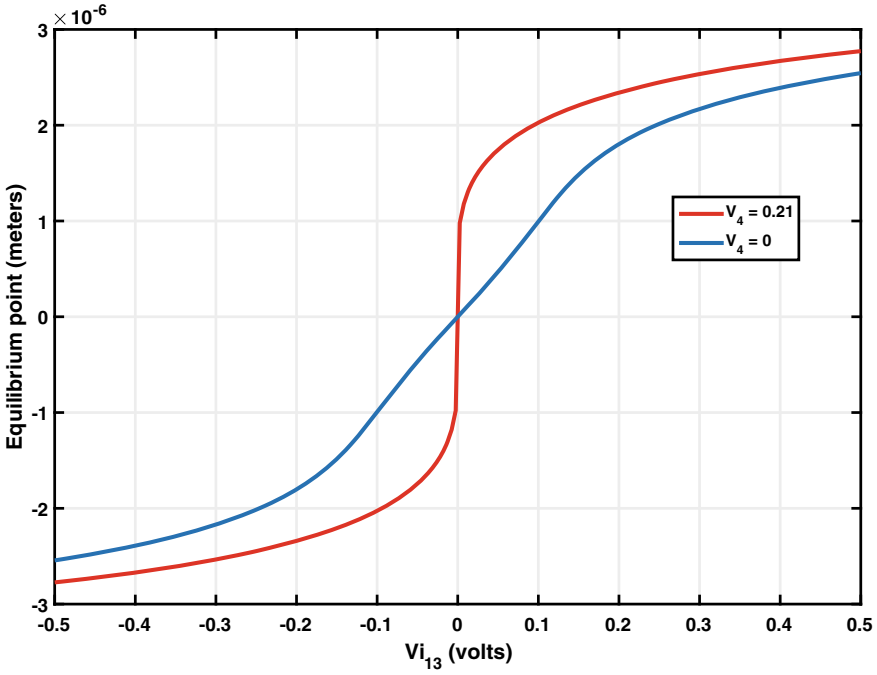


Fig. 12.9 Equilibrium position of the free end of the moving electrode as a function of $Vi_{13} = V_1 - V_3$ for the electrode configuration of Fig. 12.4 for two values of V_4. As V_4 is increased to values close to the point where the stiffness is minimized, the moving electrode becomes highly sensitive to variations in the differential drive voltage, Vi_{13}

possess an equilibrium position over a wide range for a very small range of values of the driving voltage. The electrostatic gain of the actuator is thus widely adjustable using the voltage V_4.

It is also important to note that because this repulsive actuator relies on the driving electrodes 1 and 3 being on either side of the moving electrode, one obtains an essentially linear relationship between the driving voltage, Vi_{13}, and the deflection, at least for small motions about the equilibrium position when $Vi_{13} = 0$. This can provide a significant advantage over parallel plate capacitive actuators which, along with their well-known instability problems, also result in forces and resulting electrode displacements that are proportional to the square of the bias voltage.

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Chapter 13
Parameter Identification of Acoustic Systems



There are many situations where it is desirable to estimate values of a finite set of complex parameters to describe devices and systems in acoustics. Methods for the identification of the essential parameters of dynamic systems have been pursued for numerous decades and many review articles and excellent dissertations are available on this subject [1–10]. Despite the 'maturity' of this field, it remains extremely challenging. Some of the algorithms that are available in widely-used software (such as the System Identification Toolbox in Matlab) fail to succeed in identifying the parameters of relatively simple systems [4]. Sekine et al. [11] successfully identified the two lowest modes in a positioning system for a laser but the third mode posed challenges and 'improvements of the identification accuracy of the 3rd resonance 'will be a future work.'

It is generally assumed that the system of interest can be represented in the frequency domain by the ratio of polynomials with the complex frequency as the independent variable. The identification problem consists of determining the coefficients of these polynomials. The problem of minimizing the error in the approximation becomes one of solving a nonlinear system of equations for these coefficients. An alternative to this nonlinear problem is to re-cast the error to obtain a linear least squares solution as proposed by Levy [12]. This is the approach taken by many of the successful frequency domain system identification methods. This popular approach has been adapted to multi-input/multi-output systems and has been extended to reduce the influence of measurement noise and parameter uncertainty [13].

In the following, the least squares approach is used in which the error to be minimized is defined to be the relative error in the transfer function model. This approach was apparently first suggested by Strobel [14].

Suppose the error vector (in an arbitrary number of dimensions) is a function of n complex unknowns, a_l, for $l = 1, \ldots, n$,

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/978-3-031-33009-4_13.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4_13

$$\vec{\epsilon} = \vec{\epsilon}(a_1, a_2, \ldots, a_n) \tag{13.0.1}$$

The total length of the error is

$$E = \vec{\epsilon}^* \cdot \vec{\epsilon} \tag{13.0.2}$$

where $*$ denotes the complex conjugate and $\cdot$ is the vector inner product. We would like to determine the n unknowns such that the error is a minimum. This is normally accomplished by solving the n equations obtained by setting the derivative of the total error with respect to each unknown equal to zero,

$$\frac{\partial E}{\partial a_j} = 0, \quad \text{for} \quad j = 1, 2, \ldots, n \tag{13.0.3}$$

Equations (13.0.2) and (13.0.3) give

$$\frac{\partial E}{\partial a_j} = \vec{\epsilon}^* \cdot \frac{\partial \vec{\epsilon}}{\partial a_j} + \vec{\epsilon} \cdot \frac{\partial \vec{\epsilon}^*}{\partial a_j} = 0 \tag{13.0.4}$$

The derivative of the conjugate of $\vec{\epsilon}$ with respect to a_j is troubling. We may, however, express it as a derivative with respect to the conjugate of a_j,

$$\frac{\partial \vec{\epsilon}^*}{\partial a_j} = \frac{\partial \vec{\epsilon}^*}{\partial a_j^*} \frac{da_j^*}{da_j} \tag{13.0.5}$$

The derivative, $\frac{da_j^*}{da_j}$, may not be computable unless a_j is real. It may, however, still be possible to satisfy Eq. (13.0.4). Combining Eqs. (13.0.4) and (13.0.5) gives

$$\vec{\epsilon}^* \cdot \frac{\partial \vec{\epsilon}}{\partial a_j} + \vec{\epsilon} \cdot \frac{\partial \vec{\epsilon}^*}{\partial a_j^*} \frac{da_j^*}{da_j} = 0 \tag{13.0.6}$$

Fortunately, one may determine the unknowns such that

$$\vec{\epsilon}^* \cdot \frac{\partial \vec{\epsilon}}{\partial a_j} = 0, \quad \text{for} \quad j = 1, 2, \ldots, n \tag{13.0.7}$$

Since zero is real, the conjugate of this expression is also equal to zero,

$$\vec{\epsilon}^* \cdot \frac{\partial \vec{\epsilon}}{\partial a_j} = \vec{\epsilon} \cdot \frac{\partial \vec{\epsilon}^*}{\partial a_j^*} = 0, \quad \text{for} \quad j = 1, 2, \ldots, n \tag{13.0.8}$$

Equations (13.0.4) and (13.0.6) will be then satisfied whether or not the derivative $\frac{da_j^*}{da_j}$ exists.

13.1 Least Squares Model of a Complex Transfer Function with Real Unknowns

There are many situations where a complex transfer function may be used to describe the input-output relationships of a linear system. Suppose we are given a complex transfer function $H(S)$ which might be determined experimentally or through some model. In a great many systems in engineering, this transfer function takes the form,

$$H(S) = \frac{\sum_{l=1}^{n} a_l S^{\alpha_l}}{\sum_{j=1}^{m} b_j S^{\beta_j}} \tag{13.1.1}$$

where $S = i\omega$, a_l and b_j are constants to be determined and α_l and β_j are known constants. Note that in this special case, the unknown constants are all real. We would like to determine the constants a_l and b_l such that the error between the model results of Eq. (13.1.1) and the given complex function data, $H(S)$ is minimized. The absolute error at each value of S is

$$\epsilon(S) = \frac{\sum_{l=1}^{n} a_l S^{\alpha_l}}{\sum_{j=1}^{m} b_j S^{\beta_j}} - H(S) \tag{13.1.2}$$

Attempting to determine the constants a_l and b_j to minimize this error leads to a nonlinear set of equations for the unknowns. In order to construct a linear least squares problem to obtain these coefficients, it is often beneficial to instead multiply by the denominator in Eq. (13.1.2) and express the error as [12]

$$\epsilon(S)_A = \sum_{l=1}^{n} a_l S^{\alpha_l} - H(S) \sum_{j=1}^{m} b_j S^{\beta_j} \tag{13.1.3}$$

We will refer to this error as the 'absolute' error. Unfortunately, minimizing this absolute error may not lead to the set of coefficients that gives the desired result at all frequencies.

We chose instead to minimize the relative error between the model and the data, which may be defined by

$$\epsilon(S)_R = \frac{\frac{\sum_{l=1}^{n} a_l S^{\alpha_l}}{\sum_{j=1}^{m} b_j S^{\beta_j}} - H(S)}{H(S)} = \frac{\sum_{l=1}^{n} a_l S^{\alpha_l} - H(S) \sum_{j=1}^{m} b_j S^{\beta_j}}{H(S) \sum_{j=1}^{m} b_j S^{\beta_j}} \tag{13.1.4}$$

The given transfer function, $H(S)$ is typically known at a discrete set of frequencies, S_l, for $l = 1, \ldots, N$. The relative error at frequency S_l is then

$$\epsilon_{R_l} = \epsilon(S_l)_R = \frac{\sum_{l=1}^{n} a_l S_l^{\alpha_l} - H(S_l) \sum_{j=1}^{m} b_j S_l^{\beta_j}}{H(S_l) \sum_{j=1}^{m} b_j S_l^{\beta_j}} = \frac{\epsilon(S_l)_N}{H(S_l) \sum_{j=1}^{m} b_j S_l^{\beta_j}} \tag{13.1.5}$$

where $\epsilon(S_l)_A$ is the absolute error in the numerator as in Eq. (13.1.3). The error at each frequency will generally have both real and imaginary parts. As in typical linear least squares problems, we will define an error for the model that measures the total squared length of the error. Let this error be the sum of the squared magnitude of the errors at each frequency,

$$E = \sum_{l=1}^{N} \epsilon_{R_l} \times \epsilon_{R_l}^* \tag{13.1.6}$$

where * denotes the complex conjugate.

Unfortunately, because our model transfer function in Eq. (13.1.1) depends on the unknown constants a_l and b_j in the form of the ratio of polynomials in S, it is not feasible to cast their determination as a linear least squares problem. We can, however, construct a convenient iterative solution approach. This may be accomplished by considering the denominator of Eq. (13.1.5) to comprise a weighting function with the values of the coefficients in this denominator, b_j, for $j = 1, \ldots, m$, taken to be those obtained at the previous iteration step. Let the weighting function be denoted by W_l,

$$W_l = |\frac{1}{H(S_l) \sum_{j=1}^{m} b_j S_l^{\beta_j}}|^2 \tag{13.1.7}$$

where $||$ denotes the magnitude of the complex quantity. Several approaches to defining this weighting function are surveyed in [1].

Let the total weighted error be

$$E = \sum_{l=1}^{N} \epsilon(S_l)_N \times W_l \times \epsilon(S_l)_N^* \tag{13.1.8}$$

We would like to determine the constants a_i and b_j for all i and j so as to minimize E in Eq. (13.1.8). An approximate iterative solution may be obtained by setting our initial guess for the weighting function to be unity, $W_l^0 = 1$, where the zero superscript indicates the initial value. The Pth estimate of the total error may then be approximated by

$$E^P \approx \sum_{l=1}^{N} \epsilon(S_l)_N^P \times W_l^{P-1} \times \epsilon(S_l)_N^{P*} \tag{13.1.9}$$

where, again, the superscript P indicates the iteration number. Since the denominator in Eq. (13.1.5) is taken to be a weighting W_l^{P-1}, which is estimated from the previous iteration, the task of determining the unknown coefficients, a_i for $i = 1, \ldots, n$ and b_j for $j = 1, \ldots, m$ becomes a linear least squares problem.

Determination of the unknown coefficients at the Pth iteration will be accomplished if we set

$$\frac{\partial E^P}{\partial a_I} = 0 \tag{13.1.10}$$

for $I = 1, 2, \ldots, n$ and

$$\frac{\partial E^P}{\partial b_J} = 0 \tag{13.1.11}$$

for $J = 1, 2, \ldots, m$.

Equations (13.1.10) and (13.1.11) give a total of $n + m$ equations to determine the unknown constants a_i and b_j in Eq. (13.1.2). There will be a different set of these constants for each iteration. We will simplify the notation by resisting the temptation of embellishing these constants with the superscript P. Equations (13.1.8) and (13.1.10) give

$$\sum_{l=1}^{N} \left(\frac{\partial \epsilon_l}{\partial a_I} \epsilon_l^* + \frac{\partial \epsilon_l^*}{\partial a_I} \epsilon_l \right) W_l$$
$$= 2\Re \left[\sum_{l=1}^{N} \frac{\partial \epsilon_l}{\partial a_I} \epsilon_l^* W_l \right] = 0, \quad I = 1, \ldots, n \tag{13.1.12}$$

Equations (13.1.8) and (13.1.11) give

$$\sum_{l=1}^{N} \left(\frac{\partial \epsilon_l}{\partial b_J} \epsilon_l^* + \frac{\partial \epsilon_l^*}{\partial b_J} \epsilon_l \right) W_l$$
$$= 2\Re \left[\sum_{l=1}^{N} \frac{\partial \epsilon_l}{\partial b_J} \epsilon_l^* W_l \right] = 0, \quad J = 1, \ldots, m \tag{13.1.13}$$

Because the unknown coefficients are real, note that the two terms being summed in Eqs. (13.1.12) and (13.1.13) are conjugates of each other. Adding the terms then gives twice the real part.

Substituting Eq. (13.1.5) into Eq. (13.1.12) gives

$$\sum_{l=1}^{N} \frac{\partial \epsilon_l}{\partial a_I} \epsilon_l^* W_l = \sum_{l=1}^{N} -S_l^{\alpha_I} \times \left(H(S_l)^* \sum_{j=1}^{m} b_j (-S_l)^{\beta_j} - \sum_{i=1}^{n} a_i (-S_l)^{\alpha_i} \right) W_l \quad I = 1, 2, \ldots, n \tag{13.1.14}$$

Substituting Eq. (13.1.5) into Eq. (13.1.13) gives

$$\sum_{l=1}^{N} \frac{\partial \epsilon_l}{\partial b_J} \epsilon_l^* W_l = \sum_{l=1}^{N} H(S_l) S_l^{\beta_J} \times \left(H(S_l)^* \sum_{j=1}^{m} b_j (-S_l)^{\beta_j} - \sum_{i=1}^{n} a_i (-S_l)^{\alpha_i} \right) W_l \quad J = 1, 2, \ldots, m \tag{13.1.15}$$

We should note that it is not possible to uniquely determine all of the $m + n$ constants in Eq. (13.1.1) since both the numerator and denominator could be multiplied by the same arbitrary constant without impacting the result. To express the function in a more appropriate form, let $b_m = 1$. This will give a total of $m + n - 1$ unknowns. Equations (13.1.12) and (13.1.14) may be then rearranged to give

$$\sum_{i=1}^{n} a_i \sum_{l=1}^{N} S_l^{\alpha_I} (-S_l)^{\alpha_i} W_l - \sum_{j=1}^{m-1} b_j \sum_{l=1}^{N} S_l^{\alpha_I} (-S_l)^{\beta_j} H^*(S_l) W_l = \sum_{l=1}^{N} H^*(S_l) S_l^{\alpha_I} (-S_l)^{\beta_m} W_l \quad I = 1, 2, \ldots, n \tag{13.1.16}$$

Similarly, Eqs. (13.1.13) and (13.1.15) may be rearranged to give

$$\sum_{j=1}^{m-1} b_j \sum_{l=1}^{N} (-S_l)^{\beta_j} H(S_l) H^*(S_l) S_l^{\beta_J} W_l - \sum_{i=1}^{n} a_i \sum_{l=1}^{N} (-S_l)^{\alpha_i} H(S_l) S_l^{\beta_J} W_l = -\sum_{l=1}^{N} H(S_l) H^*(S_l) S_l^{\beta_J} (-S_l)^{\beta_m} W_l \quad J = 1, 2, \ldots, m-1 \tag{13.1.17}$$

Equations (13.1.16) and (13.1.17) constitute $n + m - 1$ linear equations for the $n + m - 1$ unknowns, a_i and b_j. It may be helpful to arrange the equations in matrix form.

$$\begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \tag{13.1.18}$$

where the elements of the unknown vectors a and b are a_i for $i = 1, \ldots, n$ and b_j for $j = 1, \ldots, m - 1$ respectively. The elements of the partitioned square matrix are given by

$$D_{11}(I, i) = \Re\left[\sum_{l=1}^{N} S_l^{\alpha_I} (-S_l)^{\alpha_i} W_l\right],$$
$$I = 1, 2, \ldots, n, \quad i = 1, 2, \ldots, n \tag{13.1.19}$$

$$D_{12}(I, j) = -\Re\left[\sum_{l=1}^{N} S_l^{\alpha_I} (-S_l)^{\beta_j} H^*(S_l) W_l\right],$$
$$I = 1, 2, \ldots, n, \quad j = 1, 2, \ldots, m - 1 \tag{13.1.20}$$

$$D_{21}(J, i) = -\Re\left[\sum_{l=1}^{N} S_l^{\beta_J} (-S_l)^{\alpha_i} H(S_l) W_l\right],$$
$$i = 1, 2, \ldots, n, \quad J = 1, 2, \ldots, m - 1 \tag{13.1.21}$$

$$D_{22}(J, j) = \Re\left[\sum_{l=1}^{N} S_l^{\beta_J} (-S_l)^{\beta_j} H(S_l) H^*(S_l) W_l\right],$$
$$j = 1, 2, \ldots, m - 1, \quad J = 1, 2, \ldots, m - 1 \tag{13.1.22}$$

The elements on the right side of Eq. (13.1.18) are given by

$$c_1(I) = \Re\left[\sum_{l=1}^{N} S_l^{\alpha_I} H^*(S_l) (-S_l)^{\beta_m} W_l\right], \quad I = 1, 2, \ldots, n \tag{13.1.23}$$

and

$$c_2(J) = -\Re\left[\sum_{l=1}^{N} S_l^{\beta_J} H(S_l) H^*(S_l)(-S_l)^{\beta_m} W_l\right],$$
$$J = 1, 2, \ldots, m-1 \tag{13.1.24}$$

Equation (13.1.18) may be readily solved for the unknown constants,

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{bmatrix} D_{11} & D_{12} \\ D_{21} & D_{22} \end{bmatrix}^{-1} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \tag{13.1.25}$$

13.2 Accounting for an Unknown Time Delay in H

In the above discussion, H is assumed to be a transfer function between a measured input signal and an output signal. There are situations where the measured transfer function is obtained with an input signal from a reference microphone to sense the input sound pressure at the device under test. Because the microphone may be located at a different distance from the sound source than is the device under test, the sound might not arrive at both devices at the same time. There will then be an undesired and unknown time delay, τ, between the two measured signals. This time delay is not accounted for in the polynomial expressions in Eq. (13.1.1). It needs to be accounted for in order to properly represent the phase of the transfer function. τ then comprises an additional unknown that is not readily determined by solving a linear system of equations, as in a simple linear least squares problem.

To determine τ along with the other unknown coefficients, the partitioned vectors a and b in Eq. (13.1.25), we chose the less than elegant approach of treating the total error given in Eq. (13.1.8) as a nonlinear function of τ and searching for the value of τ that minimizes this error. In the example below in Fig. 13.2, the value of τ was determined with the help of the Matlab function fminbnd.

13.3 Examples

In the following, we present the results of applying the above linear regression approach to sets of measured and/or simulated data. The first is based on the electronic output of an electrodynamic device that is primarily intended to be used as a miniature shaker, or vibrator in portable electronic devices. This device consists of a magnet, which acts as an inertial mass, a coil, and a suspension system that has stiffness such that the magnet oscillates relative to the coil with a resonant frequency of approximately 200 Hz. The electric voltage across the coil was measured using a high input impedance voltage preamplifier. The voltage will be proportional to the velocity of the magnet relative to the coil. This device was placed on an electrody-

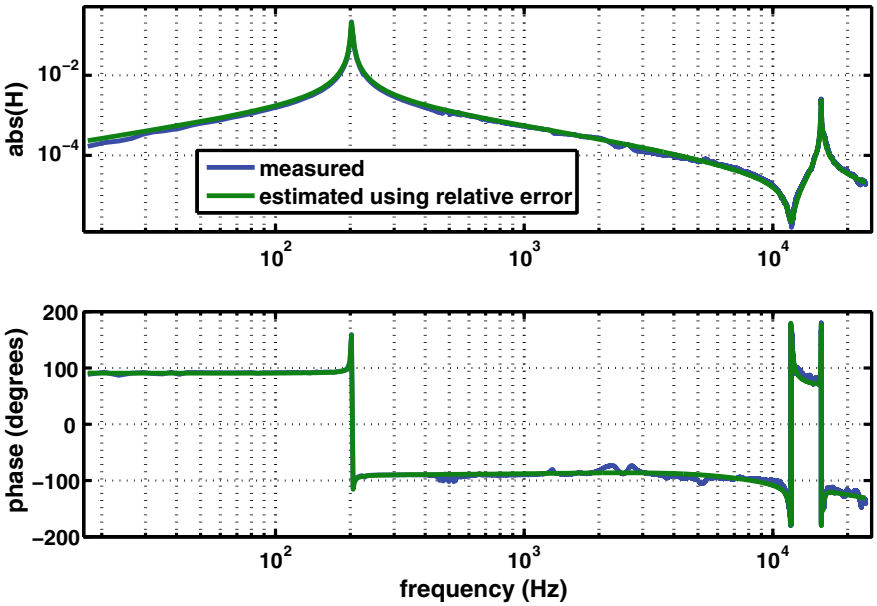


Fig. 13.1 Comparison of measured and estimated model results for a miniature electrodynamic shaker

Table 13.1 Coefficients for the data shown in Fig. 13.1

i	a_i	α_i	b_i	β_i
1	1.6421×10^{30}	0	-5.5769×10^{35}	0
2	-1.1697×10^{30}	1	2.081×10^{31}	1
3	2.5076×10^{25}	2	-3.4409×10^{29}	2
4	-3.4519×10^{20}	3	9.6336×10^{24}	3
5	6.9208×10^{15}	4	2.3901×10^{19}	4
6	$-2.3308e \times 10^{10}$	5	1.927×10^{15}	5
7	4.419×10^{5}	6	1.5823×10^{10}	6
8			96281	7
9			1	8

namic shaker to provide base excitation. The acceleration of the device was measured using a calibrated accelerometer.

The transfer function between the base acceleration and the output voltage of the device was measured with the base acceleration provided by an electrodynamic shaker driven with broadband random excitation. The results are shown in Fig. 13.1. The estimated data are nearly identical to the measured results. The identified transfer function is defined by the coefficients a_i, and b_i with exponents α_i, and β_i. The results are given in Table 13.1 and the transfer function is written explicitly in Eq. (13.3.1).

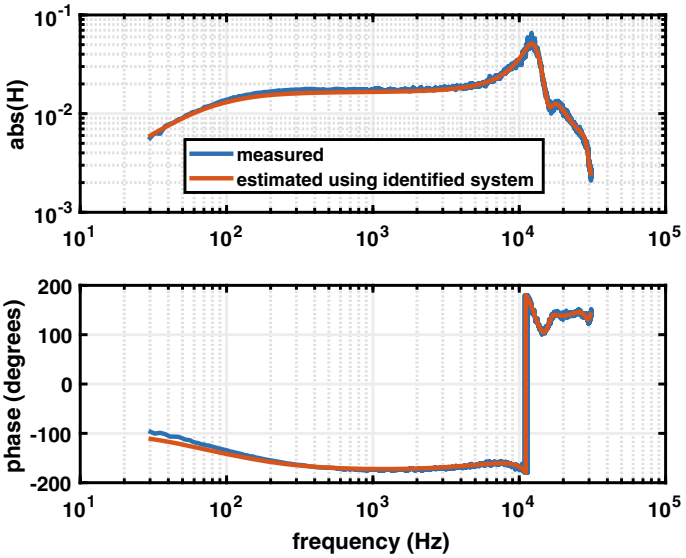


Fig. 13.2 Comparison of measured and estimated model results for a Knowles TO hearing aid microphone

$$H(S) = \frac{\begin{array}{r}[1.6421 \times 10^{30} - 1.1697 \times 10^{30} S + 2.5076 \times 10^{25} S^2 - 3.4519 \times 10^{20} S^3 \\ + 6.9208 \times 10^{15} S^4 - 2.3308e \times 10^{10} S^5 + 4.419 \times 10^{5} S^6\,]\end{array}}{\begin{array}{r}[-5.5769 \times 10^{35} + 2.081 \times 10^{31} S - 3.4409 \times 10^{29} S^2 + 9.6336 \times 10^{24} S^3 \\ + 2.3901 \times 10^{19} S^4 + 1.927 \times 10^{15} S^5 + 1.5823 \times 10^{10} S^6 + 96281 S^7 + S^8]\end{array}} \tag{13.3.1}$$

As another example, frequency response data were stored for one of the Knowles hearing aid microphones shown in Fig. 10.15. These data were stored as transfer functions between the output of the Knowles microphone and the reference microphone shown in Fig. 10.15. The results are shown in Fig. 13.2. In this case, because the input signal to the transfer function H was measured using a reference microphone which was affected by an unknown time delay, τ between it and the hearing aid microphone, the method described in Sect. 13.2 was used to estimate the unknown time delay. The numerical solution for this time delay required upper and lower bounds before the iterative solution method could begin. These bounds were estimated by assuming the reference microphone was placed within ± 2.4 cm from the hearing aid microphone. The result was an estimate of $\tau = -1.4364 \times 10^{-5}$ s. This corresponds to a position error of approximately 5 mm. The identified transfer function is defined by the coefficients a_i, and b_i with exponents α_i, and β_i. The results are given in Table 13.2.

In our next example, data were obtained for the response of a cantilever beam. The transfer function was obtained between the input force and the response at the free end of the beam. The comparison between the measured and estimated transfer functions is shown in Fig. 13.3. The data contains several small peaks which can be

Table 13.2 Coefficients for the data shown in Fig. 13.2

i	a_i	α_i	b_i	β_i
1	-3.5305×10^{28}	1	1.0411×10^{33}	0
2	-8.5282×10^{22}	2	2.1411×10^{30}	1
3	-4.5762×10^{18}	3	1.6377×10^{25}	2
4	-5.1209×10^{12}	4	6.4781×10^{20}	3
5	-9.7152×10^{7}	5	2.7231×10^{15}	4
6	-20.44	6	5.1883×10^{10}	5
7			82382	6
8			1	7

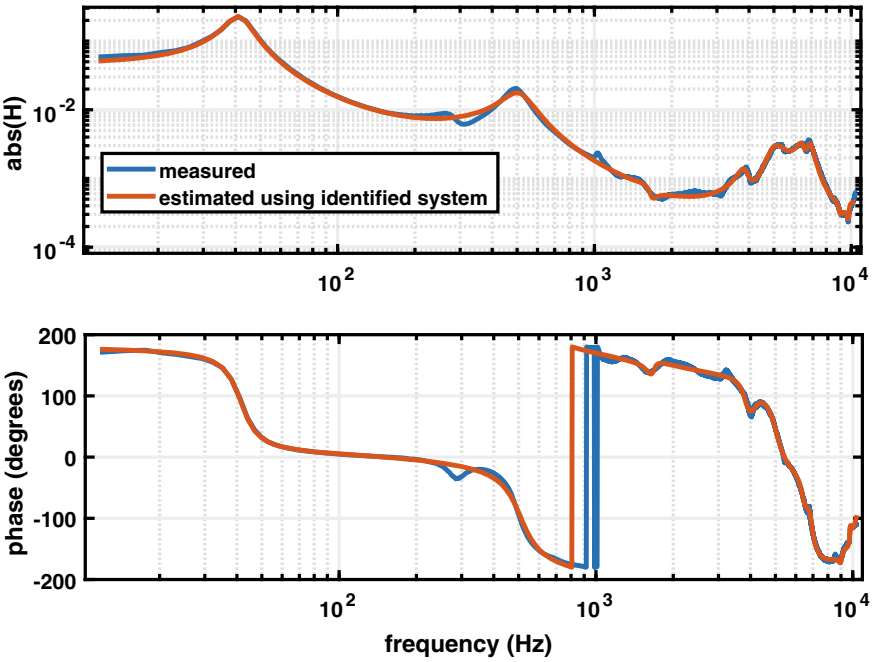


Fig. 13.3 Comparison of measured and estimated frequency response of a cantilever beam. The response of the identified system is in extremely close agreement with the measured results

accounted for only by including a significant number of terms in the polynomials. This then serves as an example of identifying a rather high order system. The figure shows that several of the small features of the response have been represented fairly well. The coefficients in the estimation polynomials are given in Table 13.3.

As an example of a much simpler system than considered in Fig. 13.3, the frequency response of a second order, high-pass electronic filter is shown in Fig. 13.4. Because the electronic filter exhibits nearly ideal, low-noise response, the identified

Table 13.3 Coefficients for the data shown in Fig. 13.3

i	a_i	α_i	b_i	β_i
1	-4.25397×10^{83}	0	8.578262×10^{84}	0
2	9.076015×10^{79}	1	-8.47875×10^{81}	1
3	4.97045×10^{77}	2	1.281758×10^{80}	2
4	2.939045×10^{73}	3	-1.3992×10^{76}	3
5	4.198826×10^{69}	4	1.435769×10^{73}	4
6	3.6550×10^{65}	5	-4.1059×10^{68}	5
7	-3.75556×10^{60}	6	1.80824×10^{65}	6
8	6.8214×10^{56}	7	-2.53165×10^{60}	7
9	-2.2515×10^{52}	8	6.44021×10^{56}	8
10	1.275×10^{47}	9	-5.41068×10^{51}	9
11	-1.8378×10^{43}	10	1.03591×10^{48}	10
12	-2.31136×10^{38}	11	-5.14410×10^{42}	11
13	-5.1656×10^{33}	12	8.860047×10^{38}	12
14	-1.00922×10^{29}	13	-2.36779×10^{33}	13
15	-4.751×10^{23}	15	4.28835×10^{29}	14
16	-1.10233×10^{19}	15	-5.1033×10^{23}	15
17			1.175084×10^{20}	16
18			-41570768694687.0	17
19			16958636219	18
20			-99.255	19
21			1	20

system is indistinguishable from the measured data. The coefficients are given in Table 13.4. From these coefficients, one can calculate the cut off frequency to be approximately 29 Hz.

Our final example consists of a Rao–Garnier benchmark [4]. The transfer function is given by

$$H(s) = \frac{-6400S + 1600}{S^4 + 5S^3 + 408S^2 + 416S + 1600} \tag{13.3.2}$$

As in [4], a Gaussian white noise input signal was simulated which was then essentially filtered by this transfer function to obtain the output signal. A time step of 10 ms was used with 7016 samples. The response in the time domain of this linear system is calculated using the FFT algorithm as discussed in Sect. 1.11. A white Gaussian random noise signal was then added to the output time domain signal such that the signal to noise ratio was 10 dB. This output signal was then used to estimate the transfer function of the white noise input to the noise-polluted output. The estimation method discussed above was then used to identify the system.

The results are shown in Fig. 13.5 and Table 13.5.

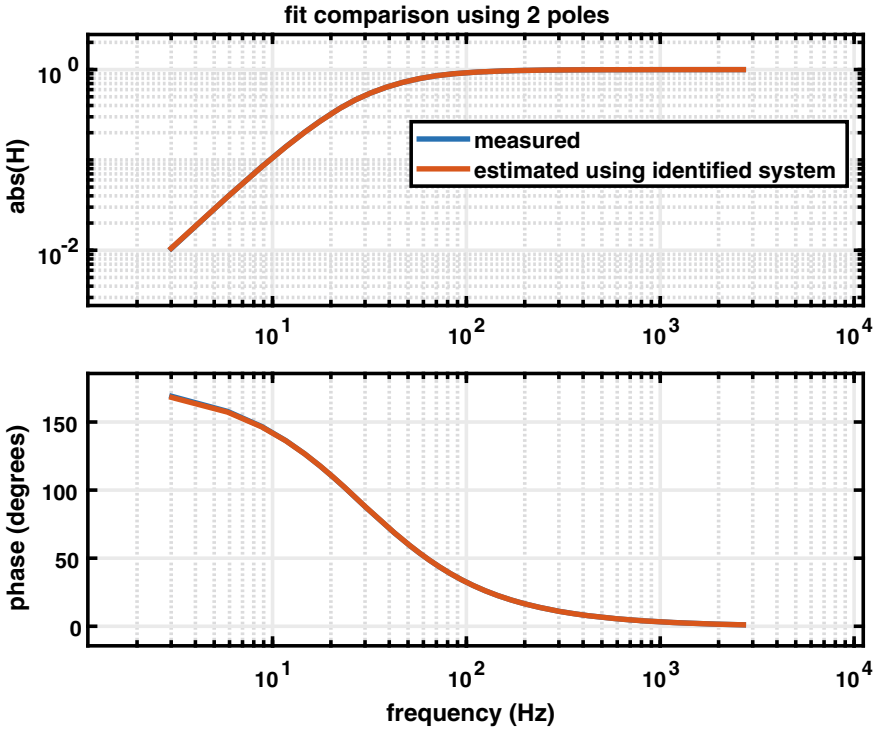


Fig. 13.4 Comparison of measured and estimated frequency response of a two-pole active high-pass filter. The response of the identified system is indistinguishable from the measured results

Table 13.4 Coefficients for the data shown in Fig. 13.4

i	a_i	α_i	b_i	β_i
1	1	2	33395	0
2			364.76	1
3			1	2

Table 13.5 Coefficients for the data shown in Fig. 13.5

i	a_i	α_i	b_i	β_i
1	1527	0	1546.2	0
2	−5127.5	1	330.05	1
3			394.75	2
4			4.0606	3
5			1	4

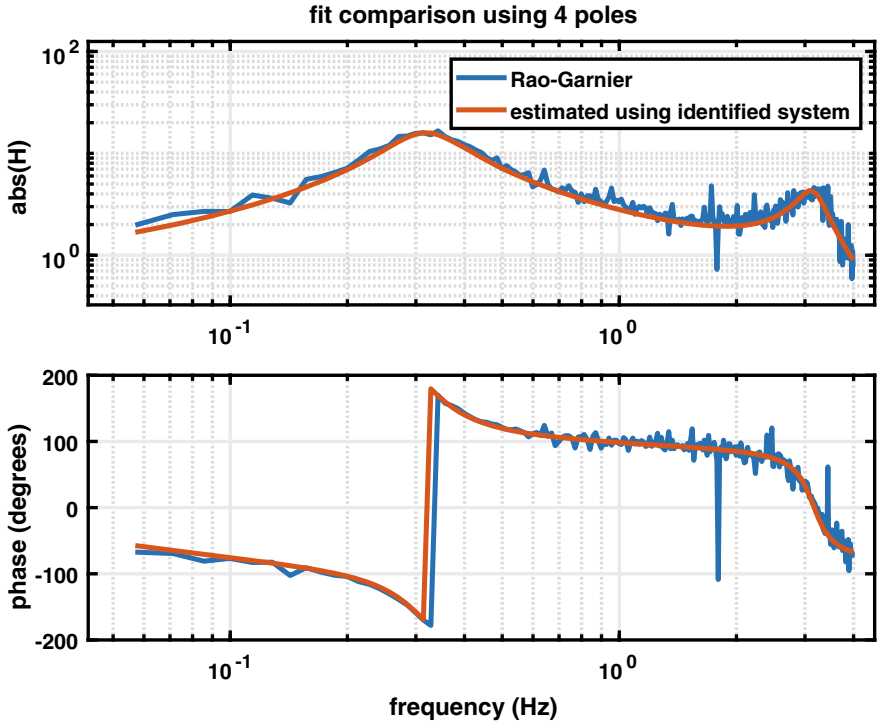


Fig. 13.5 The results obtained from the identified system are in excellent agreement with those of the RaoGarnier benchmark, used to assess the effectiveness of system identification approaches

13.4 Problems

1. An electrodynamic device has been driven with broadband noise and the transfer function between the input signal and response has been saved as "shakertest32216ForFit.mat". Click on this to load this file into the Matlab workspace. Use the Matlab script "complexregressionwithtimedelay.m" to obtain a model of the system. Note that, depending on the computer being used, this script can take a few minutes to run. The results should be as shown in Fig. 13.6. Note that you must use the correct polynomial orders and use the proper delay to obtain an accurate fit. If the delay isn't correct, the phase plot of the identified system will not agree with the phase of the measured system.

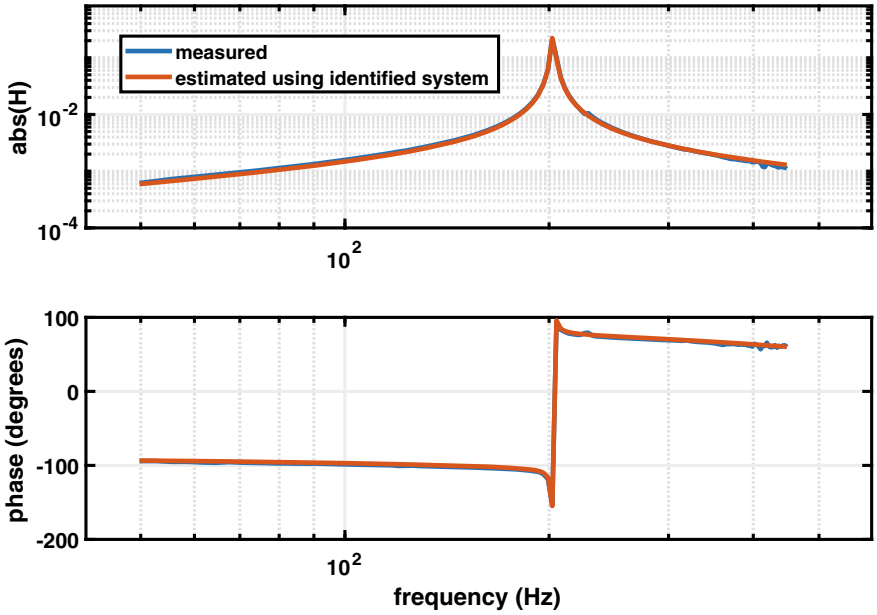


Fig. 13.6 Measured and estimated frequency response for the electrodynamic device of Problem 1

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Appendix A
The Use of Complex Notation

In acoustics, it is extremely helpful to represent harmonic signals with frequency ω through the use of complex notation where the time dependence is expressed in the form $e^{i\omega t}$ where $i = \sqrt{-1}$. The justification for this notation follows from the tremendous simplification it provides in characterizing harmonic signals. The notation must, however, be used carefully to avoid unphysical results.

To illustrate the advantages of using complex notation when solving for the response of systems undergoing harmonic excitation, in the following we compare the particular solution for the response of a simple spring/mass/damper obtained using a typical real solution approach with that obtained using complex notation. This will show that the use of a complex exponential to represent the harmonic signal saves considerable algebra. The relationship between the imaginary and real parts of the solution may be interpreted to obtain the phase, or time lag between the input to the system and the response.

A simple spring/mass/damper with harmonic excitation at the frequency ω is described by

$$m\ddot{x} + kx + c\dot{x} = f\sin(\omega t). \tag{A.1}$$

Dividing by m enables us to express this in terms of the undamped natural frequency ω_0 and the damping ratio ζ,

$$\ddot{x} + \omega_0^2 x + 2\omega_0\zeta\dot{x} = \frac{f}{m}\sin(\omega t) \tag{A.2}$$

where $\omega_0 = \sqrt{k/m}$ and $\zeta = \frac{c}{2\sqrt{km}}$.

When the effects of initial conditions have died out, the response will be dominated by the particular solution to Eq. (A.2). The particular solution may be obtained by assuming a solution in the form of entirely real quantities as

$$x(t) = A\sin(\omega t) + B\cos(\omega t). \tag{A.3}$$

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Substituting Eq. (A.3) into Eq. (A.2) leads to

$$((\omega_0^2 - \omega^2)A - 2\omega_0\zeta\omega B - f/m)\sin(\omega t) + ((\omega_0^2 - \omega^2)B + 2\omega_0\zeta\omega A)\cos(\omega t) = 0. \tag{A.4}$$

In order for Eq. (A.4) to be satisfied for at time t, a page or so of algebra leads to solutions for the unknown constants,

$$A = \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + (2\omega_0\zeta\omega)^2} f/m, \quad B = -\frac{2\omega_0\zeta\omega}{(\omega_0^2 - \omega^2)^2 + (2\omega_0\zeta\omega)^2} f/m. \tag{A.5}$$

The solution is often easier to interpret if we re-write Eq. (A.3) in terms of the amplitude X_0 and the phase ϕ,

$$x(t) = X_0 \sin(\omega t + \phi). \tag{A.6}$$

The relationship between X_0 and ϕ and A and B may be obtained using a trigonometric identity in Eq. (A.6),

$$x(t) = X_0 \sin(\omega t + \phi) = X_0 \sin(\omega t)\cos(\phi) + X_0 \cos(\omega t)\sin(\phi). \tag{A.7}$$

Comparing Eqs. (A.5) and (A.7) gives

$$X_0 = \frac{f/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\omega_0\zeta\omega)^2}} \quad \tan\phi = -\frac{2\omega_0\zeta\omega}{\omega_0^2 - \omega^2}. \tag{A.8}$$

The excitation and steady-state response can also be expressed in complex notation,

$$f(t) = fe^{i\omega t}, \quad x(t) = Xe^{i\omega t}, \tag{A.9}$$

where X is complex. The velocity is

$$u(t) = \dot{x}(t) = i\omega X e^{i\omega t} = Ue^{i\omega t}, \tag{A.10}$$

where $U = i\omega X$. The solution for X is obtained immediately by substituting into Eq. (A.1),

$$X = \frac{f}{k - \omega^2 m + i\omega c} = \frac{f/m}{\omega_0^2 - \omega^2 + i2\omega_0\zeta\omega}. \tag{A.11}$$

The complex magnitude, or modulus of X is

$$|X| = \sqrt{XX^*} = \frac{f/m}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\omega_0\zeta\omega)^2}}, \tag{A.12}$$

where * denotes the complex conjugate. This is identical to X_0 in Eq. (A.8), obtained using the real solution. The phase is obtained from the imaginary and real parts,

$$\tan\phi = -\frac{2\omega_0\zeta\omega}{\omega_0^2 - \omega^2}, \tag{A.13}$$

which is exactly as was obtained using the real solution in Eq. (A.8).

Complex notation can also be very convenient when examining damping, or energy dissipation in a damped harmonic oscillator. The rate of energy dissipated by the dashpot will equal the rate at which the applied force provides energy to the system. The work done by the applied force per cycle of period $T = 2\pi/\omega$ is

$$W_f = \int_0^T f \sin(\omega t)\dot{x}dt. \tag{A.14}$$

In terms of our complex representation, this may be shown to be

$$W_f = \frac{T}{2}\Re[UF^*] = \frac{T}{2}\Re[U^*F], \tag{A.15}$$

where * denotes the complex conjugate and $\Re[\cdot]$ denotes the real part. To see this, let $u(t) = \dot{x}(t) = \Re[Ue^{i\omega t}]$, where $U = U_r + iU_i$ and $f(t) = \Re[Fe^{i\omega t}]$ where $F = F_r + iF_i$. Equations (A.9), (A.10), and (A.14) then give

$$\begin{aligned} W_f = \int_0^T & \Re[(F_r + iF_i)(\cos(\omega t) + i\sin(\omega t))] \\ & \times \Re[(U_r + iU_i)(\cos(\omega t) + i\sin(\omega t))]dt \\ & = \frac{T}{2}(F_rU_r - F_iU_i) = \frac{T}{2}\Re[UF^*]. \end{aligned} \tag{A.16}$$

If we limit our attention to excitation at the undamped resonant frequency, $\omega = \sqrt{k/m}$, the work done by the force in one cycle becomes

$$W_f = \frac{T}{2}\frac{|F|^2}{c}. \tag{A.17}$$

The loss factor, η, which is equal to the inverse of the quality factor, Q, is defined to be

$$\eta = \frac{1}{Q} = \frac{1}{2\pi}\frac{\text{energy dissipated/cycle}}{\text{maximum energy stored/cycle}}. \tag{A.18}$$

Again, limiting our attention to the undamped resonant frequency, $\omega = \sqrt{k/m}$, the maximum energy stored in one cycle may be shown to be

$$\text{maximum energy stored/cycle} = \frac{1}{2}\frac{k|F|^2}{\omega_0^2 c^2} = \frac{|F|^2 m}{2c^2}. \tag{A.19}$$

Equation (A.17) through (A.19) give

$$\eta = \frac{c}{\sqrt{km}}, \quad Q = \frac{\sqrt{km}}{c}. \tag{A.20}$$

It is also common to quantify the damping using the damping ratio which is given by

$$\zeta = \frac{\eta}{2} = \frac{c}{2\sqrt{km}}. \tag{A.21}$$

This is, of course, the same familiar result one obtains by solving for the transient response.

Appendix B
Introduction to Probability and Random Processes

We are often concerned with the characterization of acoustic signals having an extremely complicated form and cannot be expressed explicitly. The signal will be referred to as "random" when it's precise time history cannot be reliably predicted.

Because any acoustic signal is always a function of time, say, $x(t)$, it is important to realize that any one sample (or record) is only one of a large set of possible functions that the signal could be. If one has the entire ensemble of all possible time records (or time histories) then it is possible to completely characterize the random function of time, $x(t)$. This random function is referred to as a random process.

So, given the ensemble of observed responses, $x_i(t)\ \forall i$, we can calculate various properties, or statistics to describe the process. We just can't make detailed assertions about the trajectories. For example, if $X_i(t)$ is the ith record, then the mean, $\mu_x(t_1)$, and mean square value, $\psi_x^2(t_1)$, at $t = t_1$ are given by

$$\mathrm{E}[X(t_1)] = \mu_x(t_1) = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} X_i(t_1), \tag{B.1}$$

and

$$\mathrm{E}[X^2(t_1)] = \psi_x^2(t_1) = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} X_i^2(t_1). \tag{B.2}$$

Another useful statistic is the autocorrelation function between the response at t_1 and the response at t_2,

$$\begin{aligned} R_{xx}(t_1, t_2) &= \mathrm{E}[X(t_1)X(t_2)] \\ &= \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} X_i(t_1)X_i(t_2). \end{aligned} \tag{B.3}$$

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4

If all statistics are unchanged when the time origin is shifted then the process is called **stationary**. Note that the term "stationary" refers to the statistics of the process, not the actual time histories which will obviously vary with time. Stationary random processes exist only in mathematical descriptions; all observed processes must be nonstationary. We will often assume stationarity for many problems in which the statistics remain constant over the period of observation. An example of an important nonstationary random process is the motion of the earth in an earthquake.

B.1 Ergodicity

For a stationary ergodic process, the statistics may usually be obtained from a single record of sufficient length, i.e.,

$$\mathrm{E}[X(t)] = \mathrm{E}[X] = \mu_x = \lim_{T\to\infty} \frac{1}{T} \int_0^T X_i(t)dt, \forall i, \tag{B.1}$$

and,

$$\mathrm{E}[X^2] = \psi_x^2 = \lim_{T\to\infty} \frac{1}{T} \int_0^T X_i^2(t)dt, \forall i, \tag{B.2}$$

where we have dropped the explicit time dependence in the expected value.

Since the process is stationary, the statistics must be independent of a shift in the time origin. Define a separation time $\tau = t_2 - t_1$ so that the autocorrelation function may be written as a function only of this separation time,

$$R_{xx}(t_1, t_2) = R_{xx}(t_1, t_1 + \tau) = R_{xx}(\tau), \tag{B.3}$$

since it must be independent of the absolute time, t_1. For a stationary process, the autocorrelation function may then be expressed in terms of the records making up the ensemble,

$$R_{xx}(\tau) = \lim_{N\to\infty} \frac{1}{N} \sum_{i=1}^{N} X_i(t_1)X_i(t_1 + \tau), \tag{B.4}$$

or, by using the ergodic assumption,

$$R_{xx}(\tau) = \lim_{T\to\infty} \frac{1}{T} \int_0^T X_i(t_1)X_i(t_1 + \tau)dt.\forall i. \tag{B.5}$$

In experimental situations, the ergodic assumption breaks down when there is some uncontrolled variable. Vibration measurements of a sample structure may not be representative of other structures that were manufactured in a similar manner.

B.2 Probability

Suppose we conduct an experiment N times and are interested in a particular outcome, A. If $N_{A,N}$ is the number of trials out of N in which A occurred, then the probability of A occurring is defined to be

$$\text{Prob}[A] = \mathcal{P}(A) = \lim_{N\to\infty} \frac{N_{A,N}}{N}. \tag{B.6}$$

B.3 Probability Distribution Function

Suppose that the event of interest is that the response at time t_1 is less than or equal to x, where x is some specified value (referred to as a "state variable"). The event A is then $X(t_1) \leq x$. The probability of this event is then

$$\text{Prob}[X(t_1) \leq x] = \lim_{N\to\infty} \frac{N_{A,N}}{N}. \tag{B.7}$$

$X(t_1)$ is a random variable. The result of Eq. (B.7) is clearly a function of x since if $x = \infty$ the result will equal unity and if $x = -\infty$ it will be zero. This result is referred to as a probability distribution function. Note that it will always take on values between zero and one. The probability distribution function is usually denoted by

$$F(x, t_1) = \text{Prob}[X(t_1) \leq x] = \mathcal{P}(X(t_1) \leq x), \forall t_1, \tag{B.8}$$

which will depend on the observation time t_1 for nonstationary data.

For stationary, ergodic data, $F(x, t_1) = F(x)$ and it can be obtained from a single record, $X_i(t)$, from the ensemble. For stationary data,

$$F(x) = \mathcal{P}(X_i(t_1) \leq x), \forall t_1, i. \tag{B.9}$$

The probability distribution function can also be expressed in terms of the total time in which the event was observed. If $\mathcal{T}[X_i(t) \leq x]_T$ is the total time in record i in which $X_i(t) \leq x$ in the time interval $0, T$, then

$$F(x) = \lim_{T\to\infty} \frac{\mathcal{T}[X_i(t) \leq x]_T}{T}. \tag{B.10}$$

B.4 Probability Density Functions

A more convenient probabilistic descriptor is the density function given by

$$p(x, t_1) = \frac{dF(x, t_1)}{dx}, \tag{B.11}$$

which is just the slope of the distribution function. Again, for stationary data,

$$p(x) = \frac{dF(x)}{dx}. \tag{B.12}$$

Since the slope of the distribution function is zero at $x = \pm\infty$, $p(x = \pm\infty) = 0$. Also, from Eq. (B.12),

$$F(x_2, t_1) - F(x_1, t_1) = \int_{x_1}^{x_2} p(x, t_1)dx. \tag{B.13}$$

If we let $x_2 = \infty$ and $x_1 = -\infty$, $F(\infty, t_1) = 1$, and $F(-\infty, t_1) = 0$). Equation (B.13) then shows that the area under the probability density function must be unity,

$$1 = \int_{-\infty}^{\infty} p(x, t_1)dx. \tag{B.14}$$

Now suppose that we are interested in the probability of observing the random process at time $t = t_1$, $X(t_1)$, between two levels, say x_1 and x_2. This could be written as an integral over a portion of the probability density function $p(x)$.

$$\begin{aligned} &\text{Prob}[x_1 \leq X(t_1) \leq x_2] \\ &\quad = \left(\lim_{N\to\infty} N_{X(t_1)\leq x_2, N} - N_{X(t_1)\leq x_1, N} \right) \frac{1}{N} \\ &\quad = F(x_2, t_1) - F(x_1, t_1) = \int_{x_1}^{x_2} p(x, t_1)dx. \end{aligned} \tag{B.15}$$

B.5 Expected Values in Terms of Probability Density Functions

The expected value of a function $g(x)$ is defined by

$$\text{E}[g(x)] = \int_{-\infty}^{\infty} g(x)p(x)dx, \tag{B.16}$$

where $p(x)$ is a probability density function, or a weighting function that measures the probability of x occurring. If the probability density function $p(x)$ is known for a random process $X(t)$ with sample records $X_i(t)$ then we can calculate the expected values rather easily. Note that the expected value of a function is a linear integral operator. This means that

$$\mathrm{E}[g_1(x) + g_2(x)] = \mathrm{E}[g_1(x)] + \mathrm{E}[g_2(x)], \tag{B.17}$$

and

$$\mathrm{E}[cg(x)] = c\mathrm{E}[g(x)], \tag{B.18}$$

where c is a constant. There are very important expected values in which $g(x) = x^n$ so that

$$\mathrm{E}[g(x)] = \mathrm{E}[x^n] = \int_{-\infty}^{\infty} x^n p(x)dx \tag{B.19}$$

$\mathrm{E}[x^n]$ is referred to as the nth order moment. The two most important moments are the mean given by

$$\mathrm{E}[x] = \int_{-\infty}^{\infty} xp(x)dx = \mu_x, \tag{B.20}$$

and

$$\mathrm{E}[x^2] = \int_{-\infty}^{\infty} x^2 p(x)dx = \psi_x^2 \tag{B.21}$$

is the mean square. ψ_x is the RMS response.

It is also often helpful to define "central moments" that are similar to those mentioned above but are shifted by the mean μ_x, i.e., $g(x) = (x - \mu_x)^n$. The nth central moment is usually denoted by K_n,

$$K_n = \mathrm{E}[(x - \mu_x)^n] = \int_{-\infty}^{\infty} (x - \mu_x)^n p(x)dx. \tag{B.22}$$

Note that $K_1 = 0$ and

$$K_2 = \int_{-\infty}^{\infty} (x - \mu_x)^2 p(x)dx = \sigma^2, \tag{B.23}$$

where σ^2 is the variance, the mean square deviation about the mean. The square root of the variance, σ is the standard deviation about the mean. In many important random processes, the mean is zero so that the variance is equal to the mean square response.

B.6 The Characteristic Function

The characteristic function is defined to be the inverse Fourier transform of the probability density function,

$$M_X(\theta) = \int_{-\infty}^{\infty} e^{i\theta x} p(x) dx, \tag{B.24}$$

where θ is a real parameter. Upon comparing Eqs. (B.16) and (B.24), the characteristic function may be considered to be the expected value of the exponential function, $M_X(\theta) = \mathrm{E}[e^{i\theta x}]$. A useful feature of the characteristic function is that if we expand the exponential in a Taylor's series, one can extract all of the nth order moments. Expanding the exponential and using the linearity of the expected value operation gives

$$\begin{aligned} M_X(\theta) &= \mathrm{E}[e^{i\theta x}] \\ &= \left[\sum_{n=0}^{\infty} \frac{(i\theta x)^n}{n!}\right] \\ &= \sum_{n=0}^{\infty} (i\theta)^n \mathrm{E}[x^n] \\ &= 1 + i\theta \mathrm{E}[x] + \frac{(i\theta)^2}{2} \mathrm{E}[x^2] + \ldots. \end{aligned} \tag{B.25}$$

From Eq. (28) we can note that

$$\mathrm{E}[x] = \frac{1}{i} \frac{\mathrm{d}M_X(\theta)}{\mathrm{d}\theta}|_{\theta=0} = \frac{1}{i} \int_{-\infty}^{\infty} ix e^{i\theta 0} p(x) dx, \tag{B.26}$$

and,

$$\mathrm{E}[x^n] = \frac{1}{i^n} \frac{\mathrm{d}^n M_X(\theta)}{\mathrm{d}^n \theta}|_{\theta=0} = \int_{-\infty}^{\infty} x^n p(x) dx. \tag{B.27}$$

Note that the complete characterization of the probability density in terms of the moments requires moments of all orders. That is, suppose we have an ensemble and would like to describe the probability density by measuring all of the moments, $\mathrm{E}[x^n]$. In general, it takes all of the $\mathrm{E}[x^n]$ to complete the description. An extremely fortunate feature of the Gaussian probability density (to be discussed in the following) is that it requires knowledge of only $\mathrm{E}[x]$ and $\mathrm{E}[x^2]$, i.e. only moments up to second order.

B.7 The Log Characteristic Function

Another popular descriptor of the statistics of a random process is the log characteristic function, which is simply the natural log of the characteristic function, $ln(M_X(\theta))$. Expanding the log characteristic function about $\theta = 0$ gives

$$\begin{aligned} ln(M_X(\theta)) &= \sum_{n=0}^{\infty} \frac{\theta^n}{n!} \frac{\mathrm{d}^n ln(M_X(\theta))}{\mathrm{d}^n \theta}|_{\theta=0} \\ &= \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} k_n(X), \end{aligned} \tag{B.28}$$

where

$$k_n = \frac{1}{i^n} \frac{\mathrm{d}^n ln(M_X(\theta))}{\mathrm{d}^n \theta}|_{\theta=0} \tag{B.29}$$

which is referred to as the nth cumulant. By applying Eq. (B.29) for $n = 1$ and $n = 2$ it may be shown that

$$k_1 = \mathrm{E}[x], \quad \text{and} \quad k_2 = \mathrm{E}[x^2] - \mu_x^2. \tag{B.30}$$

B.8 The Gaussian Probability Density

The vast majority of data seen in the random response of vibrating systems has a Gaussian probability density given by

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu_x)^2/(2\sigma^2)}, \tag{B.31}$$

where, as before, μ_x and σ are the mean and variance. It is often helpful to write the probability density in terms of a normalized variable, z, where

$$z = \frac{x - \mu_x}{\sigma}, \tag{B.32}$$

so that the density expressed as a function of z becomes

$$p(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}. \tag{B.33}$$

The importance of the Gaussian probability density can be appreciated by considering the central limit theorem. Most random processes come about from the superposition of a large collection of **independent** noise sources. Common examples are the noise due to air flow and thermal noise in resistors. Consider a random signal $X(t)$ to be composed of a set of independent signals $X_i(t)$, for $i = 1, 2, \ldots, n$,

$$X(t) = X_1(t) + X_2(t) + X_3(t) + X_4(t) + \cdots + X_n(t). \tag{B.34}$$

The central limit theorem states that if:

(1) the $X_i(t)$ $\forall i$ are **independent**, (we have not yet precisely defined independent but it means what is sounds like it means),
(2) $X(t)$ is not overly influenced by an individual $X_i(t)$, and
(3) the variance and higher order moments are finite,

then $X(t)$ will have a Gaussian probability density as $n \to \infty$ regardless of the probability densities of the individual $X_i(t)$. For proof and details see p. 68 of [1]. The central limit theorem is a remarkable and extremely important feature of random signals. We will employ this result later when we discuss methods of simulating random signals.

B.9 Properties of Gaussian Random Variables

From Eqs. (B.31) or (B.33) it is evident that the Gaussian probability density function depends on only two moments, or on only μ_x and σ. In terms of the normalized variable z, the nth moment is

$$\mathrm{E}[z^n] = \int_{-\infty}^{\infty} \frac{z^n}{\sqrt{2\pi}} e^{-z^2/2} dz. \tag{B.35}$$

If n is odd, $\mathrm{E}[x^n] = 0$. If n is positive and even it can be shown that

$$\mathrm{E}[x^n] = 1 \times 3 \times 5 \times \cdots (n-1) = \frac{2^{n/2}}{\sqrt{\pi}} \Gamma\left(\frac{n+1}{2}\right), \tag{B.36}$$

where $\Gamma(\cdot)$ is the gamma function.

Having Eq. (B.36) for a normalized variable, one can find the central moments, $\mathrm{E}[(x - \mu_x)^n]$. If we use Eq. (B.32),

$$x - \mu_x = z\sigma, \tag{B.37}$$

so that if n is positive and even the central moments are

$$\mathrm{E}[(z\sigma)^n] = \sigma^n \mathrm{E}[z^n] = \frac{\sigma^n 2^{n/2}}{\sqrt{\pi}} \Gamma\left(\frac{n+1}{2}\right). \tag{B.38}$$

Again, note that the central moments will be zero for n odd.

For Gaussian random variables there is a useful recursion relation that allows us to calculate higher order moments,

$$\mathrm{E}[x^n] = \mu_x \mathrm{E}[x^{n-1}] + (n-1)\sigma^2 \mathrm{E}[x^{n-2}], \quad \text{for} \quad n > 1. \tag{B.39}$$

B.10 Jointly Distributed Random Variables

It is usually necessary to describe the response of a system in terms of more than one random variable. While the probability density as discussed above is extremely important, many Gaussian random signals may have the same probability density but may differ greatly in many important ways. We will see later that a structure having a response that is a "broadband" Gaussian random process may have the same μ_x and σ as that responding as a "narrowband" Gaussian random process. However, other features of these signals may have a huge influence on fatigue or durability of the structure. We will see later that it is often necessary to describe the statistical relationship between the responses at two different times, say t_1 and t_2. Certainly, if t_1 and t_2 are close to each other then $x(t_2)$ should be fairly predictable if $x(t_1)$ is known. The predictability decays as $t_2 - t_1$ increases.

Along with considering the relationship between the responses at two times, we will need to examine the joint statistics of the displacement and derivatives of it such as the velocity and the acceleration at a given time.

To account for this higher order probability structure, we need to define the joint probability density. To do this we should return to the discussion of probability. In this case the event A of Eq. (B.6) is replaced by the intersection of two events, say A and B, $A \cap B$,

$$\mathrm{Prob}[A \cap B] = \mathcal{P}(A) = \lim_{N\to\infty} \frac{N_{A\cap B,N}}{N}. \tag{B.40}$$

Let the events A and B be $A = X_1(t) \le x_1$, and $B = X_2(t) \le x_2$. We can then define a joint probability distribution function,

$$F_{X_1X_2}(x_1, x_2) = \mathrm{Prob}[(X_1(t) \le x_1) \cap (X_2(t) \le x_2)]. \tag{B.41}$$

Note that as both $x_1, x_2 \to \infty$, the probability approaches unity and as x_1, or $x_2 \to -\infty$, the probability approaches zero. Also,

$$F_{X_1X_2}(x_1, x_2) = F_{X_1}(x_1), \quad \text{for} \quad x_2 = \infty,$$
$$= F_{X_2}(x_2), \quad \text{for} \quad x_1 = \infty, \tag{B.42}$$

where $F_{X_1}(x_1)$ and $F_{X_2}(x_2)$ are the ordinary probability distribution functions. If $x_1 = -\infty$ or $x_2 = -\infty$, $F_{X_1X_2}(x_1, x_2) = 0$.

We can define the joint probability density function by differentiating the joint probability distribution function,

$$p_{X_1X_2}(x_1, x_2) = \frac{\partial^2 F_{X_1X_2}(x_1, x_2)}{\partial x_1 \partial x_2}. \tag{B.43}$$

We can integrate the density to obtain the probability,

$$F_{X_1X_2}(x_1, x_2) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} p_{X_1X_2}(x_1, x_2) \mathrm{d}x_2 \mathrm{d}x_1. \tag{B.44}$$

If x_1 and x_2 are infinite this joint density will have unit area.

Given the joint density, $p_{X_1X_2}(x_1, x_2)$, one can obtain the individual densities by integrating,

$$p_{X_1}(x_1) = \int_{-\infty}^{\infty} p_{X_1X_2}(x_1, x_2) \mathrm{d}x_2, \tag{B.45}$$

and

$$p_{X_2}(x_2) = \int_{-\infty}^{\infty} p_{X_1X_2}(x_1, x_2) \mathrm{d}x_1. \tag{B.46}$$

The probability densities $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$ are called the "marginal" densities corresponding to the joint density $p_{X_1X_2}(x_1, x_2)$.

As an exercise, suppose we have a joint density given by

$$p_{XY}(x, y) = \frac{x(1 + 3y^2)}{4}, \quad 0 \le x \le 2 \quad 0 \le y \le 1,$$

and $p_{XY}(x, y) = 0$ elsewhere. Verify that

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} p_{XY}(x, y) \mathrm{d}x \mathrm{d}y = 1,$$

and find the marginal densities. (Answer: $p_X(x) = x/2$ and $p_Y(y) = \frac{1+3y^2}{2}$.)

If we have n jointly normal (Gaussian) random variables, X_i, $i = 1, 2, \ldots, n$, then the joint Gaussian probability density is given by

$$p(\bar{x}) = \frac{1}{(2\pi)^{n/2}|\mathcal{K}|^{1/2}} e^{-\frac{1}{2}(\bar{x}-\bar{\mu})^{\mathrm{T}}[\mathcal{K}](\bar{x}-\bar{\mu})}, \tag{B.47}$$

where

$$\bar{x} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix} \tag{B.48}$$

$$[\mathcal{K}] = \mathrm{E}[(\bar{x} - \bar{\mu})^{\mathrm{T}}(\bar{x} - \bar{\mu})], \tag{B.49}$$

is the "covariance matrix" and

$$|\mathcal{K}| = \det[\mathcal{K}]. \tag{B.50}$$

B.11 Conditional Probability Density

Knowing the joint probability density, we can examine the probability of having a response, say x_2 at time t_2 on the condition that the response at t_1 be x_1. Returning to our earlier definition of probability, this conditional probability can be viewed as the number of times both events occurred out of the total number of times one event, say event A occurred. The probability of the event B occurring conditionally on the occurence of the event A is then

$$\begin{aligned} \mathrm{Prob}[B|A] &= \lim_{N\to\infty} \frac{N_{A\cap B,N}}{N_{A,N}} \\ &= \lim_{N\to\infty} \frac{(N_{A\cap B,N})/N}{(N_{A,N})/N} \\ &= \frac{\mathrm{Prob}[A \cap B]}{\mathrm{Prob}[A]}. \end{aligned} \tag{B.51}$$

We say that the events A and B are **independent** if

$$\mathrm{Prob}[B|A] = \mathrm{Prob}[B]. \tag{B.52}$$

That is, the probability of B occuring doesn't depend on the probability of A. This also implies that for independent events A and B,

$$\mathrm{Prob}[A \cap B] = \mathrm{Prob}[A] \times \mathrm{Prob}[B]. \tag{B.53}$$

Analogous to our development of the joint probability density, we can define a conditional probability density for two variables x and y as

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x, y)}{p_Y(y)}, \tag{B.54}$$

where $p_{X,Y}(x, y)$ is the joint probability density.

If X and Y are independent random variables then

$$p_{X|Y}(x|y) = p_X(x), \tag{B.55}$$

so that Eq. (B.54) gives

$$p_{X,Y}(x, y) = p_X(x)p_Y(y). \tag{B.56}$$

B.12 Functions of Random Variables

There are numerous situations where we may know the probability density, $p(x)$ of a random variable X and would like to know the probability density, $p(y)$ of a random variable Y where $y = f(x)$ with $f(\cdot)$ being some reasonably well-behaved function. The existence of this probability density is not guaranteed but if we again return to our discussion of probability and imagine an event $Y \leq y$, the probability of this event is given by the probability distribution function,

$$\begin{aligned} F_Y(y) = \text{Prob}(Y \leq y) &= \text{Prob}(f(X) \leq y) \\ &= \int_{f(x)\leq y} p_X(x)\mathrm{d}x, \end{aligned} \tag{B.57}$$

where the domain of integration is only over the region in which $f(x) \leq y$. If there is a one to one mapping from x to y then we can find a function $g(y)$ which is the inverse of $f(x)$ so that $x = g(y)$. If we could somehow express the integral over x in Eq. (B.57) as an integral over y in the form

$$F_Y(y) = \int_{-\infty}^{y} p_Y(y)\mathrm{d}y, \tag{B.58}$$

then $p_Y(y)$ would be the probability density of the function of x.

The integral in Eq. (B.58) can be set up if $f(\cdot)$ is a monotonic function of x. First consider the case where it is monotonically increasing. In this case the domain in which $f(x) \leq y$ can be described by $x \leq g(y)$. So, for $f(x)$ monotonically increasing in x,

$$F_Y(y) = \int_{-\infty}^{x} p_X(x)\mathrm{d}x = \int_{-\infty}^{g(y)} p_X(x)\mathrm{d}x. \tag{B.59}$$

If we use the fact that

$$\mathrm{d}x = \frac{\mathrm{d}g}{\mathrm{d}y}\mathrm{d}y, \tag{B.60}$$

then Eq. (B.59) becomes

$$F_Y(y) = \int_{-\infty}^{x} p_X(x)\mathrm{d}x = \int_{-\infty}^{y} p_X(g(y))\frac{\mathrm{d}g}{\mathrm{d}y}\mathrm{d}y, \tag{B.61}$$

where we have changed the upper limit of integration because we have changed the variable of integration from x to y.

Comparing Eqs. (B.58) and (B.61) we can see the relation between the probability densities of x and y,

$$p_Y(y) = p_X(g(y))\frac{\mathrm{d}g}{\mathrm{d}y}. \tag{B.62}$$

Note that if $f(x)$ is monotonically decreasing, then $\frac{\mathrm{d}g}{\mathrm{d}y} \leq 0$. To account for this we must modify the result of Eq. (B.62) by taking the absolute value,

$$p_Y(y) = p_X(g(y))|\frac{\mathrm{d}g}{\mathrm{d}y}|. \tag{B.63}$$

As an example, consider the relation between two random variables given by

$$Y = aX + b, \tag{B.64}$$

so that $f(x) = ax + b$. We are given $p_X(x)$ and would like to know the probability density for y. Inverting $f(x)$ gives $g(y) = \frac{y-b}{a}$ and $\frac{\mathrm{d}g}{\mathrm{d}y} = 1/a$. The probability density in terms of y is then obtained by applying Eq. (B.63),

$$p_Y(y) = p_X\left(\frac{y-b}{a}\right)|1/a|. \tag{B.65}$$

B.13 Estimating Probability Densities from Discrete Data

Since knowing the probability density for a given signal, $X(t_j)$, for $j = 1, 2, \ldots, n$ can permit the calculation of numerous statistics we need a reasonably efficient means of estimating it given discrete data. To estimate the probability density we should

divide the entire range of levels we expect the signal to lie in into a set of n_{bin} number of bins. If p_i is the estimate of the probability density corresponding to the ith bin, then Eq. (B.14) (for a stationary process where we can drop the time dependence) becomes

$$1 = \int_{-\infty}^{\infty} p(x)dx = \lim_{n_{\text{bin}} \to \infty} \sum_{i=1}^{n_{bin}} p_i \Delta x_{\text{bin}}, \tag{B.66}$$

where

$$\Delta x_{\text{bin}} = \frac{x_{\max} - x_{\min}}{n_{\text{bin}}}, \tag{B.67}$$

and $x_{\max}$ and $x_{\min}$ are the maximum and minimum values of the data. Our discretization of the range of values of x means that we consider

$$p(x_i) = p_i, \tag{B.68}$$

where

$$x_i = \Delta x_{\text{bin}}(i-1) + x_{\min}. \tag{B.69}$$

From Eqs. (B.67) and (B.69) we can see that $x_1 = x_{\min}$ and $x_{n_{\text{bin}}+1} = x_{\max}$.

As in Eq. (B.15), the probability of finding a value of $X(t_j)$ within one of the bins is

$$\text{Prob}[x_i \leq X \leq x_{i+1}] = \int_{x_i}^{x_{i+1}} p(x)dx = p_i \Delta x_{\text{bin}}, \tag{B.70}$$

for $i = 1, \ldots, n_{\text{bin}}$. This probability can be estimated as in Eq. (B.6) where the event A is

$$A = x_i \leq X(t_j) \leq x_{i+1}. \tag{B.71}$$

One could construct a loop (or set of loops) through the data to count the number of times the events occurred. It may be more efficient, however, to define an array, $bin(i)$ in which we accumulate the number of times the event occurred. If $bin(i)$ is initialized to zero, let

$$bin(i) \to bin(i) + 1, \tag{B.72}$$

where the index i is obtained from Eq. (B.69) with the discrete data $X(t_j)$ substituted for x_i,

$$i = (X(t_j) - x_{\min})/\Delta x_{\text{bin}} + 1. \tag{B.73}$$

The index i must be an integer which can be ensured in Matlab using the "round" function. Equations (B.72) and (B.73) are evaluated for $j = 1, 2, \ldots, n$. Once the array $bin(i)$ has been used to count the number of times the signal took on values corresponding to each bin, the probability corresponding to each bin level will be the ratio of $bin(i)$ to the number of observations, n. Equation (B.70) will then give,

$$p_i \Delta x_{\text{bin}} = \frac{bin(i)}{n}, \tag{B.74}$$

so that

$$p_i = \frac{bin(i)}{n \Delta x_{\text{bin}}}. \tag{B.75}$$

Appendix C
The Mean Square Response of a Spring/Mass/Damper

In this appendix, we calculate the mean square response of a spring/mass/damper that is driven by weakly stationary white random excitation.

C.1 Response of a One Degree of Freedom System

Consider a single degree of freedom system governed by

$$m\ddot{x} + kx + c\dot{x} = f(t), \tag{C.1}$$

where m, k, and c are constants, $x(t)$ is the response, and $f(t)$ is the applied force. The initial conditions may be expressed as $x(0) = x_n$, and $\dot{x}(0) = v_0$, where x_0 and v_0 are given constants. The complete solution to Eq. (1) for $x(t)$ can be obtained by a variety of methods. A convenient approach is to take the Laplace transform of Eq. (C.1), where the Laplace transform of $x(t)$ is defined to be

$$\mathcal{L}[x(t)] = X(s) = \int_0^\infty x(t)e^{-st}dt. \tag{C.2}$$

It is helpful to first re-write Eq. (1) by dividing by m

$$\ddot{x} + \omega_0^2 x + 2\omega_0\zeta\dot{x} = f(t)/m, \tag{C.3}$$

where $\omega_0 = \sqrt{k/m}$, and $2\omega_0\zeta = c/m$. ζ is called the damping ratio of the system. If $\zeta < 1$ the system is underdamped (oscillatory response), if $\zeta = 1$ it is critically damped (maximum decay rate), and if $\zeta > 1$ it is overdamped (non-oscillatory response). The Laplace transform of Eq. (C.3) is

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4

$$s^2X(s) - sx_0 - v_0 + \omega_0^2 X(s) + 2\omega_0\zeta sX(s) - 2\omega_0\zeta x_0 = F(s), \tag{C.4}$$

where

$$F(s) = \mathcal{L}[f(t)/m] = \int_0^\infty \frac{f(t)}{m} e^{-st} dt. \tag{C.5}$$

Solving Eq. (C.4) for $X(s)$ gives

$$X(s) = \frac{F(s)}{s^2 + \omega_0^2 + s2\omega_0\zeta} + \frac{(s + 2\omega_0\zeta)x_0}{s^2 + \omega_0^2 + s2\omega_0\zeta} + \frac{v_0}{s^2 + \omega_0^2 + s2\omega_0\zeta}. \tag{C.6}$$

The solution for the response in the time domain, $x(t)$, is obtained by taking the inverse Laplace transform of Eq. (C.6). Since the Laplace transform is a linear integral operator, it is possible to obtain the inverse transforms of each of the three terms in Eq. (C.6) individually and then add the results together. This gives

$$x(t) = \int_0^t \frac{f(\tau)}{m} h(t - \tau) d\tau + \frac{x_0}{\sqrt{1 - \zeta^2}} e^{-\zeta\omega_0 t} \sin(\omega_d t + cos^{-1}(\zeta)) + v_0 h(t), \tag{C.7}$$

where

$$h(t) = \frac{e^{-\zeta\omega_0 t}}{\omega_d} \sin(\omega_d t) \tag{C.8}$$

is the impulse response and

$$\omega_d = \omega_0\sqrt{1 - \zeta^2} \tag{C.9}$$

is the damped natural frequency. Note that in Eq. (C.7) we have used the fact that the impulse response is zero for negative arguments, $h(t) = 0$ for $t < 0$.

If the system is subjected to an ideal impulse at $t = 0$ and is initially at rest so that $x_0 = v_0 = 0$ then we can let $f(t) = \delta(t)$, where $\delta(t)$ is the Dirac delta function. Equation (C.7) then gives

$$x(t) = h(t). \tag{C.10}$$

Note that we obtain the same result if the system is unforced so that $f(t) = 0$ and let the initial conditions be $x_0 = 0$ and $v_0 = 1$.

In the case were the parameters of the system, k, m, and c, in Eq. (C.1), are simple constants, the exact solution for the response can always be obtained as Eq. (C.7) above. In cases where the excitation $f(t)$, takes on a complicated form, the convolution integral in Eq. (C.7) can be very difficult to evaluate. In this situation, it is advantageous to employ a numerical solution technique such as the central difference method.

We will often be interested in the response long after the effects of the initial conditions have died out so that the terms involving the initial conditions in Eq. (C.7) may be neglected, giving

$$x(t) = \int_0^t \frac{f(\tau)}{m} h(t-\tau) d\tau. \tag{C.11}$$

C.2 Mean Square Response by Time Domain Integration

We often need to know the mean square response due to weakly stationary white noise excitation. The mean square response is the average of the squared response, $E[x(t)^2]$, where $E[\cdot]$ denotes the expected value. White noise is defined to be a random process where the autocorrelation function is a Dirac delta function,

$$R_{ff}(\tau) = E[f(t_1) f(t_2)] = 2\pi K \delta(t_2 - t_1), \tag{C.12}$$

where K is a constant that is equal to the double-sided power spectral density of the force with units of newtons^2/radian. This means that the signal at any instant of time is totally uncorrelated with the signal at any other instant of time. The mean square of $x(t)$ may be computed from Eq. (C.11),

$$\begin{aligned} E[x(t)^2] &= E\left[\left(\int_0^t \frac{f(\tau_1)}{m} h(t-\tau_1) d\tau_1\right)\left(\int_0^t \frac{f(\tau_2)}{m} h(t-\tau_2) d\tau_2\right)\right] \\ &= \int_0^t \int_0^t E[f(\tau_1) f(\tau_2)] \frac{1}{m^2} h(t-\tau_1) h(t-\tau_2) d\tau_1 d\tau_2. \end{aligned} \tag{C.13}$$

Equation (C.12) enables us to carry out one of the integrals in Eq. (C.13),

$$E[x(t)^2] = \frac{2\pi K}{m^2} \int_0^t h^2(t-\tau) d\tau. \tag{C.14}$$

The integration can be simplified by making the substitution, $u = t - \tau$,

$$E[x(t)^2] = \frac{2\pi K}{m^2}\int_0^t h^2(u)du = \frac{2\pi K}{m^2}\int_0^t \frac{e^{-2\zeta\omega_0 u}}{\omega_d^2}\sin^2(\omega_d u)du$$
$$= -\frac{2\pi K}{m^2}\frac{\mathrm{e}^{-2\omega_0\zeta t}\left({\omega_d}^2 + \omega_0^2{\zeta}^2 - \omega_0^2{\zeta}^2\cos(2\omega_d t) - {\omega_d}^2\mathrm{e}^{2\omega_0\zeta t} + \omega_d\omega_0\zeta\sin(2\omega_d t)\right)}{4\omega_d^2\omega_0\zeta\left({\omega_d}^2 + \omega_0^2{\zeta}^2\right)}. \tag{C.15}$$

Taking the limit as $t \to \infty$ gives

$$E[x(t)^2] = \frac{\pi K}{2m^2\omega_0^3\zeta}. \tag{C.16}$$

C.3 Frequency Domain Approach

The mean square response given above may also be calculated by integrating the power spectral density of the response in the frequency domain. The frequency response is obtained by taking the Fourier transform of the impulse response. This becomes

$$H(\omega) = \frac{1}{\omega_0^2 - \omega^2 + 2\omega_0 i\omega}. \tag{C.17}$$

Note that this is equivalent to replacing the Laplace transform variable s with $i\omega$. The mean square response may be obtained by integrating the power spectral density of $x(t)$ over all frequencies,

$$E[x^2(t)] = \frac{1}{m^2}\int_{-\infty}^{\infty} H(\omega)H^*(\omega)K\,d\omega, \tag{C.18}$$

where * denotes the complex conjugate.

The integration in Eq. (C.18) is easiest to carry out with help from the residue theorem. If we have a complex function F of a complex variable z, the integral around a closed contour encompassing the upper half plane is equal to $2\pi i$ multiplied by the sum of the residues of F in the upper half plane,

$$\oint F(z)dz = 2\pi i \times \sum_i Res_i[F(z), z_i]. \tag{C.19}$$

In our case, Eq. (C.18) is a real integral over the real variable ω but this can be converted to a complex integral where ω is taken to be the real part of the complex variable z. The integration around the entire upper half plane consists of the integral along the real axis along with an integral around the entire plane at infinity. Fortu-

nately, in this problem, the integrand vanishes for infinite values of the integration variable so only the integral over the real axis contributes.

If we replace ω with the complex variable z, our contour integral may then be written as

$$\frac{1}{m^2}\oint H(z)H^*(z)K\,dz. \tag{C.20}$$

To determine the residues, we must first find the poles of $H(z)H^*(z)$ that are located in the upper half plane. First, we consider $H(z)$,

$$H(z) = \frac{1}{\omega_0^2 - z^2 + 2\omega_0 i z} = \frac{1}{(z - z_1)(z - z_2)}, \tag{C.21}$$

where z_1 and z_2 are the poles of $H(z)$. The poles are the roots of the denominator which are given by

$$z = \omega_0(\zeta i \pm \sqrt{1 - \zeta^2}), \tag{C.22}$$

so that

$$z_1 = \omega_0(-\sqrt{1 - \zeta^2} + \zeta i) \quad \text{and} \quad z_2 = \omega_0(\sqrt{1 - \zeta^2} + \zeta i). \tag{C.23}$$

Note that these are the only poles of interest in our integrand since the poles of $H^*(z)$ will be in the lower half plane. Having the poles of $H(z)$, we can write its conjugate in terms of its poles,

$$H^*(z) = \frac{1}{(z - z_1^*)(z - z_2^*)}. \tag{C.24}$$

The right hand side of Eq. (C.19) depends on the two poles z_1 and z_2,

$$\begin{aligned}
&\frac{2\pi i}{m^2} \times \sum_i Res_i[K H(z)H^*(z), z_i] = \\
&\frac{2\pi i K}{m^2}\left(\frac{z - z_1}{(z - z_1)(z - z_2)(z - z_1^*)(z - z_2^*)}|_{z=z_1}\right. \\
&\left. + \frac{z - z_2}{(z - z_1)(z - z_2)(z - z_1^*)(z - z_2^*)}|_{z=z_2}\right) \\
&= \frac{2\pi i K}{m^2}\left(\frac{1}{(z - z_2)(z - z_1^*)(z - z_2^*)}|_{z=z_1}\right. \\
&\left. + \frac{1}{(z - z_1)(z - z_1^*)(z - z_2^*)}|_{z=z_2}\right).
\end{aligned} \tag{C.25}$$

Substituting Eqs. (C.23) and simplifying gives

$$E[x^2(t)] = \frac{\pi K}{m^2 2\omega_0^3 \zeta} \tag{C.26}$$

as we obtained from the time domain integration in Eq. (C.16).

Appendix D
Analysis of Circuit Noise

In the following, we describe the analysis of electronic noise in a capacitive microphone preamplifier circuit. The circuit comprises a charge amplifier and utilizes the OPA657 operational amplifier. The power spectrum of the output noise is estimated based on the resistance and capacitance connected to either the plus or negative inputs and the output of the amplifier. The thermal noise of each resistor is included along with the input current and voltage noise of the operational amplifier.

D.1 General Operational Amplifier Noise Analysis

In order to construct a model of the output noise from a given amplifier design for microphone applications, we will first describe a reasonably general analysis of noise in operational amplifier circuits. This general model will then be used to estimate the output noise for the specific circuit values used in our design.

Figure D.1 shows a circuit that can be used to estimate the output noise from a fairly general linear operational amplifier circuit. In this circuit, the subscript f denotes the feedback loop connecting V_o to the minus input. The subscript m denotes the input components connected to the minus input and the subscript p denotes components connected to the positive input. The voltage sources, V_{tm}, V_{tp}, and V_{tf}, represent the equivalent thermal noise voltage of the resistors. These are uncorrelated, Gaussian white noise sources. The square root of the power spectral densities of these noise signals are given by

$$e_{tp} = \sqrt{4K_B T R_p}, \quad e_{tm} = \sqrt{4K_B T R_m},$$
$$e_{tf} = \sqrt{4K_B T R_f} \tag{D.1}$$

with units of volts/$\sqrt{\text{Hz}}$. $K_B = 1.38 \times 10^{-23}$ is Boltzmann's constant in joule/kelvin and T is the absolute temperature. Note that at room temperature $4K_B T \approx 1.6 \times 10^{-20}$ J. The nomenclature is listed in Table D.1.

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4

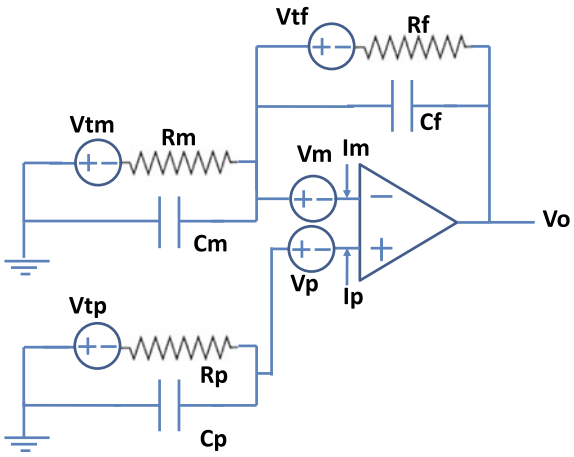


Fig. D.1 Circuit used to model sources of noise in a typical operational amplifier circuit. This model includes voltage and current noise at the inputs of the amplifier and voltage noises created by thermal noise in the resistors

Table D.1 Nomenclature

e_{no}^2	spectrum of V_o
Z_f	Thevenin impedance of the feedback circuit
Z_m	Thevenin impedance at the minus input
Z_p	Thevenin impedance at the plus input
e_{ep}^2	Noise spectrum due to V_{ep}
e_{em}^2	Noise spectrum due to V_{em}
e_{ef}^2	Noise spectrum due to V_{ef}
e_i^2	Amplifier current noise spectrum
e_v^2	Amplifier voltage noise spectrum

The circuit shown in Fig. D.1 also shows voltage and current sources, V_m, V_p, I_m, and I_p at the inputs of the operational amplifier. These represent the equivalent input noise signals for the amplifier. These are typically taken to be uncorrelated, weakly stationary random signals. The power spectra of these noise signals are normally specified by the manufacturer of the operational amplifier. Let e_{vm} and e_{vp} be the power spectra (in volts/$\sqrt{\text{Hz}}$) of V_m, and V_p and let e_{im} and e_{ip} be the power spectra (in amps/$\sqrt{\text{Hz}}$) of I_m, and I_p. The voltage noise spectra are normally specified as the total, $e_v^2 = e_{vm}^2 + e_{vp}^2$. The current noise spectra are assumed to be equal, $e_i = e_{im} = e_{ip}$. In order to obtain a model for the noise output of the circuit of Fig. D.1, it is helpful to use Thevenin equivalent circuits for each resistor/voltage source/capacitor combination.

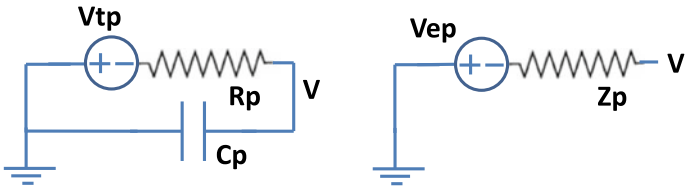


Fig. D.2 Thevenin equivalent circuit

The Thevenin equivalent circuit for the components connected to the positive input of the amplifier are shown in Fig. D.2. The Thevenin equivalent voltage source can be found by determining the voltage V in Fig. D.2. The impedance Z_p is determined by setting the voltage source V_{tp} to zero and determining the impedance of the parallel combination of R_p and C_p. These give

$$V_{ep} = \frac{V_{tp}}{1 + i\omega C_p R_p} \quad Z_p = \frac{R_p Z_{cp}}{R_p + Z_{cp}}, \tag{D.2}$$

where the impedance of the capacitor is

$$Z_{cp} = \frac{1}{i\omega C_p}. \tag{D.3}$$

Equivalent voltage sources and impedances may be determined for the other resistor/voltage source/capacitor combinations in Fig. D.1 to obtain,

$$V_{em} = \frac{V_{tm}}{1 + i\omega C_m R_m} \quad Z_m = \frac{R_m Z_{cm}}{R_m + Z_{cm}}, \tag{D.4}$$

$$Z_{cm} = \frac{1}{i\omega C_m}, \tag{D.5}$$

and

$$V_{ef} = \frac{V_{tf}}{1 + i\omega C_f R_f} \quad Z_f = \frac{R_f Z_{cf}}{R_f + Z_{cf}}. \tag{D.6}$$

$$Z_{cf} = \frac{1}{i\omega C_f} \tag{D.7}$$

Note that while the equivalent thermal noise of each resistor, as given in Eq. (D.1), increases with the resistance, the parallel capacitor causes the Thevenin equivalent

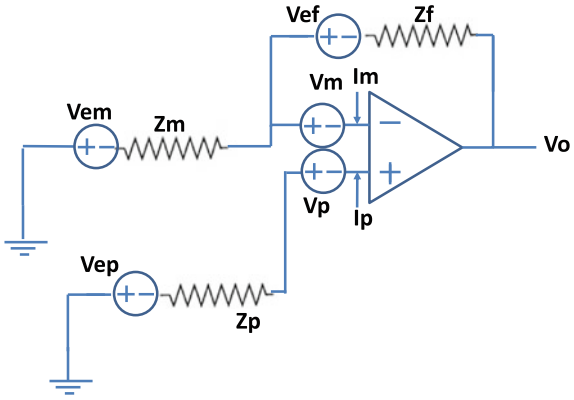


Fig. D.3 Circuit of Fig. D.1 simplified using Thevenin equivalent noise voltage sources and impedances

voltages in Eqs. (D.2), (D.4), and (D.6) to reduce as the resistance gets large. Substituting the first of Eq. (D.1) into Eq. (D.2) and assuming that R_p is large gives

$$V_{ep} = \frac{V_{tp}}{1 + i\omega C_p R_p} \approx \frac{\sqrt{4K_B T R_p}}{i\omega C_p R_p} = \frac{\sqrt{4K_B T}}{i\omega C_p \sqrt{R_p}}. \tag{D.8}$$

Equation (D.8) shows that the Thevenin equivalent voltage is reduced as R_p becomes large. The influence of each resistor decreases as its value increases.

By using Eqs. (D.2) through (D.7), the circuit of Fig. D.1 may be simplified to that shown in Fig. D.3.

Equations (D.2) through (D.8) provide the Thevenin equivalent voltages that would result from the thermal noise voltages due to the resistors in the circuit. These thermal noise voltages are weakly stationary, white random processes with root power spectra given in Eq. (D.1). We would like to express the power spectral density of the random output voltage of the entire circuit. Because the noise voltages due to the resistors and the amplifier voltage and current noise are all uncorrelated, the power spectral density of the total output noise is obtained by adding the power spectra of the individual noise sources. The resulting power spectral density, e_{no}^2, of the output voltage, V_o, may be shown to be

$$\begin{aligned} e_{no}^2 = &\left|1 + \frac{Z_f}{Z_m}\right|^2 (e_v^2 + e_{ep}^2 + e_i^2 |Z_p|^2) \\ &+ \left|\frac{Z_f}{Z_m}\right|^2 e_{em}^2 + e_{ef}^2 + e_i^2 |Z_f|^2. \end{aligned} \tag{D.9}$$

This result is a modification to Eq. (3–6) of Motchenbacher and Connelly[2]. Combining the above expressions gives the noise power spectral density in terms of the circuit components of Fig. D.1 and the given input voltage and current noise power spectral densities, e_v^2 and e_i^2,

$$\begin{aligned} e_{no}^2 = & \left(\frac{(R_m + R_f)^2 + (R_m R_f \omega (C_f + C_m))^2}{R_m^2 + (R_f R_m \omega C_f)^2} \right) \\ & \times \left(e_v^2 + \frac{4K_B T R_p}{1 + (R_p \omega C_p)^2} + e_i^2 \frac{R_p^2}{1 + (R_p \omega C_m)^2} \right) \\ & + \left(\frac{R_f}{R_m} \right)^2 \times \frac{1 + (R_m \omega C_m)^2}{1 + (R_f \omega C_f)^2} \times \frac{4K_B T R_m}{1 + (R_m \omega C_m)^2} \\ & + \frac{4K_B T R_f}{1 + (R_f \omega C_f)^2} + e_i^2 \frac{R_f^2}{1 + (R_f \omega C_f)^2}. \end{aligned} \tag{D.10}$$

D.2 Noise Prediction for a Specific Design

The circuit of Fig. D.1 has been used to construct a preamplifier for a microphone by configuring it to operate as a charge amplifier. The positive amplifier input has been shorted to ground by setting $R_p = 0$. The feedback has been set so that the feedback capacitor is on the order of the capacitance of the microphone, $C_f = 1.0 \times 10^{-12}$ farad. The feedback resistor is needed to stabilize the DC gain and has been taken to be $R_f = 10 \times 10^9$ ohms. The design parameters are listed in Table D.2.

The predicted output noise spectrum (e_{no}) as predicted by Eq. (D.9) is shown in Fig. D.4 along with the results of two measurements. One set of measured data (shown in red) was obtained for the circuit without the microphone (C_m) connected. There will always be some capacitance at the negative input to the amplifier due either to the input capacitance of the amplifier or to the parasitic capacitance of the circuit board. The output noise was also measured with the microphone chip connected. As

Table D.2 Design parameters

R_m	1×10^{12}	ohms
C_m	2.7×10^{-12}	farad
R_p	0	ohm
C_p	1.0×10^{-12}	farad
R_f	10×10^9	ohms
C_f	1.0×10^{-12}	farad
V_n	5.0×10^{-9}	volt
I_n	1.3×10^{-15}	amp

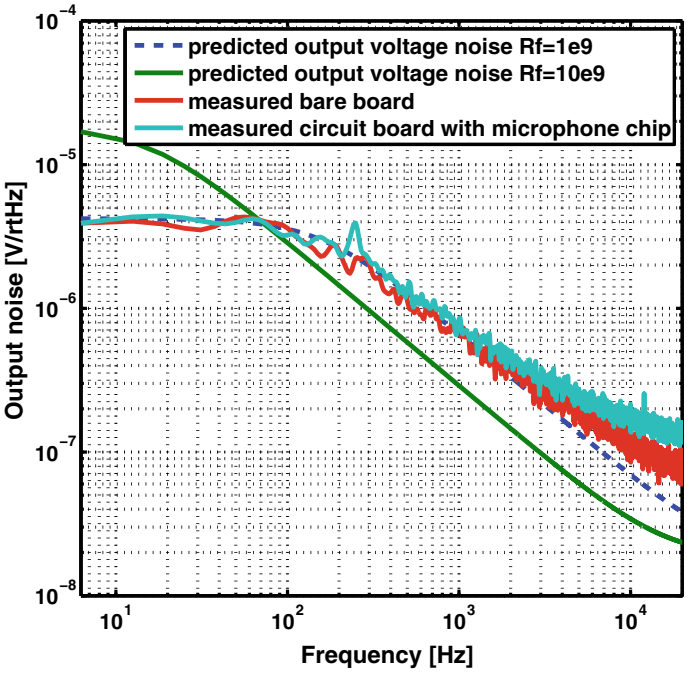


Fig. D.4 The predicted circuit noise is in close agreement with measurements. The use of a feedback resistor with resistance $R_f = 10 \times 10^9$ ohms shows lower noise at high frequencies than when a resistance $R_f = 1 \times 10^9$ ohms is used

shown in the figure, the presence of the microphone affected the output noise only at frequencies above about 2 kHz.

Predictions of the circuit noise are shown for two values of the feedback resistor, R_f. One is the actual value used in the circuit design, $R_f = 10 \times 10^9$ ohms and the other corresponds to $R_f = 1 \times 10^9$ ohms. The two curves indicate that the value of this resistor has a significant influence on the output noise, as is expected. However, it is clear that the prediction with the lower value of R_f than was actually used in the circuit gives much better agreement with measurements. This may be because of leakage resistance in the circuit board, which, being made of fiberglass, was not composed of the appropriate materials for high impedance circuits. High impedance circuits are normally constructed on ceramic substrates which have much higher resistance than fiberglass. Ceramic substrates are, unfortunately, much more expensive than the circuit boards used in this study.

Appendix E
Some Useful Formulas

Equation (2.5.3):

$$
\begin{aligned}
&< I(t) >= \frac{1}{T}\int_0^T p\cos(\omega t+\phi)u\cos(\omega t+\psi)dt \\
&= \frac{1}{T}\int_0^T p\cos(\omega t+\phi)u\cos(\omega t+\phi+\beta)dt \\
&= \frac{1}{T}\int_0^T p\cos(\omega t+\phi) \\
&u(\cos(\omega t+\phi)\cos(\beta)-\sin(\omega t+\phi)\sin(\beta))dt \\
&= \frac{1}{T}\int_0^T pu\cos^2(\omega t+\phi)\cos(\beta)dt \\
&= \frac{pu\cos(\beta)}{2} = \frac{pu\cos(\psi-\phi)}{2} = \frac{pu\cos(\phi-\psi)}{2},
\end{aligned}
\tag{E.1}
$$

where the period of the oscillation is $T = 2\pi/\omega$. This also leads to

$$
< P^2(t) >= \frac{1}{T}\int_0^T p^2\cos^2(\omega t+\phi)dt = \frac{p^2}{2}. \tag{E.2}
$$

Equation (1.7.5):

$$
G_{pp}(f) = 4\pi S_{pp}(2\pi f). \tag{E.3}
$$

Equation (1.4.9):

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R. N. Miles, *Physical Approach to Engineering Acoustics*, Mechanical Engineering Series, https://doi.org/10.1007/978-3-031-33009-4

$$SPL_A = 10\log_{10}(G_{pp}\int_0^{\infty}\frac{1}{P_{ref}^2}H(f)H^*(f)df)$$
$$\approx 10\log_{10}(G_{pp}) + 135.2423. \quad \text{(E.4)}$$

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Index

A
A-Weighted SPL, 3, 5, 33, 34
Absolute error, 341, 342
Absorption coefficient, 188, 190–192
Acceleration sensitivity, 265
Acoustic impedance, 44, 48, 50, 51
Acoustic power, 188, 192
Air density, 37–39, 42, 47, 48, 50, 51
Air gap, 66, 68–71, 73, 79–81
Air molecules, 249–251, 258
Arbitrary waves, 45, 47, 51
Atmospheric pressure, 197, 240
Autocorrelation, 359, 360, 377
Autocorrelation function, 5–9, 20, 311

B
Back volume, 237–240, 243, 244, 246–248, 250–252, 254, 291, 292
Bias voltage, 293, 295, 300, 302, 304–306, 312
Bistable electrostatic pendulum, 333
Boltzmann's constant, 249, 308, 381
Boundary element, 141, 168
Boundary element solution, 182
Broadband random process, 367
Buffer amplifier, 295, 296, 307, 312

C
Capacitance, 315, 318, 325–329, 334
Capacitance matrix, 317, 320, 321
Cardioid directivity, 268, 269
Central limit theorem, 366
Characteristic function, 364, 365
Charge, 315–323, 325, 329, 330, 333–335
Charge amplifier, 298, 299, 302, 303, 308, 309, 326, 333, 334
Charge density, 321
Circuit noise, 386
Coincidence, 57, 60, 61, 81
Collapse voltage, 304
Computer Aided Design (CAD), 161, 165
Conditional probability, 369, 370
Conductor, 315, 318
Conical duct, 94
Constitutive equation, 193, 195
Convolution integral, 19, 22, 23, 25
Couette ow, 198
Coulombs's law, 315
Covariance matrix, 369
Co-vibrating mass, 247
Critically damped, 375
Cut-off frequency, 242, 247, 248

D
D'Alembert, 44
D'Alembert solution, 186
Dashpot constant, 245, 257
dBA, 3–5, 7, 13, 33, 34
Decaying sound fields, 190
Differential amplifier, 333
Dirac delta function, 142, 148, 153, 181
Directional microphone noise, 279
Directivity factor, 273, 274, 278, 284, 285
Directivity index, 272, 274, 275, 278, 282, 283, 292
Double wall resonance, 74
Double walls, 70, 71, 74, 76, 79–81
Ducts, 83, 86–88, 92, 94–96, 98, 100–102
Dynamic viscosity, 193, 198, 211, 245, 292

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E
Effect of boundary impedance on sound field, 151
Effect of reflection on frequency response, 118
Eigenfunctions, 175, 176, 179, 180
Electric potential, 315–318, 321
Electrical sensitivity, 293–295, 310, 314
Electrodynamic microphones, 312, 313
Electrostatic force, 315, 317, 321–324, 331, 332, 337
Electrostatic stiffness, 331, 332, 336
Energy dissipation, 357
Equation of state, 35, 38, 49, 194, 195, 238
Equipartition theorem, 249, 252, 255
Equivalent Input Noise (EIN), 251, 252, 281
Equivalent pressure due to acceleration, 266
Equivalent stiffness, 303–306
Ergodicity, 360
Euler equation, 37, 39, 40, 43, 47
Expansion muffler, 83, 84, 86, 101
Expected value, 5, 7, 8, 12, 19, 20, 360, 362–364, 377
Exponential horn, 98–100

F
Fast Fourier Transform (FFT), 7, 9, 11, 12, 15, 21, 23–30
FFT algorithm, 350
Fiberglass, 71, 72, 76–78, 81, 82
Field incidence, 61, 62
Field incidence transmission loss, 61, 62
Finite Fourier transform, 7–10, 12, 34
First-order array, 266, 270, 279
Flow through a slit, 198
Fluctuating air density, 238
Force due to constant charge, 300
Fourier transform, 5–12, 15, 19–29, 34
Fredholm integral equation, 149
Functions of random variables, 370

G
Gamma function, 366
Gaussian probability density, 365
Gaussian white noise, 350, 381
Gauss' theorem, 143, 146, 148, 194
Generalized coordinate, 322, 323
Green's function for a half-space, 144
Green's functions, 141, 142, 144–147, 149, 159, 160
Ground potential, 318

H
Hamilton's principle, 295, 297, 300, 301, 303, 322, 323
Harmonic sound fields, 242
Harmonic waves, 105, 120
Helmholtz equation, 175
Helmholtz resonator, 86–89, 101, 243, 244, 248
High-pass filter, 242

I
Ideal diaphragm properties, 244
Ideal microphone thermal noise, 252
Image source- sound reflection from a wall, 114
Impulse response, 376, 378
Independence, 360, 366, 369, 370
Infinite tube, 47, 48
Instantaneous sound intensity, 41
Integral equation, 318

J
Joint probability density, 370

K
Kinematic viscosity, 194
Knowles microphone, 348

L
Laplace transform, 375, 376, 378
Laplace's equation, 160
Least squares, 15–18, 339, 341–343, 346
Loudspeaker in a vented box, 131, 136
Loudspeaker-simple model, 109, 132

M
Marginal probability density, 368
Mass conservation, 195, 196, 200, 203
Mass law, 53, 55, 74
Mean square, 359, 363, 377
Mean square response, 250, 251, 255, 256, 259–261, 311, 363, 375, 377, 378
Measure particle velocity using pressure microphones, 251, 285, 287
Mechanical sensitivity, 237–240, 242–244, 247, 248, 271, 292
Microphone diaphragm, 237–240, 242, 246, 247, 249, 252, 262, 263, 265, 287, 291, 292

Microphone directivity pattern, 264, 276, 279, 287, 292
Microphone package, 243, 265
Microphone sensitivity, 237, 244, 248
Modal approach, 180, 182
Momentum conservation, 194, 196, 197, 218
Mufflers, 83–86, 88, 89, 91, 101, 102

N
Narrowband levels, 34
Narrowband random process, 367
Navier–Stokes equations, 193
Newton's second law, 35–37
Nominal capacitance, 294, 295, 298, 299, 301
Normal incidence, 56, 58, 61, 62, 81, 82
Numerical differentiation, 26, 27
Numerical integration, 27, 33

O
One dimensional waves, 41, 47, 48
One-third octave band SPL, 3, 5, 13
Overdamped, 375

P
P_{ref}, 1–3, 14, 31–33
Parallel electrode, 301
Parallel plate, 315, 325–328, 333, 334, 337
Parasitic capacitance, 295, 298, 299, 307
Pass-band sensitivity, 243
Permittivity of free space, 315, 335
Poiseuille flow, 198
Polysilicon, 246, 247
Porous layers, 71, 76
Potential energy, 316, 317, 321–325, 327, 329, 330, 332
Power spectral density, 5–7, 9, 10, 12–14, 18–20, 33, 34, 250–253, 255–262, 281, 288, 289, 307, 311, 312
Pressure gradient, 266, 268, 273, 274, 285–288
Pressure microphone, 197
Probability, 361–373
Probability density function, 362
Probability distribution function, 361, 370
Pulsating sphere, 107, 108, 121–123, 136

R
Radiation impedance, 111–113
Radiation load, 246
Random incidence, 56, 61, 62, 80–82
Random incidence transmission loss, 56, 61, 80, 81
Random pressure, 250
Readout circuit, 334
Rectangular box, 175
Reference intensity, 42
Relative error, 339, 341
Residue theorem, 378
Ribbon microphone, 270, 287

S
Second-order array, 266
Second-order directional microphones, 269
Separation of variables, 175, 176
Side-branch muffler, 88, 101
Simple source, 107, 111, 121, 137
Singular integrals, 162
Slow Fourier transform, 15
Softening electrode, 325
Sound absorption coefficient, 41–44, 50
Sound energy density for a diffuse field, 185, 187
Sound energy density for a plane wave, 186
Sound field due to point source in an enclosure, 152
Sound field in a rectangular tube, 163
Sound field inside a sphere, 166
Sound in a tube, 51
Sound intensity, 185, 187–189
Sound intensity level, 42
Sound Pressure Level (SPL), 1–3, 5, 7, 31–34, 185, 188–192
Sound radiation from a dipole, 114
Sound radiation from a line source, 114
Sound radiation from a sphere, 167
Sound radiation from a surface, 137
Sound radiation from a vibrating piston, 124
Sound radiation from multiple pistons, 129
Sound reverberation time, 190
Sound speed, 239, 281
Sound transmission loss, 53, 55, 56, 61, 62, 71, 79, 80, 82
Spherical waves, 104, 106
Spring/mass/damper, 26, 325, 355
Squeeze film damping, 199
Stationary process, 360, 372
STL file, 161–163, 320, 321
Stray capacitance, 307

T
Tensor notation, 193, 194, 196

Thermal noise of velocity-driven mass, 251
Thermal-mechanical noise, 249, 252
Thevenin equivalent voltage, 383, 384
Time averaged sound intensity, 41
Time delay, 346, 348
Transfer function, 339, 341, 342, 346–348, 350
Transfer matrices, 53, 62, 63, 65, 66, 69, 70, 72, 73, 76–78, 81
Tube with open ends, 51
Tube with piston at end, 40
Tube with spring/mass/damper at end, 40
Tubes with varying cross sectional area, 93
Two microphone sound intensity measurement, 288

U
Underdamped, 375

V
Vent, 237, 240–242, 244, 245, 247, 248, 251, 252, 265, 292
Vent damping, 242, 244, 245
Viscous damping, 238, 241, 245, 255, 257, 270
Viscous damping coefficient, 198
Voltage amplifier, 334

W
Walls with mechanical coupling, 76
Wave number, 177
Webster horn equation, 93
White noise, 250, 259, 261